Tópicos Especiais sobre Margens Oceânicas

Métodos de analise de isótopos estáveis e aplicações na oceanografia

Prof. Dr. Christian Millo (millo@usp.br)

(Umas das) Áreas de aplicação da geoquímica dos isótopos estáveis:

1. Petrologia
2. Hidrologia
3. Climatologia e Paleoclimatologia
4. Oceanografia e Paleoceanográfia
5. Biologia e Química Orgânica
6. Botânica e Agronomia
7. Bromatologia (Ciências dos Alimentos)
8. Medicina
9. Criminalística

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9. Criminalística

Sumario do curso

1. Analise isotópica de oxigênio em água do mar
2. Analise isotópica de hidrogênio em água do mar
3. Analise isotópica de carbonato de cálcio
4. Analise isotópica de carbono inorgânico dissolvido (DIC)
5. Analise isotópica de material orgânico

Leituras sugeridas:

Sharp (2007) *Principles of Stable Isotope Geochemistry*. Prentice Hall 344 pp.

Hoefs (2009) *Stable Isotope Geochemistry*. Springer.

De Groot (2009) *Handbook of Stable Isotope Analytical Techniques* Vol. II. Elsevier. 1398 pp.

Zeebe & Wolf-Gladrow (2001) *CO_2 in Seawater: Equilibrium, Kinetics, Isotopes*. Elsevier. 360 pp.

Leituras sugeridas:

Finnigan GasBench II *Operating Manual*.

Finnigan Kiel Carbonate Device *Operating Manual*.

isótopo

iso = mesmo

topo = lugar

PERIODIC TABLE OF THE ELEMENTS

Table of Selected Radioactive Isotopes

| | | |

ISÓTOPOS do mesmo elemento:

1) Possuem o mesmo numero de prótons (mesma carga)

então

tem o mesmo comportamento químico (salvo em alguns casos)

2)Possuem numero de nêutrons diferentes (massas diferentes)

então

tem comportamento físico diferente

ISÓTOPOS do mesmo elemento:

1) Possuem o mesmo numero de prótons (mesma a carga)

então

tem o mesmo comportamento químico (salvo em alguns casos)

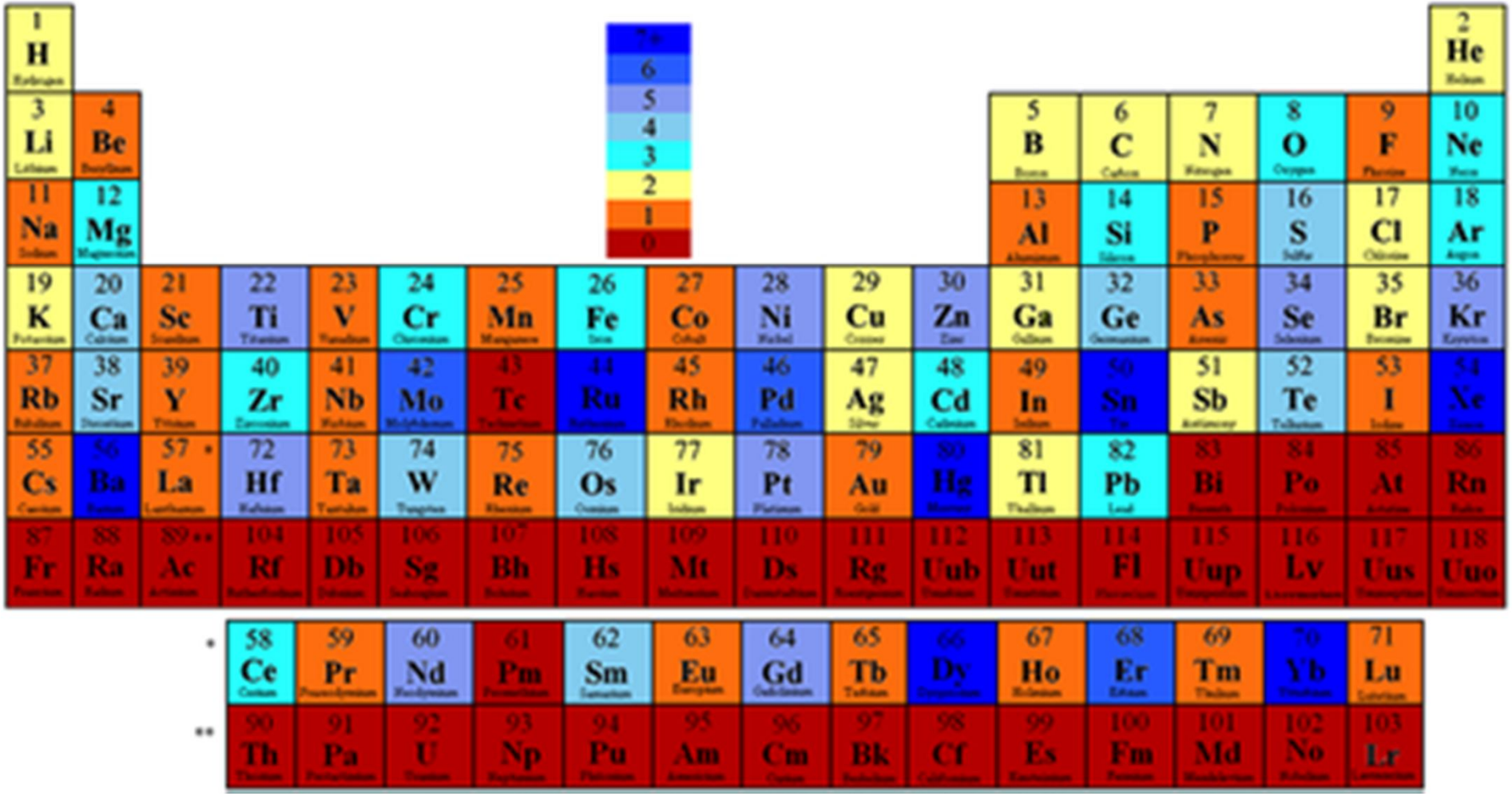
2)Possuem numero de nêutrons diferentes (massas diferentes)

então

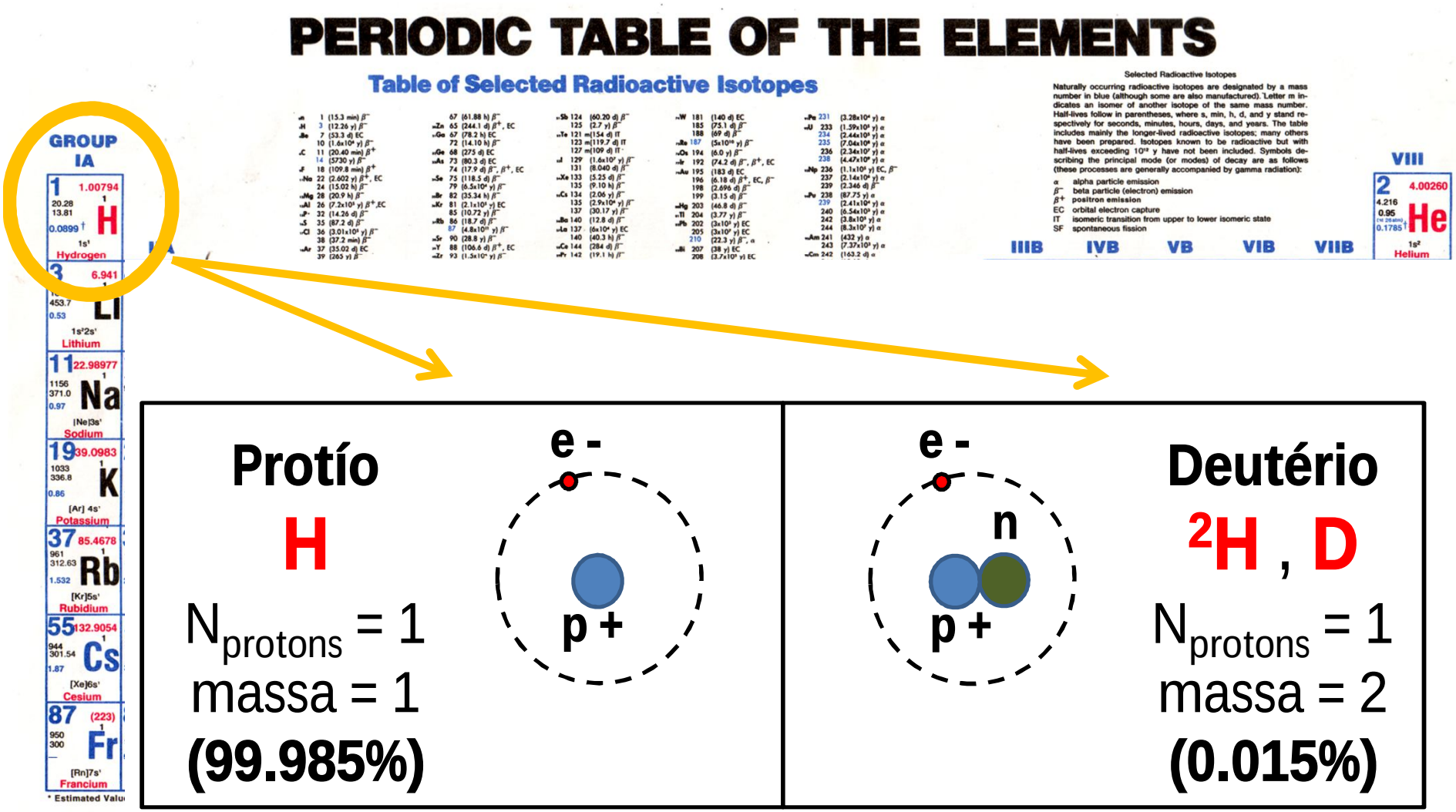
tem comportamento físico diferente

PODEM SER USADOS COMO MARCADORES NATURAIS graça ao processo de fracionamento isotópico

Número de Isótopos Estáveis por Elemento



Fonte: Wikipedia



Déuterio: descoberto em 1934 para Harold Urey (Premio Nobel da química)

H_2O HDO

H_2O

HDO

Isotopicamente leve

Isotopicamente pesada

ISÓTOPOS ESTÁVEIS DO OXIGÊNIO

$N_{prótons} = 8$

^{16}O (99.76%)

^{18}O (0.20%)

^{17}O (0.04%)

H_2O

HDO

$H_2{}^{18}O$

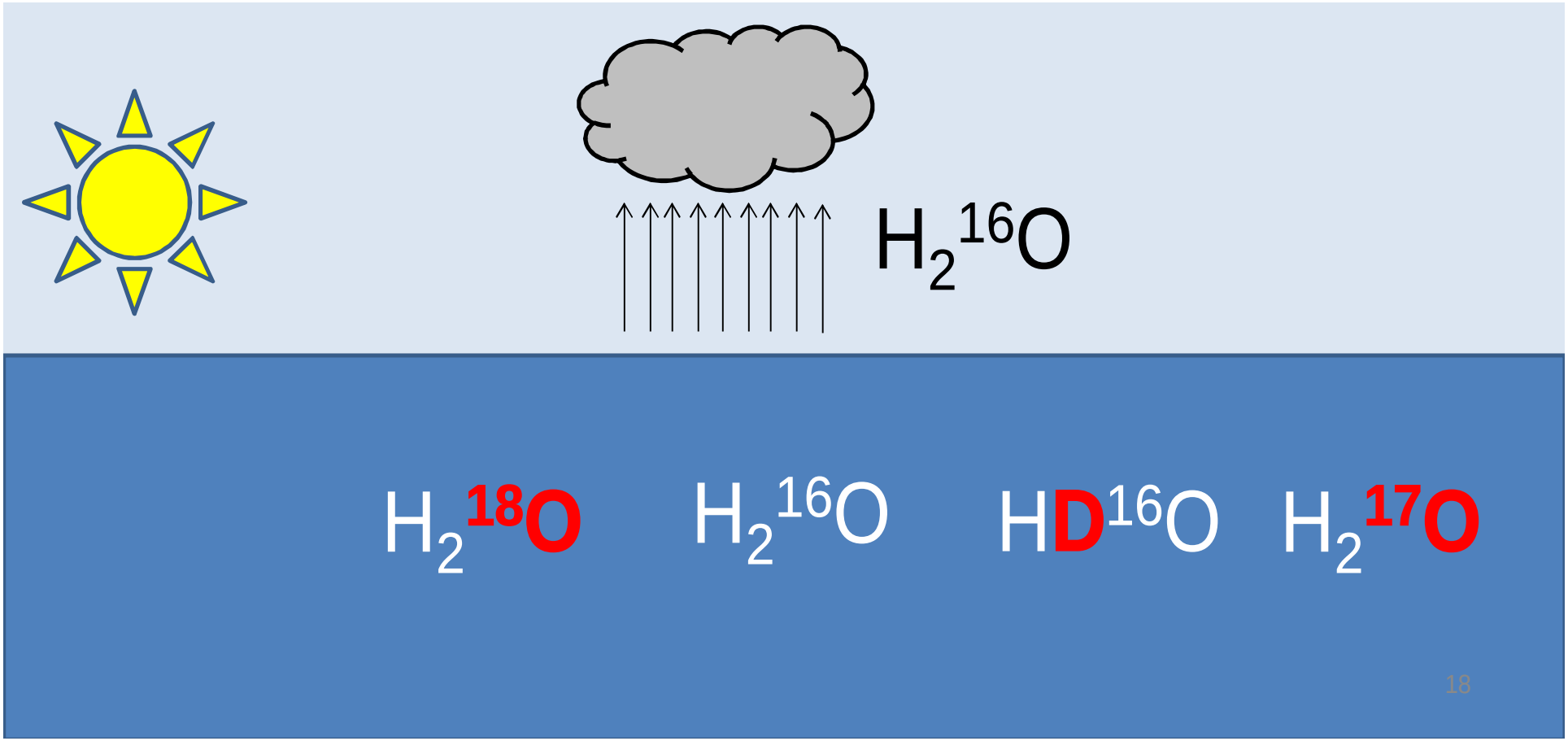
Agua isotopicamente leve e pesada

H_2O	Abundancia (%)
$H_2{}^{16}O$	99.73098
$H_2{}^{18}O$	0.199978
$H_2{}^{17}O$	0.037888
$HD^{16}O$	0.031460
$HD^{18}O$	0.0000006
$HD^{17}O$	0.0000001
$D_2{}^{16}O$	0.00000002
$D_2{}^{17}O$	0.00000000001
$D_2{}^{18}O$	0.000000000005

(Sharp, 2007)

Diferença de massa = comportamento físico diferente

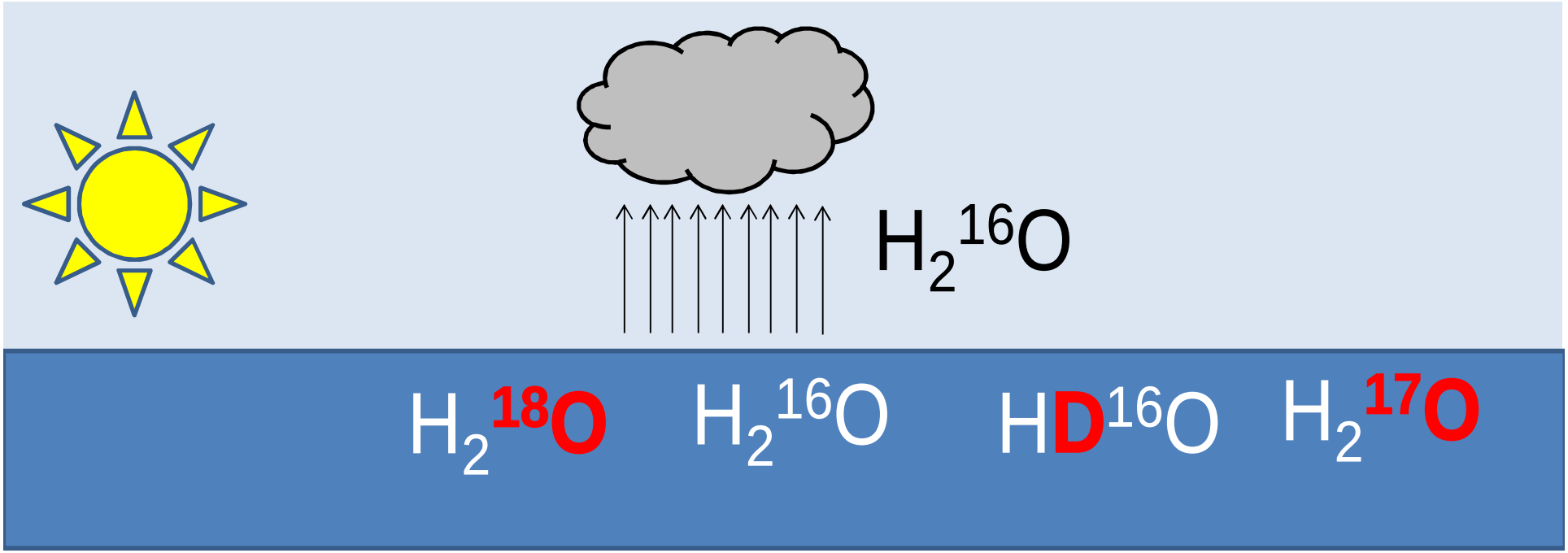
Exemplo: processo de evaporação



Diferença de massa = comportamento físico diferente

Exemplo: processo de evaporação

Fase vapor enriquecida em agua isotopicamente leve



Fase liquida enriquecida em agua isotopicamente pesada

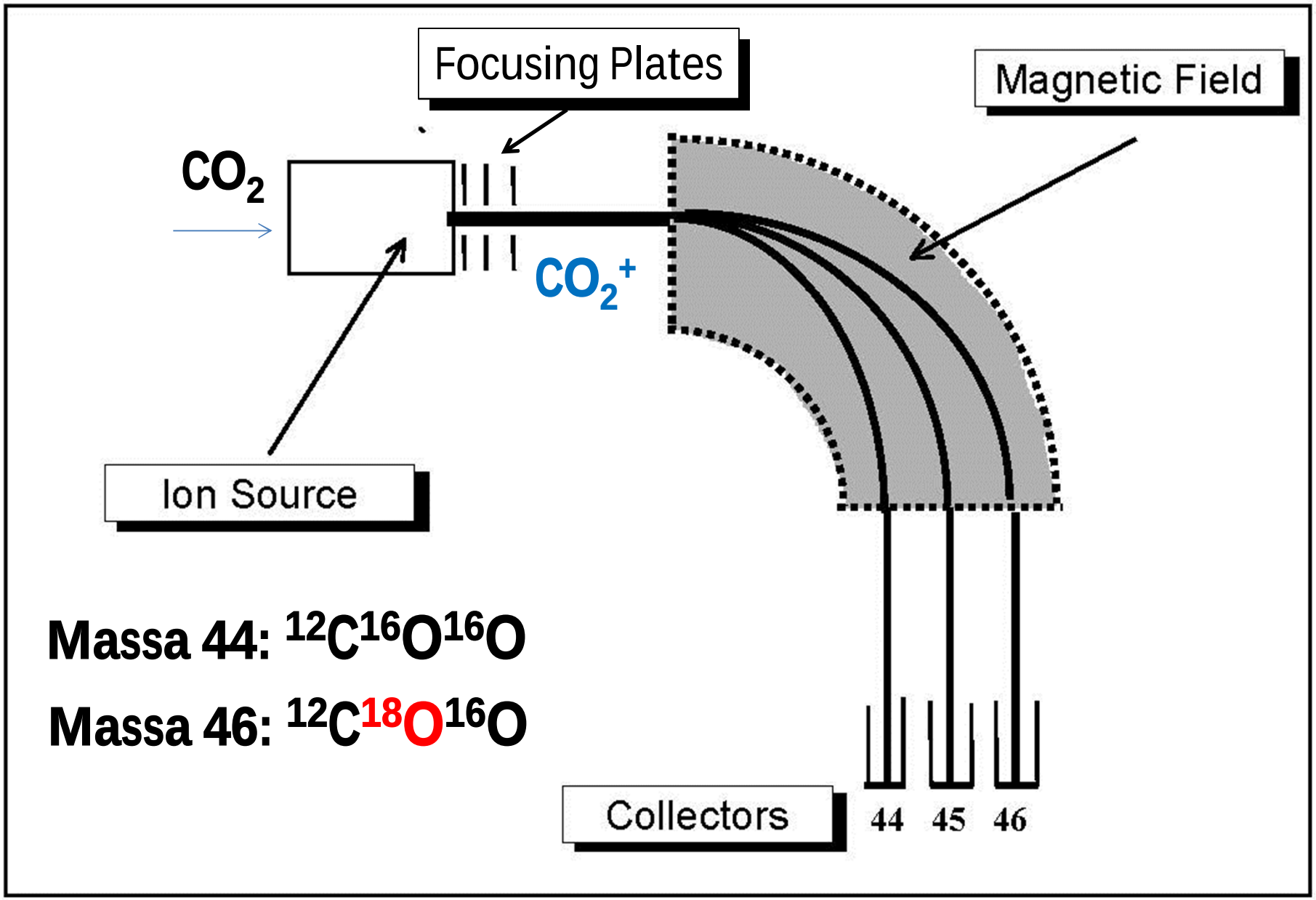
Como se exprime a abundancia de um isótopo em uma amostra

Razão isotópica do ^{18}O

$$^{18}R = \frac{^{18}O}{^{16}O}$$

Espectrometria de massa de fase gasosa

Modified from Analytical Precision AP2003 *Operating Manual*.



Razão isotópica do ^{18}O

$$^{18}R = \frac{^{18}O}{^{16}O}$$

Notação delta

Delta ≡ *a diferença entre a razão isotópica da amostra e a razão isotópica de um padrão internacional, exprimida em por mil.*

$$\delta^{18}O \equiv \frac{\left(\frac{^{18}O}{^{16}O}\right)_{amostra} - \left(\frac{^{18}O}{^{16}O}\right)_{padrão\ internacional}}{\left(\frac{^{18}O}{^{16}O}\right)_{padrão\ internacional}} \times 1000$$

Notação delta

$$\delta^{18}O \equiv \frac{\left(\frac{^{18}O}{^{16}O}\right)_{amostra} - \left(\frac{^{18}O}{^{16}O}\right)_{padrão\ internacional}}{\left(\frac{^{18}O}{^{16}O}\right)_{padrão\ internacional}} \times 1000$$

Na literatura:

$$\delta^{18}O \equiv \left\{ \frac{\left(\frac{^{18}O}{^{16}O}\right)_{amostra}}{\left(\frac{^{18}O}{^{16}O}\right)_{padrão\ internacional}} - 1 \right\} \times 1000$$

Notação delta:
$\delta^{18}O$

$$\frac{\left(\frac{^{18}O}{^{16}O}\right)_{amostra}}{\left(\frac{^{18}O}{^{16}O}\right)_{padrão\ internacional}} = 1 + \delta^{18}O \quad \text{por mil}$$

Craig & Gordon (1965) modificado

- $\delta^{18}O$ = **+**5‰ : a amostra é 5‰ **enriquecida** em ^{18}O em relação ao padrão internacional
- $\delta^{18}O$ = **-**5‰ : a amostra é 5‰ **empobrecida** em ^{18}O em relação ao padrão internacional

- Perda de ^{18}O $\rightarrow$ o valor de $\delta^{18}O$ torna-se menor (e vice-versa)

Teste:

com base na definição

de $\delta^{18}O$, qual é a

definição de δD?

δD

$$\delta D \equiv \frac{\left(\frac{D}{^1H}\right)_{amostra} - \left(\frac{D}{^1H}\right)_{padrão\ internacional}}{\left(\frac{D}{^1H}\right)_{padrão\ internacional}} \times 1000$$

Padrão isotópico internacional
(Agencia Internacional da Energia Atômica IAEA, Viena)

para a composição isotópica de água o padrão é

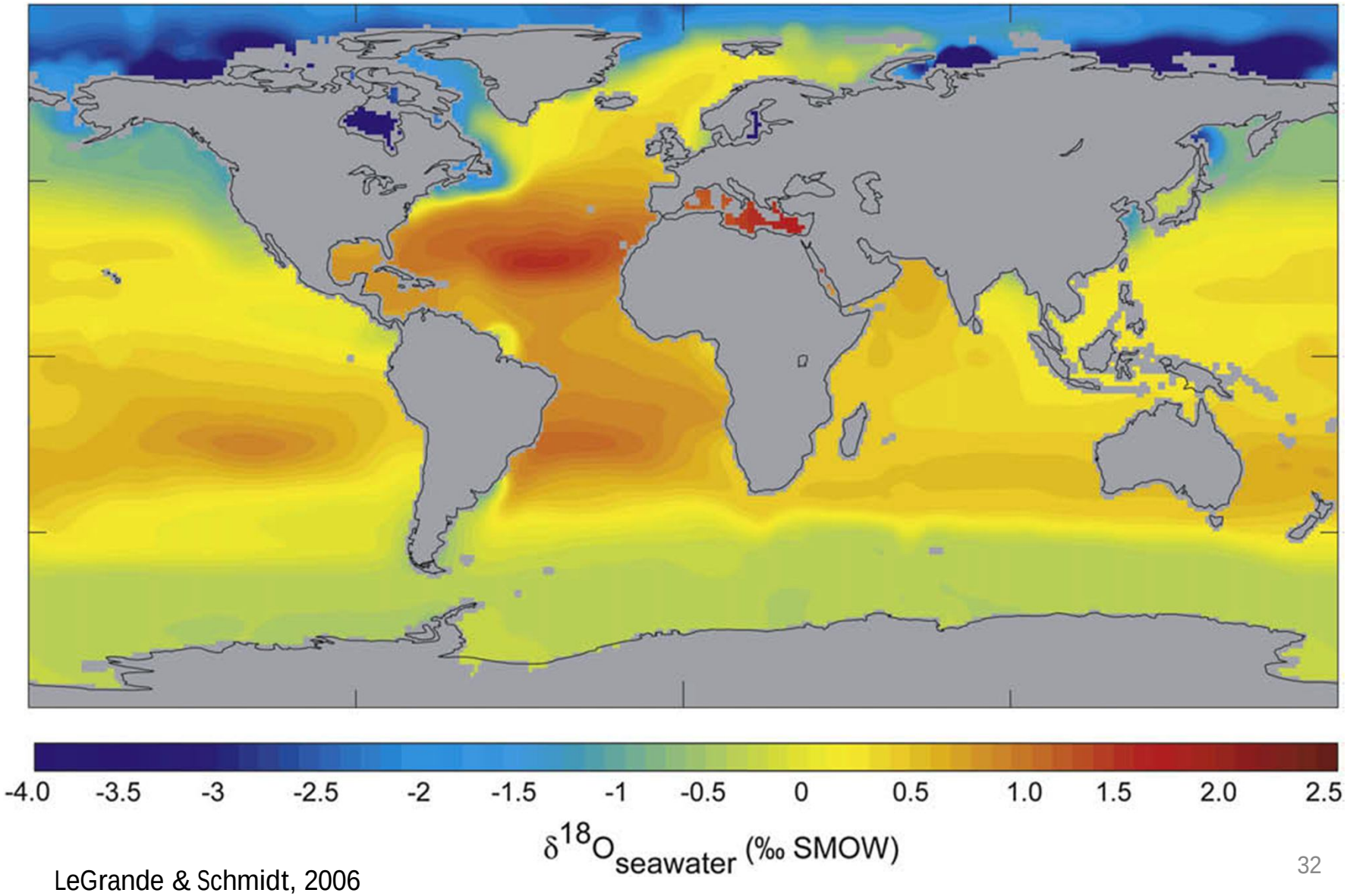
SMOW: Standard Mean Ocean Water (H_2O)

$\delta^{18}O = 0‰$ (por definição)

$\delta D = 0‰$ (por definição)

V-SMOW = Vienna SMOW

$\delta^{18}O$ da superfície do oceano atual



LeGrande & Schmidt, 2006

Teste:

Água 1: $\delta^{18}O$ = - 2,0 ‰

Água 2: $\delta^{18}O$ = + 0,5 ‰

Água 3: $\delta^{18}O$ = 0,0 ‰

qual dessas águas é

isotopicamente mais pesada?

Teste:

Água 1: $\delta^{18}O$ = - 2,0 ‰

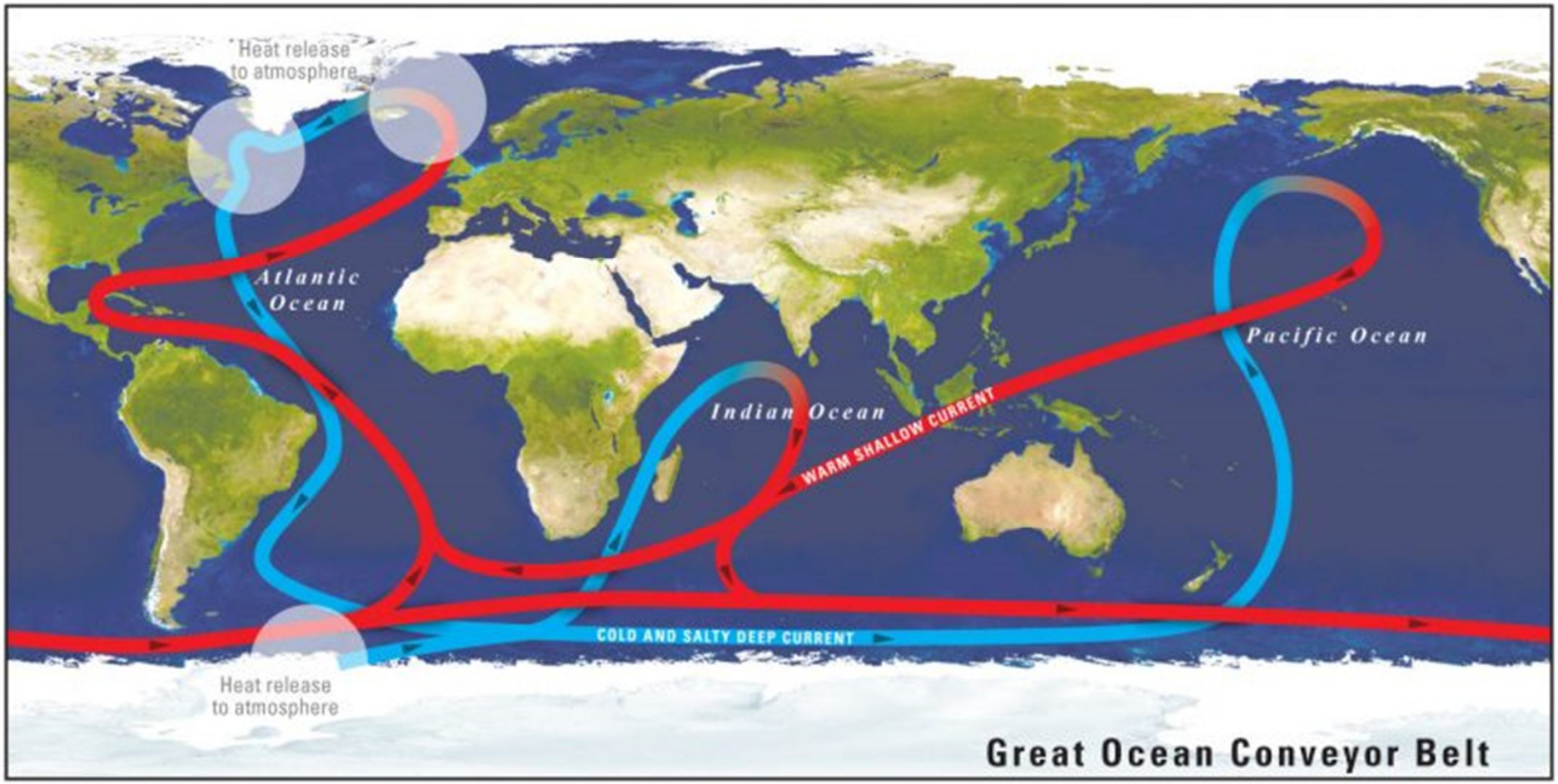
Água 2: $\delta^{18}O$ = **+ 0,5** ‰

Água 3: $\delta^{18}O$ = 0,0 ‰

ÁGUA

Analise isotópica de água e aplicações

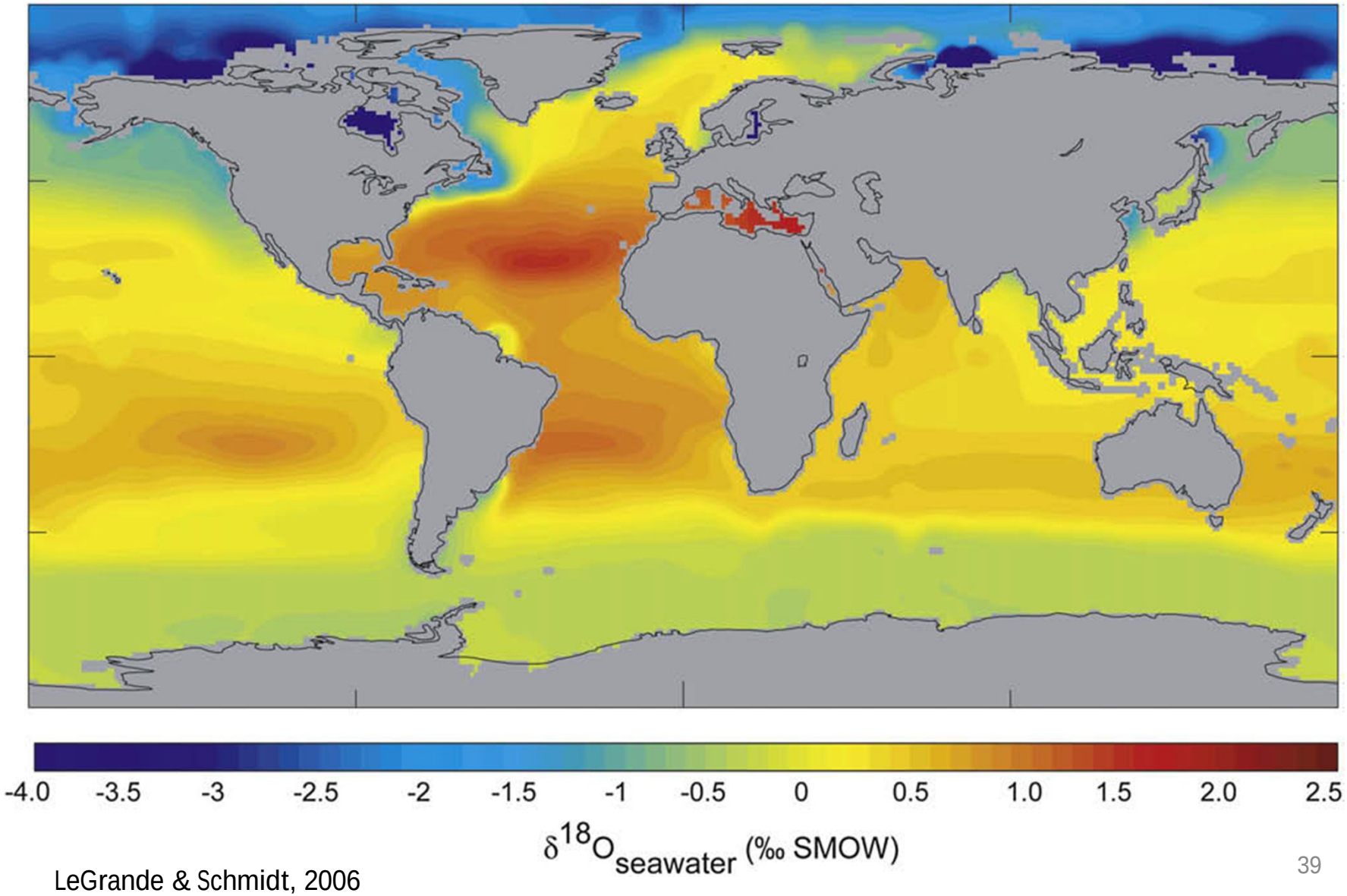
Estudo da circulação oceânica



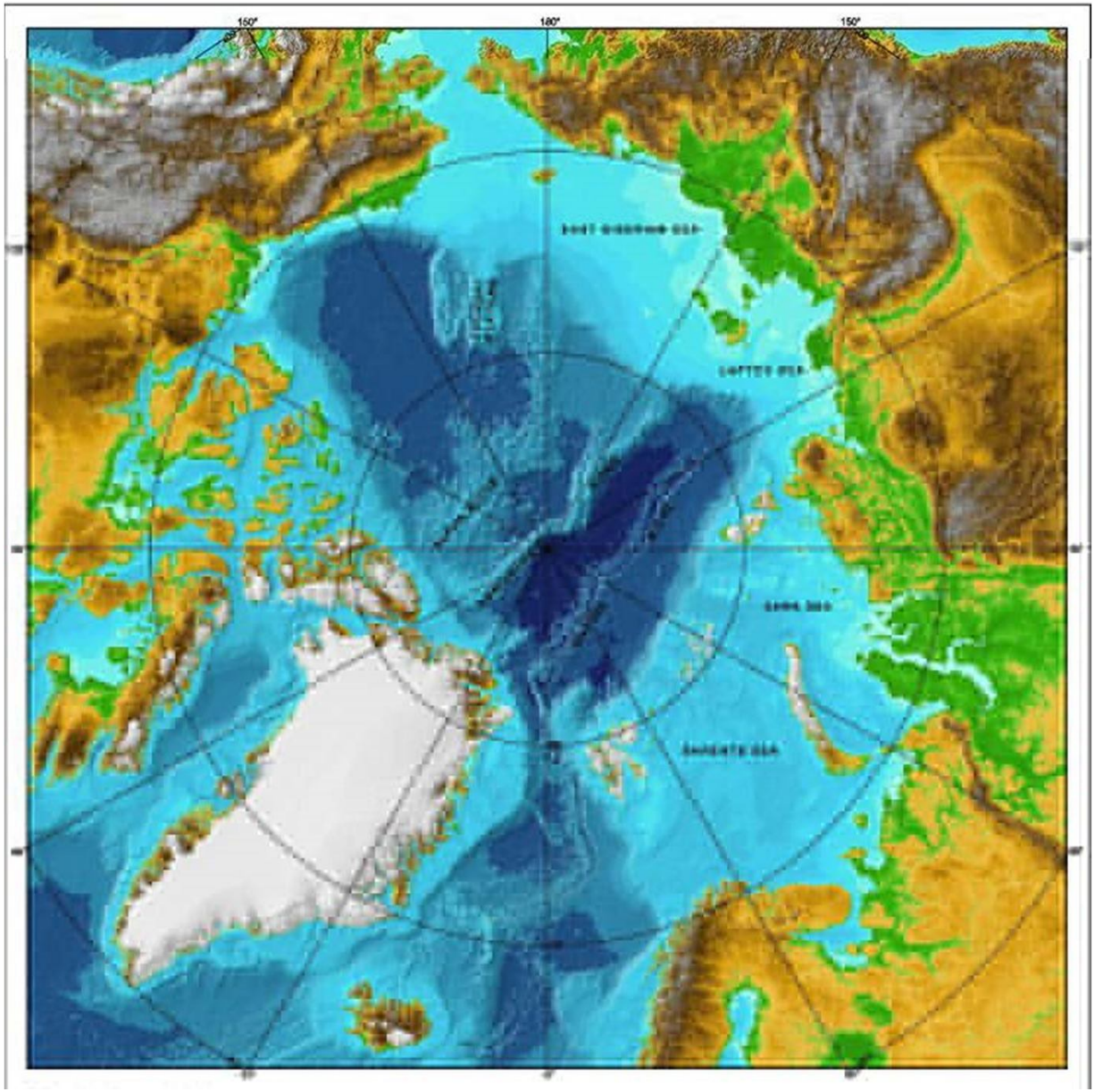
Source: Shireen Gonzaga in EARTH, October 12, 2016
http://earthsky.org/earth/warm-ocean-currents-are-slowing-down

Circulação oceânica superficial

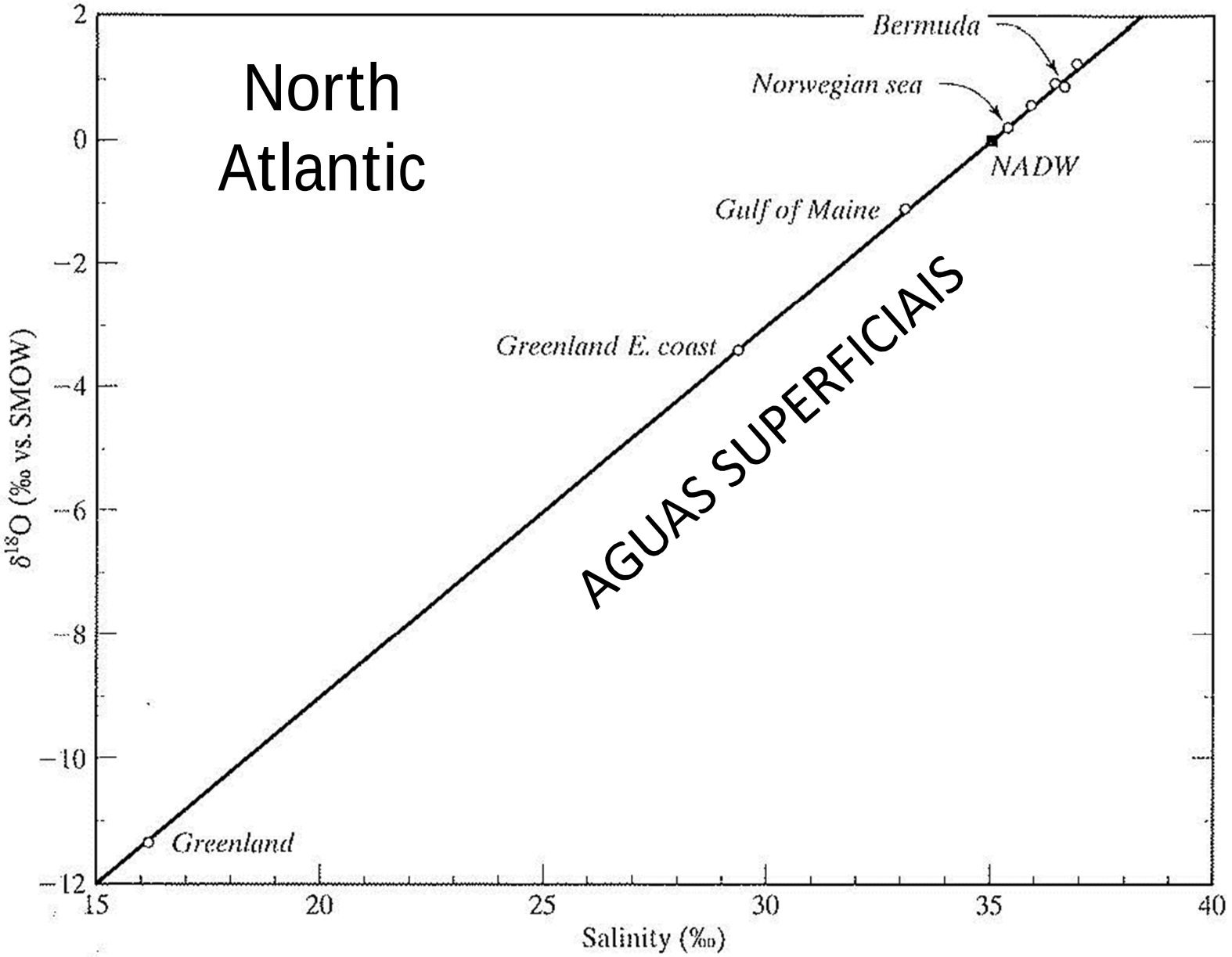
$\delta^{18}O$ da superfície do oceano atual



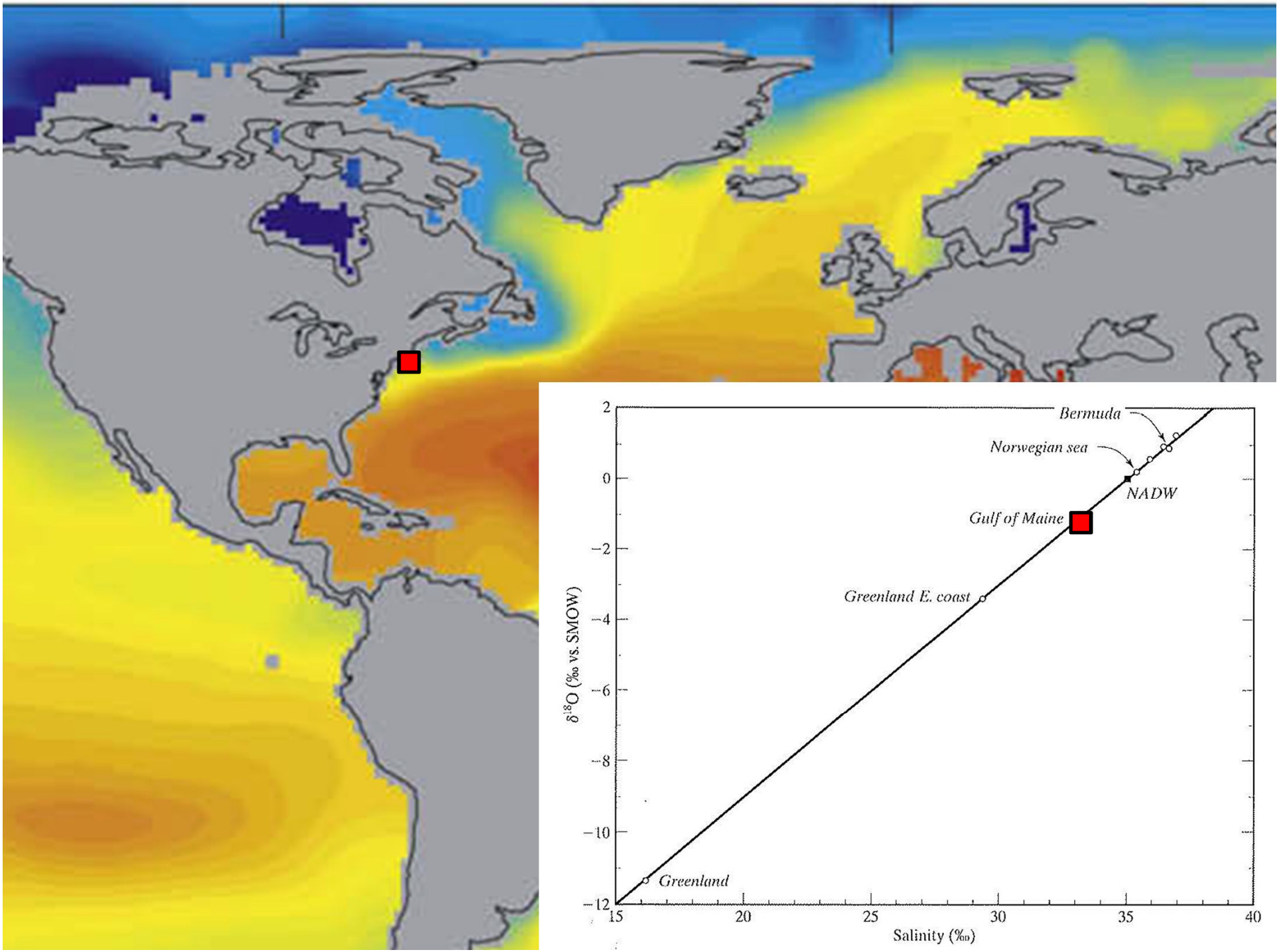
LeGrande & Schmidt, 2006

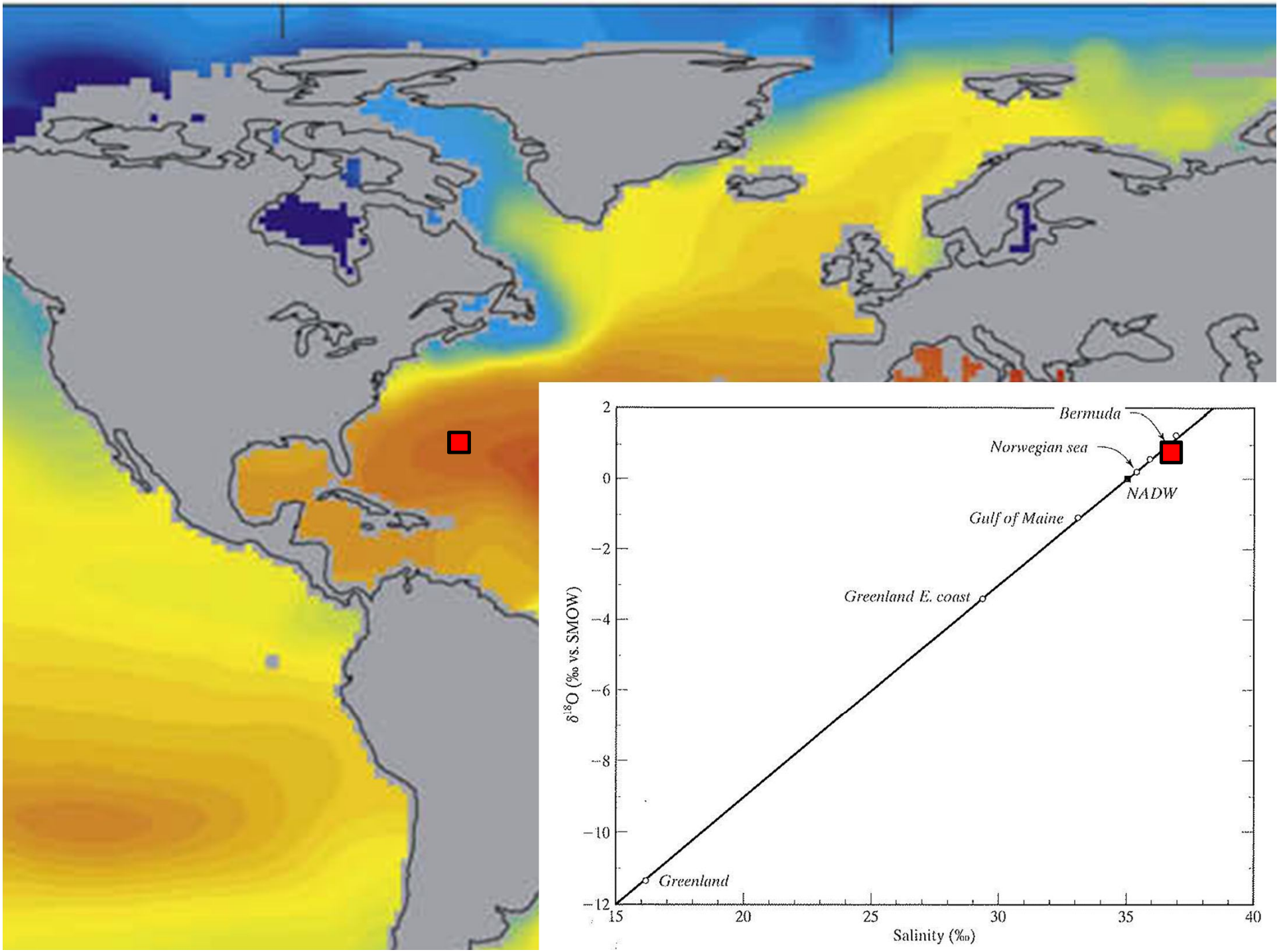

150°
180°
150°

Relação entre $\delta^{18}O$ do oceano e salinidade



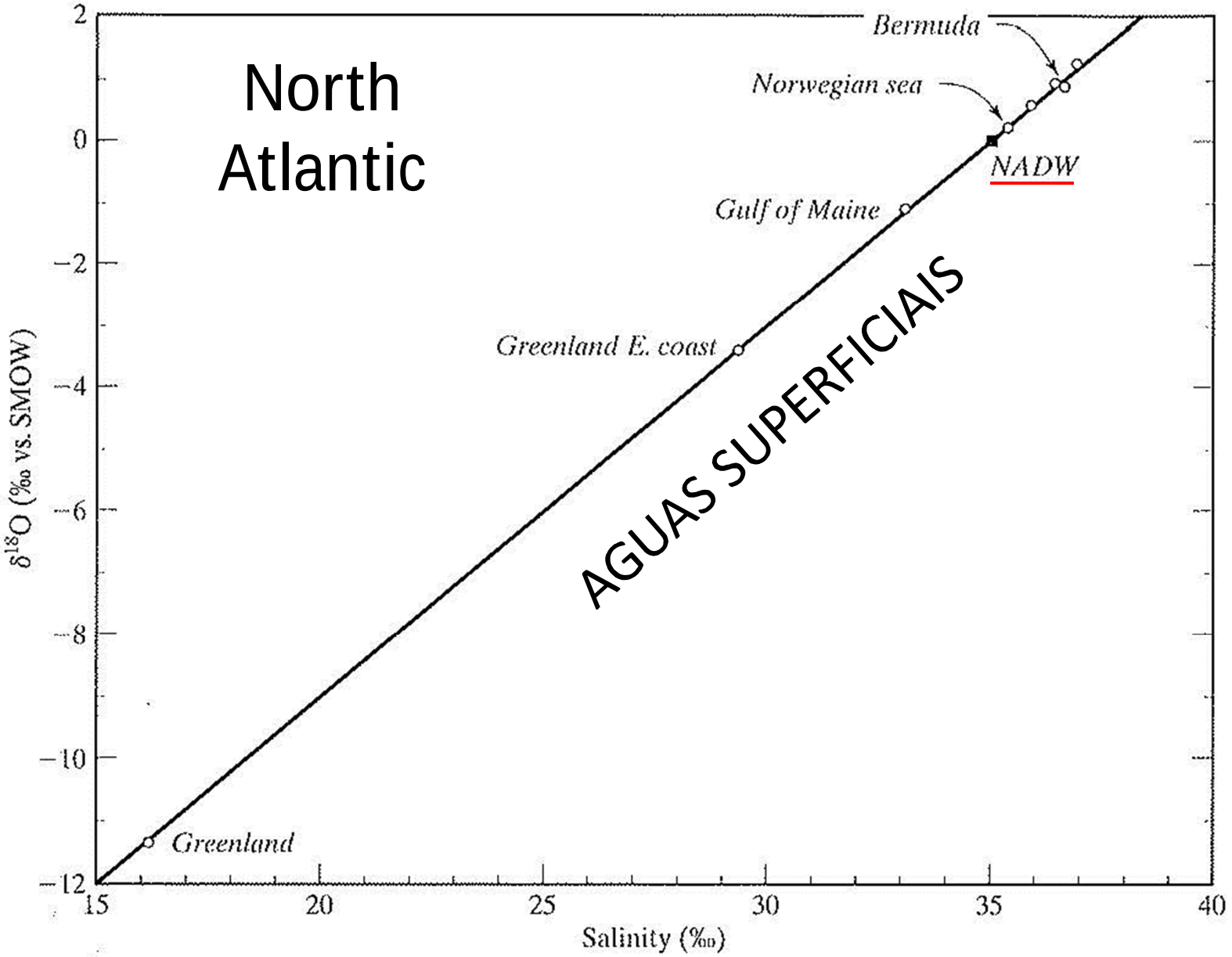
(Craig & Gordon 1965, modified in Sharp, 2007)


Bermuda
Norwegian sea
NADW
Gulf of Maine
Greenland E. coast
Greenland
2
0
−2
−4
−6
−8
−10
−12
15
20
25
30
35
40
$\delta^{18}O$ (‰ vs. SMOW)
Salinity (‰)

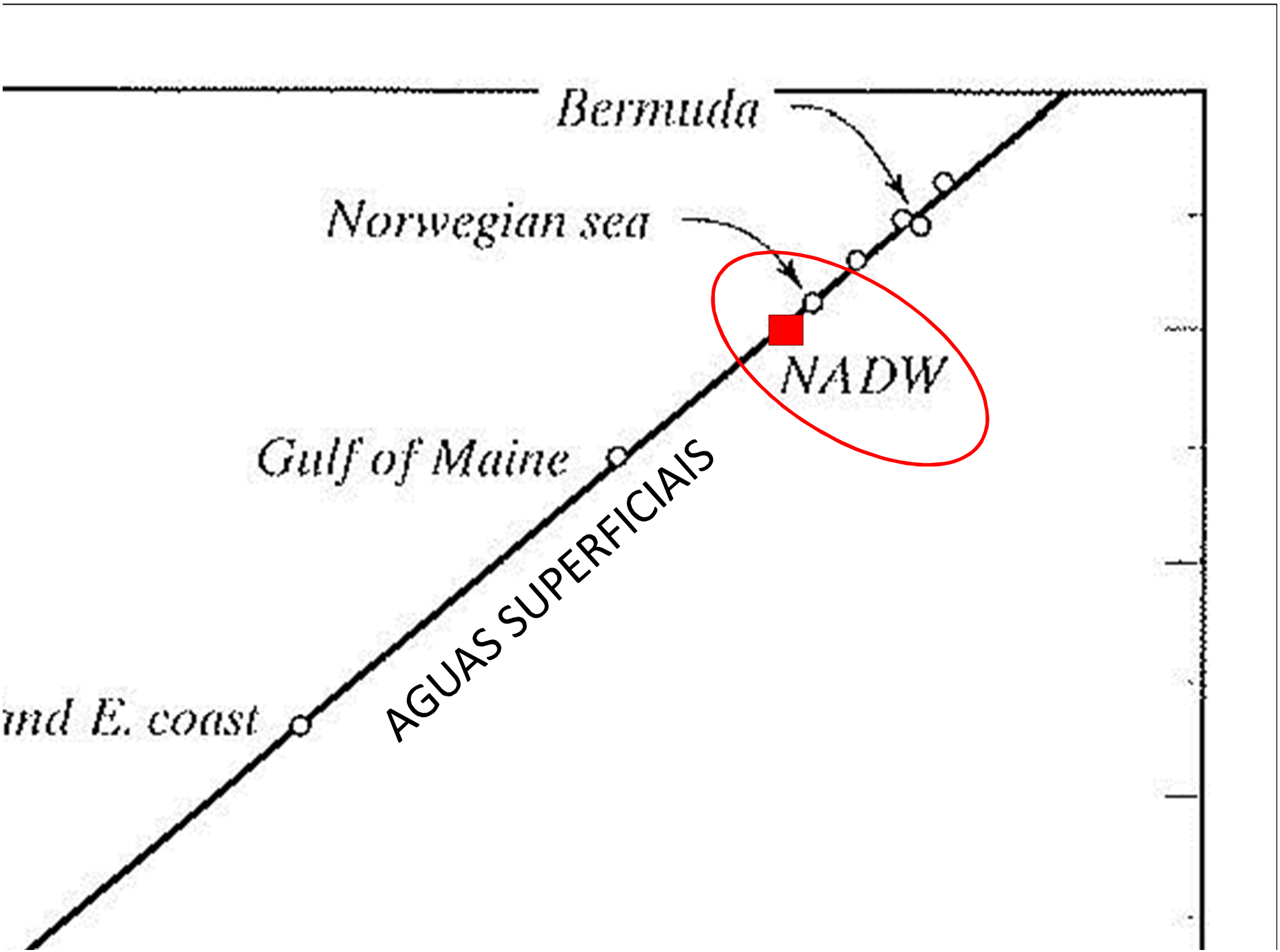

Bermuda
Norwegian sea
NADW
Gulf of Maine
Greenland E. coast
Greenland
2
0
−2
−4
−6
−8
−10
−12
15
20
25
30
35
40
$\delta^{18}O$ (‰ vs. SMOW)
Salinity (‰)

Aplicações no estudo da circulação oceânica profunda

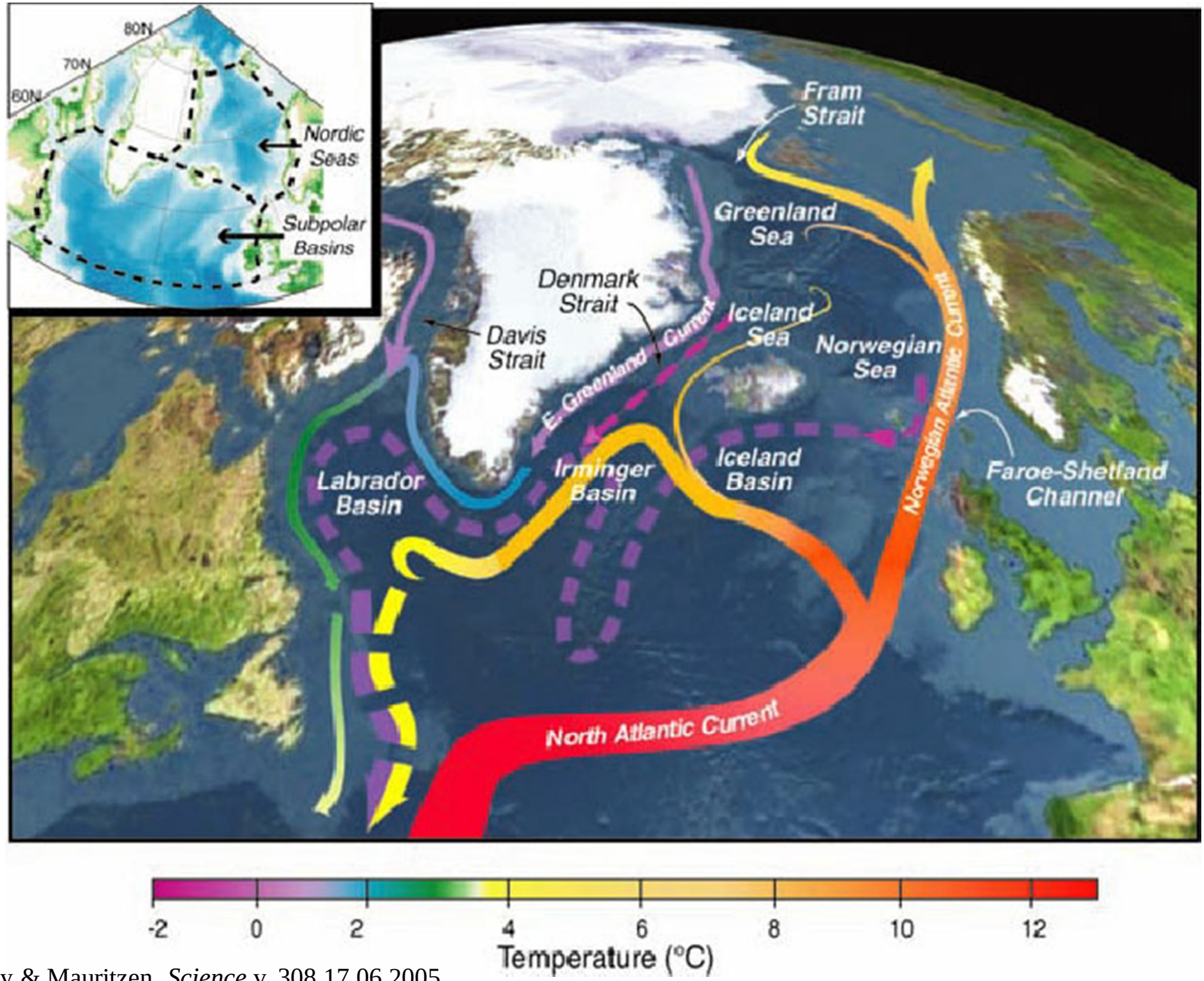
Relação entre $\delta^{18}O$ do oceano e salinidade



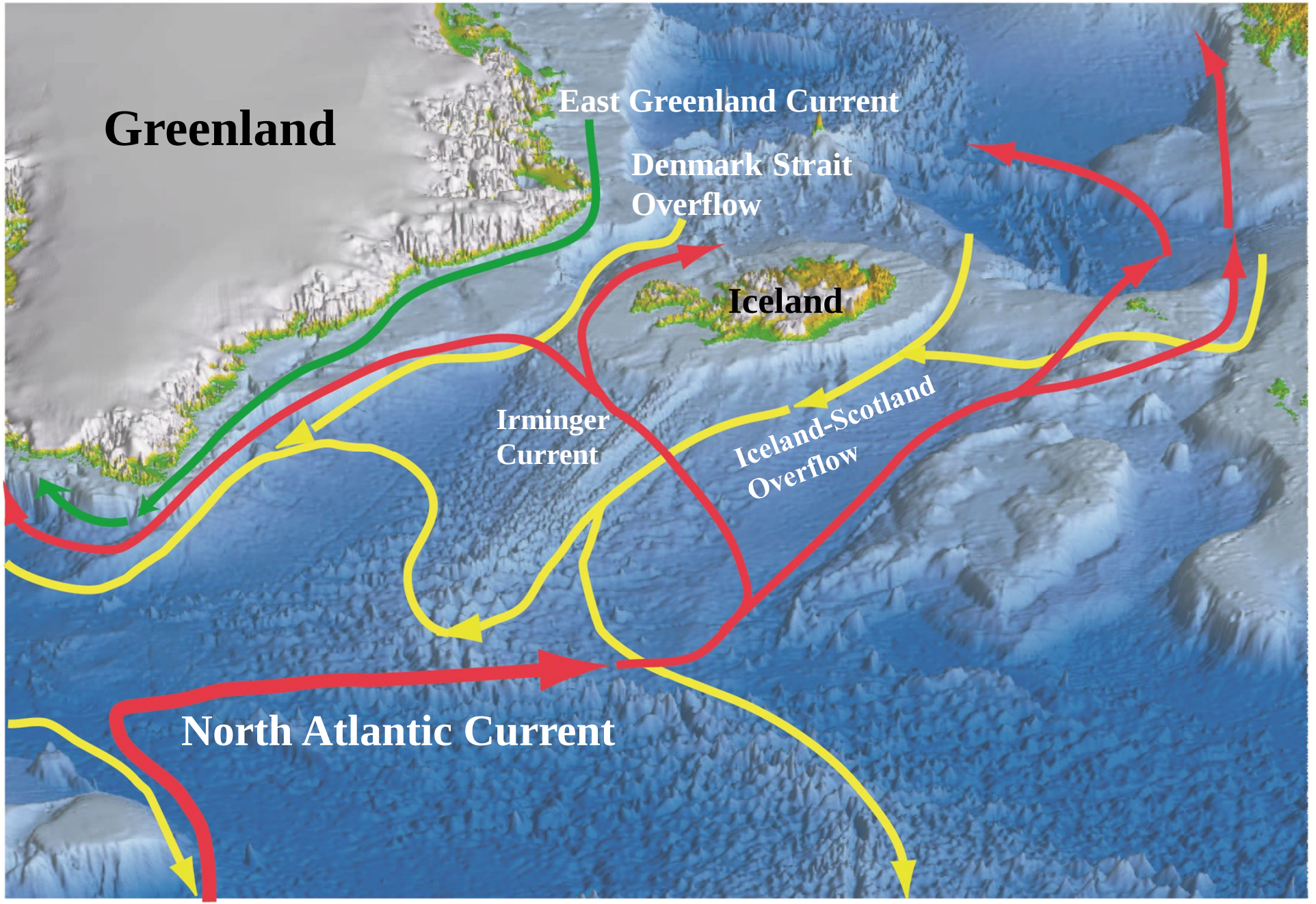
(Craig & Gordon 1965, modified in Sharp, 2007)


Bermuda
Norwegian sea
NADW
Gulf of Maine
AGUAS SUPERFICIAIS
nd E. coast

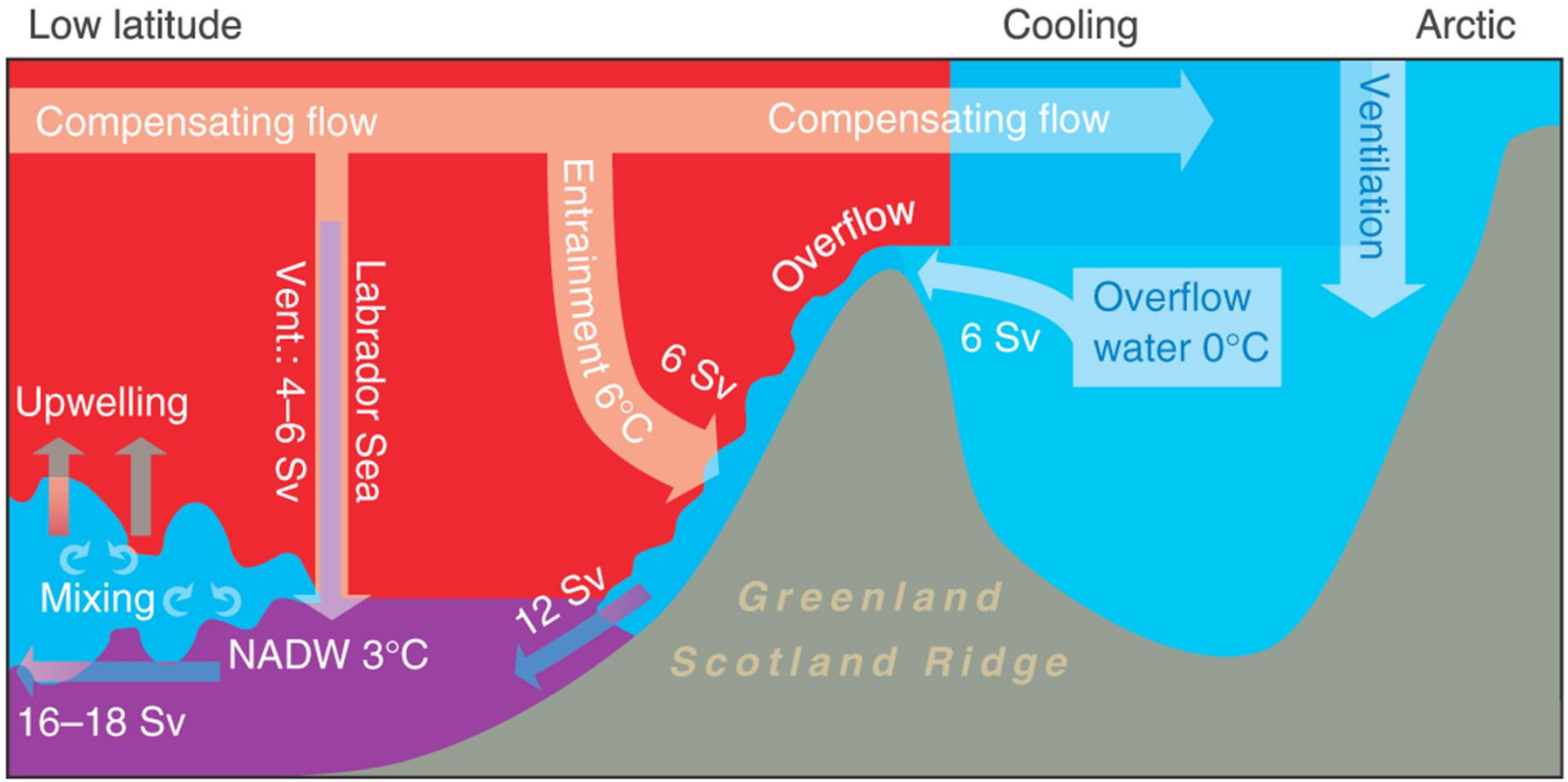
Formação de NADW



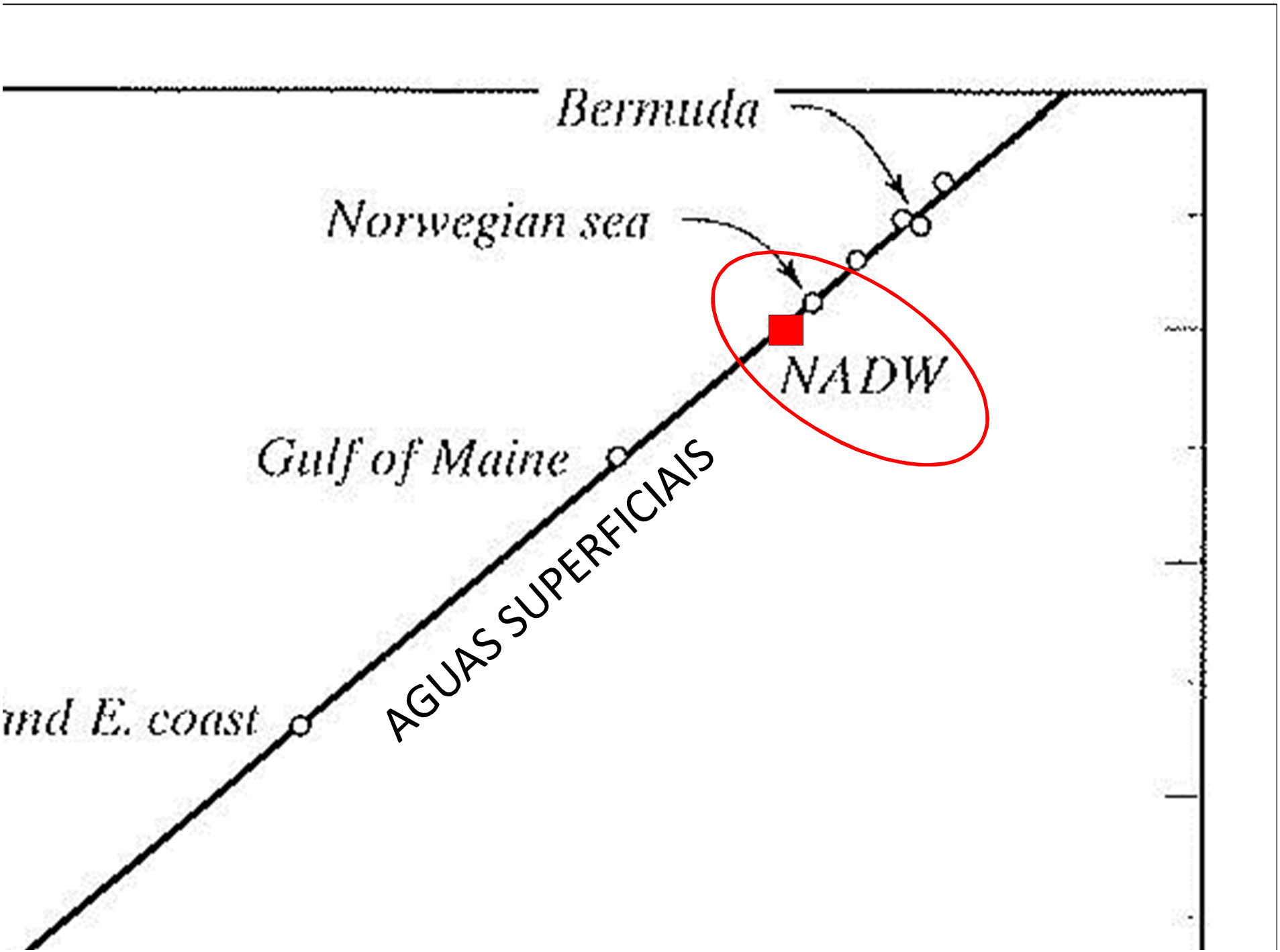
Curry & Mauritzen, *Science* v. 308 17.06.2005


Greenland
East Greenland Current
Denmark Strait
Overflow
Iceland
Irminger
Current
Iceland-Scotland
Overflow
North Atlantic Current

The Denmark Strait Overflow

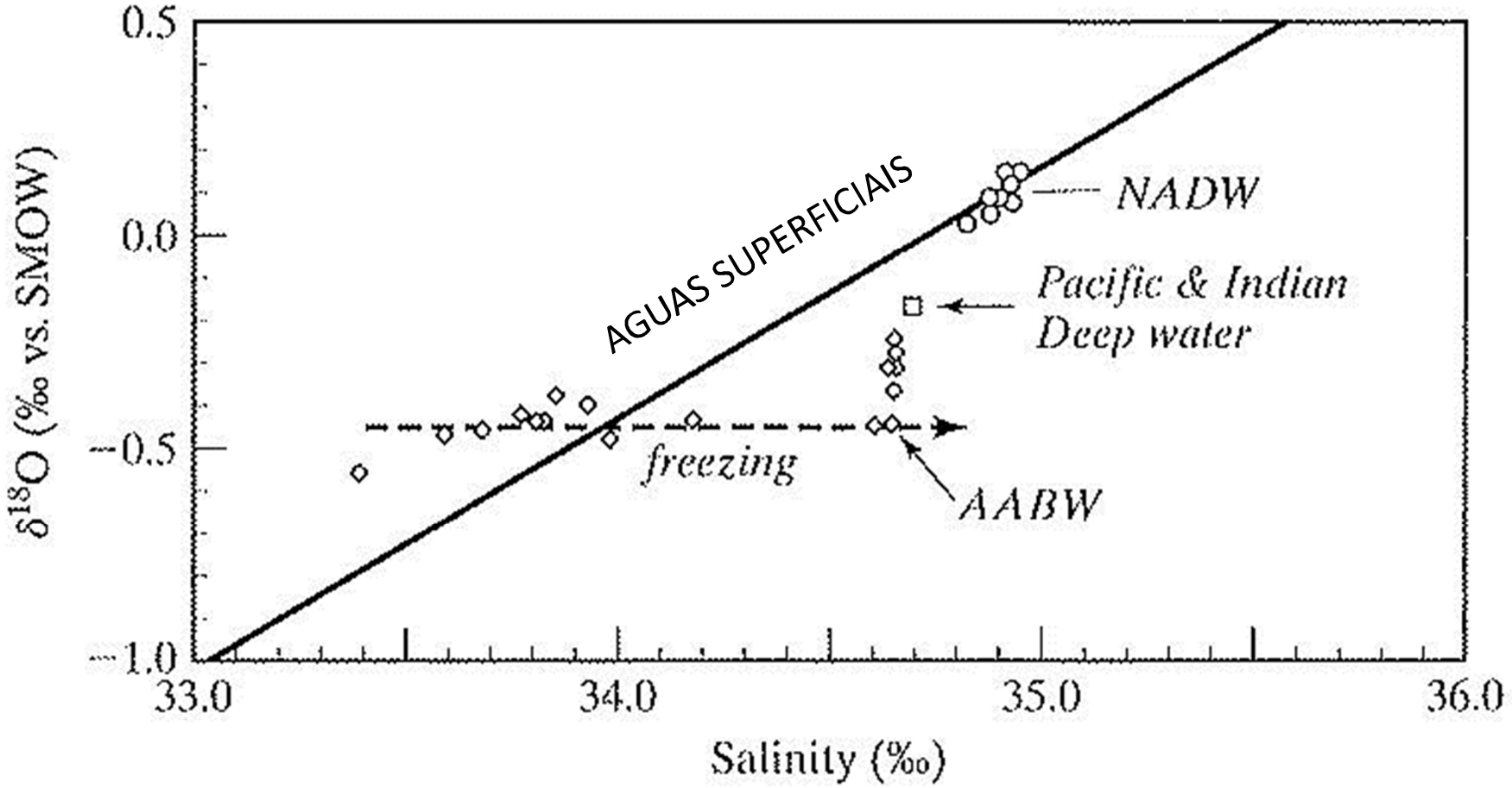


After Hansen et al., Science v. 305 (2004) p. 953-954.


Bermuda
Norwegian sea
NADW
Gulf of Maine
AGUAS SUPERFICIAIS
nd E. coast

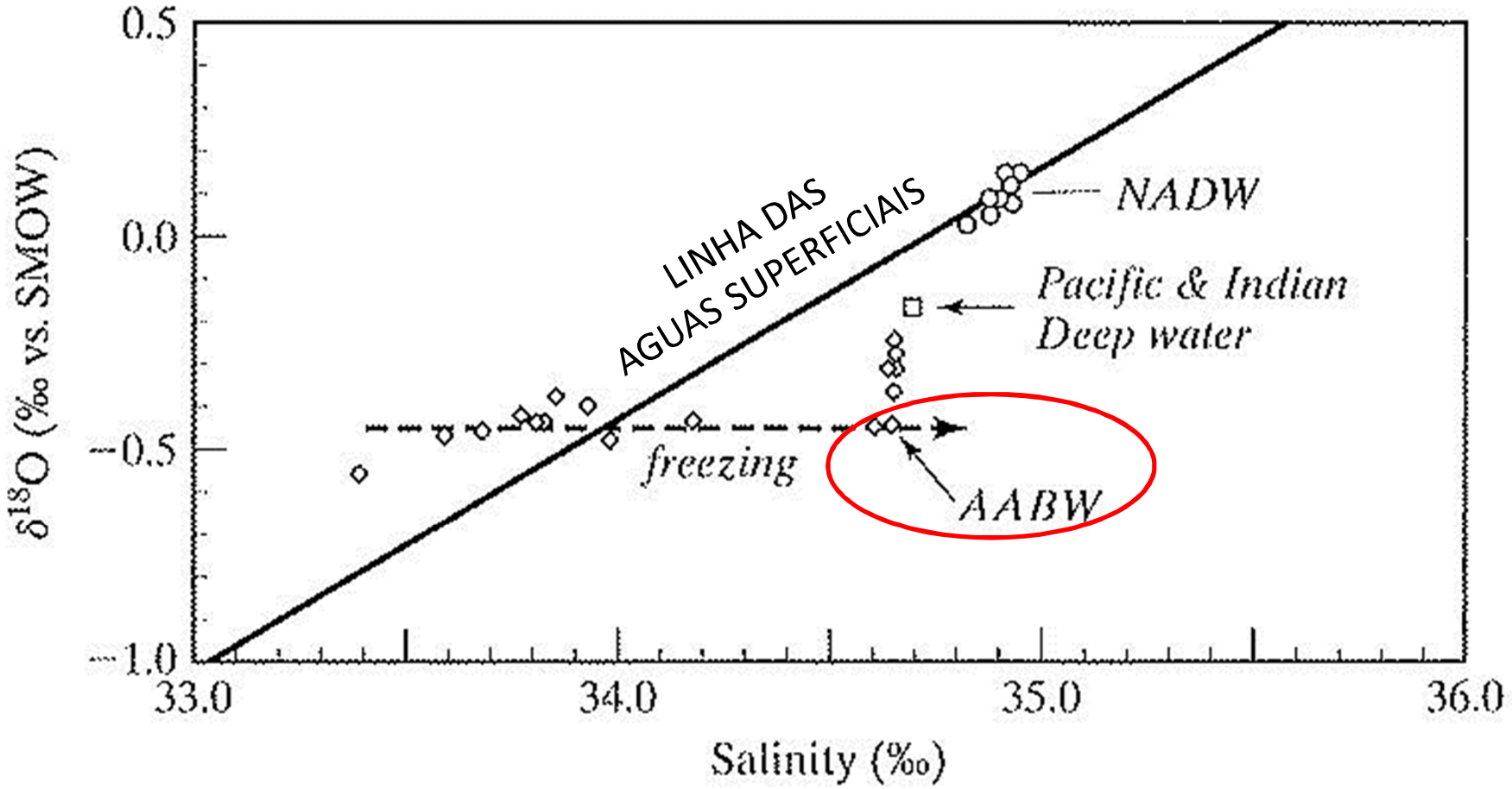
Formação de Antarctic Bottom Water AABW

Relação entre $\delta^{18}O$ do oceano e salinidade



(Craig & Gordon 1965, modified in Sharp, 2007)

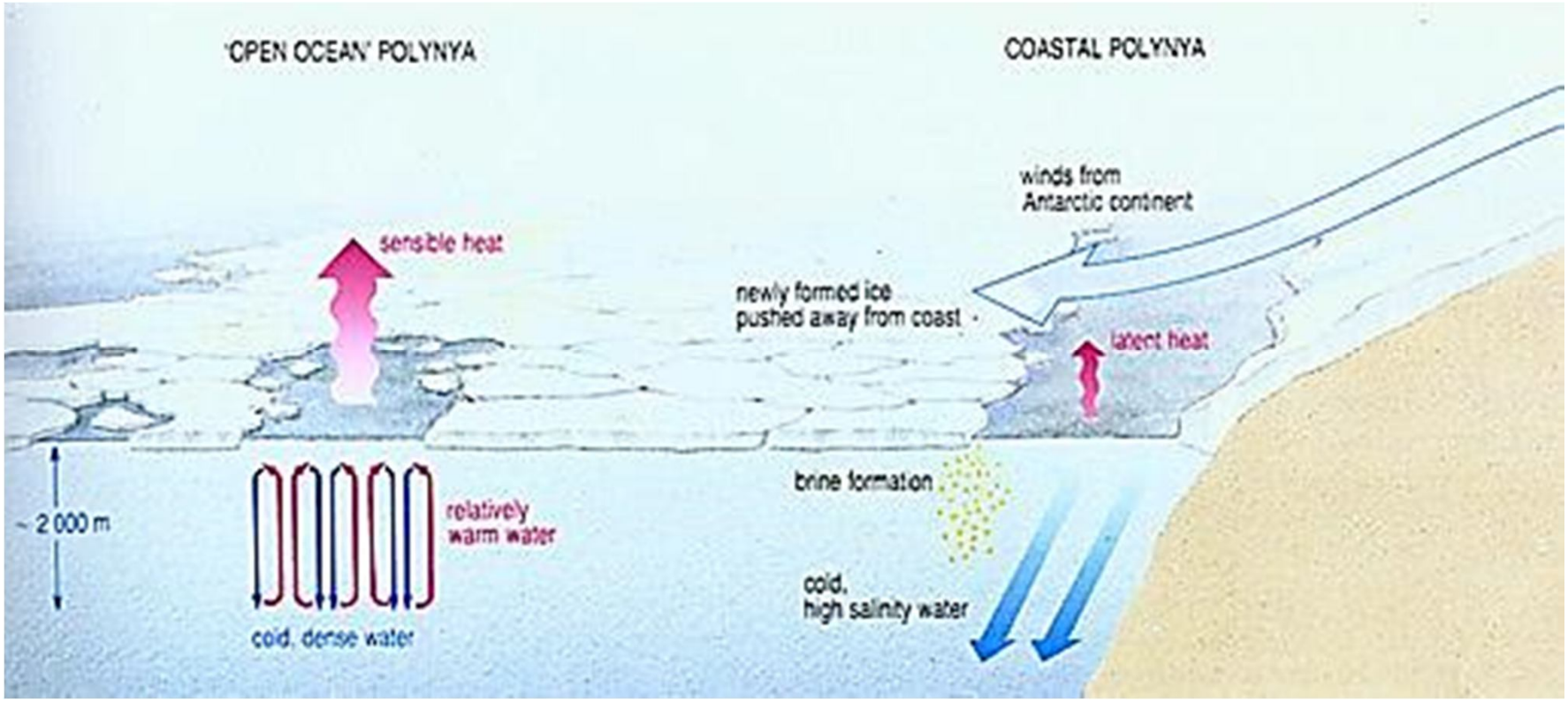
Relação entre $\delta^{18}O$ do oceano e salinidade



(Craig & Gordon 1965, modified in Sharp, 2007)

Formação de água profunda na Antártida

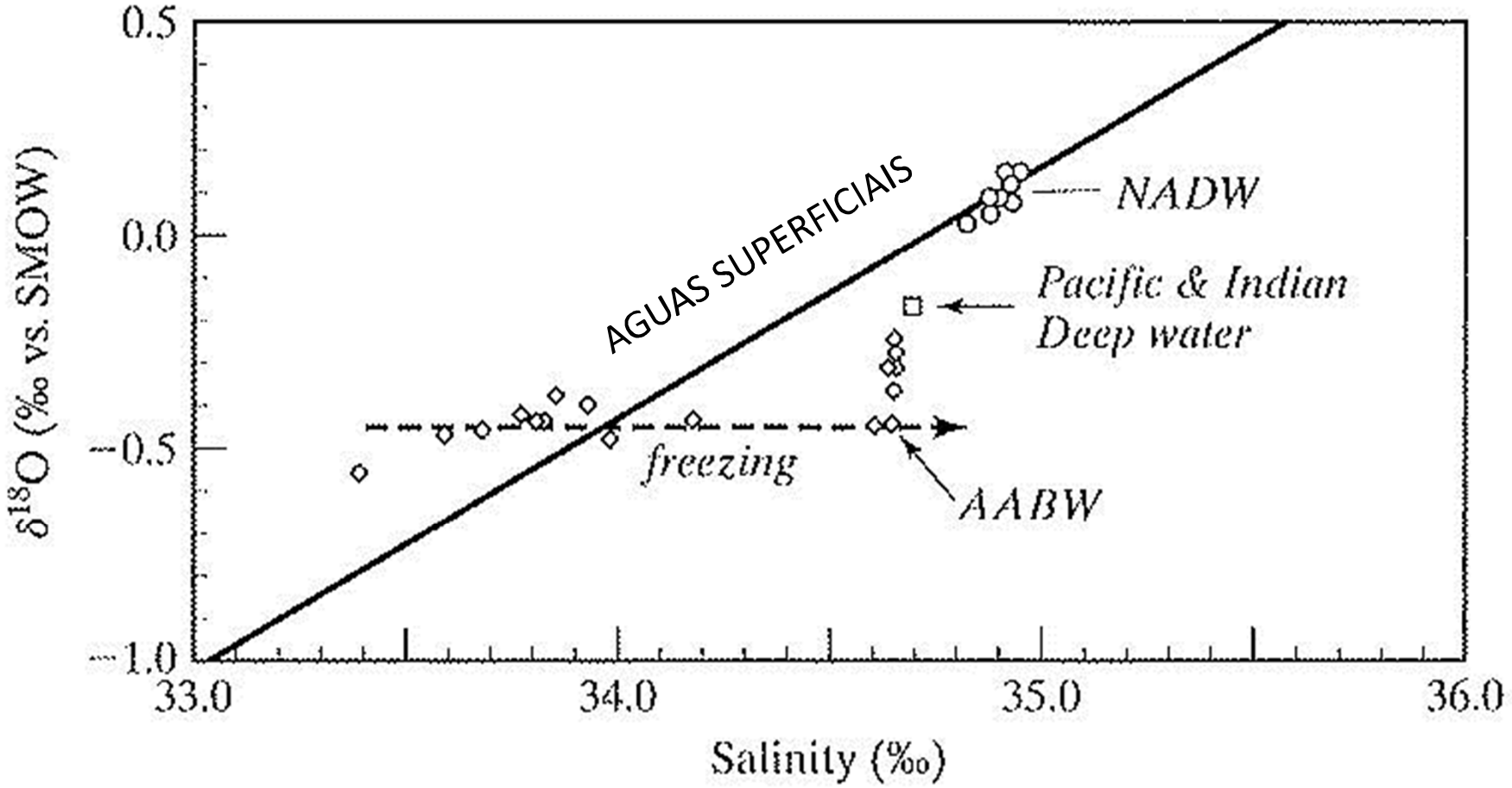
Brine formation



www.ldeo.columbia.edu

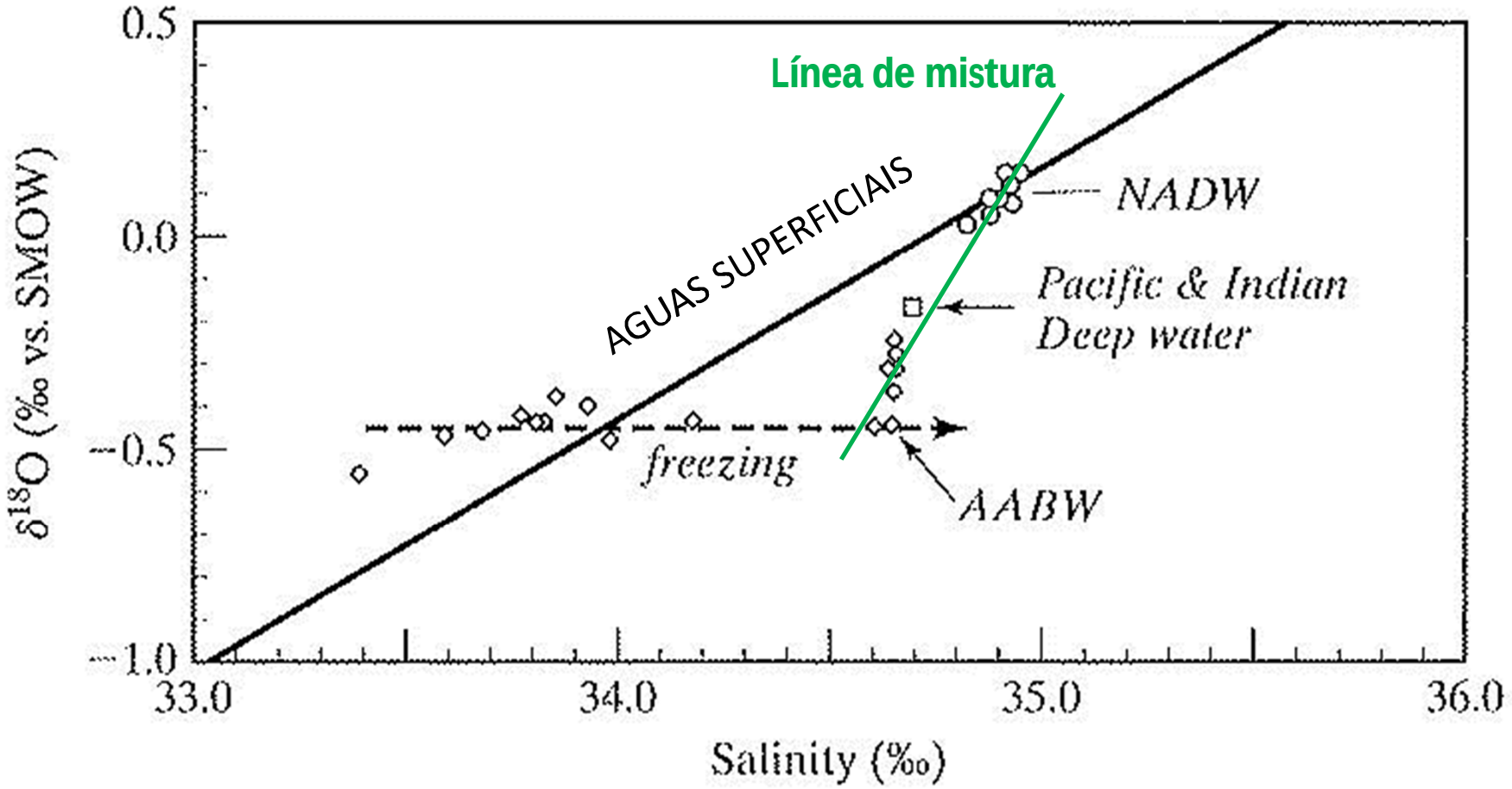
Águas profundas no Pacifico e Índico

Águas profundas no Pacífico e Índico



(Craig & Gordon 1965, modified in Sharp, 2007)

Águas profundas no Pacífico e Índico



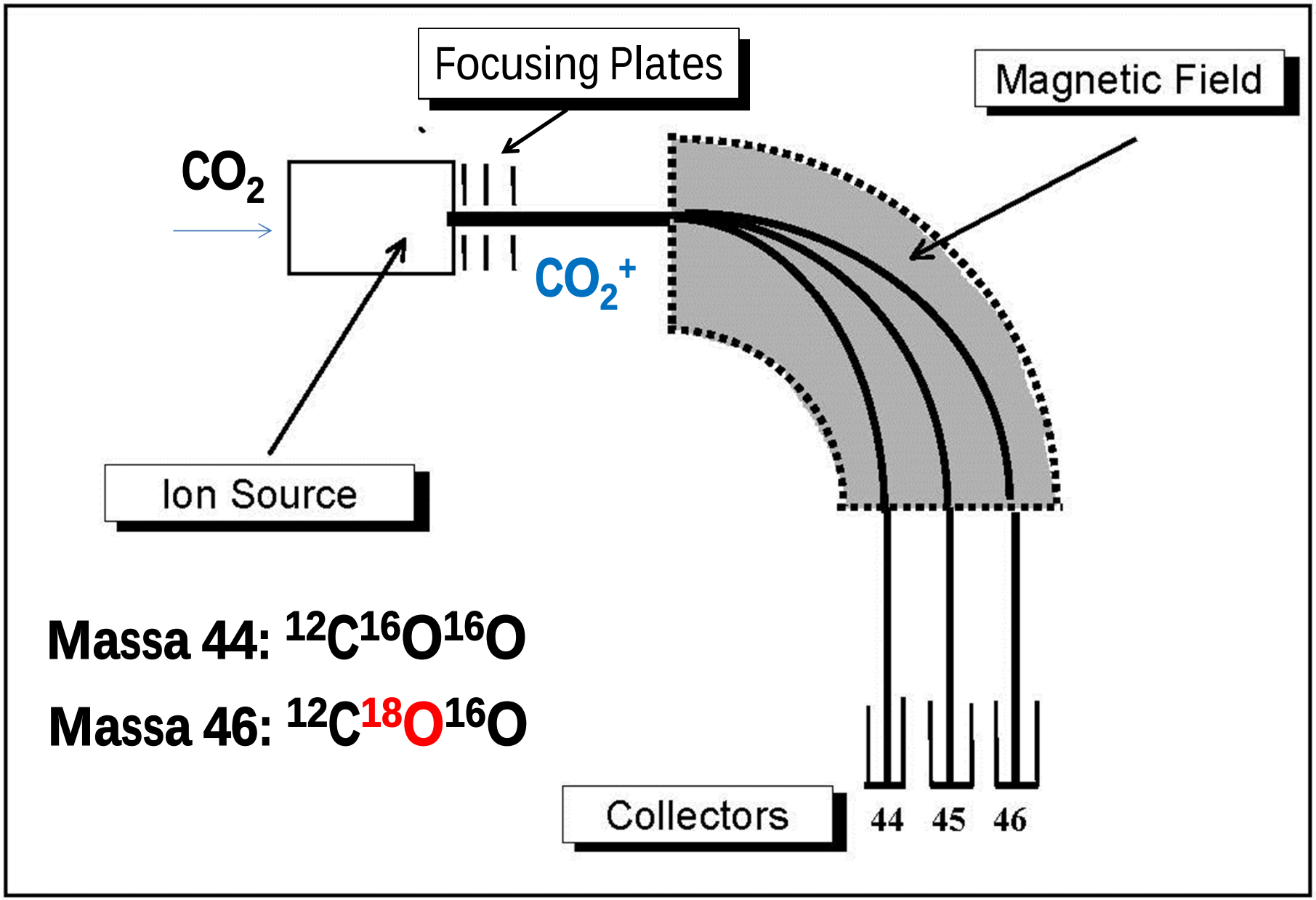
(Craig & Gordon 1965, modified in Sharp, 2007)

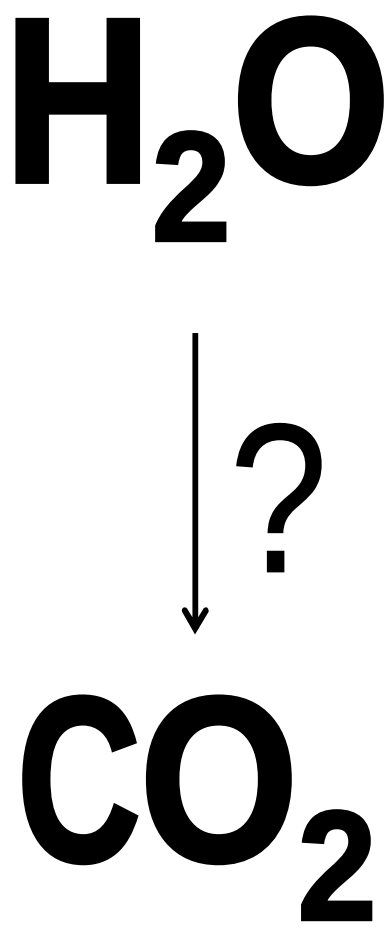
Isótopos do oxigênio

Método de analise de água (do mar)

Espectrometria de massa de fase gasosa

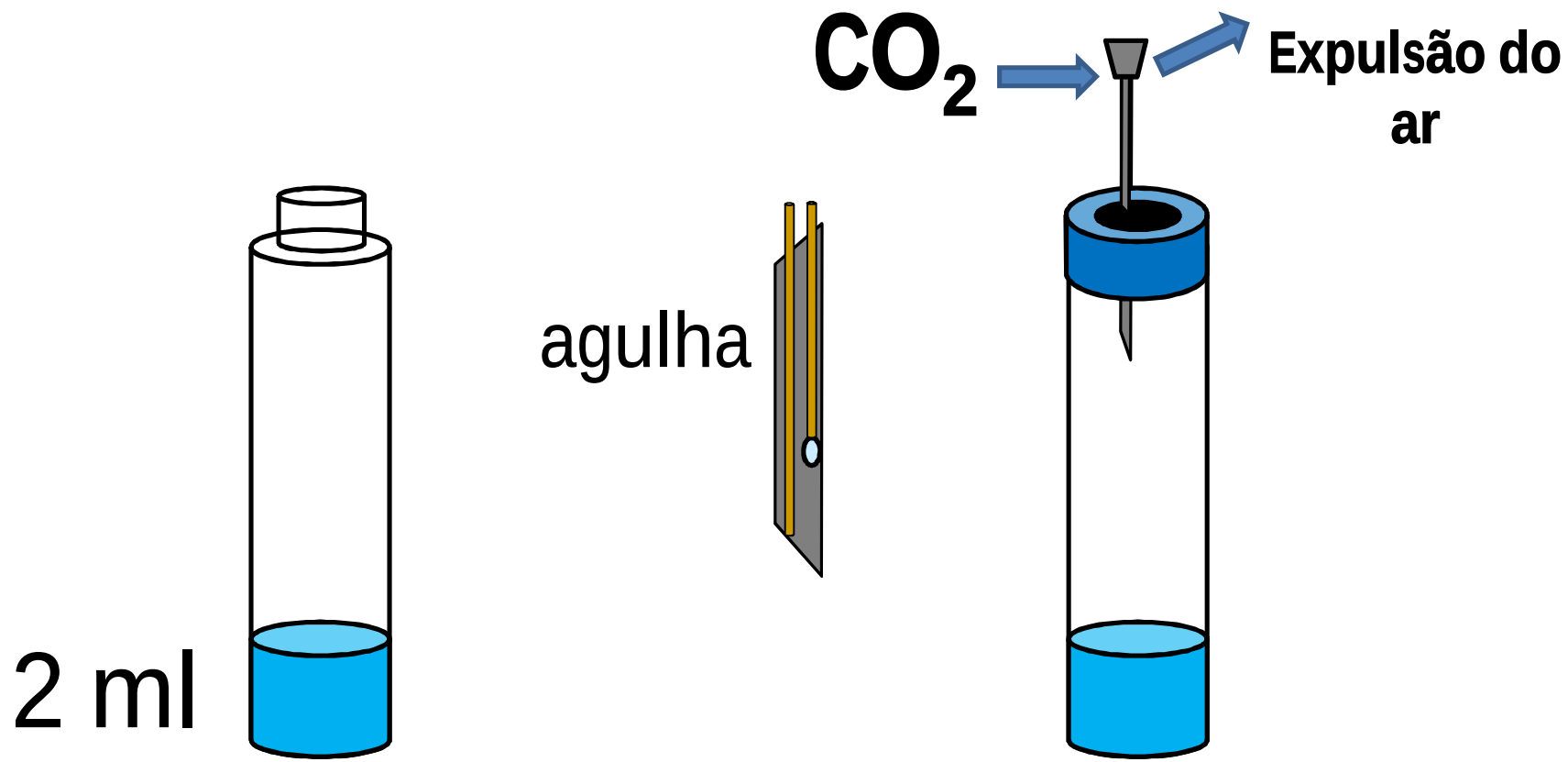
Modified from Analytical Precision AP2003 *Operating Manual*.




H_2O
?
CO_2

Preparação de amostras de água para analise isotópica do oxigênio

Método do equilíbrio (Epstein & Mayeda, 1953)



$$C^{16}O^{16}O + H_2{}^{18}O \rightleftharpoons C^{16}O^{18}O + H_2{}^{16}O$$

Equilíbrio isotópico

T constante (25 °C)

Fator de fracionamento de isótopos do oxigênio entre H_2O e $CO_{2(g)}$

Isótopos do oxigênio

Método de analise de água (do mar)

Equipamento usado

Finnigan GasBench II *Operating Manual*.

Finnigan GasBench II

Finnigan Delta V Isotope Ratio Mass Spectrometer

Finnigan GasBench II

Finnigan GasBench II *Operating Manual*.

Figure 2-8. Non-Thermostated Sample Tray - Side View (Part No. 111 2780)

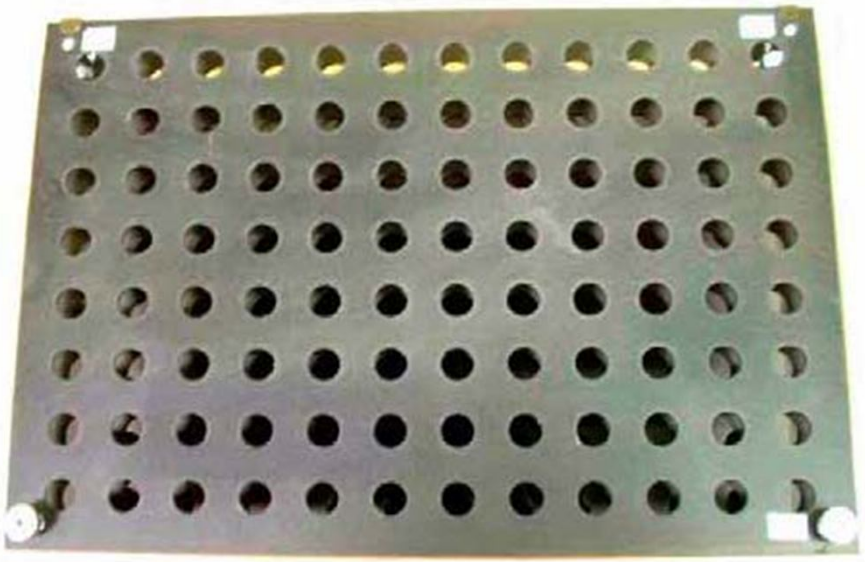
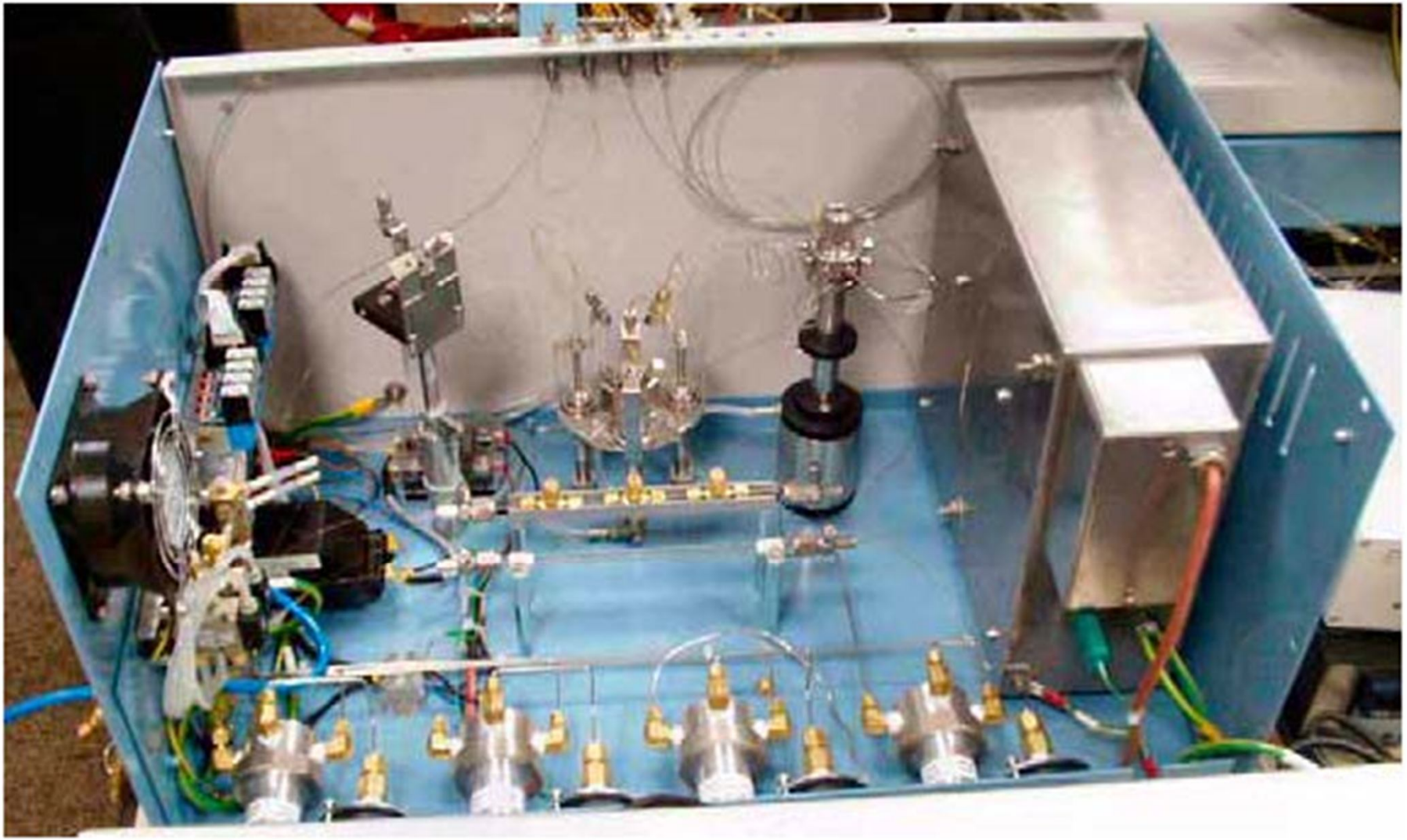


Figure 2-9. Non-Thermostated Sample Tray - Top View (Part No. 111 2780)

Finnigan GasBench II

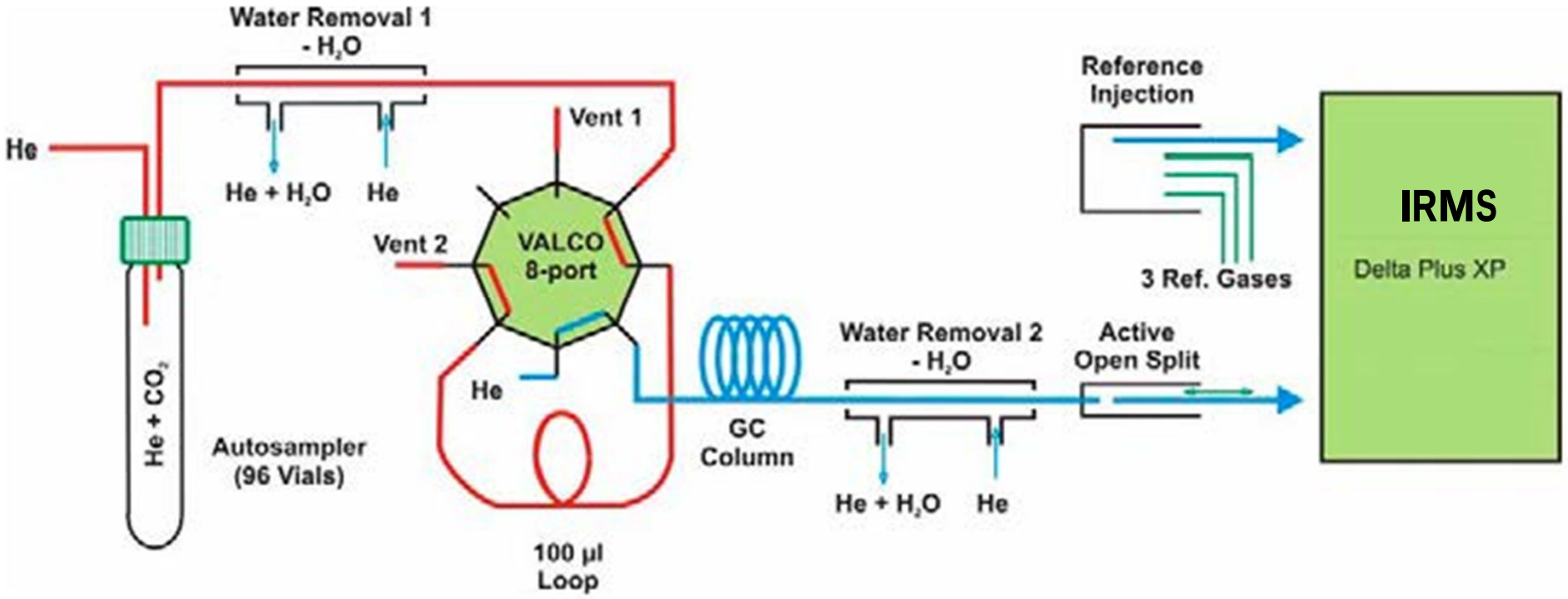
sistema a fluxo contínuo de He

Finnigan GasBench II



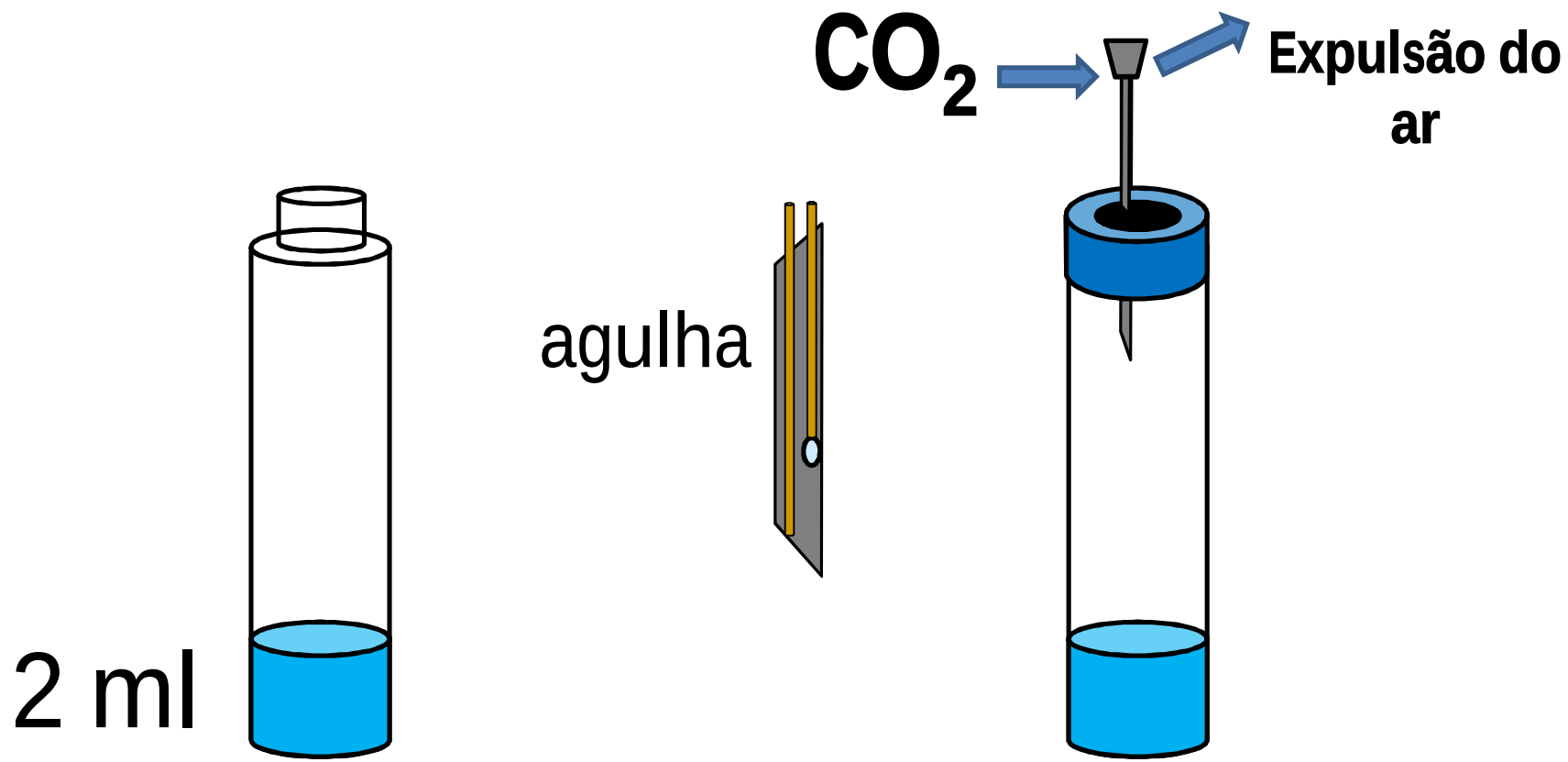
Finnigan GasBench II *Operating Manual*.

GasBench II



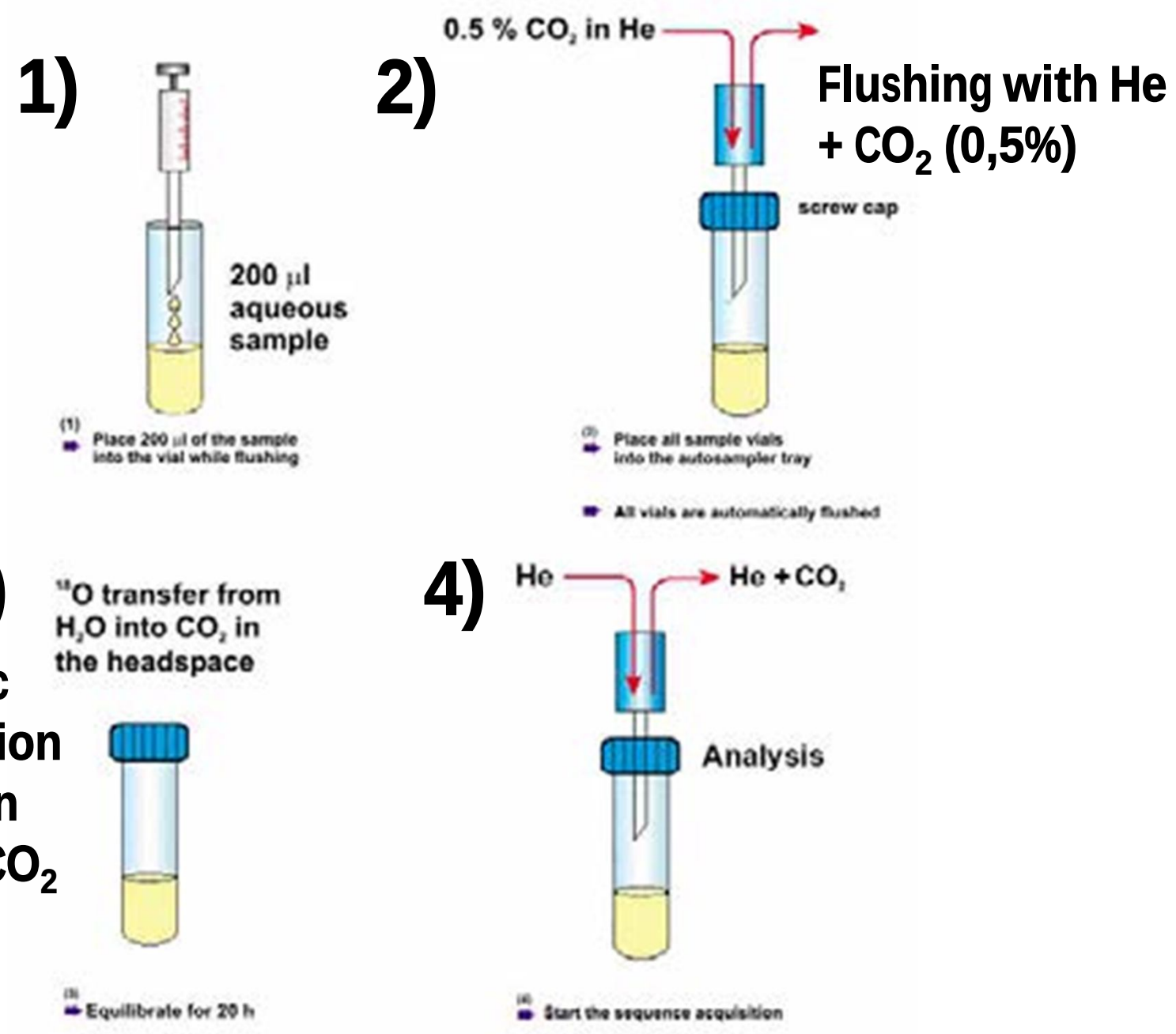
Finnigan GasBench II *Operating Manual*.

Método do equilíbrio (Epstein & Mayeda, 1953)



$$C^{16}O^{16}O + H_2{}^{18}O \rightleftharpoons C^{16}O^{18}O + H_2{}^{16}O$$

Equilíbrio isotópico



Finnigan GasBench II *Operating Manual*.

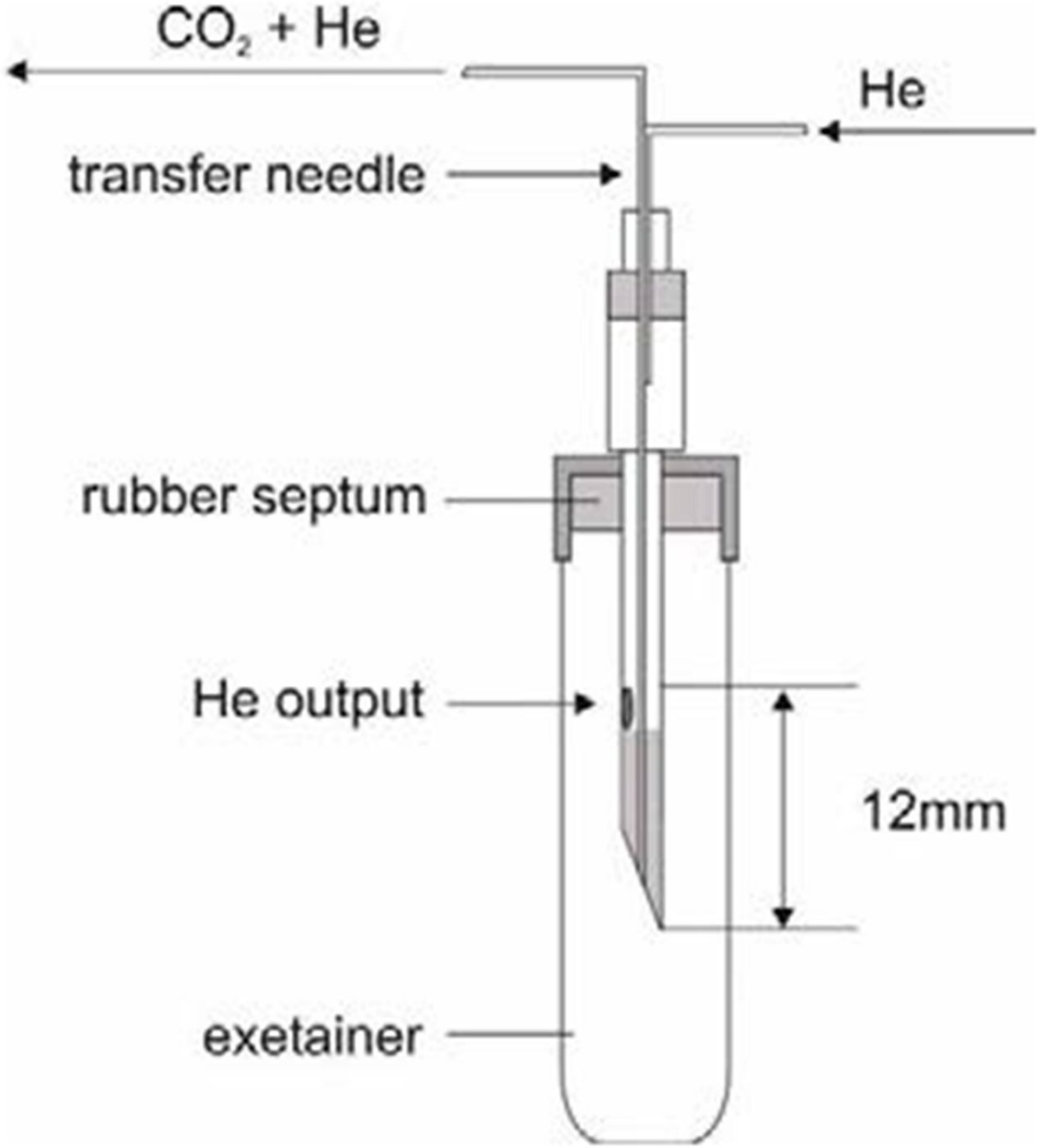
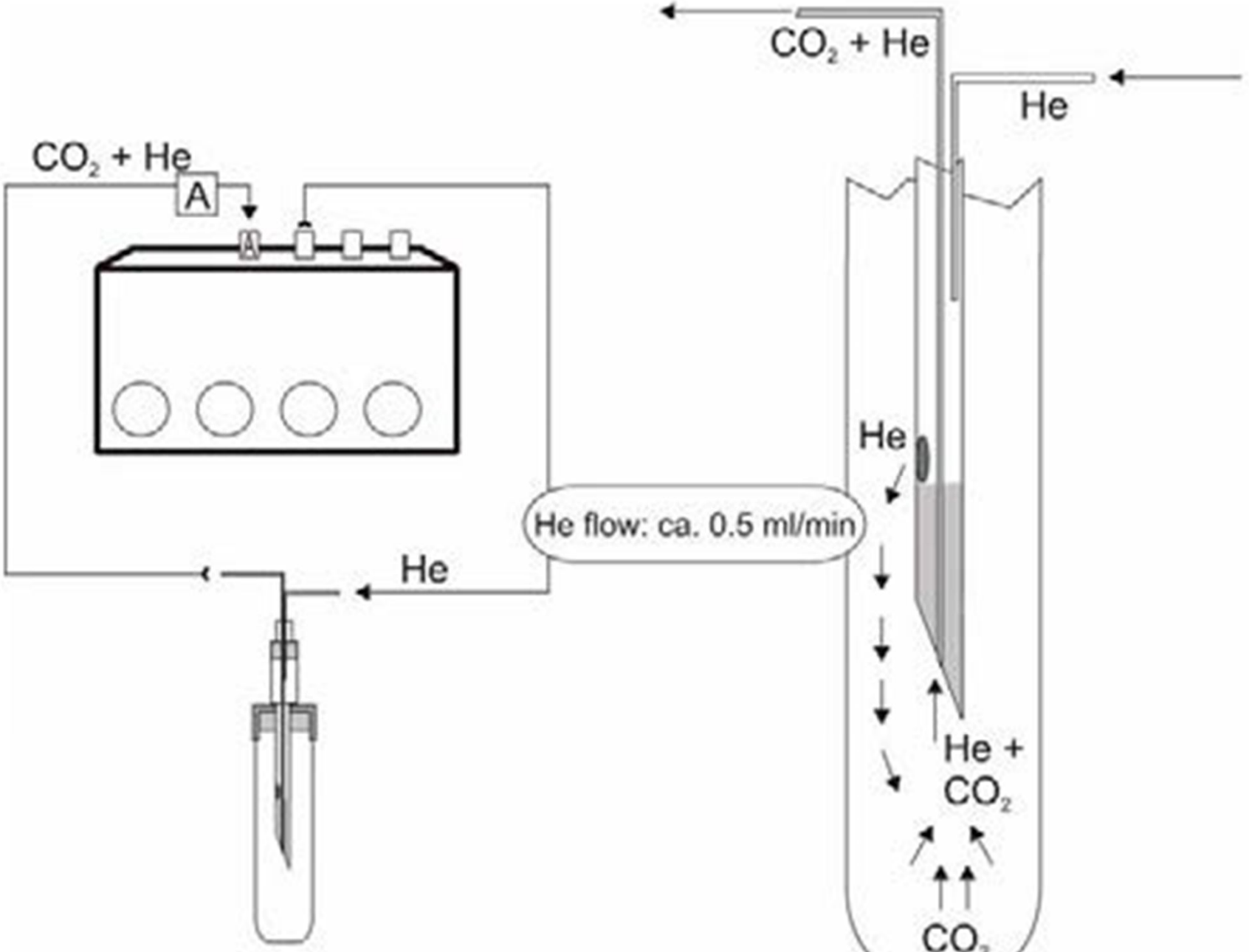
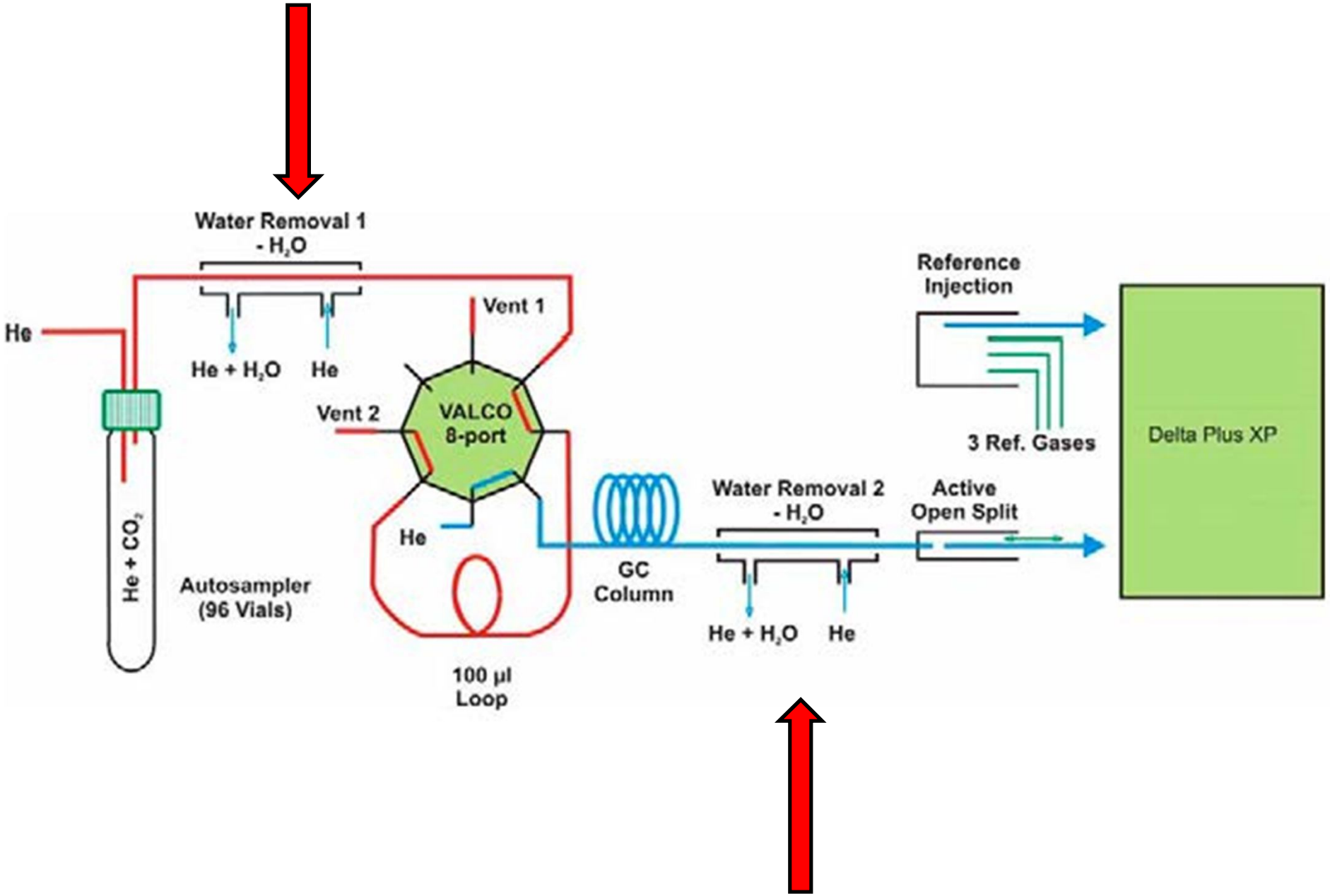


Figure 2-15. Measurement Needle

Finnigan GasBench II *Operating Manual*.

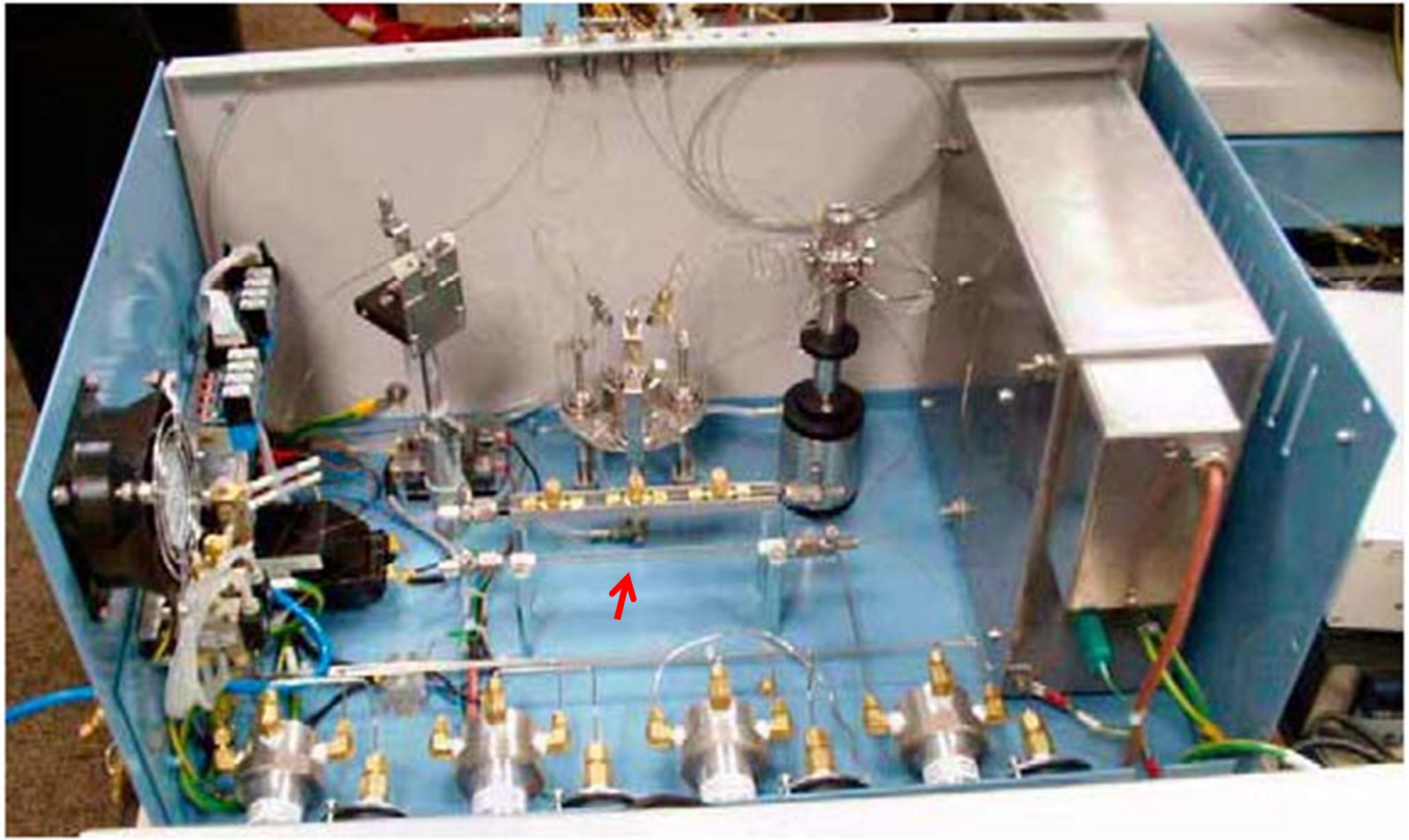


Finnigan GasBench II *Operating Manual*.



Finnigan GasBench II *Operating Manual*.

Finnigan GasBench II



Finnigan GasBench II *Operating Manual*.

NAFION® Water Trap

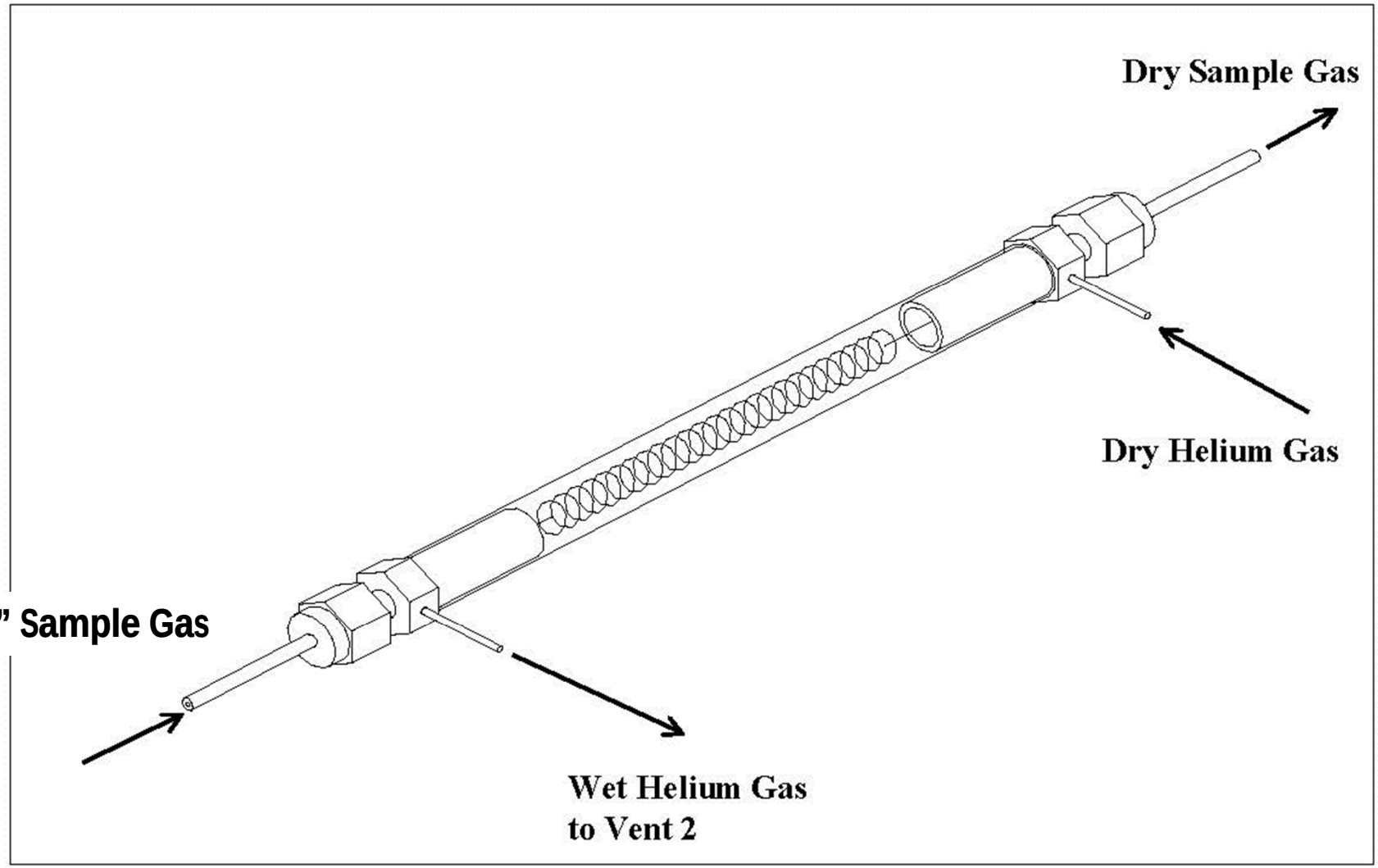
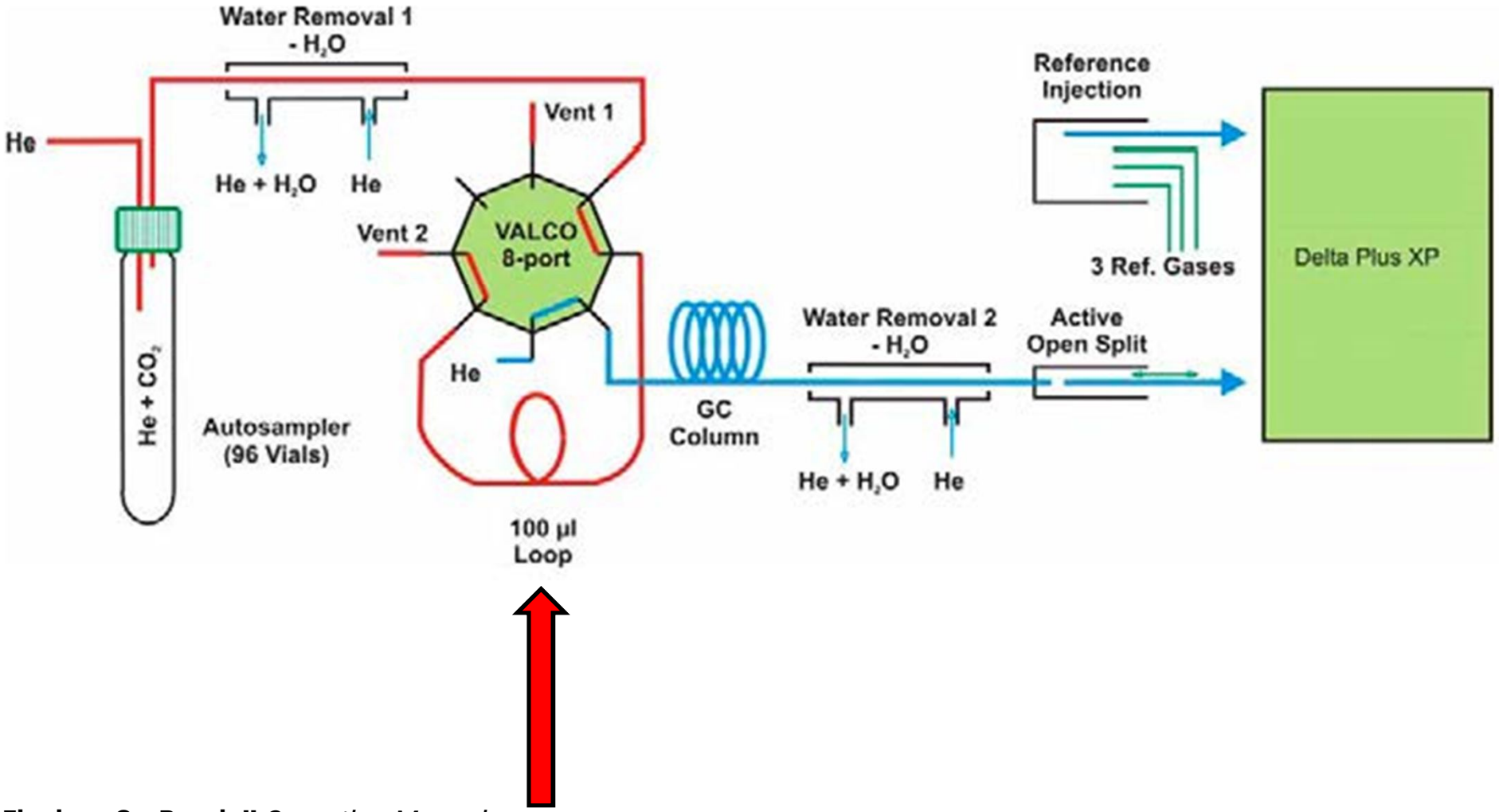


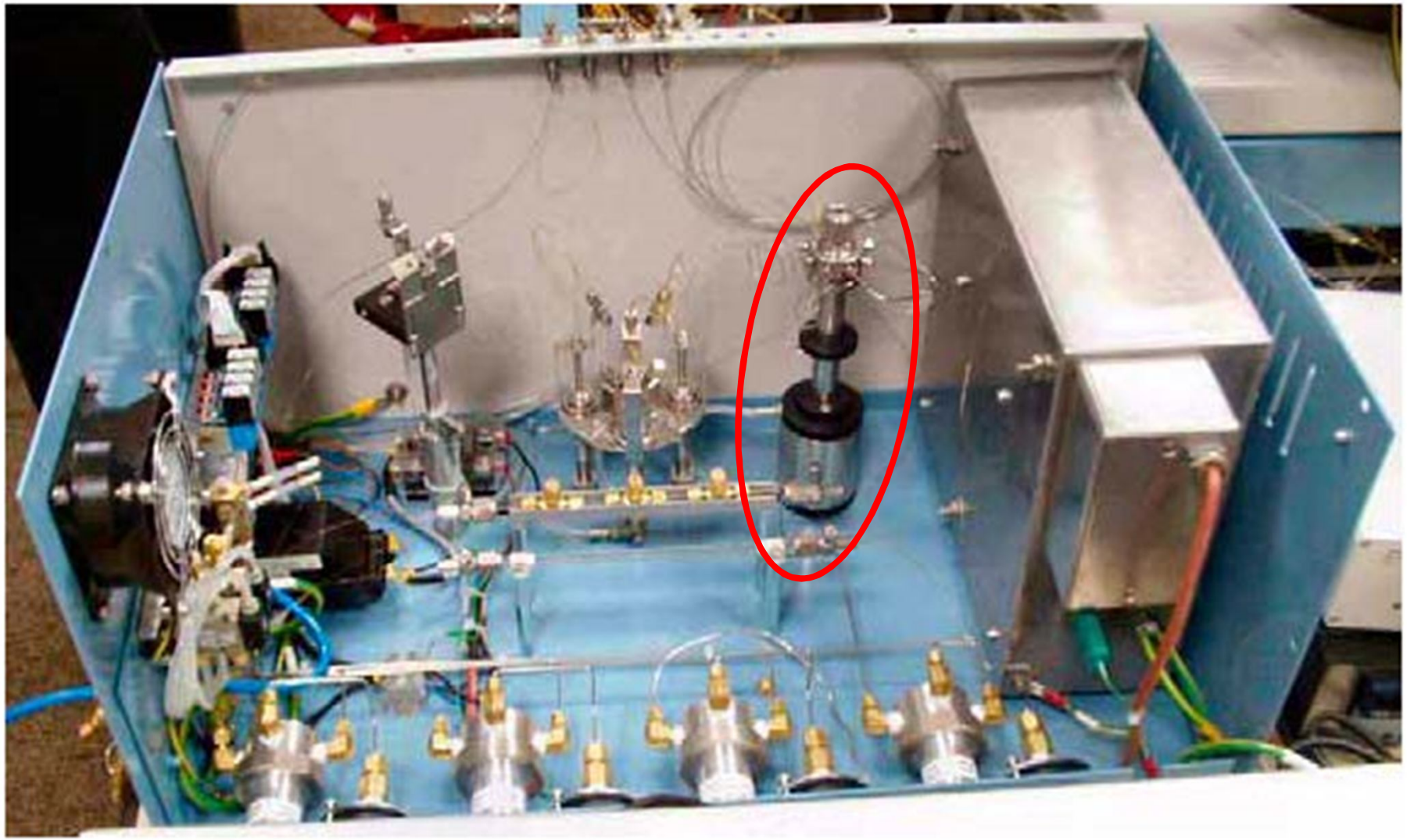
Figure 38 : Schematic of the Water Removal Device

Modified from Analytical Precision AP2003 *Operating Manual*.



Finnigan GasBench II *Operating Manual*.

Finnigan GasBench II



Finnigan GasBench II *Operating Manual*.

VALCO® valve

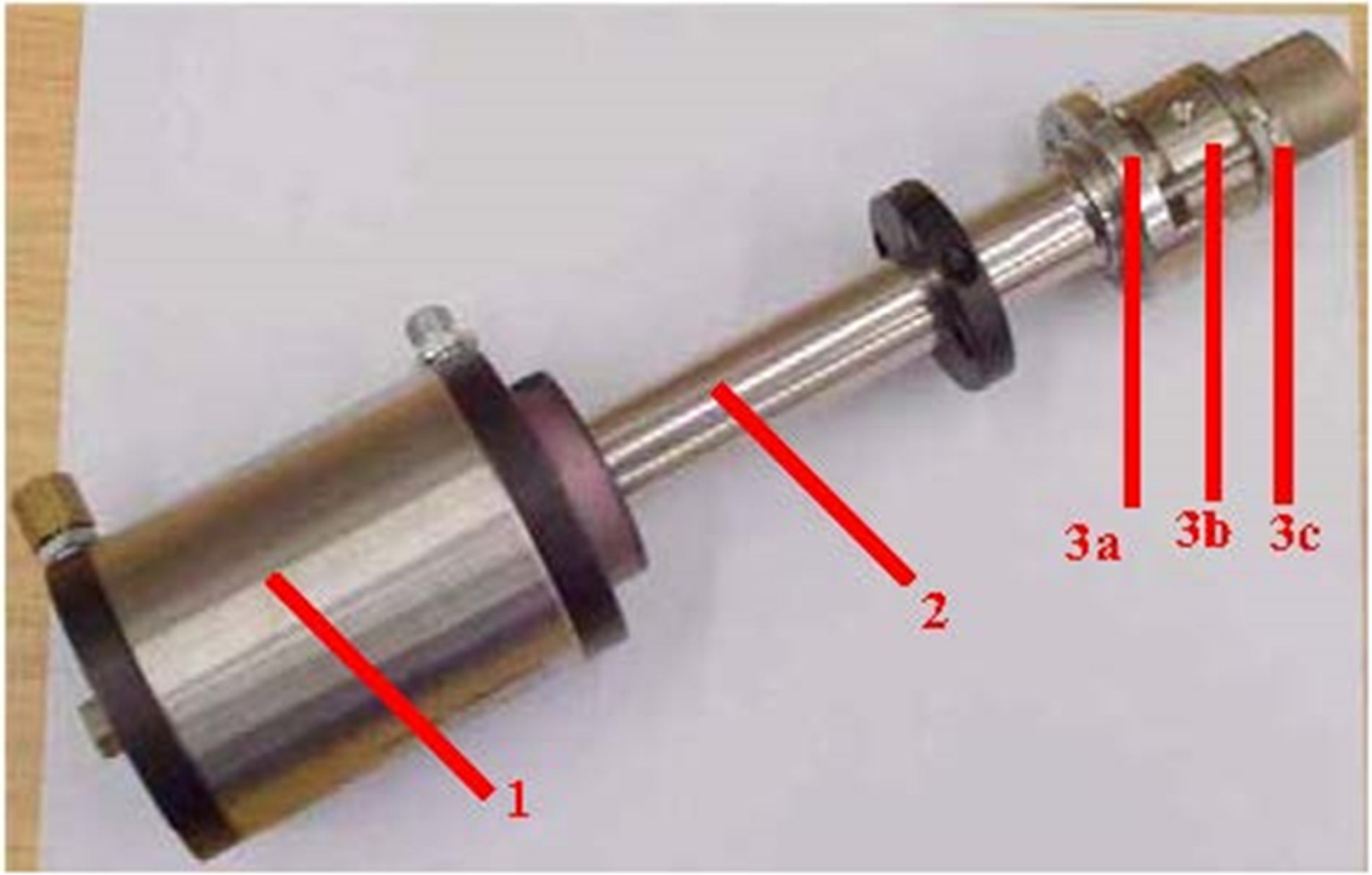


Figure 2-22. Valco Eight Port Valve - Side View

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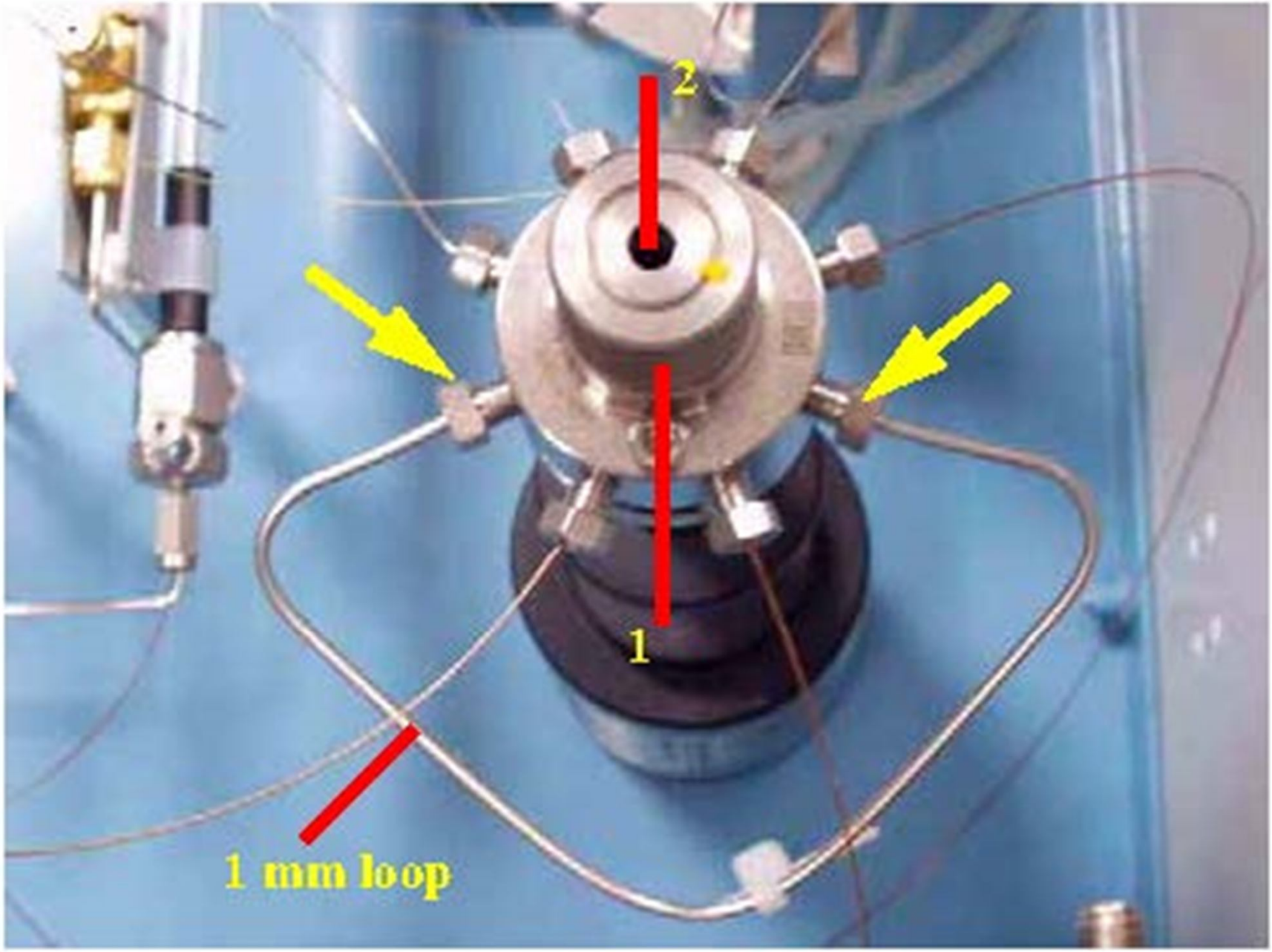
VALCO® valve



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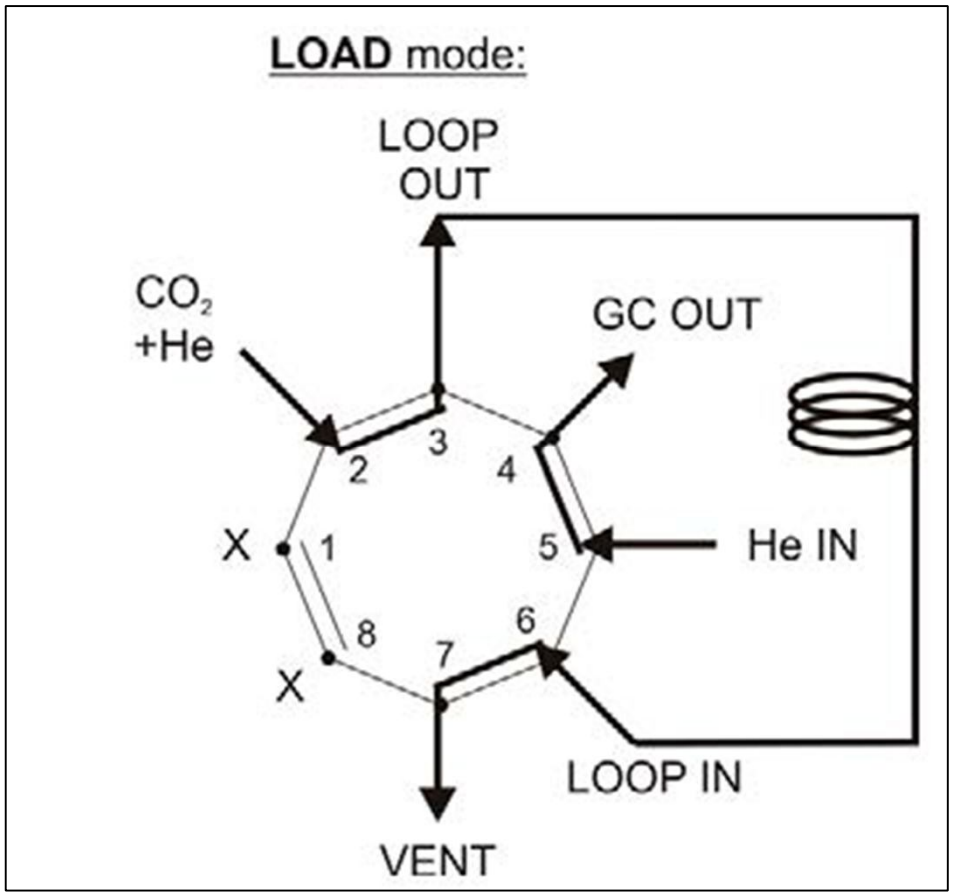
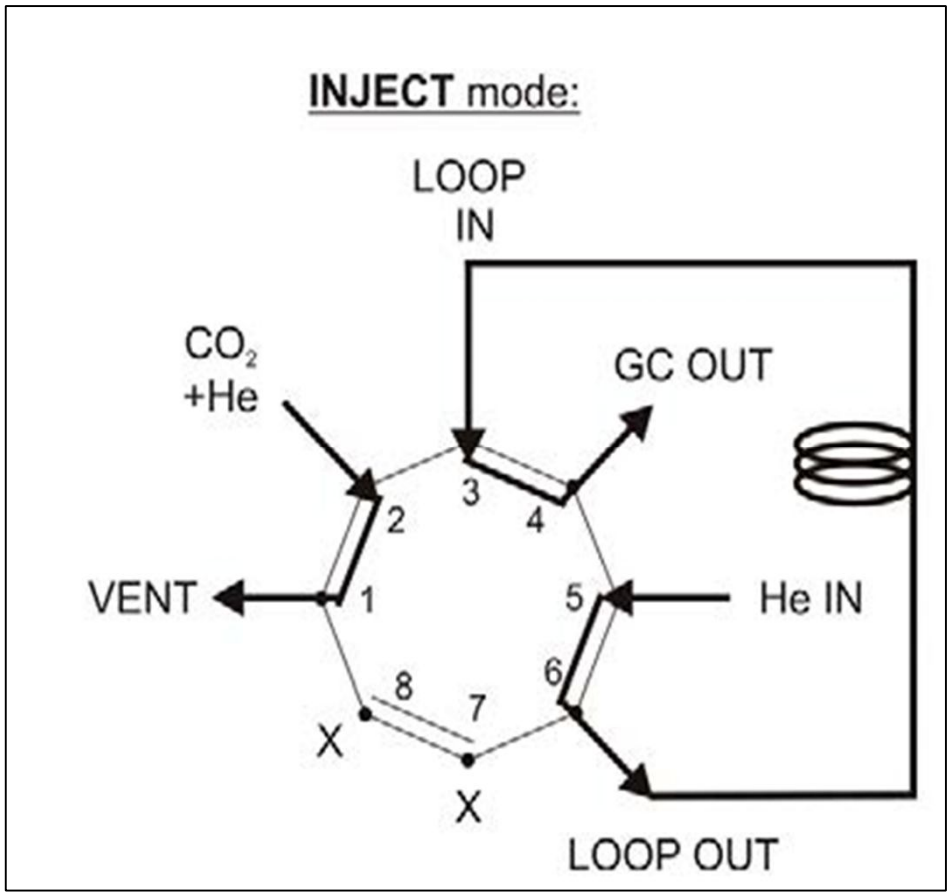
Como funciona uma válvula VALCO®





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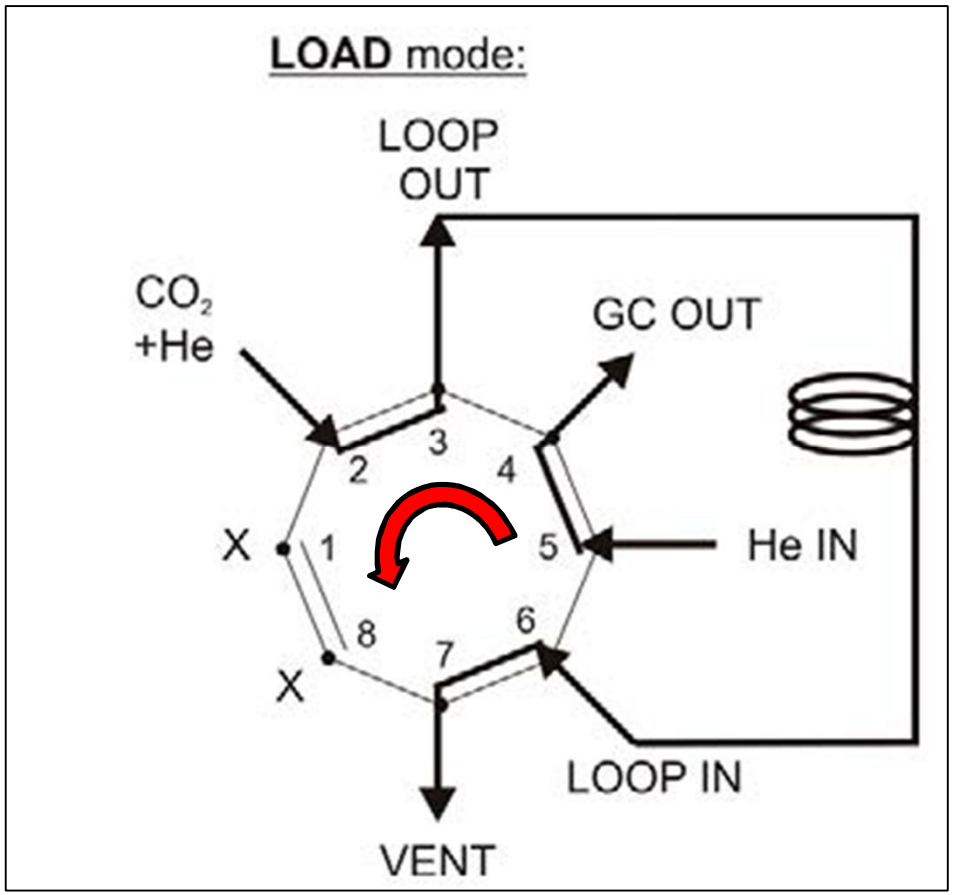
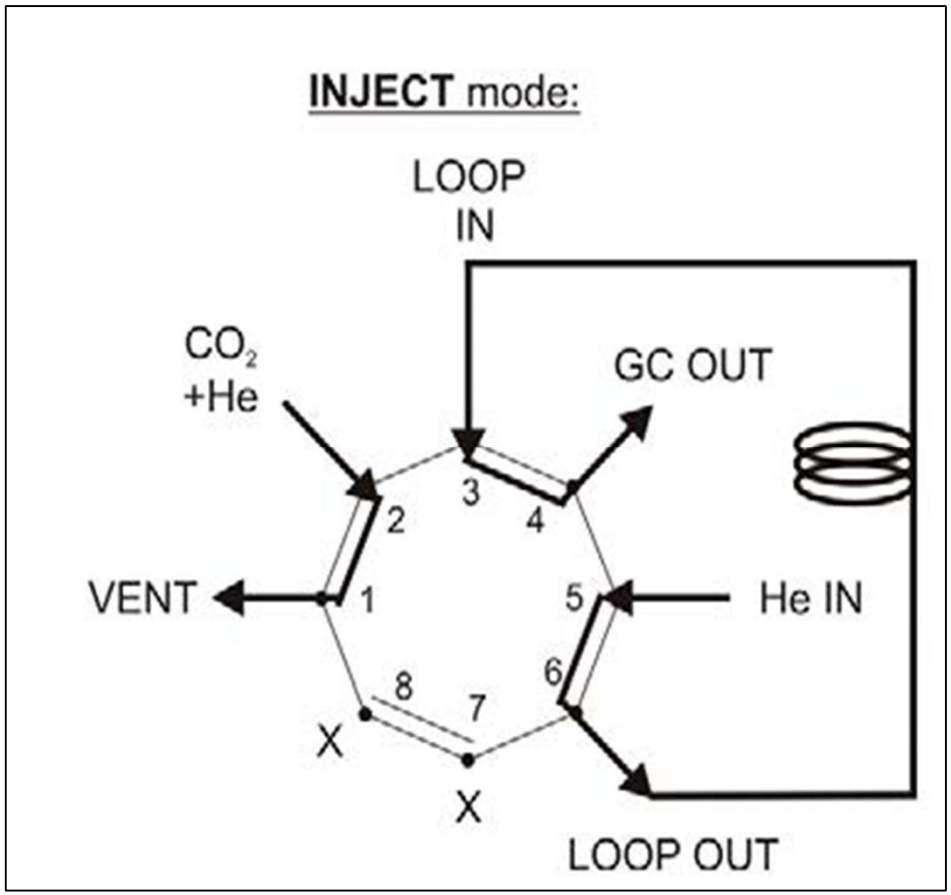
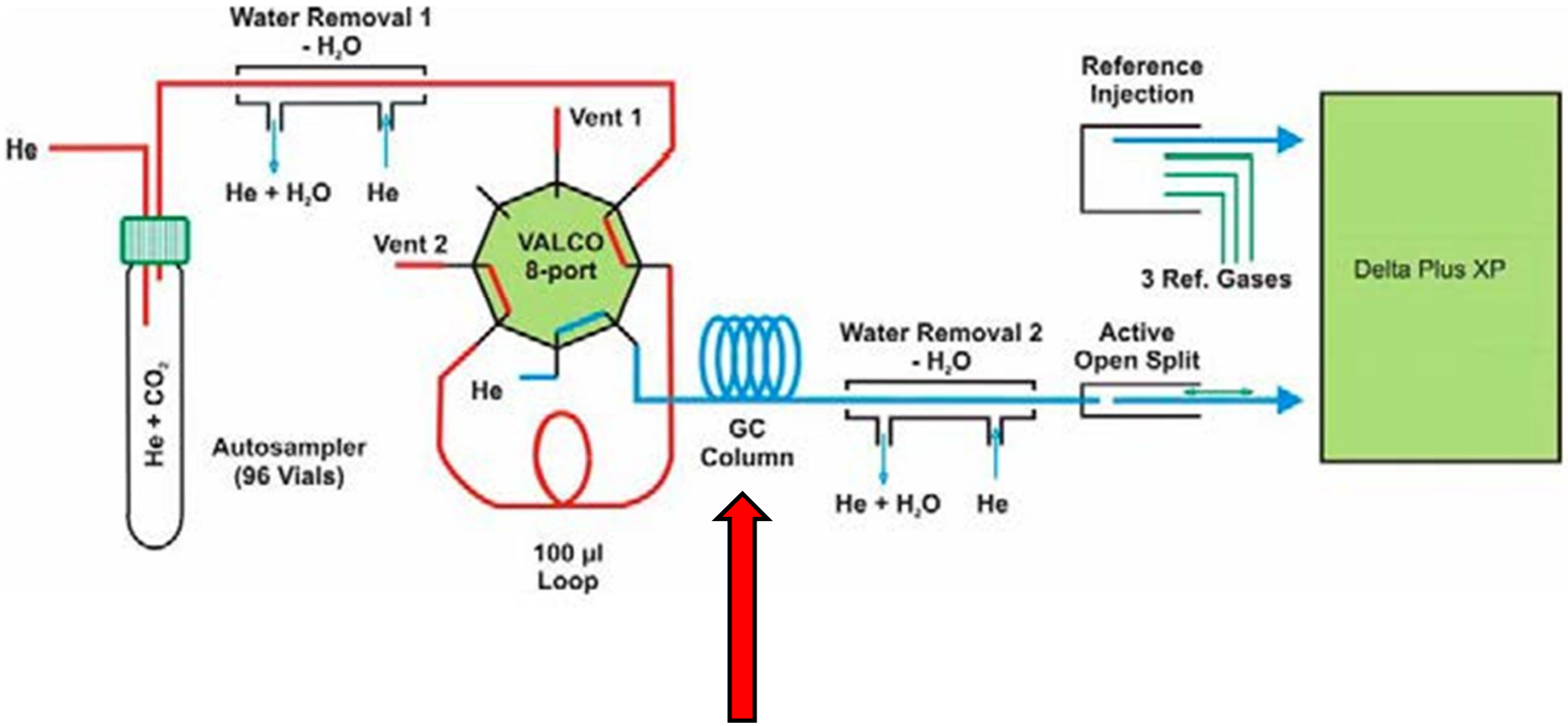
Como funciona uma válvula VALCO®





Finnigan GasBench II *Operating Manual*.



Finnigan GasBench II *Operating Manual*.

Cromatographic Column

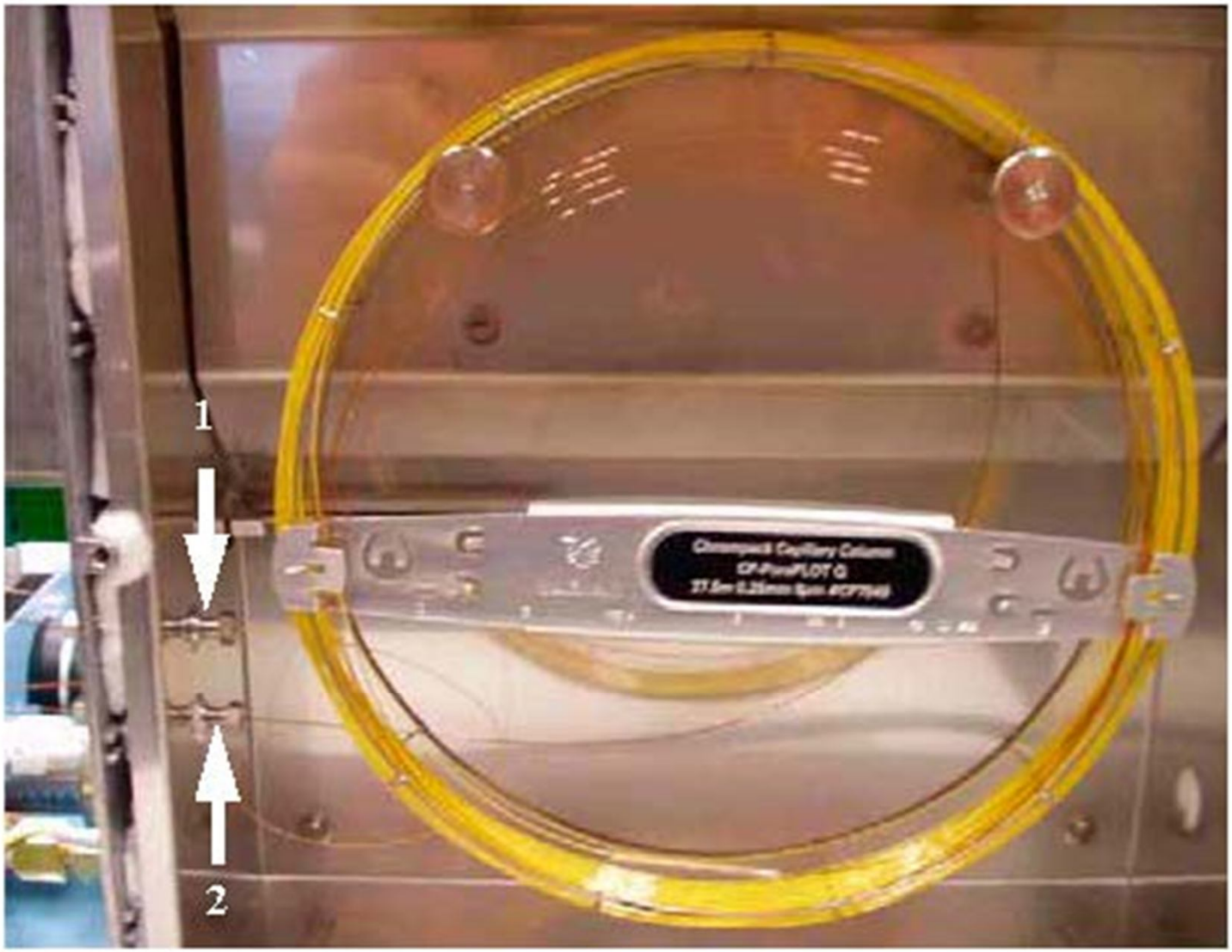


Figure 2-32. Column (Installed)

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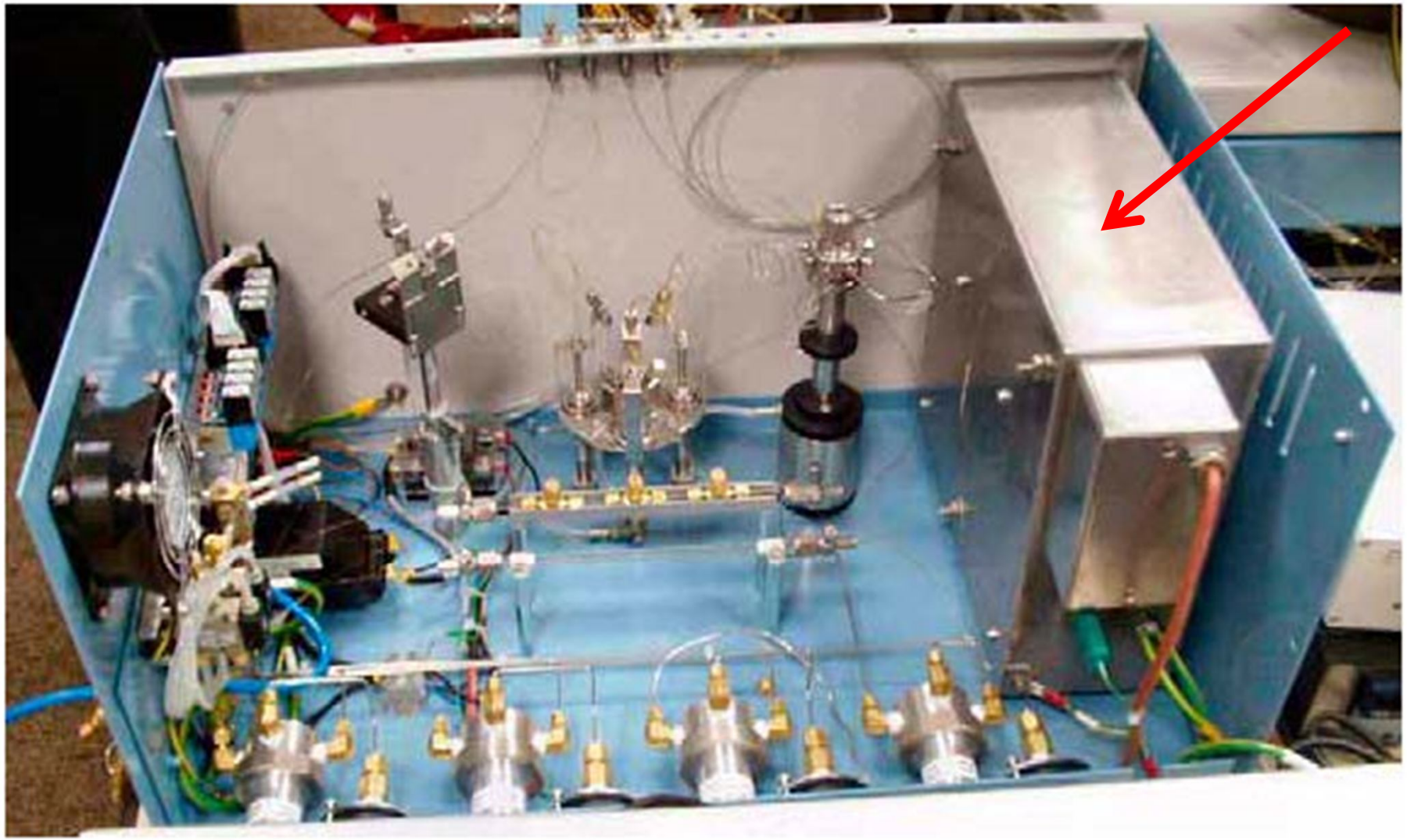
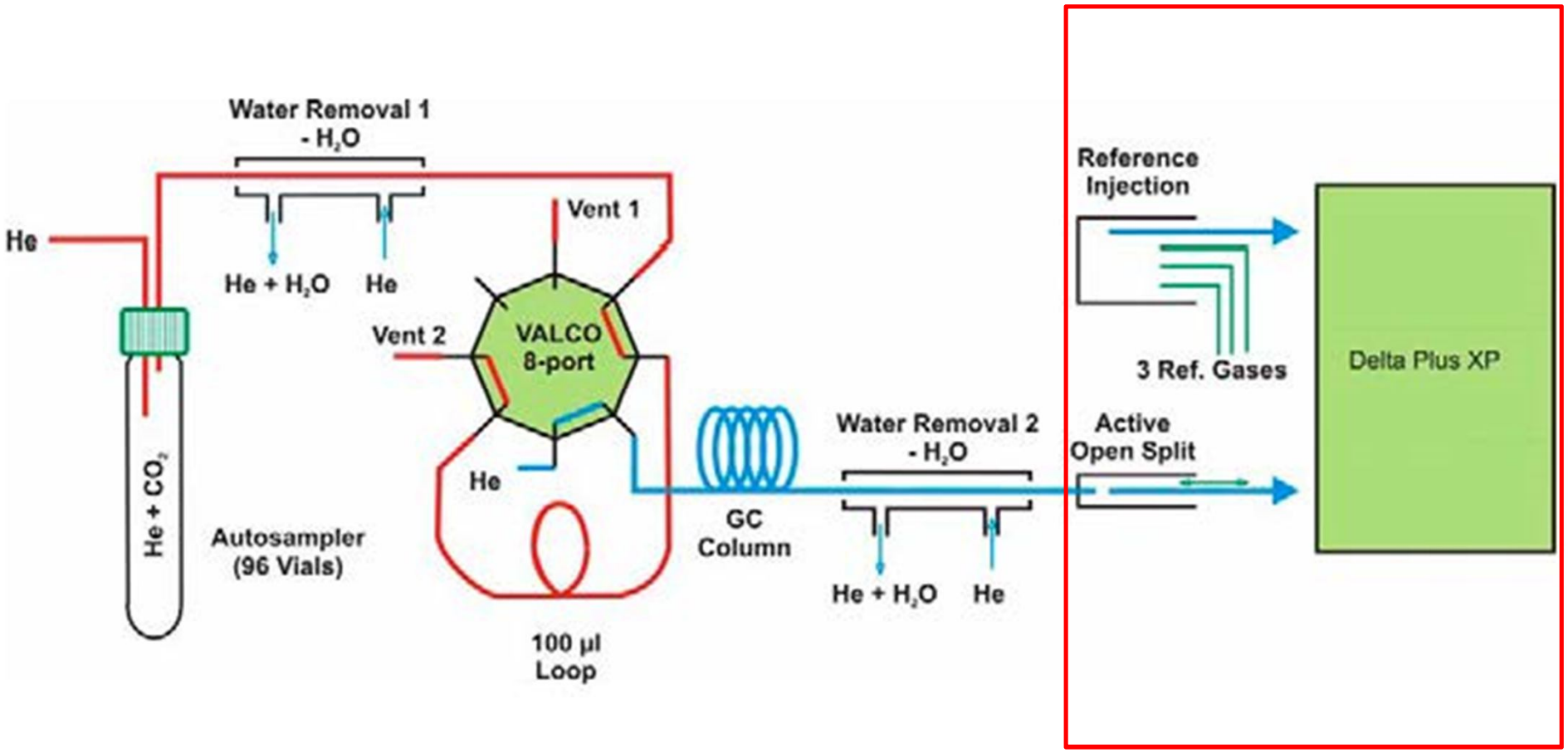
Finnigan GasBench II

72 °C

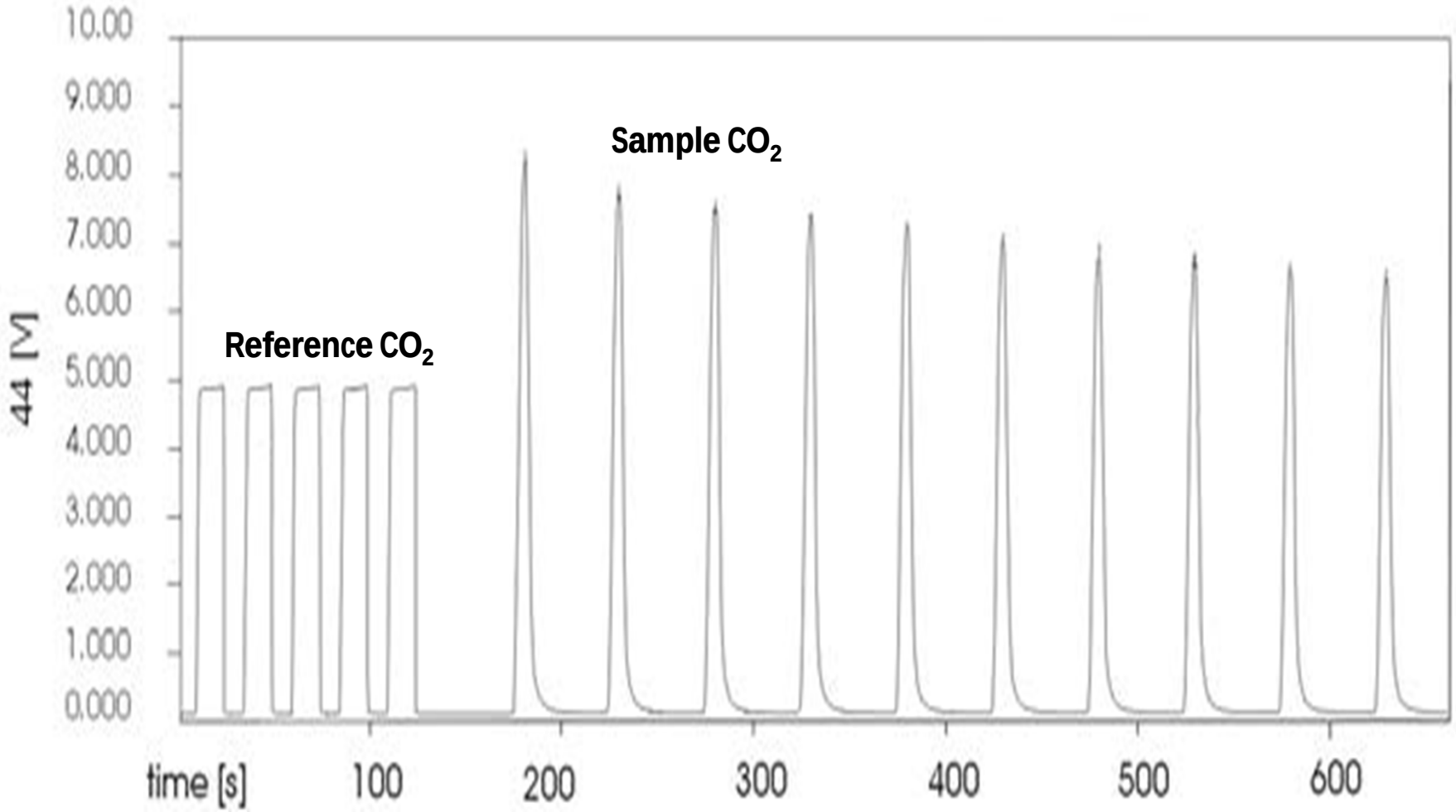


Finnigan GasBench II *Operating Manual*.



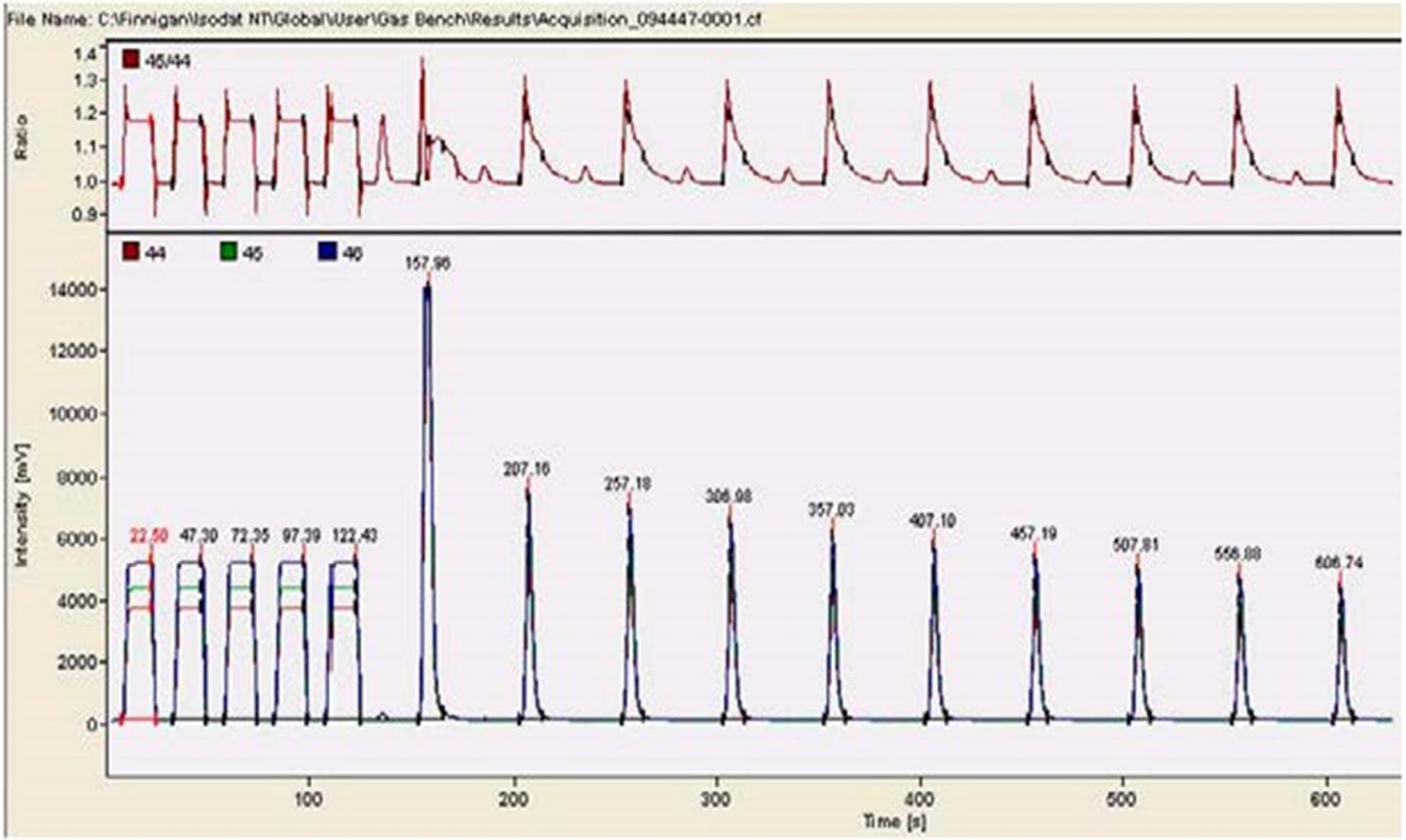
Finnigan GasBench II *Operating Manual*.

Cromatograma



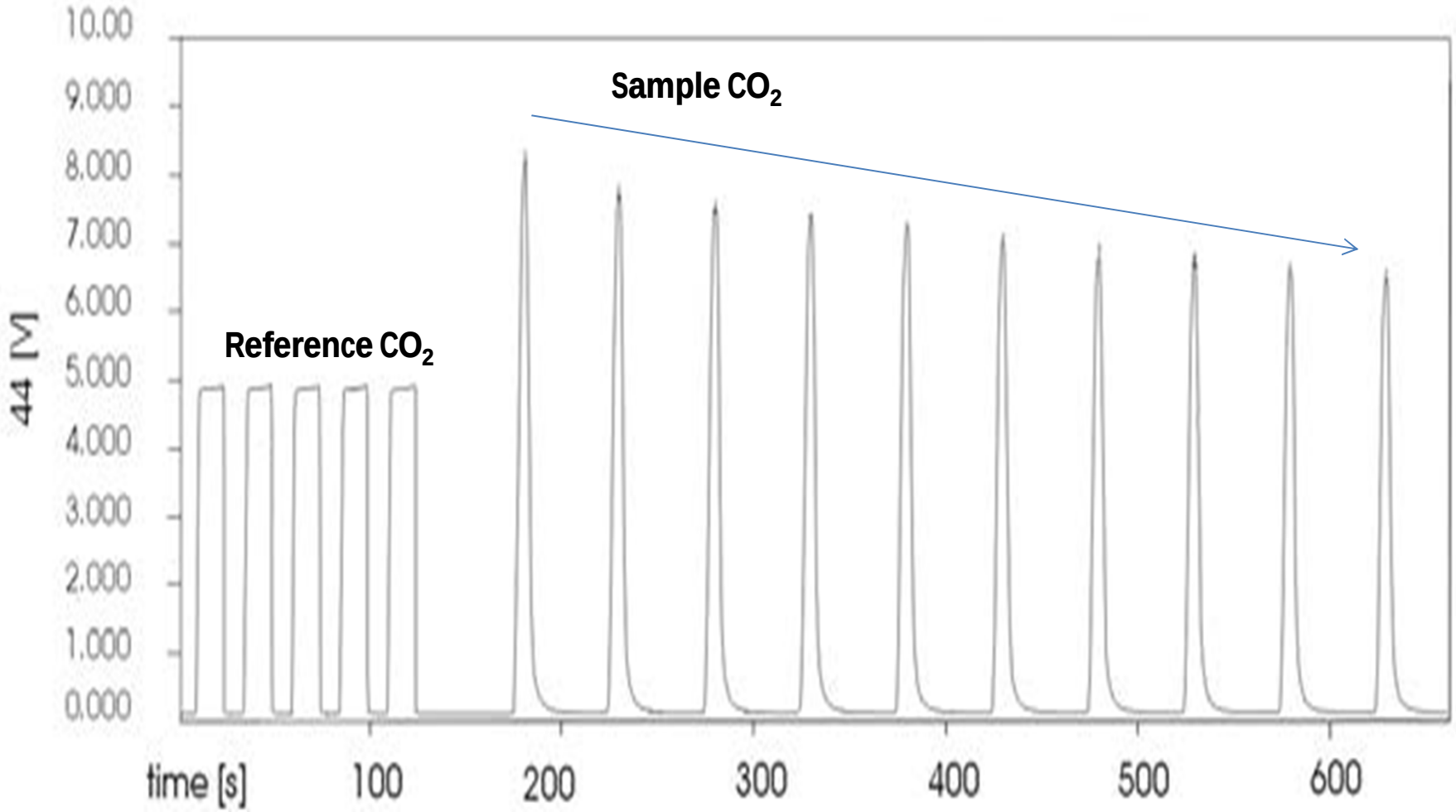
Finnigan GasBench II *Operating Manual*.

Cromatograma



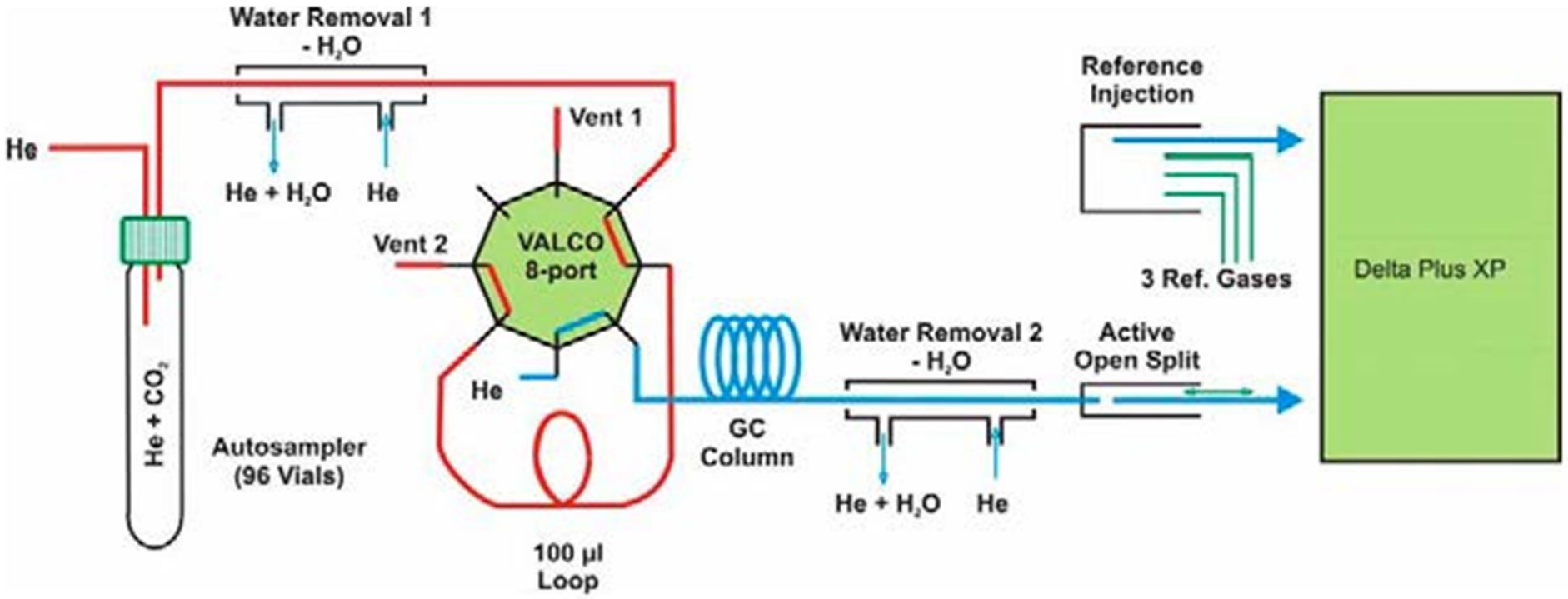
Finnigan GasBench II *Operating Manual*.

Cromatograma



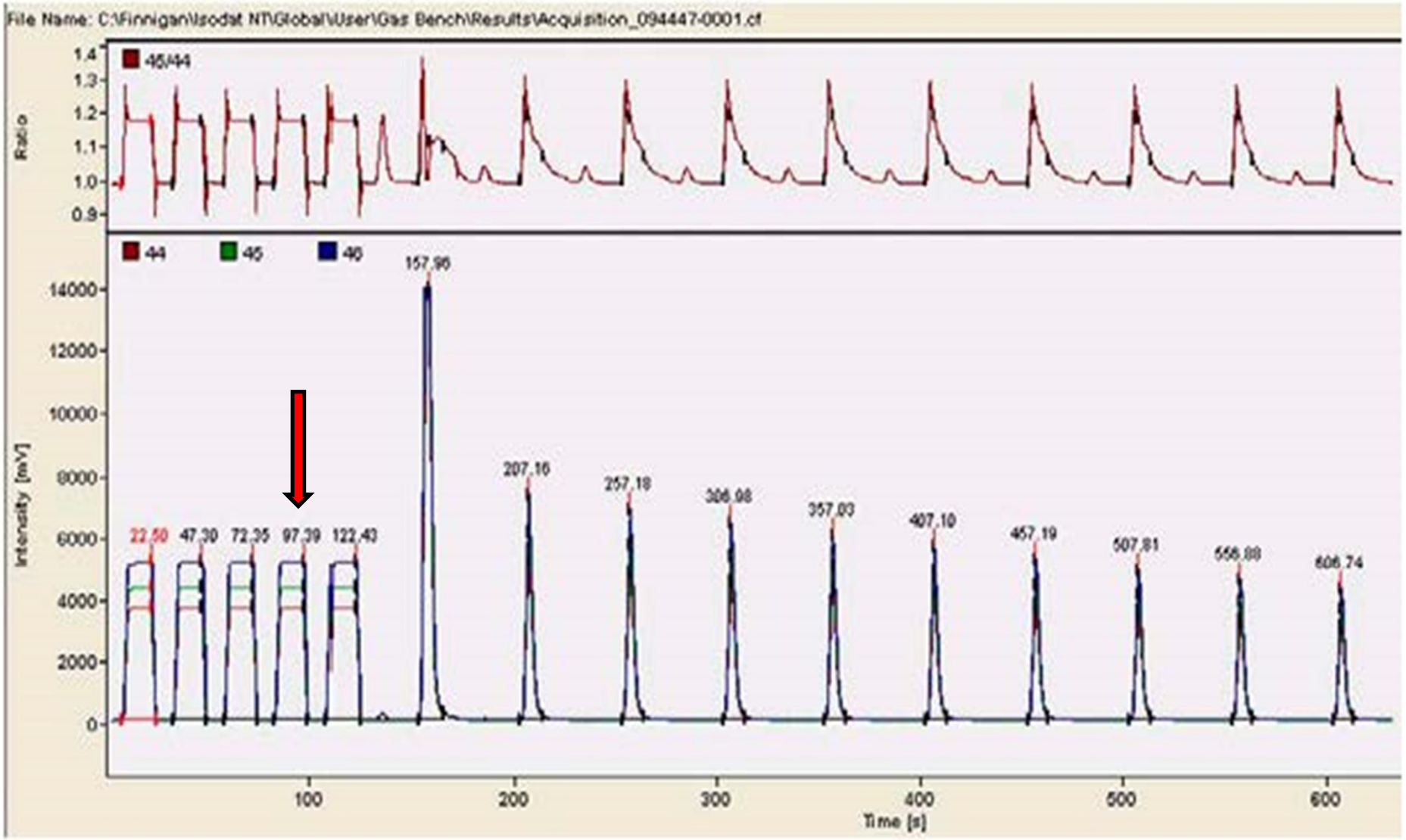
Finnigan GasBench II *Operating Manual*.

GasBench II

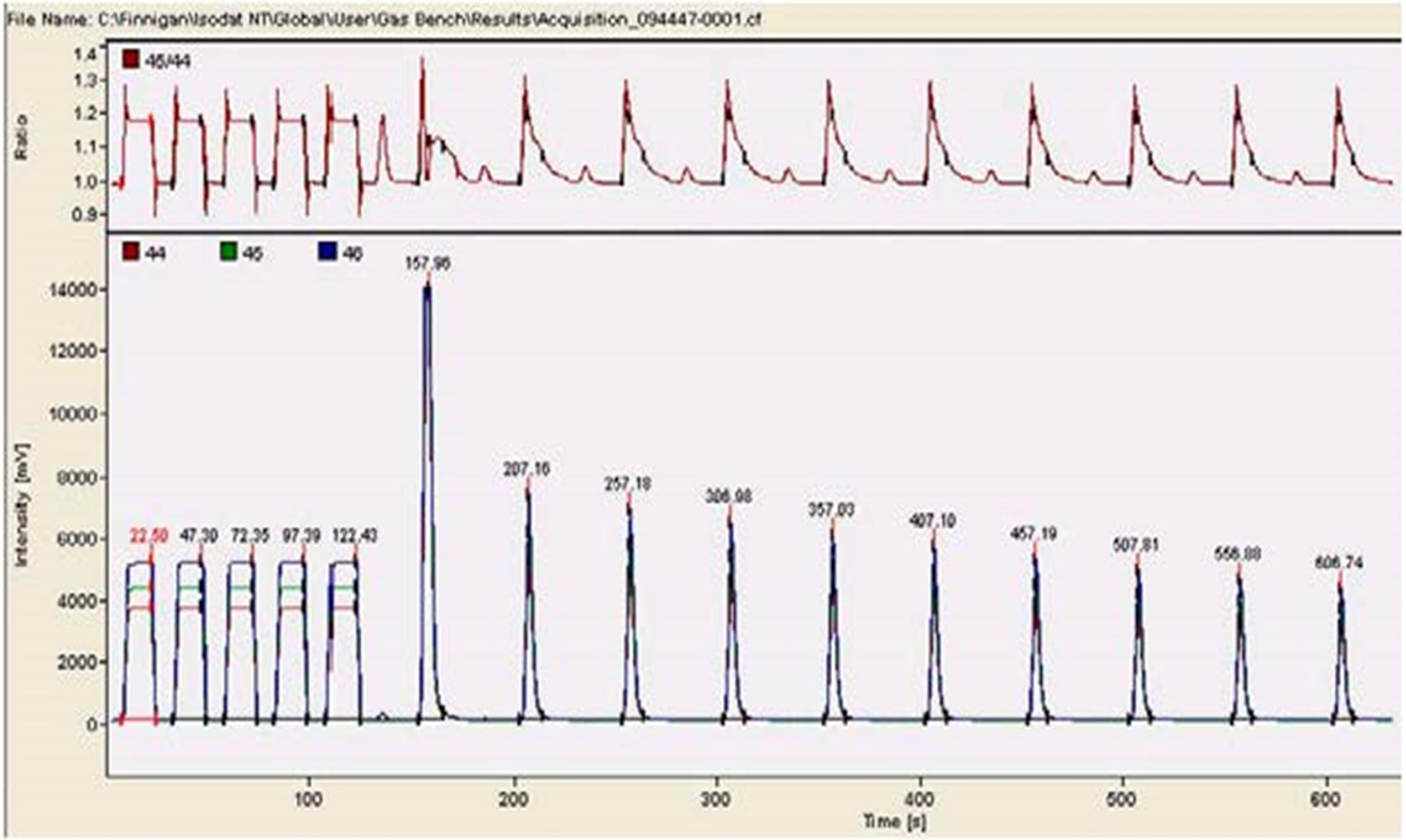


Finnigan GasBench II *Operating Manual*.

Como é calculado o valor de $\delta^{18}O$ do CO_2



Finnigan GasBench II *Operating Manual*.



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Massa 44 = $^{12}C\ ^{16}O\ ^{16}O$

Massa 46 = $^{12}C\ ^{16}O\ ^{18}O$

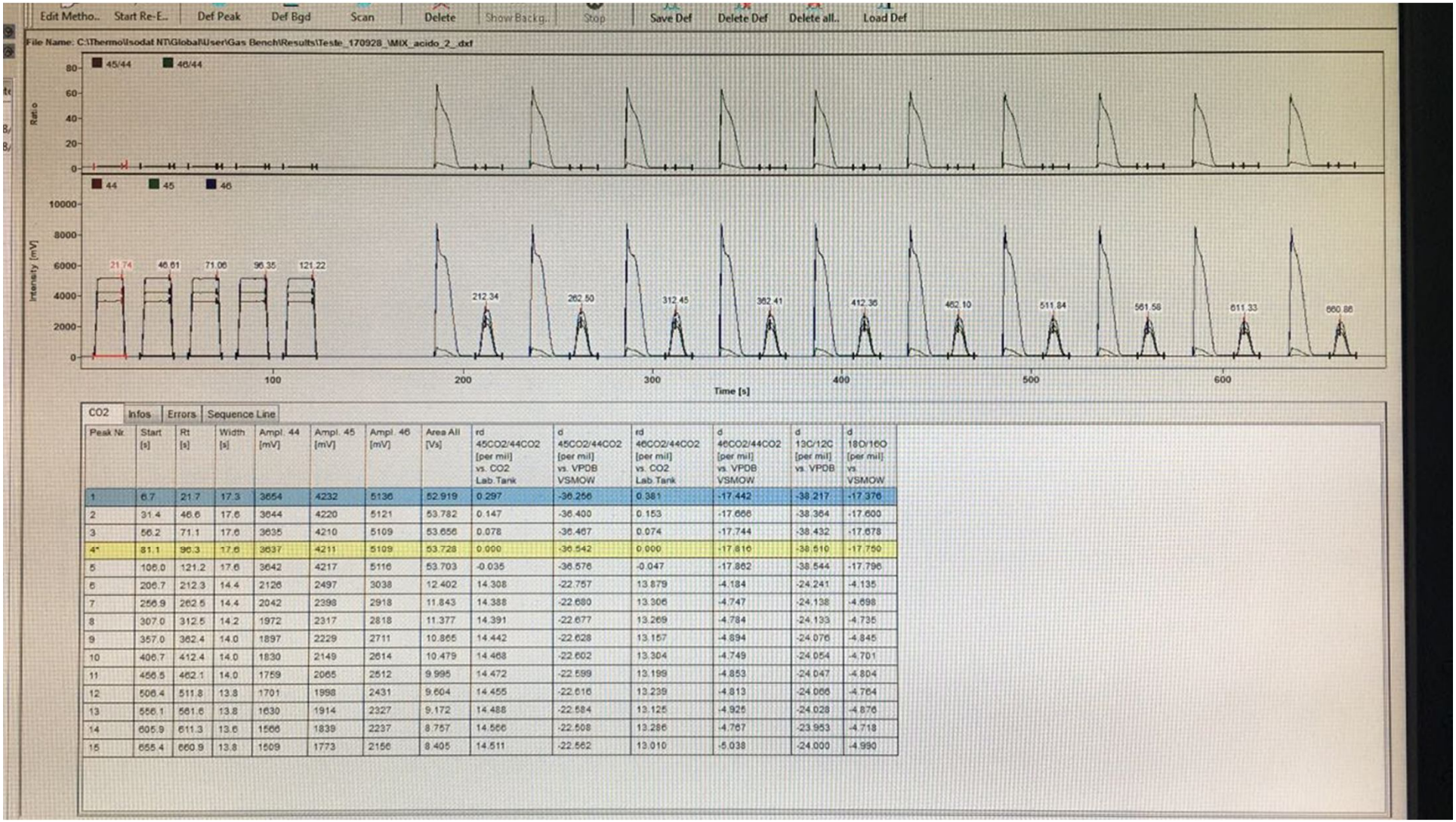
$$\frac{\mathbf{m\ 46}}{\mathbf{m\ 44}} = \frac{\mathbf{^{18}O}}{\mathbf{^{16}O}}$$

Calculo do valor de $\delta^{18}O$ da amostra

$$\frac{\left(\frac{\text{Int m 46}}{\text{Int m 44}}\right)_{\text{AMOSTRA}} - \left(\frac{\text{Int m 46}}{\text{Int m 44}}\right)_{\text{GAS DE REFERÊNCIA}}}{\left(\frac{\text{Int m 46}}{\text{Int m 44}}\right)_{\text{GAS DE REFERÊNCIA}}} \times 10^3$$

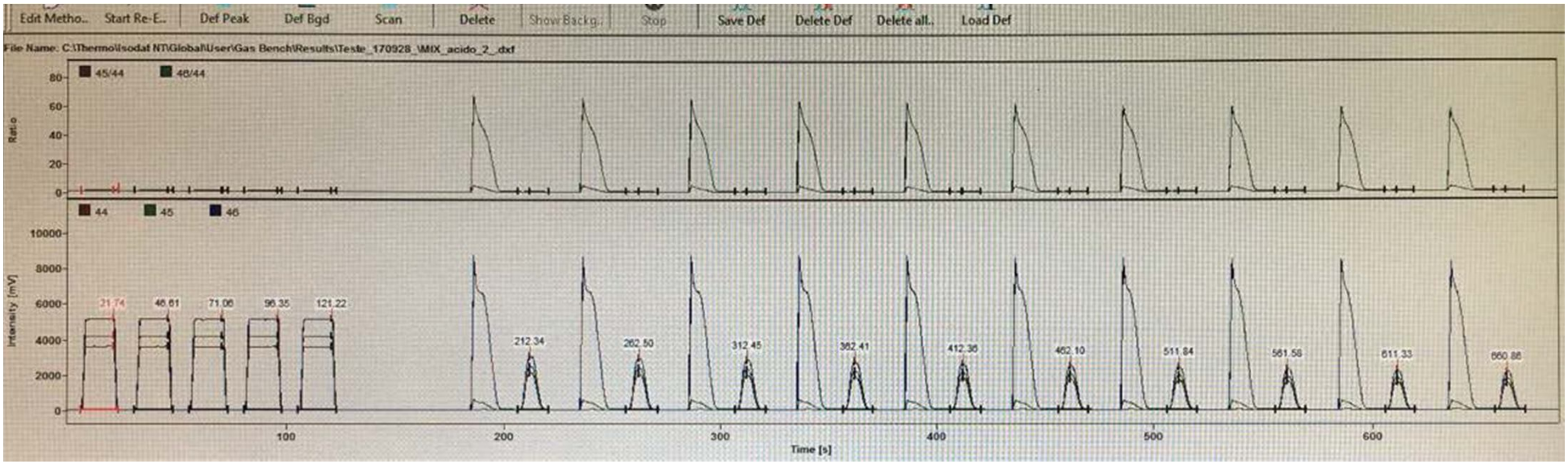
Gás de referência
não é
padrão internacional

Interferência Isobárica



CO2 | Infos | Errors | Sequence Line

Peak Nr.	Start [s]	Rt [s]	Width [s]	Ampl. 44 [mV]	Ampl. 45 [mV]	Ampl. 46 [mV]	Area All [Vs]	rd 45CO2/44CO2 [per mil] vs. CO2 Lab.Tank	d 45CO2/44CO2 [per mil] vs. VPDB VSMOW	rd 46CO2/44CO2 [per mil] vs. CO2 Lab.Tank	d 46CO2/44CO2 [per mil] vs. VPDB VSMOW	d 13C/12C [per mil] vs. VPDB	d 18O/16O [per mil] vs. VSMOW
1	6.7	21.7	17.3	3654	4232	5136	52.919	0.297	-36.256	0.381	-17.442	-38.217	-17.376
2	31.4	46.6	17.6	3644	4220	5121	53.782	0.147	-36.400	0.153	-17.666	-38.364	-17.600
3	56.2	71.1	17.6	3635	4210	5109	53.656	0.078	-36.467	0.074	-17.744	-38.432	-17.678
4*	81.1	96.3	17.6	3637	4211	5109	53.728	0.000	-36.542	0.000	-17.816	-38.510	-17.750
5	106.0	121.2	17.6	3642	4217	5116	53.703	-0.035	-36.576	-0.047	-17.862	-38.544	-17.796
6	206.7	212.3	14.4	2126	2497	3038	12.402	14.308	-22.757	13.879	-4.184	-24.241	-4.135
7	256.9	262.5	14.4	2042	2398	2918	11.843	14.388	-22.680	13.306	-4.747	-24.138	-4.698
8	307.0	312.5	14.2	1972	2317	2818	11.377	14.391	-22.677	13.269	-4.784	-24.133	-4.735
9	357.0	362.4	14.0	1897	2229	2711	10.865	14.442	-22.628	13.157	-4.894	-24.076	-4.845
10	406.7	412.4	14.0	1830	2149	2614	10.479	14.468	-22.602	13.304	-4.749	-24.054	-4.701
11	456.5	462.1	14.0	1759	2065	2512	9.995	14.472	-22.599	13.199	-4.853	-24.047	-4.804
12	506.4	511.8	13.8	1701	1998	2431	9.604	14.455	-22.616	13.239	-4.813	-24.066	-4.764
13	556.1	561.6	13.8	1630	1914	2327	9.172	14.488	-22.584	13.125	-4.925	-24.028	-4.876
14	605.9	611.3	13.6	1566	1839	2237	8.757	14.566	-22.508	13.286	-4.767	-23.953	-4.718
15	655.4	660.9	13.8	1509	1773	2156	8.405	14.511	-22.562	13.010	-5.038	-24.000	-4.990


Edit Metho..
Start Re-E..
Def Peak
Def Bgd
Scan
Delete
Show Backg..
Stop
Save Def
Delete Def
Delete all..
Load Def
File Name: C:\Thermo\Isodat NT\Global\User\Gas Bench\Results\Teste_170928_\MIX_acido_2_.dxf
45/44
46/44
Ratio
44
45
46
Intensity (mV)
21.74
46.61
71.06
96.35
121.22
212.34
262.50
312.45
362.41
412.36
462.10
511.84
561.58
611.33
660.88
Time [s]

Testes para verificar o bom funcionamento do IRMS antes da analise

Testes antes da analise

- **Stability test** (chamado também *Zero-Enrichment Test* ou *Standard on-off*)

- **Linearity test**

Stability Test

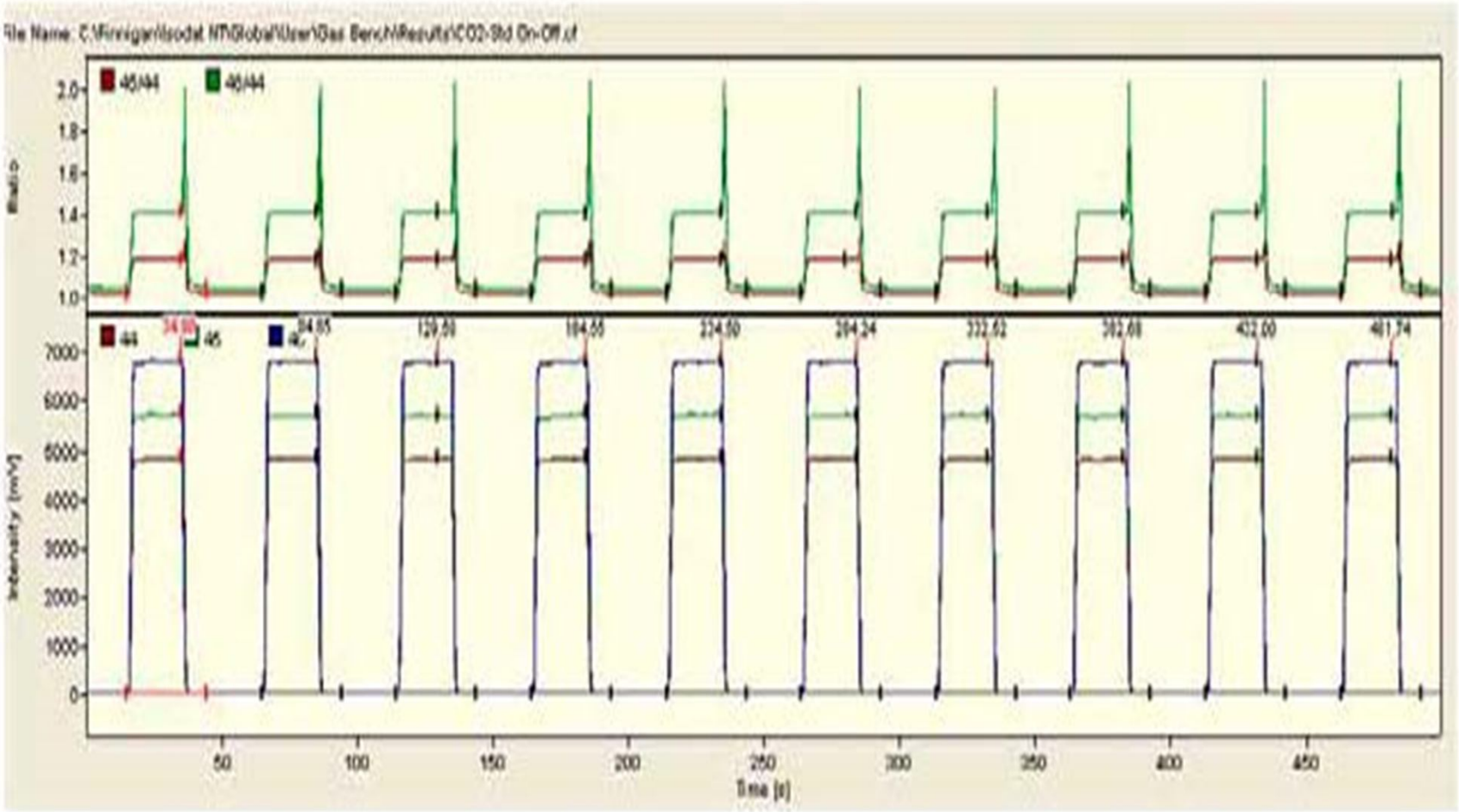


Figure 4-3. Zero Enrichment - Chromatogram

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Linearity Test

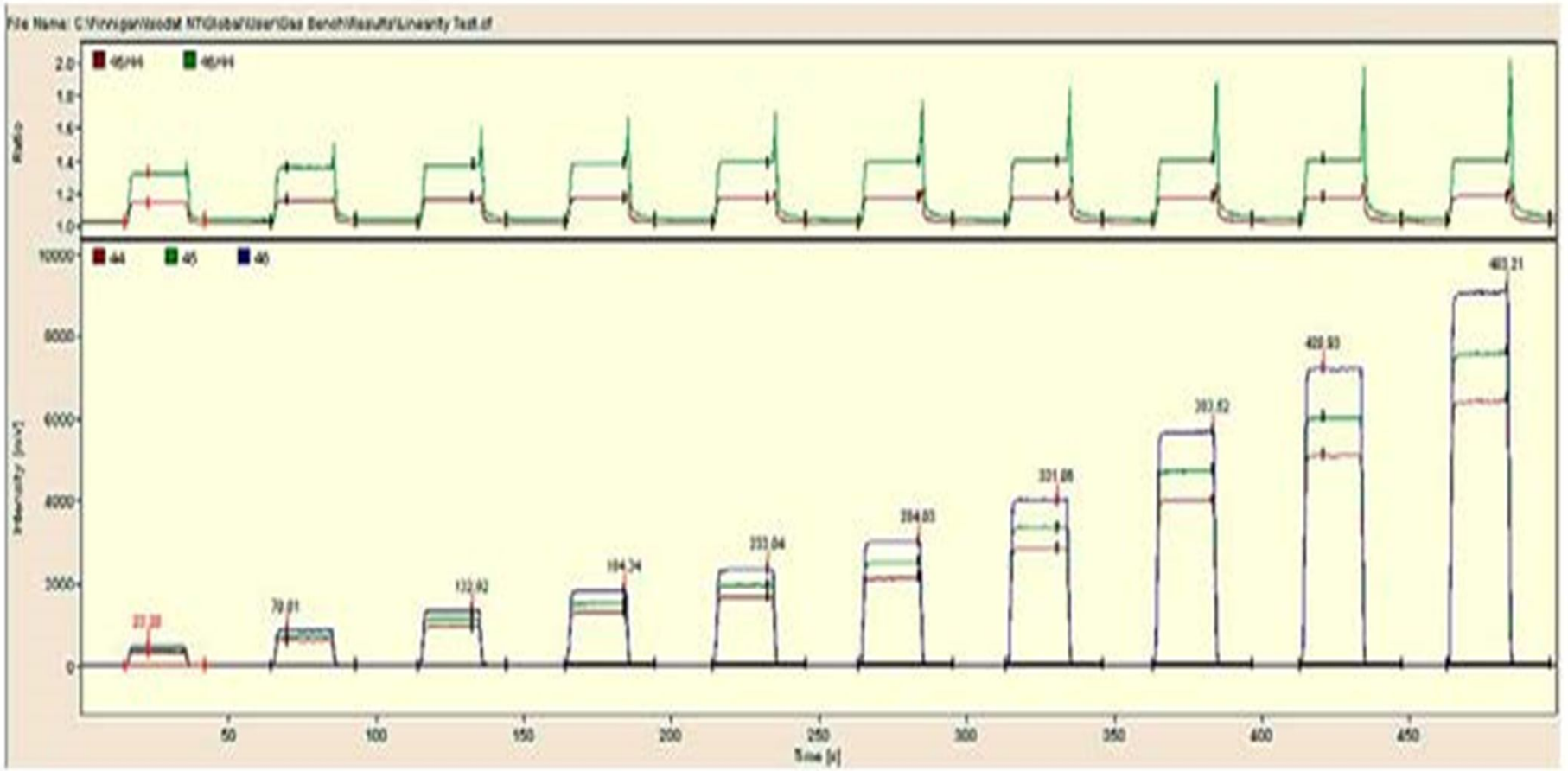


Figure 4-5. Linearity Test - Chromatogram

Finnigan GasBench II *Operating Manual*.

Testes antes da analise

- **Stability test:** desvio padrão admitido = ±0,06 por mil para $\delta^{18}O$ e $\delta^{13}C$

- **Linearity test:** pendência admitida = ±0,5 por mil/V para $\delta^{18}O$ e $\delta^{13}C$

Gás de referência
não é
padrão internacional

Uso dos padrões internos e internacionais

Uso dos padrões internos e internacionais

1) Correção dos valores de $\delta^{18}O$ medidos (*raw data*), de forma que sejam exprimido em relação ao padrão internacional (V-SMOW)

2) Correção da deriva instrumental

3) Correção do efeito de non linearidade (quando o valor de $\delta^{18}O$ depende da intensidade do sinal em Volts)

Uso dos padrões internos e internacionais

1) Correção dos valores de $\delta^{18}O$ medidos (*raw data*), de forma que sejam exprimido em relação ao padrão internacional (V-SMOW)

2) Correção da deriva instrumental

3) Correção do efeito de non linearidade (quando o valor de $\delta^{18}O$ depende da intensidade do sinal em Volts)

Calculo do valor de $\delta^{18}O$ da amostra

RAW $\delta^{18}O$

$$\frac{\left(\frac{\text{Int m 46}}{\text{Int m 44}}\right)_{\text{AMOSTRA}} - \left(\frac{\text{Int m 46}}{\text{Int m 44}}\right)_{\text{GAS DE REFERÊNCIA}}}{\left(\frac{\text{Int m 46}}{\text{Int m 44}}\right)_{\text{GAS DE REFERÊNCIA}}} \times 10^3$$

Correção de dados brutos:

- com um padrão
- com mais do que um padrão

Correção com um padrão

Correção = $\delta^{18}O$ medido - $\delta^{18}O$ certificado

logo:

$\delta^{18}O$ corrigido = $\delta^{18}O$ medido + Correção

Note que para o padrão

$\delta^{18}O$ corrigido deveria ser = $\delta^{18}O$ certificado

(acurácia)

Correção com um padrão

Correção = $\delta^{18}O$ medido - $\delta^{18}O$ certificado

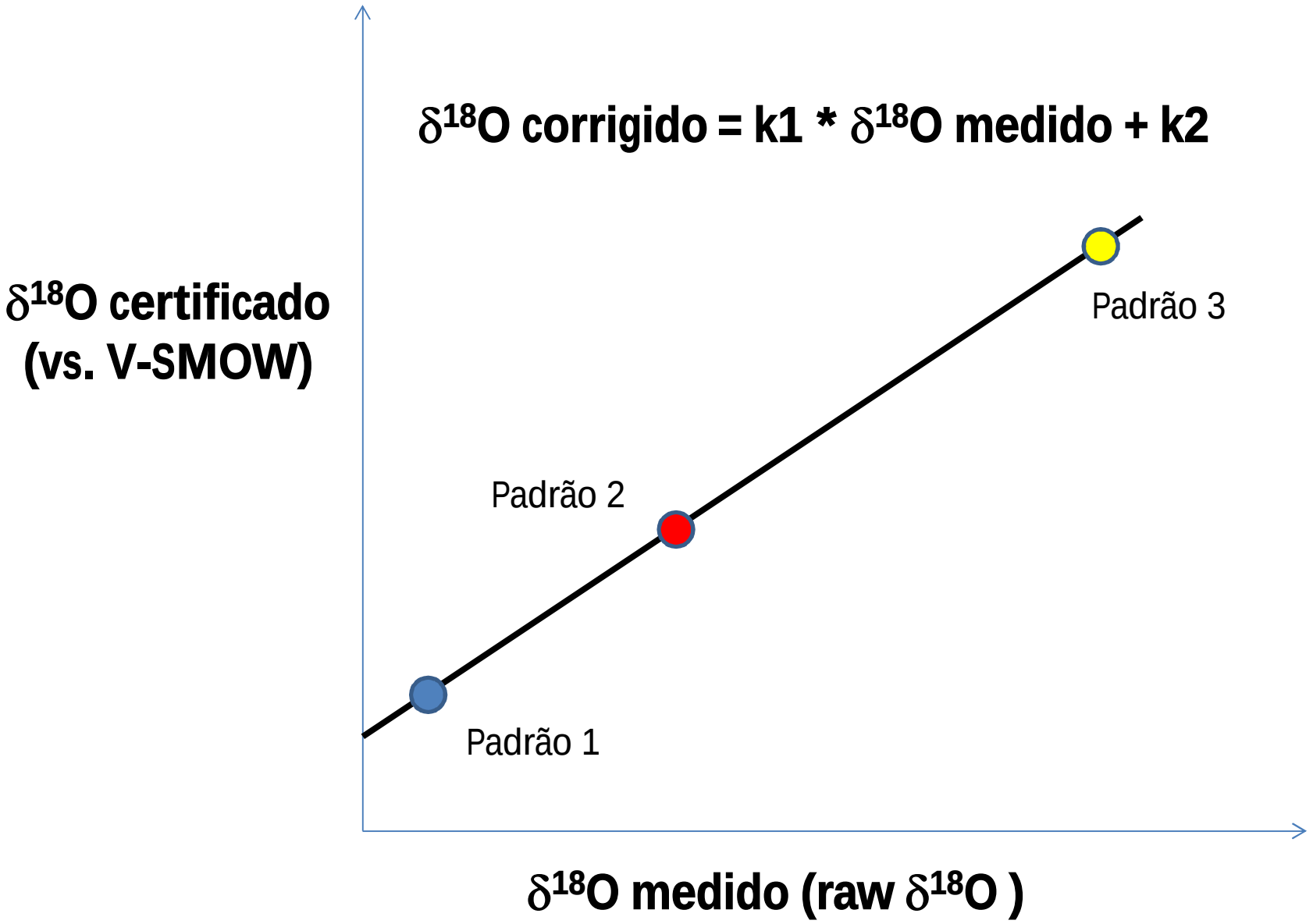
VANTAGEM: simplicidade

DESVANTAGEM: se o valor isotópico do padrão é muito diferente do da amostra a precisão cai

Correção de dados brutos:

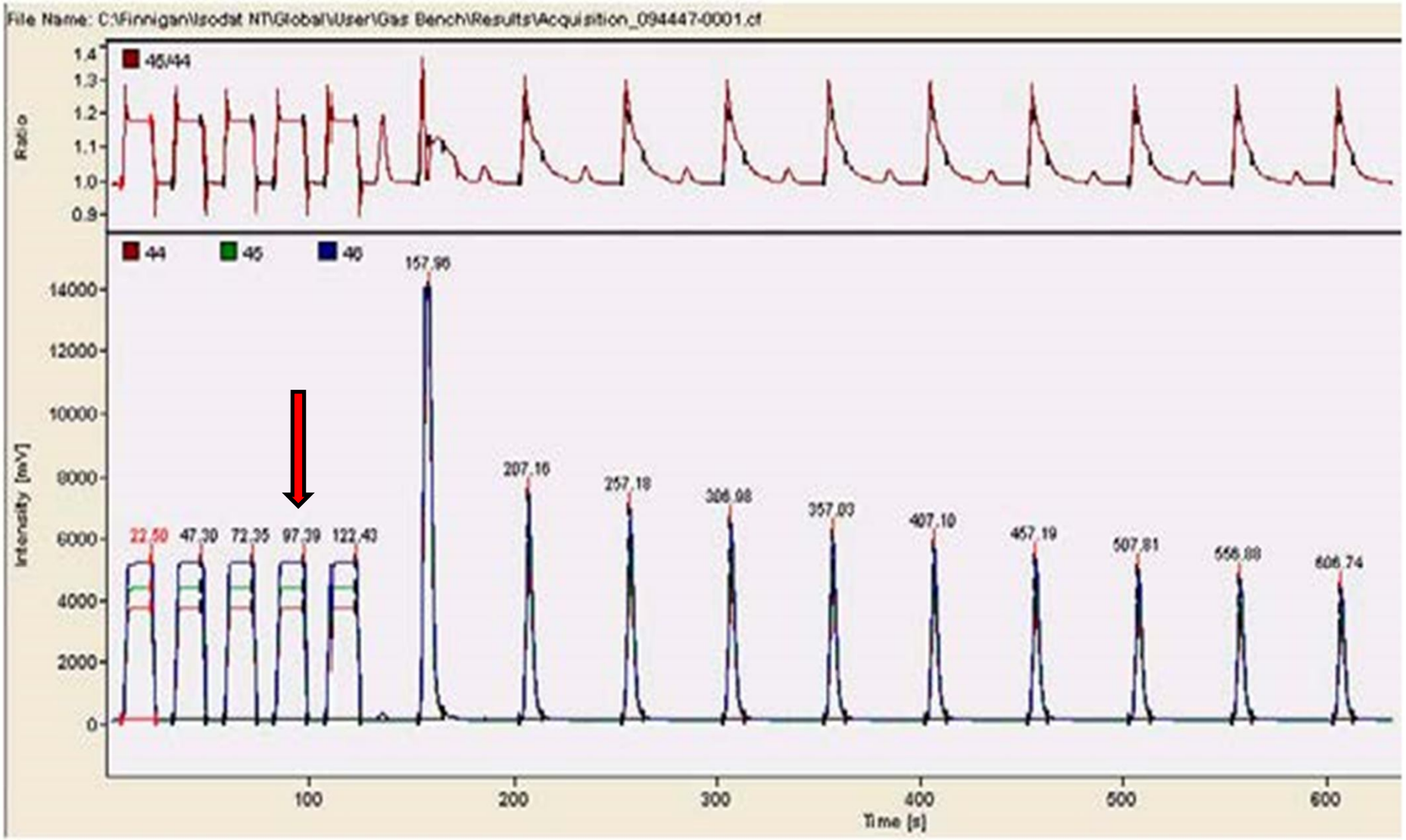
- **com um padrão**
- **com mais do que um padrão**

Correção com um conjunto de padrões
(3 á 4, analisados em triplicata)


δ18O corrigido = k1 * δ18O medido + k2
δ18O certificado (vs. V-SMOW)
Padrão 3
Padrão 2
Padrão 1
δ18O medido (raw δ18O)

Idealmente, por cada padrão o valor de $\delta^{18}O$ corrigido deveria resultar igual ao $\delta^{18}O$ certificado (acurácia da medição)

Precisão vs. acurácia



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Correção com 3 ou mais padrões

$\delta^{18}O$ corrigido = k1 * $\delta^{18}O$ medido + k2

VANTAGEM: dá para analisar amostras com valores isotópicos bem heterogêneos

DESVANTAGEM: muito mais trabalhoso. Além disso, os padrões podem apresentar dispersão nos valores

Uso dos padrões internos e internacionais

1) Correção dos valores de $\delta^{18}O$ medidos (*raw data*), de forma que sejam exprimido em relação ao padrão internacional (V-SMOW)

2) Correção da deriva instrumental

3) Correção do efeito de non linearidade (quando o valor de $\delta^{18}O$ depende da intensidade do sinal em Volts)

Figure 2-8. Non-Thermostated Sample Tray - Side View (Part No. 111 2780)

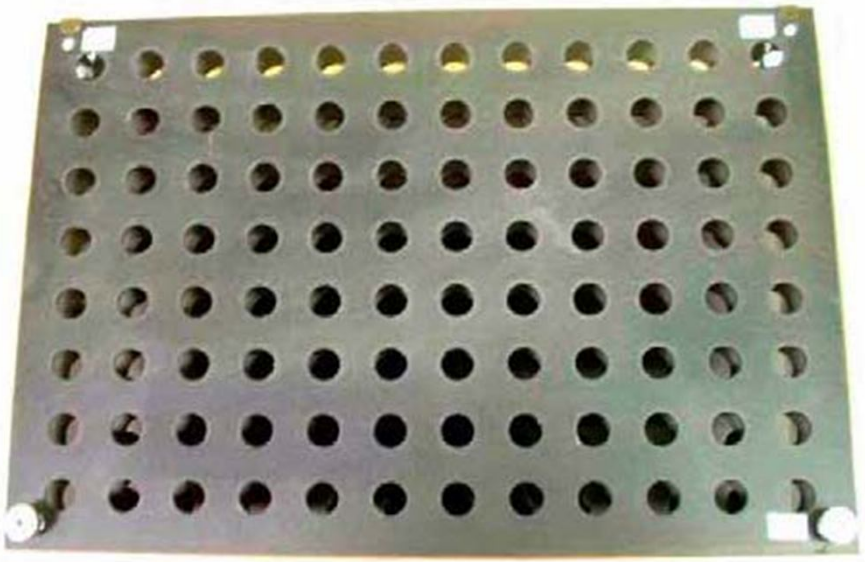
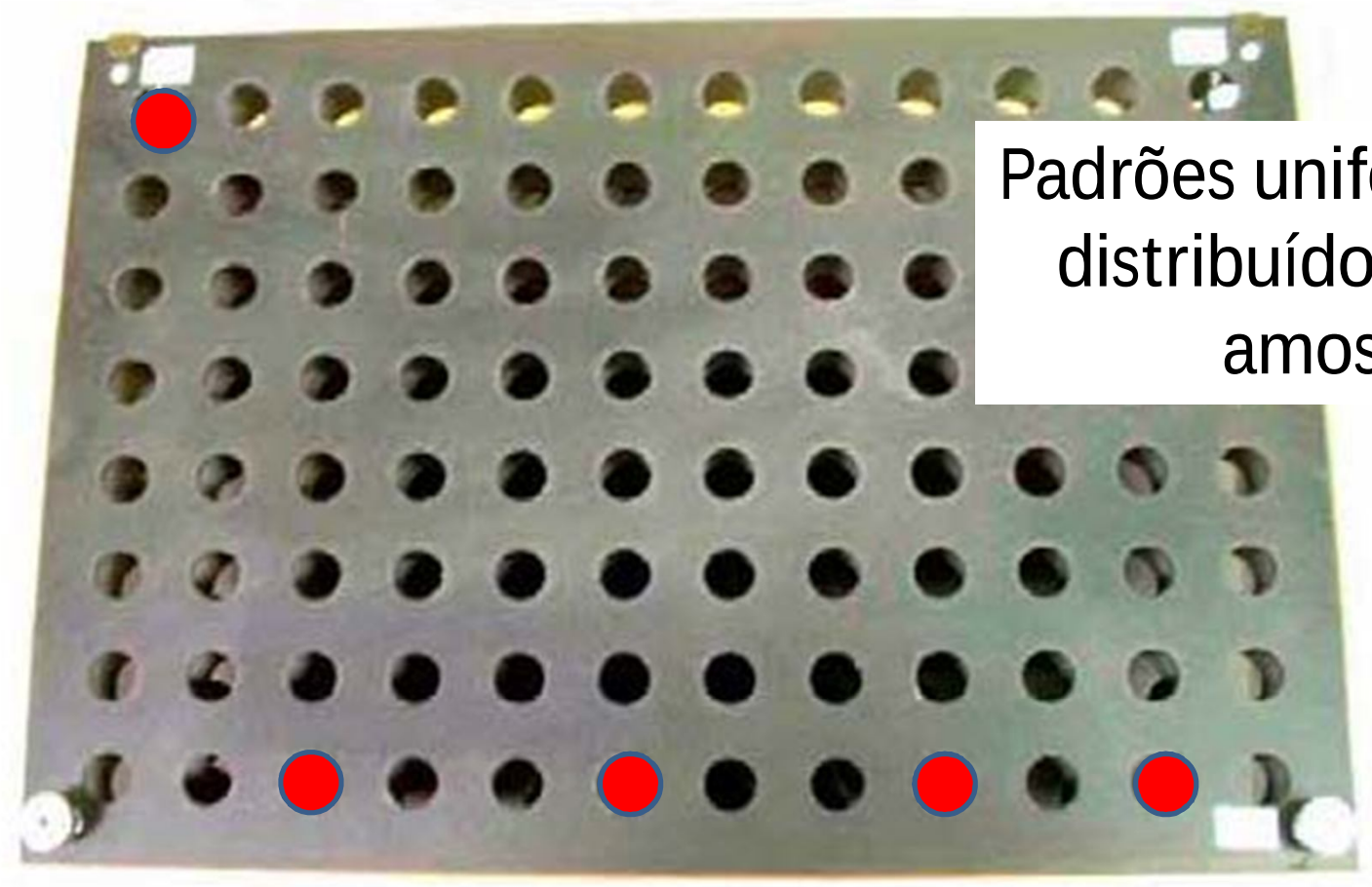


Figure 2-9. Non-Thermostated Sample Tray - Top View (Part No. 111 2780)



Padrões uniformemente distribuídos entre as amostras

Figure 2-9. **Non-Thermostated Sample Tray - Top View (Part No. 111 2780)**

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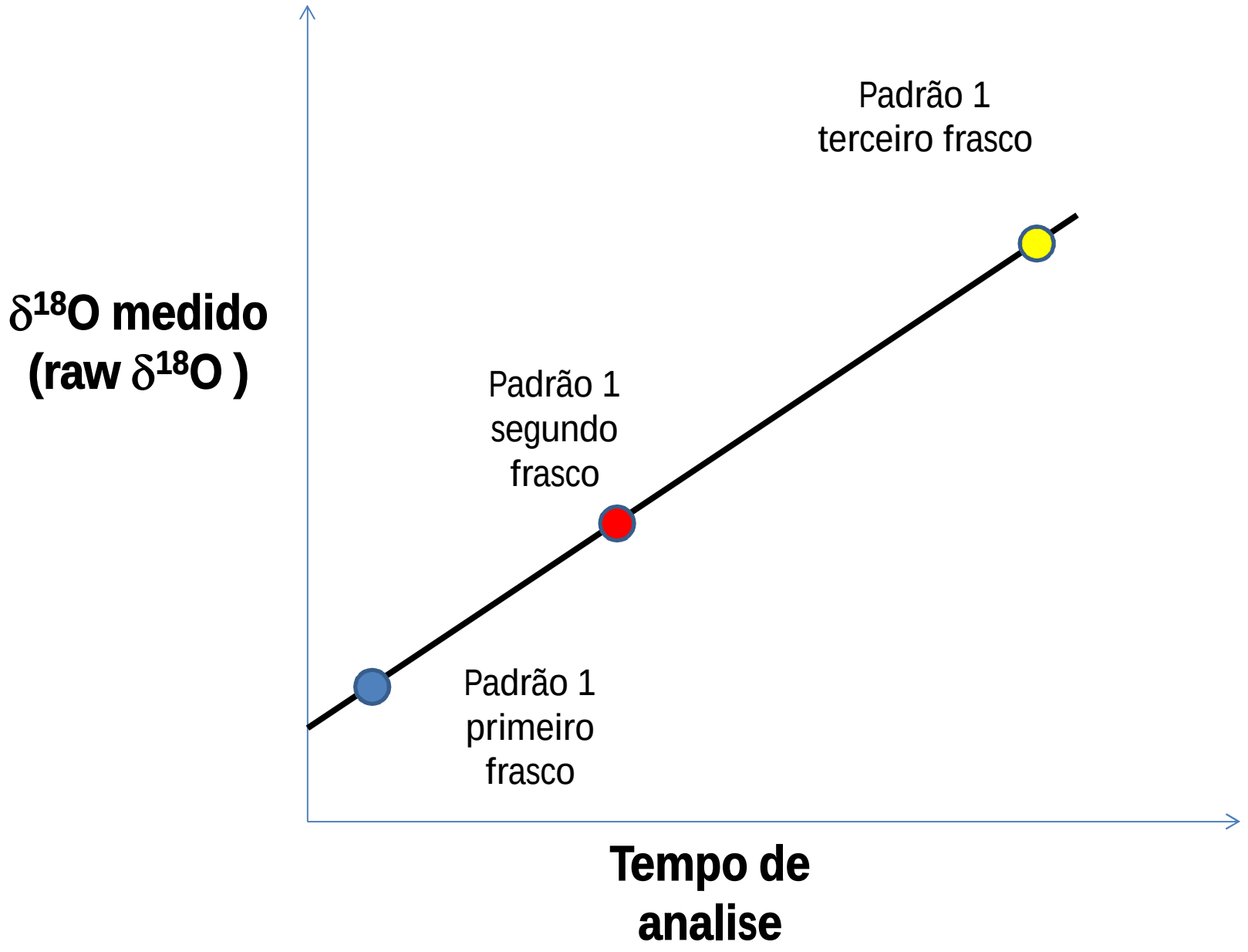
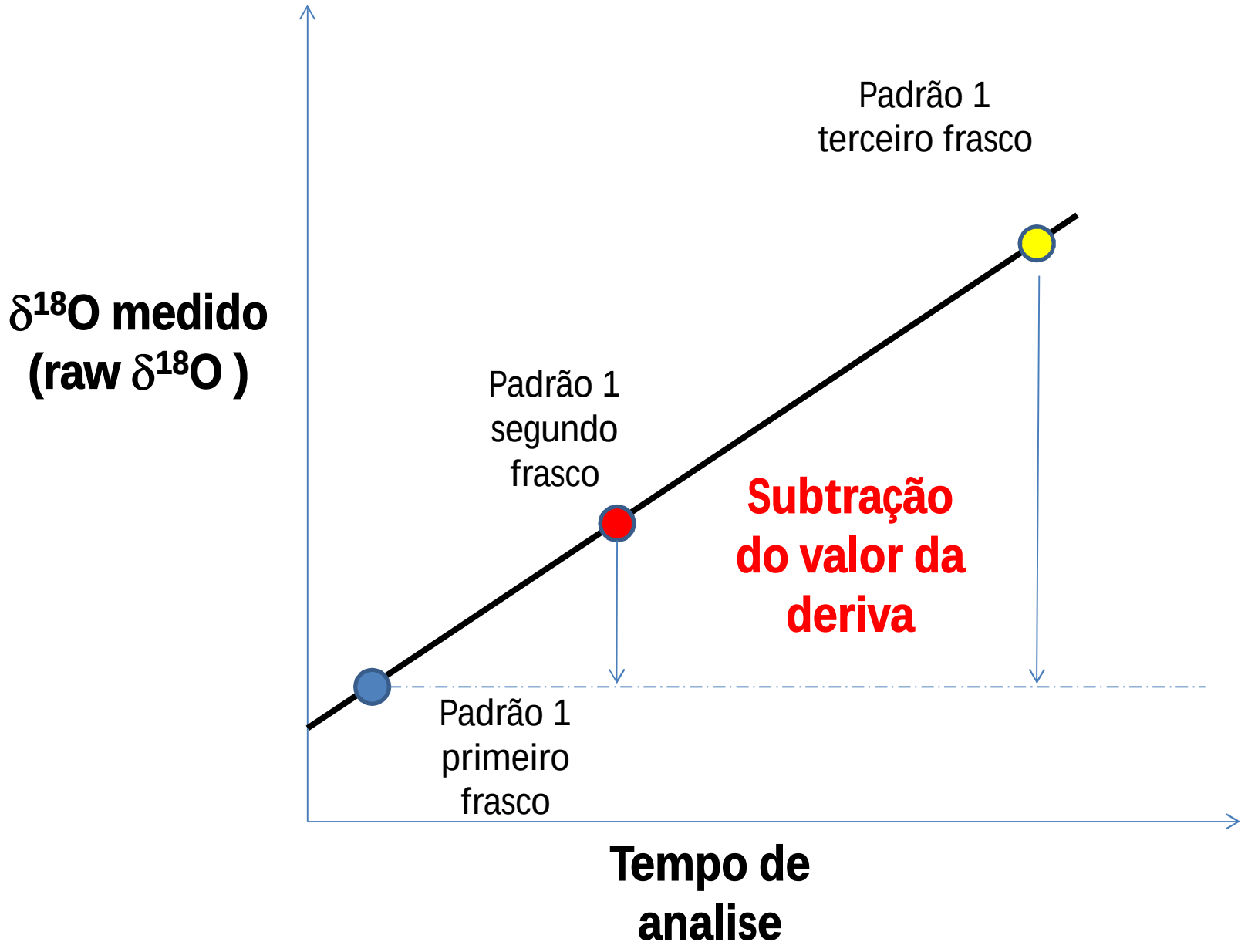
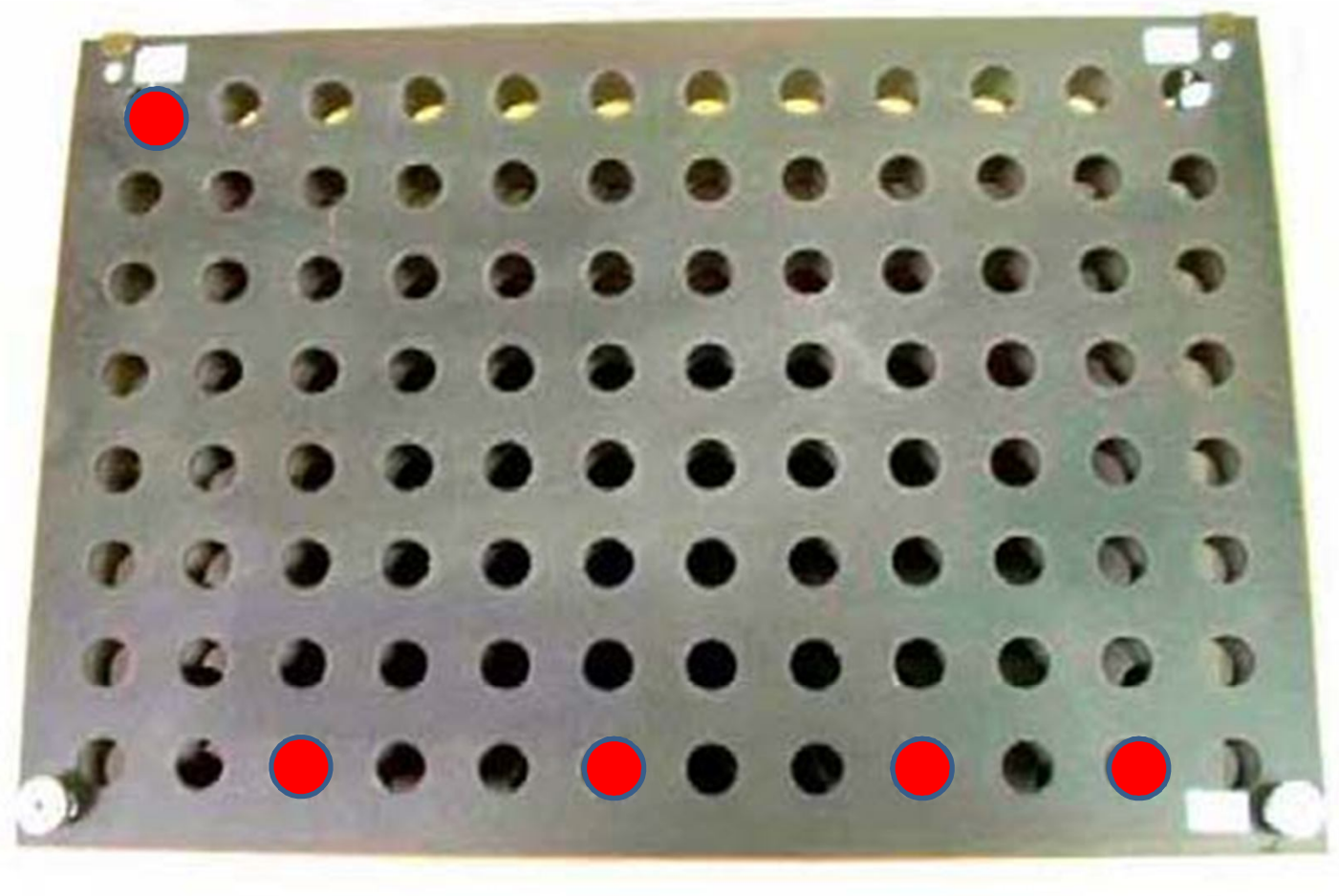
Deriva instrumental



Como corrigir a deriva instrumental

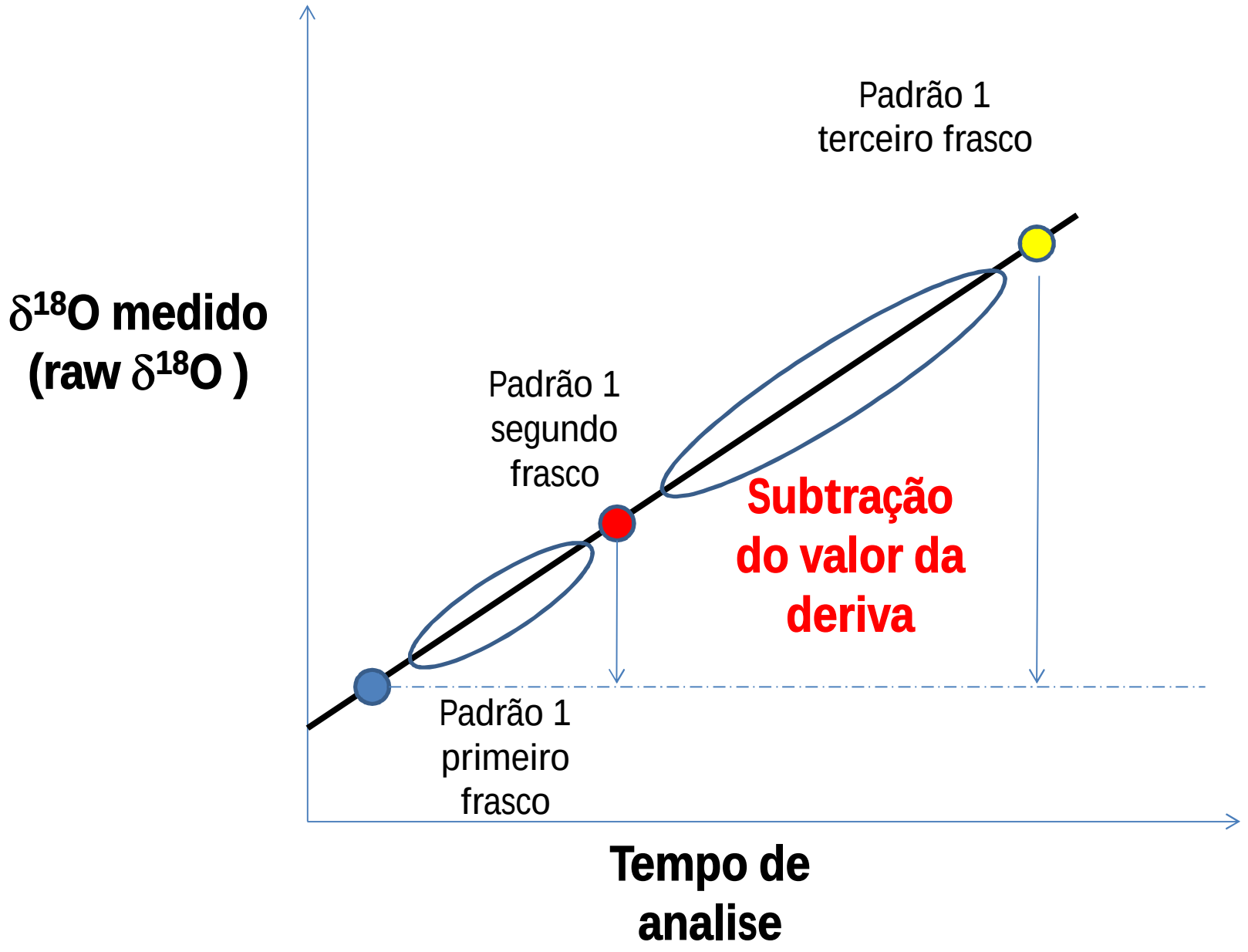




Esta correção deve ser aplicada para todas as amostras, conforme o tempo de analise de cada um

Finnigan GasBench II *Operating Manual*.

Como corrigir a deriva instrumental



Analise isotópica de hidrogênio em água (do mar)

H Device

Finnigan GasBench II *Operating Manual*.

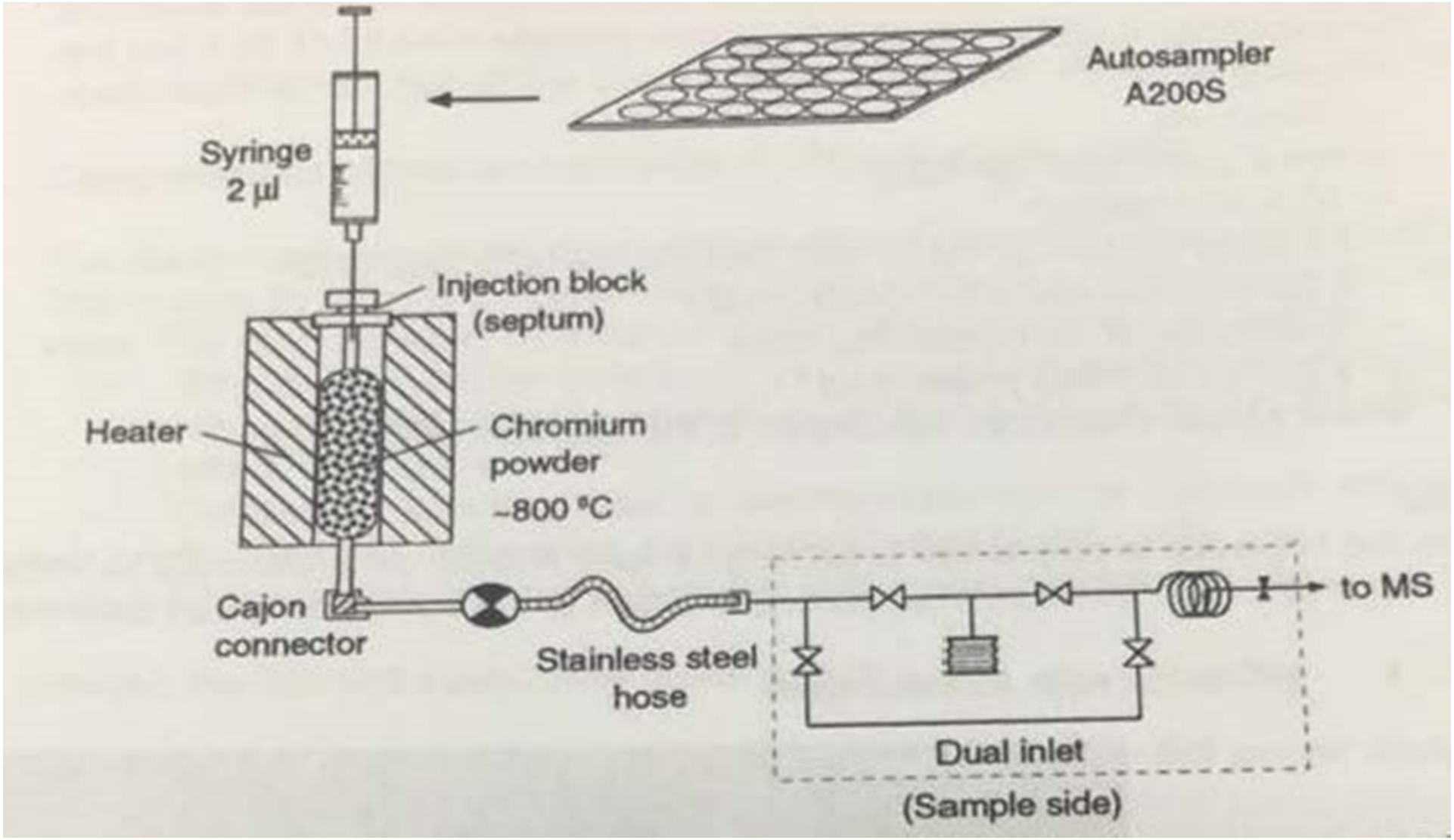
Finnigan H/Device

Finnigan Delta V Isotope Ratio Mass Spectrometer

NSU 011-00A
Volume 10ul
Scale 54 mm
ID 01

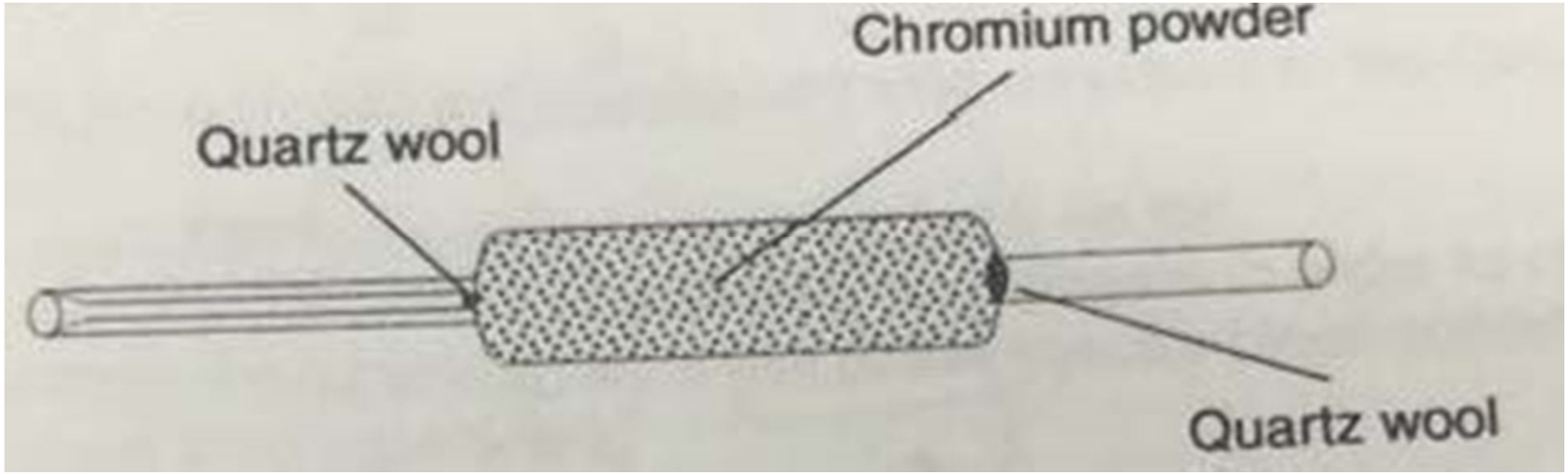
SGE
MADE IN AUSTRALIA
1 mL

H Device : princípio de funcionamento



Finnigan H/Device *Operating Manual*.

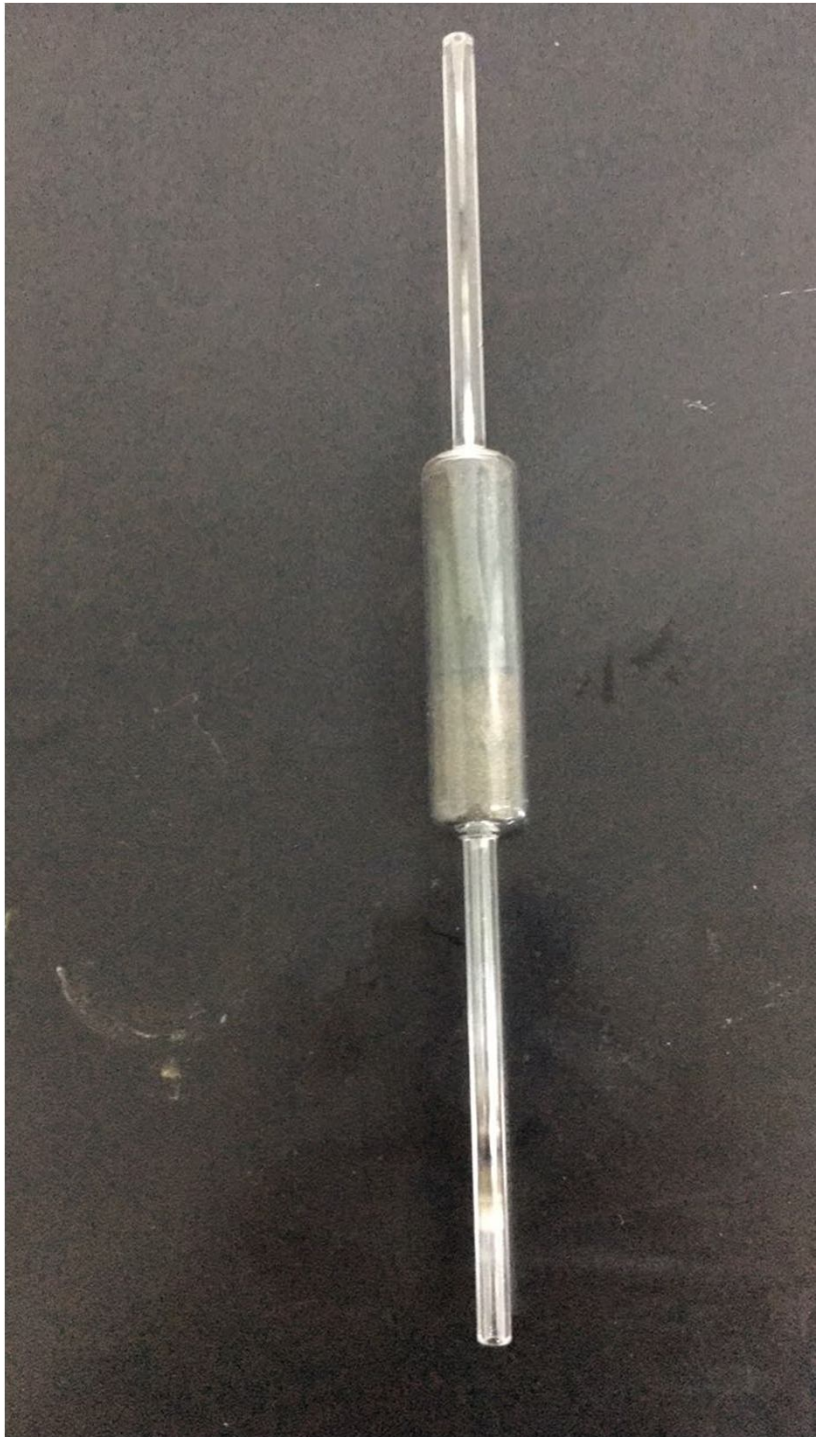
H Device : princípio de funcionamento

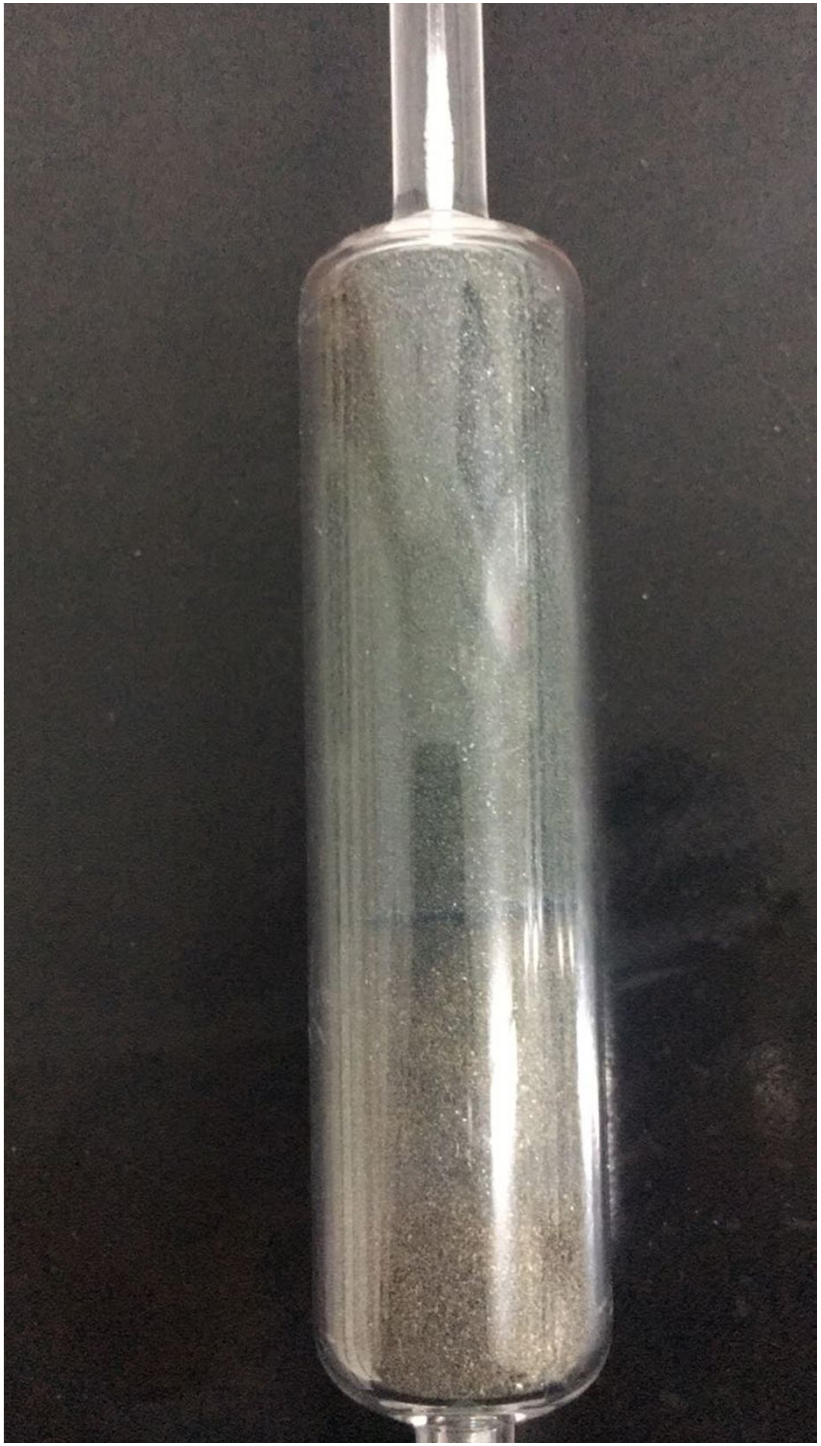


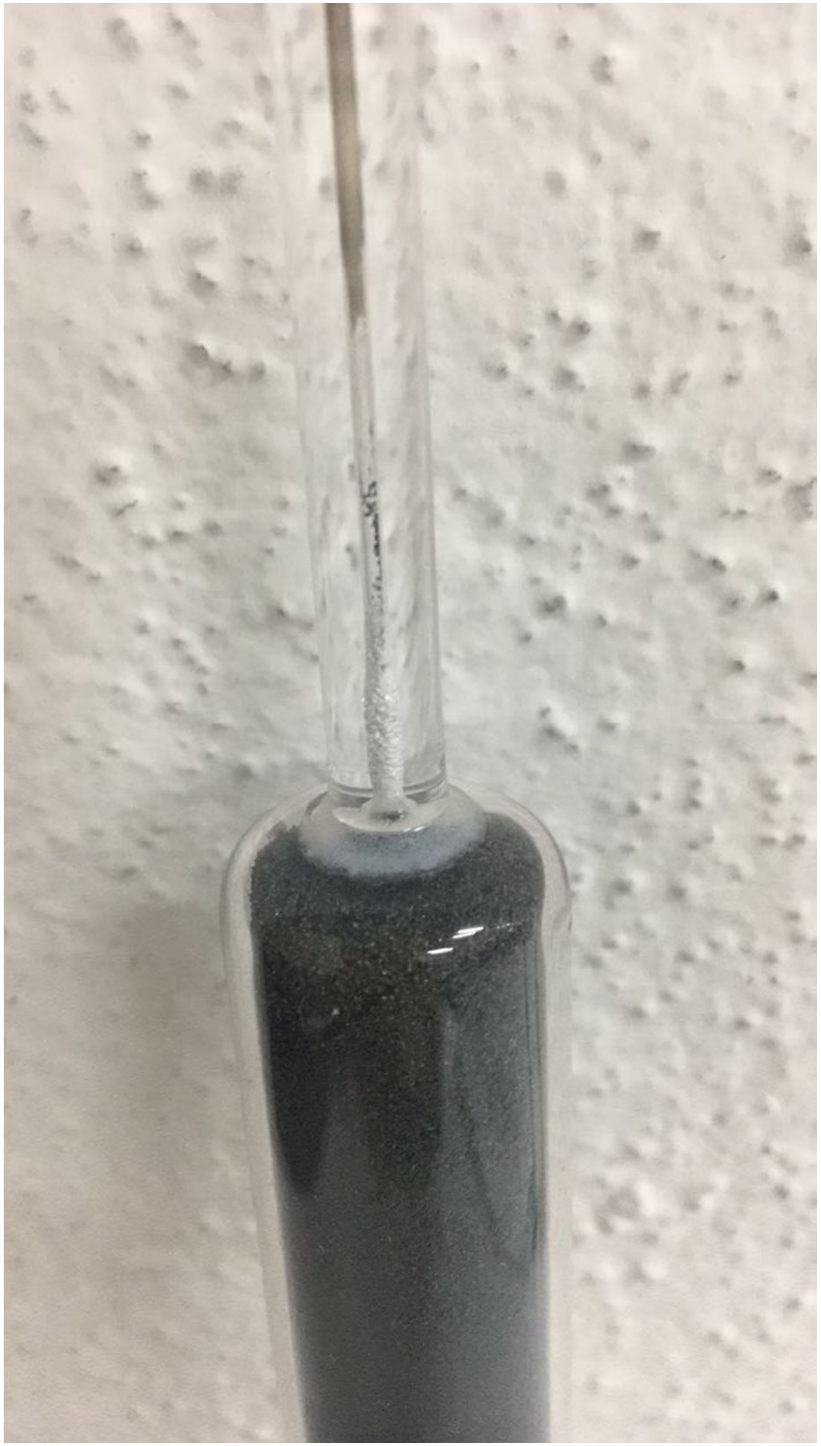
$$2\,Cr + 3\,H_2O \rightarrow Cr_2O_3 + 3\,\mathbf{H_2}$$

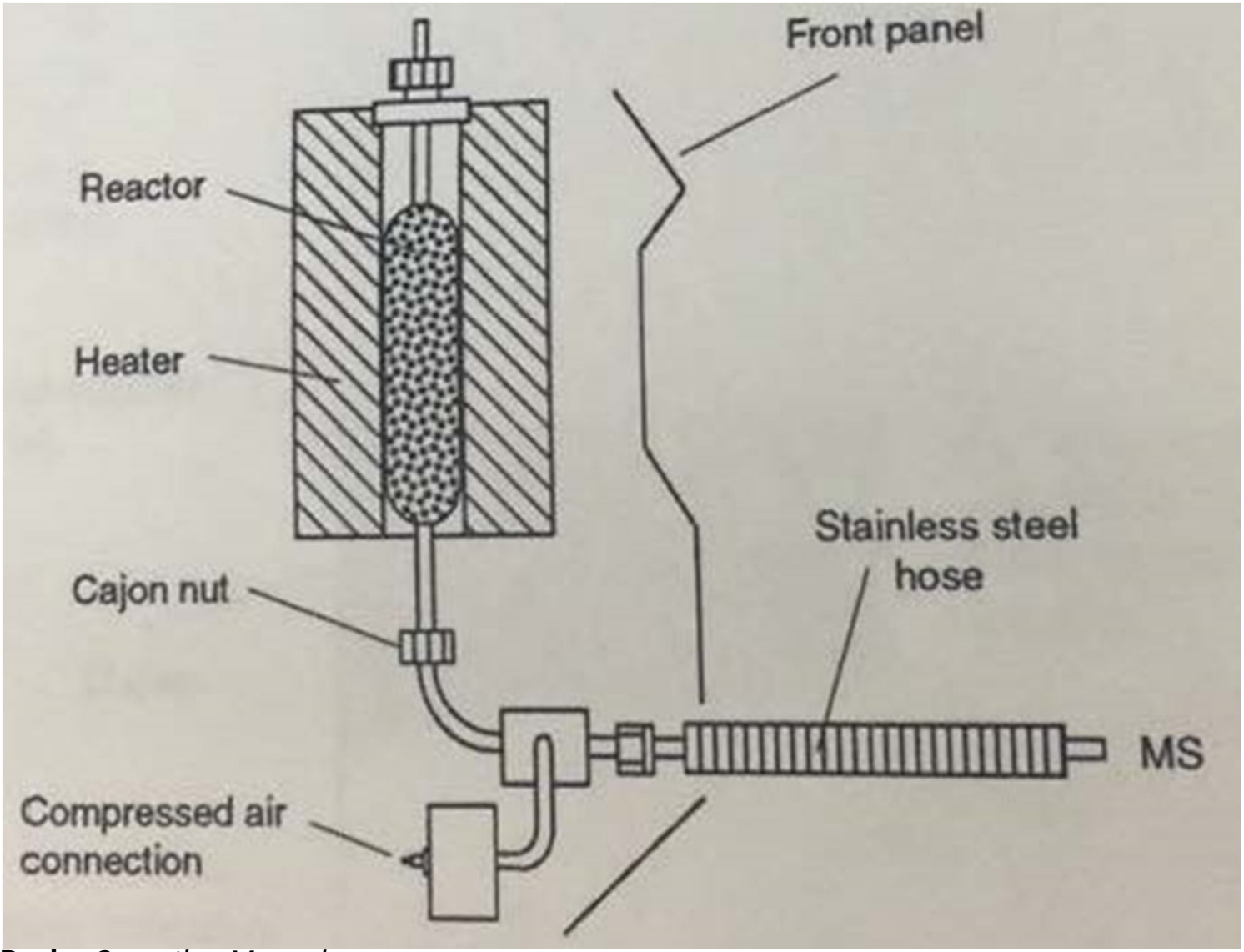
800 °C

Finnigan H/Device *Operating Manual*.



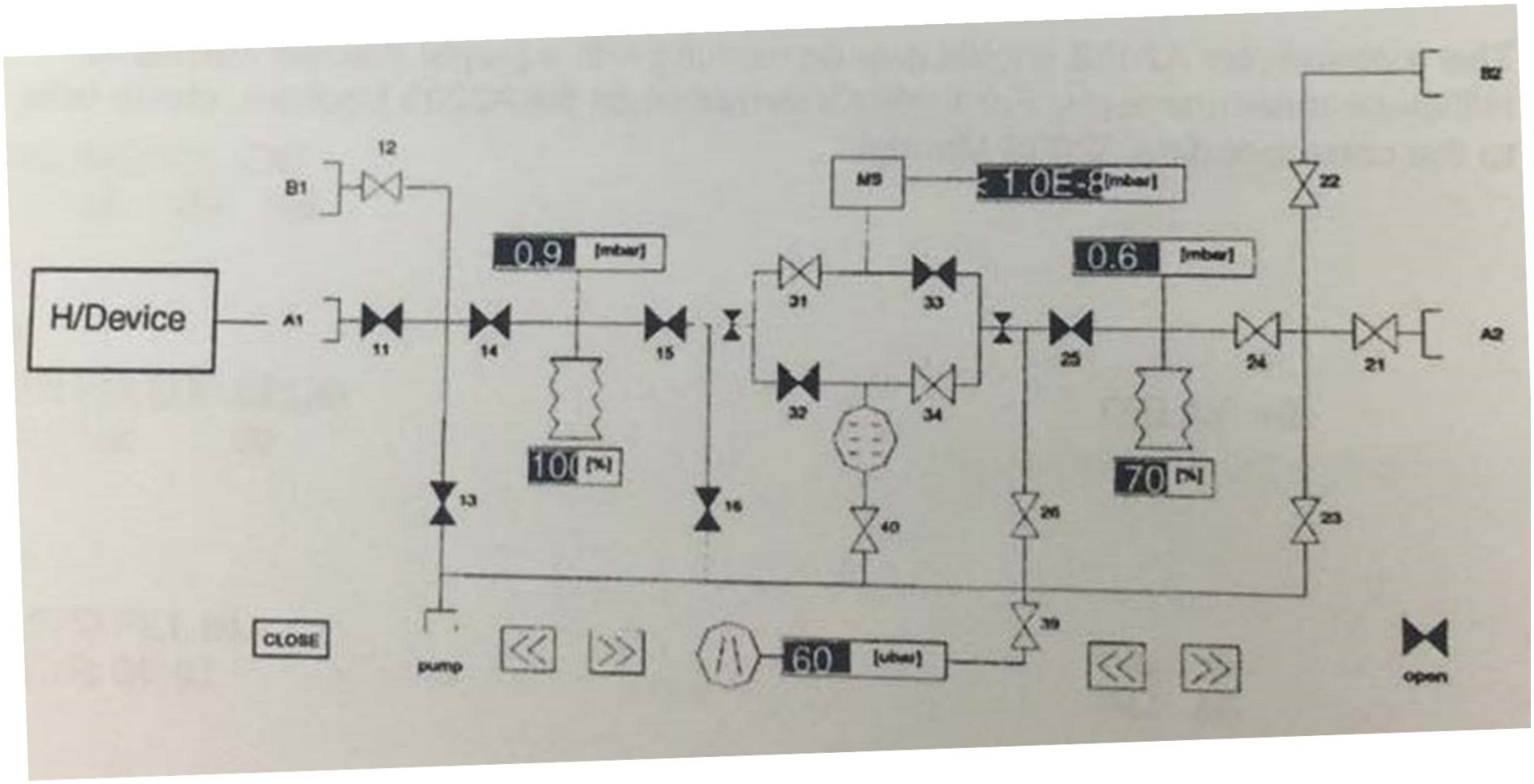






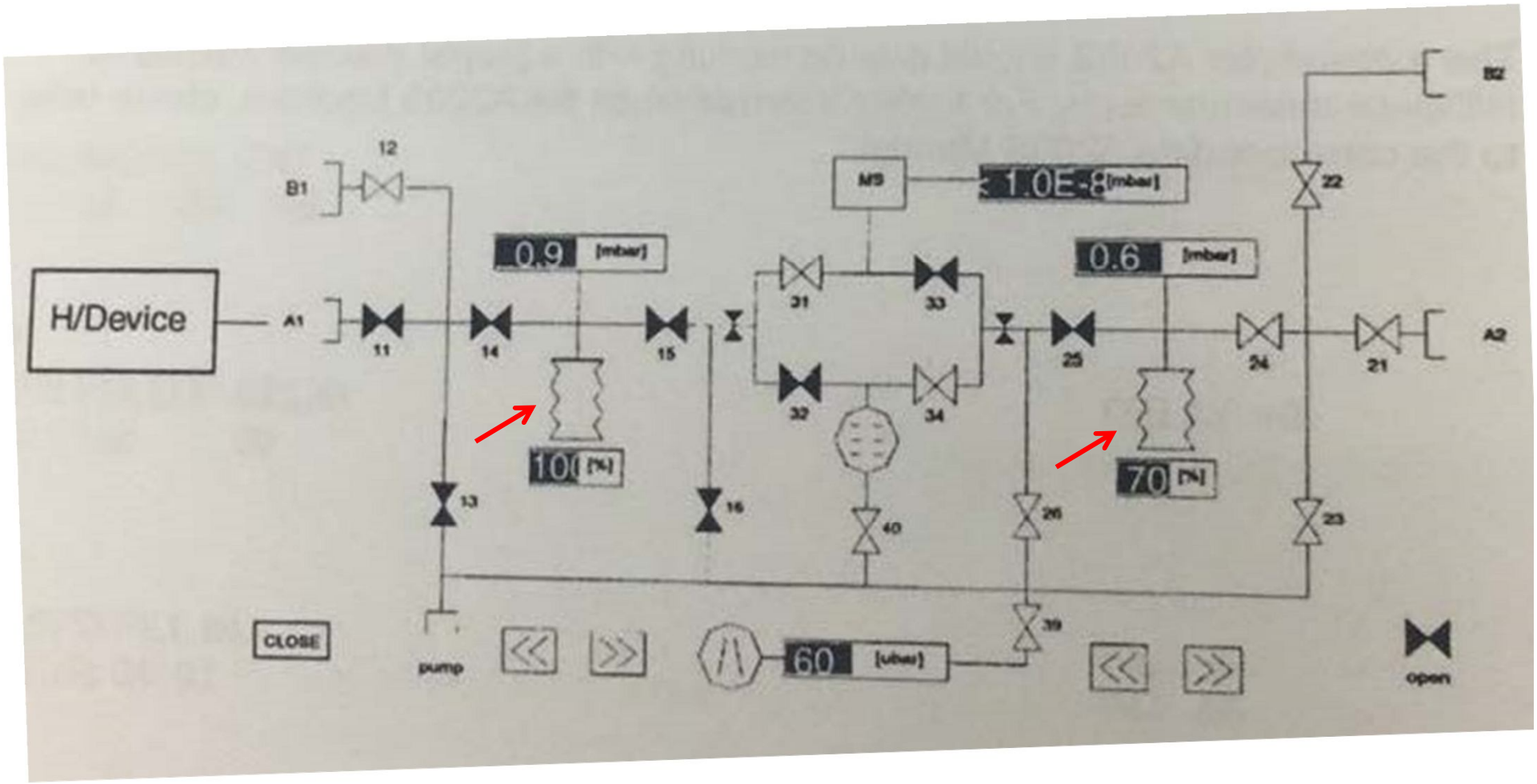
Finnigan H/Device *Operating Manual*.

Dual Inlet System



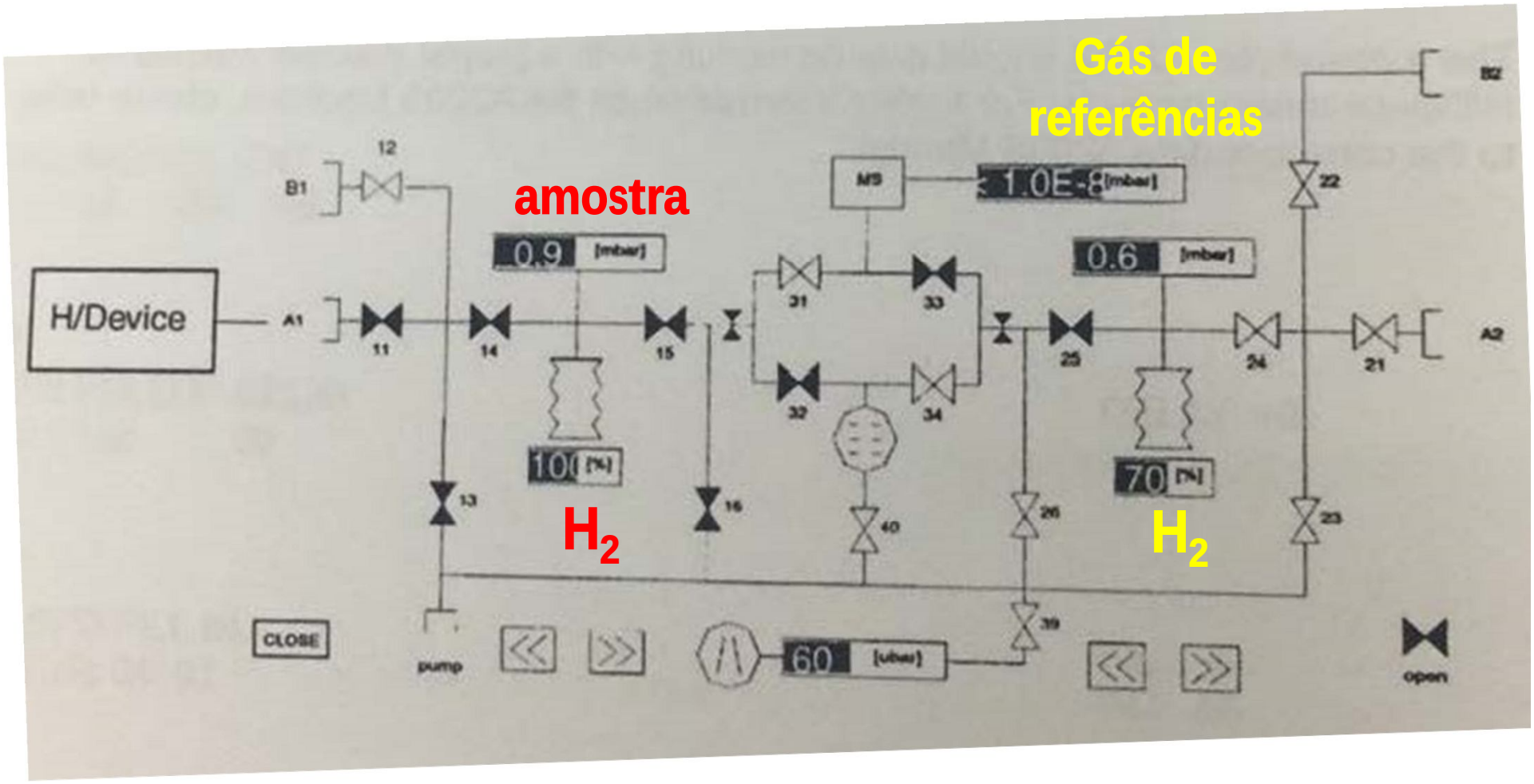
Finnigan H/Device *Operating Manual*.

Dual Inlet System



Dois *Bellows* = volumes variáveis

Dual Inlet System

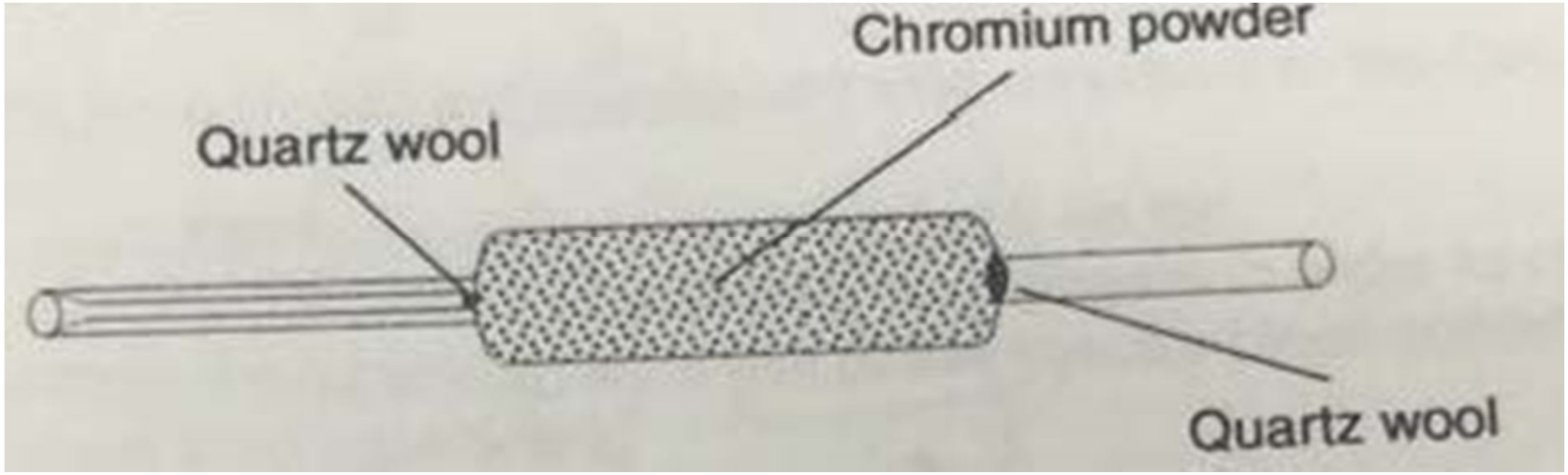


Finnigan H/Device *Operating Manual*.

H Device

- **Analise de água**
- **Analise de gases voláteis (por ex. CH_4)**

H Device : princípio de funcionamento



$$2\,Cr + 3\,H_2O \rightarrow Cr_2O_3 + 3\,\mathbf{H_2}$$

800 °C

Finnigan H/Device *Operating Manual*.

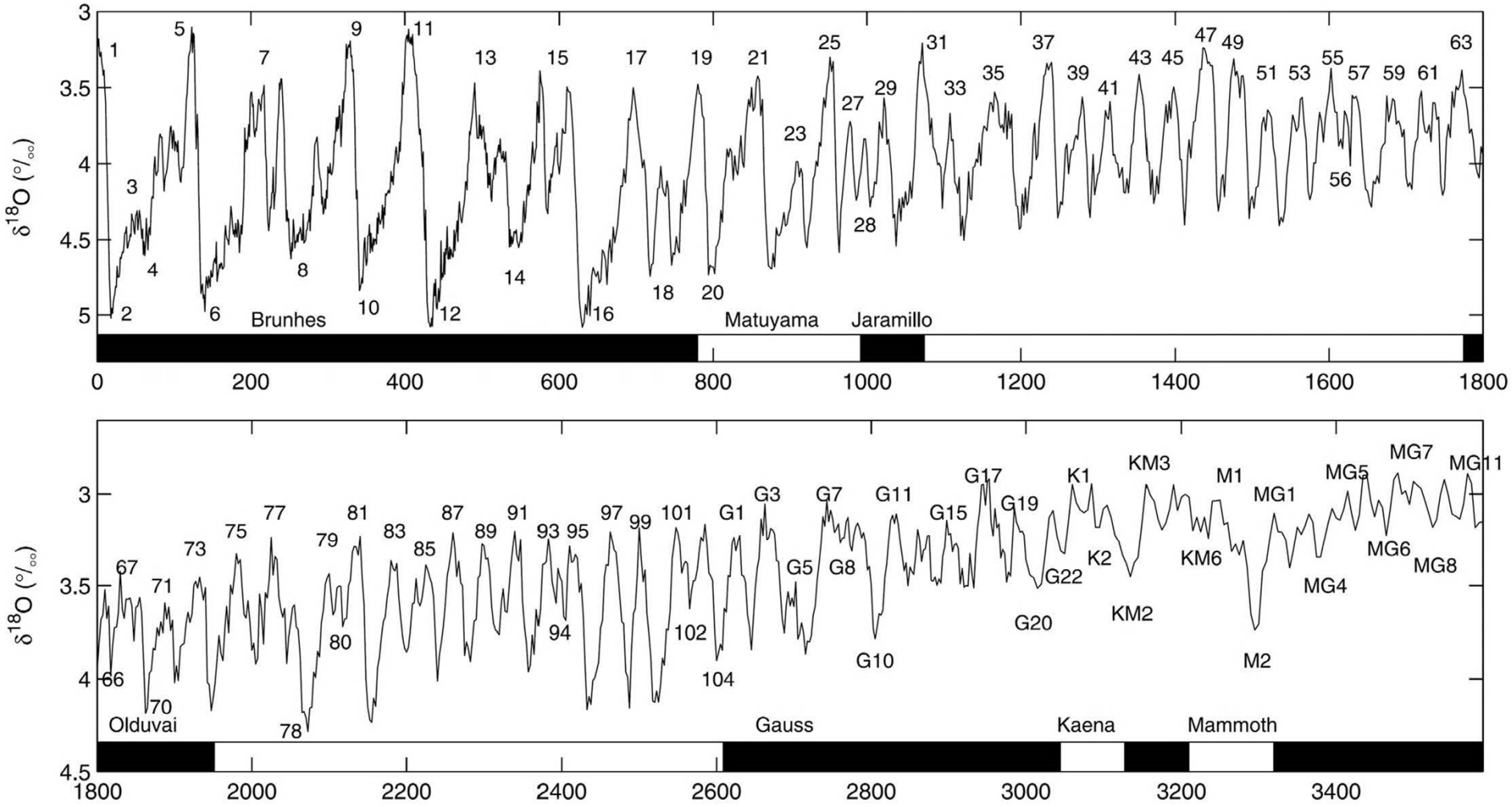
CARBONATO DE CÁLCIO

Analise isotópica de carbonato de cálcio

1) Isótopos do oxigênio e aplicações na paleoceanografia e paleoclimatologia

Lisiecki and Raymo, 2005

ESTÁGIOS ISOTÓPICOS



$$\delta^{18}O_{CaCO_3} = f\,(\delta^{18}O_{água}\ ,T)$$

$$\delta^{18}O_{CaCO_3} = f\,(\delta^{18}O_{água}\ ,T)$$

$$\delta^{18}O_{água}\uparrow\ ,\ \delta^{18}O_{carbonato}\uparrow$$

$$\delta^{18}O_{CaCO_3} = f\,(\delta^{18}O_{água}\ , T)$$

$\delta^{18}O_{CaCO_3}$

$T \uparrow,\ \delta^{18}O_{carbonato} \downarrow$

aumenta ao diminuir da T
(por causa do fracionamento ao equilíbrio isotópico)

Estudos iniciais

experimento de Epstein (1953)

Equação de Paleotemperatura

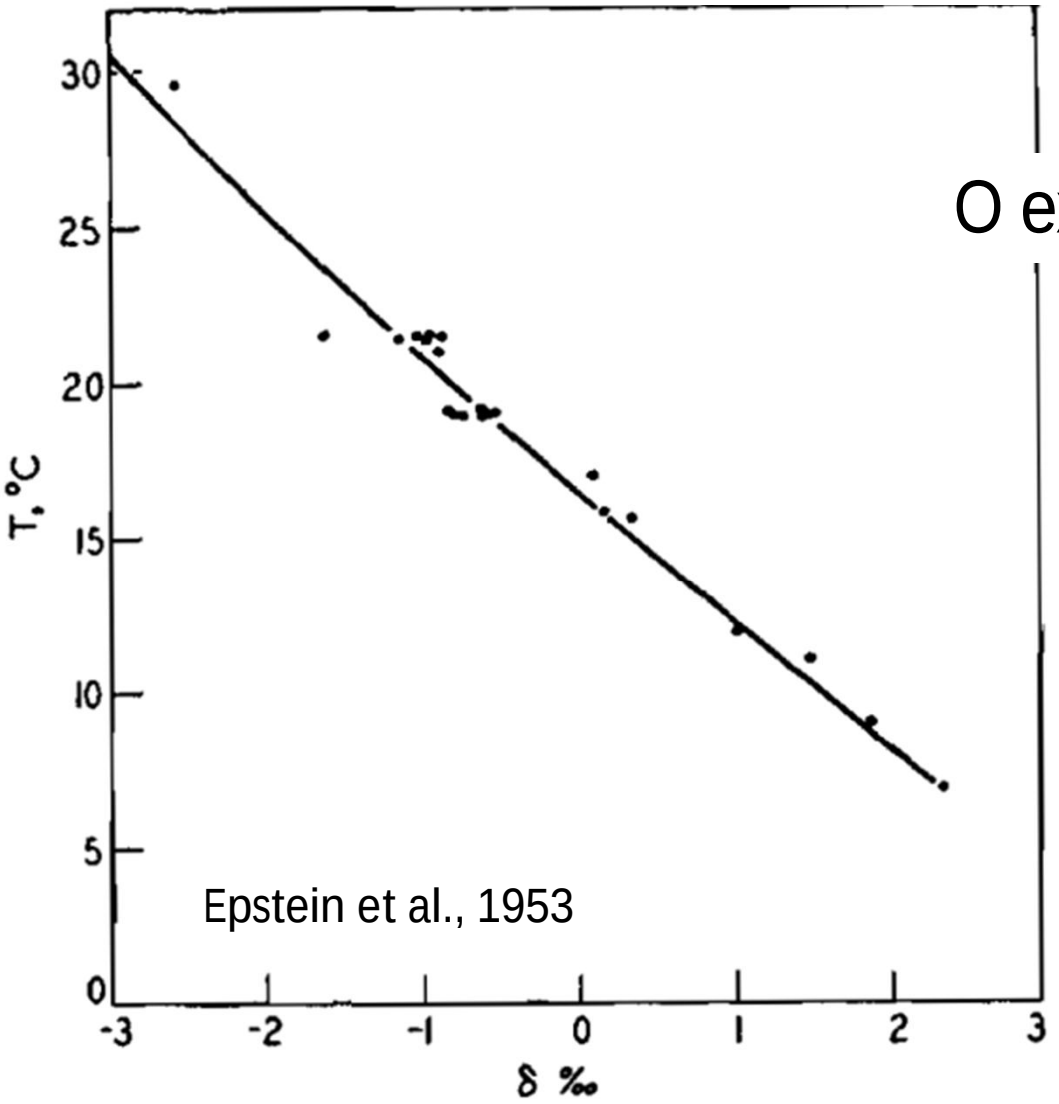
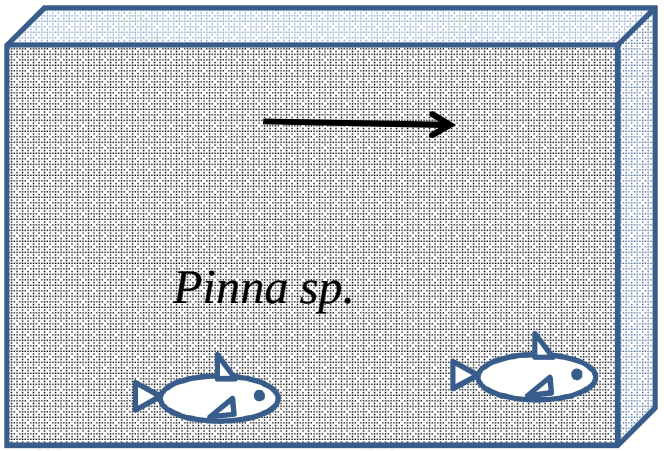


FIGURE 9.—ISOTOPIC TEMPERATURE SCALE

O experimento de Epstein:



T controlada

$\delta^{18}O_{H_2O}$ conhecido

Least squares calculations of the data in Table 7 gave the following equation:

$$t = 16.5 - 4.3\delta + 0.14\delta^2,$$

where t is the temperature in °C, δ is the permil differences between the ratio of masses 46 and 44 of the sample and of the working gas. The

Epstein et al., 1953

Equação de Paleotemperatura

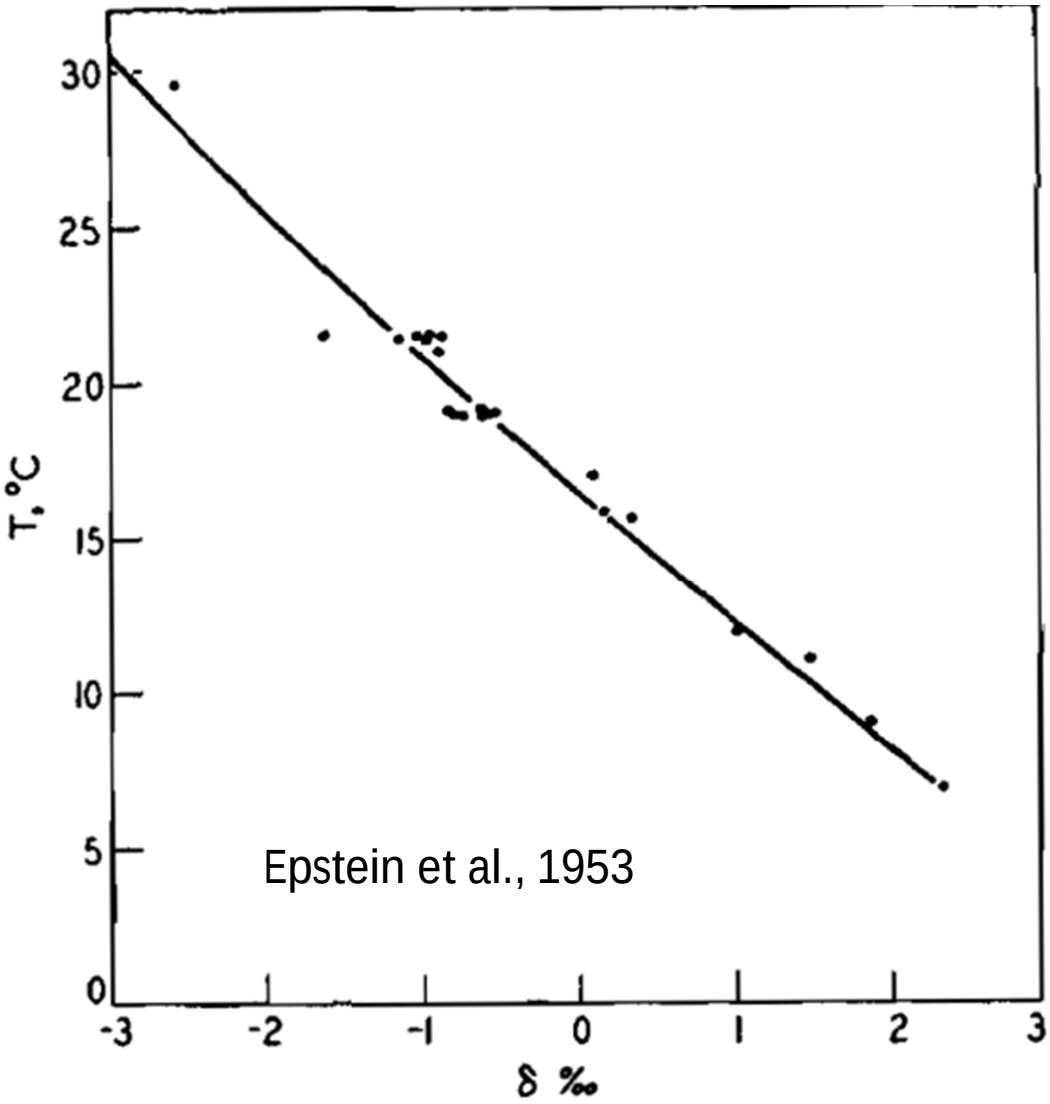
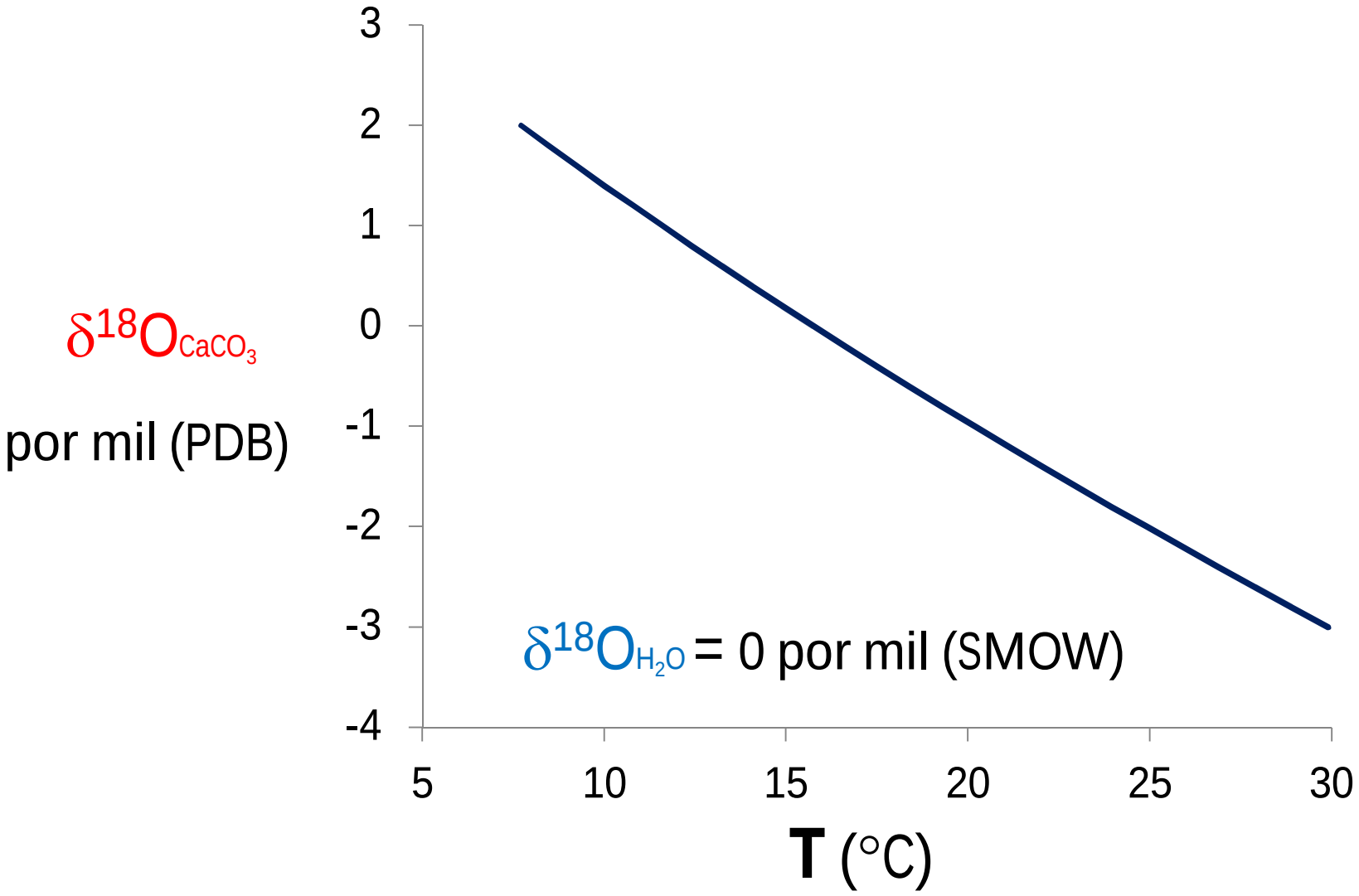


FIGURE 9.—ISOTOPIC TEMPERATURE SCALE

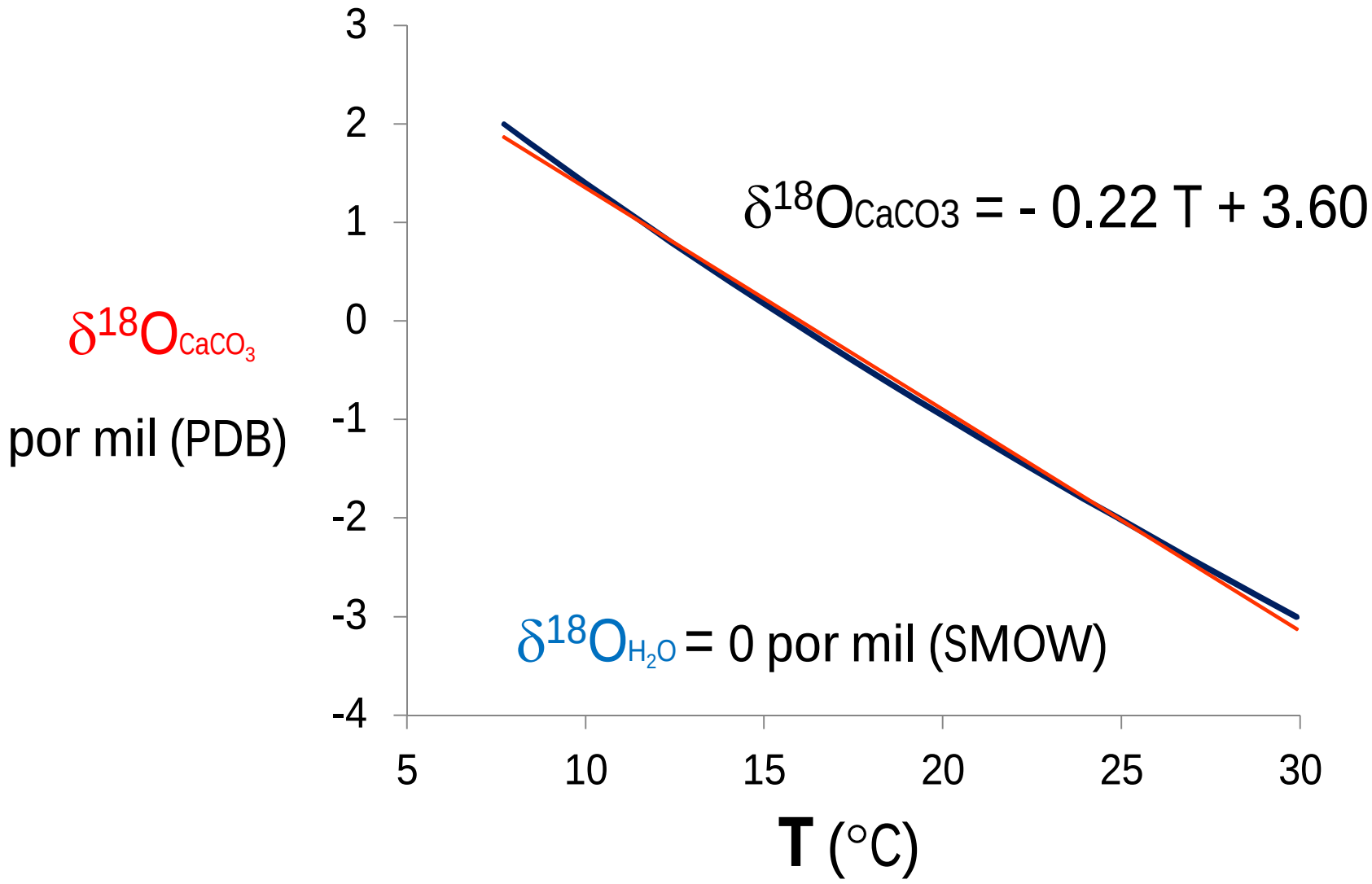
O valor de $\delta^{18}O$ de $CaCO_3$ **diminui ao aumentar da *temperatura***

$$\mathbf{T} = 16.9 - 4.2(\delta^{18}O_{CaCO_3} - \delta^{18}O_{H_2O}) + 0.13(\delta^{18}O_{CaCO_3} - \delta^{18}O_{H_2O})^2$$

$$T = 16.9 - 4.2(\delta^{18}O_{CaCO_3} - \delta^{18}O_{H_2O}) + 0.13(\delta^{18}O_{CaCO_3} - \delta^{18}O_{H_2O})^2$$



$$T = 16.9 - 4.2(\delta^{18}O_{CaCO_3} - \delta^{18}O_{H_2O}) + 0.13(\delta^{18}O_{CaCO_3} - \delta^{18}O_{H_2O})^2$$



Equação de Paleotemperatura

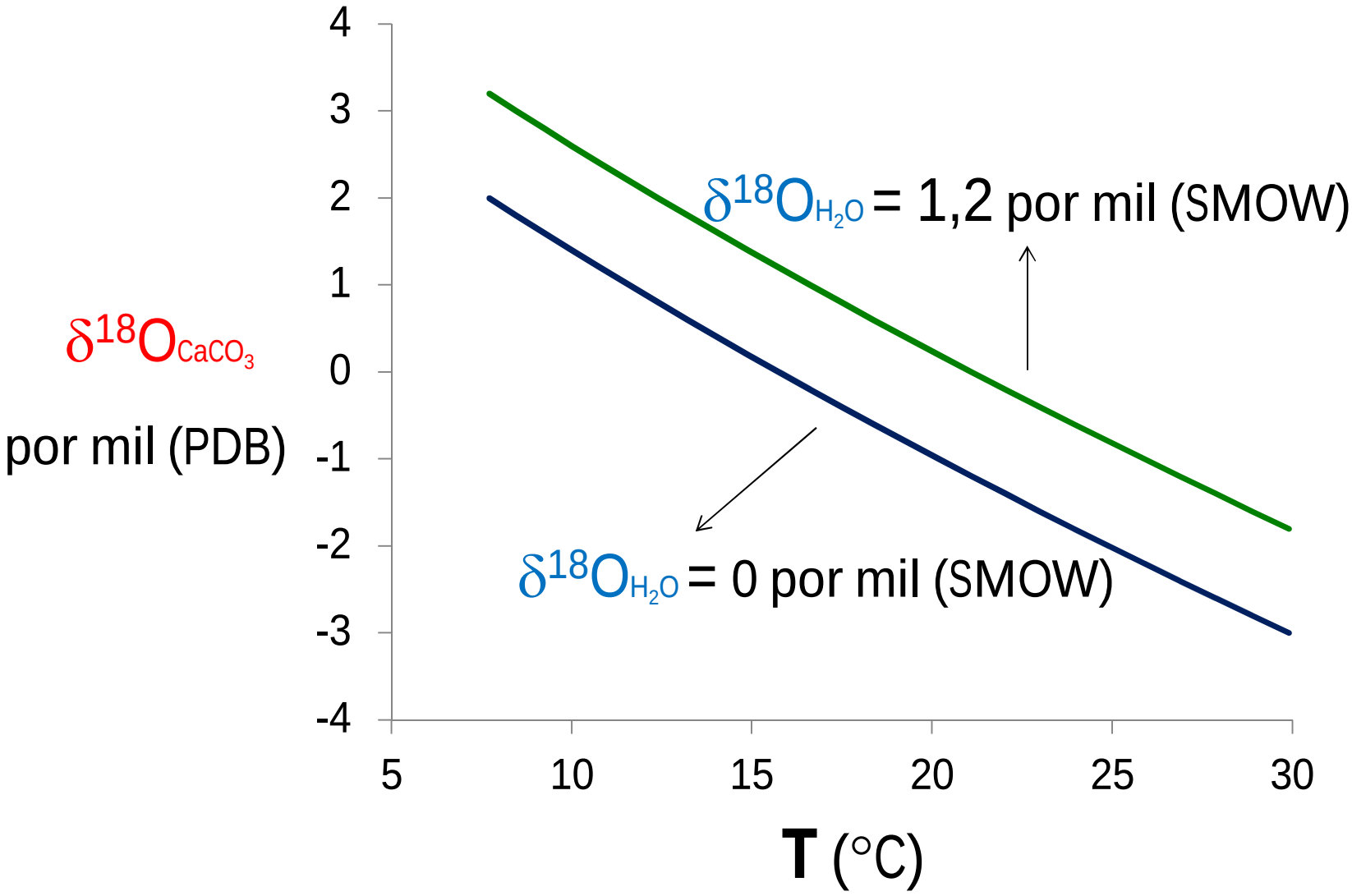
$$\mathbf{T} = 16.9 - 4.2(\delta^{18}O_{CaCO_3} - \delta^{18}O_{H_2O}) + 0.13(\delta^{18}O_{CaCO_3} - \delta^{18}O_{H_2O})^2$$

O valor de $\delta^{18}O$ do $CaCO_3$ depende da

Temperatura da agua (**inversamente proporcional**)

$\delta^{18}O$ do $CaCO_3$ diminui de cerca de **0,25** por mil
por cada aumento de temperatura de **1 °C**

$$\mathbf{T} = 16.9 - 4.2(\delta^{18}O_{CaCO_3} - \delta^{18}O_{H_2O}) + 0.13(\delta^{18}O_{CaCO_3} - \delta^{18}O_{H_2O})^2$$



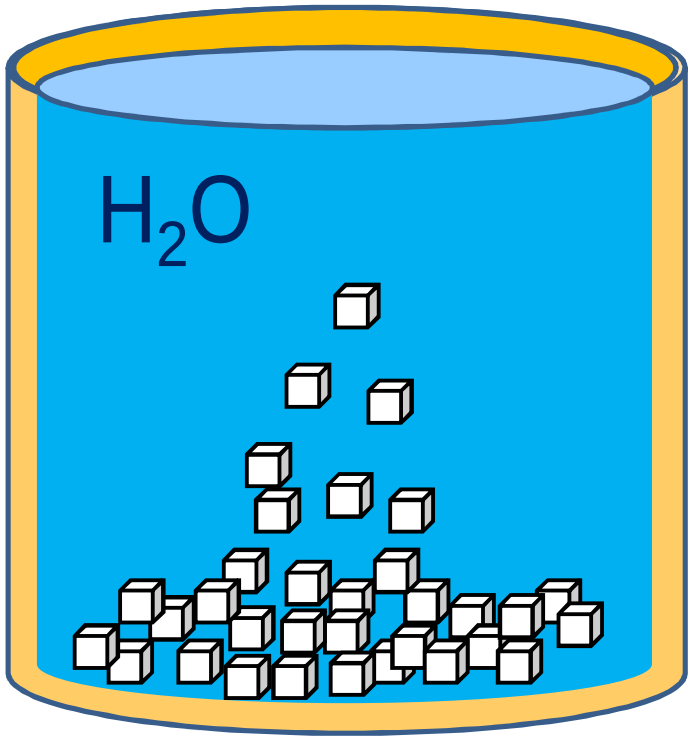
$\delta^{18}O_{CaCO_3}$

$T \uparrow,\ \delta^{18}O_{carbonato} \downarrow$

aumenta ao diminuir da T

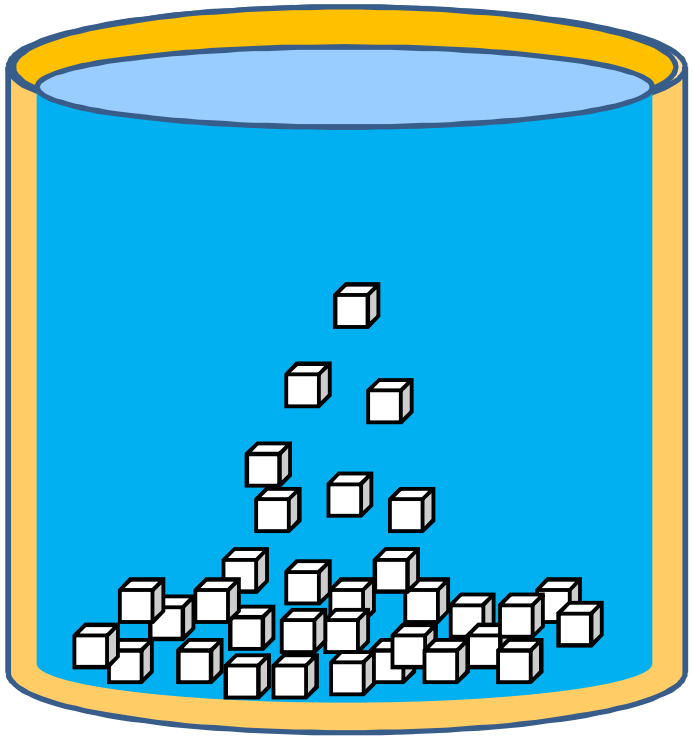
por que ?

Fracionamento isotópico do oxigênio ao equilíbrio



Equilíbrio isotópico

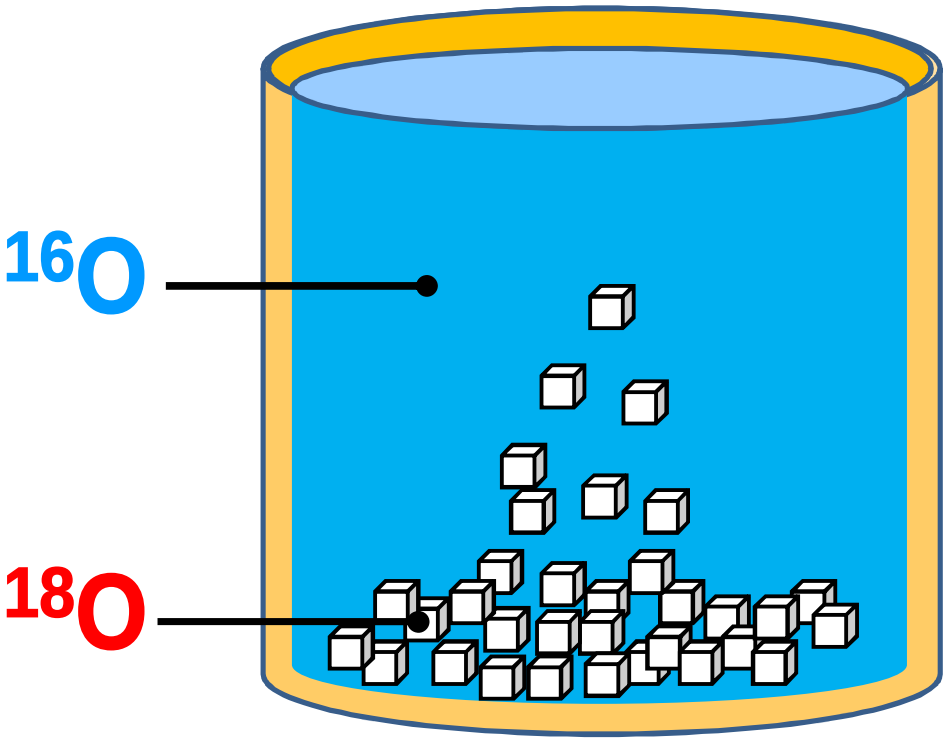
H_2O $CaCO_3$



Equilíbrio isotópico

H_2O $CaCO_3$

$H_2{}^{16}O$ $CaC^{16}O^{16}O^{18}O$



Equilíbrio isotópico

H_2O $CaCO_3$

$CaCO_3$ fica enriquecido em ^{18}O em relação á H_2O

Equilíbrio isotópico entre $CaCO_3$ e H_2O

$$CO_2 + H_2O \rightleftharpoons H^+ + HCO_3^-$$

$$CO_2 + OH^- \rightleftharpoons HCO_3^-$$

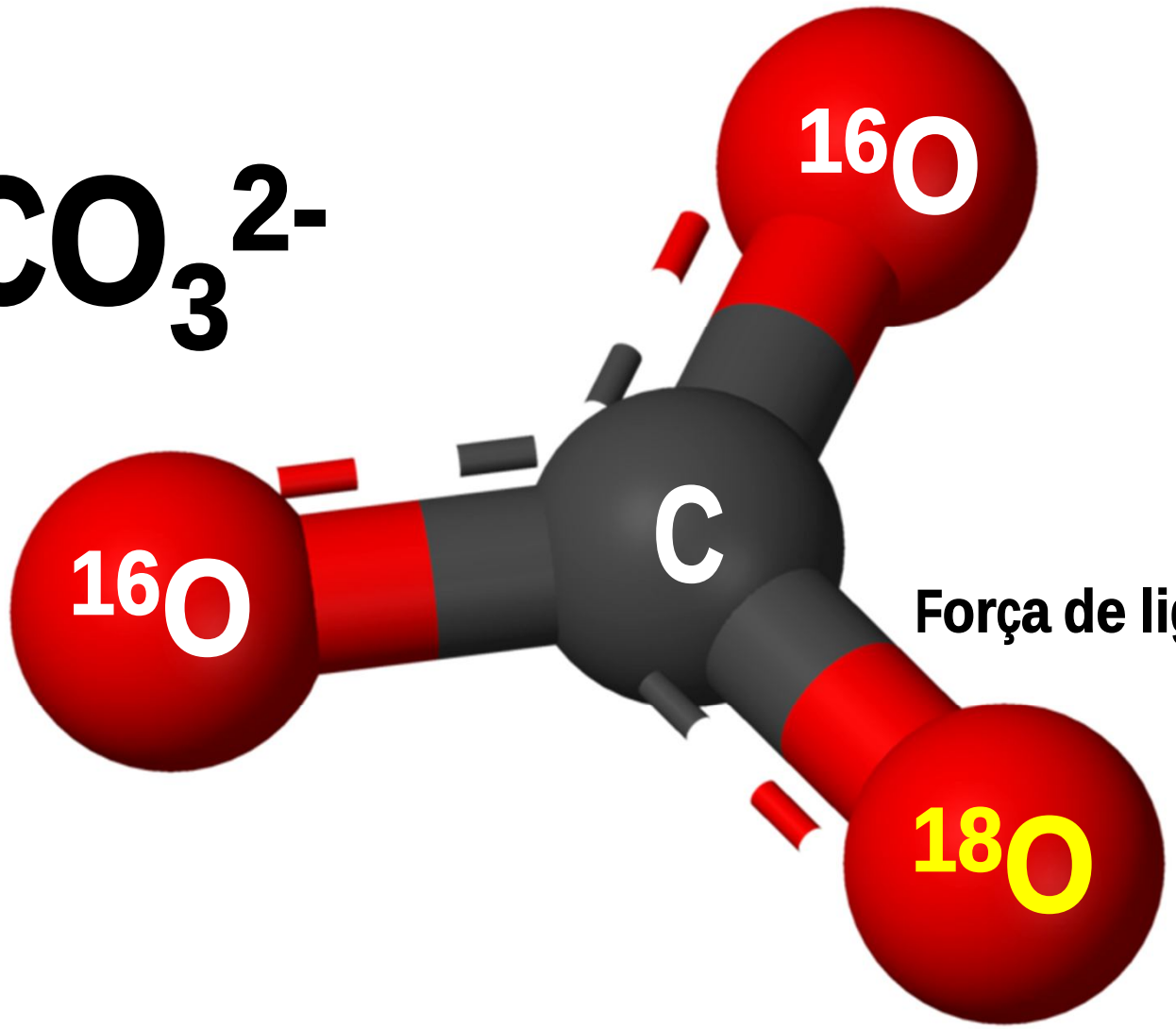
$$CO_3^{2-} + H^+ \rightleftharpoons HCO_3^-$$

$$\downarrow$$

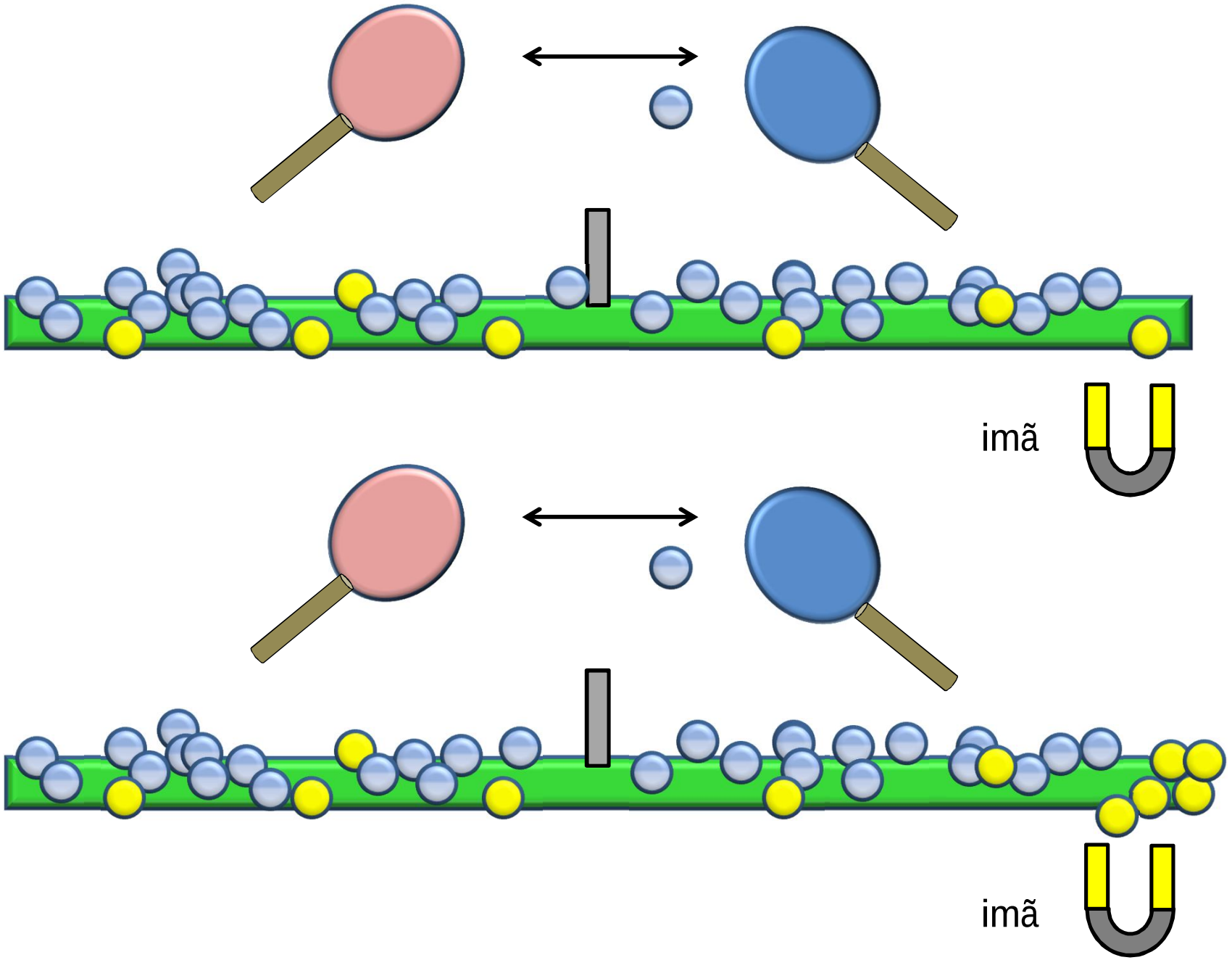
$$CO_3^{2-} + Ca^{2+} \rightarrow CaCO_3(s)$$

íon carbonato

CO_3^{2-}



Força de ligação maior


imã
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Fracionamento isotópico ao equilíbrio entre $CaCO_3$ e H_2O

$$CO_2 + H_2O \rightleftharpoons H^+ + H\overset{18}{C}O_3^-$$

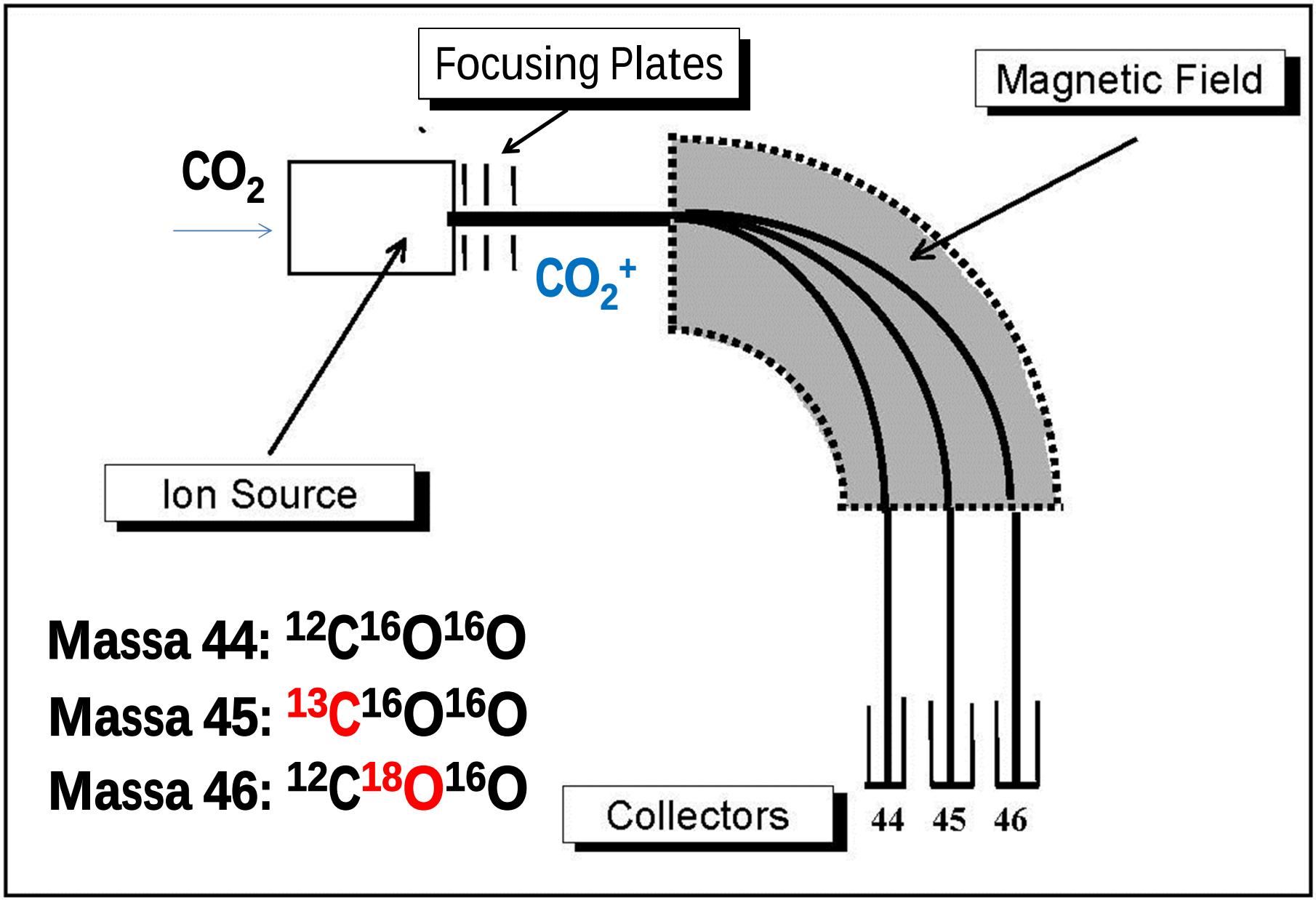
$$CO_2 + OH^- \rightleftharpoons H\overset{18}{C}O_3^-$$

$$CO_3^{2-} + H^+ \rightleftharpoons HCO_3^-$$

$$\downarrow$$

$$CO_3^{2-} + Ca^{2+} \rightarrow Ca\overset{18}{C}O_3(s)$$

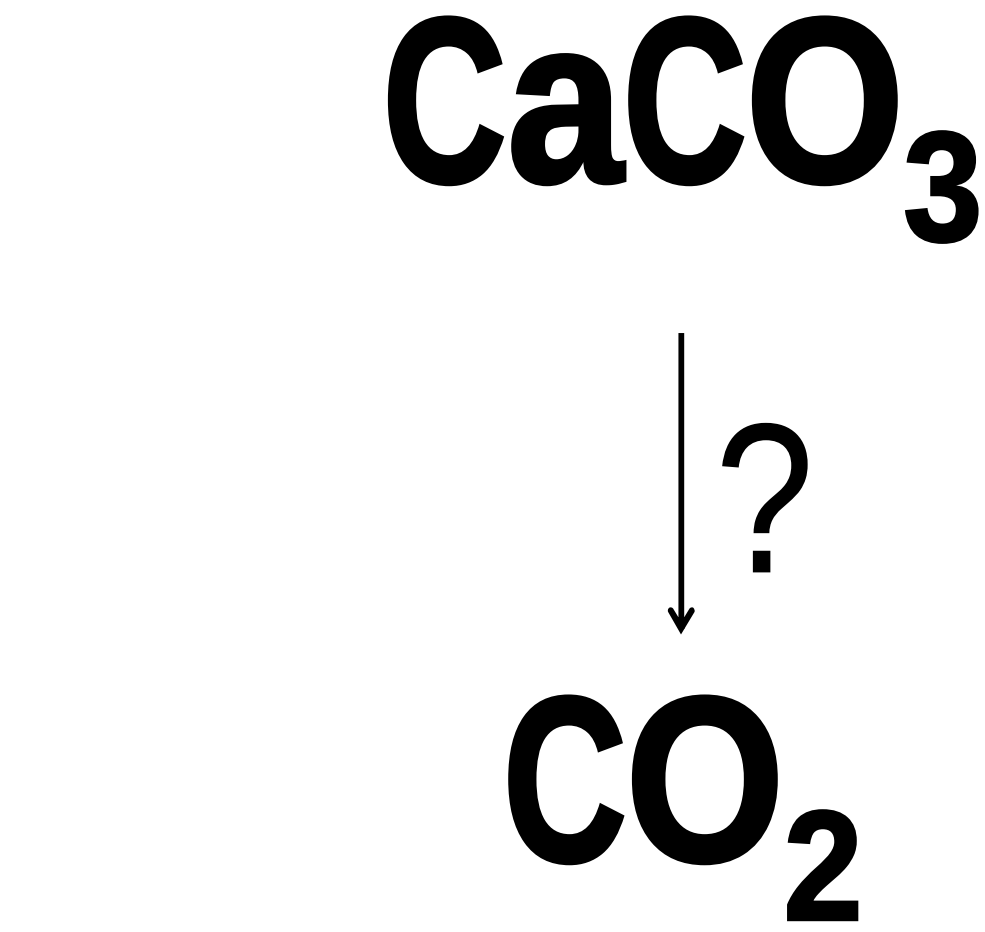
Preparação de amostras de $CaCO_3$ para analise isotópica de oxigênio e carbono

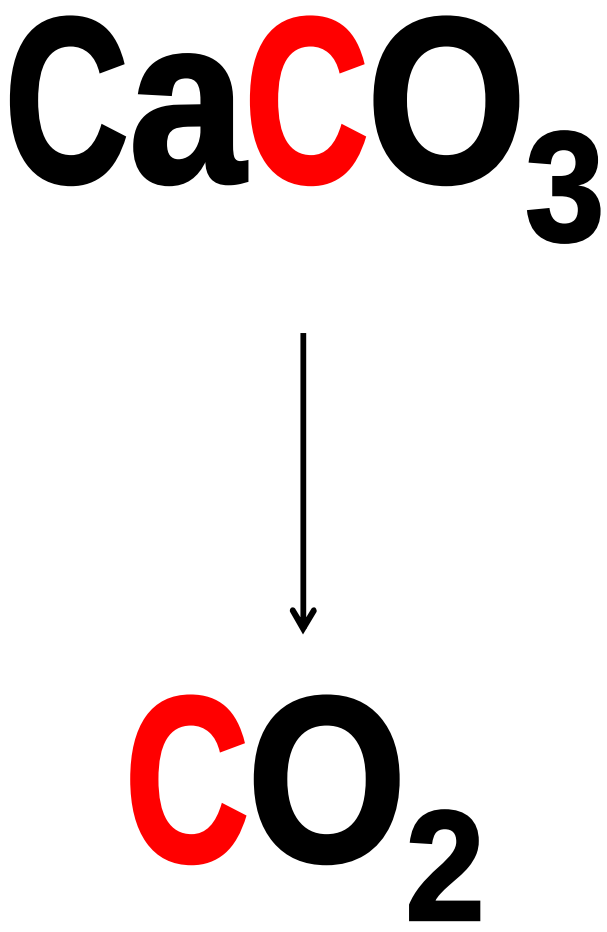


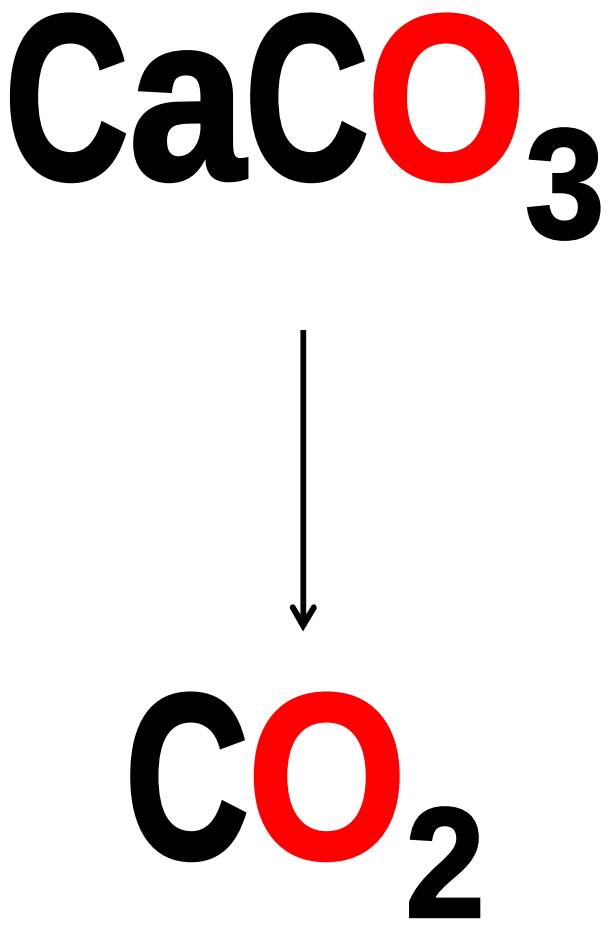
Massa 44: $^{12}C^{16}O^{16}O$

Massa 45: $^{13}C^{16}O^{16}O$

Massa 46: $^{12}C^{18}O^{16}O$


$CaCO_3$
?
CO_2


CaCO3
CO2


CaCO3
CO2

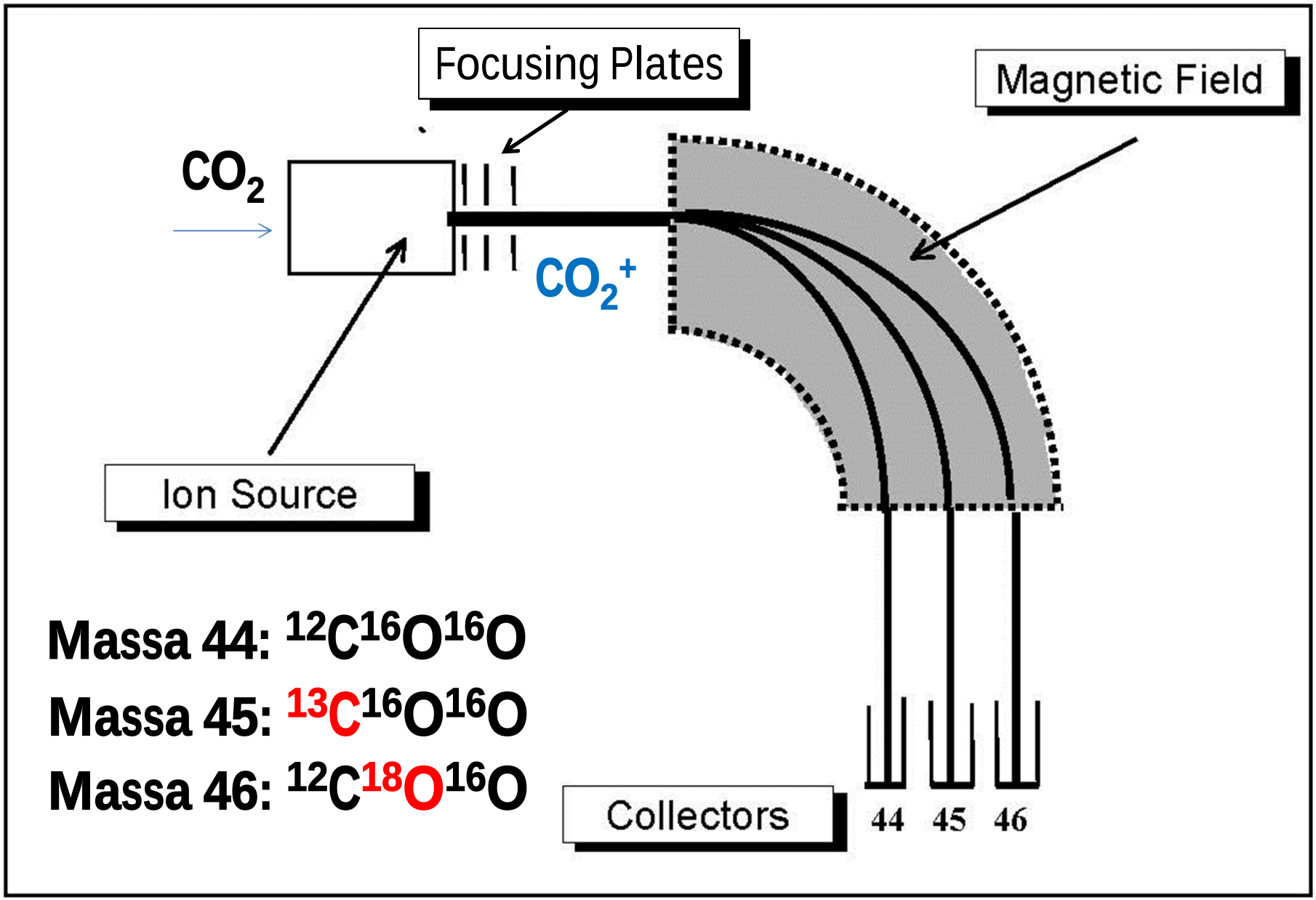
Método de McCrea (1950)

$$3\ CaCO_3 + 2\ H_3PO_4 \rightarrow 3\ CO_2 + Ca_3(PO_4)_2 + 3\ H_2O$$

+ impuridades:
O_2, N_2, SO_2 se o
carbonato não é puro

Método de McCrea (1950)

$$3\ CaCO_3 + 2\ H_3PO_4 \rightarrow 3\ CO_2 + Ca_3(PO_4)_2 + 3\ H_2O$$



Massa 44: $^{12}C^{16}O^{16}O$

Massa 45: $^{13}C^{16}O^{16}O$

Massa 46: $^{12}C^{18}O^{16}O$

Massa 44 = $^{12}C\ ^{16}O\ ^{16}O$

Massa 45 = $^{13}C\ ^{16}O\ ^{16}O$

Massa 46 = $^{13}C\ ^{16}O\ ^{18}O$

$$\frac{^{13}C}{^{12}C} = \frac{m\ 45}{m\ 44}$$

$$\frac{^{18}O}{^{16}O} = \frac{m\ 46}{m\ 44}$$

Calculo de $\delta^{13}C$ da amostra:

$$\frac{\left(\frac{\text{Int m 45}}{\text{Int m 44}}\right)_{\text{AMOSTRA}} - \left(\frac{\text{Int m 45}}{\text{Int m 44}}\right)_{\text{GAS DE REFERÊNCIA}}}{\left(\frac{\text{Int m 45}}{\text{Int m 44}}\right)_{\text{GAS DE REFERÊNCIA}}} \times 10^3$$

Calculo de $\delta^{18}O$ da amostra:

$$\frac{\left(\frac{\text{Int m 46}}{\text{Int m 44}}\right)_{\text{AMOSTRA}} - \left(\frac{\text{Int m 46}}{\text{Int m 44}}\right)_{\text{GAS DE REFERÊNCIA}}}{\left(\frac{\text{Int m 46}}{\text{Int m 44}}\right)_{\text{GAS DE REFERÊNCIA}}} \times 10^3$$

Equipamentos para a analise isotópica de carbonatos

Finnigan GasBench II *Operating Manual*.

Finnigan GasBench II

Finnigan Delta V Isotope Ratio Mass Spectrometer

GasBench

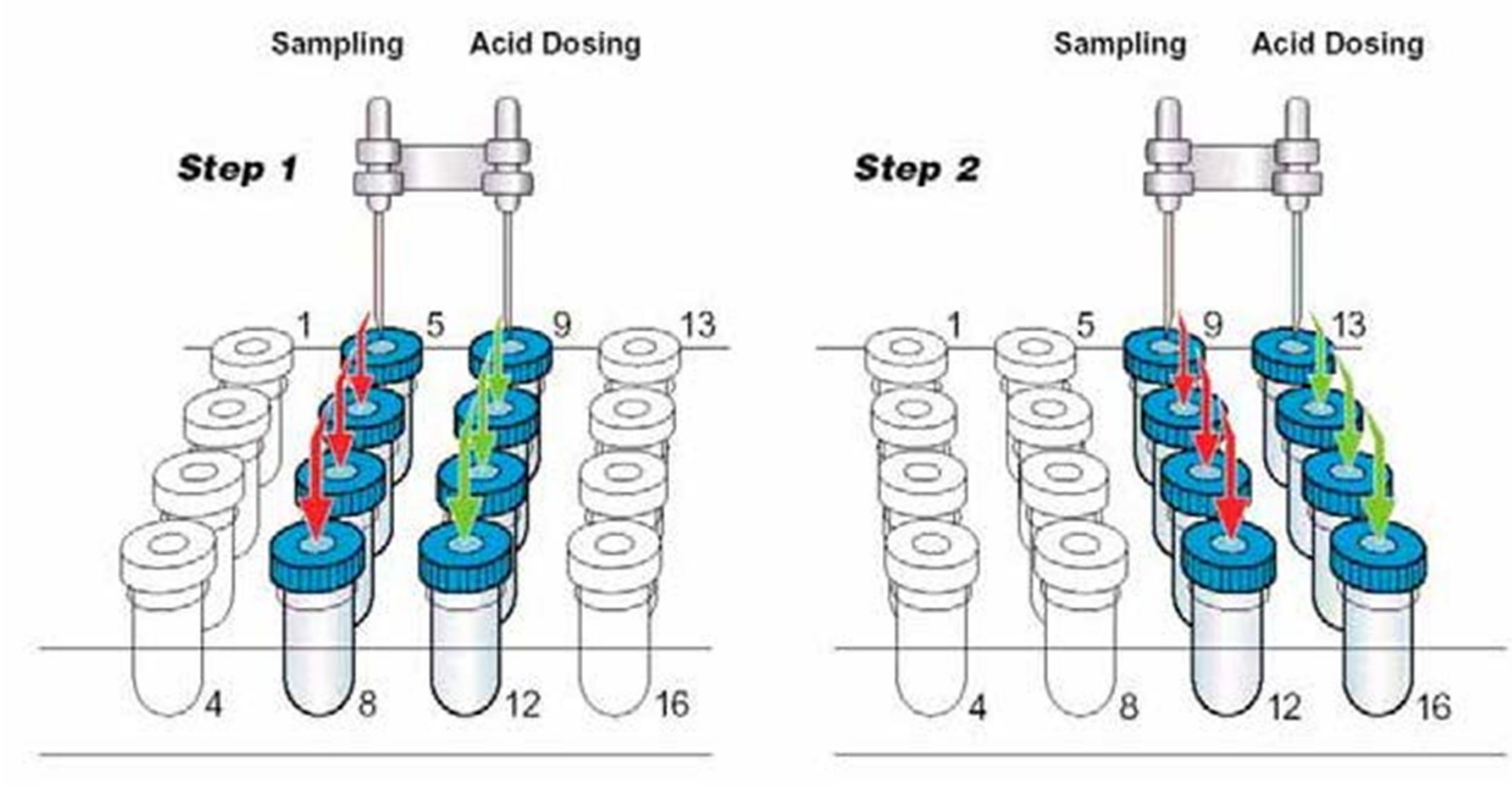
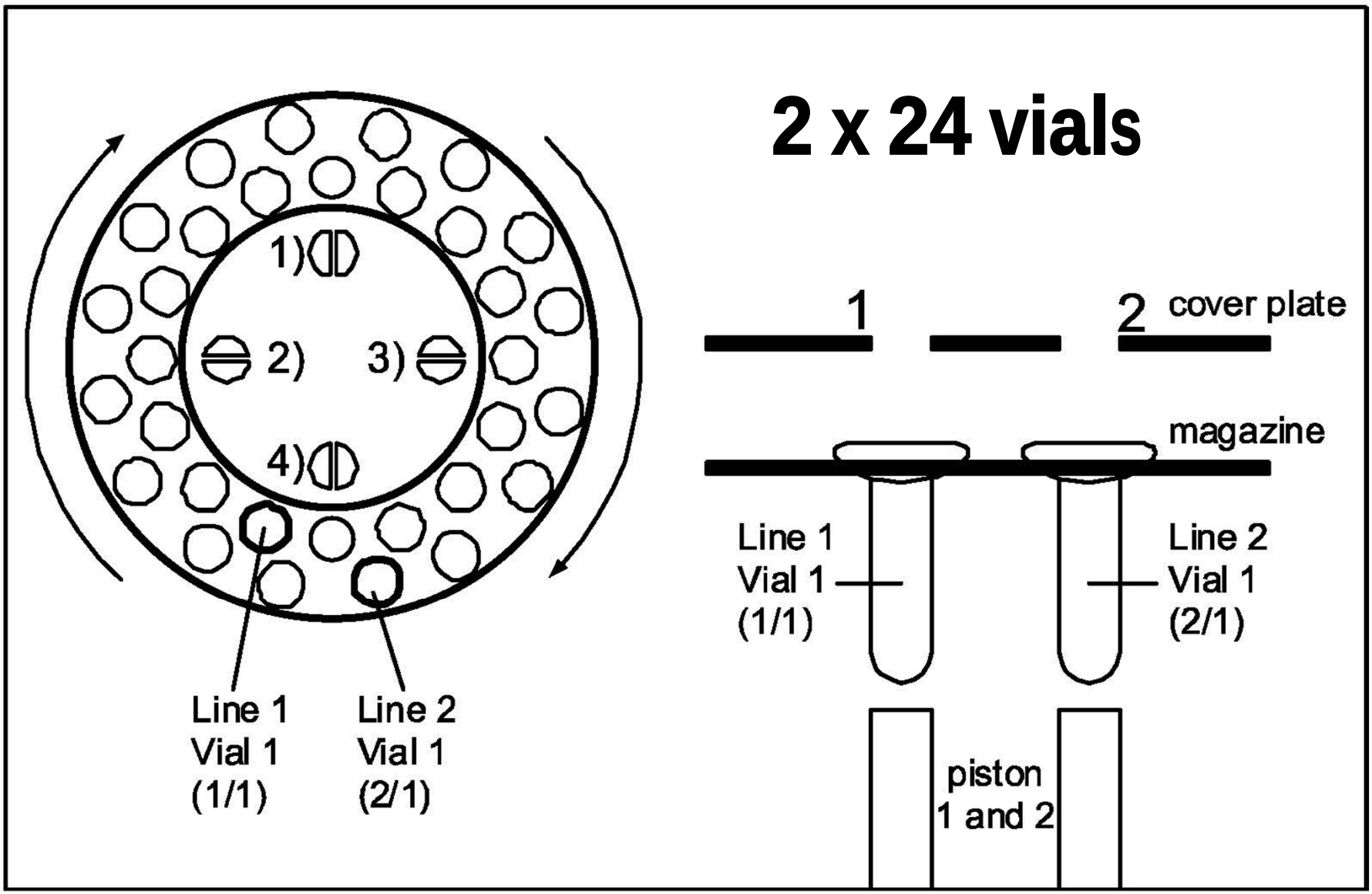


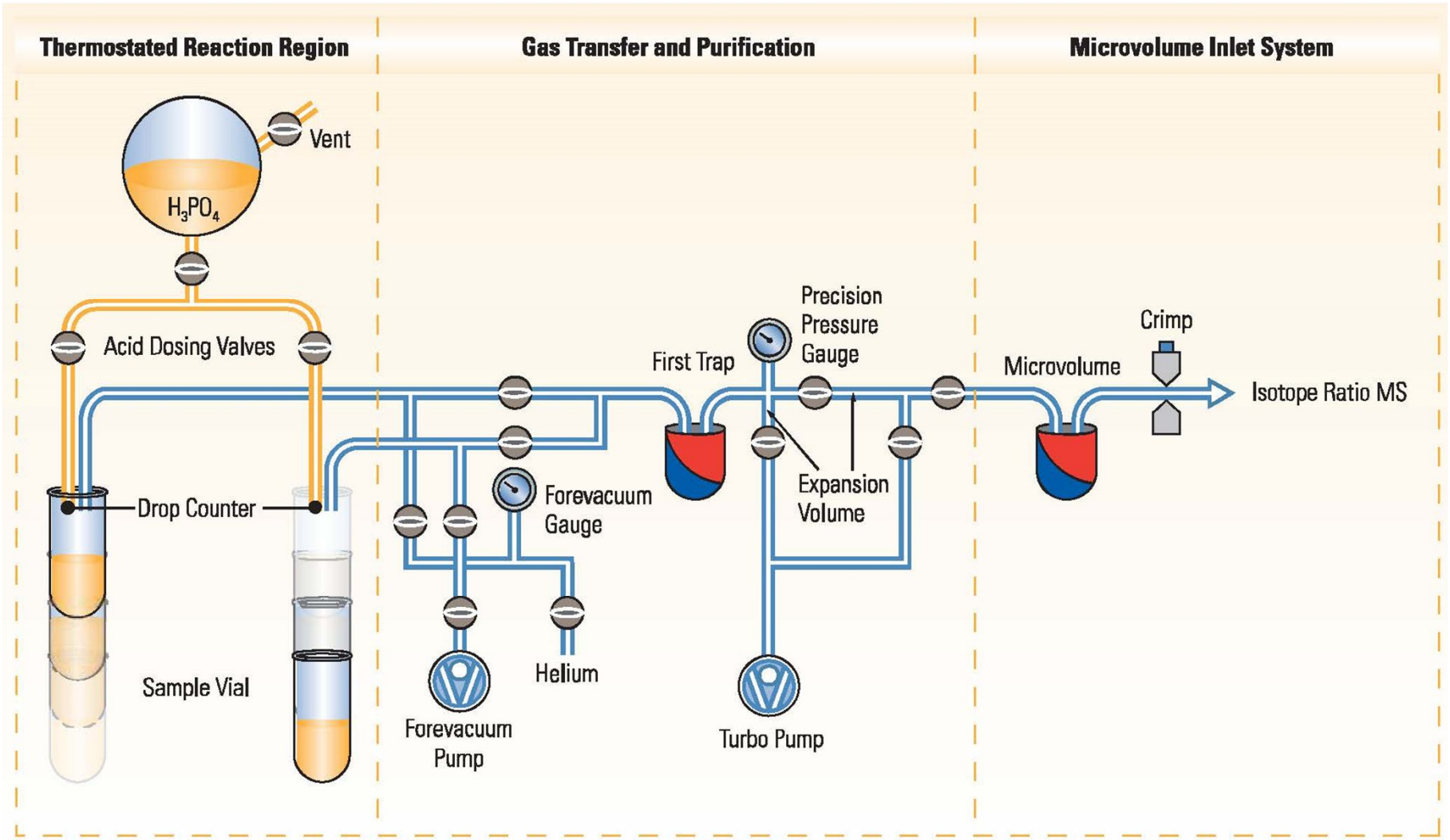
Figure 5-2. Defining the Sequence - Double Needle Setup

Kiel Device

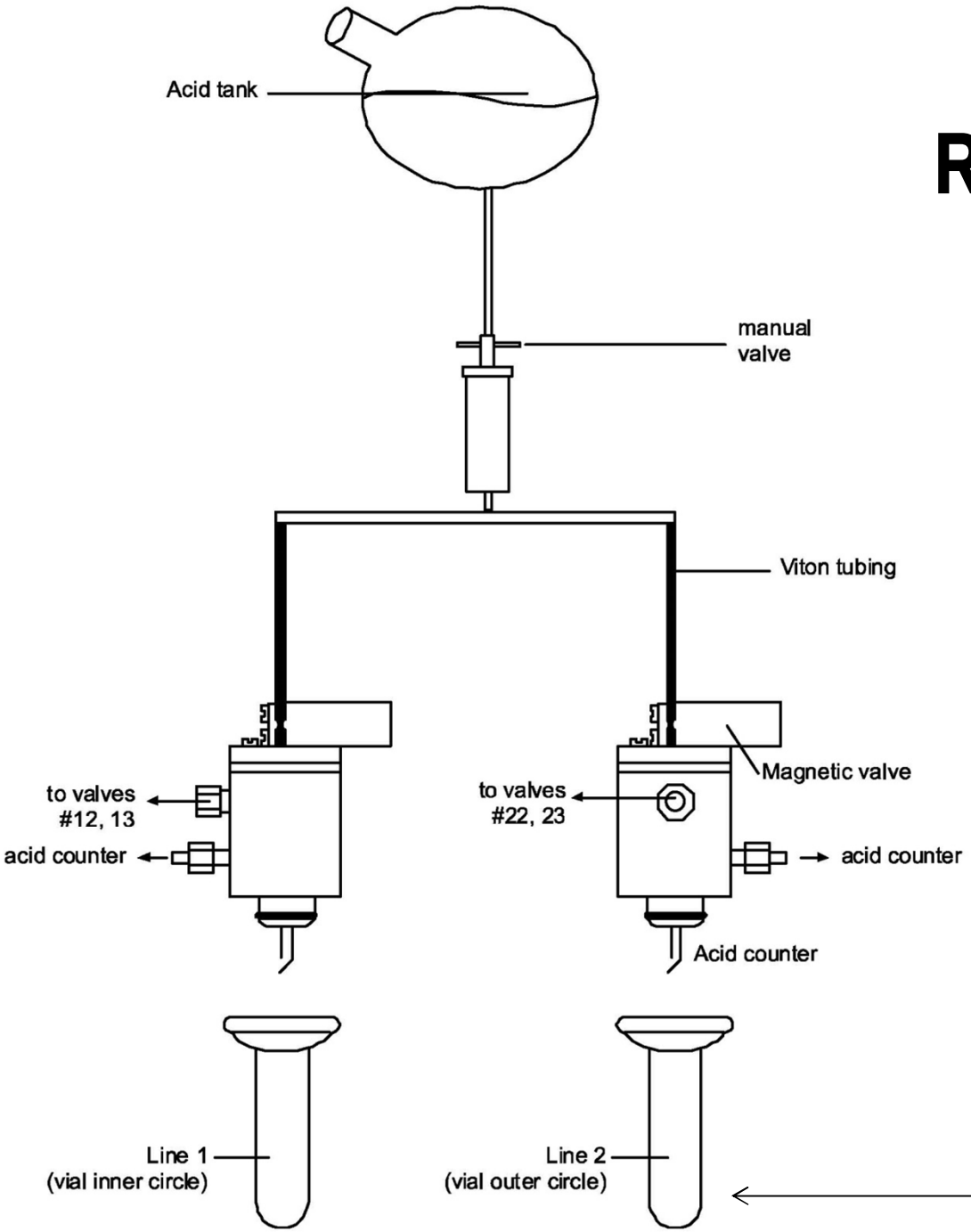
KIEL CARBONATE DEVICE


2 x 24 vials
1)
2)
3)
4)
Line 1
Vial 1
(1/1)
Line 2
Vial 1
(2/1)
1
2
cover plate
magazine
Line 1
Vial 1
(1/1)
Line 2
Vial 1
(2/1)
piston
1 and 2

Kiel Device: sistema sob vácuo



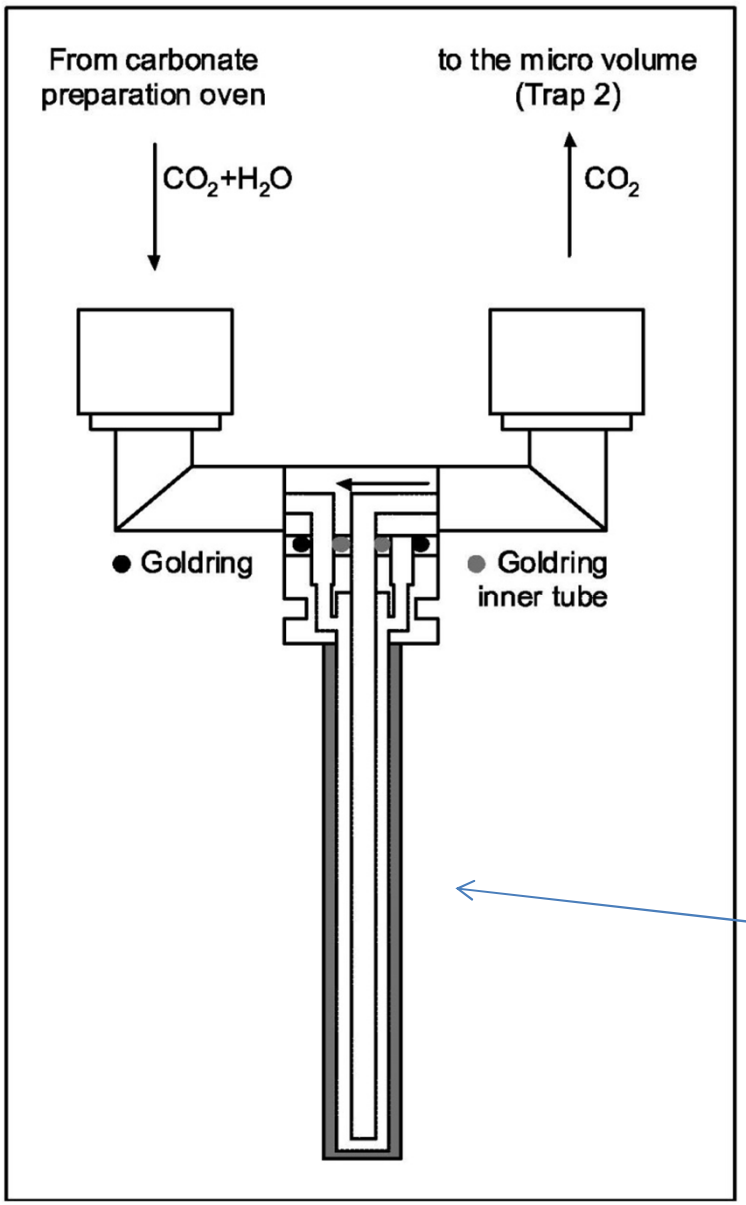
Schematic of the KIEL IV Carbonate Device



Reação com H_3PO_4
75 °C

$CaCO_3$
min 10 µg

Kiel Device: separação criogênica



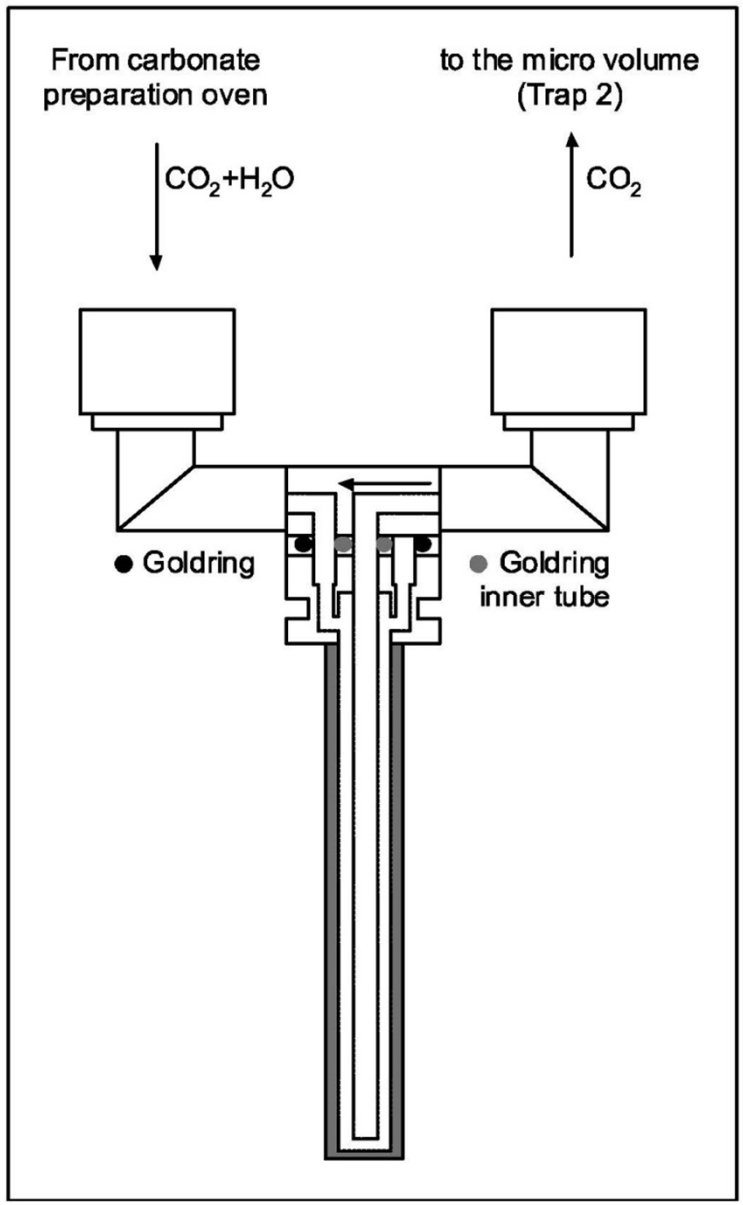
N_2 líquido (-196 °C)

Método de McCrea (1950)

$$3\ CaCO_3 + 2\ H_3PO_4 \rightarrow 3\ CO_2 + Ca_3(PO_4)_2 + 3\ H_2O$$

+ impuridades:
O_2, N_2, SO_2 se o
carbonato não é puro

Kiel Device: separação criogênica



FASES DA SEPARAÇÃO:

1) -196 °C: captura H_2O e CO_2

O_2 e N_2 são incondensaveis e são eliminados;

2) -120 °C: CO_2 liberado para o micro volume, H_2O fica retida;

3) CO_2 novamente retido no micro volume (N_2 líquido, -196 °C) antes da entrada no IRMS

GasBench x Kiel Device

GasBench

1) Sob fluxo contínuo de He

2) Separação mediante GC

3) min 100 µg

4) 60 amostra/dia

5) Gasto de He

6) Não é possível ajustar o tamanho do pico de gás de referência em relação ao de amostra

7) Pode analisar H_2O, $CaCO_3$, carbono inorgânico dissolvido e CO_2 atmosférico ou humano

Kiel Device

1) Sob vácuo

2) Separação criogênica

3) min 10 µg

4) 46 amostra/dia

5) Gasto de N_2 líquido

6) É possível ajustar o tamanho do pico de gás de referência em relação ao de amostra (melhora a precisão)

7) Pode analisar só $CaCO_3$ mas é geralmente mais preciso

Padrões internacionais de referência

$CaCO_3$

$\delta^{18}O$ e $\delta^{13}C$: padrão V-PDB

(valores = zero por definição)

$\delta^{18}O$: padrão V-SMOW quando comparado com $\delta^{18}O$ da água

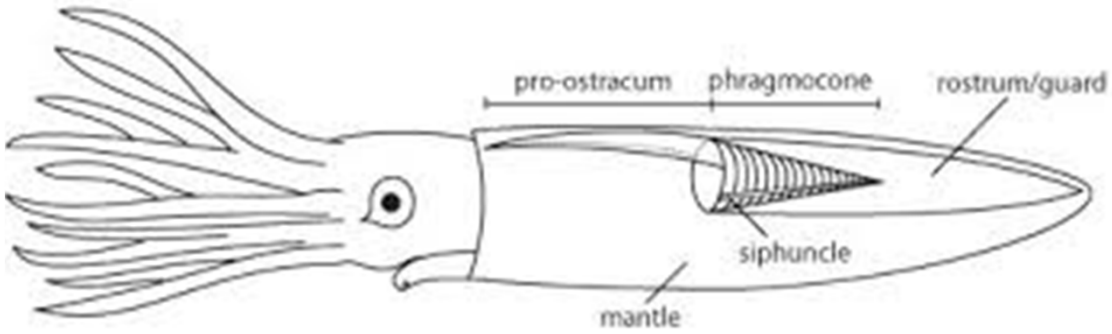
Padrões isotópicos internacionais
(Agencia Internacional da Energia Atômica IAEA, Viena)

PDB: Pee Dee Belemnite (Calcita)

$\delta^{18}O$ = 0‰ (por definição)

$\delta^{13}C$ = 0‰ (por definição)

Belemnitella americana

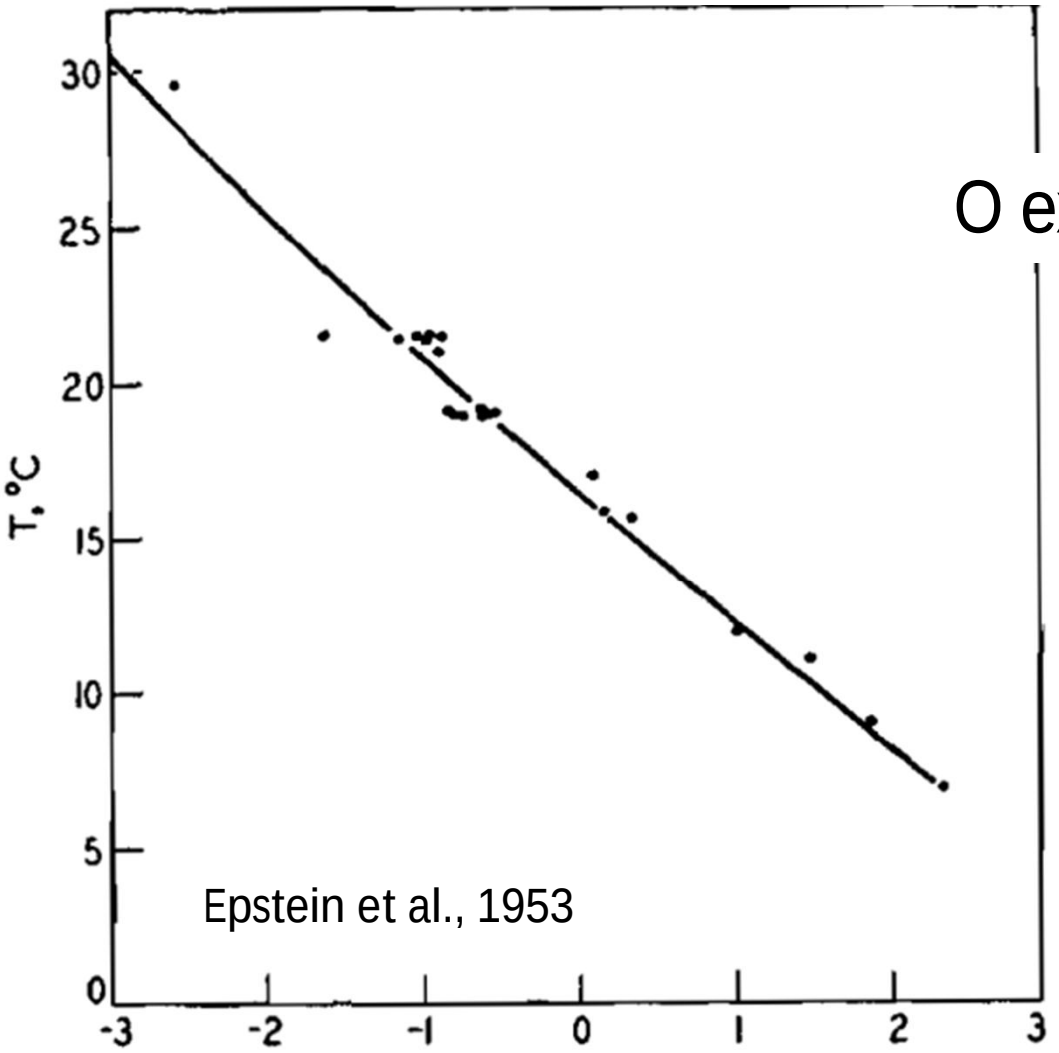


Lateral view of the internal shell of a generalized belemnite. ©Liz Shea, DMNH.

www.thefossilforum.com

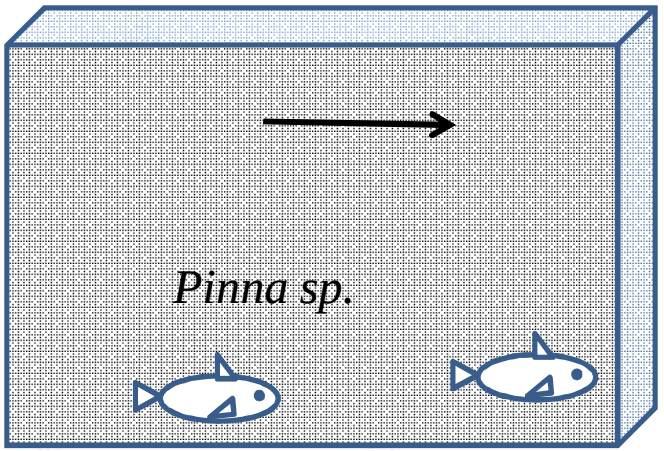
Mais sobre as equações de (paleo) temperatura

Equação de Paleotemperatura



$\delta^{18}O_{CaCO_3} - \delta^{18}O_{H_2O}$

O experimento de Epstein:



T controlada

$\delta^{18}O_{H_2O}$ controlado

$$\mathbf{T} = 16.9 - 4.2(\delta^{18}O_{CaCO_3} - \delta^{18}O_{H_2O}) + 0.13(\delta^{18}O_{CaCO_3} - \delta^{18}O_{H_2O})^2$$

na verdade era $\delta^{18}O$ de CO_2 produzido por reação com ácido

na verdade era $\delta^{18}O$ de CO_2 equilibrado com H_2O

Portanto, no caso o usuário tenha valores de $\delta^{18}O$ de $CaCO_3$ e H_2O (verdadeiros) é preciso usar uma outra equação (corrigida)

$$T = 15.75 - 4.3(\delta^{18}O_{CaCO_3} - \delta^{18}O_{H_2O}) + 0.14(\delta^{18}O_{CaCO_3} - \delta^{18}O_{H_2O})^2$$

$CaCO_3$ vs. V-PDB

H_2O vs. V-SMOW

Sharp (2007) *Principles of Stable Isotope Geochemistry*. Prentice Hall 344 pp.

Porem, existem varias equações especificas para organismos diferentes (moluscos, otólitos de peixes, foraminíferos de diferentes bacias oceânicas) ou para carbonatos inorgânicos (espeleotemas, etc). Portanto, as vezes é preciso **converter** os valores exprimidos vs. V-PDB para V-SMOW (ou vice-versa)

Equation of Kim & O'Neil (1997)
(carbonato inorgânico precipitado em laboratório)

$$10^3 \ln \alpha_{calcite\text{-}water} = 18.03\,(10^3\,T^{-1}) - 32.42$$

$$(\sim \delta^{18}O_{cal} - \delta^{18}O_{water})$$

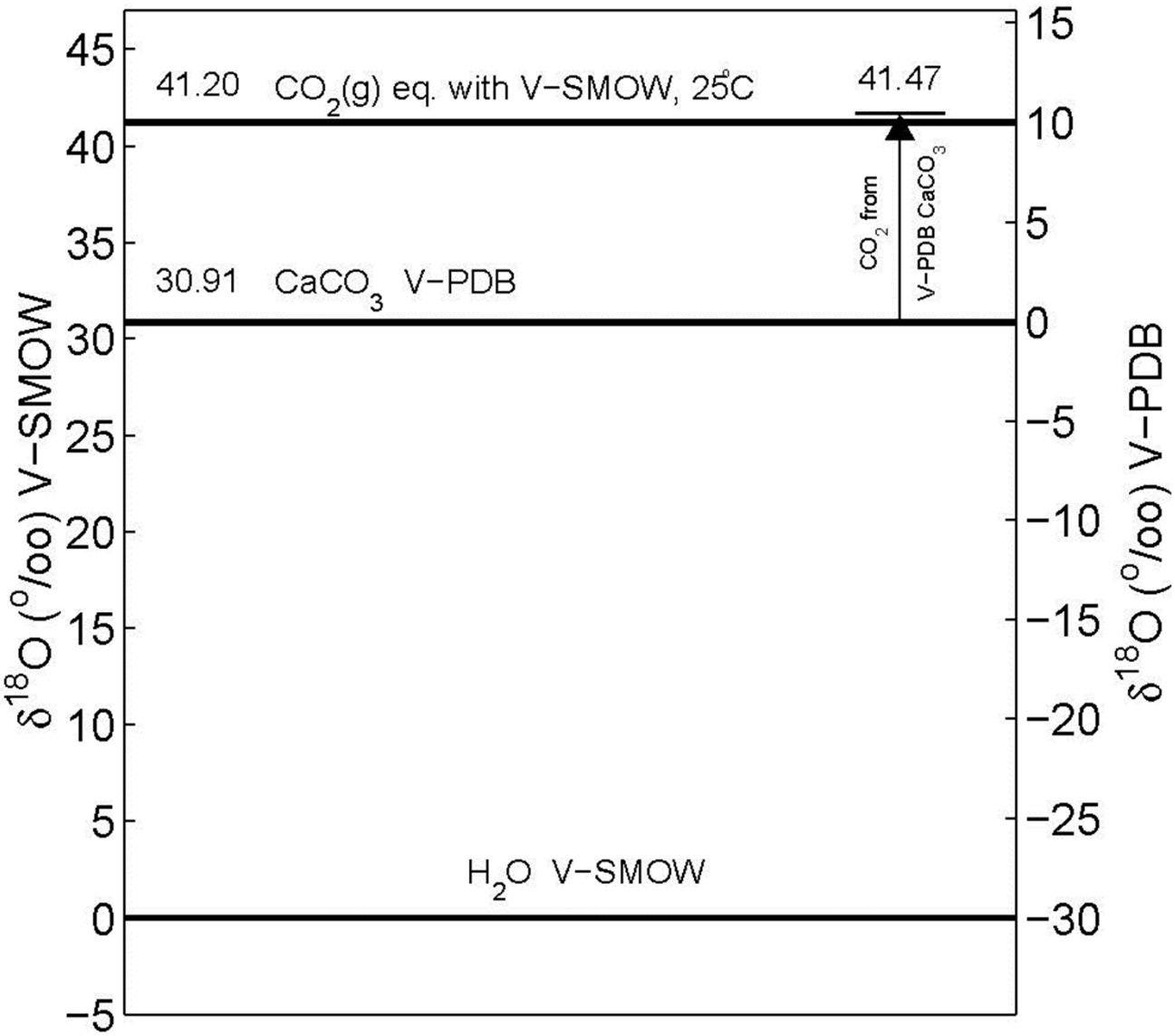
Essa relação vale **só** se a precipitação acontece

ao **equilíbrio isotópico** entre $CaCO_3$ e agua

Kim & O'Neil , 1997
Geochim. Cosmochim. Acta (61), 3461-3475.

Conversão de escala entre SMOW e PDB

SMOW vs. PDB



Zeebe & Wolf-Gladrow (2006)

Conversão de escala entre SMOW e PDB

$$\delta^{18}O_{SMOW} = 1{,}03091\ (\delta^{18}O_{PDB}) + 30{,}91$$

$$\delta^{18}O_{PDB} = 0{,}97002\ (\delta^{18}O_{SMOW}) - 29{,}98$$

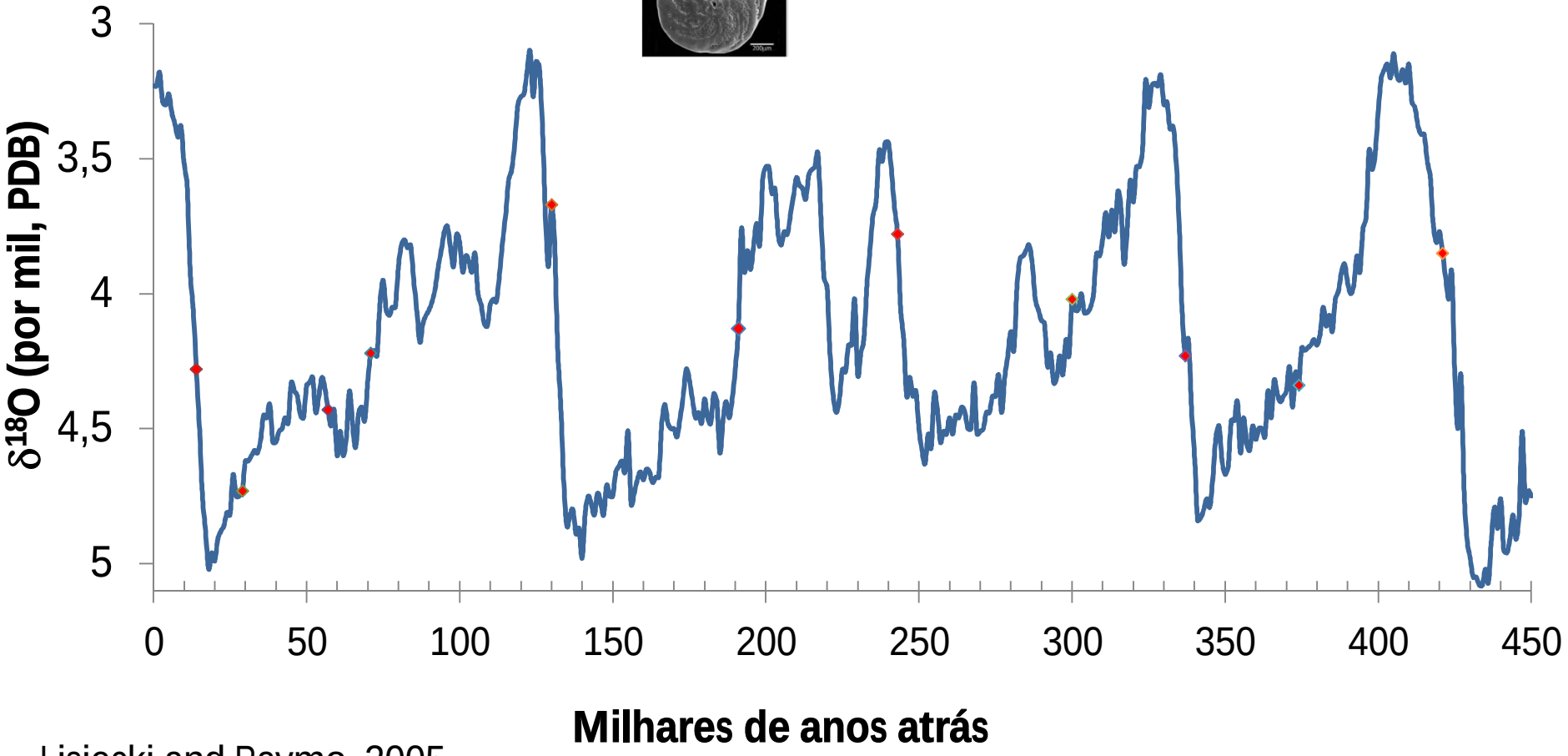
(Coplen et al. 1983)

Estratigrafia isotópica

Marine Isotope Stages (MIS)

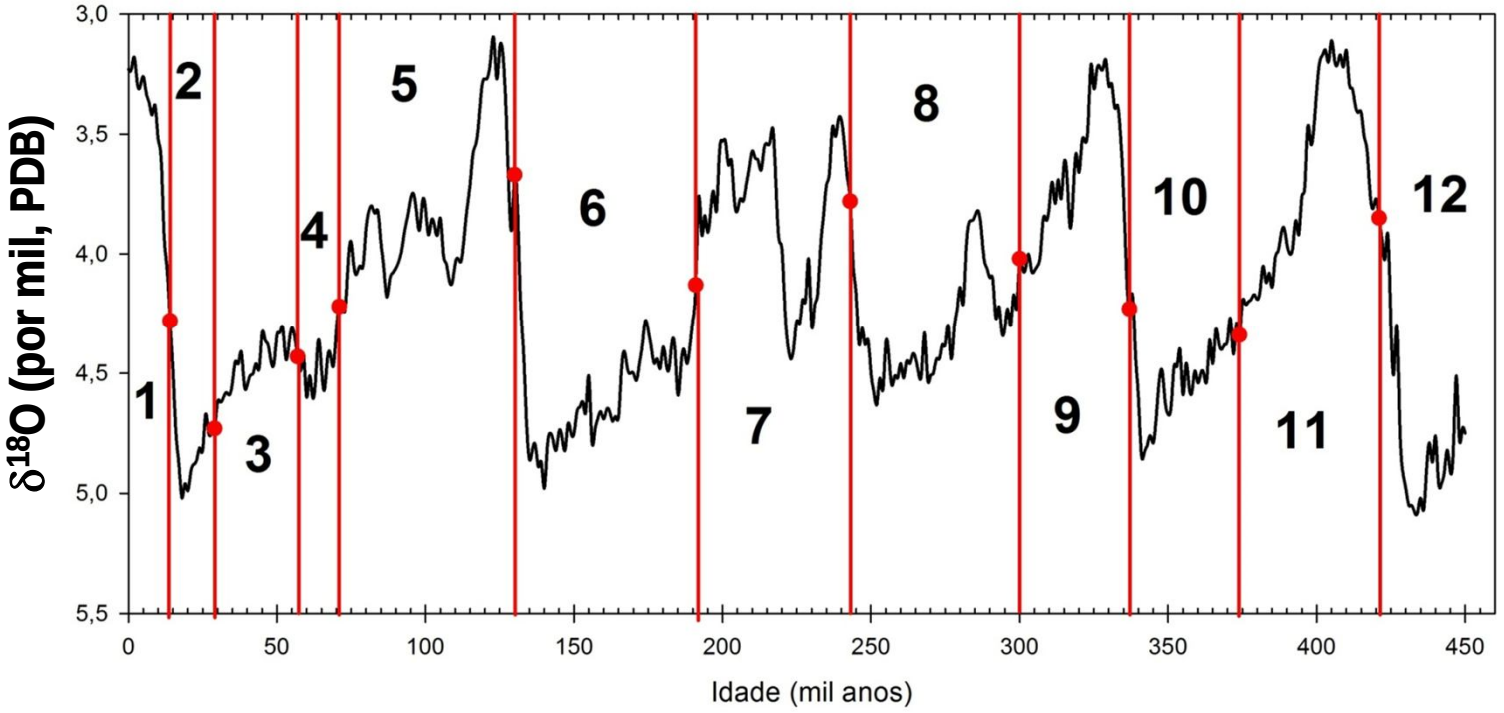
Curva padrão de $\delta^{18}O$ do oceano global (LR04): Estratigrafia isotópica dos ciclos glaciais

Bentônicos epifaunais



Lisiecki and Raymo, 2005

Curva padrão LR04
Estratigrafia isotópica

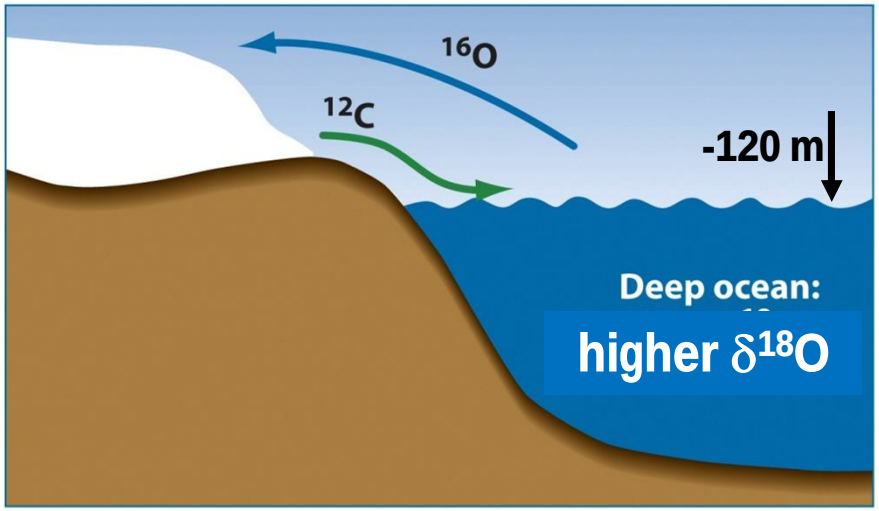


MIS limites	LR04 idade(mil anos)
1/2	14
2/3	29
3/4	57
4/5	71
5/6	130
6/7	191
7/8	243
8/9	300
9/10	337
10/11	374
11/12	421
12/13	478

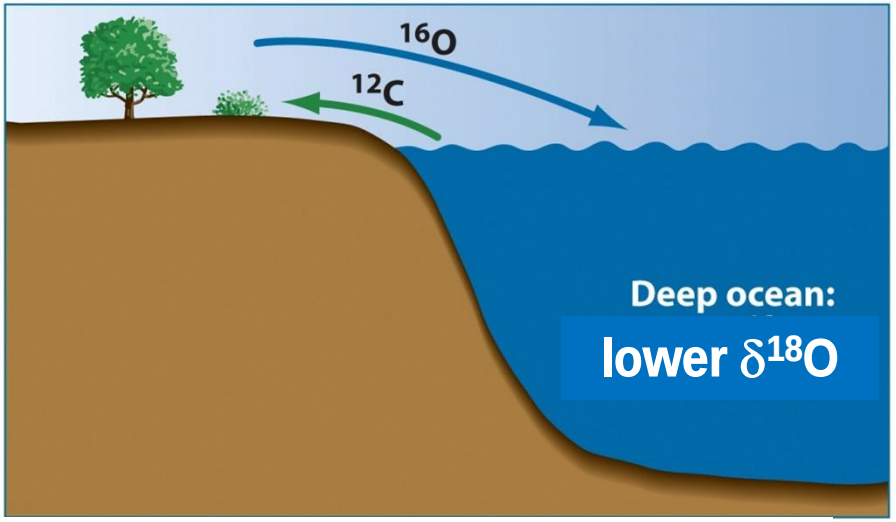
$\delta^{18}O$ do oceano

ice volume effect

GLACIAL

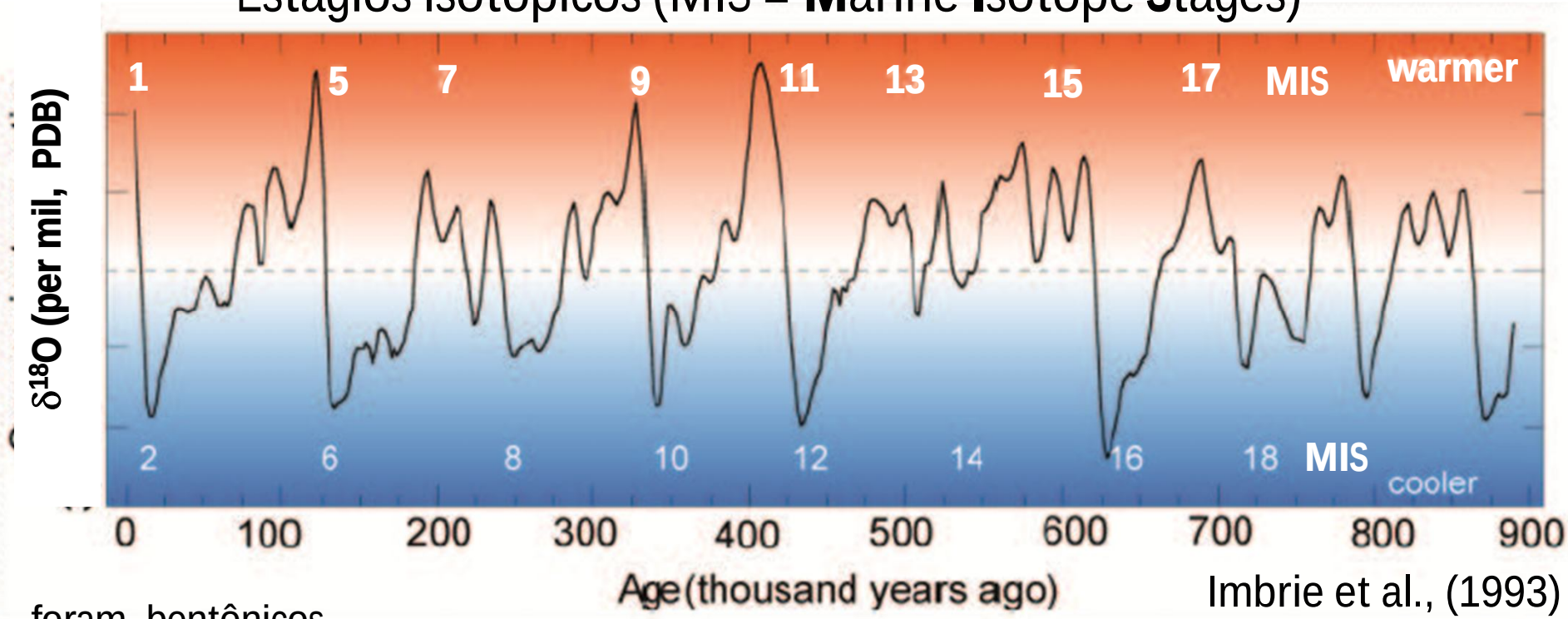


INTERGLACIAL



Estratigrafia isotópica dos ciclos glaciais

Estágios isotópicos (MIS = **M**arine **I**sotope **S**tages)



Imbrie et al., (1993)

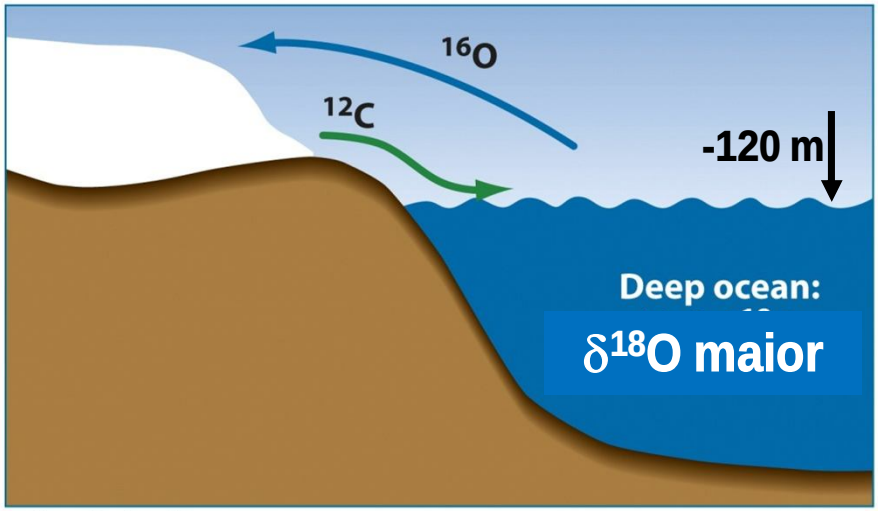
foram. bentônicos

Curva obtida por sobreposição (***stack***) de varias curvas (testemunhos coletados in todos oceanos).

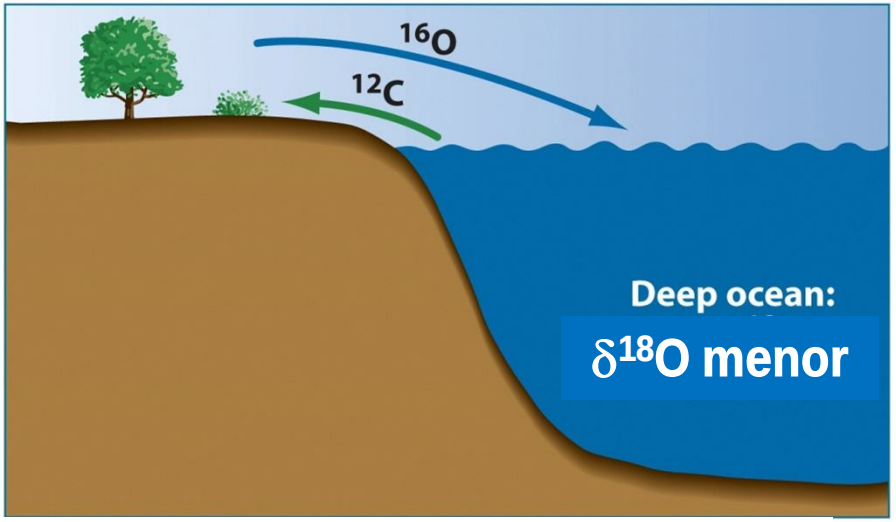
$\delta^{18}O$ do oceano

ice volume effect

GLACIAL



INTERGLACIAL

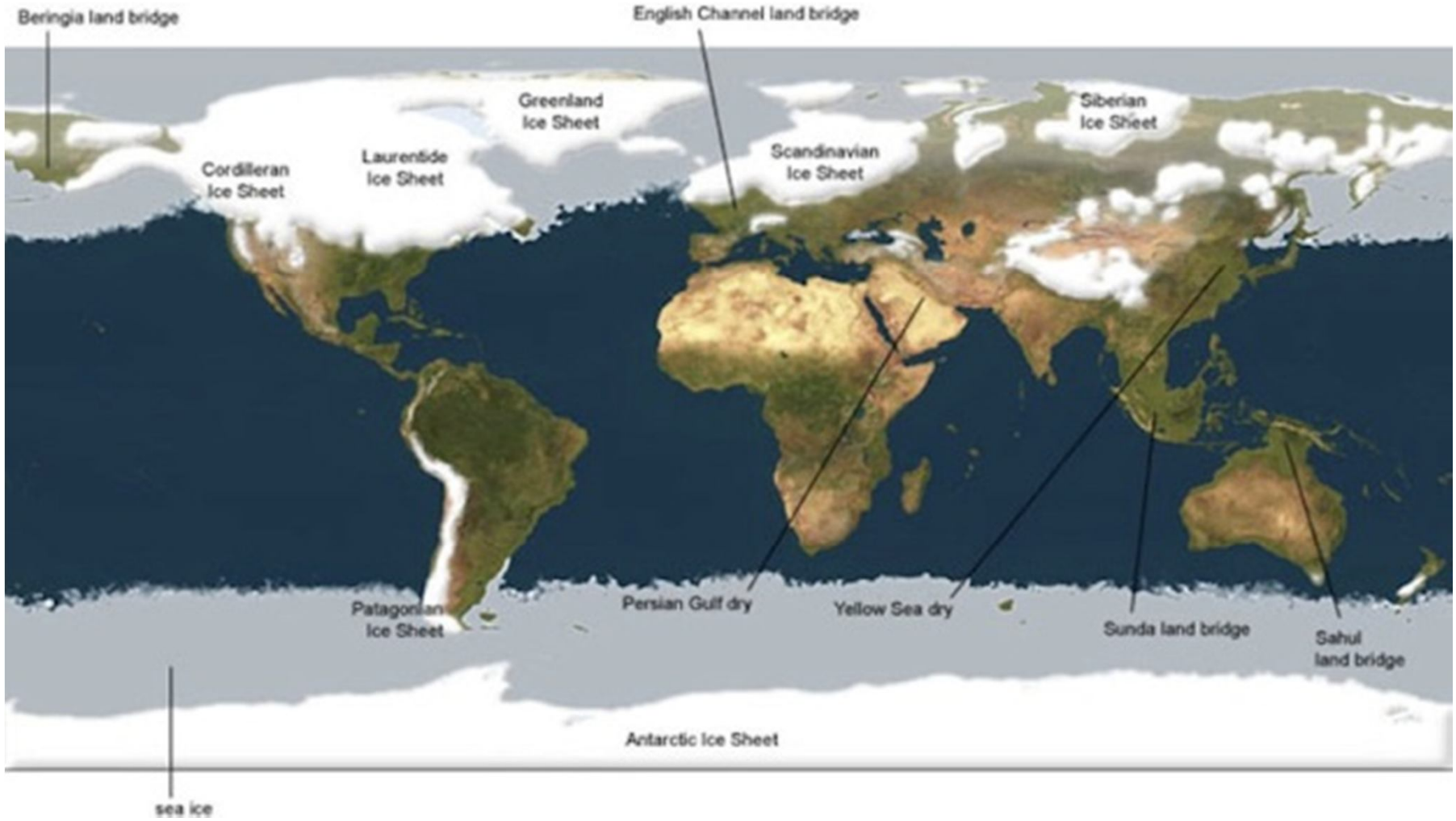


Períodos glaciais:

Água do oceano global era enriquecida em ^{18}O

$\delta^{18}O$ do oceano era **1,2‰** maior do que hoje

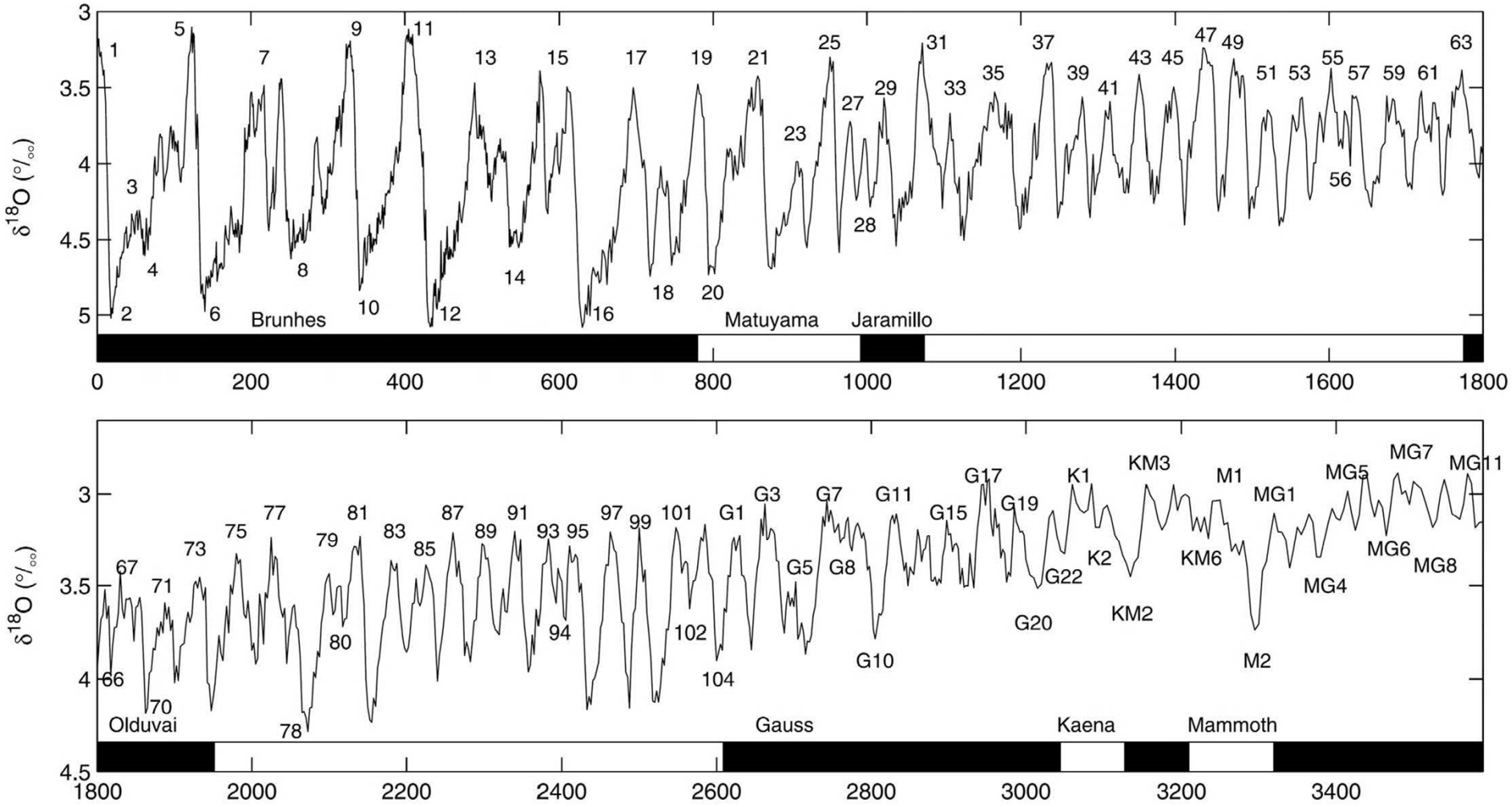
Estudo de glaciações: quantas?



www.pinterest.com

Lisiecki and Raymo, 2005

ESTÁGIOS ISOTÓPICOS



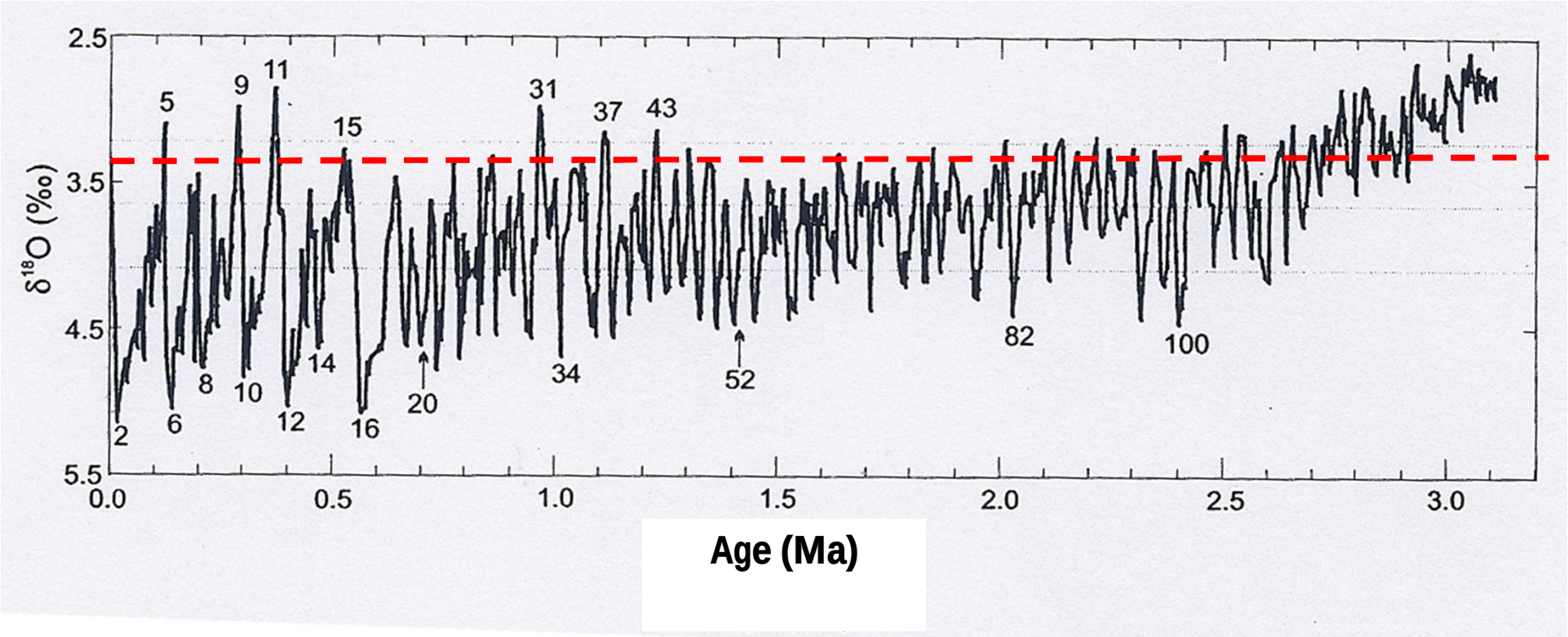
Principais feições climáticas do Pleistoceno (e do Quaternário todo i.e., os últimos ~2.5 Ma)

Períodos com valores de δ^{18}**O menores**

que durante o Holoceno (o presente interglacial)

indicam que existiram ocasiões em que havia

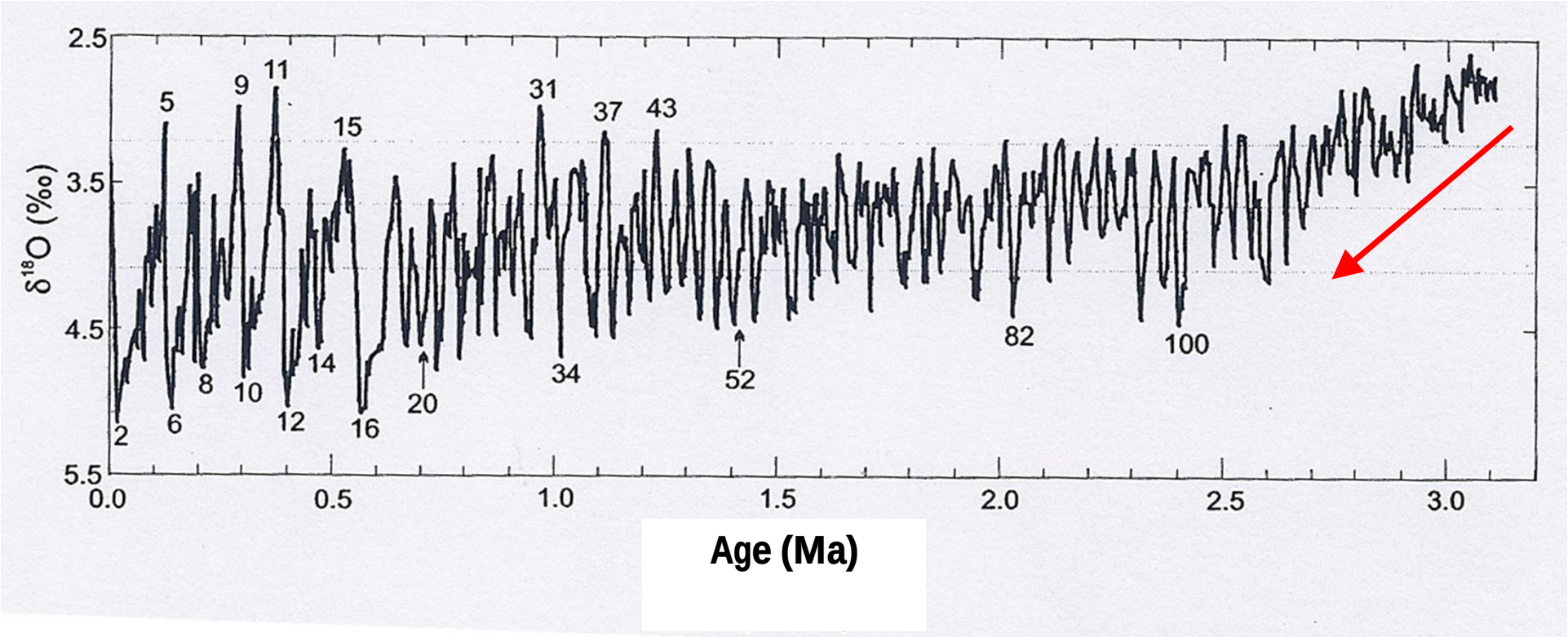
menos gelo continental do que hoje



De 3.1 a 2.6 Ma: aumento de $\delta^{18}O$

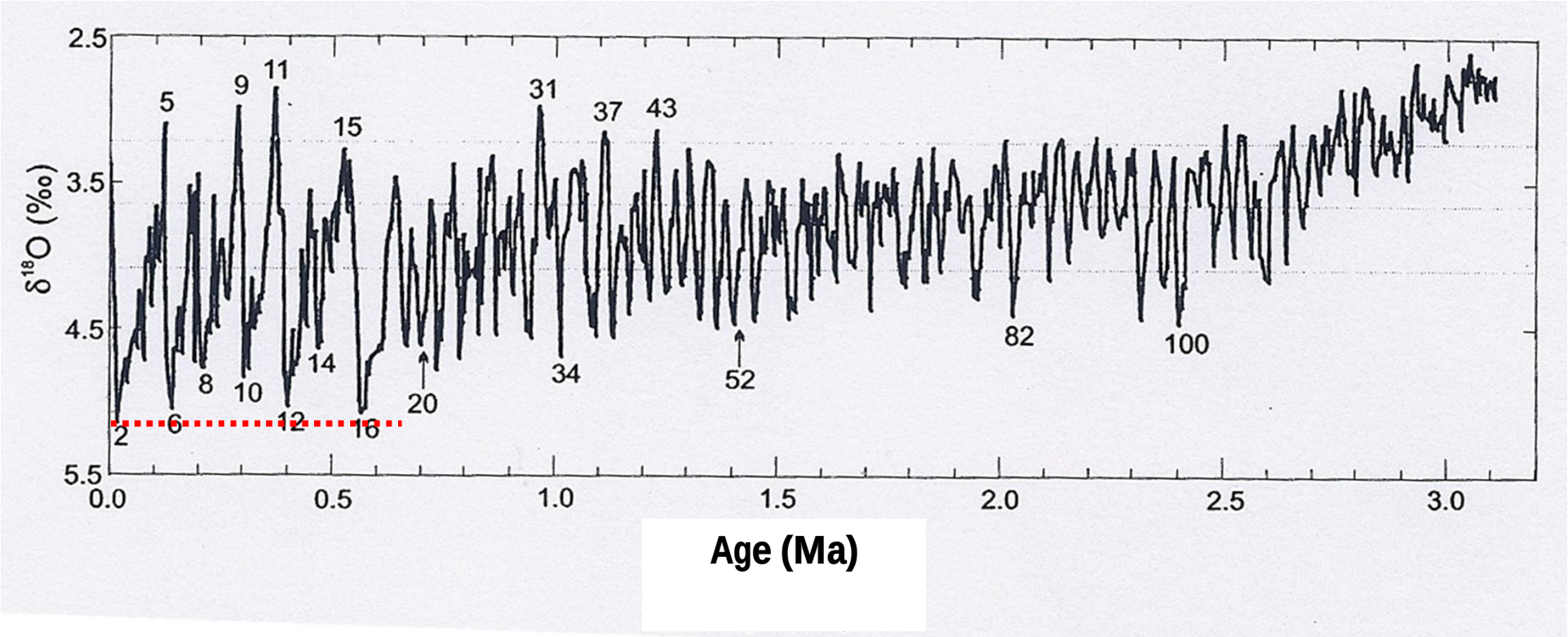
indica **progressivo resfriamento**, provavelmente associado com **crescimento da calota Antártica**

e das geleiras continentais no hemisfério norte



Somente a partir de ~0.6 Ma (MIS 16) os episódios de glaciação continental são comparáveis em magnitude com o período glacial mais recente

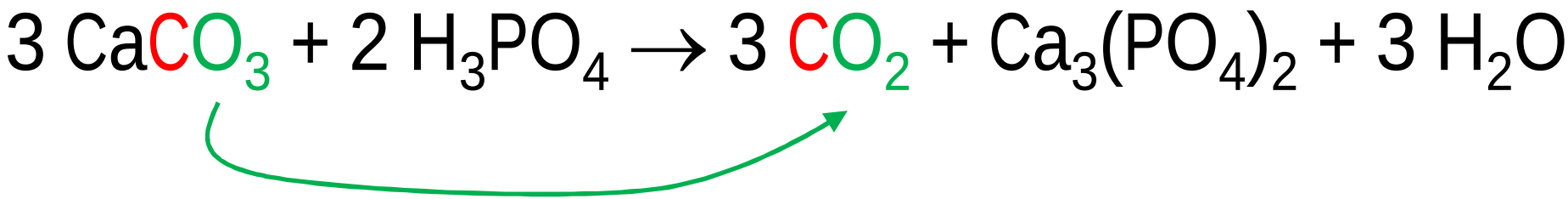
(Last Glacial Max. = MIS 2)



Questões importantes ao respeito da analise isotópica de $CaCO_3$

Fator de fracionamento devido ao ácido (*acid fractionation factor*)

Método de McCrea (1950)



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ScienceDirect
Chemical Geology 246 (2007) 135–146
CHEMICAL GEOLOGY
INCLUDING
ISOTOPE GEOSCIENCE
www.elsevier.com/locate/chemgeo

Phosphoric acid fractionation factors for calcite and aragonite between 25 and 75 °C: Revisited

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[b] *Geological Survery of Canada, Ottawa, Ontario, Canada K1A OE8*

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Editor: D. Rickard

Abstract

This paper reports the results of an investigation to re-determine the phosphoric acid fractionation factor for calcite and aragonite in order to improve the accuracy and limit the uncertainty of this very important quantity. The $^{18}O/^{16}O$ ratio of 100% of the oxygen in calcite and aragonite was determined directly by combining data from a two-step procedure involving thermal decomposition of the carbonate, followed by fluorination of the residual CaO using BrF_5.

The oxygen isotope composition of the CO_2 released upon the thermal decarbonation of calcite and aragonite is enriched in ^{18}O relative to that of the remaining CaO. Reproducibility of the oxygen isotope composition of the CO_2 derived from each of the decarbonation and fluorination steps was 0.55‰ (1σ). Nevertheless, statistically reliable acid fractionation factors for calcite ($s_e=0.07$) and aragonite ($s_e=0.10$) were determined on the basis of a large set of combined data from conventional phosphoric acid reactions at three different temperatures, plus the decarbonation/fluorination procedure. The acid fractionation factors ($\alpha_{CO_2(ACID)\text{-carbonate}}$) determined in this study for calcite ($\alpha_{CO_2(ACID)\ Calcite}=1.01030$) and aragonite ($\alpha_{CO_2(ACID)\ Aragonite}=1.01063$) at 25 °C, differ slightly from previously published values.

On the basis of our new determinations at 25, 50, and 75 °C, revised expressions for the temperature dependence of the acid

$$1000\ln\alpha_{CO_2(ACID)-Calcite} = 3.59\left(10^3/T\right) - 1.79$$

$$1000\ln\alpha_{CO_2(ACID)-Aragonite} = 3.39\left(10^3/T\right) - 0.83$$

$$10^3 \ln\alpha_{(CO2\text{-}CaCO3)} \cong \delta^{18}O_{CO2} - \delta^{18}O_{CaCO3}$$

Fracionamento de ^{18}O entre água e carbonato de cálcio não é o mesmo para calcita e aragonita

Equation of Kim & O'Neil (1997)
(carbonato inorgânico precipitado em laboratório)

$$10^3 \ln \alpha_{calcite\text{-}water} = 18.03\,(10^3\,T^{-1}) - 32.42$$

$$(\sim \delta^{18}O_{cal} - \delta^{18}O_{water})$$

Essa relação vale **só** se a precipitação acontece

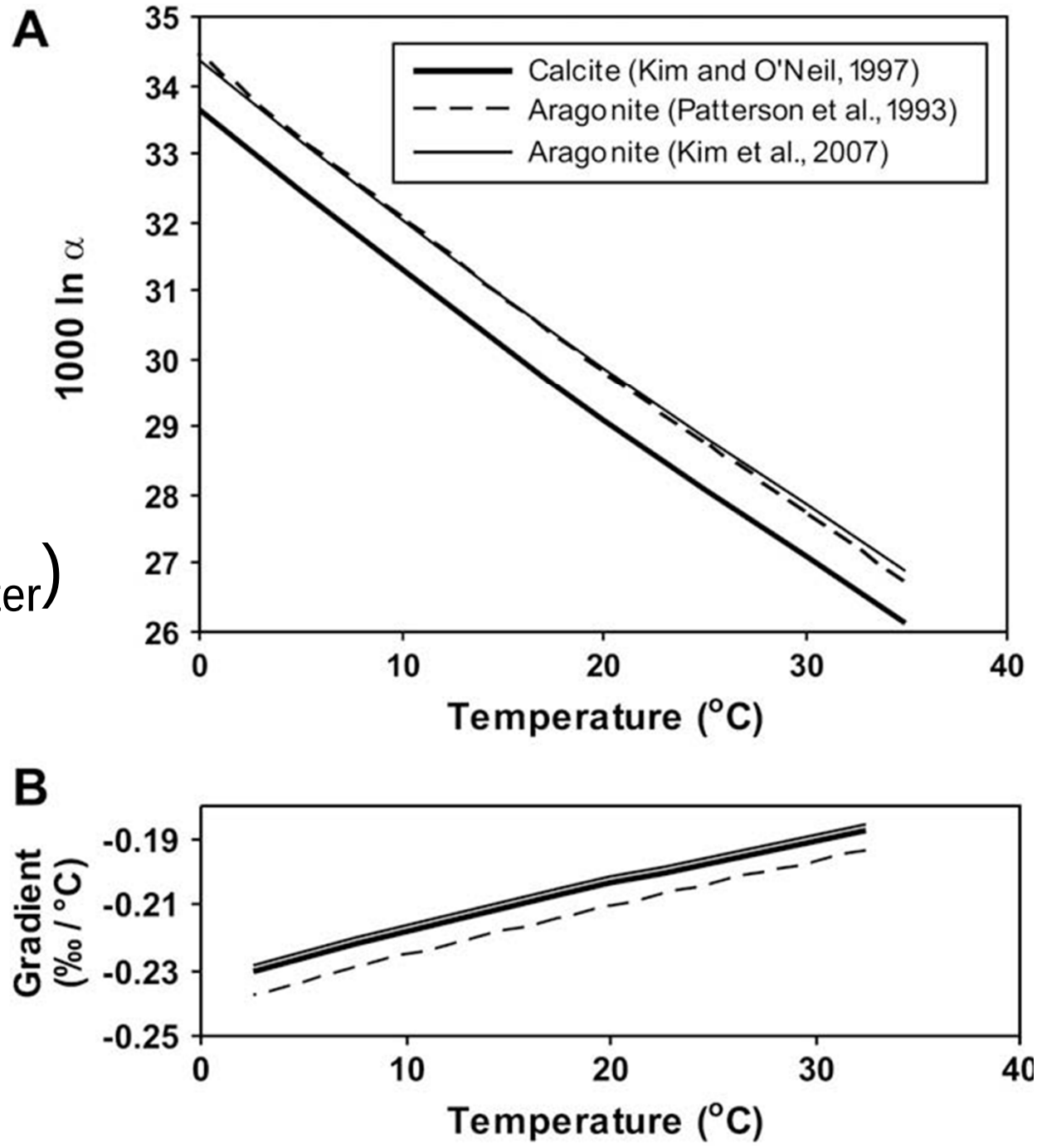
ao **equilíbrio isotópico** entre $CaCO_3$ e agua

Kim & O'Neil , 1997
Geochim. Cosmochim. Acta (61), 3461-3475.

EFEITO DA TEMPERATURA

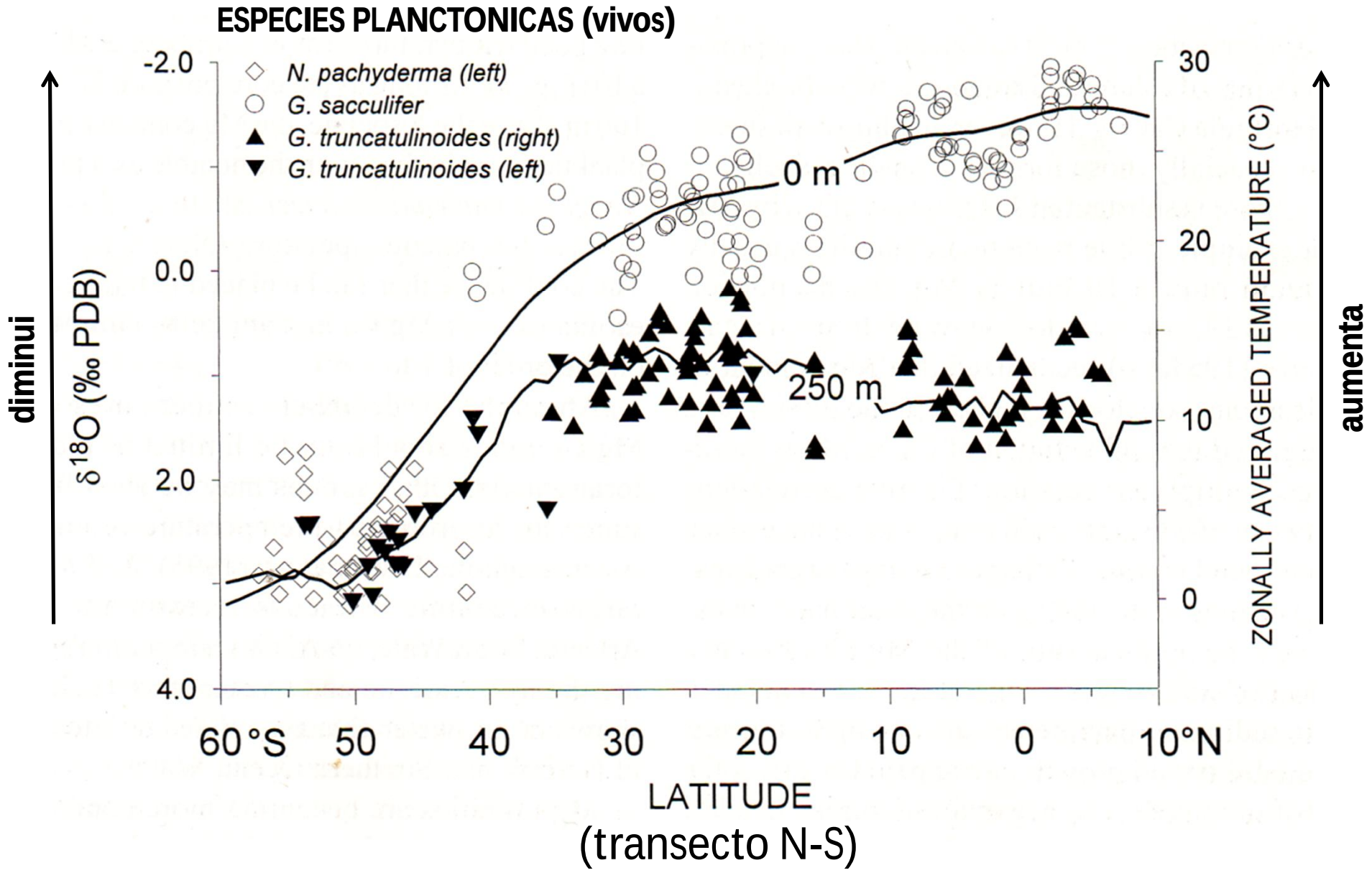
$10^3 \ln \alpha_{CaCO3\text{-}water}$

$(\sim \delta^{18}O_{carb} - \delta^{18}O_{water})$

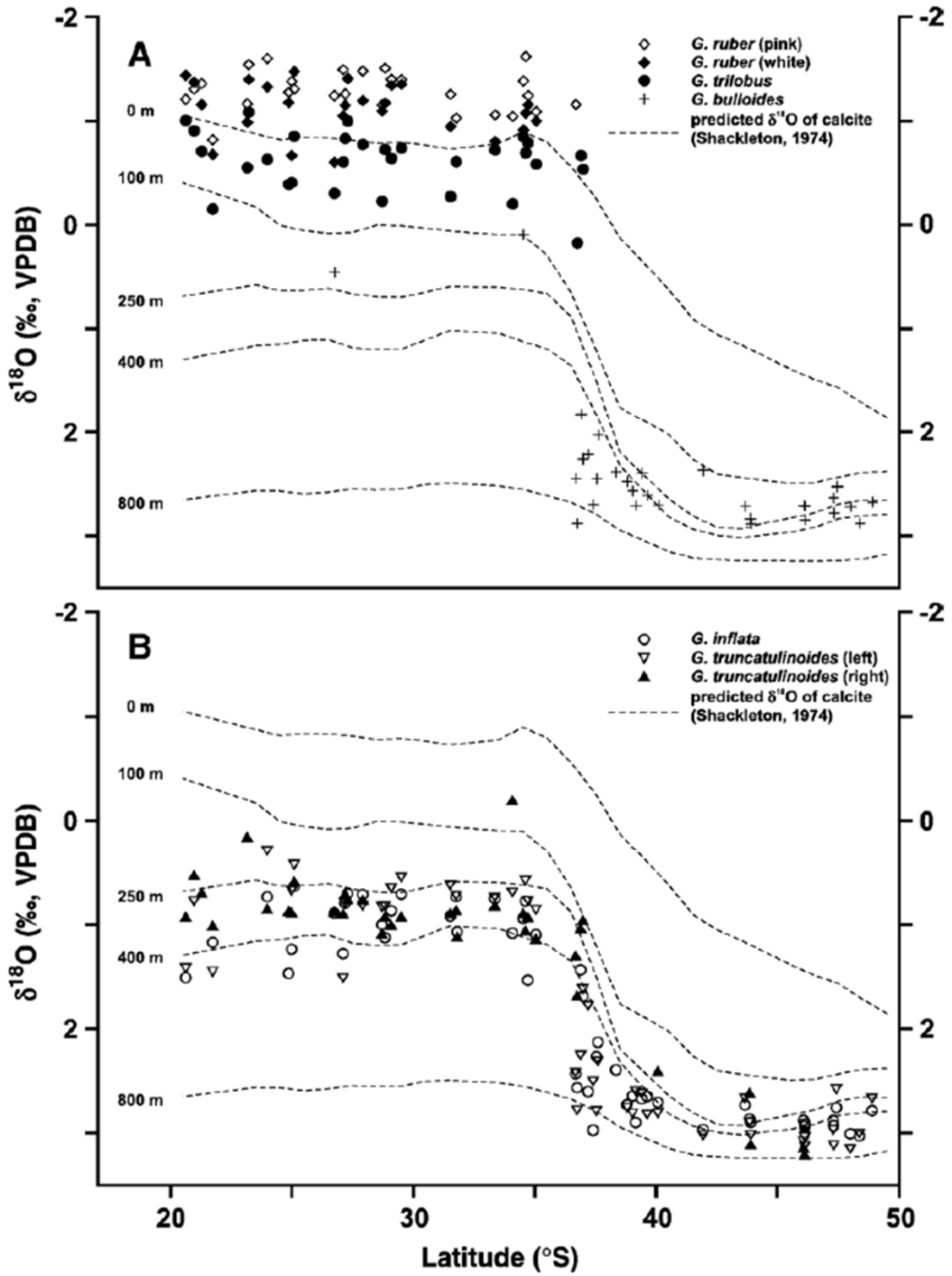


Lachniet et al. 2009

Reconstrução de paleoestratificação



G. Fisher & G. Wefer eds. (1999): *Use of proxies in paleoceanography* (Springer), 735 pp.



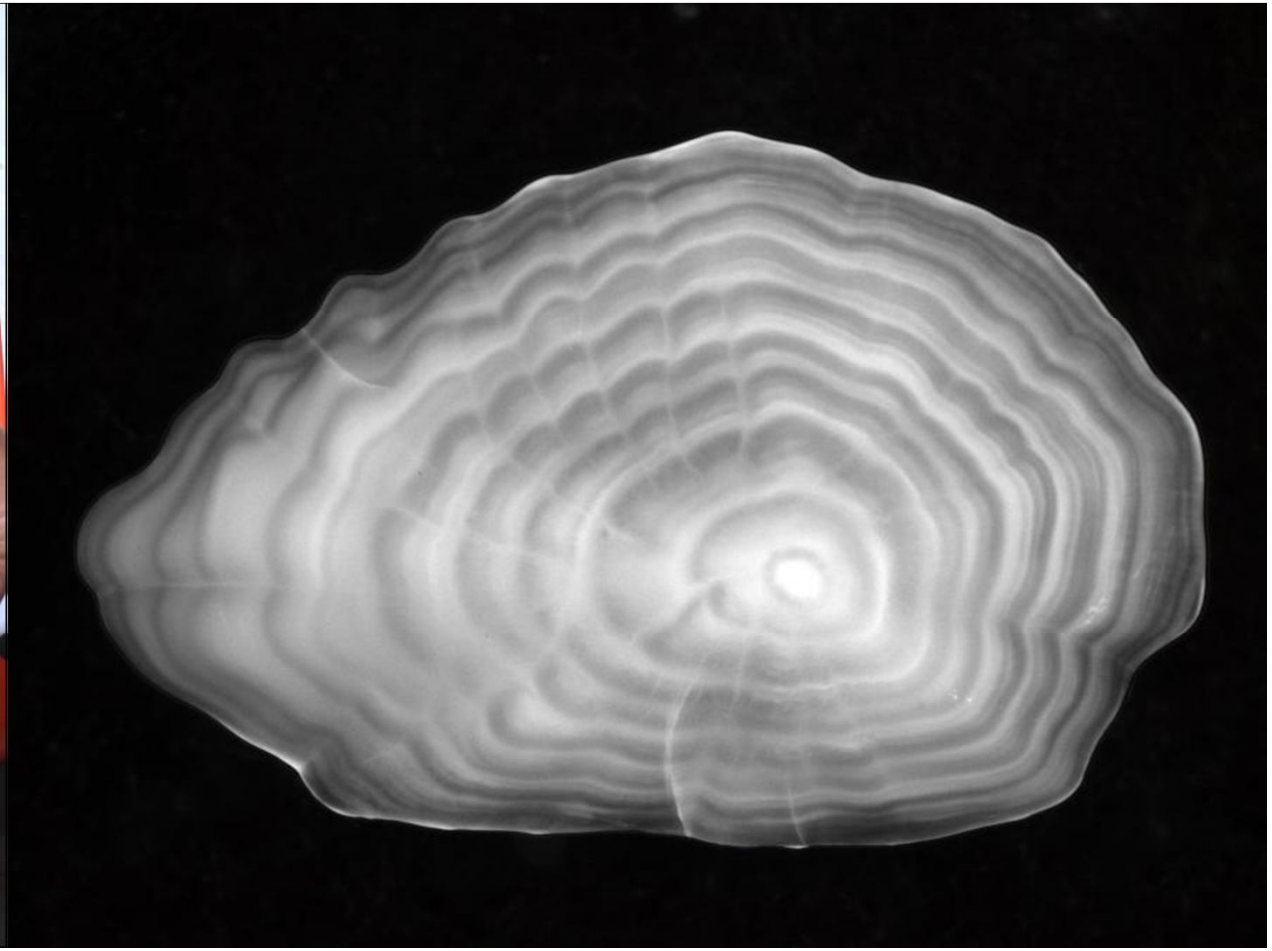
Chiessi et al. 2007
Marine Micropaleontology (64)
pp. 52–66

$$\delta^{18}O_{CaCO_3} = f\,(\delta^{18}O_{água}\ ,T)$$

$$\delta^{18}O_{CaCO_3} = f\,(\delta^{18}O_{água}\ ,T)$$

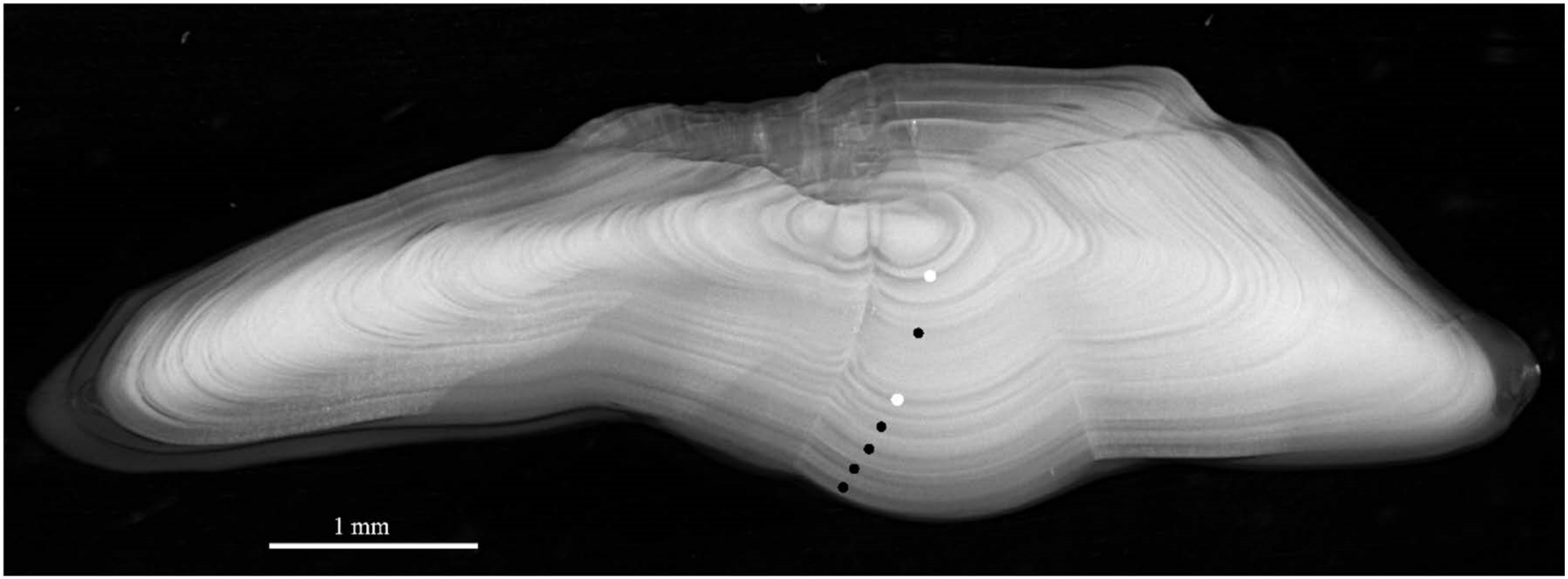
$$\text{densidade} = f\,(S\ ,T)$$

$\delta^{18}O$ em otólitos de peixes



Determinação de idade de peixes

Camadas de crescimento diário em otólitos de peixes



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Full length article

Age validation of Pacific cod (*Gadus macrocephalus*) using high-resolution stable oxygen isotope ($\delta^{18}O$) chronologies in otoliths

Craig R. Kastelle[a,*], Thomas E. Helser[a], Jennifer L. McKay[b], Chris G. Johnston[a,1], Delsa M. Anderl[a], Mary E. Matta[a], Daniel G. Nichol[a]

[a] Resource Ecology and Fisheries Management Division, Alaska Fisheries Science Center National Marine Fisheries Service, National Oceanic and Atmospheric Administration, 7600 Sand Point Way, Seattle, WA 98115, USA
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Age validation
Stable oxygen isotopes
Ageing bias
Temperature
Otolith

ABSTRACT

The Pacific cod (*Gadus macrocephalus*) fishery in Alaska is large and economically important; in 2014 its ex-vessel value was $203.8 million. Management relies on integrated assessments that employ survey abundance trends and fishery catches as well as age compositions from assessment surveys. However, Pacific cod age determination based on otolith growth zone counts has historically been difficult, adding to the uncertainty in biological reference points. Further, a mismatch exists where modes in fish length frequencies are larger than mean lengths at otolith-based age. To address ageing inaccuracy, we conducted an age validation study using stable oxygen isotopes ($\delta^{18}O$). This approach is based upon the principle that variability in marine carbonate $\delta^{18}O$ is inversely related to water temperature, which we independently verified in Pacific cod otoliths ($r^2 = 0.74$). We sequentially microsampled 40 Pacific cod otoliths from the core to the margin and measured the aragonite $\delta^{18}O$ by isotope ratio mass spectrometry. This provided a detailed $\delta^{18}O$ life history chronology for each specimen. First, we identified seasonal variation (the cyclical pattern of otolith $\delta^{18}O$ values) and determined whether the number of $\delta^{18}O$ maxima, considered to represent "true fish age," was consistent with the age estimated from growth zone counts. Second, we estimated the probability of bias in the ages determined from growth zone counts. Overall, the probability of assigning an age (based on the number of counted growth zones) equal to the true age (number of $\delta^{18}O$ maxima) was approximately 61%. However, the probabilities of over- or underestimating the age by 1 year were 25% and 13%, respectively. The probability of over- or underestimating true age by 2 or more years was very low (<2%). The probability of age misclassification was used to correct bias in mean length at age.

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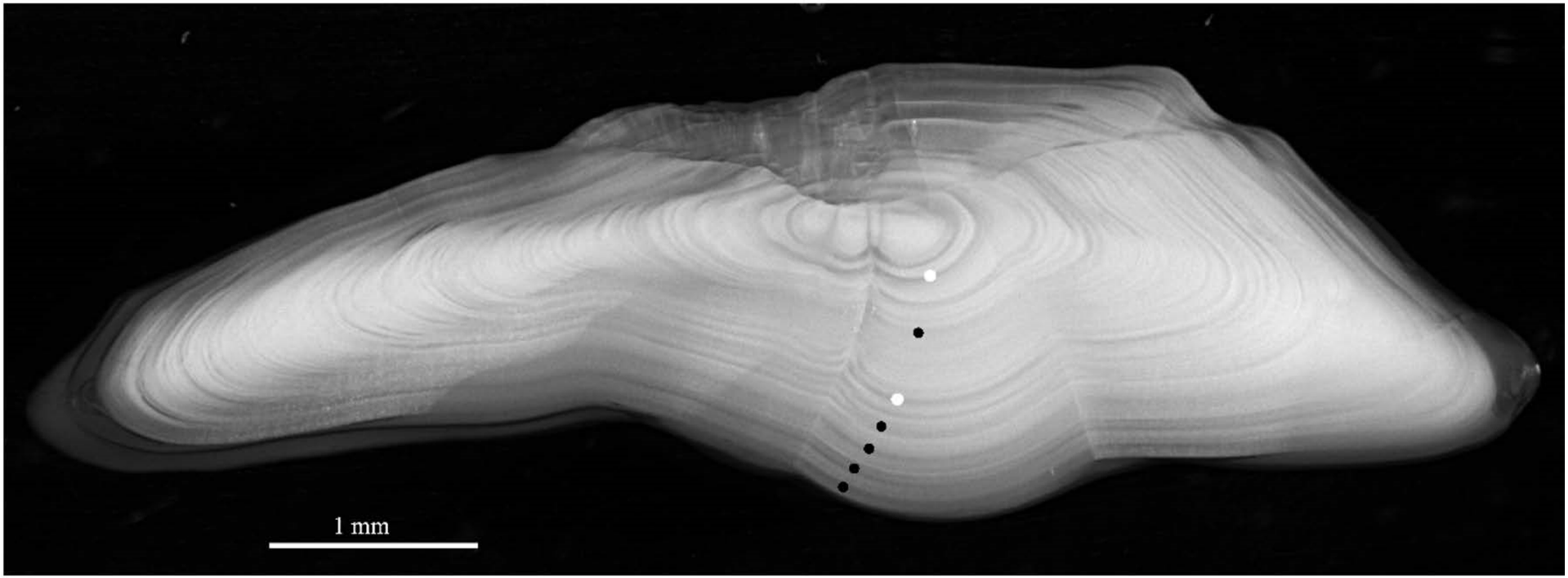
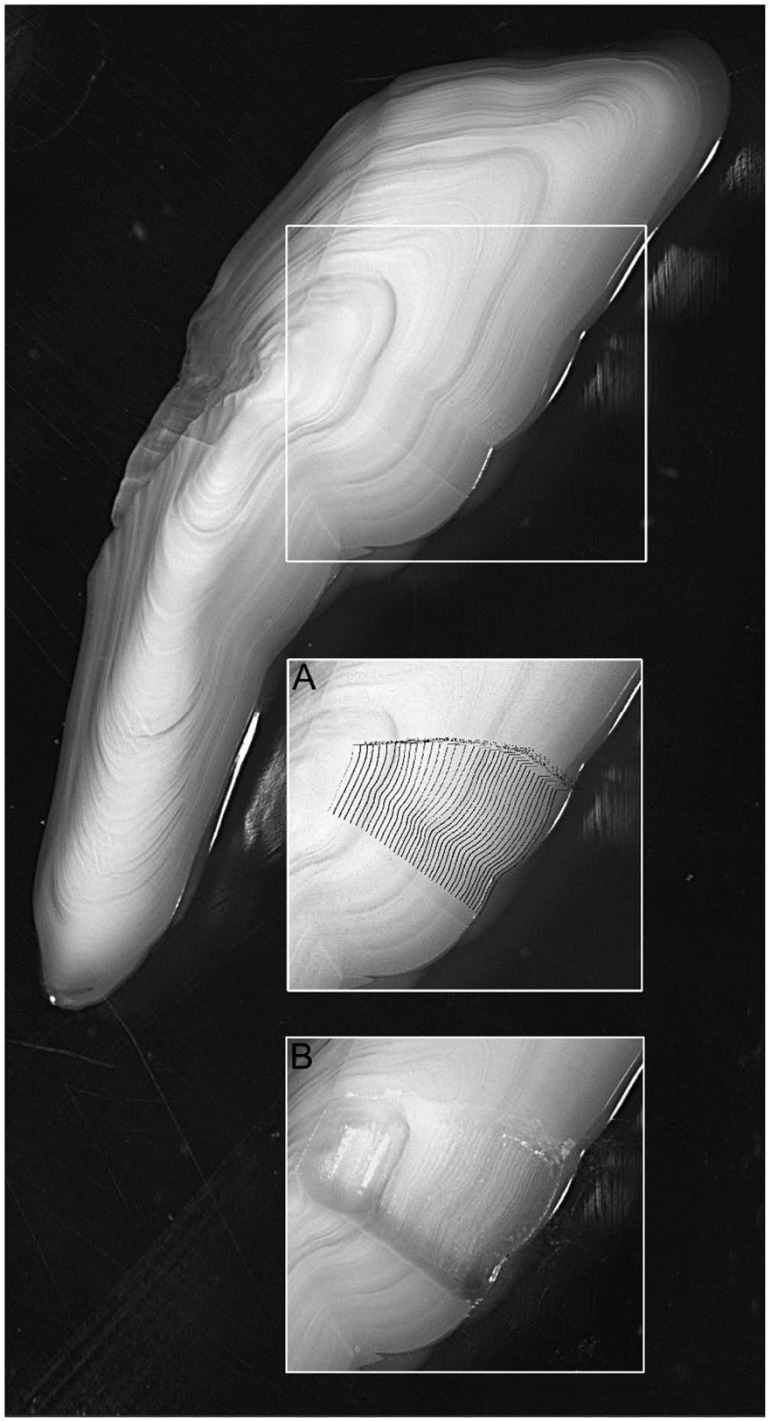
Camadas de crescimento diário em otólitos de peixes



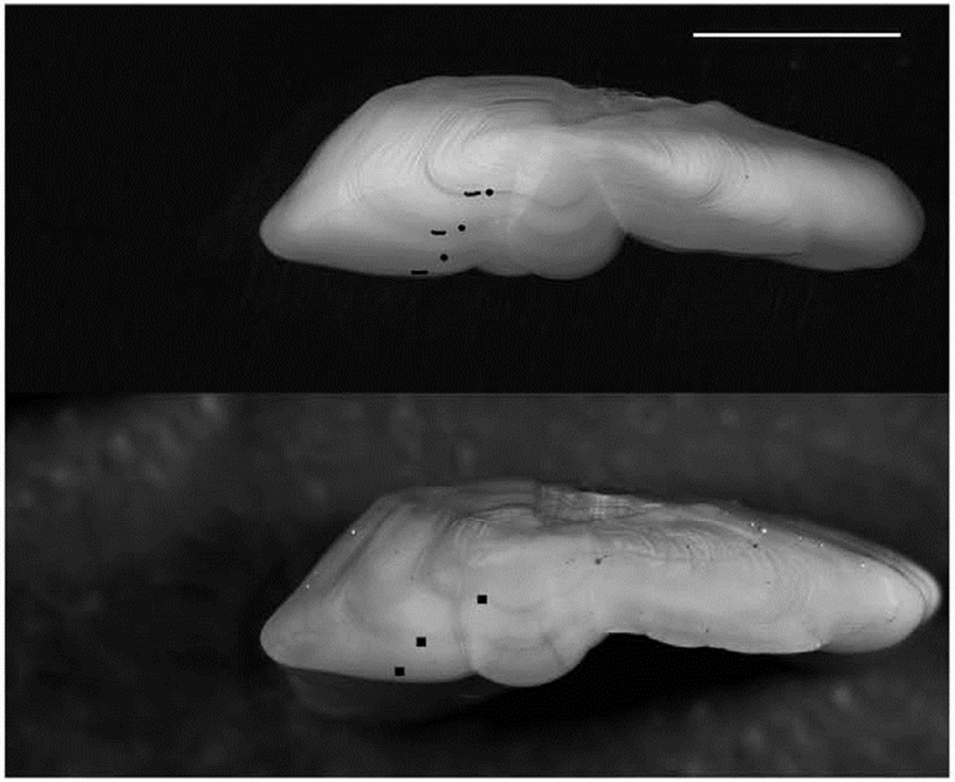
Questão cientifica:

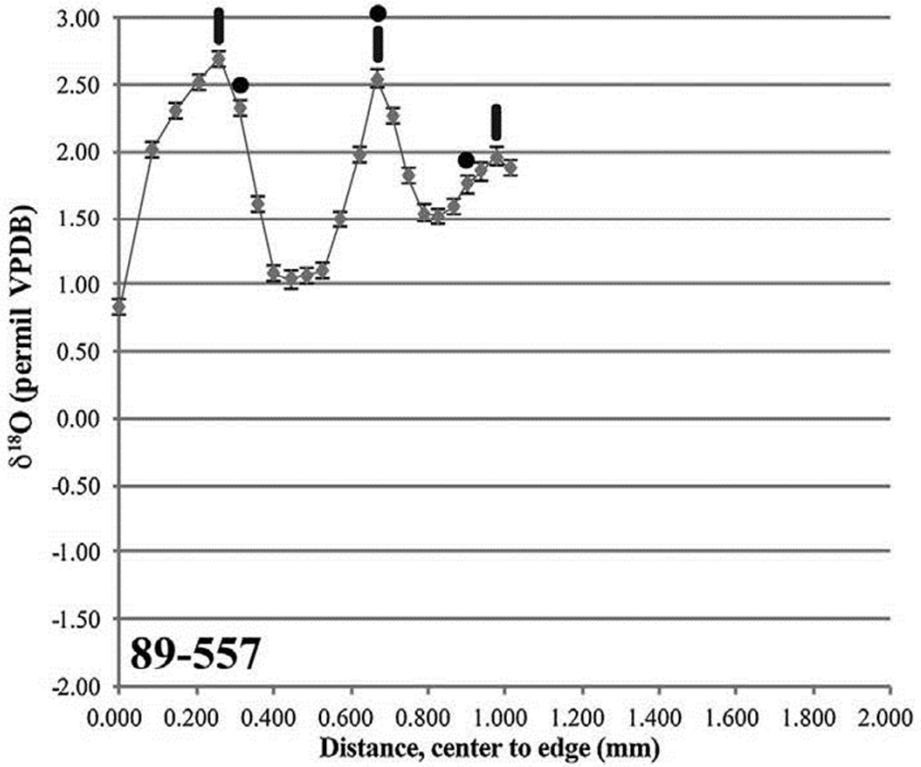
como identificar os anos de idade?



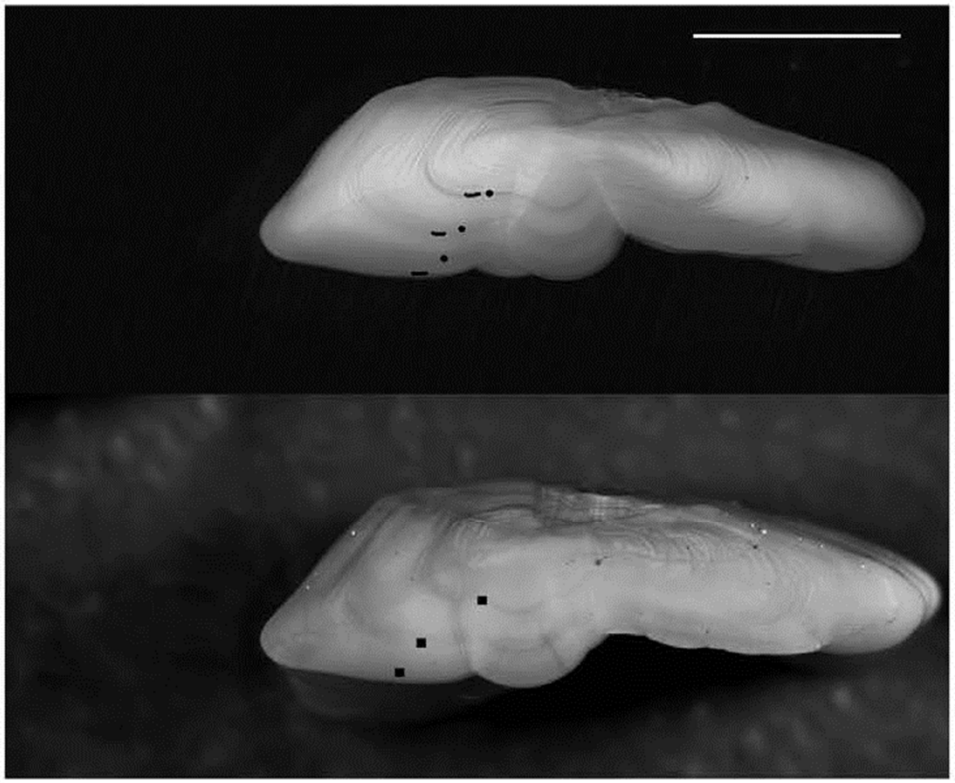
Micro-amostragem para obter perfis isotópicos

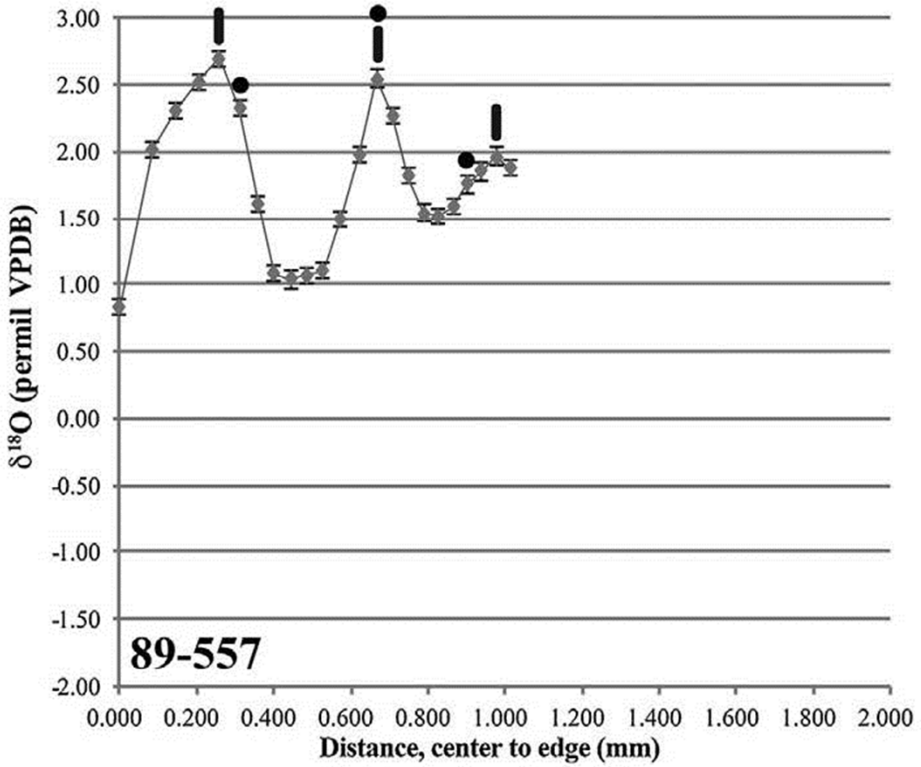
Como identificar os anos de idade?



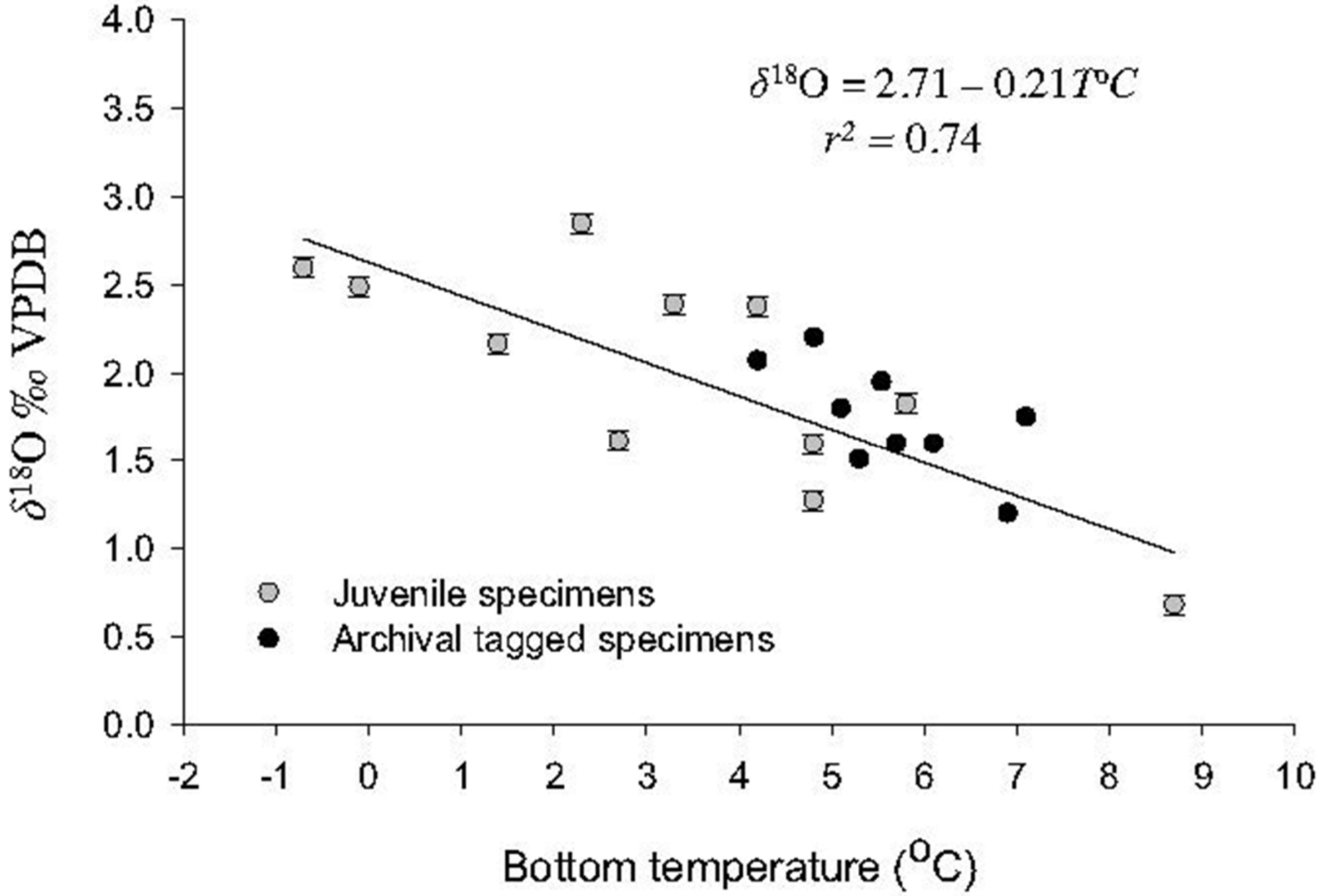


Picos de máximo $\delta^{18}O$ = invernos





O que precisamos verificar
antes de afirmar
que de fato
os picos de $\delta^{18}O$ = invernos ?


$\delta^{18}O = 2.71 - 0.21T°C$
$r^2 = 0.74$
$\delta^{18}O$ ‰ VPDB
Juvenile specimens
Archival tagged specimens
Bottom temperature (°C)
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
-2
-1
0
1
2
3
4
5
6
7
8
9
10

Uma relação inversa é encontrada

entre $\delta^{18}O_{oto}$ e temperatura,

o que confirma que

valores máximos de $\delta^{18}O$ = invernos

Estudo da mudança de profundidade de habitat de enguias de mar profundo

Estudo da mudança de profundidade de habitat de enguias de mar profundo

Questão cientifica:

Fase de vida inicial: camada superior

Fase adulta: camada profunda

?

Journal of FISH BIOLOGY

fsbi An International Society for Fish Biology

Journal of Fish Biology (2015) **86**, 845–853
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Vertical habitat shift of viviparous and oviparous deep-sea cusk eels revealed by otolith microstructure and stable-isotope composition

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**Institute of Oceanography, National Taiwan University, 1, Sec. 4, Roosevelt Rd, Taipei, Taiwan, Republic of China and †National Museum of Marine Science and Technology, No. 367, Pei-Ning Rd., Zhongzheng Dist., Keelung City, Taiwan, Republic of China*

(Received 22 July 2014, Accepted 12 November 2014)

Otolith stable-oxygen-isotope composition and microstructure were analysed in order to investigate the vertical habitat shift of deep-sea cusk eels (Ophidiiformes). Otolith $\delta^{18}O$ profiles suggested that both viviparous blind cusk eels and oviparous cusk eels experienced a pelagic larval stage and then settled to the deep-sea floor over a vertical distance that ranged among individuals from 200 to >1000 m. This result shows that the larvae of viviparous *Barathronus maculatus* undertake an ontogenetic vertical migration after a period of larval drift that may facilitate their wide distribution on the sea floor.

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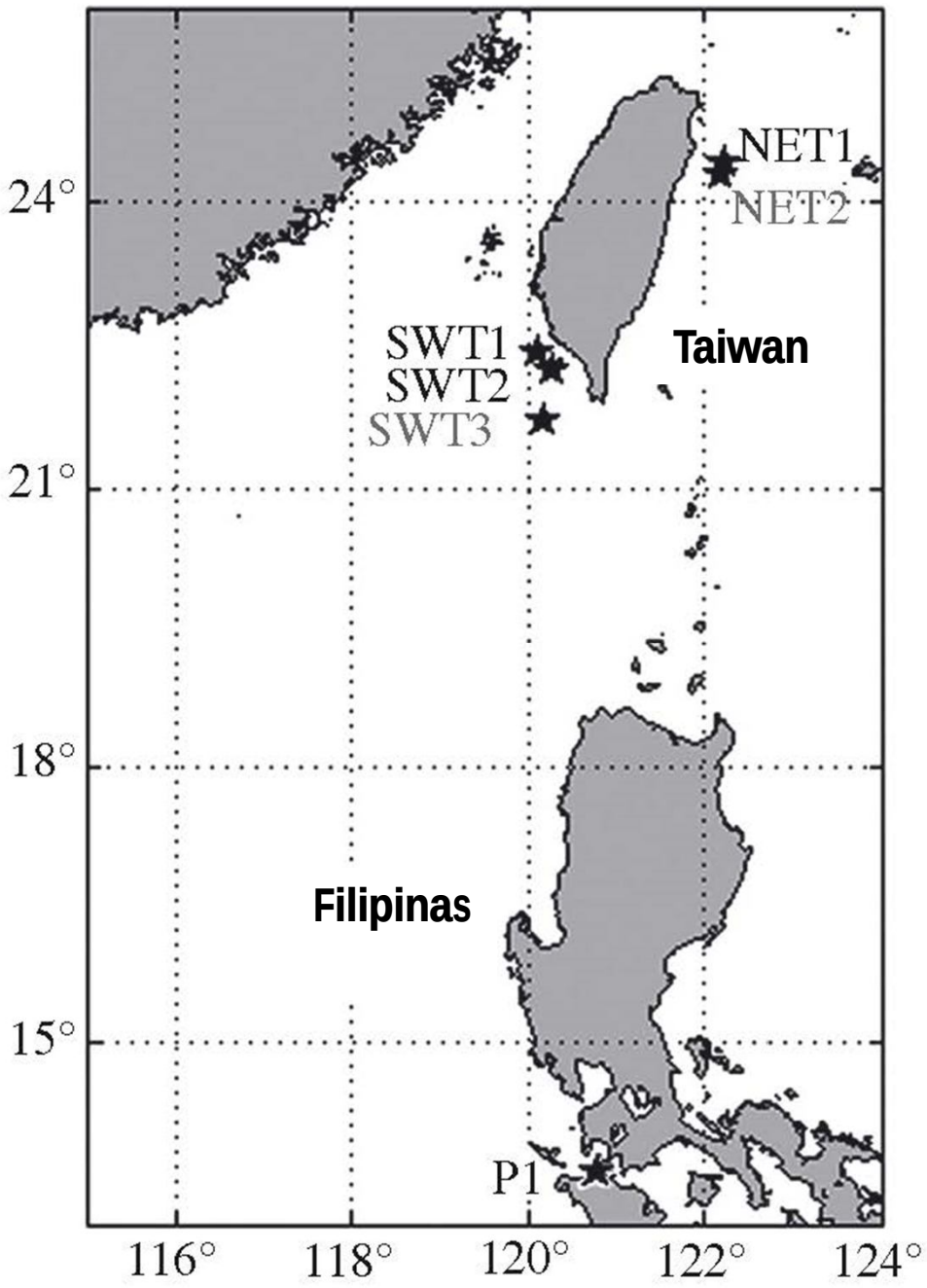
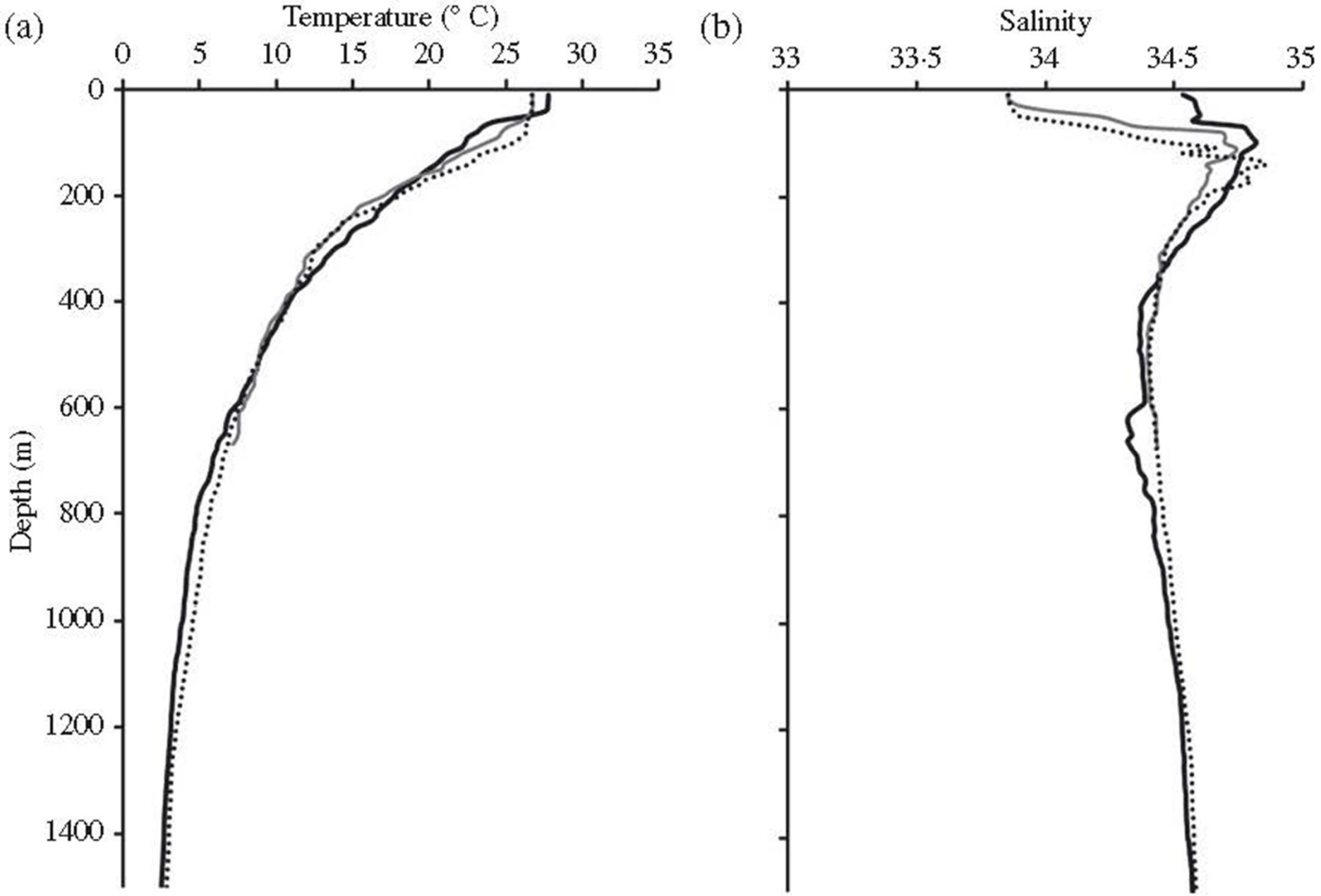
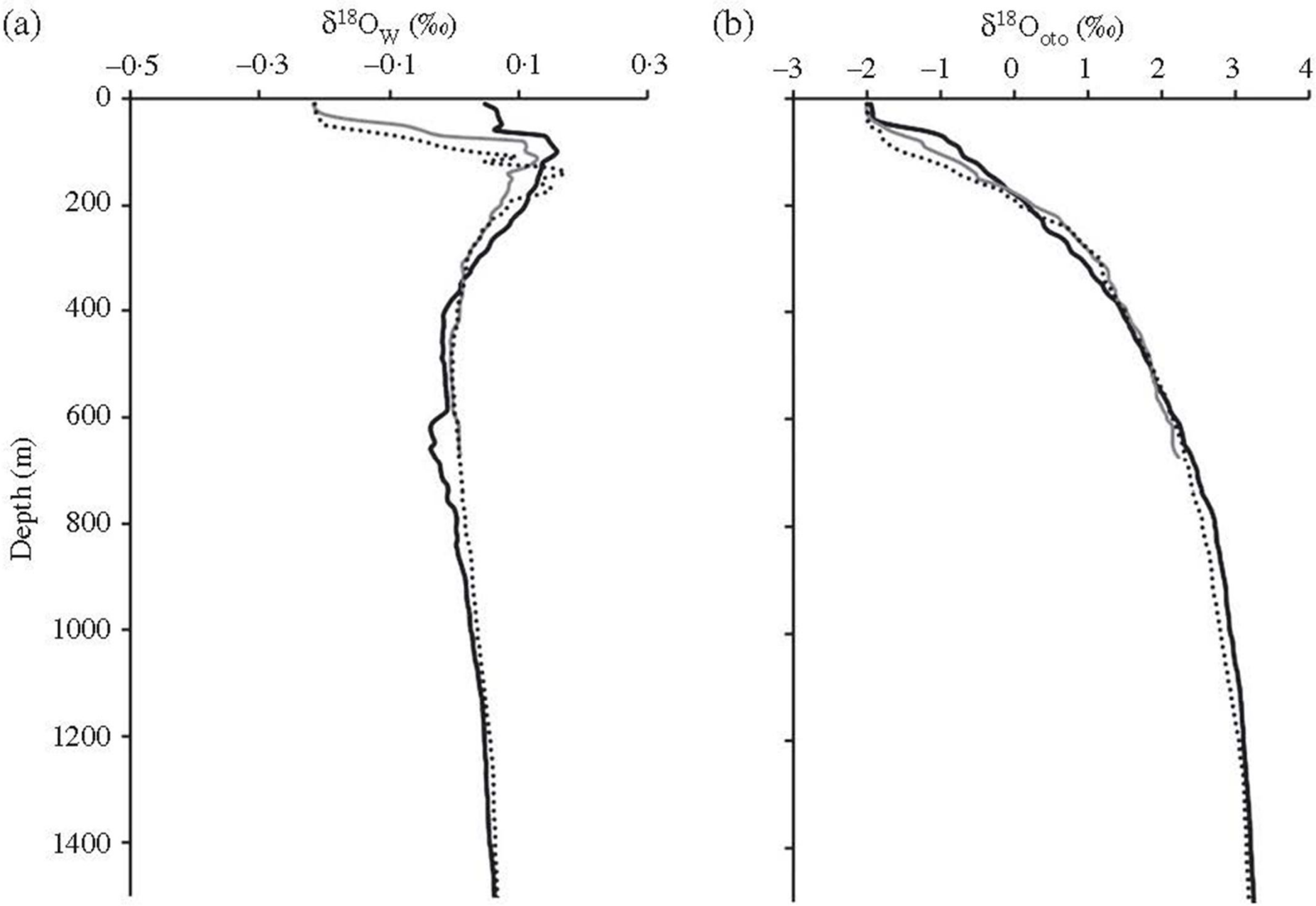
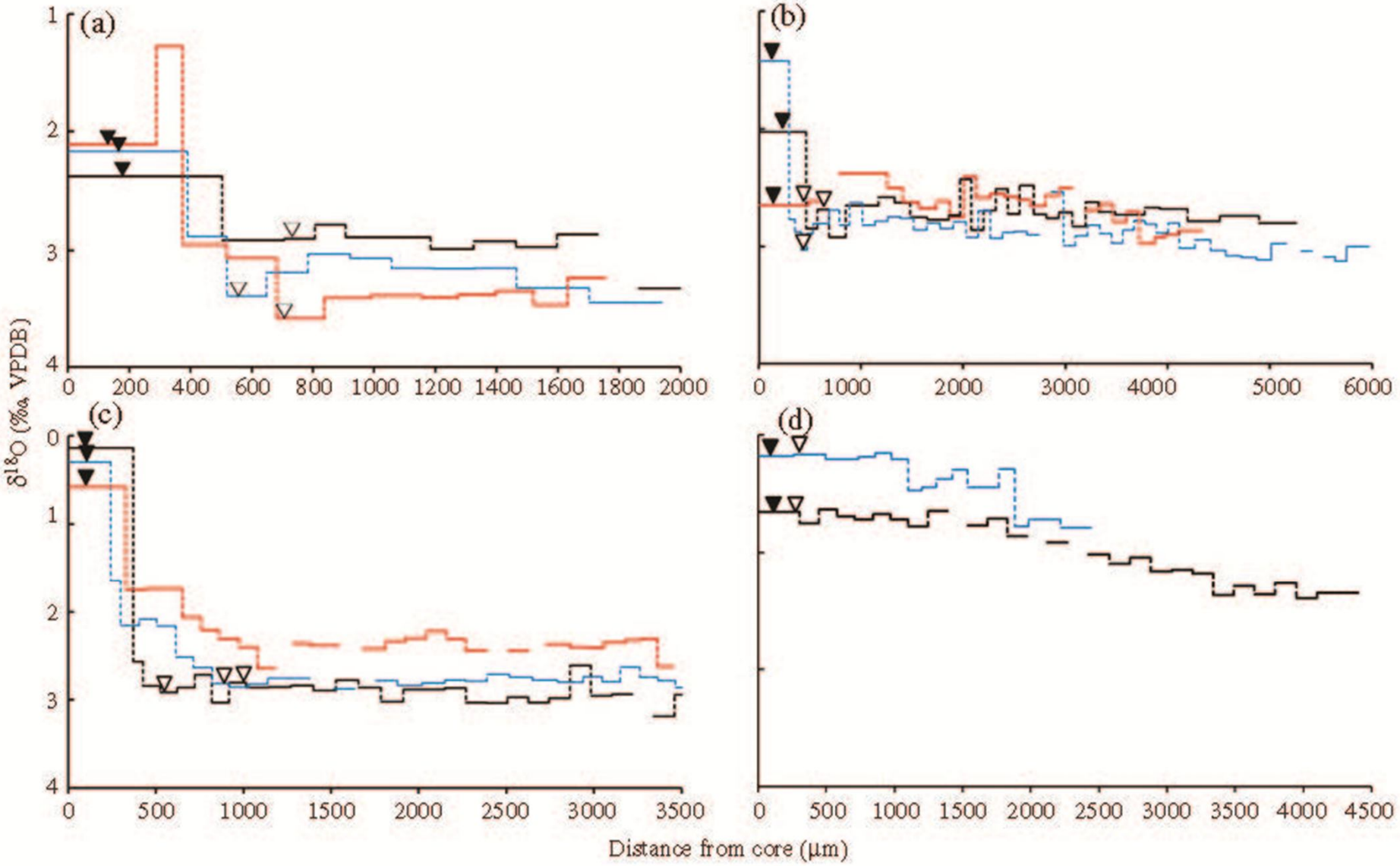
Key words: *Barathronus maculatus*; ontogenetic vertical migration; Ophidiidae; oxygen isotope; vertical distribution.


NET1
NET2
SWT1
SWT2
SWT3
Taiwan
Filipinas
P1
24°
21°
18°
15°
116°
118°
120°
122°
124°

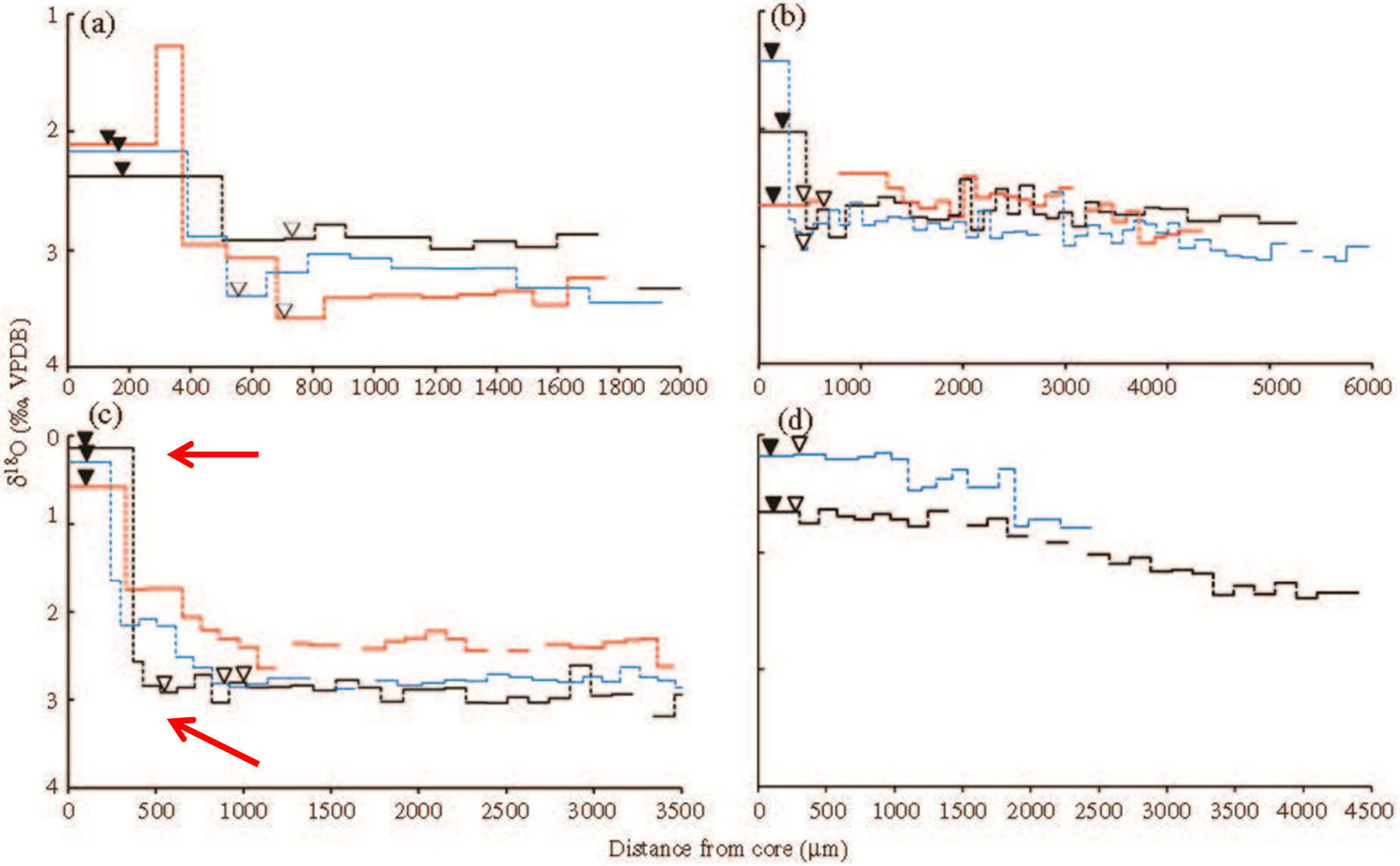

(a)
Temperature (° C)
0
5
10
15
20
25
30
35
Depth (m)
0
200
400
600
800
1000
1200
1400
(b)
Salinity
33
33·5
34
34·5
35

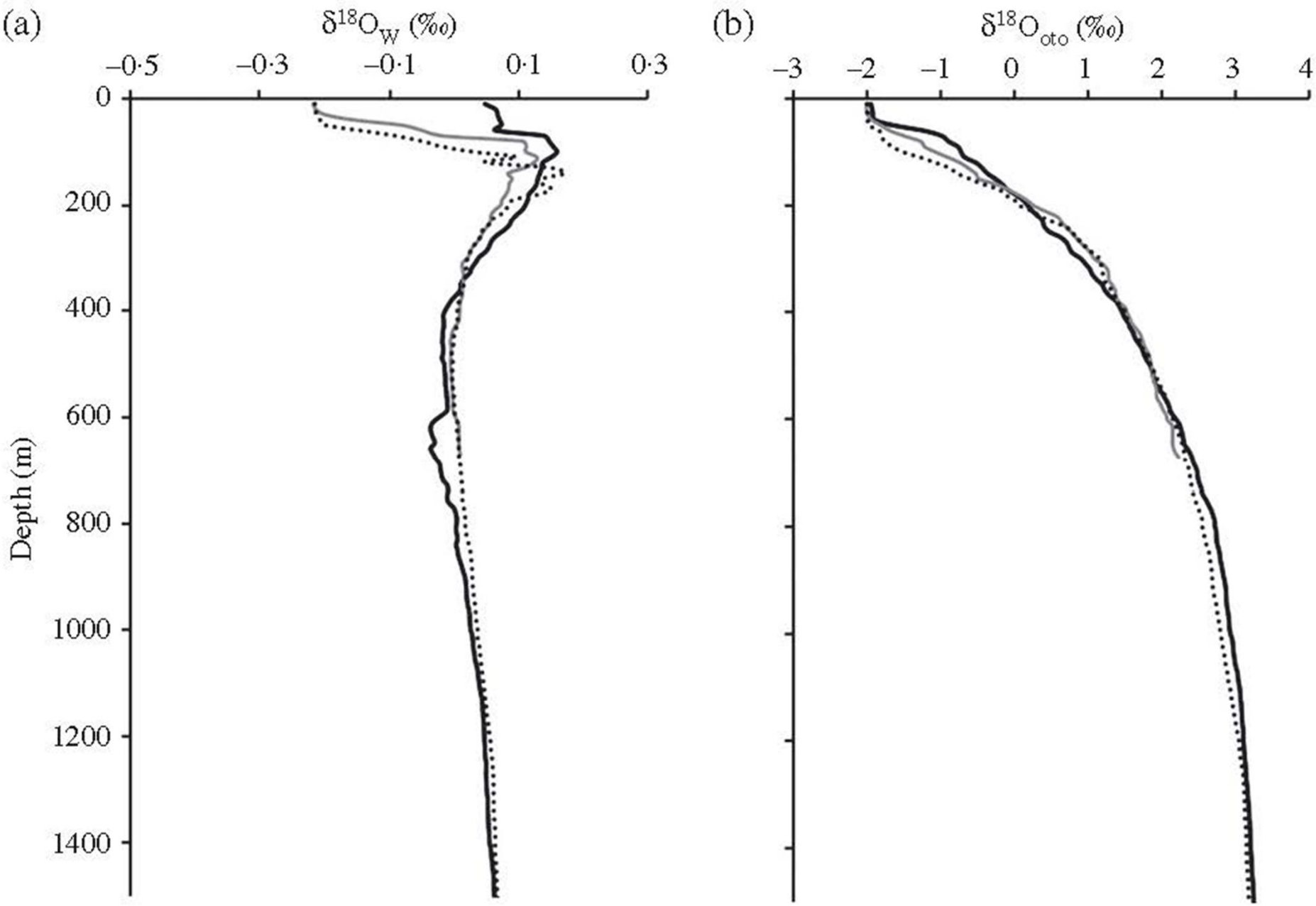

(a)
$\delta^{18}O_W$ (‰)
−0·5
−0·3
−0·1
0·1
0·3
0
200
400
600
800
1000
1200
1400
Depth (m)
(b)
$\delta^{18}O_{oto}$ (‰)
−3
−2
−1
0
1
2
3
4

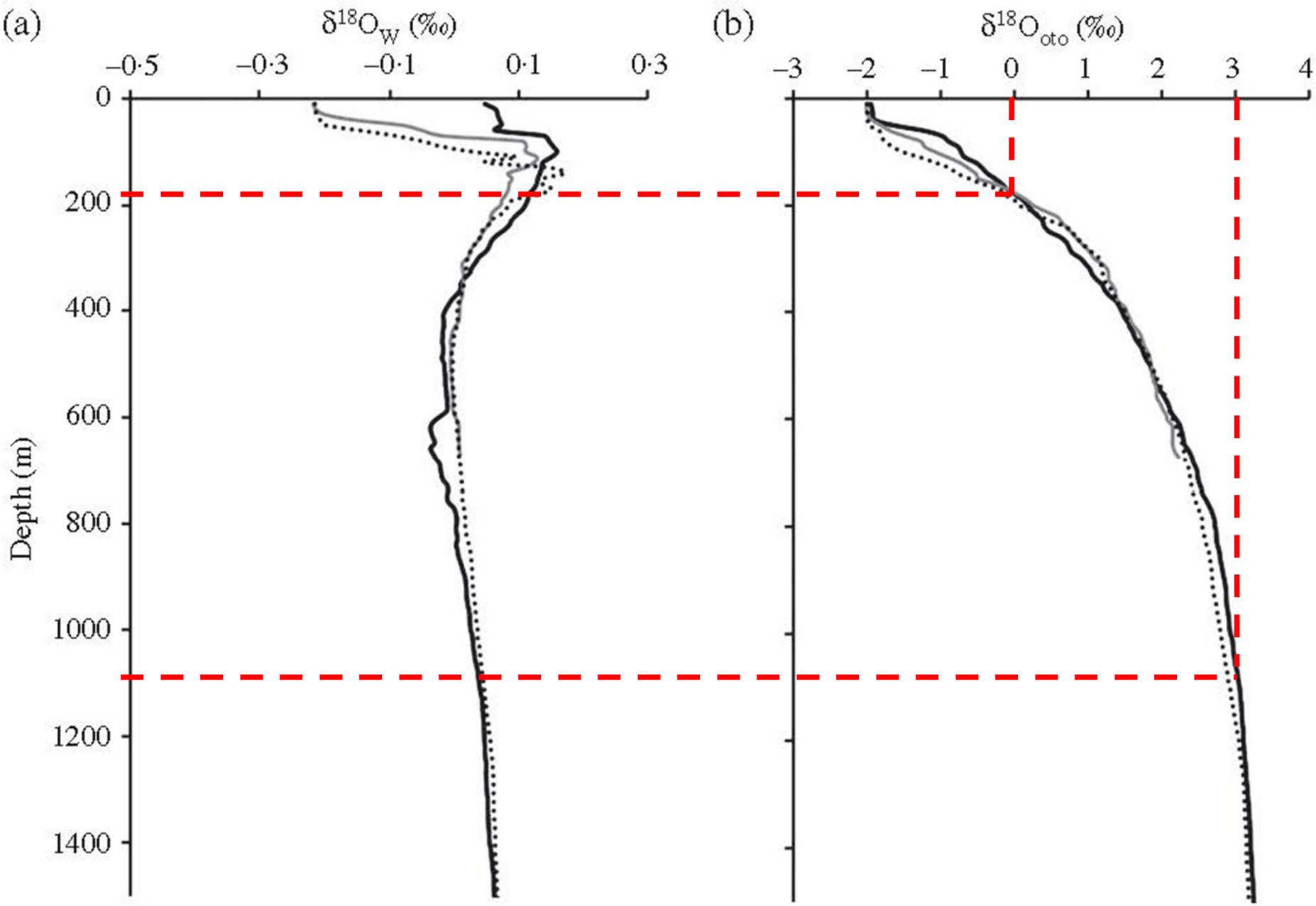
Perfis isotópicos de otólitos



Perfis isotópicos de otólitos




(a)
$\delta^{18}O_W$ (‰)
−0·5
−0·3
−0·1
0·1
0·3
0
200
400
600
800
1000
1200
1400
Depth (m)
(b)
$\delta^{18}O_{oto}$ (‰)
−3
−2
−1
0
1
2
3
4


(a)
δ18Ow (‰)
−0·5
−0·3
−0·1
0·1
0·3
(b)
δ18Ooto (‰)
−3
−2
−1
0
1
2
3
4
Depth (m)
0
200
400
600
800
1000
1200
1400

Estudo da mudança de profundidade de habitat de enguias de mar profundo

Conclusão:

Fase de vida inicial: **200 m**

Fase adulta: **> 1000 m**

Estimativa da distribuição vertical de espécies de peixe em ambiente lacustre

Estimativa da distribuição vertical de espécies de peixe em ambiente lacustre

Questão cientifica:

Existe distribuição vertical das espécies de peixe no lago estudado?

Knowledge and Management of Aquatic Ecosystems (2015) 416, 35
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DOI: 10.1051/kmae/2015033

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Knowledge &
Management of
Aquatic
Ecosystems

Journal fully supported by Onema

Life history traits of the fish community in Lake Annecy: evidence from the stable isotope composition of otoliths

D. Gerdeaux[(1),*], E. Dufour[(2)]

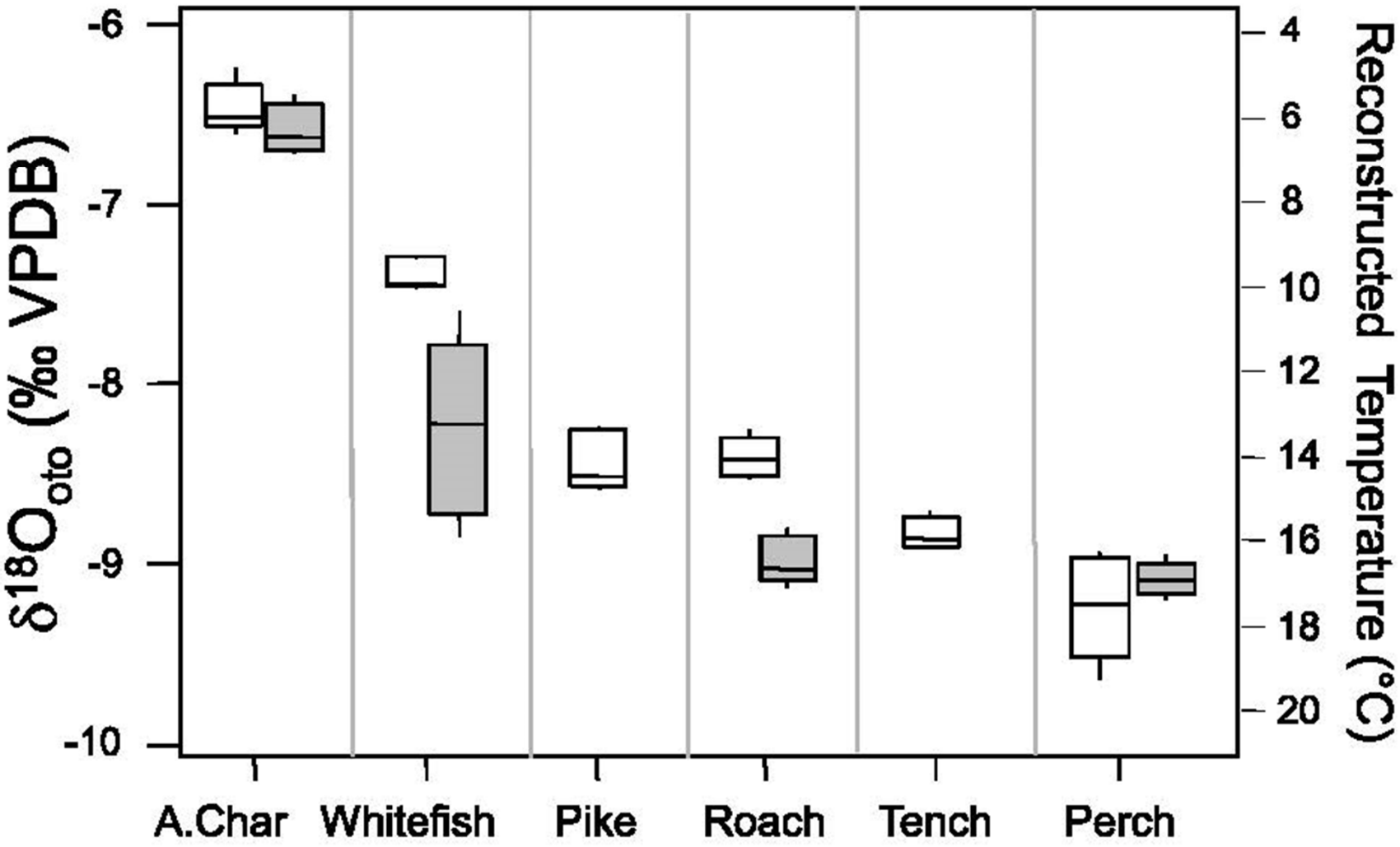
Received September 13, 2015
Revised October 30, 2015
Accepted October 31, 2015

ABSTRACT

Key-words:
stable isotopes, oxygen, carbon, temperature, metabolism

Stable isotope analysis (SIA) of fish otoliths shows great potential for exploring the ecology of fish, but this method has not been applied to the study of lacustrine fish communities. Both sequential and whole-otolith SIAs were performed on six fish species of Lake Annecy and their results compared to muscle SIA. The first purpose of these investigations was to test the use of $\delta^{18}O_{oto}$ values for reconstructing fish thermal history and delimiting spatial distribution in a stratified lake. Comparison of species-specific fractionation equations and the general equation developed for freshwater fishes showed that the general one was the best suited for thermal estimation of Lake Annecy fishes and suggested that inter-specific differences or specific "vital effects" are not the only reason for apparent difference in fractionation. Thermal estimations based on SIA were consistent with descriptions of thermal habitats in the literature, except in the case of roach (*Rutilus rutilus*). Based on the current results, roach appears to live in a colder habitat than do perch (*Perca fluviatilis*). The high water transparency and thermal stratification of Lake Annecy could explain this distribution. Moreover, perch juveniles were found to live in two different thermal niches. This finding highlights the great plasticity of the species. Second, the potential use of $\delta^{13}C_{oto}$ values to reconstruct variation in diet and metabolism was assessed. The proportion of metabolic carbon (*M*) contributing to otolith carbon varies markedly among species. Comparison with $\delta^{13}C_{muscle}$ values shows no direct relationship between $\delta^{13}C_{oto}$ and diet at either the intra-individual or the inter-specific level. A strong linear relationship between either *M* or the isotopic offset between otolith and muscle ($\Delta\delta^{13}C_{oto\text{-}muscle}$) and $\delta^{18}O_{oto}$ values was found; this relationship reveals the dependence of *M* on the ambient temperature at which the species occurs. This relationship might be general and could be used to explore variation in fish metabolism in the future.

D. Gerdeaux and E. Dufour: Knowl. Manag. Aquat. Ecosyst. (2015) 416, 35



Estimativa da distribuição vertical de espécies de peixe em ambiente lacustre

Conclusão:

Sim, existe distribuição vertical das espécies de peixe

Estudo da separação geográfica entre populações de peixes da mesma espécie

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Geochimica et Cosmochimica Acta 71 (2007) 87–94
Geochimica
www.elsevier.com/locate/gca

Oxygen and carbon stable isotopes in otoliths record spatial isolation of Patagonian toothfish (*Dissostichus eleginoides*)

Julian Ashford *, Cynthia Jones

Center for Quantitative Fisheries Ecology, Old Dominion University, Norfolk, Virginia, VA 23529, USA

Received 25 April 2006; accepted in revised form 23 August 2006

Abstract

Strong contrasts in ambient isotope ratios and in diet suggest stable isotopes in the otoliths of oceanic fish can resolve water masses and geographic areas, promising a powerful multivariate approach for examining population structure and provenance. To test this, whole otoliths were taken from Patagonian toothfish (*Dissostichus eleginoides*) sampled off the Patagonian Shelf and South Georgia, on either side of a population boundary, and otolith $\delta^{18}O$ and $\delta^{13}C$ values were measured to see if they could distinguish South American-caught fish from those taken in the Antarctic. Values of otolith $\delta^{18}O$ and $\delta^{13}C$ predicted capture area with 100% success, validating their use for distinguishing provenance and corroborating the prior evidence of population isolation. Values of $\delta^{18}O$ in the otoliths reflected ambient values as well as seawater temperature: low values in Patagonian Shelf fish were consistent with exposure to Antarctic Intermediate Water (AAIW), and high values in South Georgia fish were consistent with exposure to Circumpolar Deep Water (CDW). In contrast, differences in otolith $\delta^{13}C$ appeared to reflect diet: relative depletion of otolith $\delta^{13}C$ at South Georgia compared to the Patagonian Shelf were most likely linked to differences in sources of metabolic carbon, as well as $\delta^{13}C$ in dissolved inorganic carbon (DIC) of seawater. These contrasting properties strongly suggest that stable isotopes can resolve the provenance of toothfish from Antarctic sampling areas that hitherto have been difficult to separate. These results show that, by using the chemistry recorded in otoliths, researchers can exploit biogeochemical variation in fully marine environments to examine the spatial ecology of oceanic fish.
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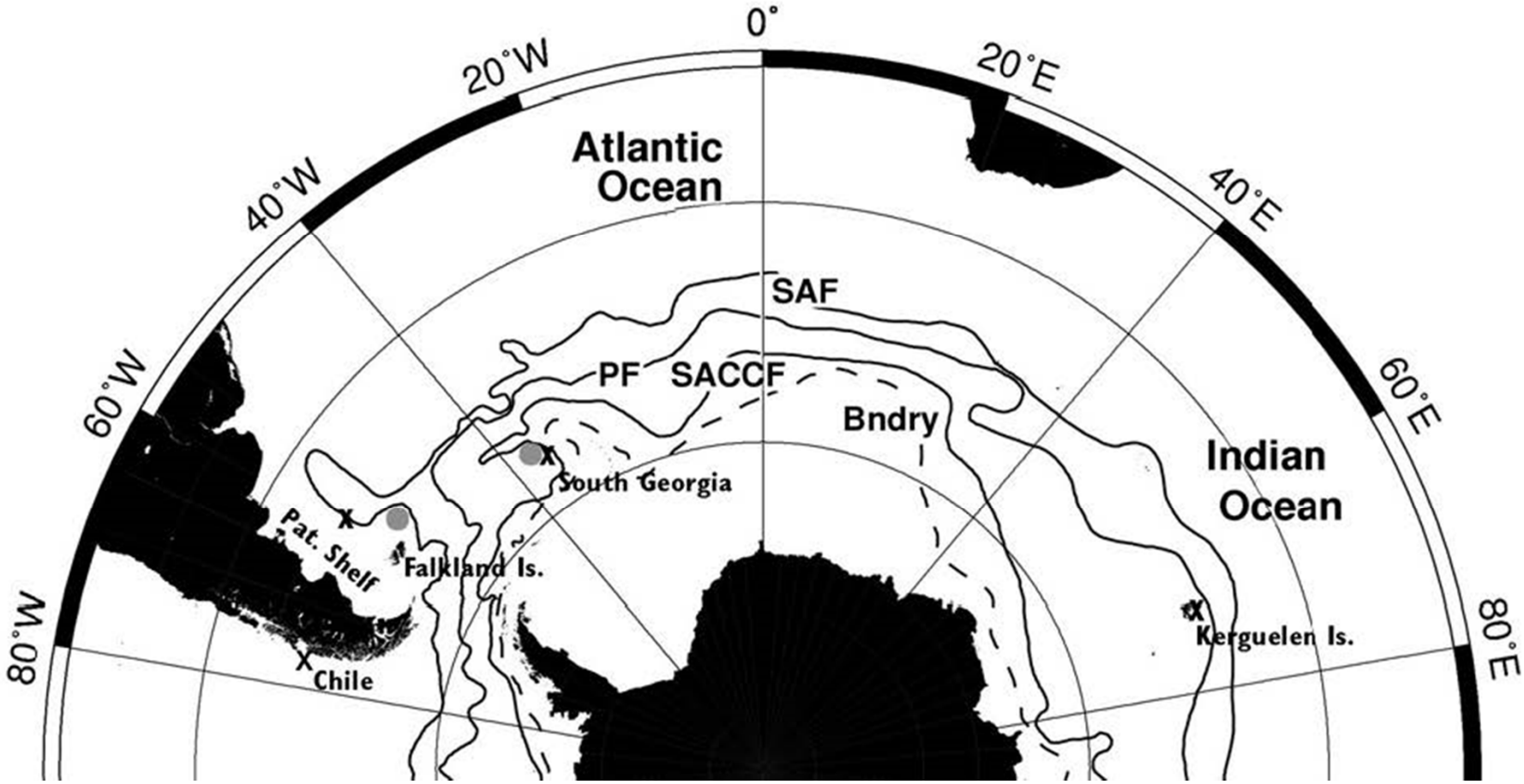
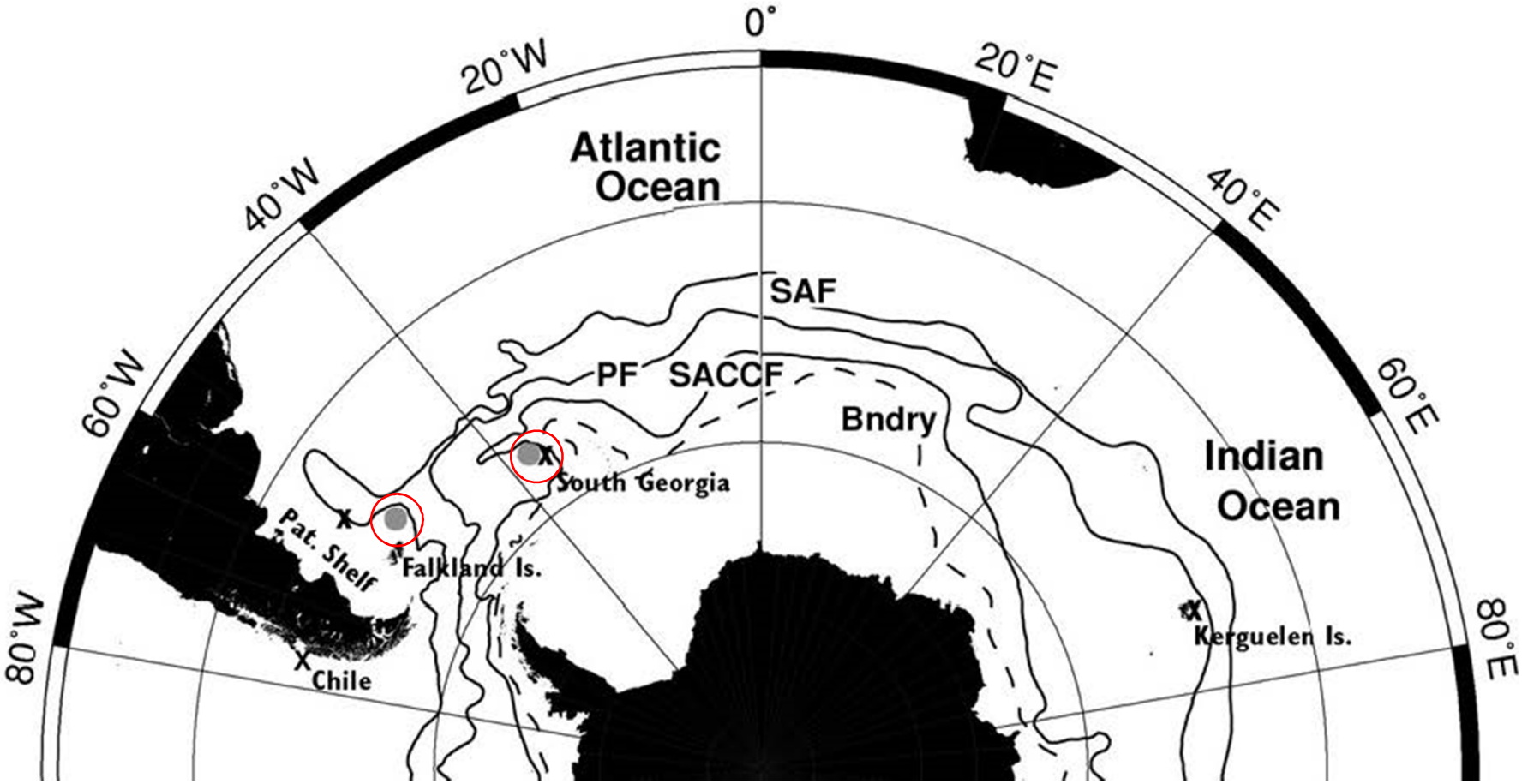


SAF = Subantarctic Front

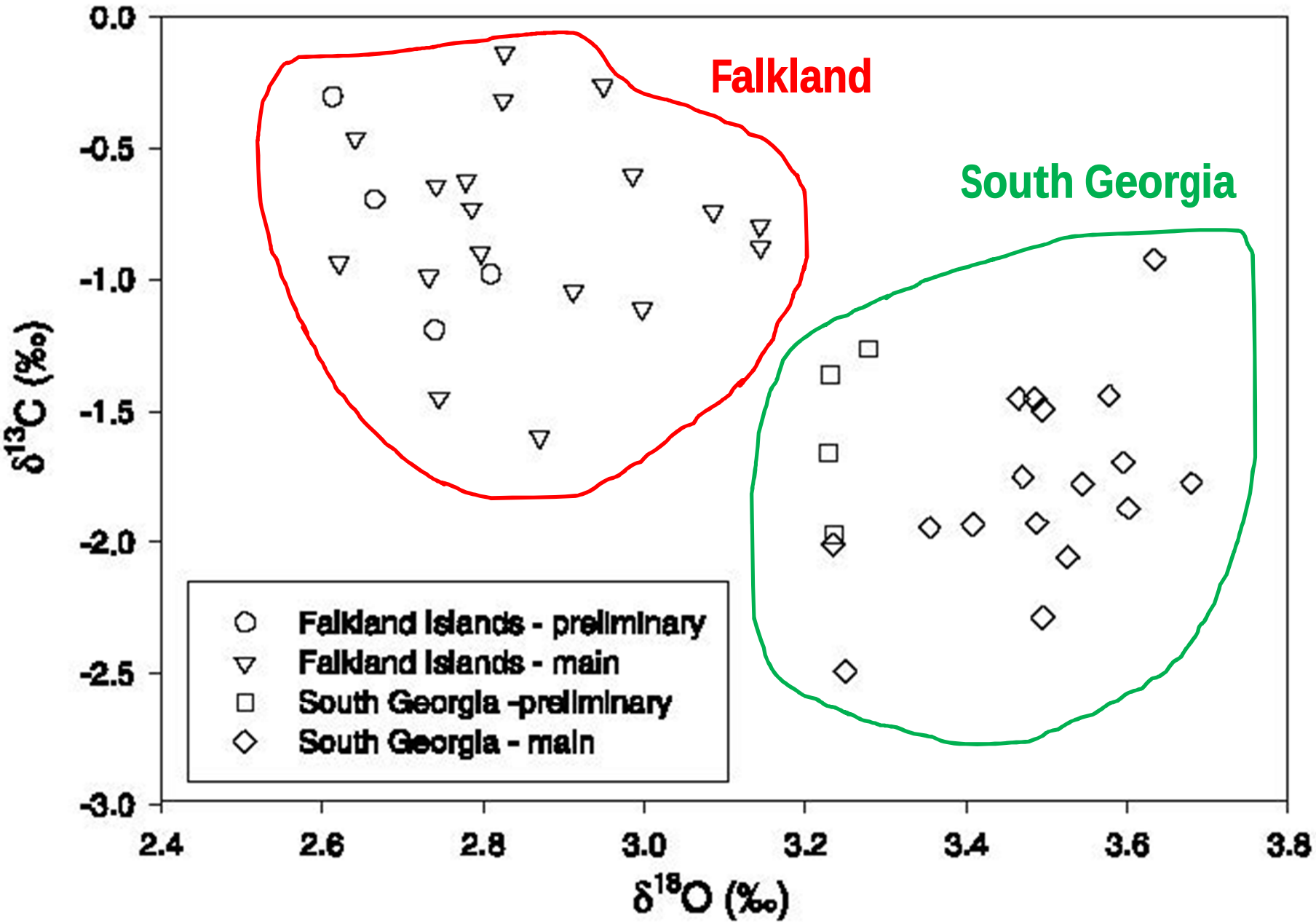


SAF = Subantarctic Front

Estudo da separação geográfica entre populações de peixes da mesma espécie

Questão cientifica:

Os grupo de peixes estudados pertencem á mesma população ou a duas populações distintas ?


Falkland
South Georgia
Falkland Islands - preliminary
Falkland Islands - main
South Georgia -preliminary
South Georgia - main
0.0
-0.5
-1.0
-1.5
-2.0
-2.5
-3.0
2.4
2.6
2.8
3.0
3.2
3.4
3.6
3.8
δ13C (‰)
δ18O (‰)

Sumario do curso

1. Analise isotópica de oxigênio em água do mar
2. Analise isotópica de hidrogênio em água do mar
3. Analise isotópica de carbonato de cálcio
4. Analise isotópica de carbono inorgânico dissolvido (DIC)
5. Analise isotópica de material orgânico

Carbono inorgânico dissolvido

DIC

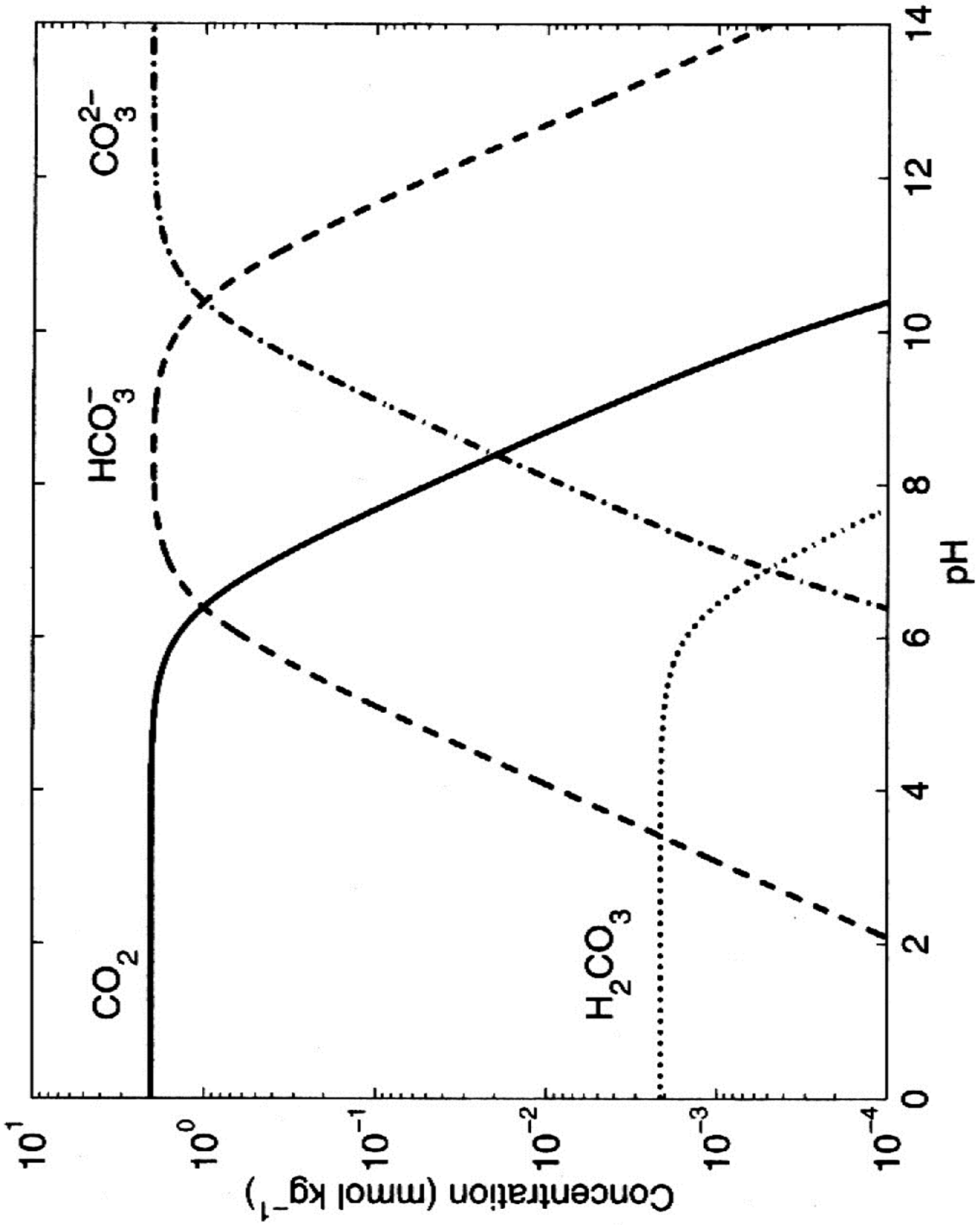
(ΣCO_2)

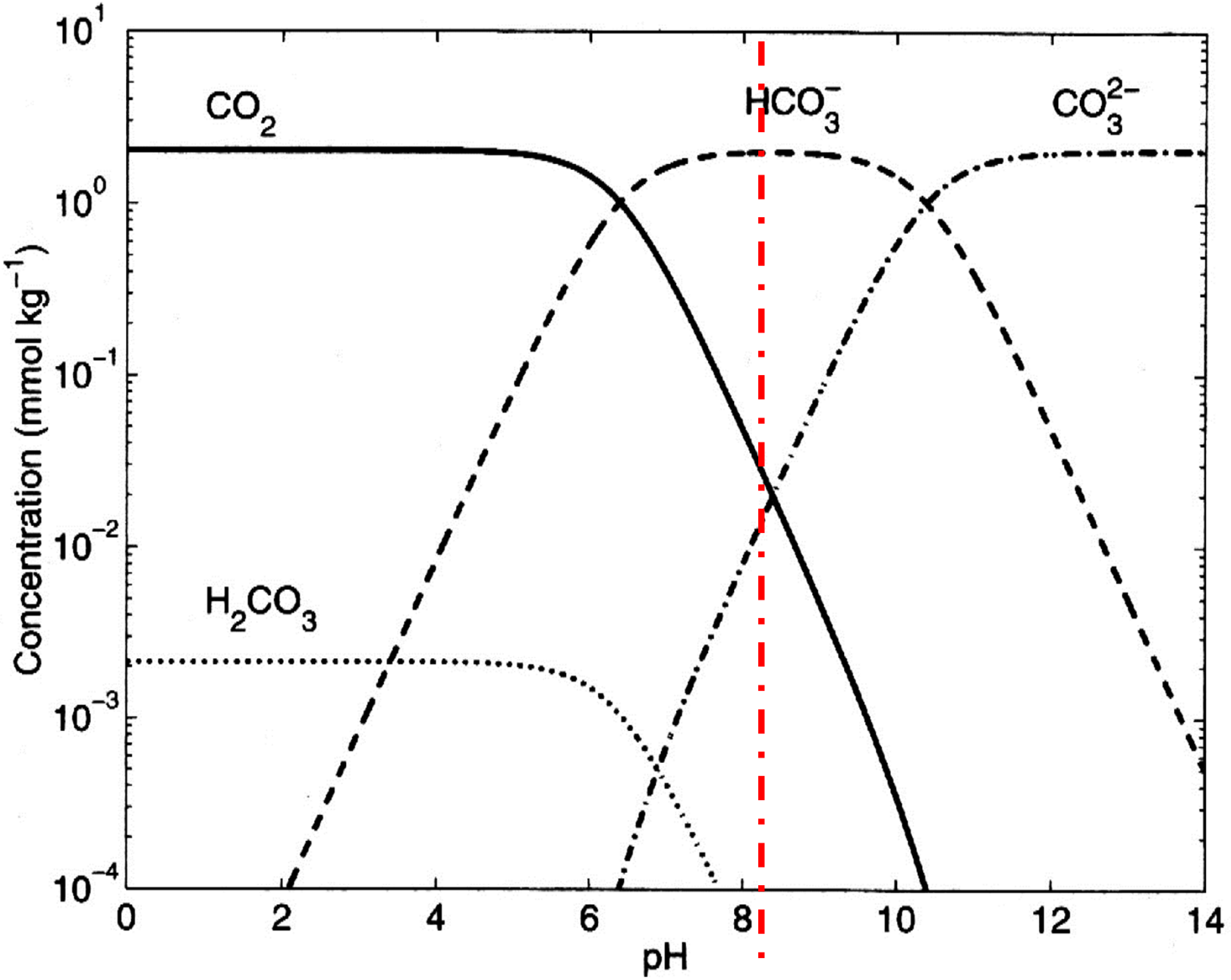
(CO_2 tot)

- HCO_3^- (91%)
- CO_3^{2-} (8%)
- CO_2 (1%)
- H_2CO_3 (3×10^{-3}%)

A abundância de cada espécie está ligada ao pH

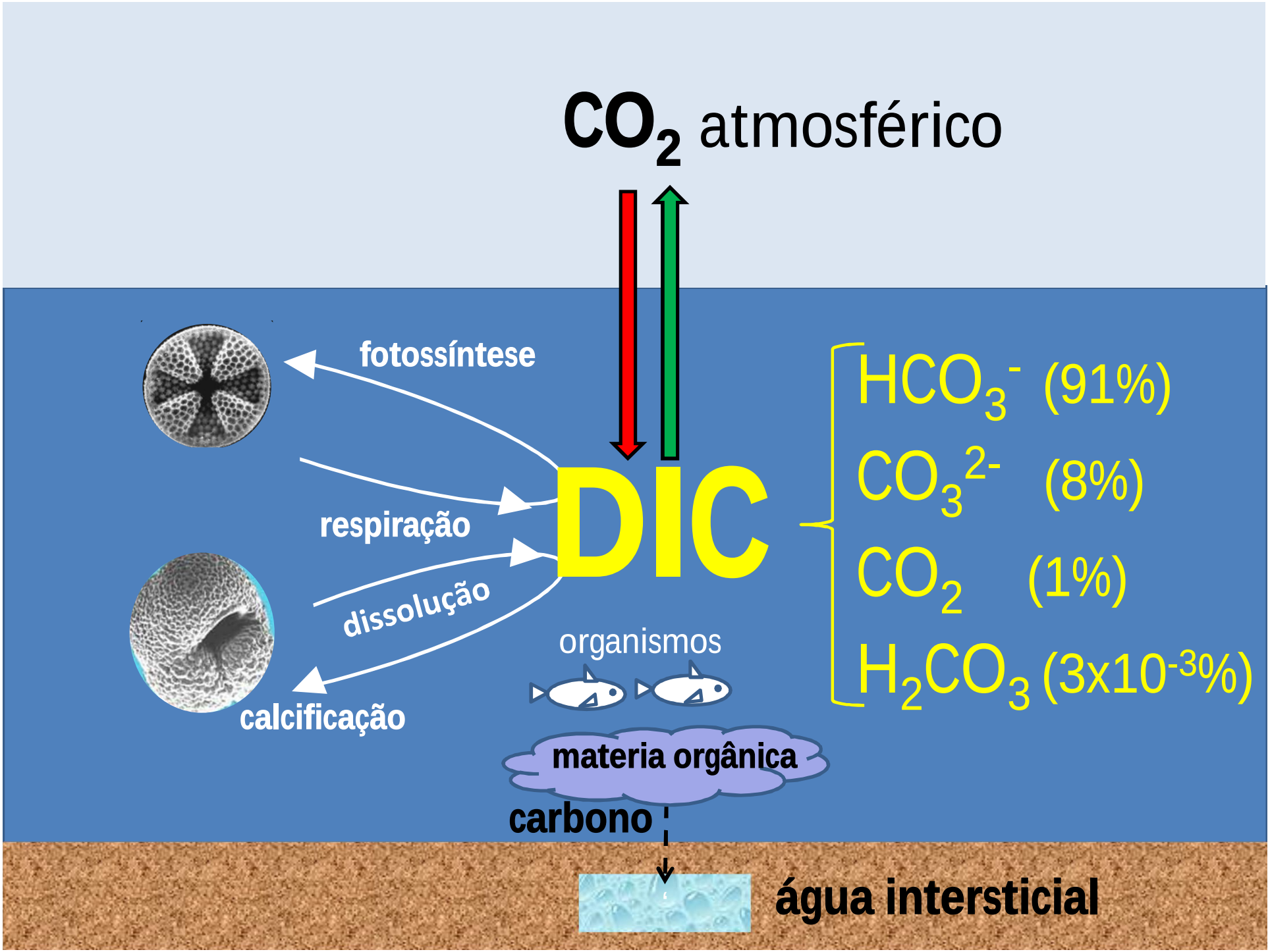
Source: Zeebe G.C.A. Vol. 63 (nr.13-14) (1999) 2001 -2007





Surface Seawater: 8.1

Source: Zeebe G.C.A. Vol. 63 (nr.13-14) (1999) 2001 -2007


CO_2 atmosférico
fotossíntese
respiração
dissolução
calcificação
DIC
HCO_3^- (91%)
CO_3^{2-} (8%)
CO_2 (1%)
H_2CO_3 (3x10^{-3}%)
organismos
materia orgânica
carbono
água intersticial

ISÓTOPOS ESTÁVEIS DO CARBONO

^{12}C

(98.9%)

CARBONO ISOTOPICAMENTE LEVE

^{13}C

(1.1%)

CARBONO ISOTOPICAMENTE PESADO

Razão isotópica

$$^{13}R = \frac{^{13}C}{^{12}C}$$

Notação delta

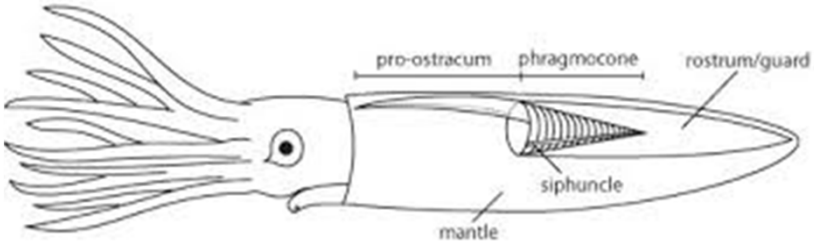
Delta ≡ *a diferença entre a razão isotópica da amostra e a razão isotópica de um padrão internacional, exprimida em por mil.*

$$\delta^{13}C \equiv \frac{\left(\frac{^{13}C}{^{12}C}\right)_{\text{amostra}} - \left(\frac{^{13}C}{^{12}C}\right)_{\text{padrão internacional}}}{\left(\frac{^{13}C}{^{12}C}\right)_{\text{padrão internacional}}} \times 1000$$

Padrões isotópicos internacionais
(Agencia Internacional da Energia Atômica IAEA, Viena)

PDB: Pee Dee Belemnite (Calcita)

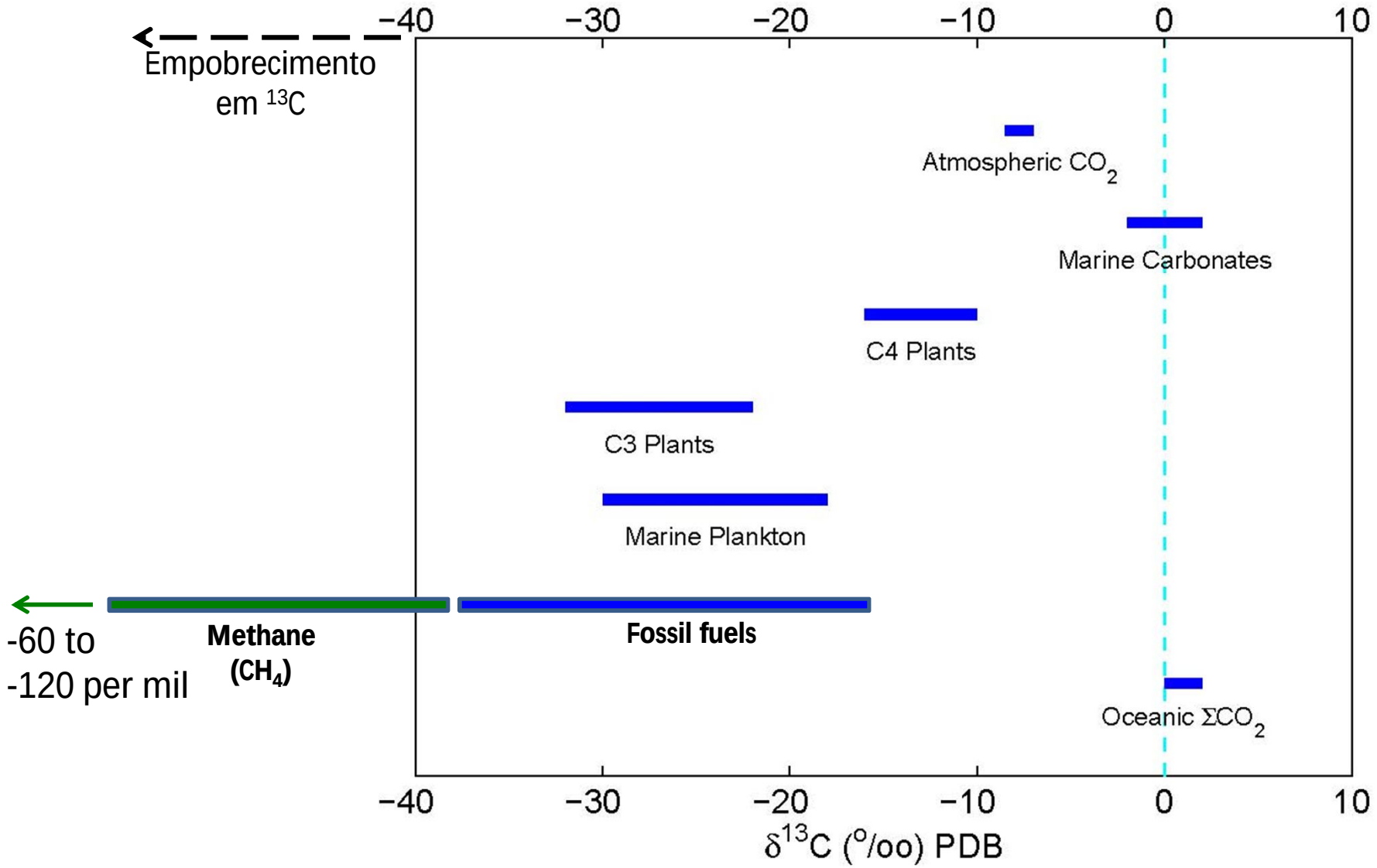
- Carbonatos
- Compostos de C
- **$\delta^{13}C = 0‰$ (por definição)**



Lateral view of the internal shell of a generalized belemnite. ©Liz Shea, DMNH.

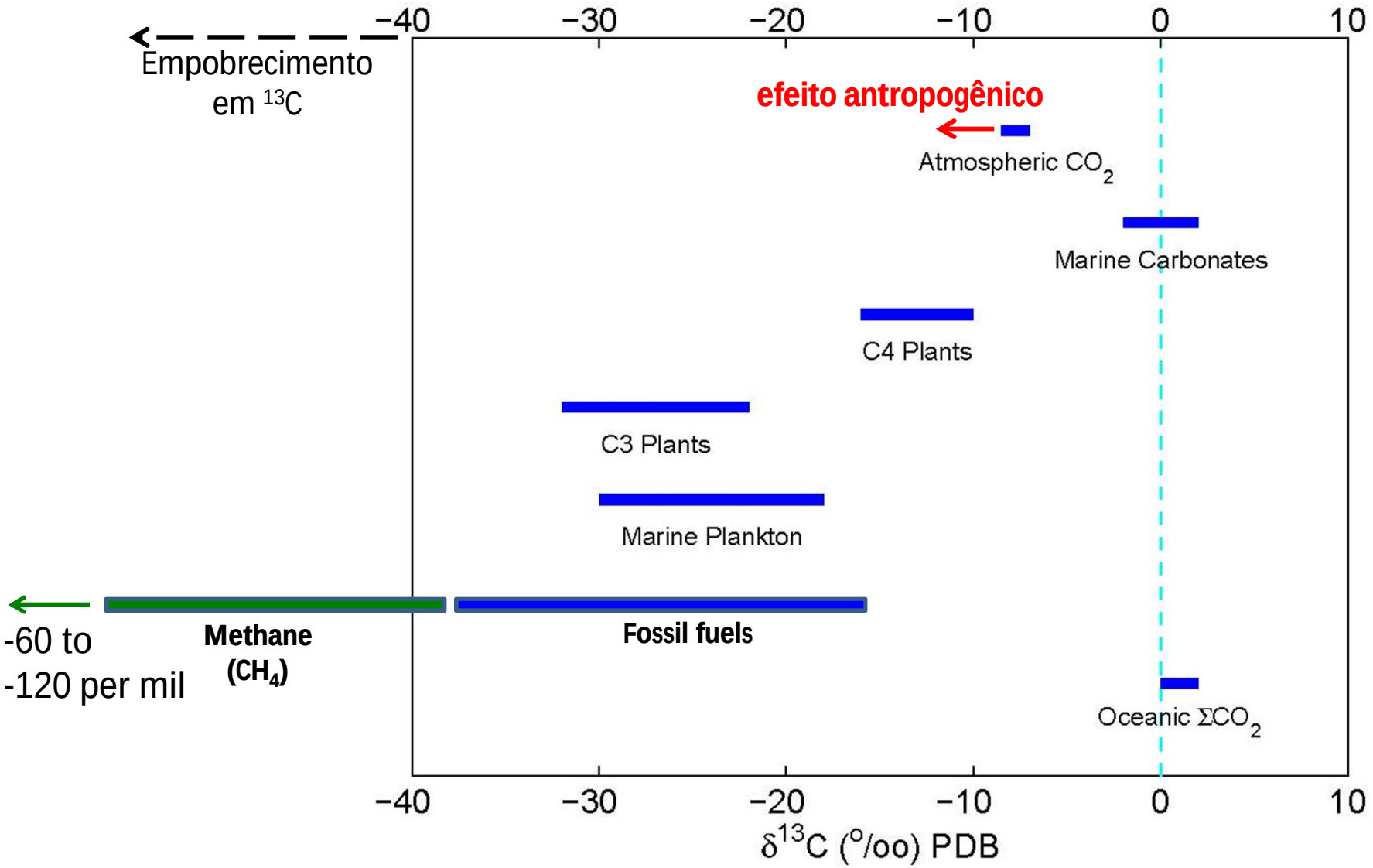
www.thefossilforum.com

Valores de $\delta^{13}C$ em substancias naturais



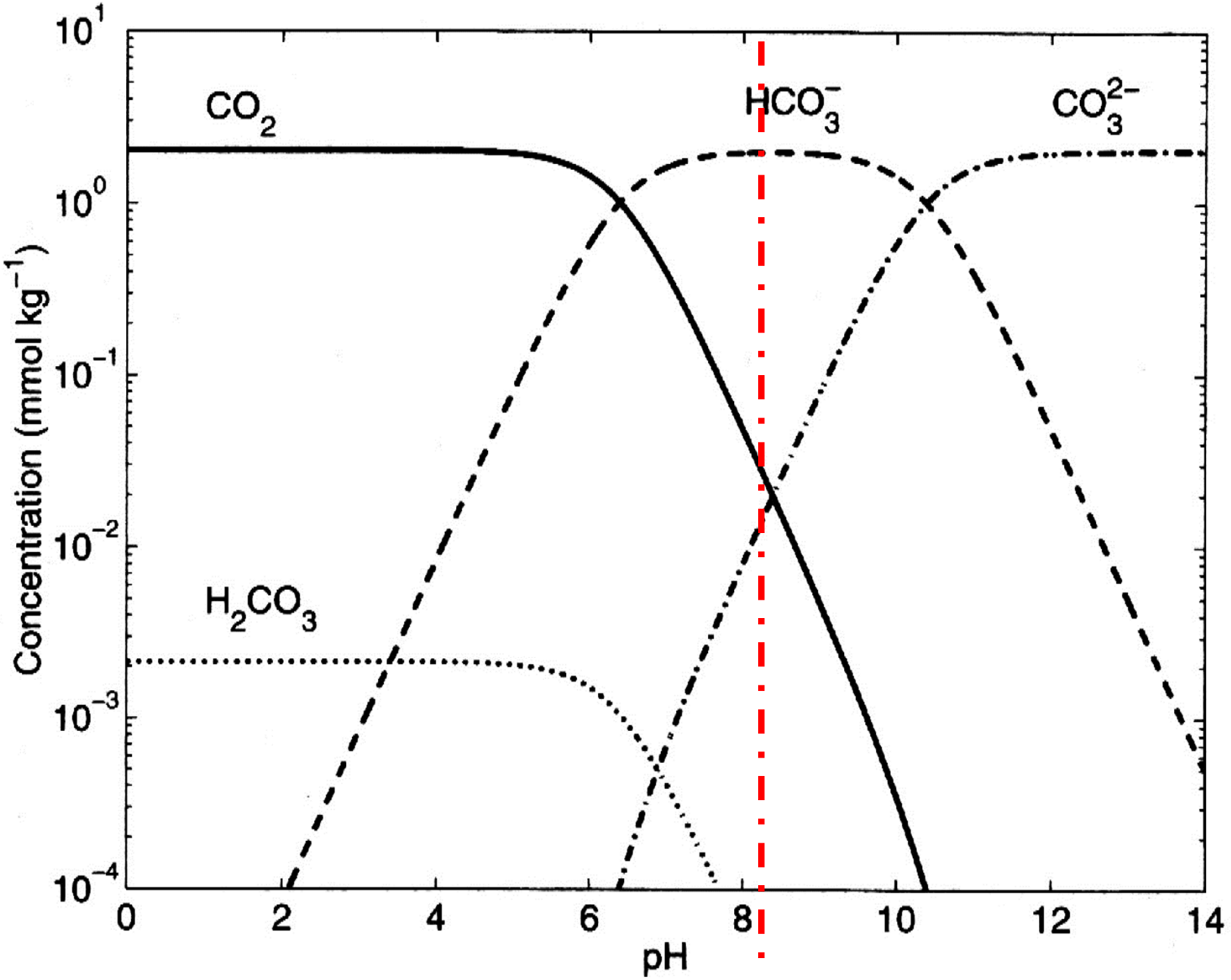
Zeebe & Wolf-Gladrow (2001)

Valores de $\delta^{13}C$ em substancias naturais



Zeebe & Wolf-Gladrow (2001)

Como é medido o valor de $\delta^{13}C$ do DIC



Surface Seawater: 8.1

Source: Zeebe G.C.A. Vol. 63 (nr.13-14) (1999) 2001 -2007

Carbono inorgânico dissolvido

DIC

(ΣCO_2)

(CO_2 tot)

- HCO_3^- (91%)
- CO_3^{2-} (8%)
- CO_2 (1%)
- H_2CO_3 (3×10^{-3}%)

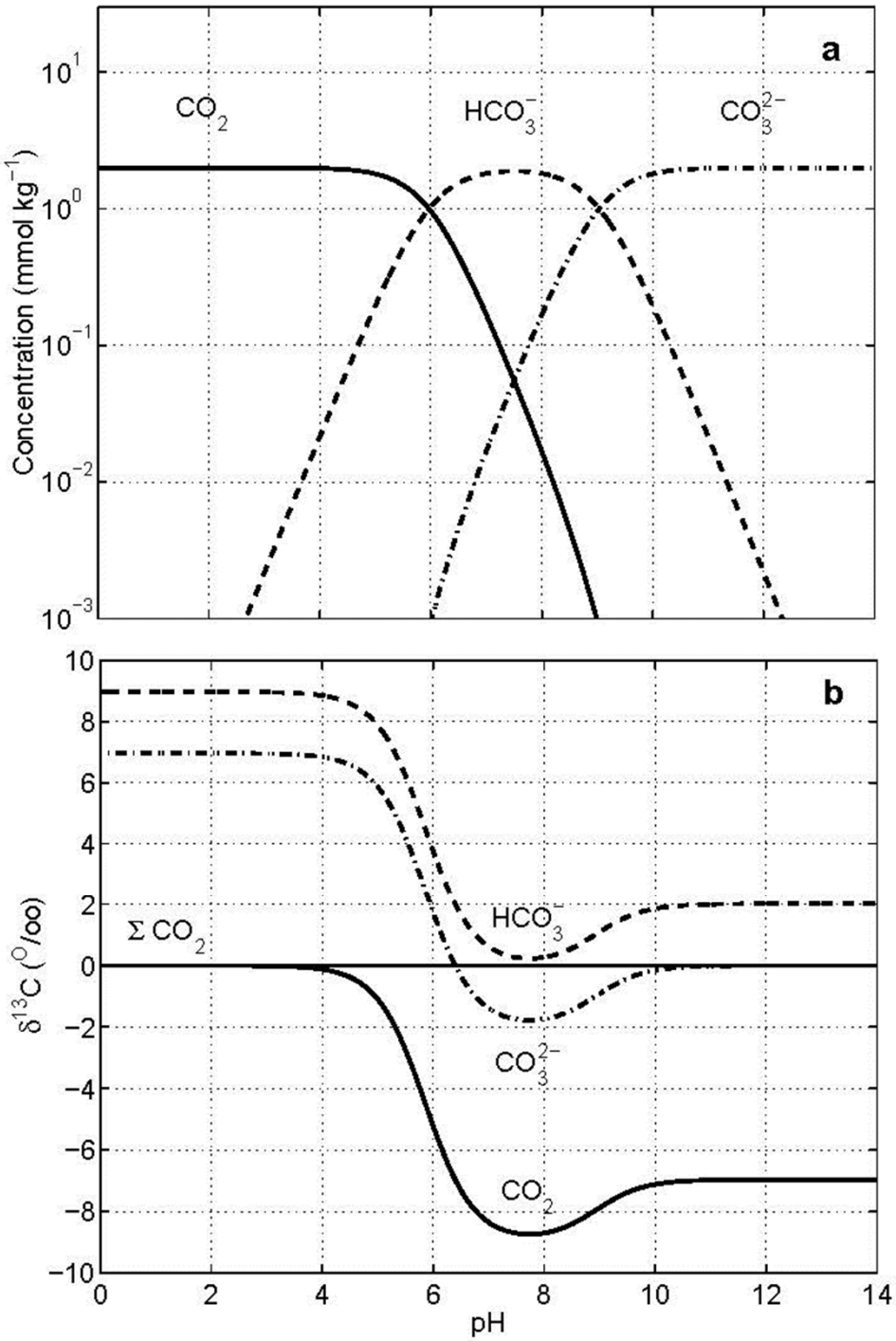
A abundância de cada espécie está ligada ao pH

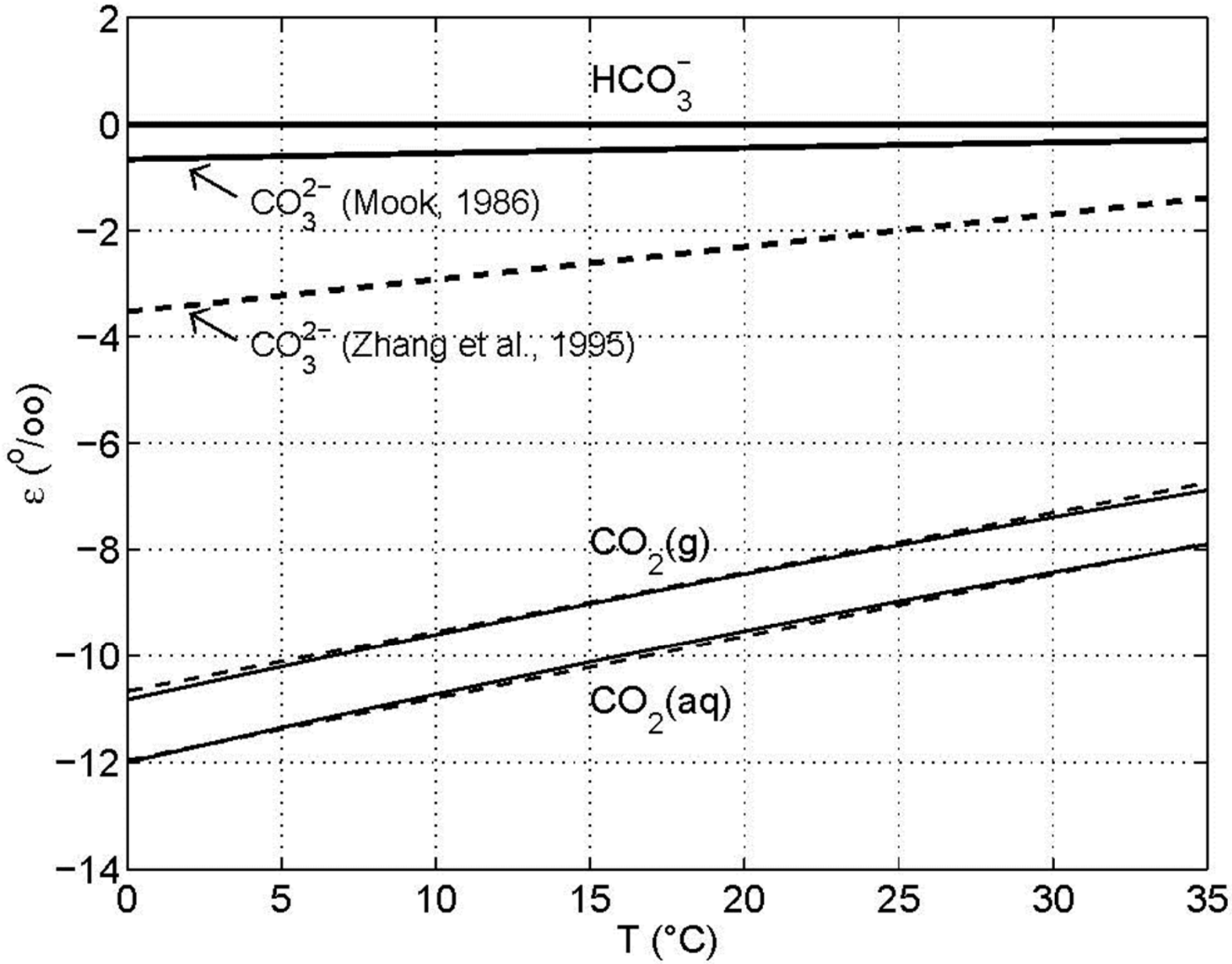
Equilíbrio no sistema carbonato

$$CO_2 + H_2O \underset{k_{-1}}{\overset{k_{+1}}{\rightleftharpoons}} H^+ + HCO_3^-$$

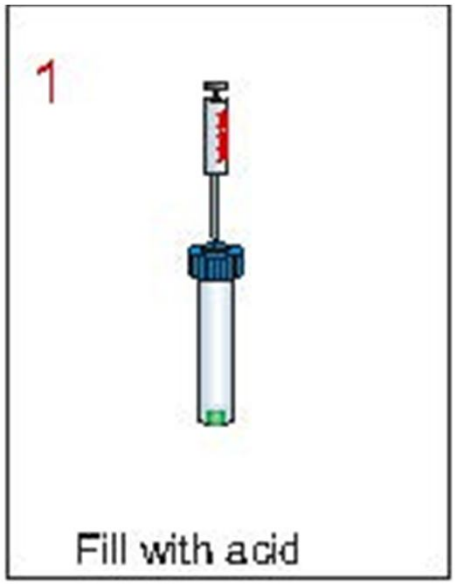
$$CO_2 + OH^- \underset{k_{-4}}{\overset{k_{+4}}{\rightleftharpoons}} HCO_3^-$$

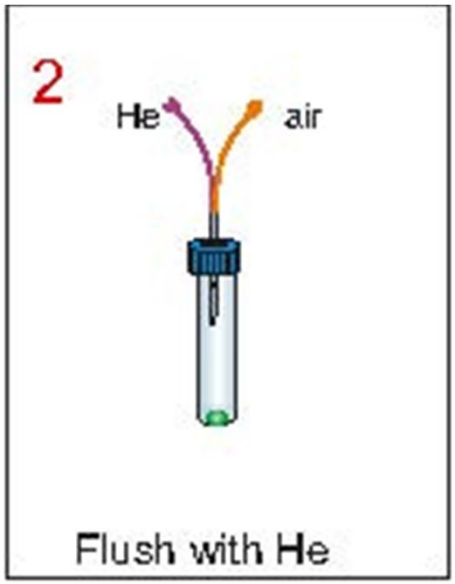
$$CO_3^{2-} + H^+ \underset{k_{-5}}{\overset{k_{+5}}{\rightleftharpoons}} HCO_3^-$$

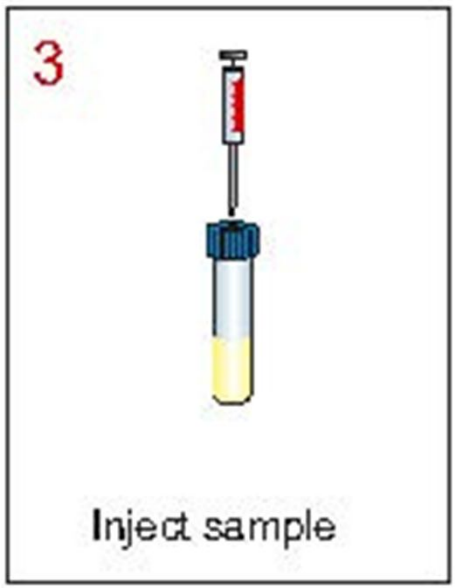

a
CO_2
HCO_3^-
CO_3^{2-}
Concentration (mmol kg^{-1})
10^1
10^0
10^{-1}
10^{-2}
10^{-3}
b
ΣCO_2
HCO_3^-
CO_3^{2-}
CO_2
$\delta^{13}C$ ($^O/oo$)
10
8
6
4
2
0
−2
−4
−6
−8
−10
0
2
4
6
8
10
12
14
pH

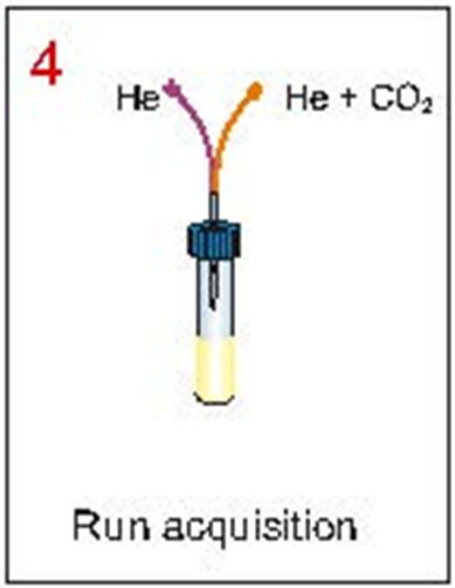

2
0
−2
−4
−6
−8
−10
−12
−14
0
5
10
15
20
25
30
35
T (°C)
ε (°/oo)
HCO3−
CO32− (Mook, 1986)
CO32− (Zhang et al., 1995)
CO2(g)
CO2(aq)

Método analítico bem estabelecido





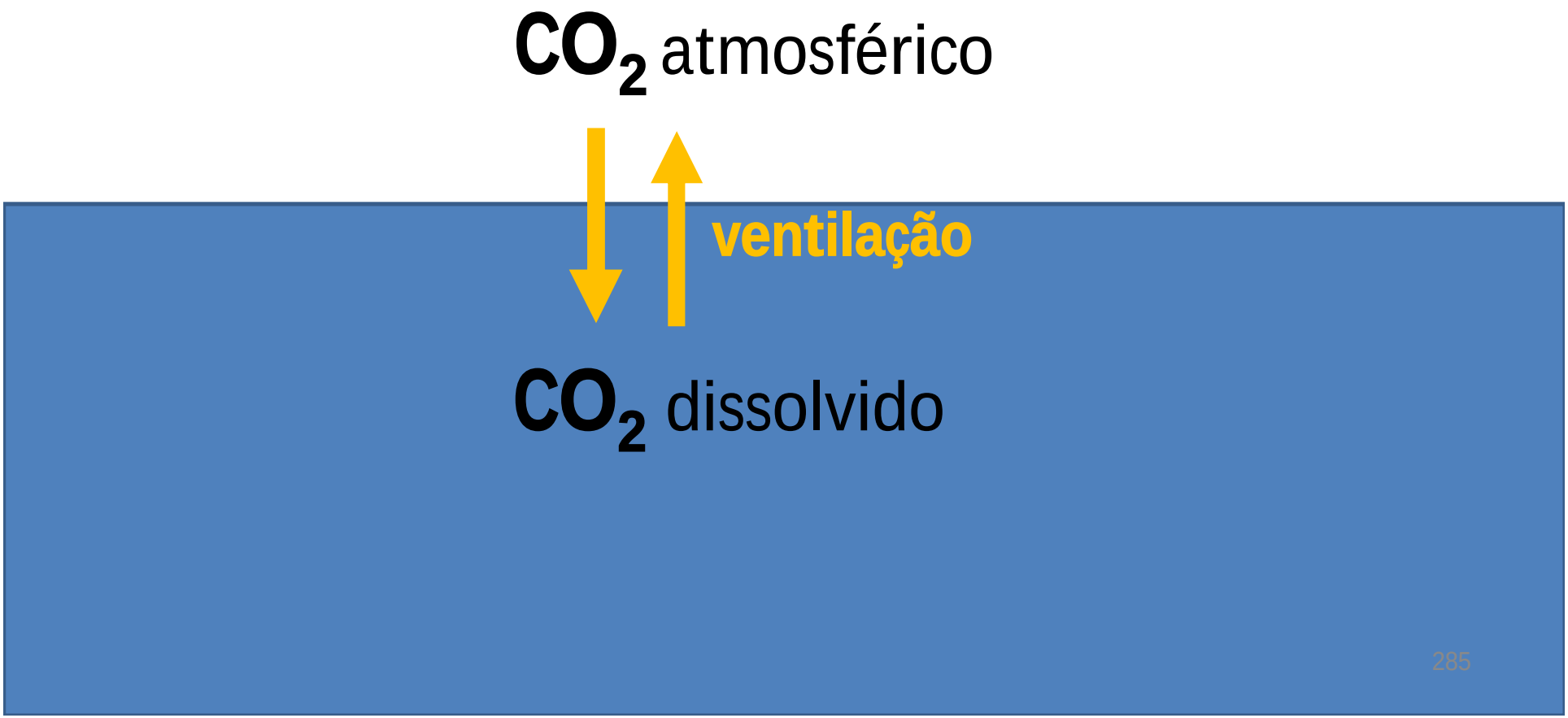




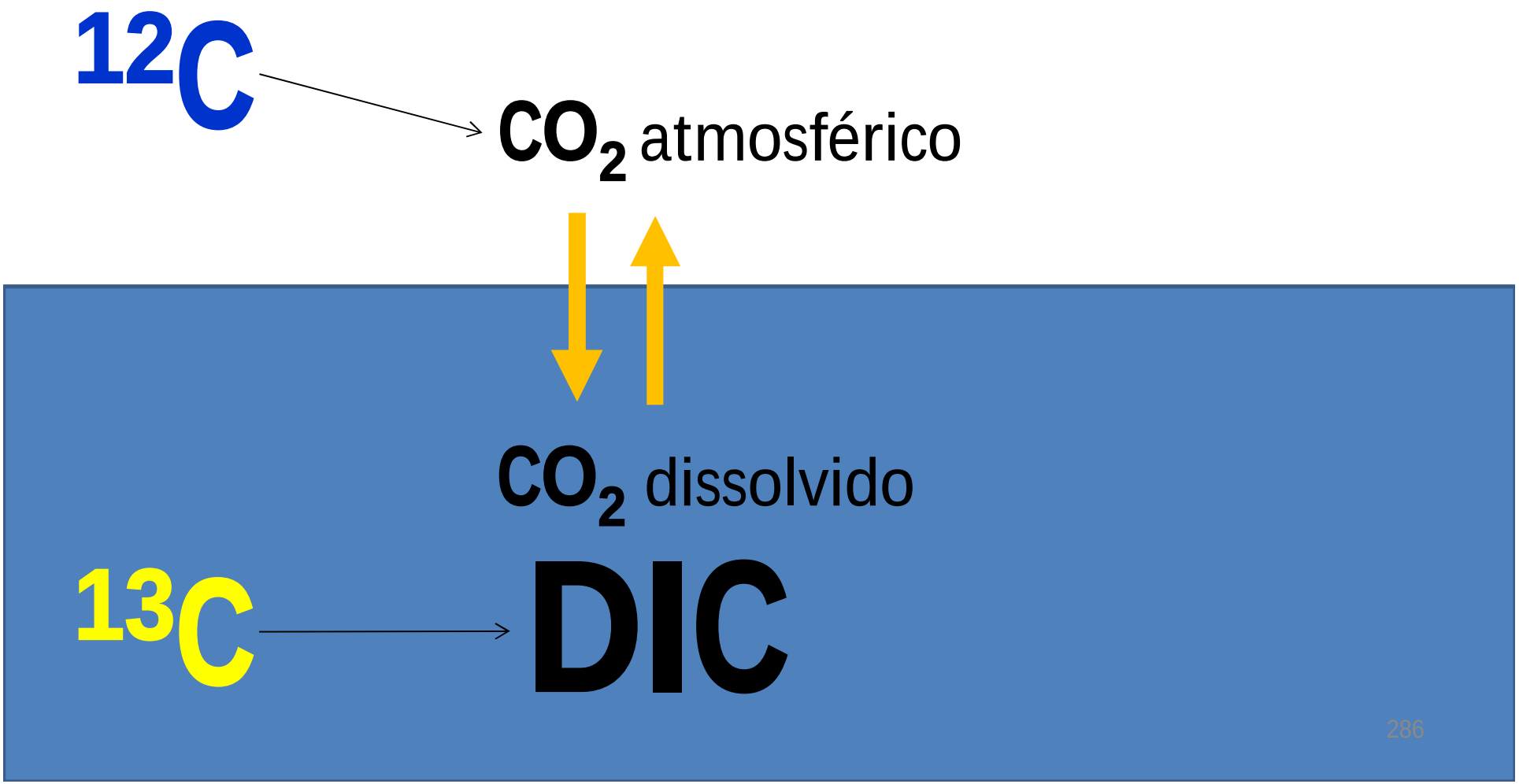
(Finnigan™ Gas Bench II Operating Manual)

$\delta^{13}C$ do DIC como marcador oceanográfico

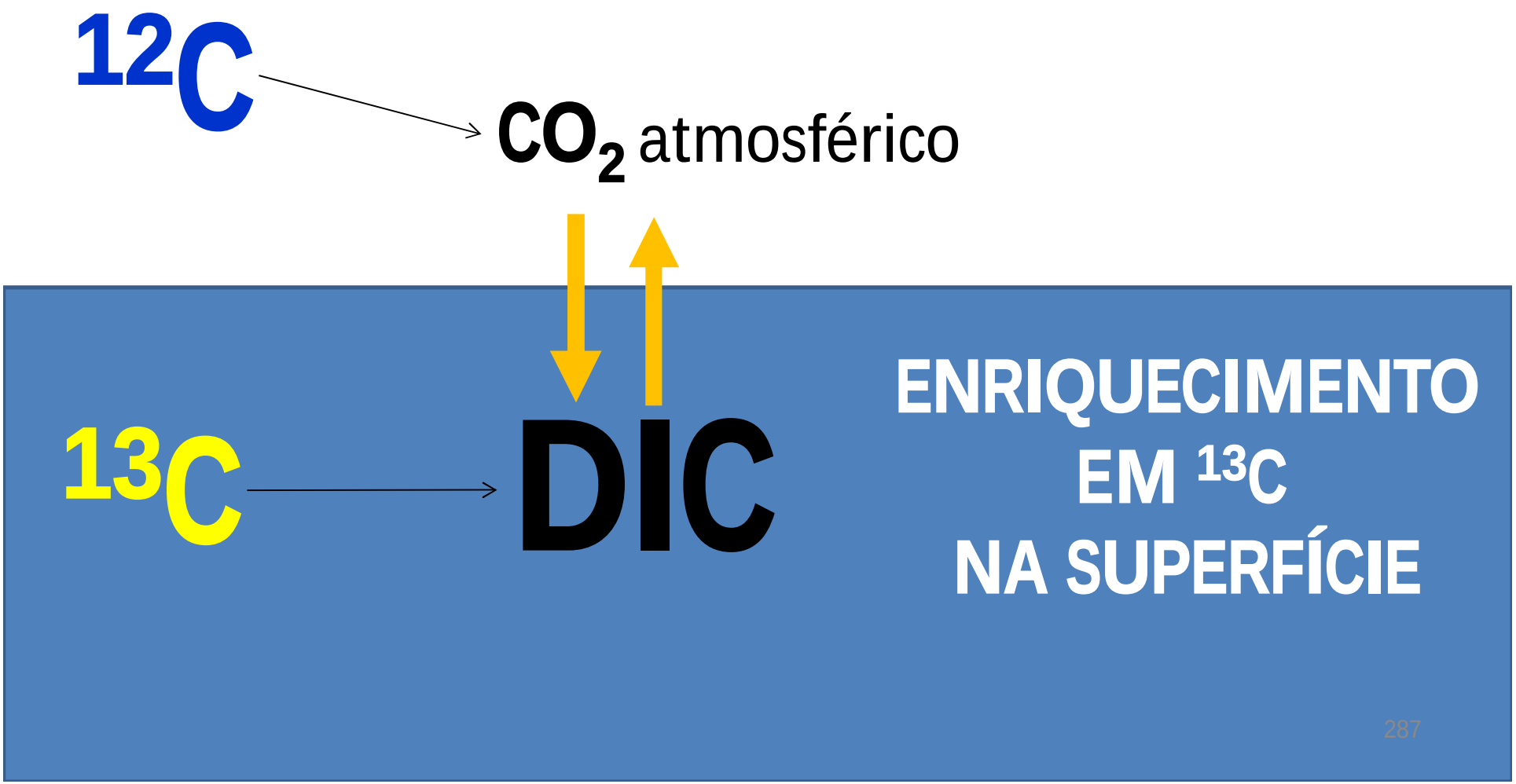
1) Fracionamento de ^{12}C e ^{13}C entre atmosfera e superfície do oceano


CO_2 atmosférico
ventilação
CO_2 dissolvido

Fracionamento de ^{12}C e ^{13}C entre atmosfera e superfície do oceano



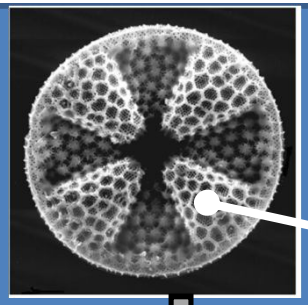
Fracionamento de ^{12}C e ^{13}C entre atmosfera e superfície do oceano



Isótopos estáveis do carbono como marcadores:

2) Fracionamento de ^{12}C e ^{13}C devido ao ciclo biológico

Fracionamento de ^{12}C e ^{13}C devido ao ciclo biológico



^{12}C concentrado nos tecidos orgânicos

enriquecimento em ^{13}C na superfície

Degradação da M.O. libera ^{12}C na água

enriquecimento em ^{12}C da água intermediaria

Resumo: o teor em ^{13}C no oceano depende de

troca de CO_2 atmosfera-oceano

atividade biológica

enriquecimento em ^{13}C
(aumento da $^{13}C/^{12}C$)
no **DIC** da **superfície** do mar

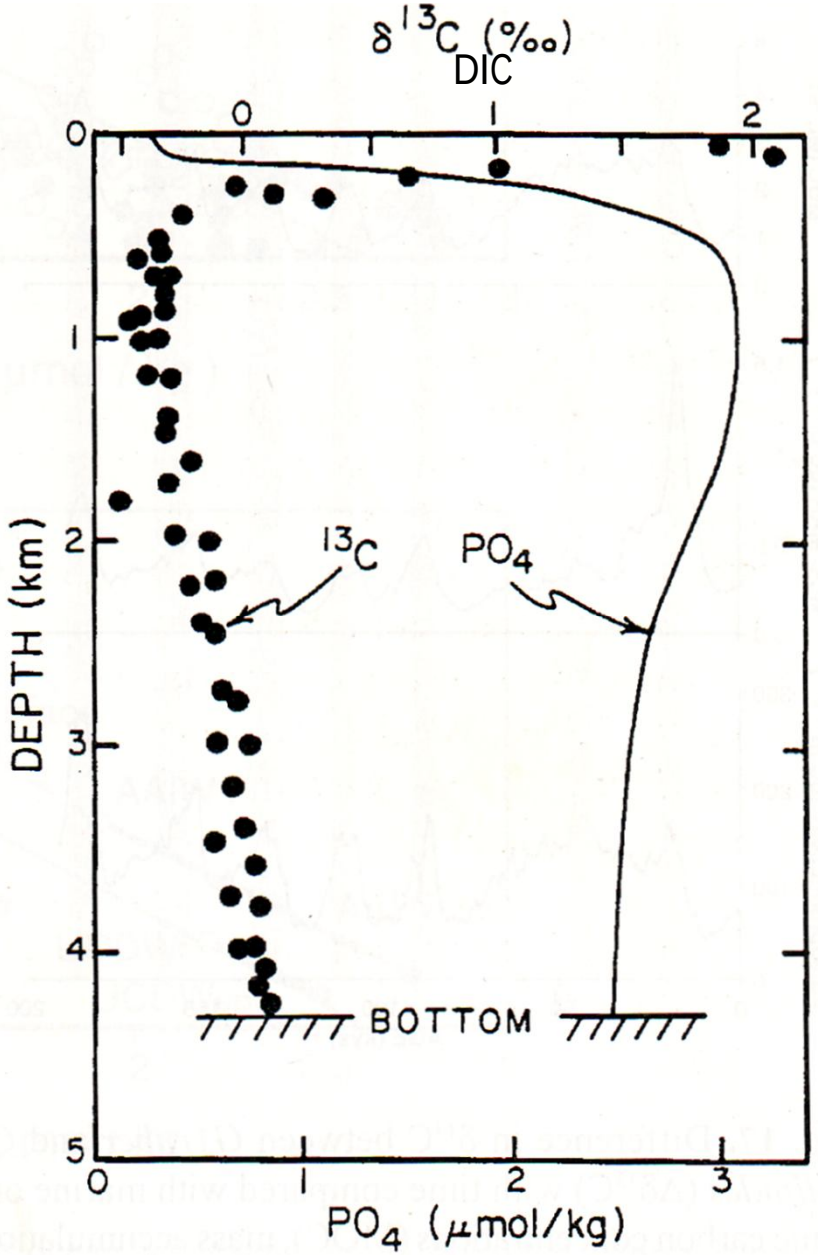
Resumo

Decaimento da materia orgânica

enriquecimento em ^{12}C no **DIC**
(diminuição da $^{13}C/^{12}C$)
em **agua intermediaria**
e intersticial

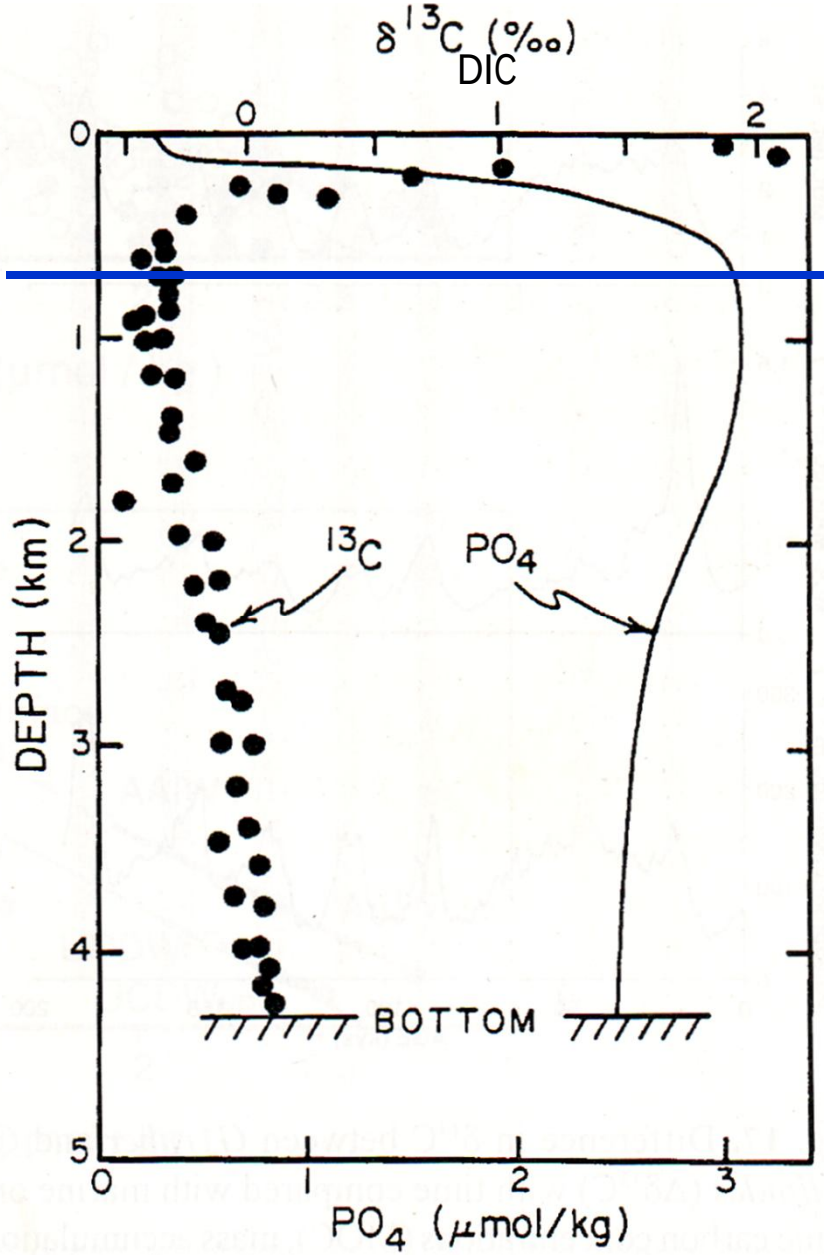
Relação entre **teor em ^{13}C**
e concentração de **nutrientes** no oceano

Teor de ^{13}C e concentração de fosfato na coluna d'agua



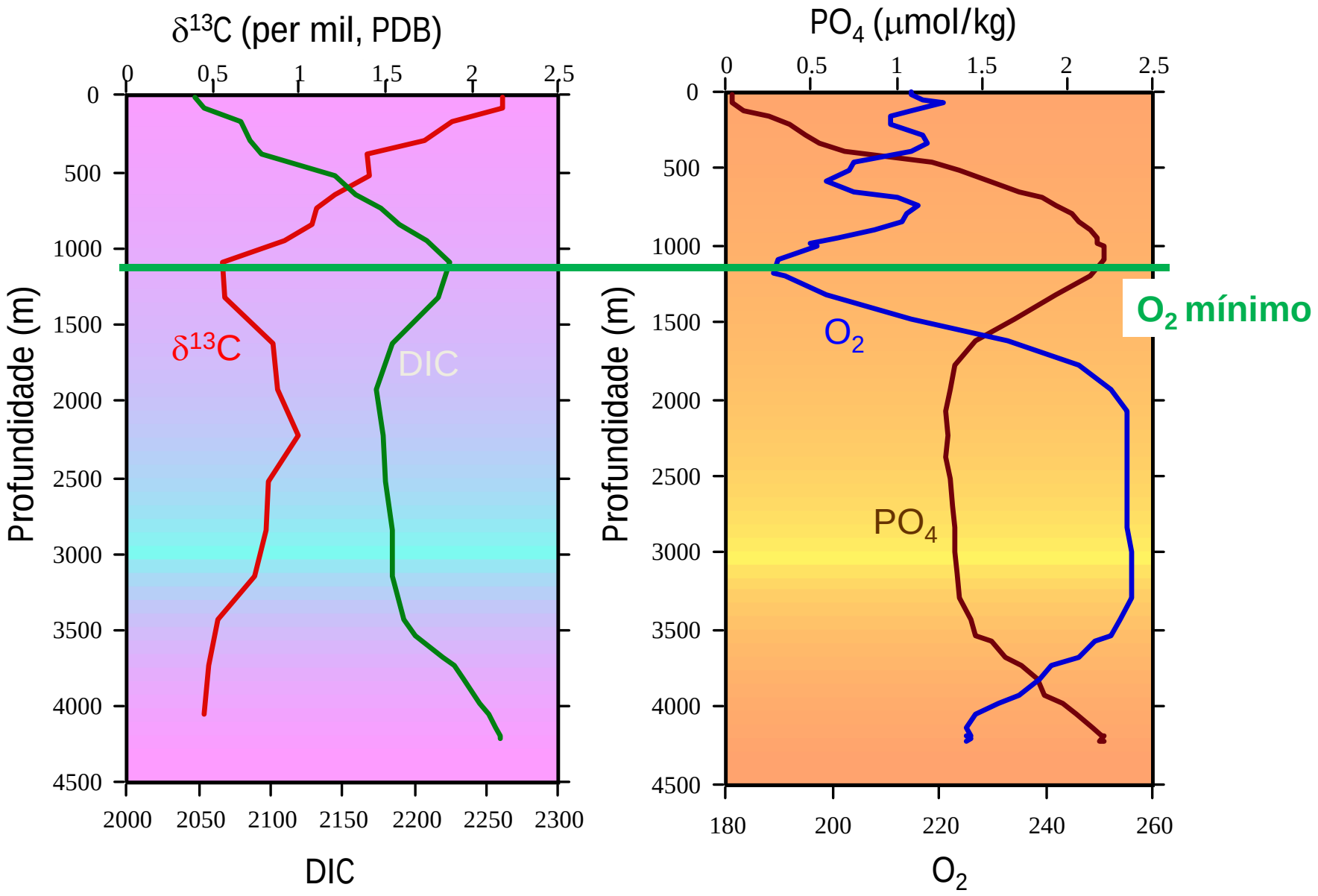
G. Fisher & G. Wefer eds. (1999)

Teor de ^{13}C e concentração de fosfato na coluna d'agua



Decaimento da materia orgânica

G. Fisher & G. Wefer eds. (1999)


δ13C (per mil, PDB)
0
0.5
1
1.5
2
2.5
Profundidade (m)
0
500
1000
1500
2000
2500
3000
3500
4000
4500
δ13C
DIC
2000
2050
2100
2150
2200
2250
2300
DIC
PO4 (µmol/kg)
O2
PO4
180
200
220
240
260
O2
O2 mínimo

Remineralização da materia orgânica

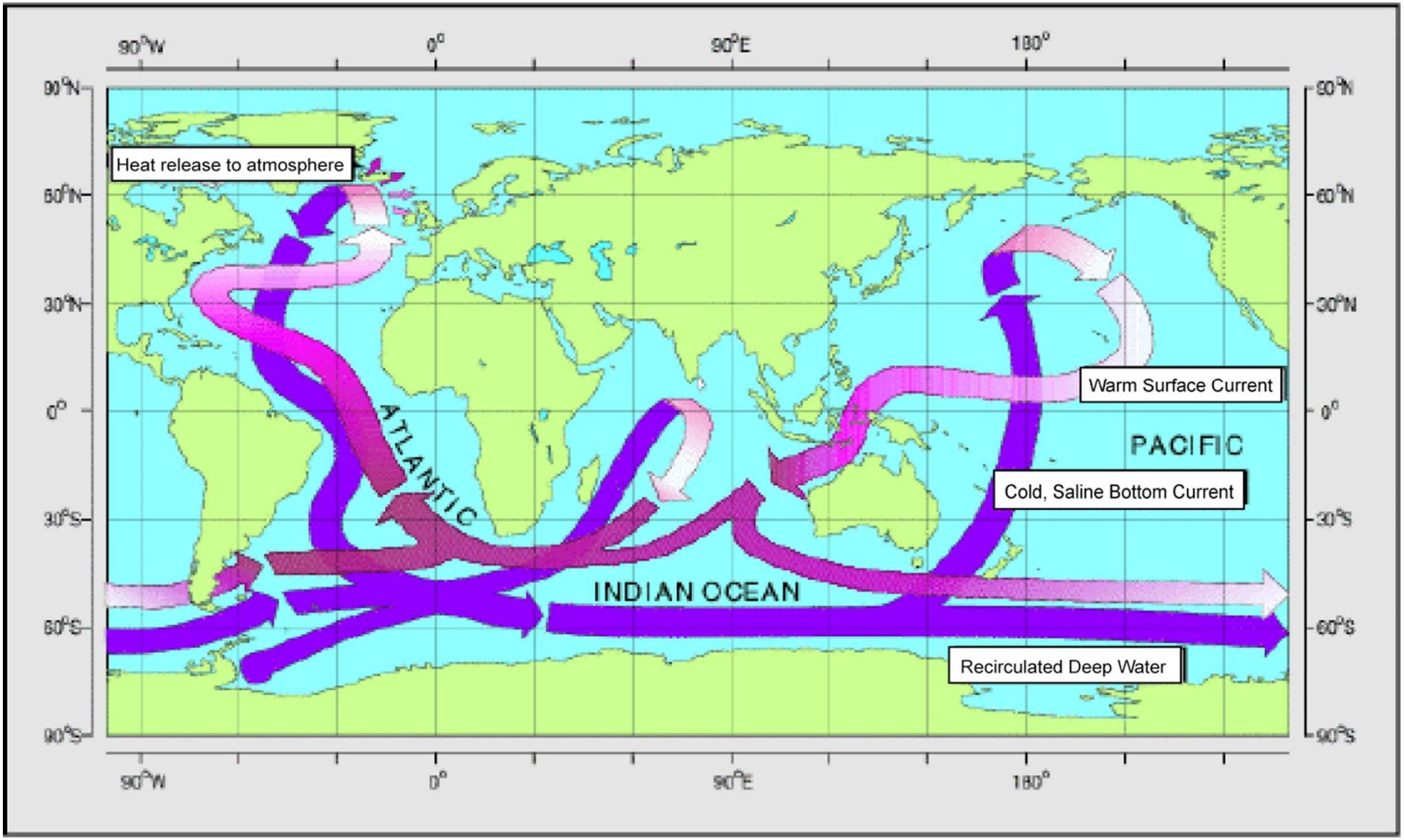
implica **empobrecimento** em ^{13}C
(e enriquecimento em ^{12}C) no **DIC**
= diminuição do valor de $\delta^{13}C_{DIC}$

que está **correlacionada** com

- **diminuição** da **[O_2]**
- **aumento** da **[nutrientes]**
- **aumento** da **[DIC]**

Uso do $\delta^{13}C$ do DIC como marcador oceanográfico

A circulação termoalina global



after W. Broecker, modified by E. Maier-Reimer

Projeto GEOSECS

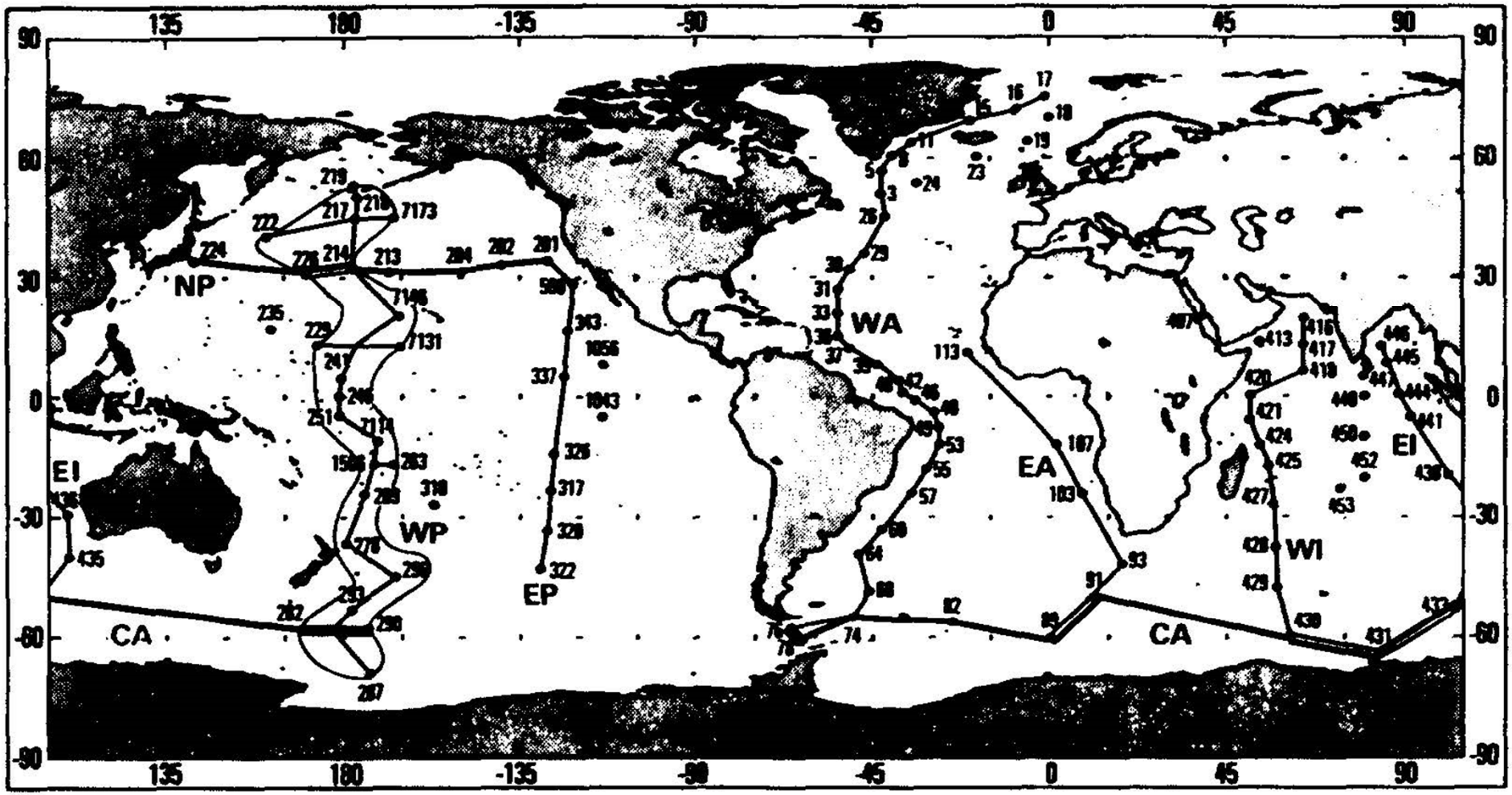
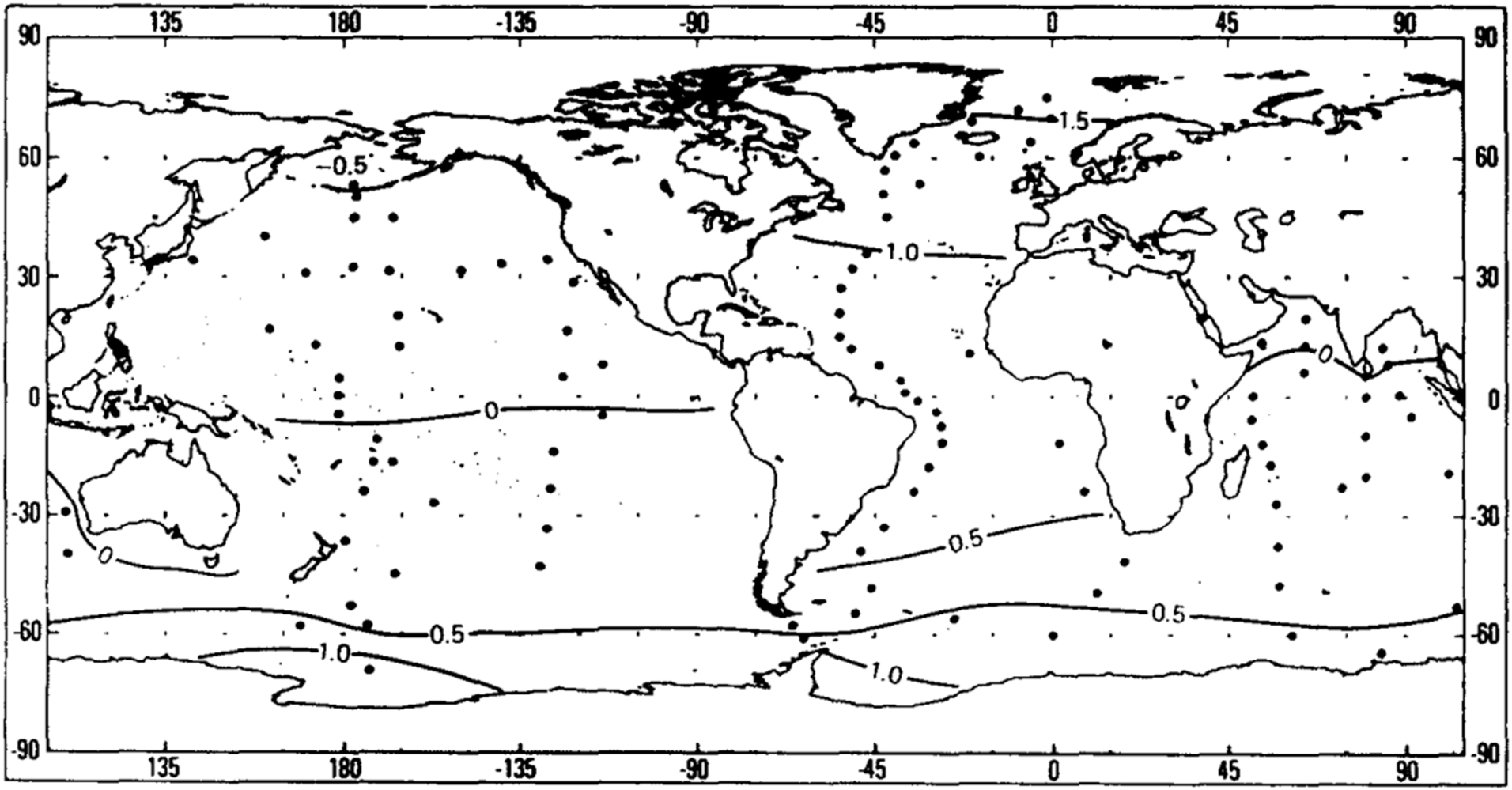


Fig. 1. Map of the world showing the GEOSECS stations at which samples were collected for $\delta^{13}C$ analyses. The lines indicate stations that have been combined to produce vertical sections and are designated as NP, North Pacific; EP, East Pacific; etc.

$\delta^{13}C$ do DIC no oceano



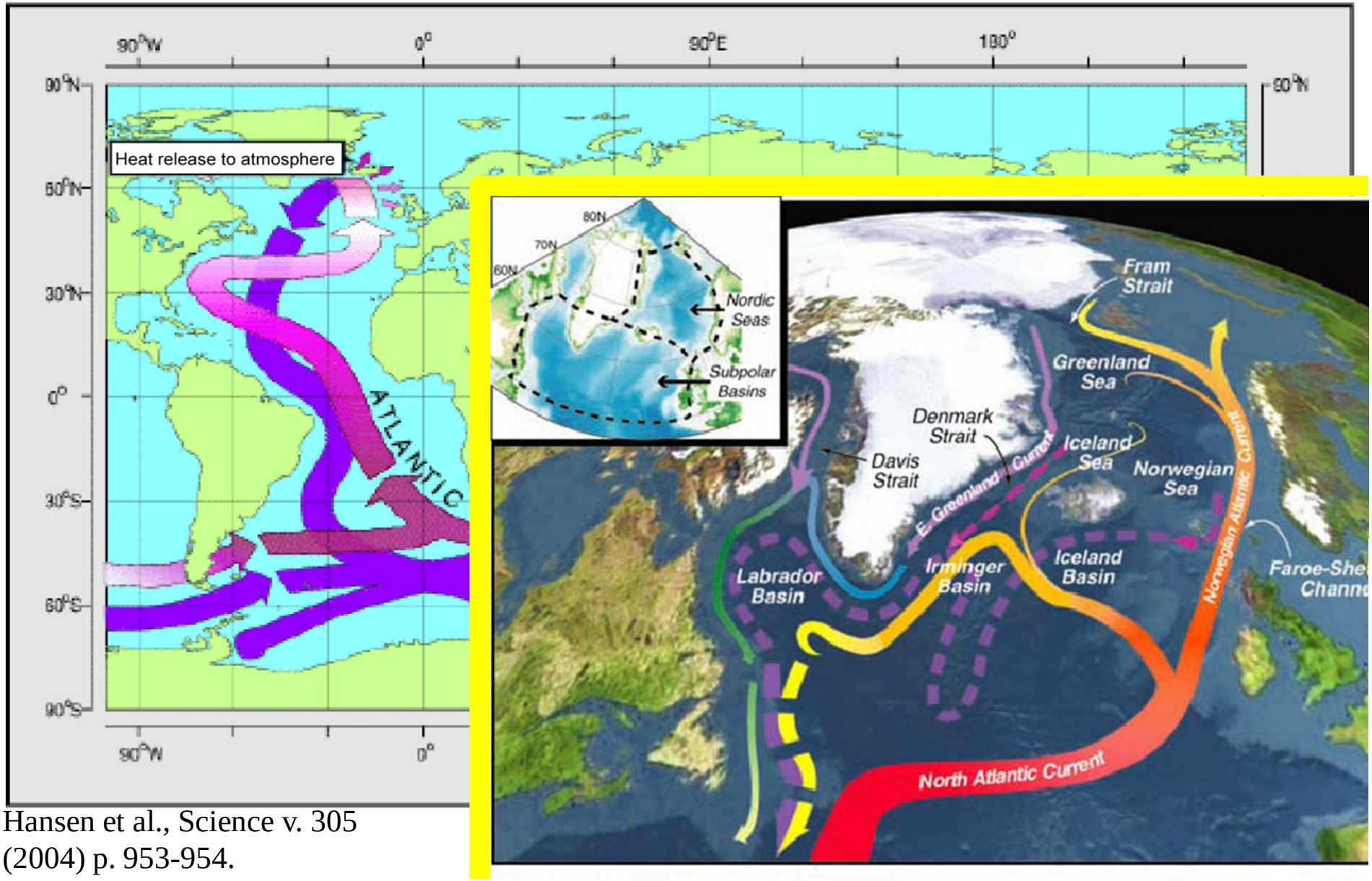
Uso do $\delta^{13}C$ do DIC como marcador oceanográfico:

Duas massas d'água profunda: NADW e AABW

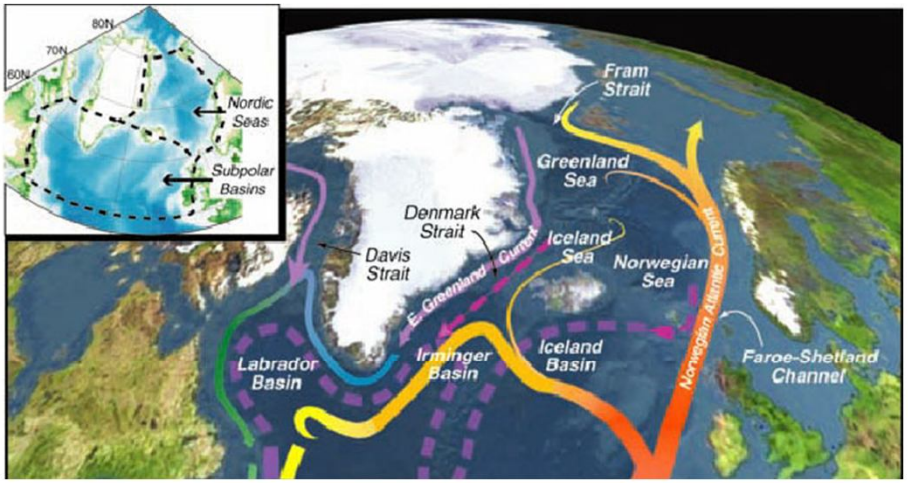
"ENDMEMBERS"

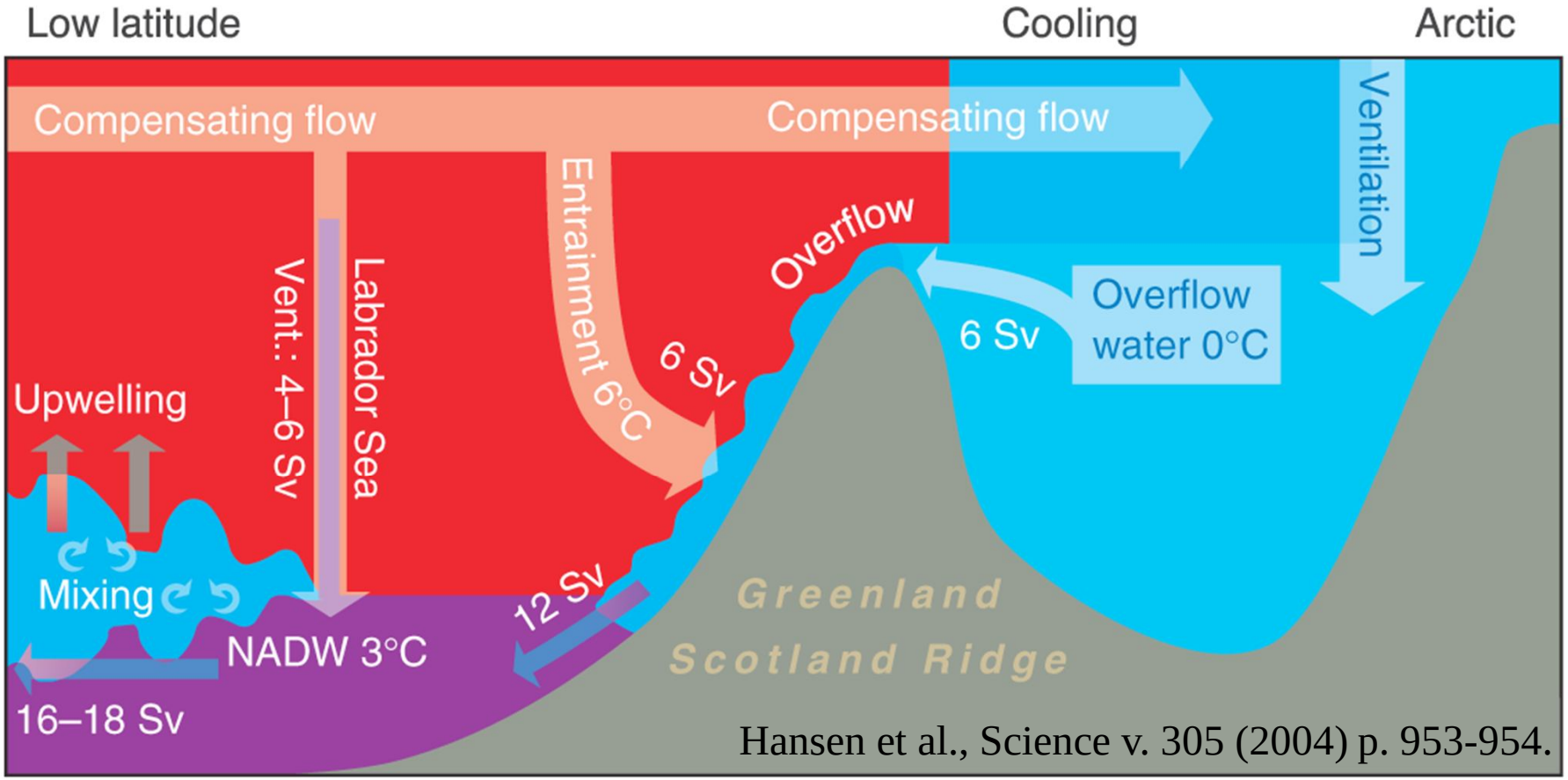
dois componentes extremos nos processos de misturamento

Formação de NADW



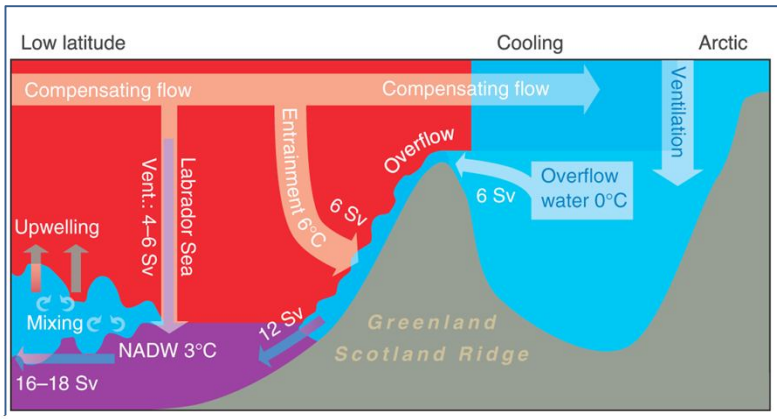
Hansen et al., Science v. 305 (2004) p. 953-954.

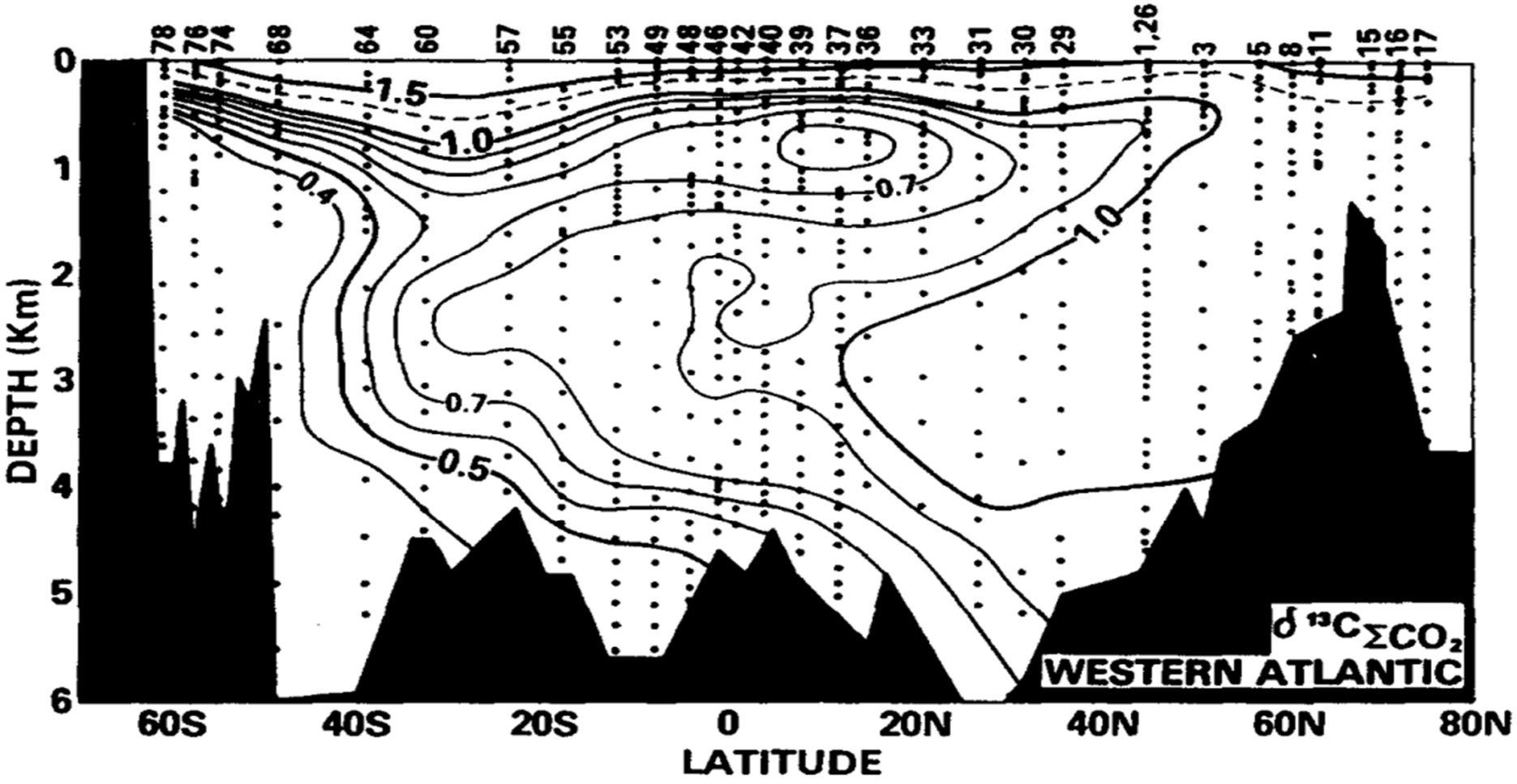




Hansen et al., Science v. 305 (2004) p. 953-954.

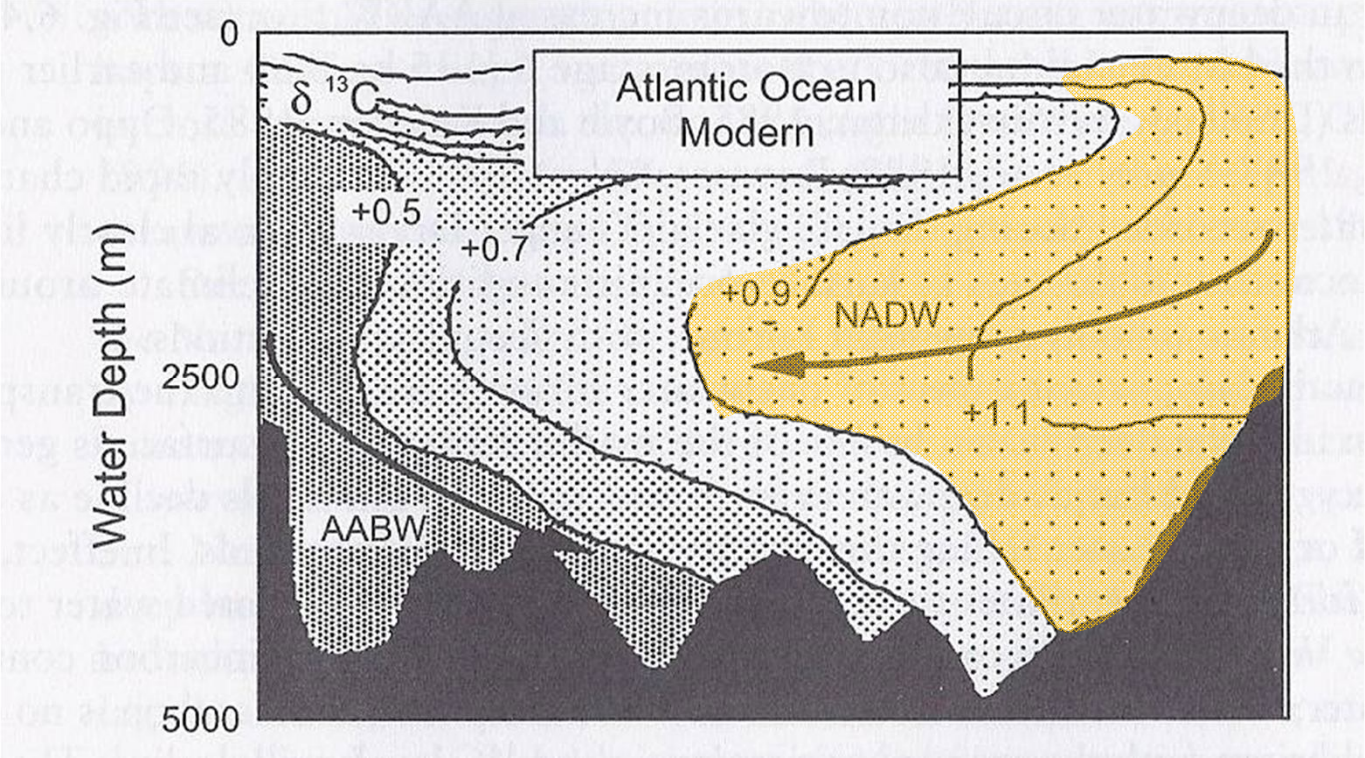
$\delta^{13}C_{DIC}$ no Atlântico Oeste





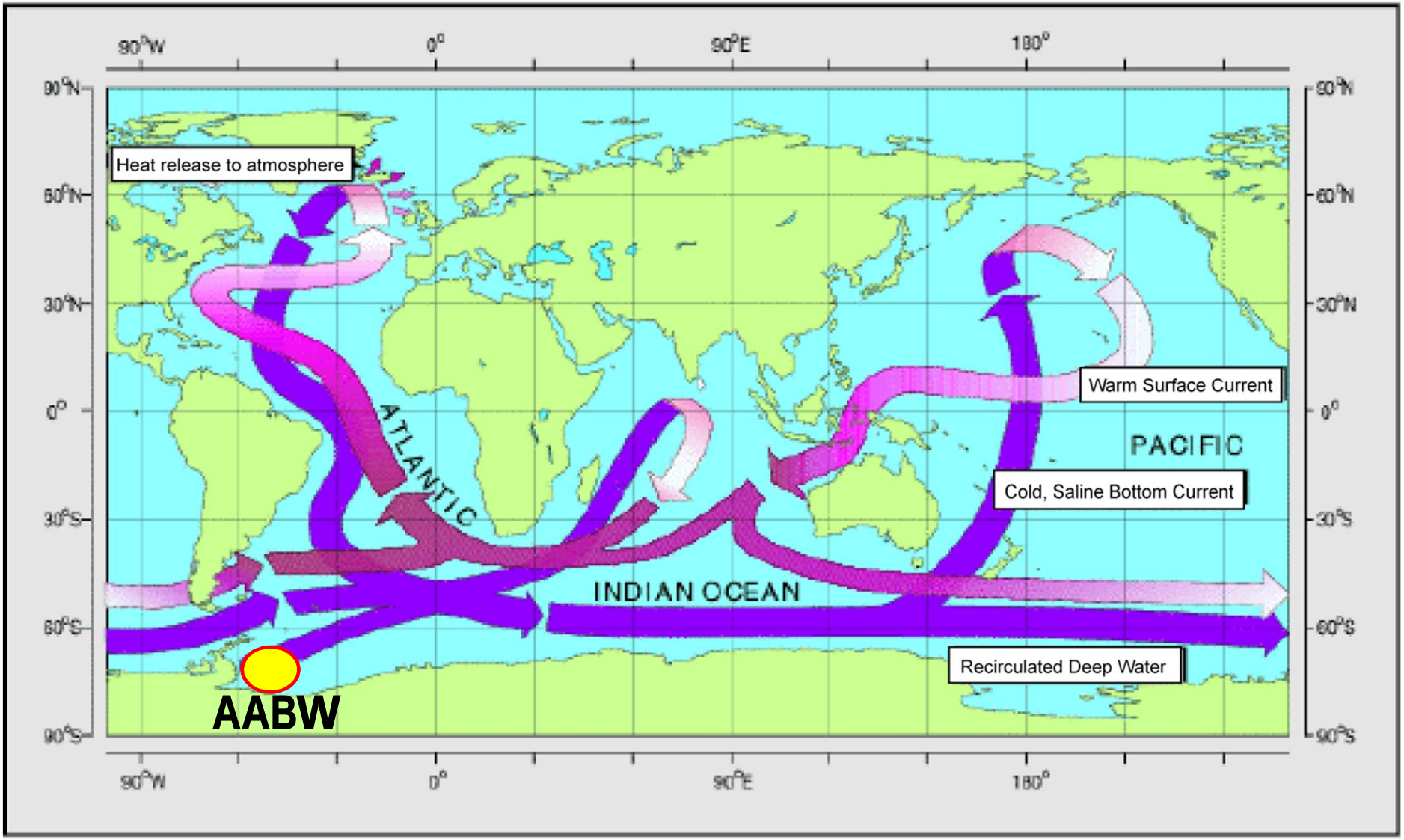
North Atlantic Deep Water (NADW)

- concentração de fosfato baixa (< 1.0 μmol/kg)
- enriquecida em ^{13}C
- $\delta^{13}C_{DIC}$ relativamente elevado (1.0 a 1.4‰)



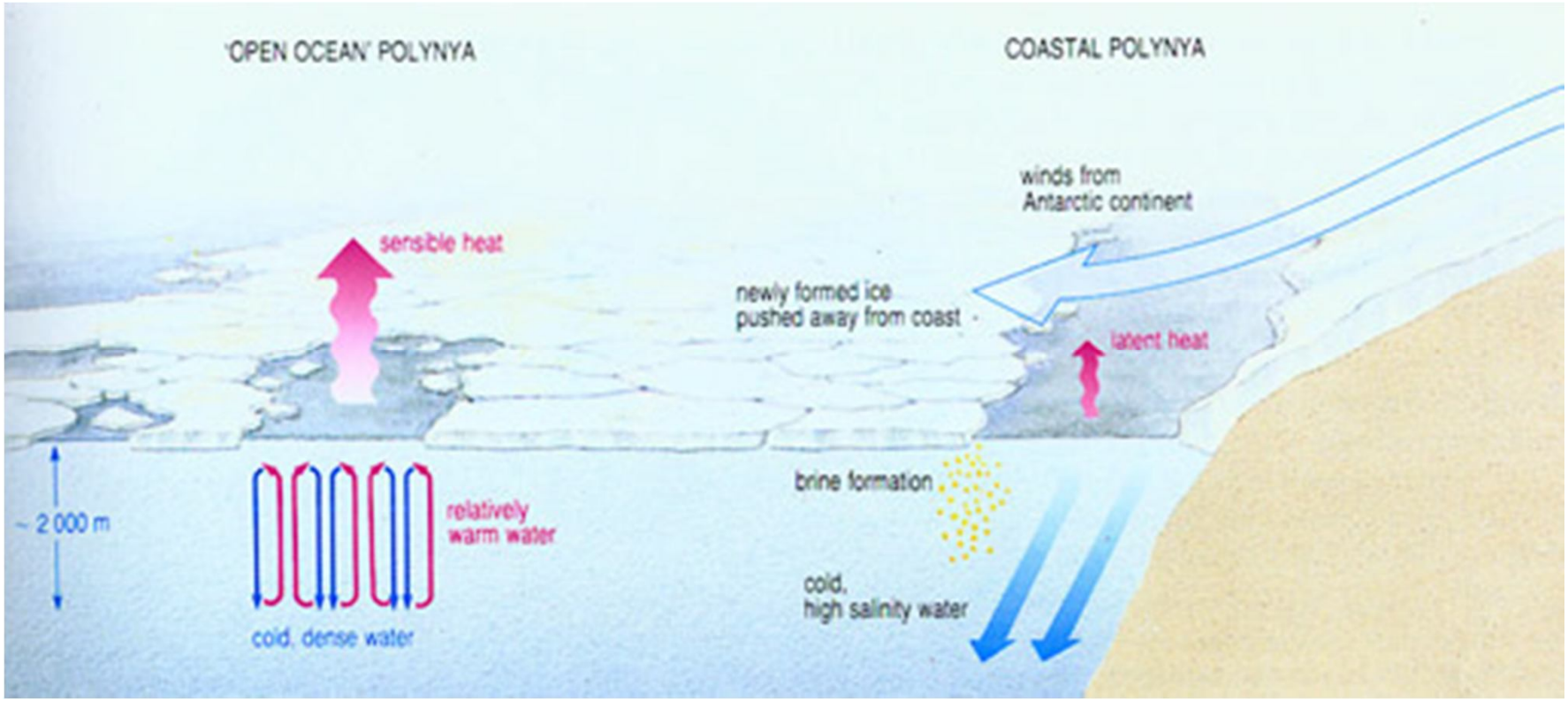
The Atlantic Thermohaline Circulation

- A key Element of the Global Oceanic Circulation -



after W. Broecker, modified by E. Maier-Reimer

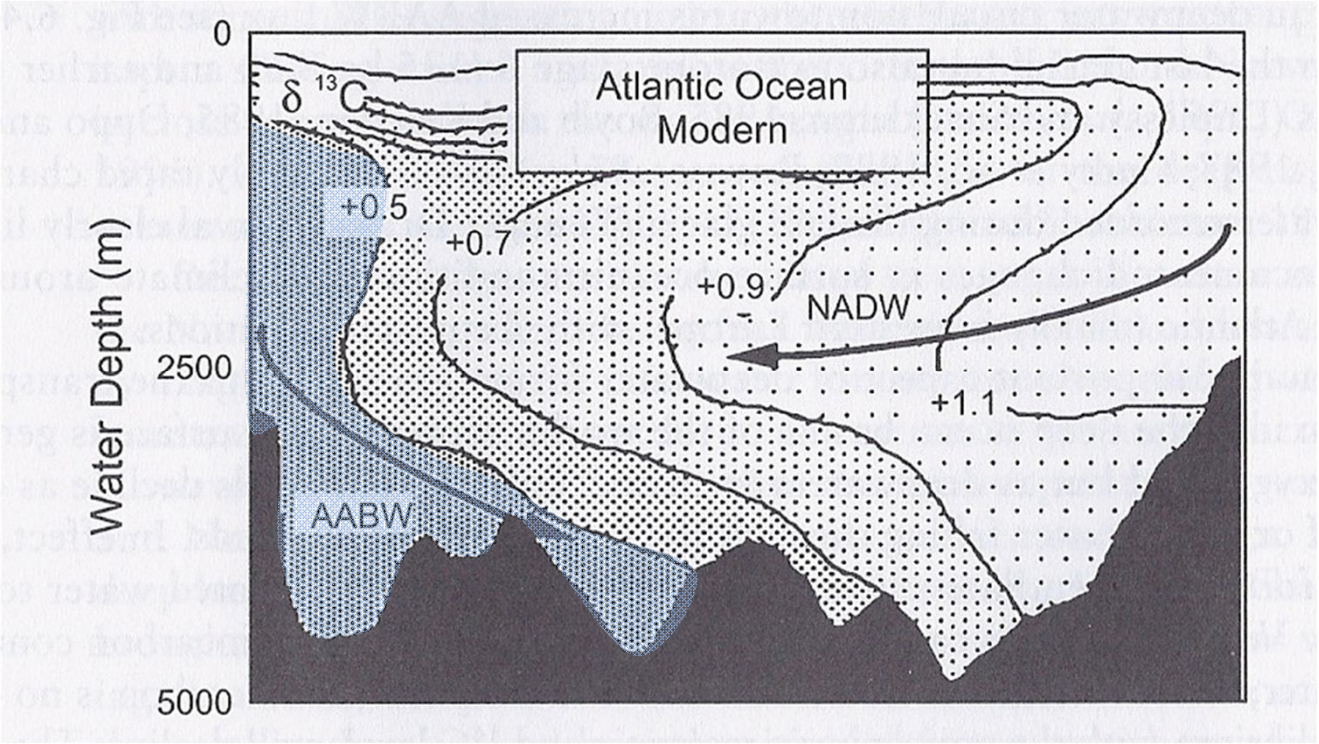
Formação de água profunda em Antártica
AABW



www.ldeo.columbia.edu

Antarctic Bottom Water (AABW)

- mais rica em fosfato em sua área fonte do que a NADW
- valores de $\delta^{13}C_{DIC}$ são geralmente menores (~0.5‰).



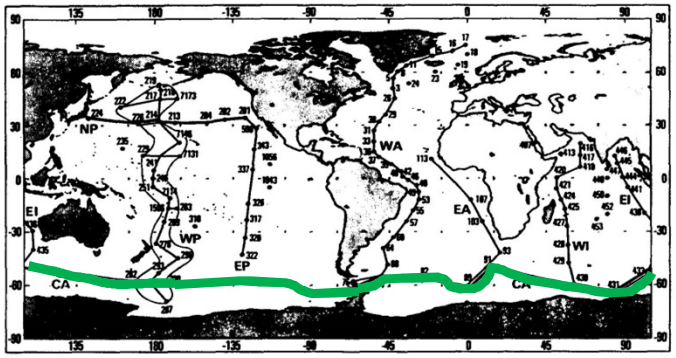
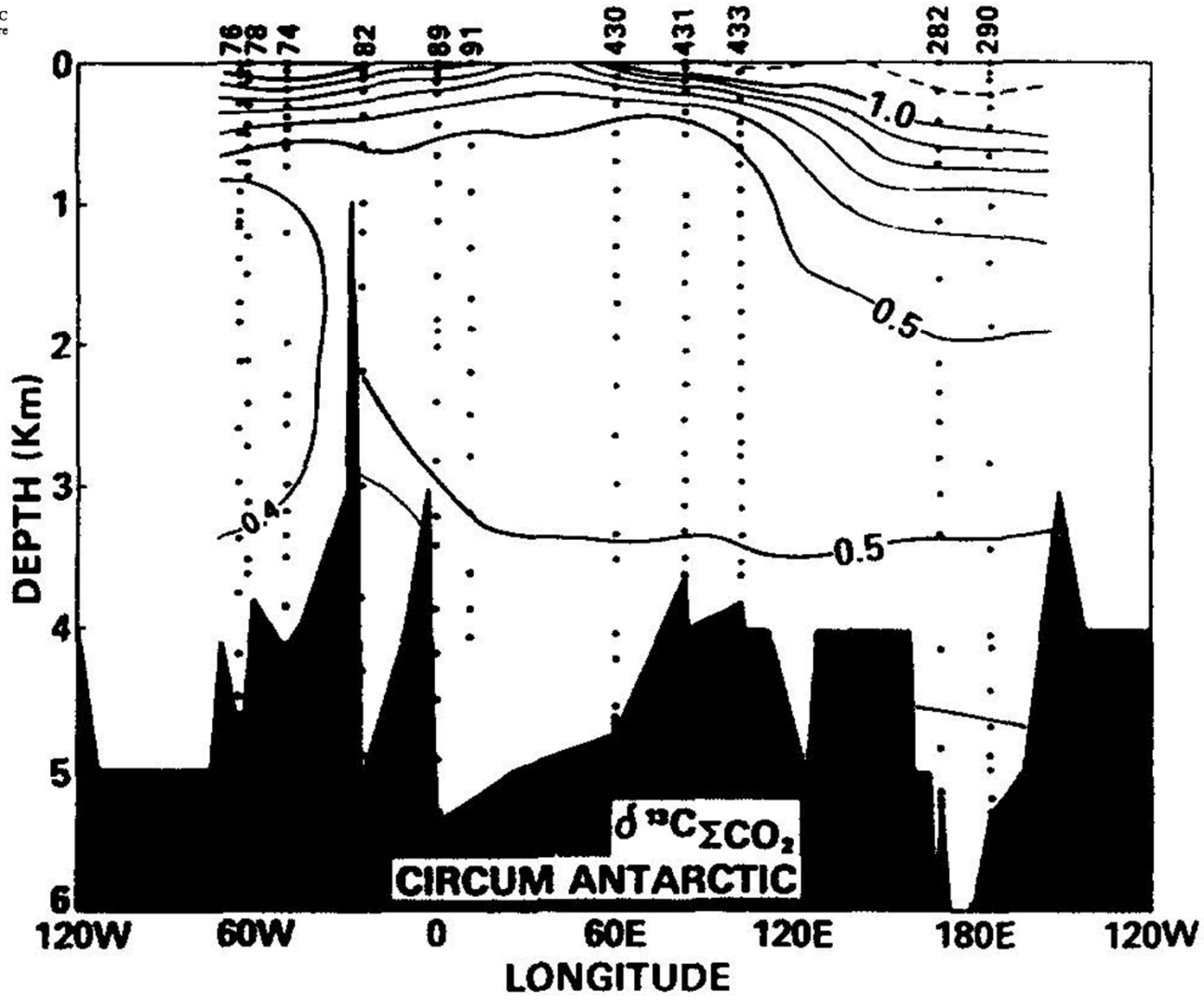


Fig. 1. Map of the world showing the GEOSECS stations at which samples were collected for $\delta^{13}C$ analyses. The lines indicate stations that have been combined to produce vertical sections and are designated as NP, North Pacific; EP, East Pacific; etc.

$\delta^{13}C_{DIC}$ na corrente profunda Circumpolar Antarctica

onde está a AABW ?

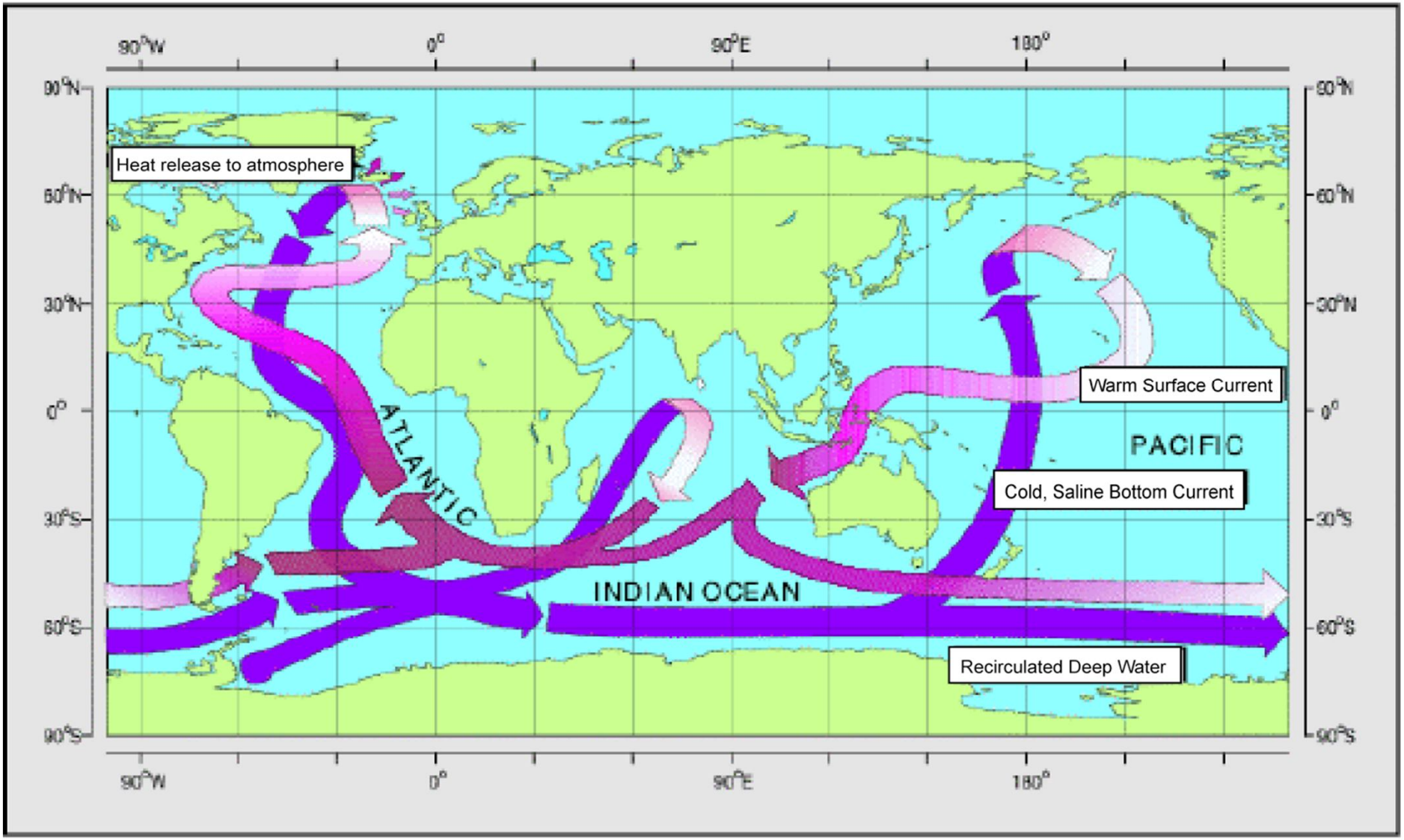


Depois de deixar as camadas superficiais dos oceanos...

os valores de $\delta^{13}C$ das massas d'água profundas sofrem alteração através de dois processos:

- remineralização e oxidação do carbono orgânico
- mistura entre massas d'água com diferentes composições isotópicas

A circulação termoalina global



after W. Broecker, modified by E. Maier-Reimer

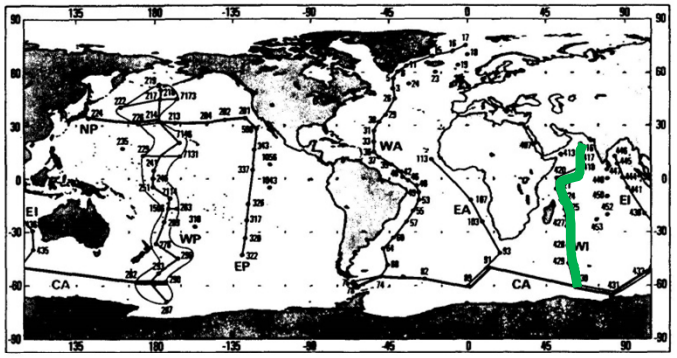
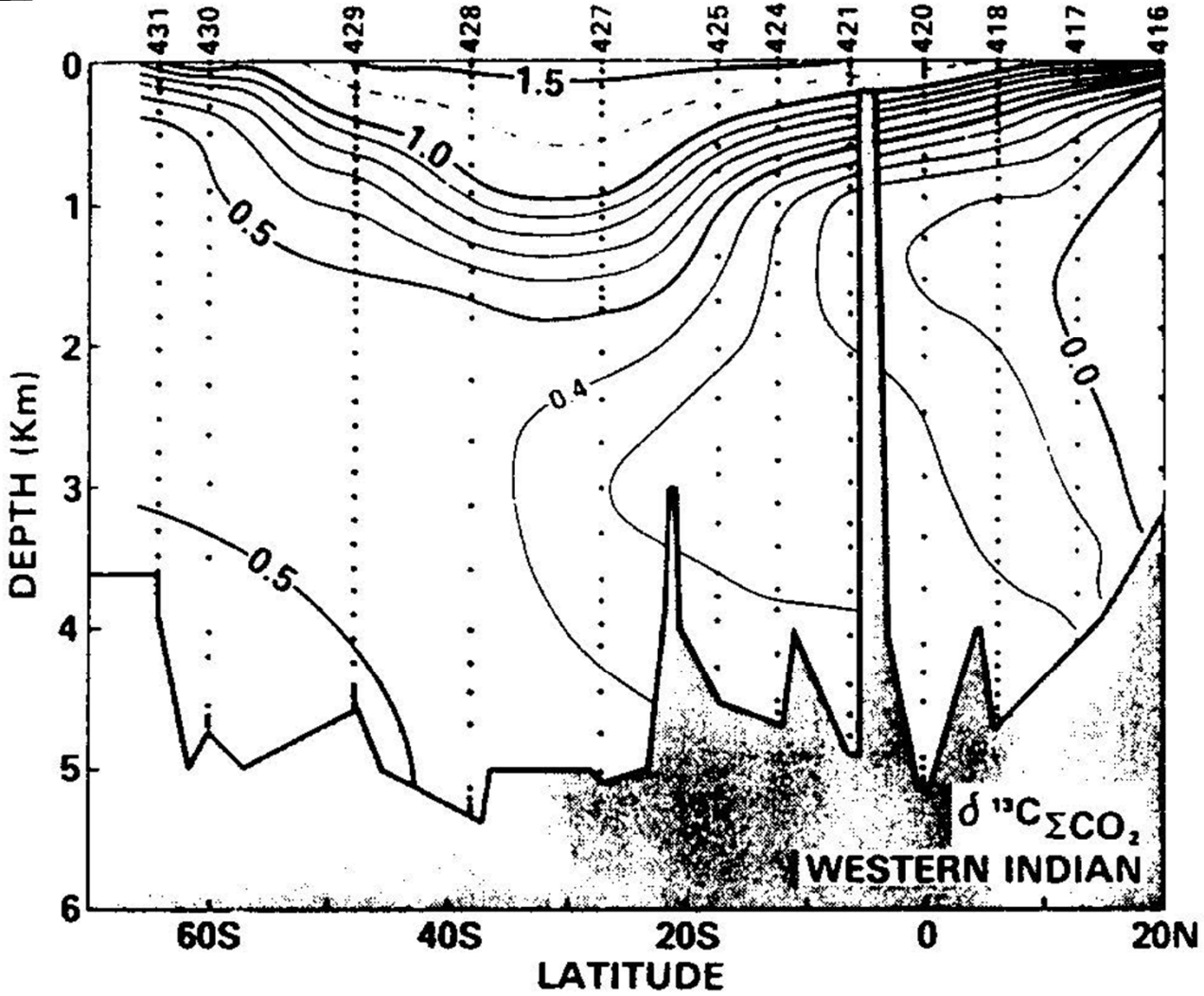


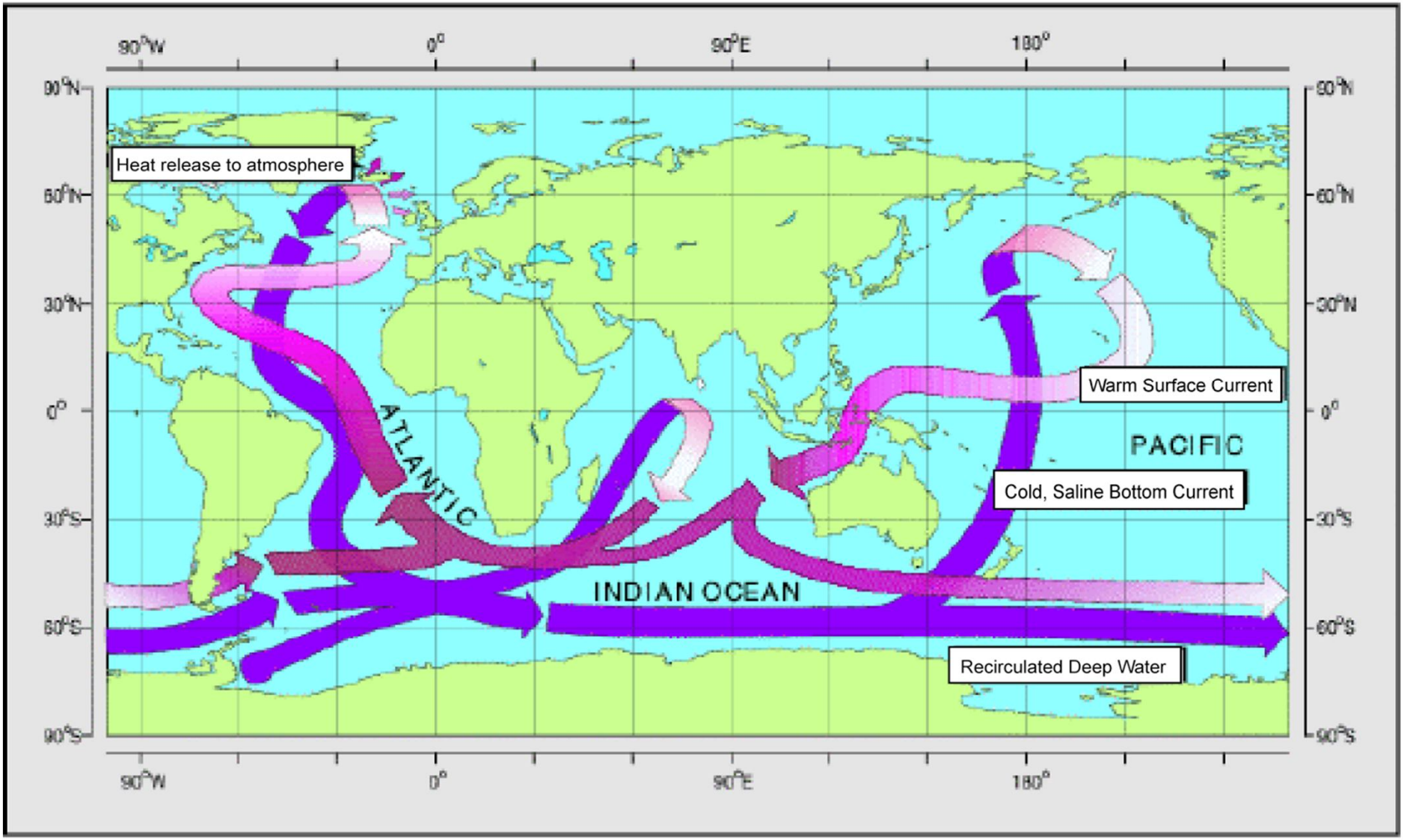
Fig. 1. Map of the world showing the GEOSECS stations at which samples were collected for $\delta^{13}C$ analyses. The lines indicate stations that have been combined to produce vertical sections and are designated as NP, North Pacific; EP, East Pacific; etc.

$\delta^{13}C_{DIC}$ no Indico Oeste

Que massa de água é essa ?



A circulação termoalina global



after W. Broecker, modified by E. Maier-Reimer

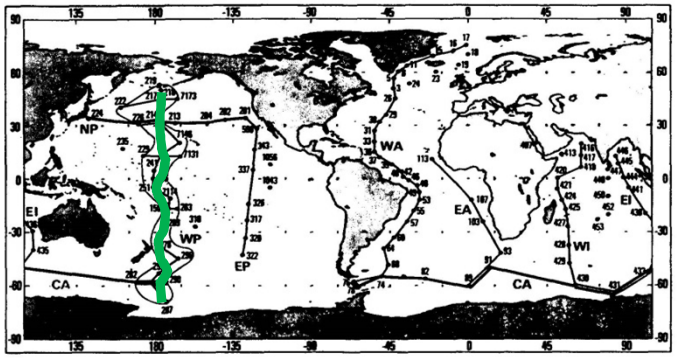
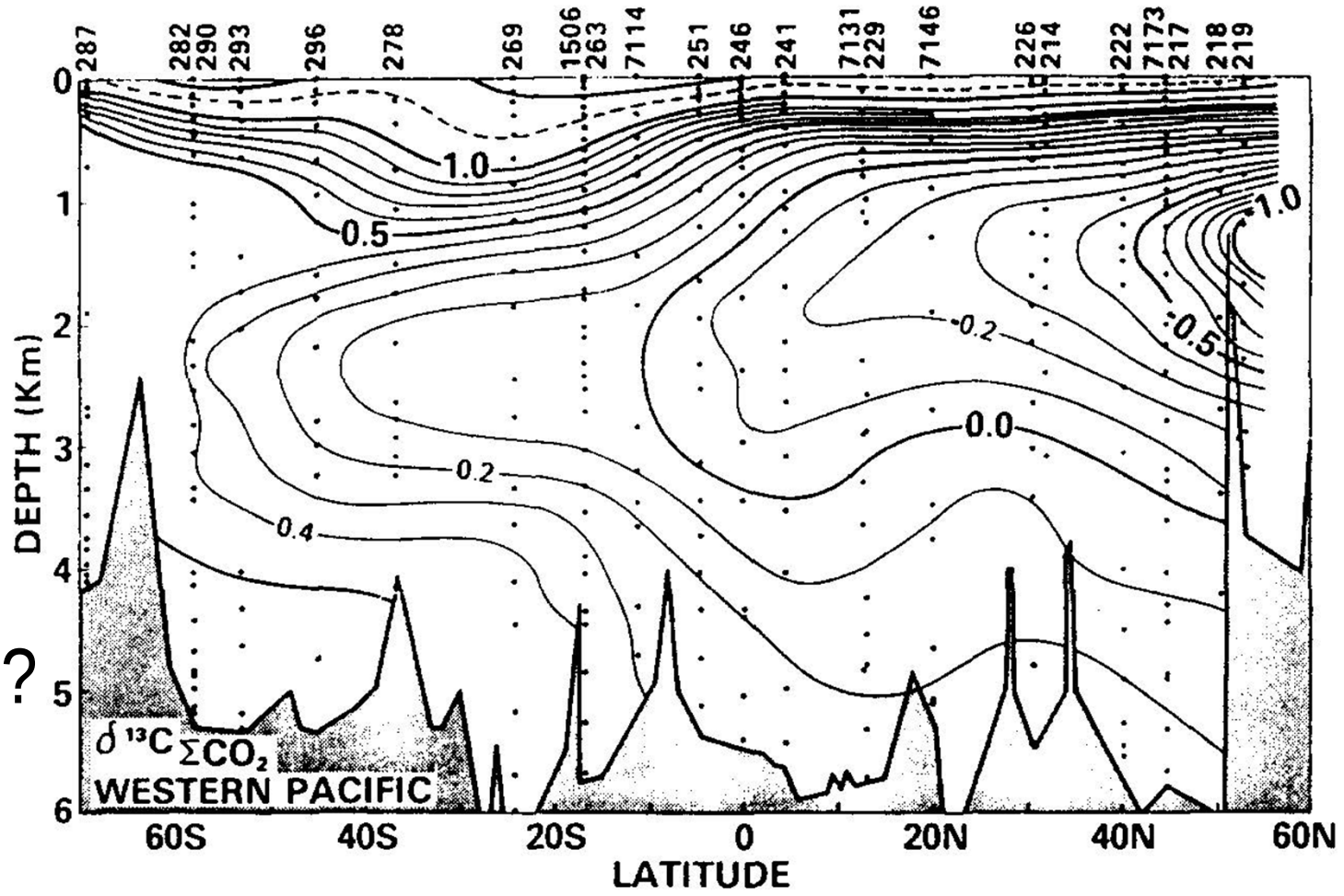


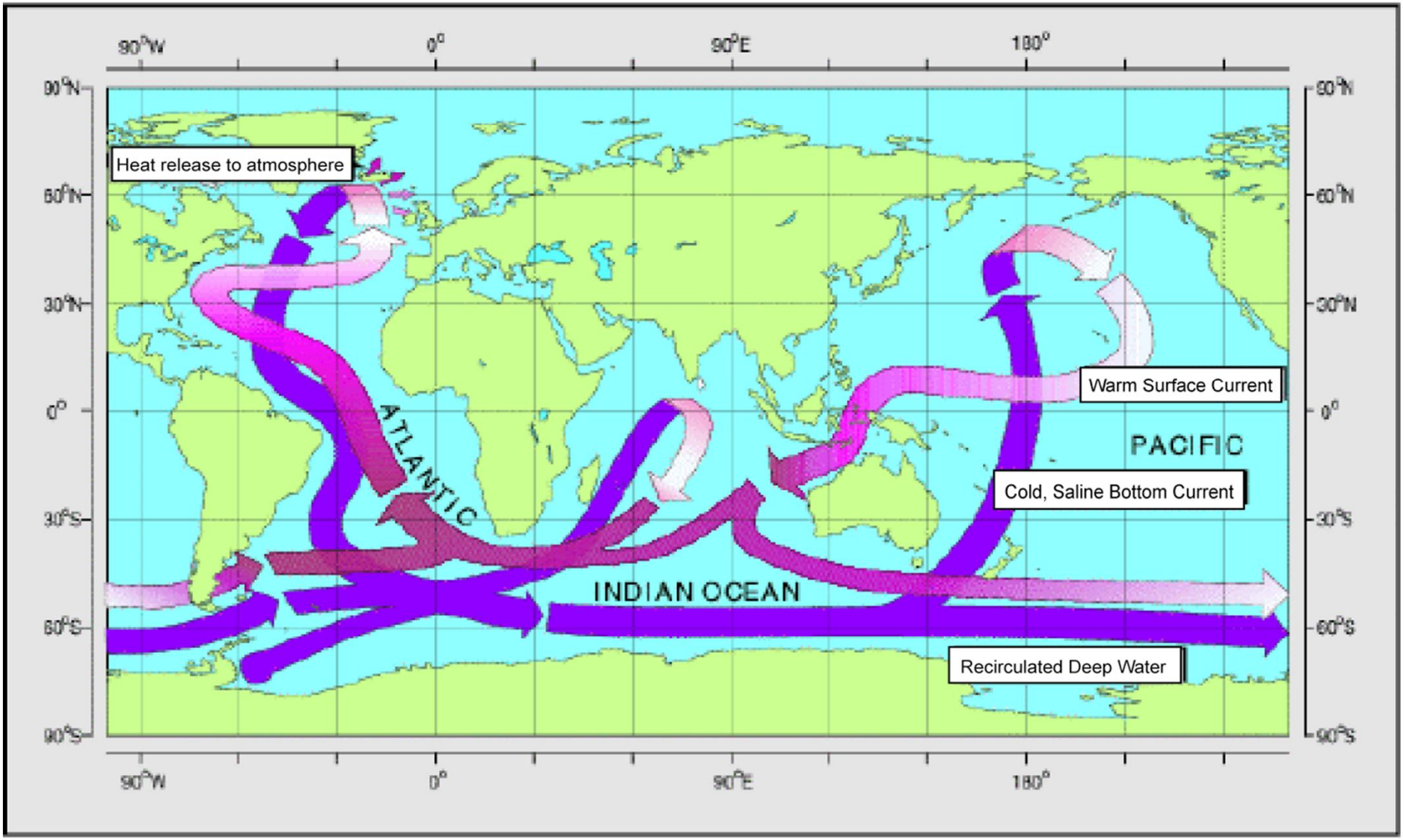
Fig. 1. Map of the world showing the GEOSECS stations at which samples were collected for $\delta^{13}C$ analyses. The lines indicate stations that have been combined to produce vertical sections and are designated as NP, North Pacific; EP, East Pacific; etc.

$\delta^{13}C_{DIC}$ no Pacifico Oeste

Por que valores mais negativos para o norte ?



A circulação termoalina global



after W. Broecker, modified by E. Maier-Reimer

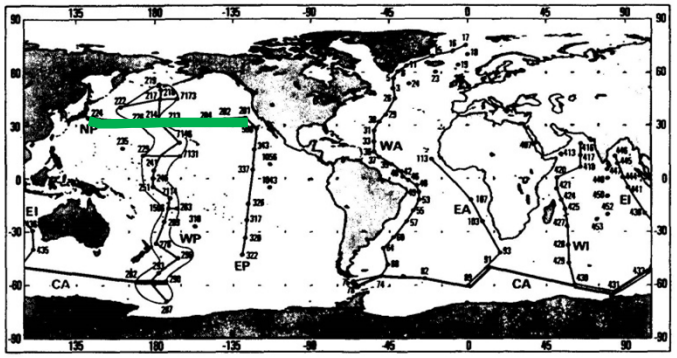
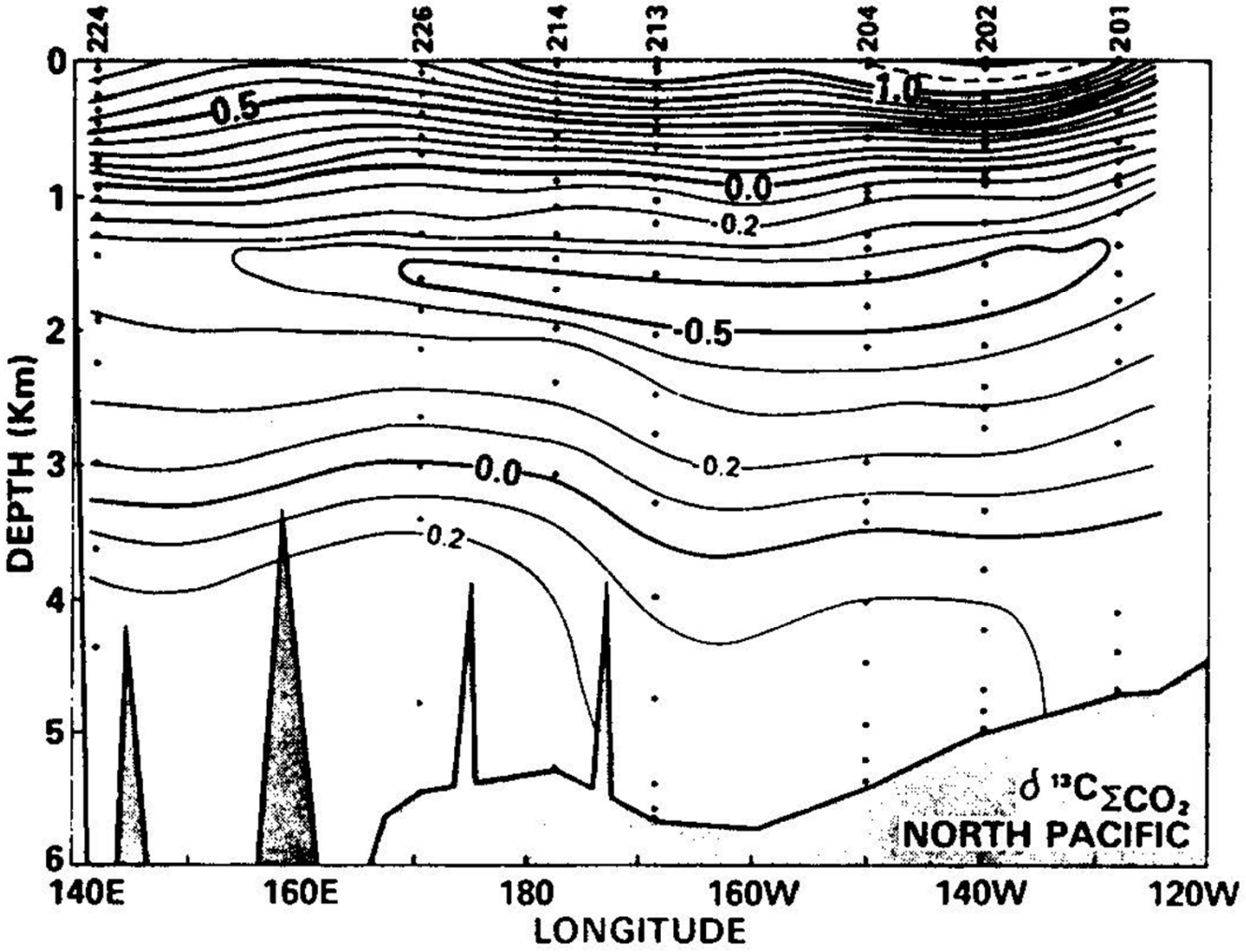


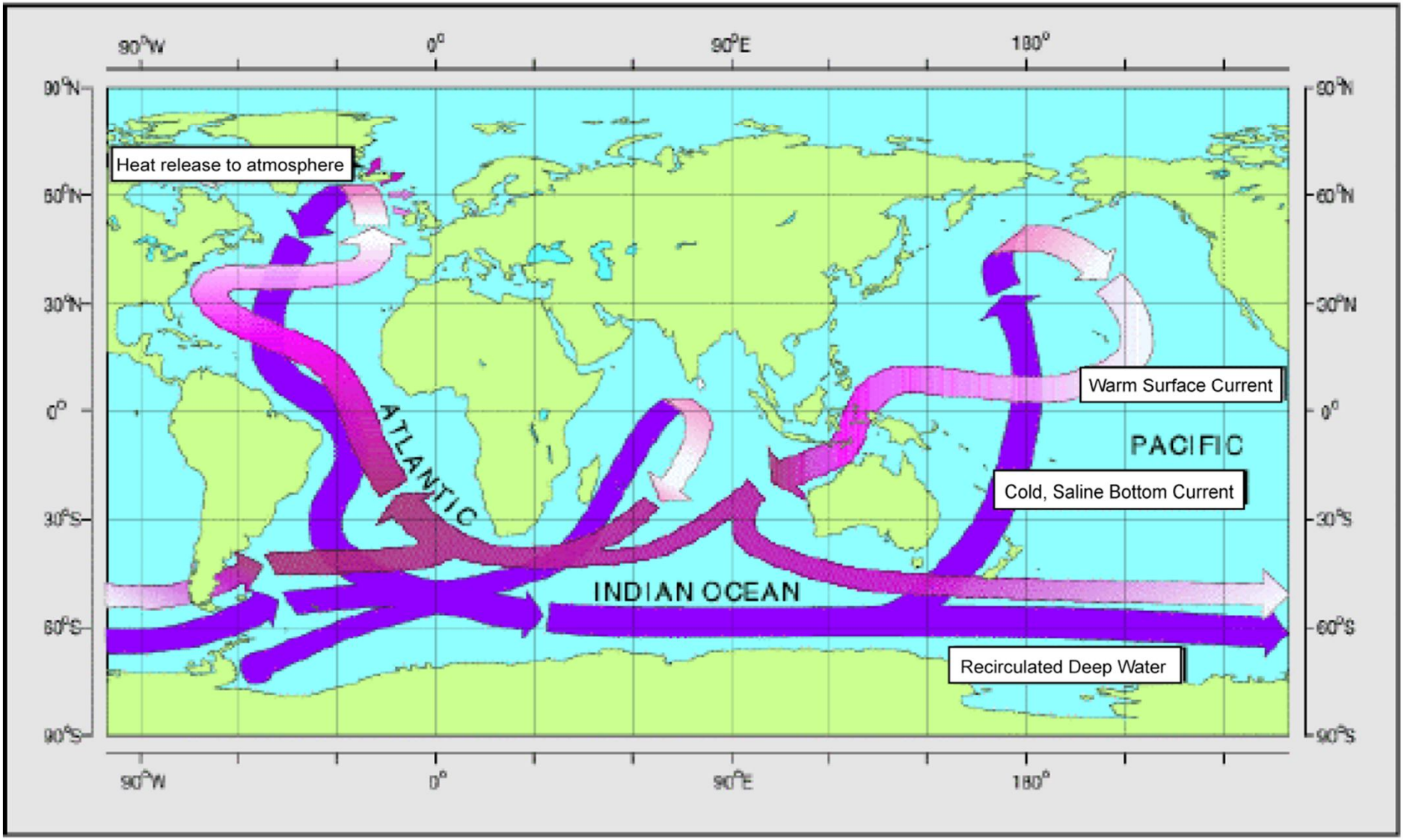
Fig. 1. Map of the world showing the GEOSECS stations at which samples wer analyses. The lines indicate stations that have been combined to produce vertic designated as NP, North Pacific; EP, East Pacific; etc.

$\delta^{13}C_{DIC}$ no Pacifico Norte

Por que valores mais negativos "para cima" ?



A circulação termoalina global



after W. Broecker, modified by E. Maier-Reimer

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journal homepage: www.elsevier.com/locate/jmarsys

Continuous shipboard measurements of oceanic $\delta^{18}O$, δD and $\delta^{13}C_{DIC}$ along a transect from New Zealand to Antarctica using cavity ring-down isotope spectrometry

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[b] Research Institute for the Environment and Livelihoods, Charles Darwin University, Darwin, Australia
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Sea-ice
Southern ocean
Polar front
Sub-tropical front
Sub-Antarctic front

ABSTRACT

Cavity ring-down spectrometers, with automated sampling interfaces, were deployed to allow measurements of water isotopes ($\delta^{18}O$, δD) and dissolved inorganic carbon ($\delta^{13}C_{DIC}$) stable isotope ratios at high temporal resolution along a transect from New Zealand to the Antarctic continental shelf. Measurements every 10 min for $\delta^{18}O$ and δD, 15 min for DIC yielded 2499 and 2289 discrete measurements respectively. High resolution data enabled the delineation of water mass boundaries as well as revealing insights into surface hydrological and biological processes. $\delta^{18}O$, δD, and $\delta^{13}C_{DIC}$ decreased southwards, dropping by approximately 1.0‰, 7.0‰, and 0.5‰, respectively. Though the decline in $\delta^{13}C_{DIC}$ with latitude was generally linear, the drop in $\delta^{18}O$ and δD was punctuated by areas of rapid, significant change corresponding to the Sub-Tropical, Sub-Antarctic and Polar Fronts. North of the Sub-Antarctic Front (approx. 54.5°S) the dominant control on water and DIC isotopes was the precipitation–evaporation balance and the contribution of upwelling waters, respectively. Further south, in close proximity to the sea ice and on the Antarctic shelf, water isotope values were more variable and predominantly influenced by the melting/freezing of sea-ice coupled to inputs from glacial/snow melt water. Local increases in $\delta^{13}C_{DIC}$ were likely due to photosynthetic enrichment of the DIC pool. Using this new instrumentation has provided one of the most comprehensive oceanic transect data sets yet achieved and illustrates the potential of these methods to delineate discrete water masses and advance our knowledge of both water and inorganic carbon cycling processes in the ocean. This methodology, combining high-resolution isotopic measurements with hydrographic data, has significant benefits in modelling water mixing in locations with multiple sources and controlling processes.

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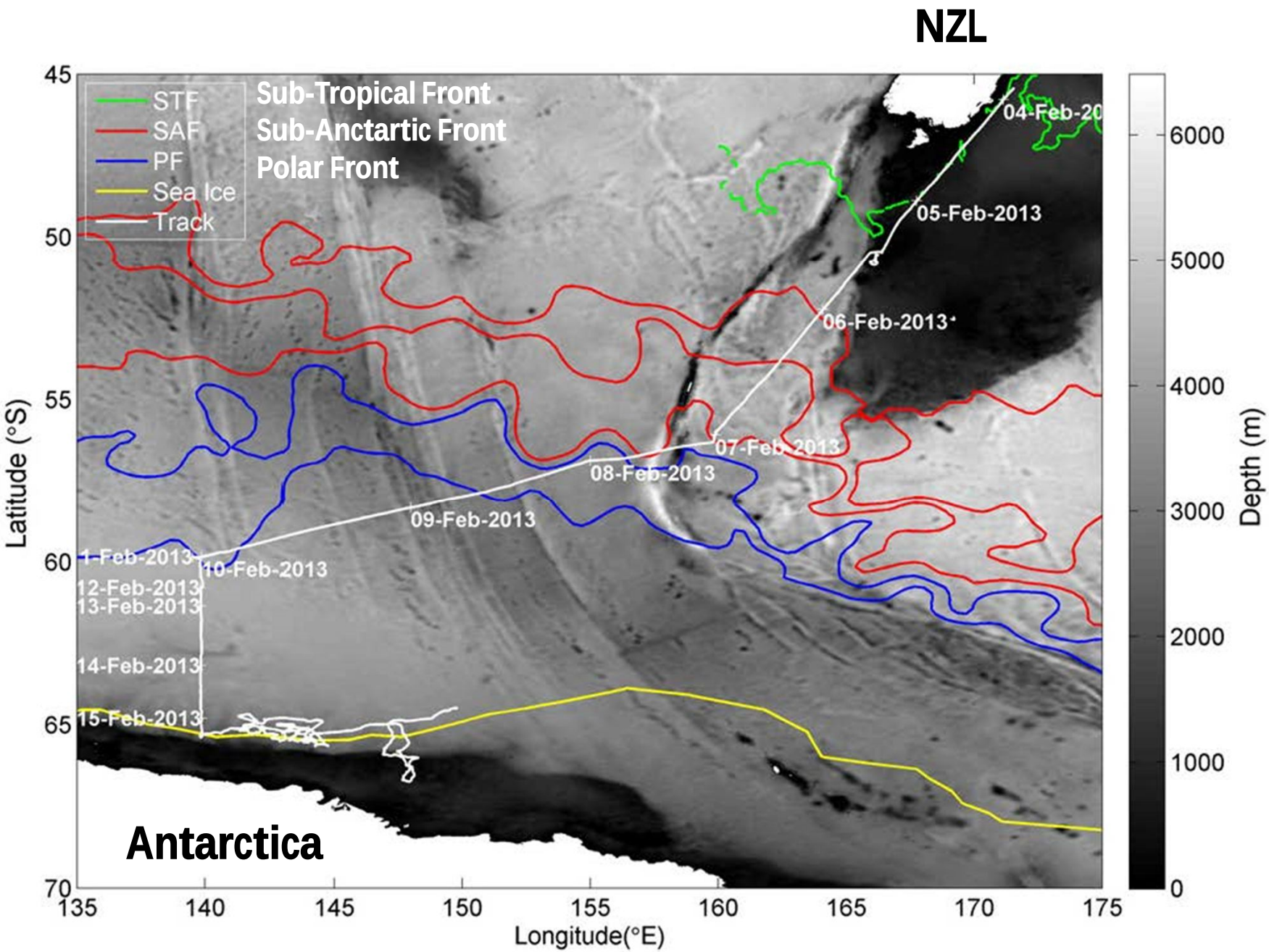
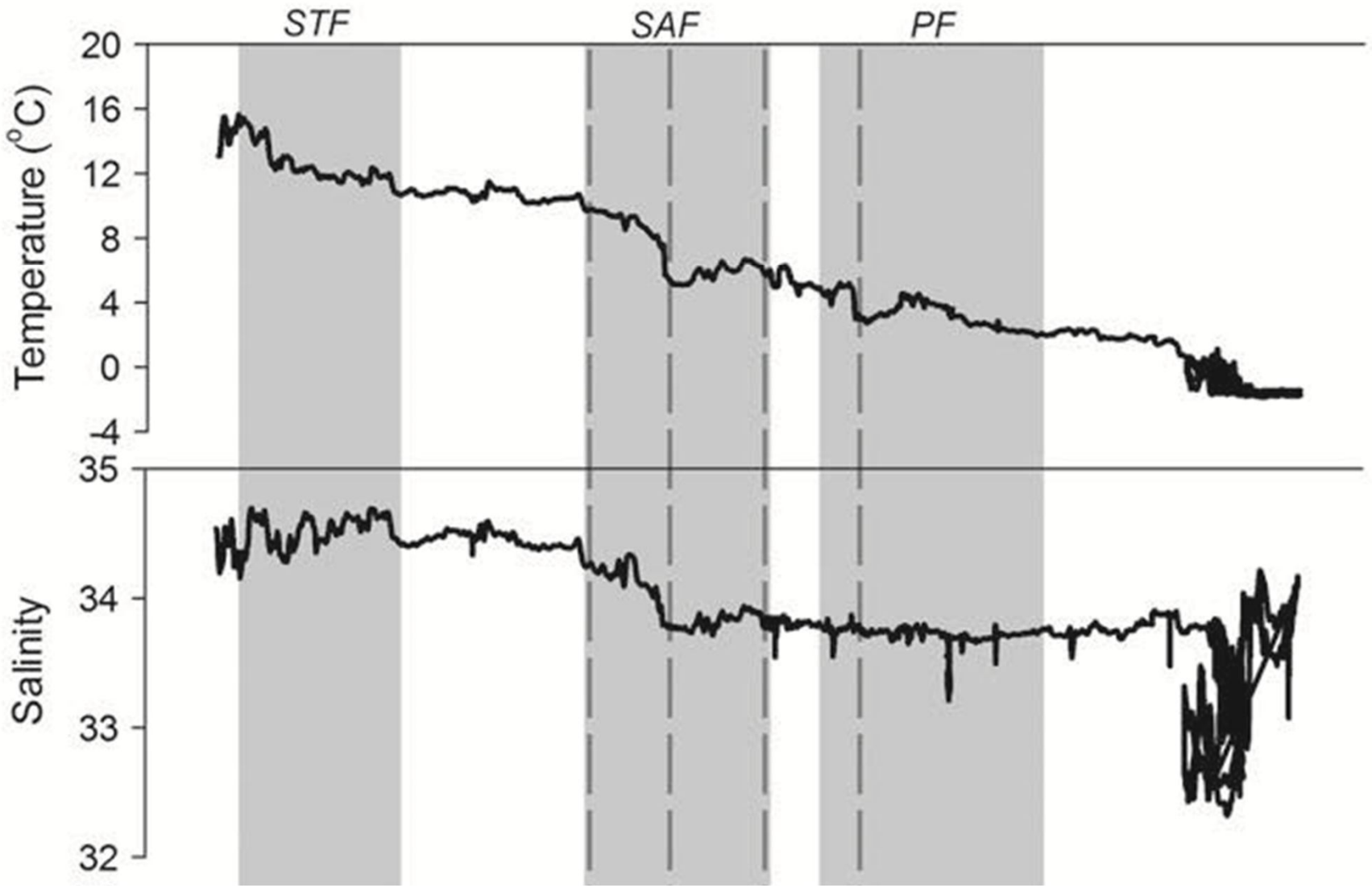
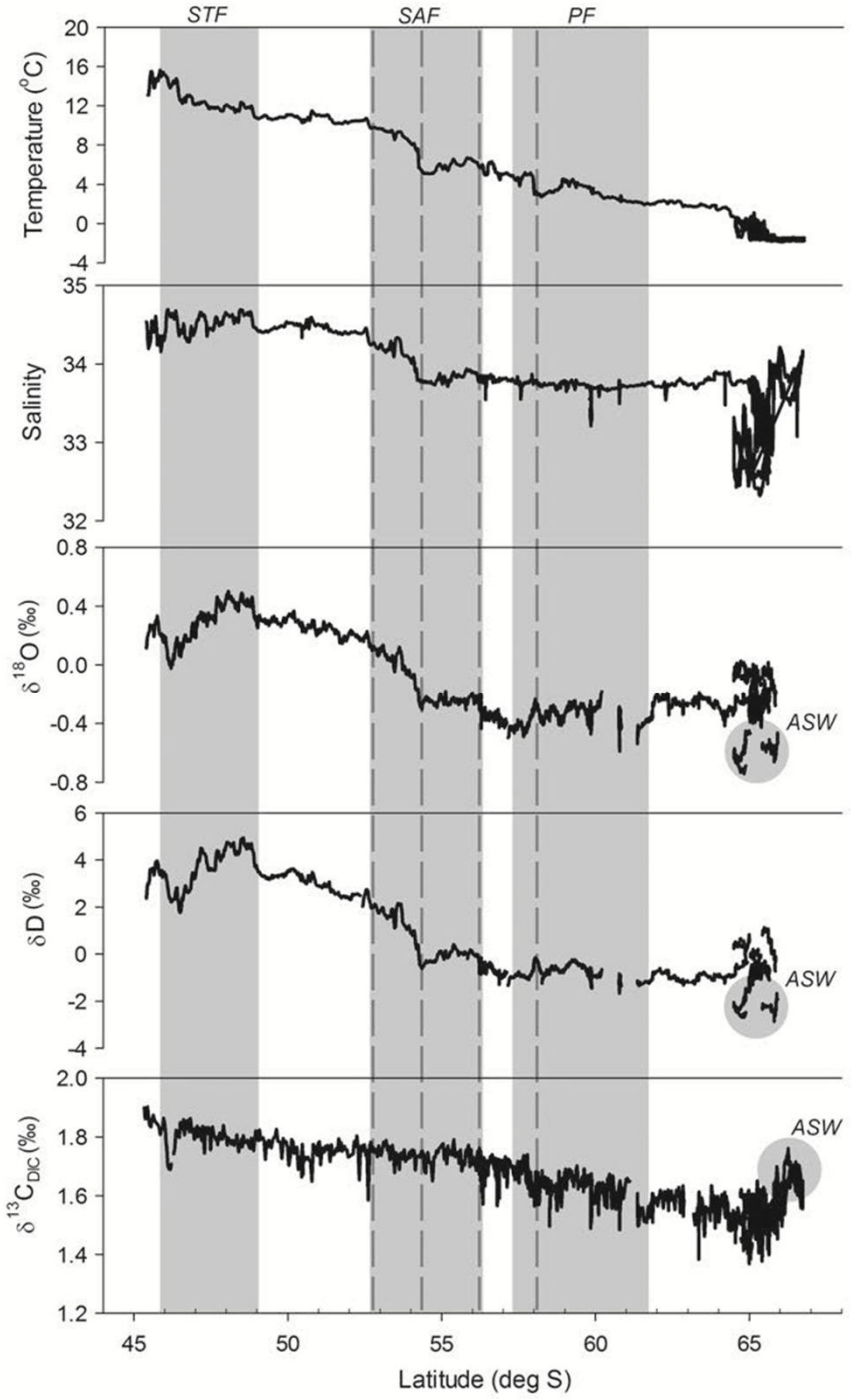
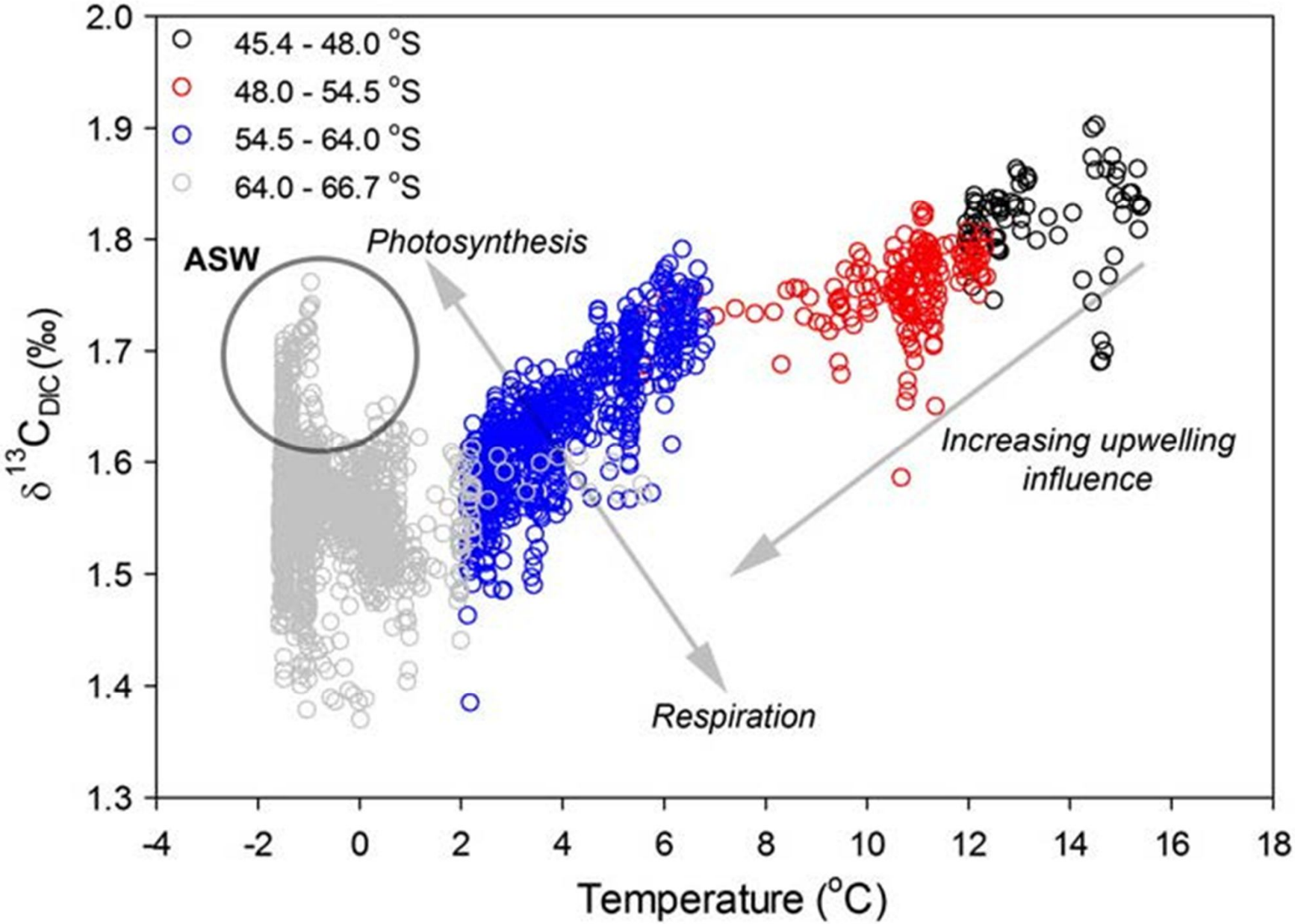

NZL
STF
SAF
PF
Sea Ice
Track
Sub-Tropical Front
Sub-Anctartic Front
Polar Front
04-Feb-20
05-Feb-2013
06-Feb-2013
07-Feb-2013
08-Feb-2013
09-Feb-2013
10-Feb-2013
1-Feb-2013
12-Feb-2013
13-Feb-2013
14-Feb-2013
15-Feb-2013
Antarctica
Latitude (°S)
45
50
55
60
65
70
Longitude(°E)
135
140
145
150
155
160
165
170
175
Depth (m)
6000
5000
4000
3000
2000
1000
0


STF
SAF
PF
Temperature (°C)
20
16
12
8
4
0
-4
Salinity
35
34
33
32

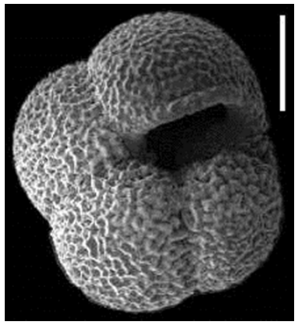


Antarctic Shelf Water

Processos que afetam $\delta^{13}C$ DIC na água



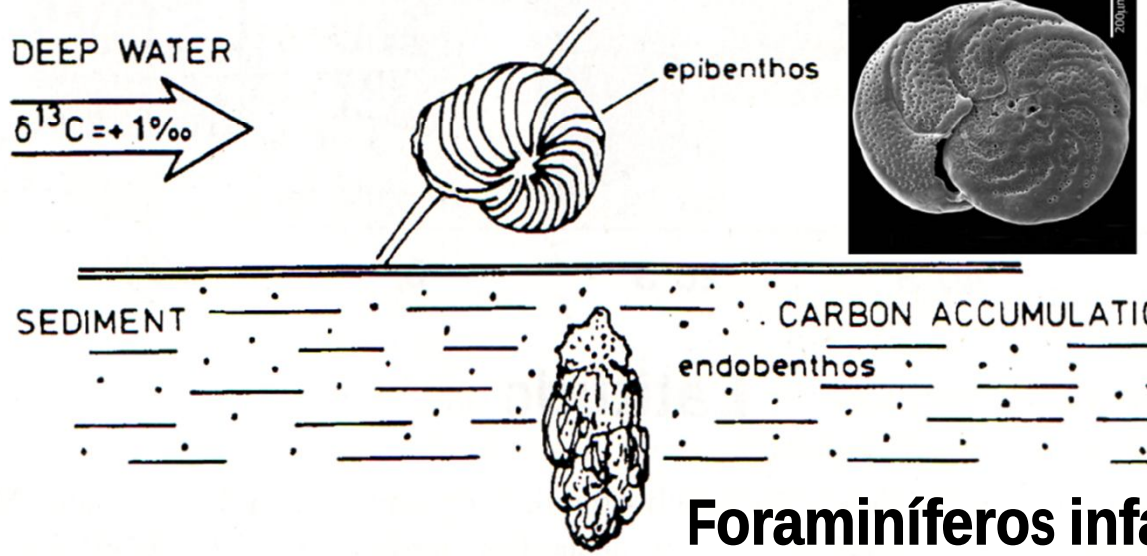
Uso do $\delta^{13}C$ de foraminíferos como marcador paleoceanográfico (massas de água e nutrientes)



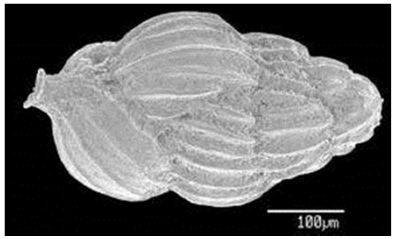
Foraminíferos planctónicos

testas:

$CaCO_3$



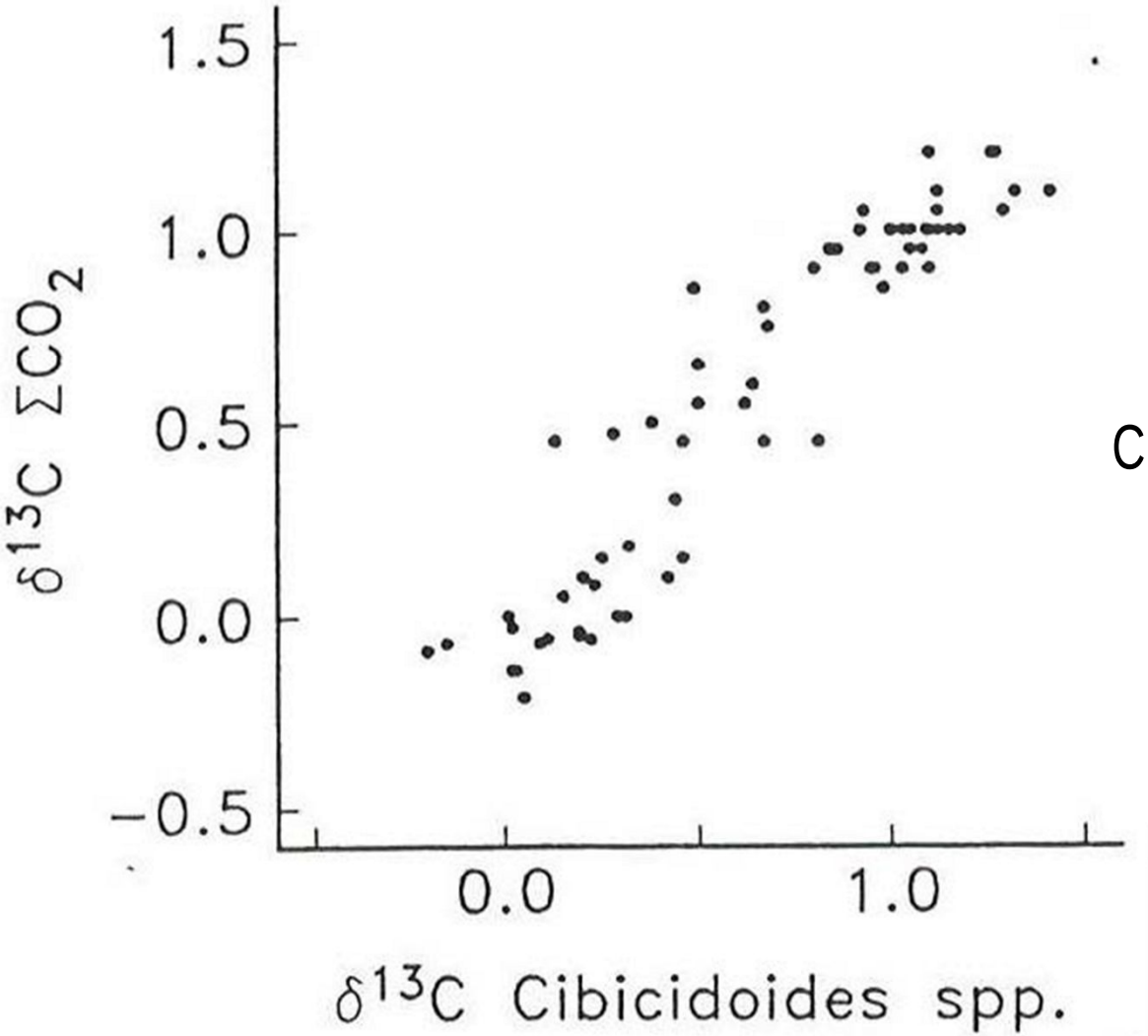
Foraminíferos bentônicos epifaunais



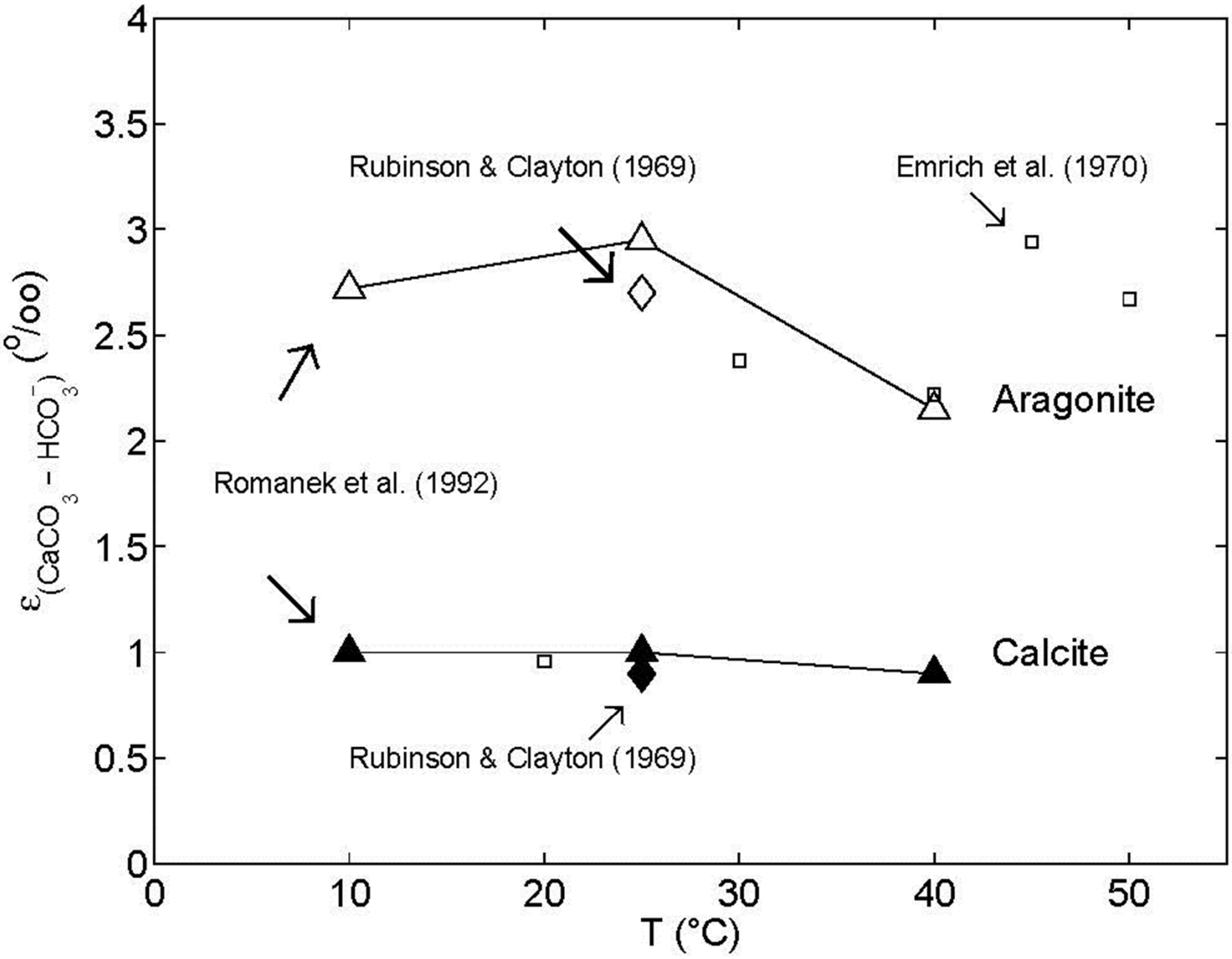
Foraminíferos infaunais

G. Fisher & G. Wefer eds. (1999)

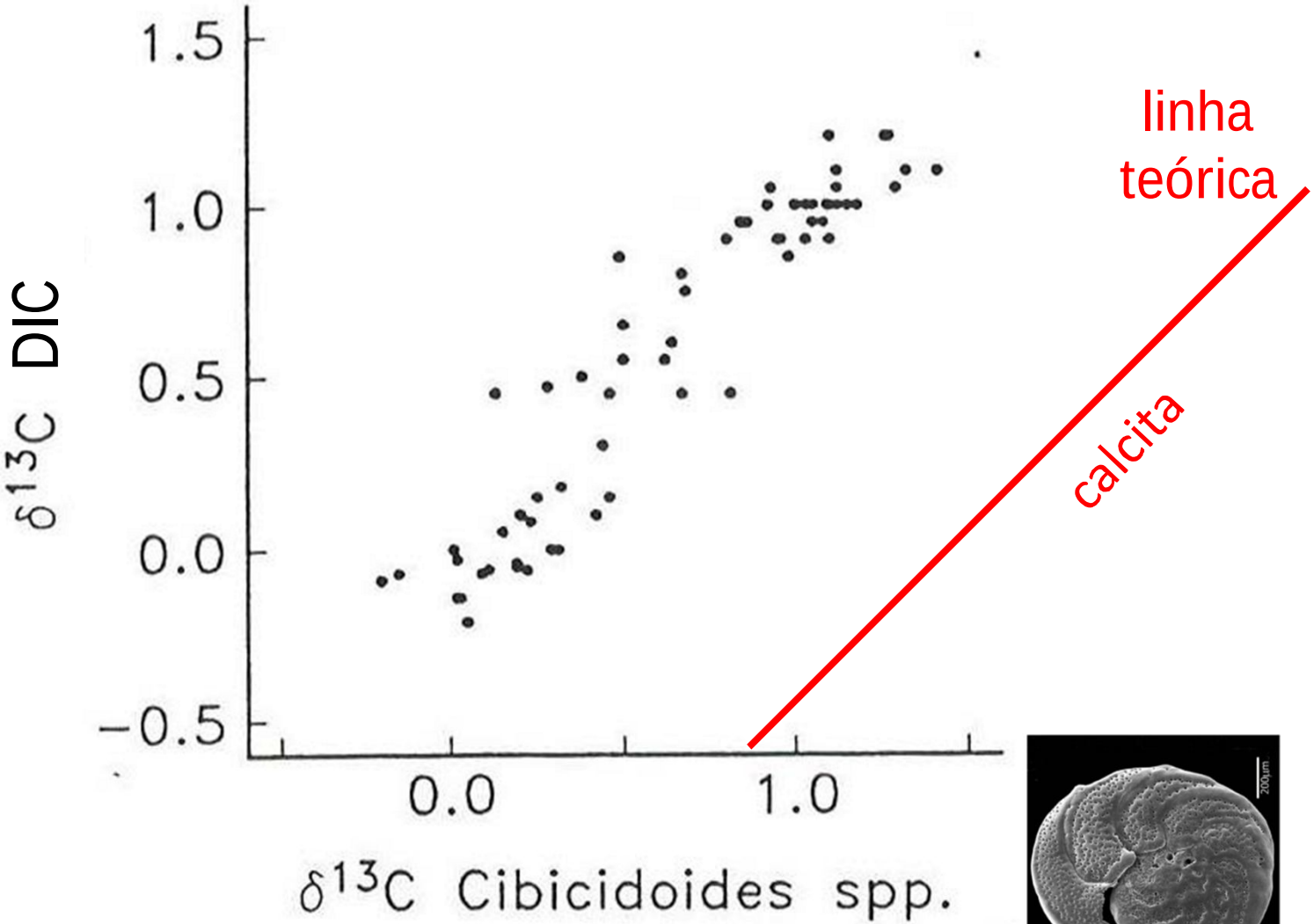
Correlação entre $\delta^{13}C$ do DIC e $\delta^{13}C$ do $CaCO_3$ de foraminíferos bentônicos atuais



Curry et al. (1988)

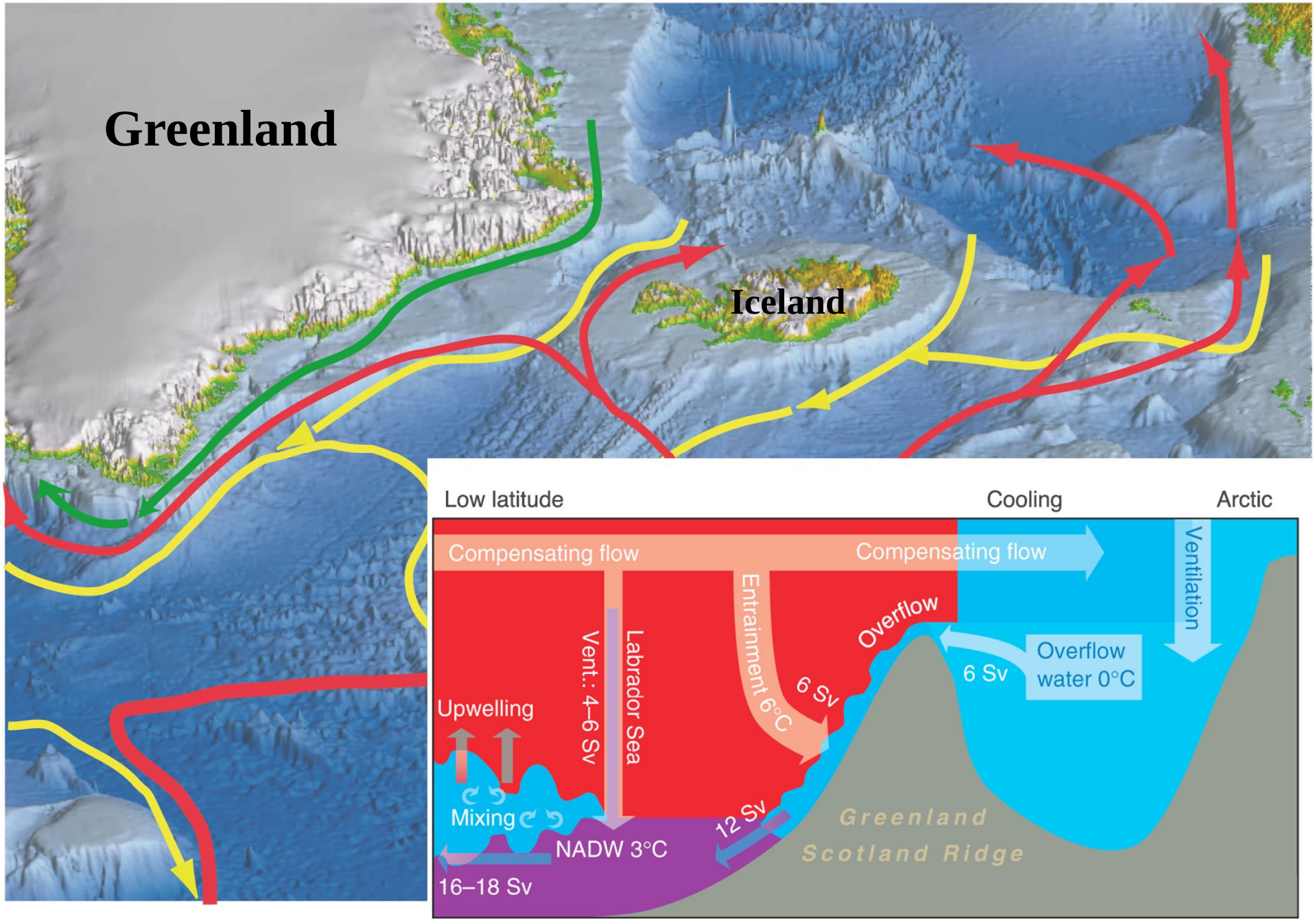

Rubinson & Clayton (1969)
Emrich et al. (1970)
Aragonite
Romanek et al. (1992)
Calcite
Rubinson & Clayton (1969)
$\varepsilon_{(CaCO_3 - HCO_3^-)}$ (°/oo)
T (°C)
0
0.5
1
1.5
2
2.5
3
3.5
4
0
10
20
30
40
50

Correlação entre $\delta^{13}C$ do DIC e $\delta^{13}C$ do $CaCO_3$ de foraminíferos bentônicos atuais

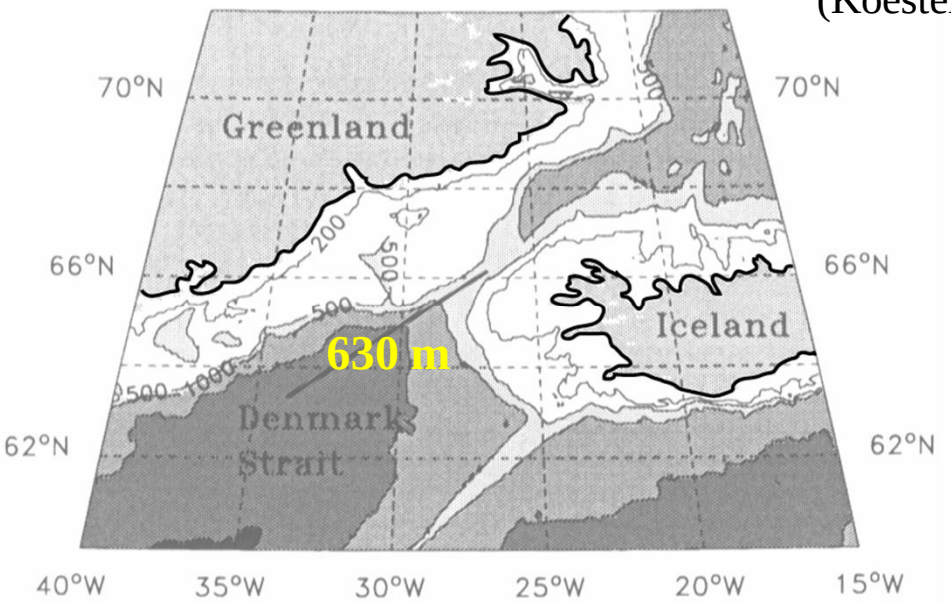


Curry et al. (1988)

Exemplo de aplicação em paleoceanográfia:
formação de NADW durante o LGM

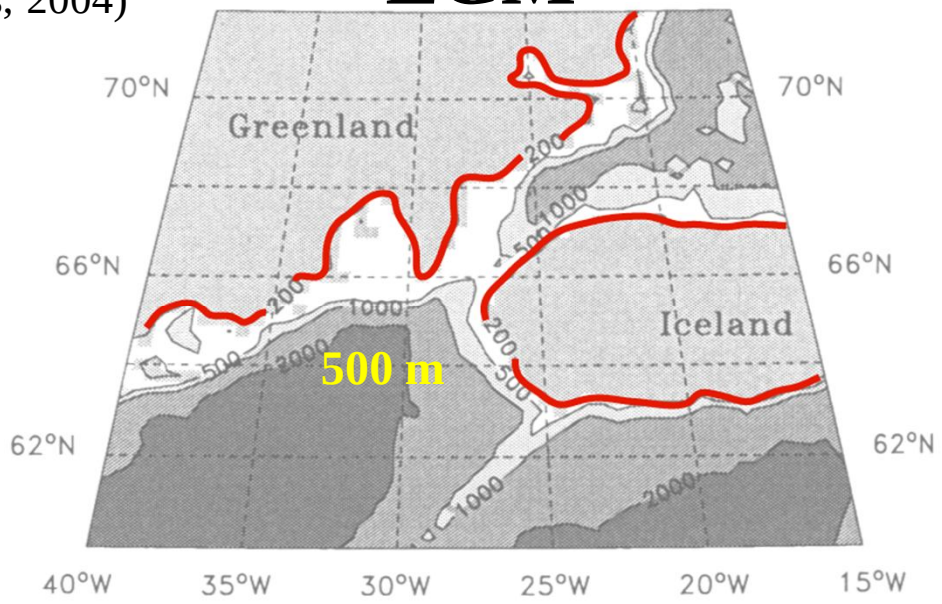

Greenland
Iceland
Low latitude
Cooling
Arctic
Compensating flow
Compensating flow
Ventilation
Overflow
Entrainment 6°C
6 Sv
Overflow water 0°C
6 Sv
Upwelling
Vent.: 4–6 Sv
Labrador Sea
Mixing
12 Sv
NADW 3°C
16–18 Sv
Greenland Scotland Ridge

Atual



(Koesters, 2004)

LGM

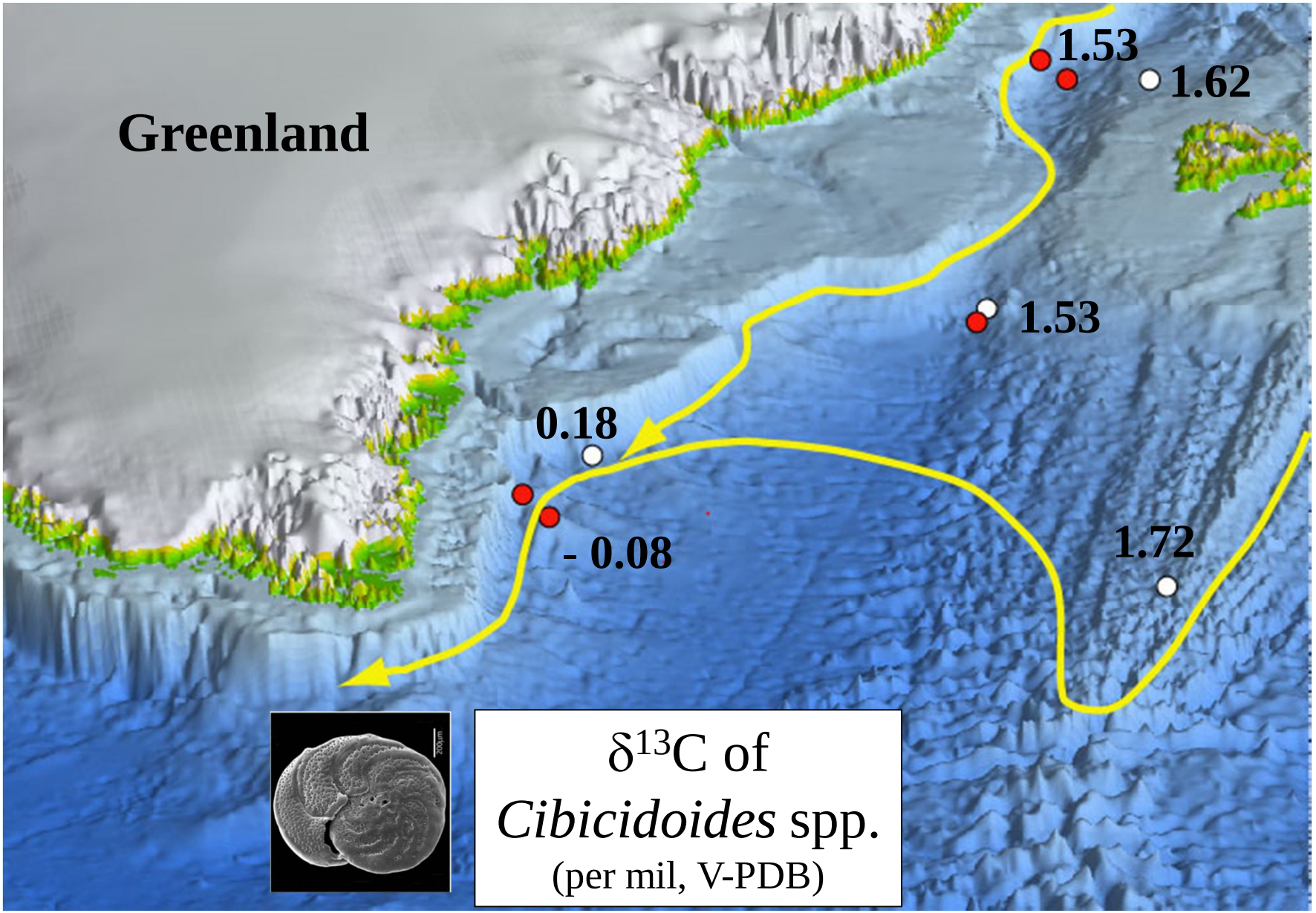


Durante o LGM:

- DK Strait **estava restrito** (banquisas de Islandia e Groenlandia)
- Nivel do mar **~ 130 m** mais baixo (eustasia + glacio-isostasia)
- Redução do gradiente de densidade N-S (pulsos de agua doce)

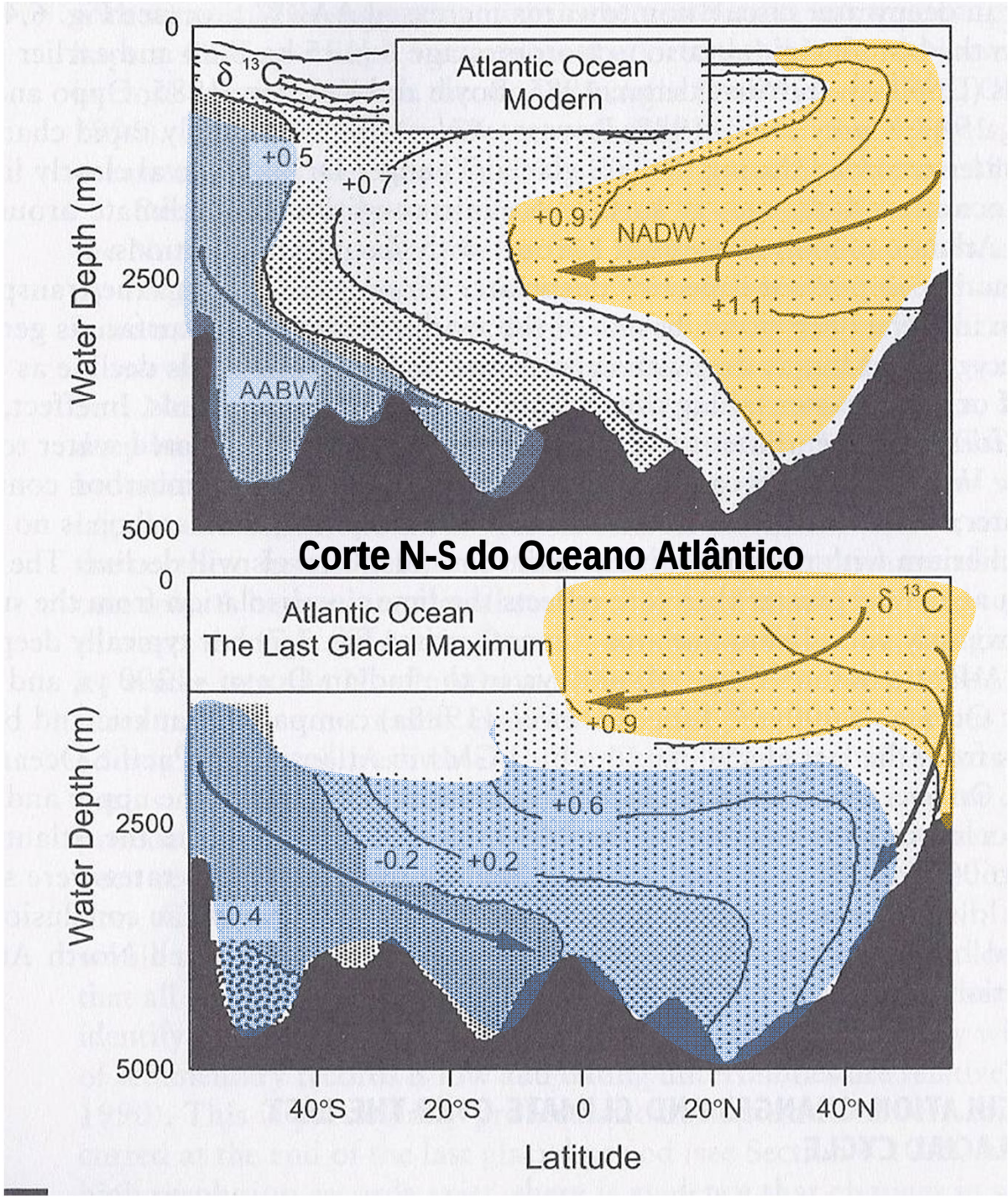
=> Menor produção de água profunda N-Atlântica?

Formação d'agua profunda N-Atlântica no LGM



HOJE

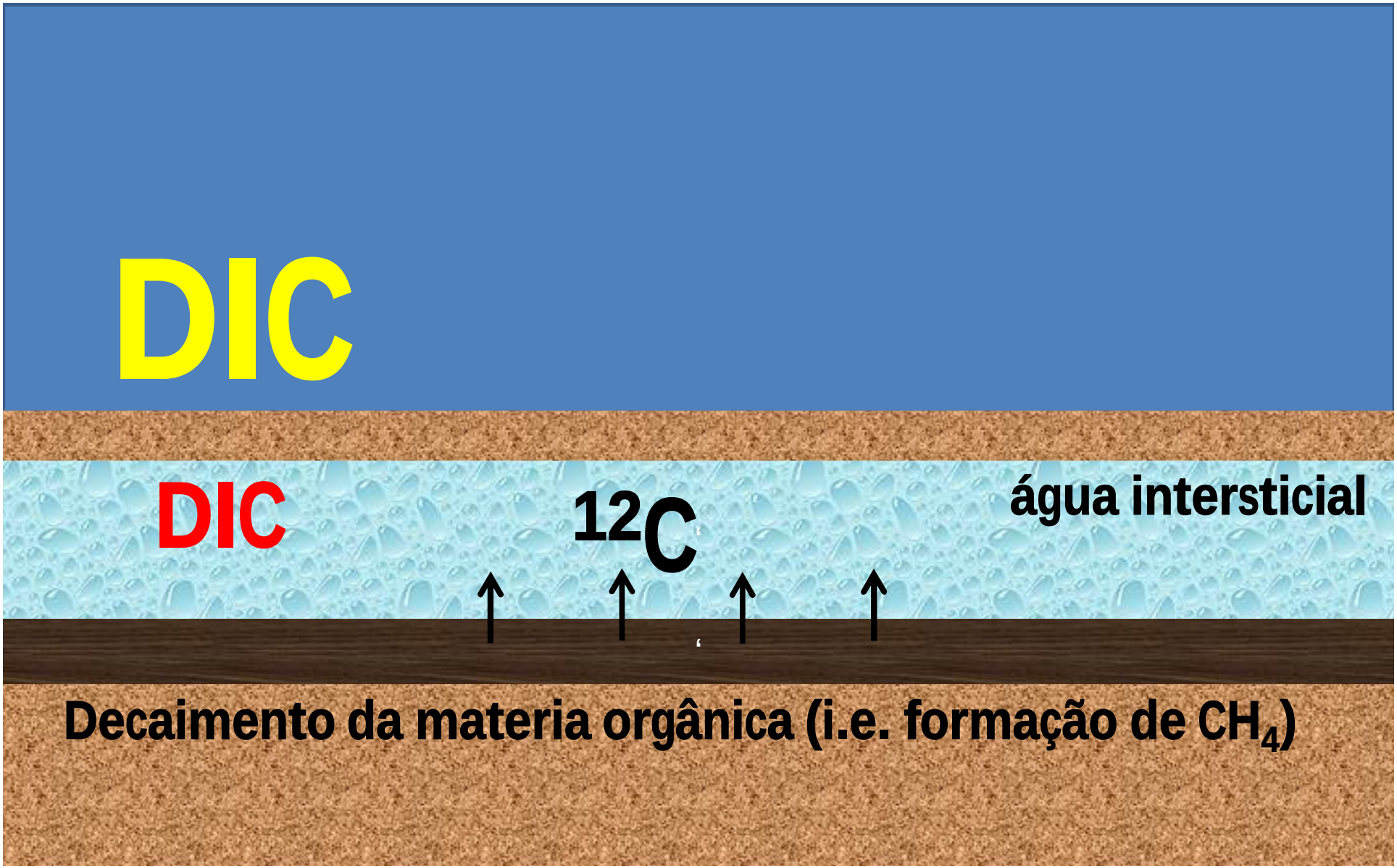
LGM



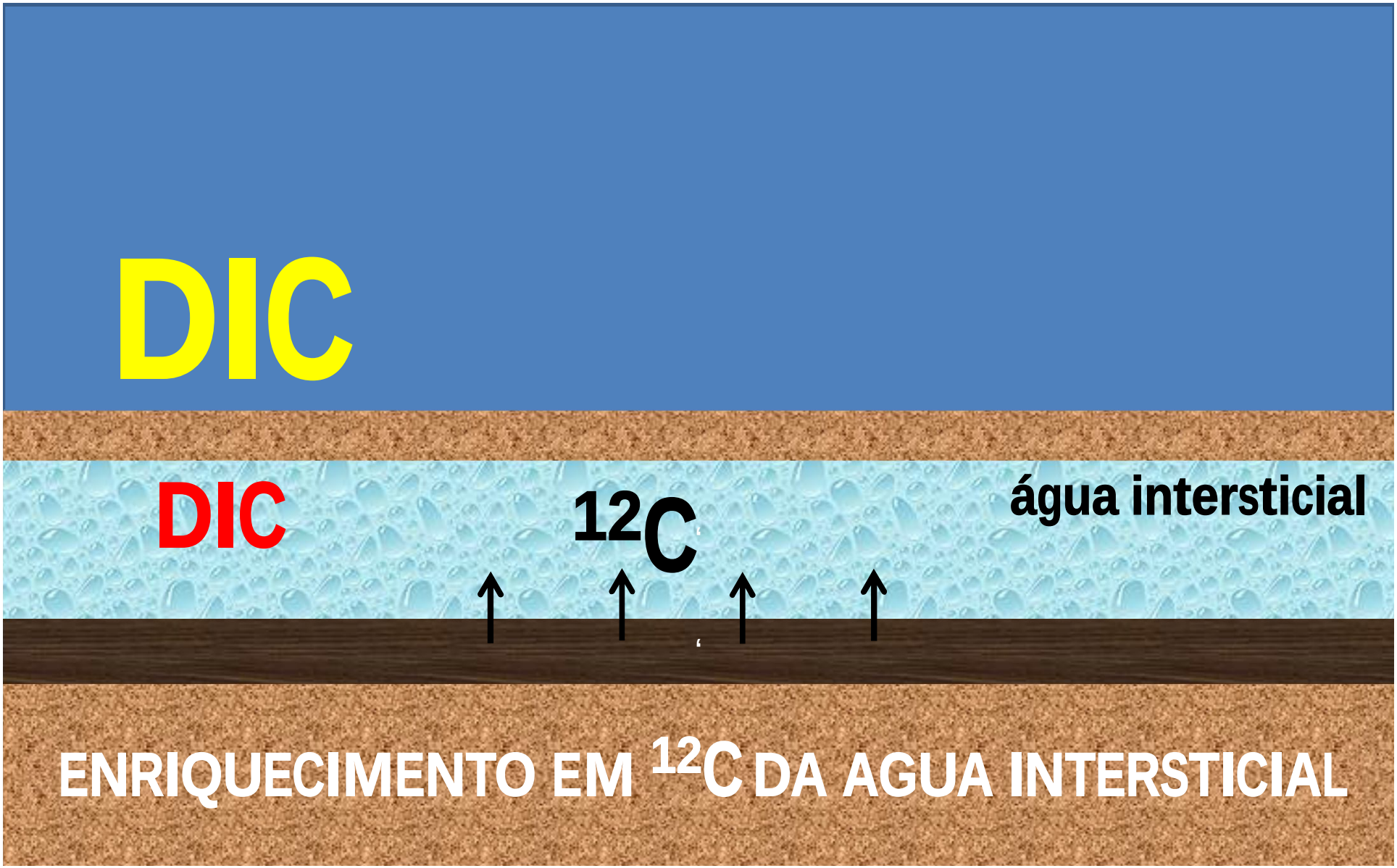
Corte N-S do Oceano Atlântico

Efeito da degradação da materia orgânica na água intersticial

Efeito do decaimento da materia orgânica na água intersticial



Efeito do decaimento da materia orgânica na água intersticial



Journal of Oceanography, Vol. 64, pp. 303 to 310, 2008

Dissolved Inorganic Carbon (DIC) and Its Carbon Isotopic Composition in Sediment [redacted] from the Shenhu Area, Northern South China Sea

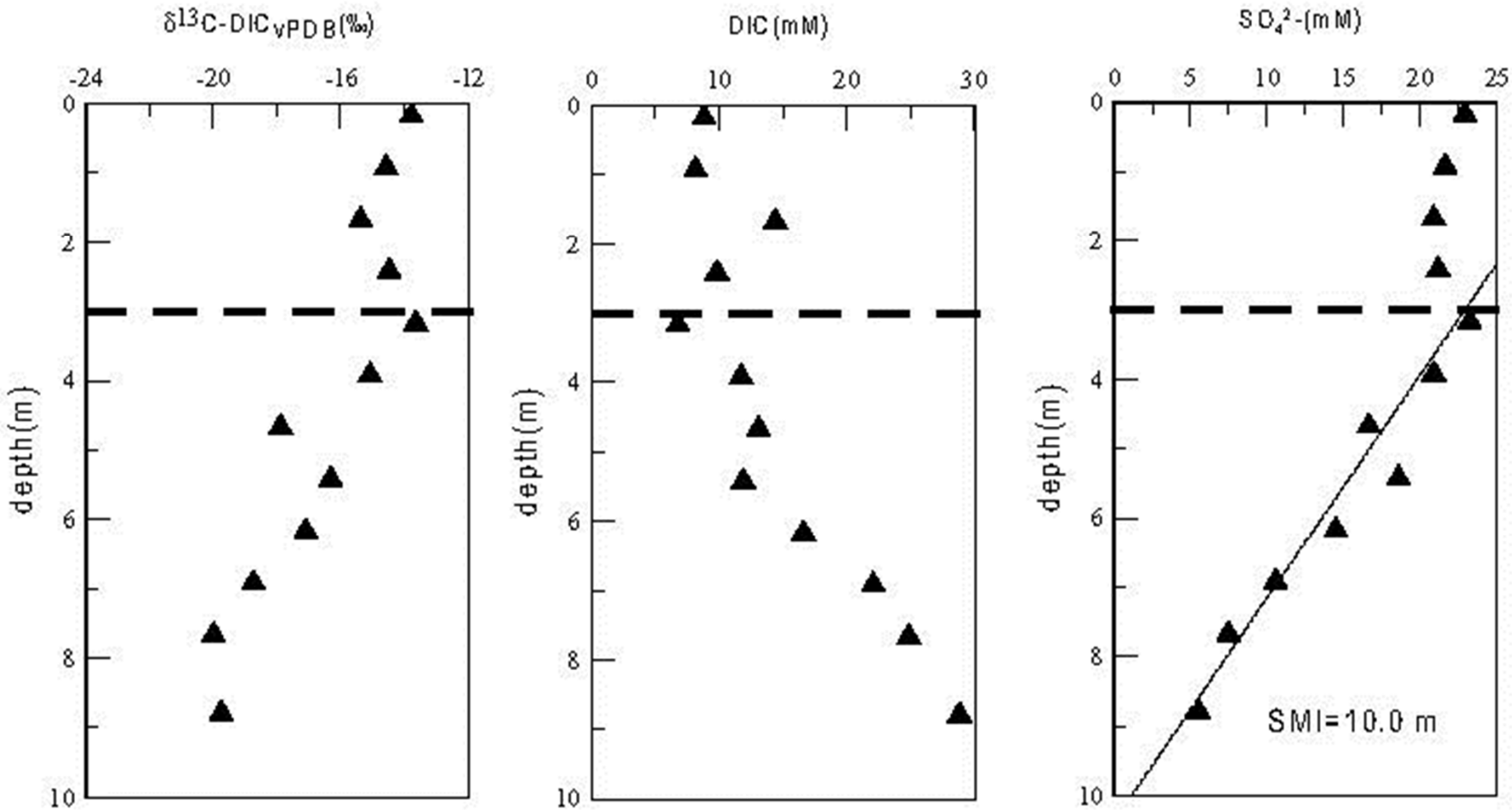
TAO YANG[1,2], SHAO-YONG JIANG[1,2]*, JING-HONG YANG[1,2], GE LU[1,2], NENG-YOU WU[3], JIAN LIU[3] and DAO-HUA CHEN[3]

[1]*State Key Laboratory for Mineral Deposits Research, Department of Earth Sciences, Nanjing University, Nanjing 210093, China*
[2]*Center for Marine Geochemistry Research, Nanjing University, Nanjing 210093, China*
[3]*Guangzhou Marine Geological Survey, China Geological Survey, Guangzhou 510760, China*

(Received 27 April 2007; in revised form 16 September 2007; accepted 20 October 2007)

The Shenhu area is one of the most favorable places for the occurrence of gas hydrates in the northern continental slope of the South China Sea. Pore water samples were collected in two piston cores (SH-A and SH-B) from this area, and the concentrations of sulfate and dissolved inorganic carbon (DIC) and its carbon isotopic composition were measured. The data revealed large DIC variations and very negative δ^{13}C-DIC values. Two reaction zones, 0–3 mbsf and below 3 mbsf, are identified in the sediment system. At site SH-A, the upper zone (0–3 mbsf) shows relatively constant sulfate and DIC concentrations and δ^{13}C-DIC values, possibly due to bioturbation and fluid advection. The lower zone (below 3 mbsf) displays good linear gradients for sulfate and DIC concentrations, and δ^{13}C-DIC values. At site SH-B, both zones show linear gradients, but the decreasing gradients for δ^{13}C-DIC and SO_4^{2-} in the lower zone below 3 mbsf are greater than those from the upper zone, 0–3 mbsf. The calculated sulfate-methane interface (SMI) depths of the two cores are 10.0 m and 11.1 m, respectively. The depth profiles of both DIC and δ^{13}C-DIC showed similar characteristics as those in other gas hydrate locations in the world oceans, such as the Blake Ridge. Overall, our results indicate an anaerobic methane oxidation (AMO) process in the sediments with large methane flux from depth in the studied area, which might be linked to the formation of gas hydrates in this area.

Keywords:
· Dissolved inorganic carbon (DIC),
· δ^{13}C-DIC,
· sulfate gradient,
· gas hydrate,
· South China Sea.


δ13C-DICvPDB(‰)
-24
-20
-16
-12
0
2
4
6
8
10
depth(m)
DIC(mM)
0
10
20
30
0
2
4
6
8
10
depth(m)
SO4 2-(mM)
0
5
10
15
20
25
0
2
4
6
8
10
depth(m)
SMI=10.0 m

[1]*State Key Laboratory for Mineral Deposits Research, Department of Earth Sciences, Nanjing University, Nanjing 210093, China*
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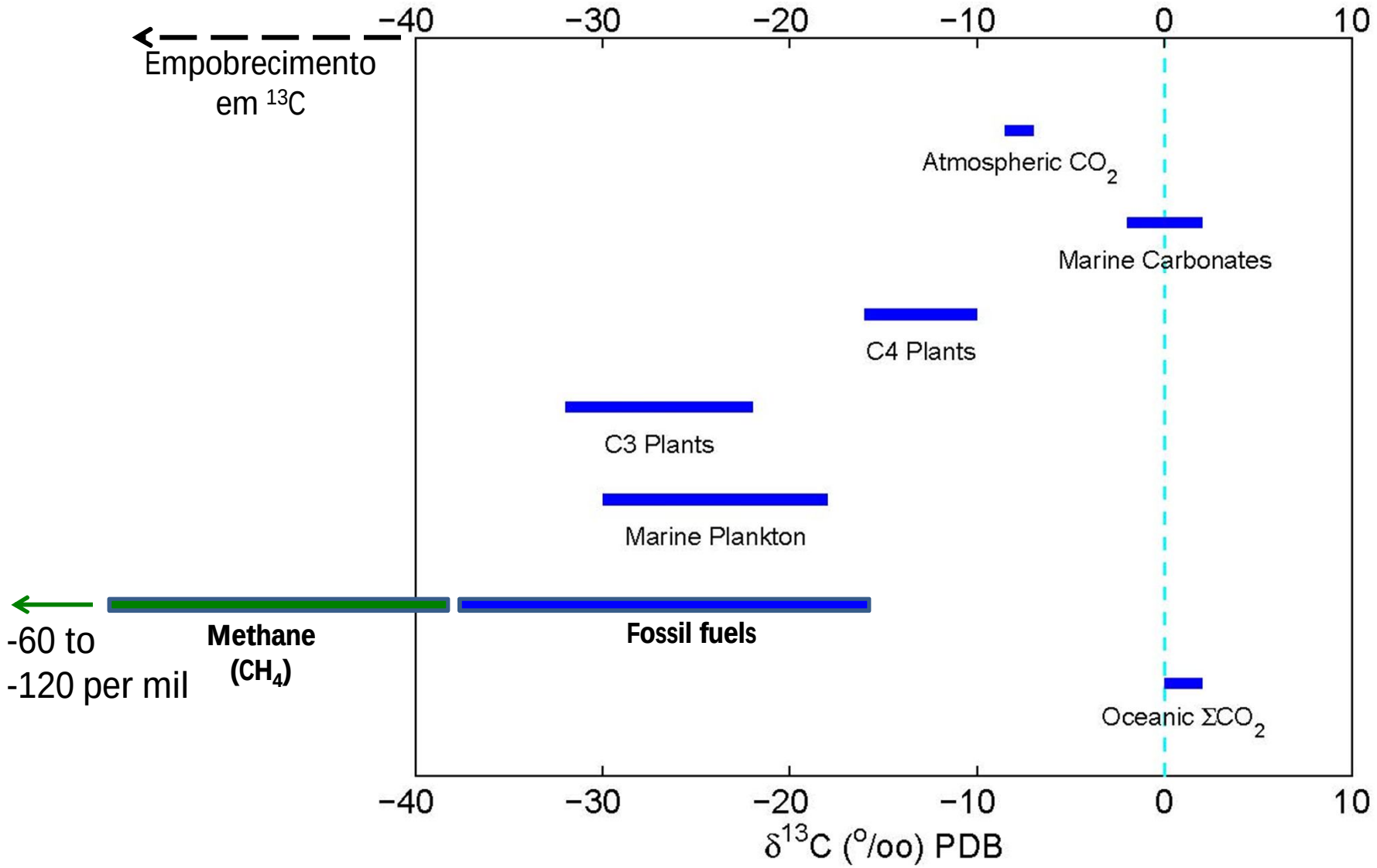
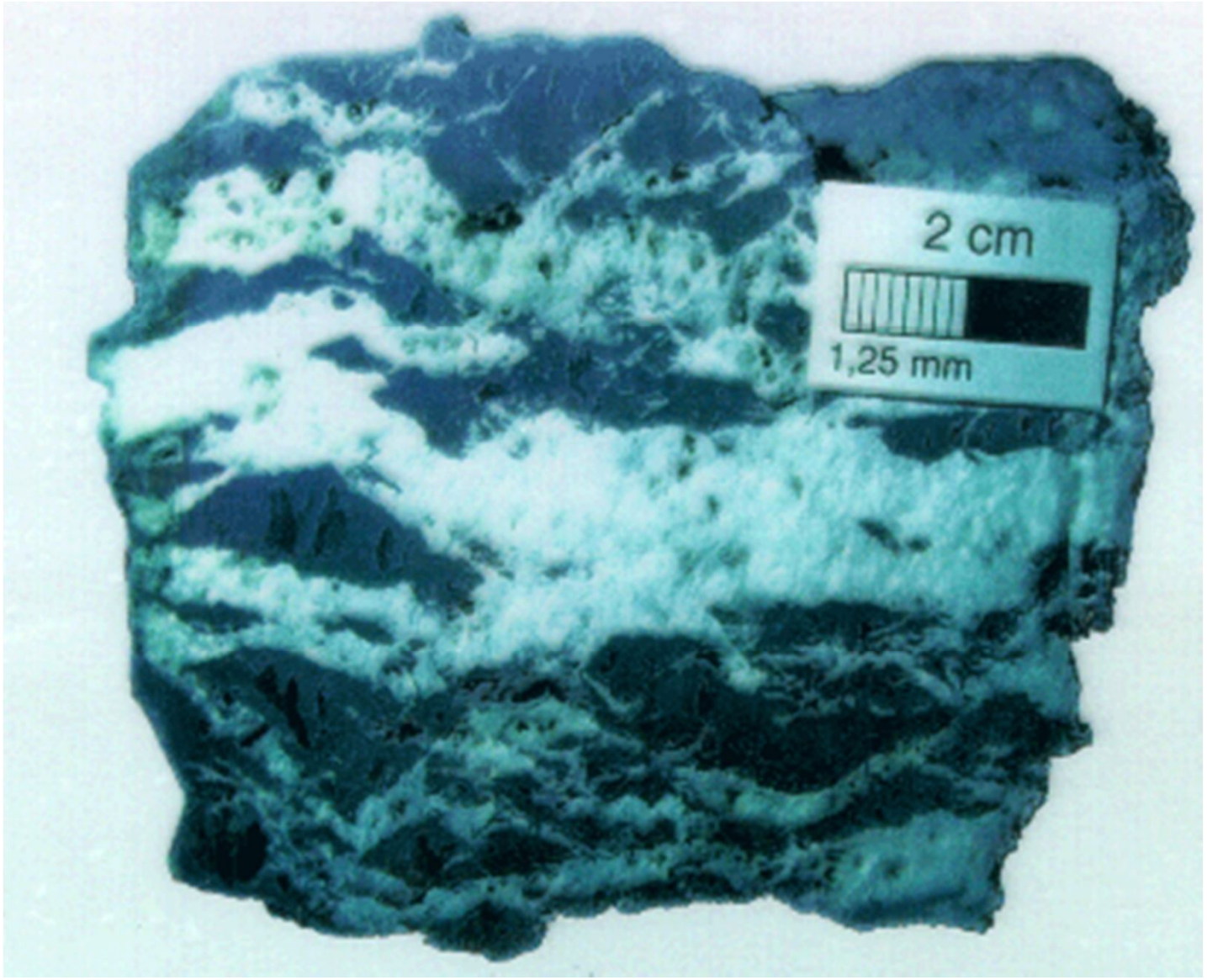
Valores de $\delta^{13}C$ em substancias naturais



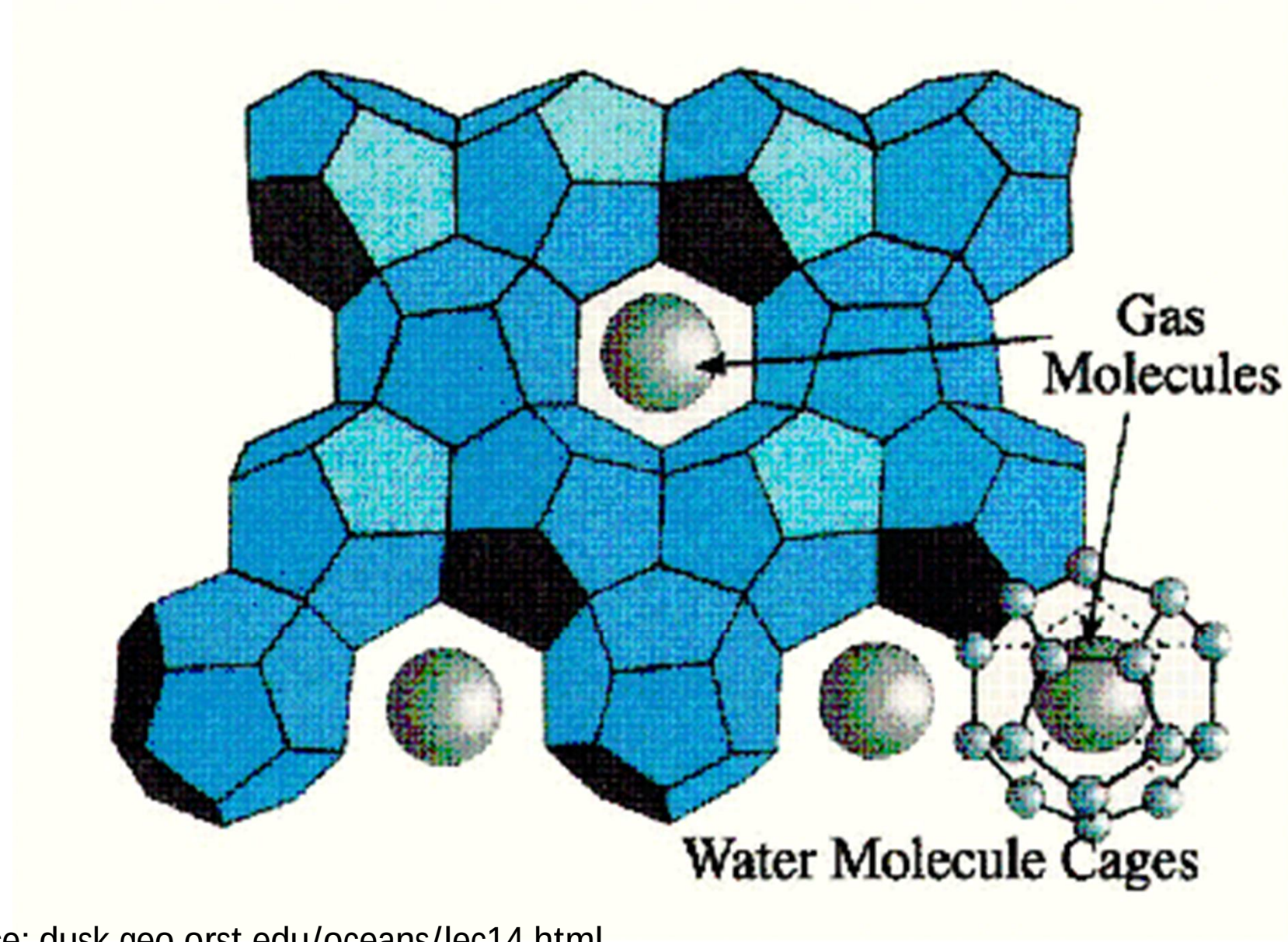
Zeebe & Wolf-Gladrow (2001)

Hidrato de metano (ou Clatrato)



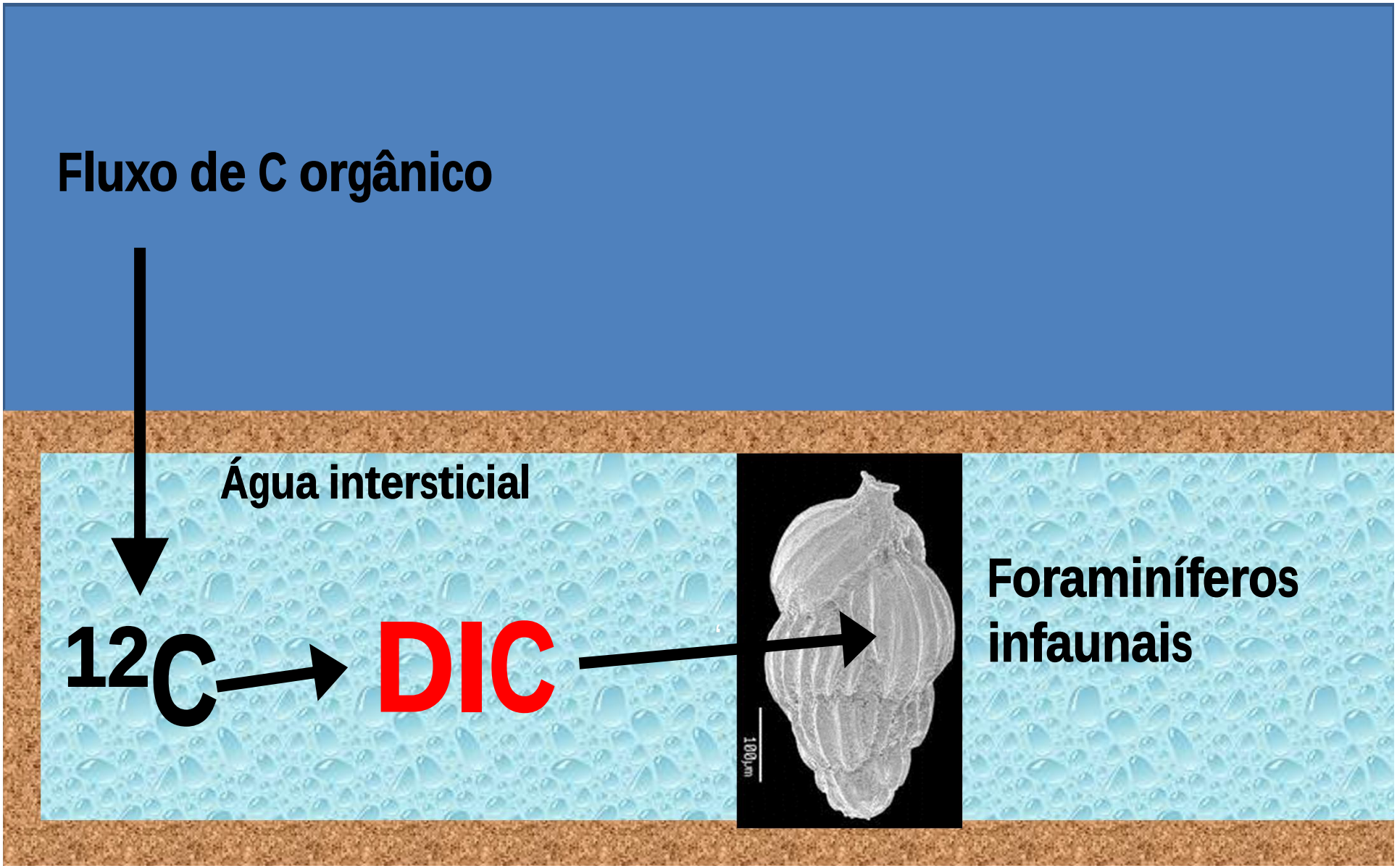
Source: dusk.geo.orst.edu/ oceans/lec14.html

Structure of Gas Hydrates:

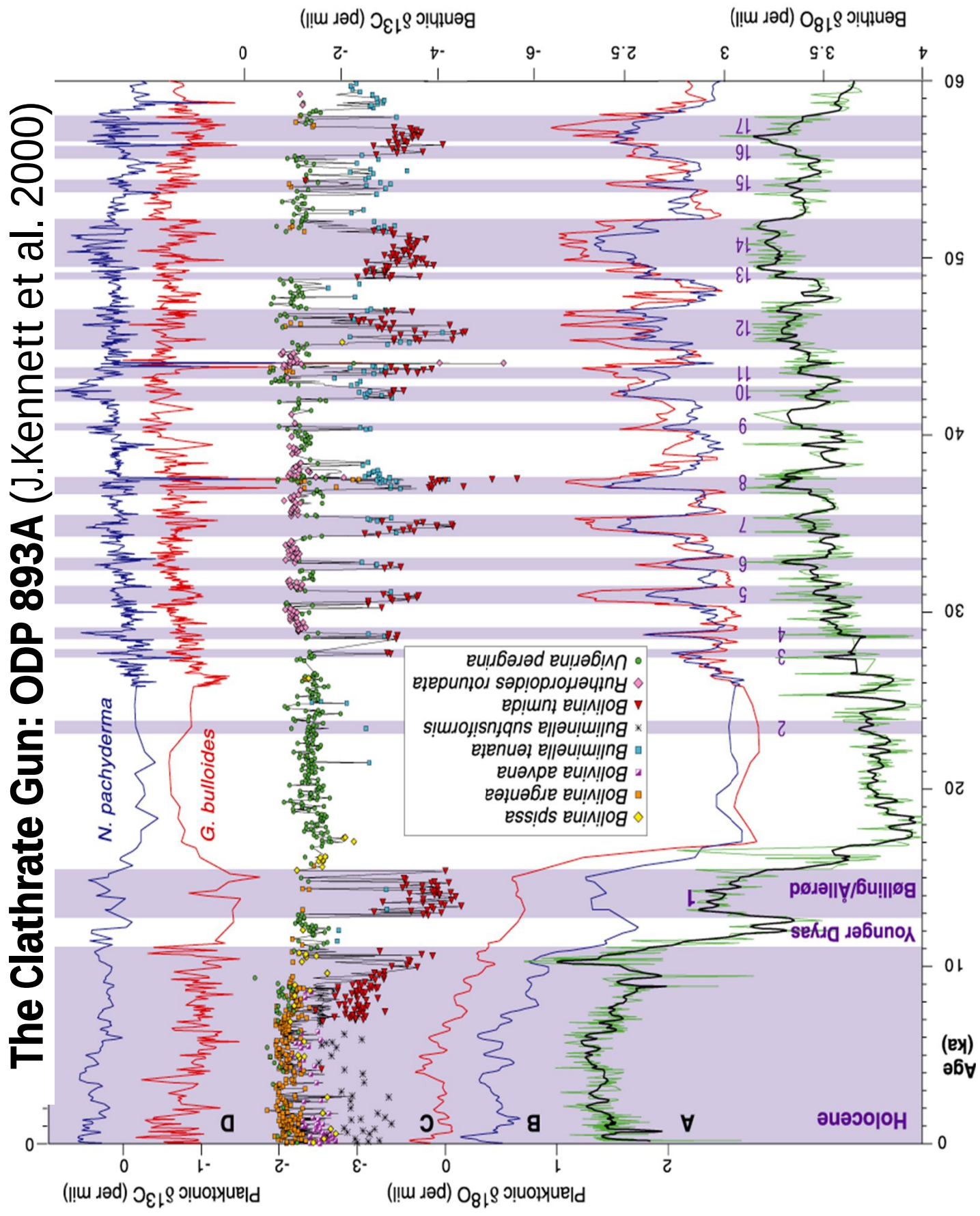


Source: dusk.geo.orst.edu/oceans/lec14.html

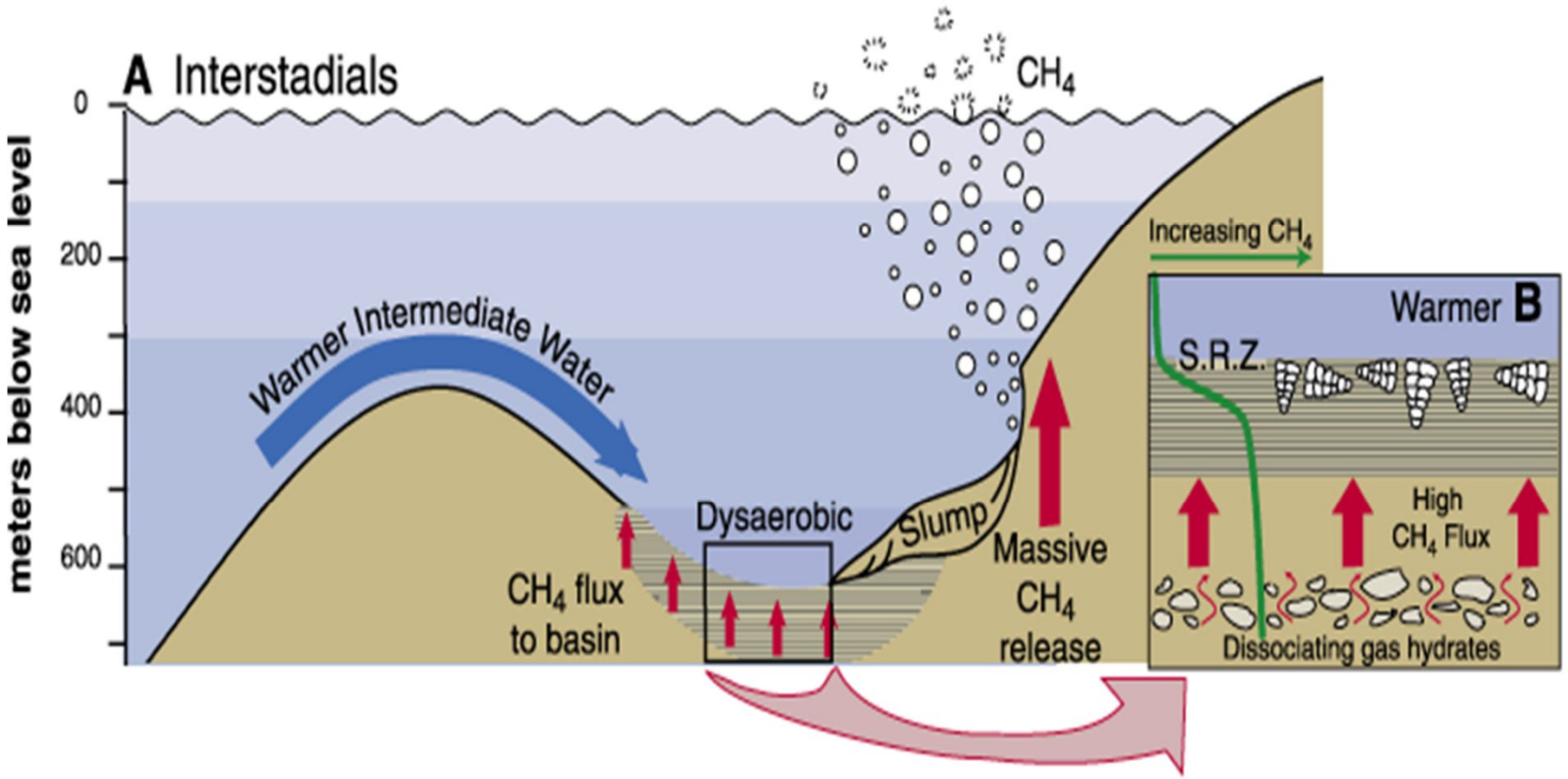
Paleo-taxa de acumulação de carbono orgânico



Source: J.Kennett et al. Science Vol. 288 (2000) 128-133



Clathrate Gun Hypothesis



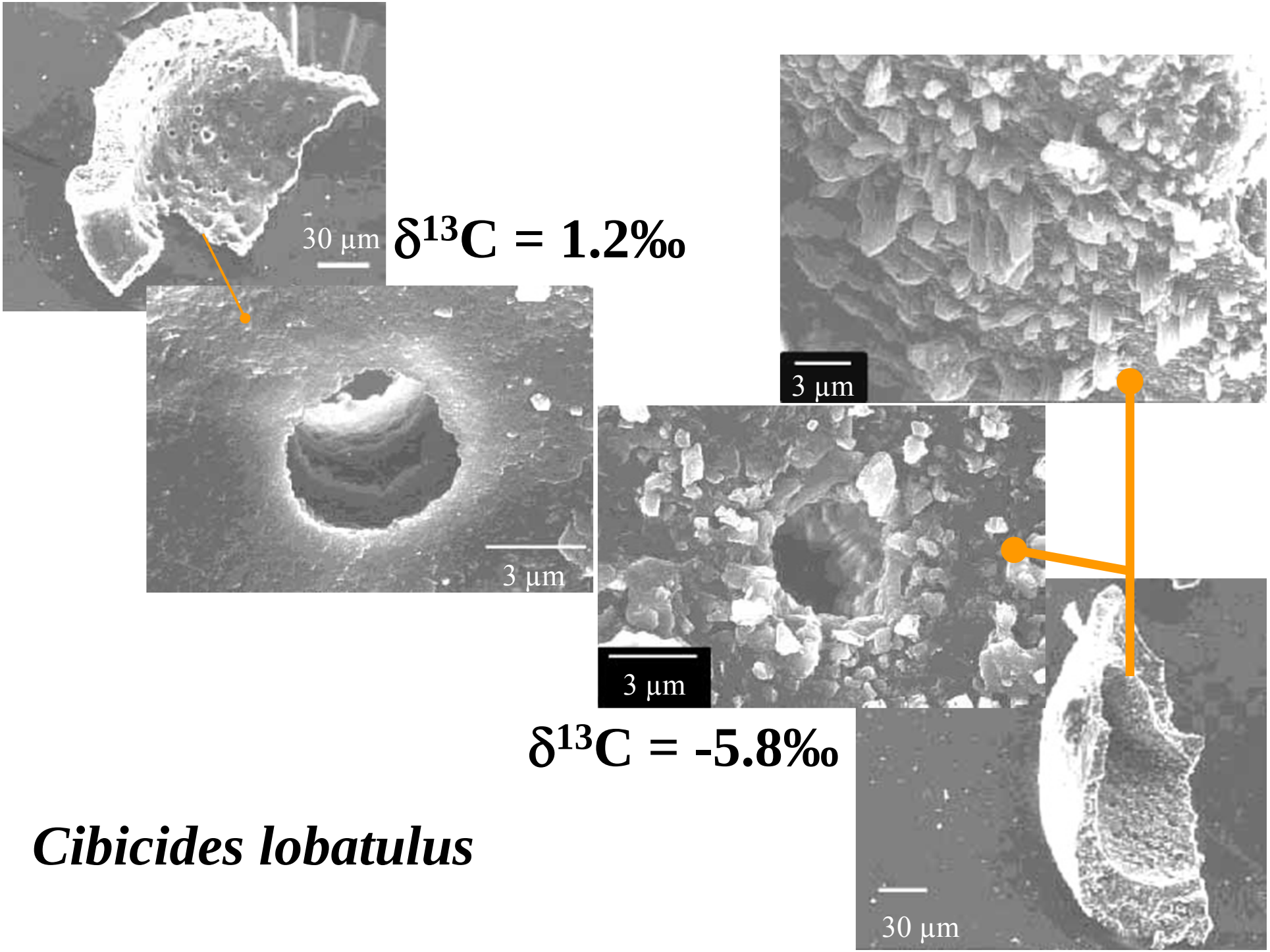
Kennett et al., Science, Vol. 288 (2000), p.128-133

Complicações que afetam as reconstituições baseadas no $\delta^{13}C$ de foraminíferos

- Sinal sazonal
- Migração de foraminíferos na coluna de água
- Diagênese (contaminação por $CaCO_3$ secundário)
- Efeitos vitais

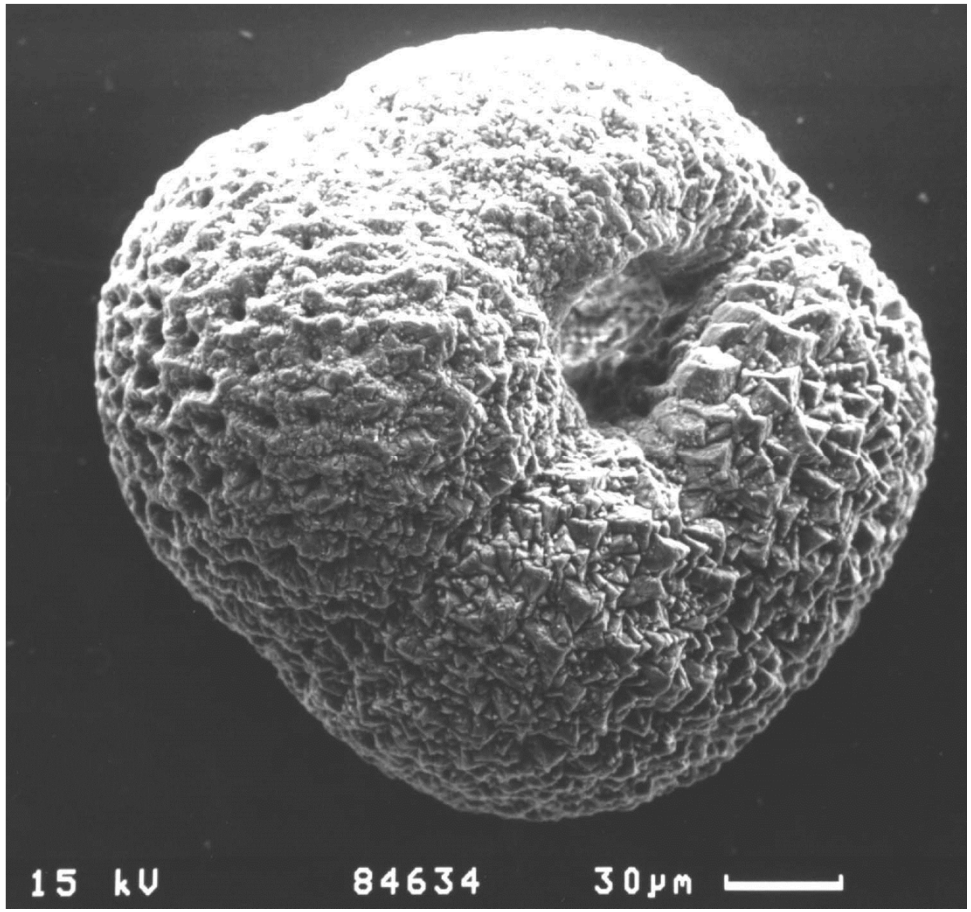
Diagênese

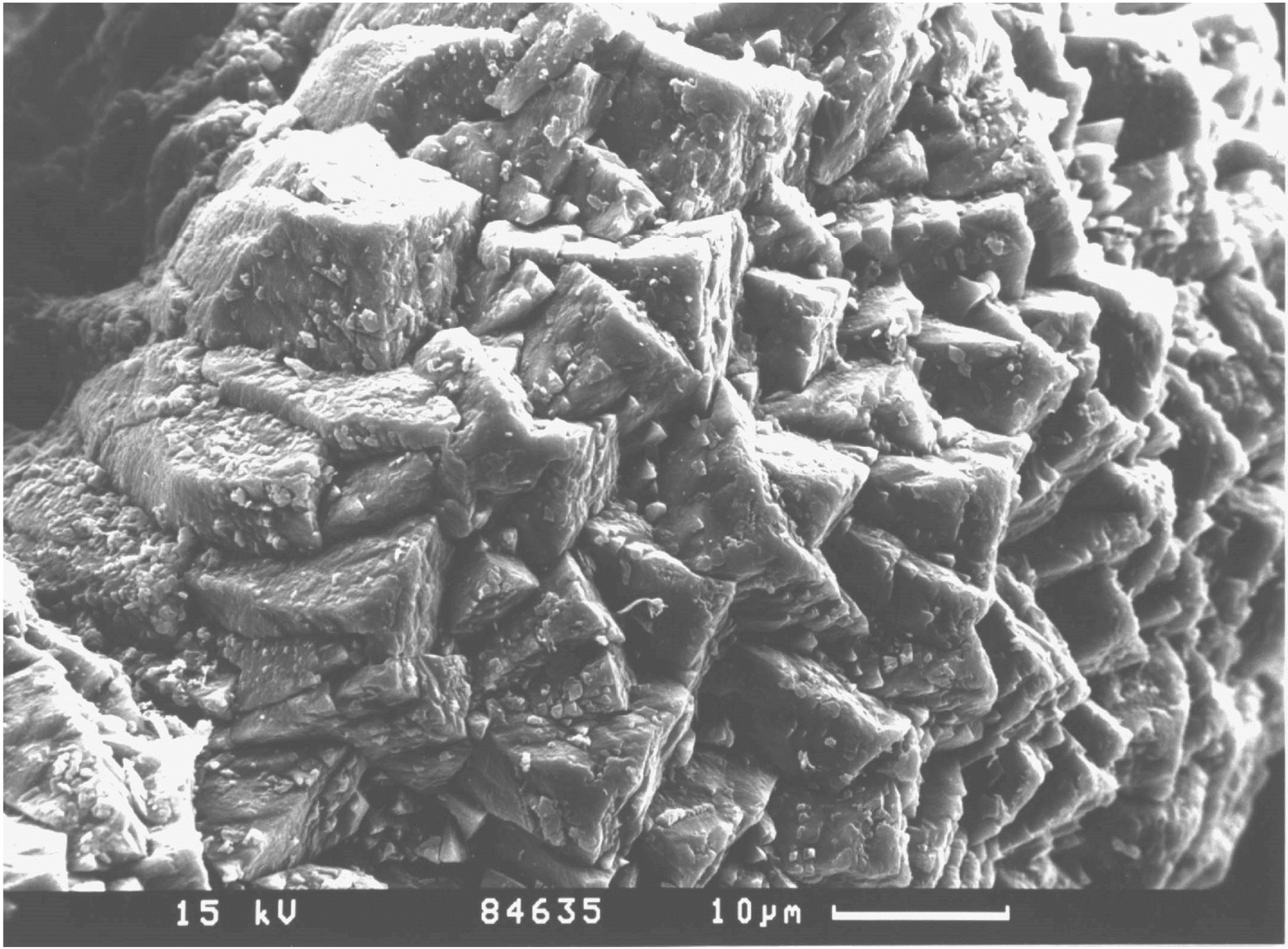
(contaminação por $CaCO_3$ secundário)

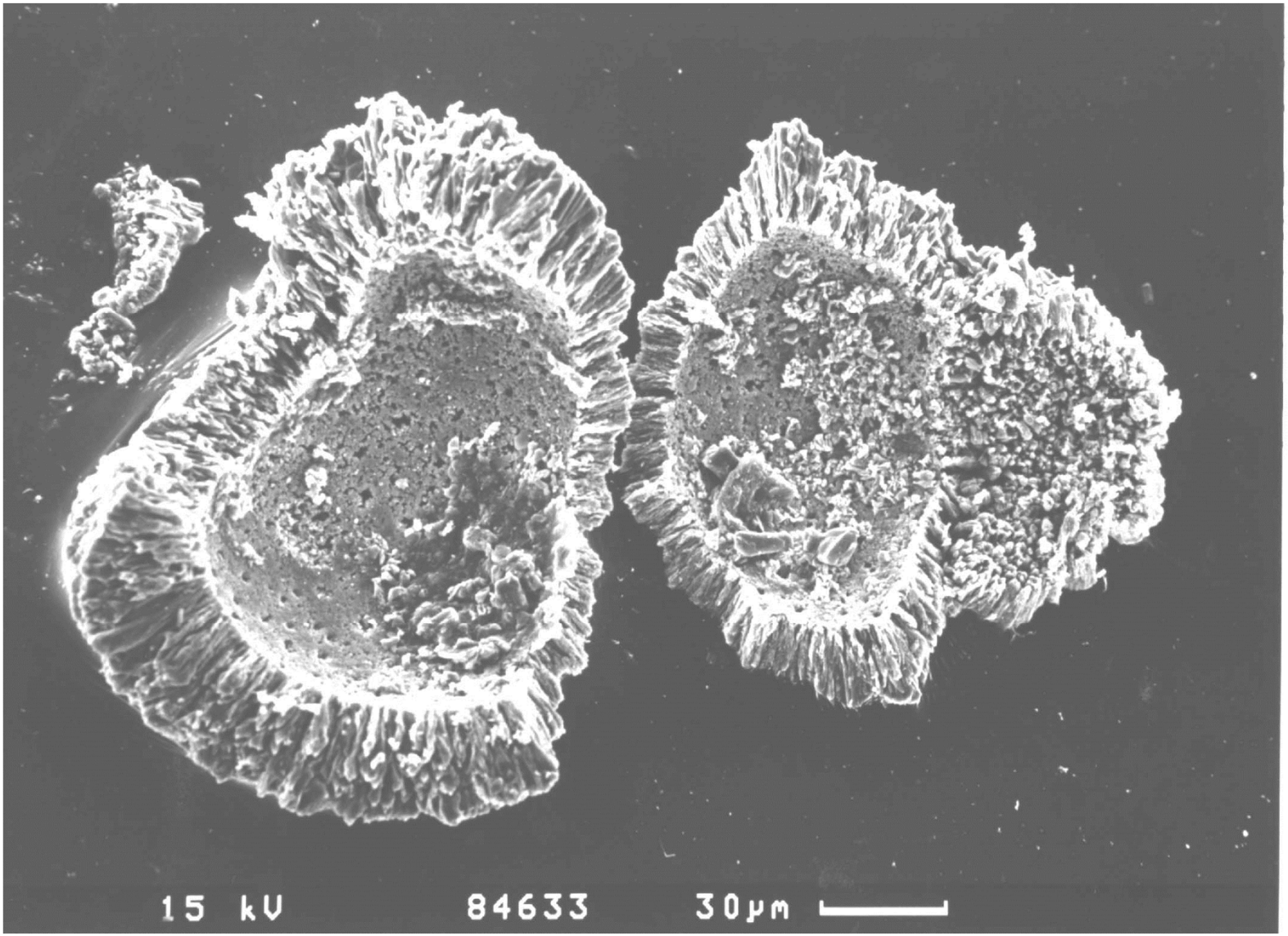


Cibicides lobatulus

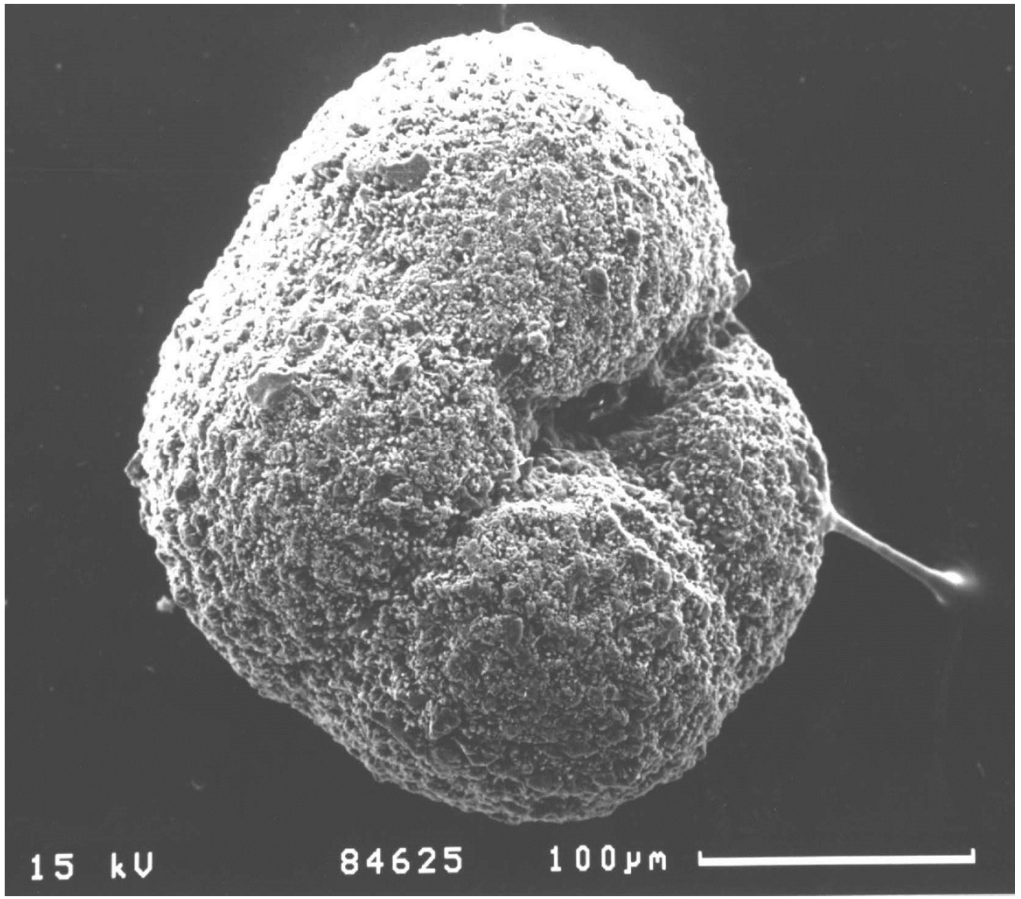
N. pachyderma sinistral
$\delta^{13}C$ = 0 per mil

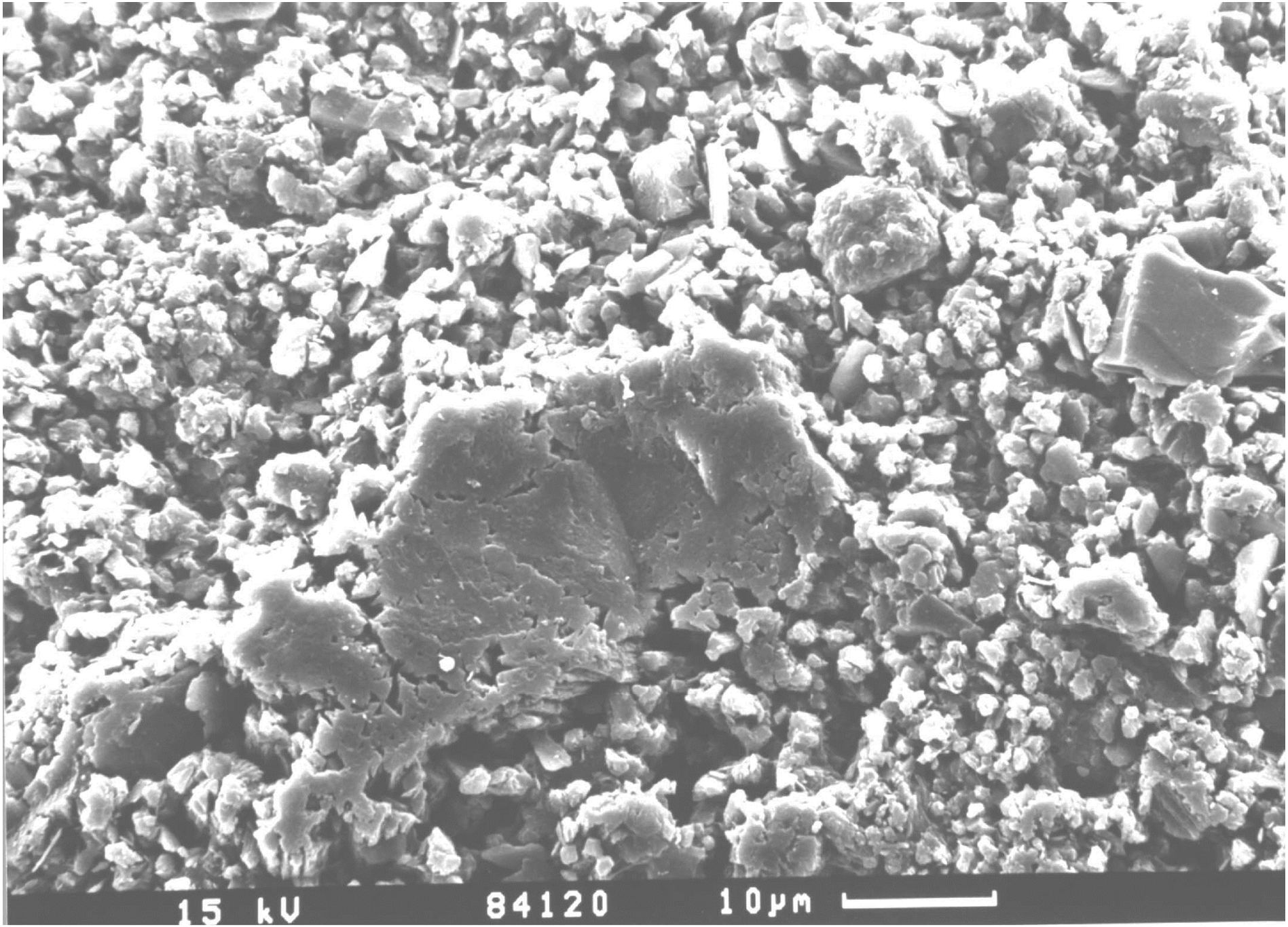


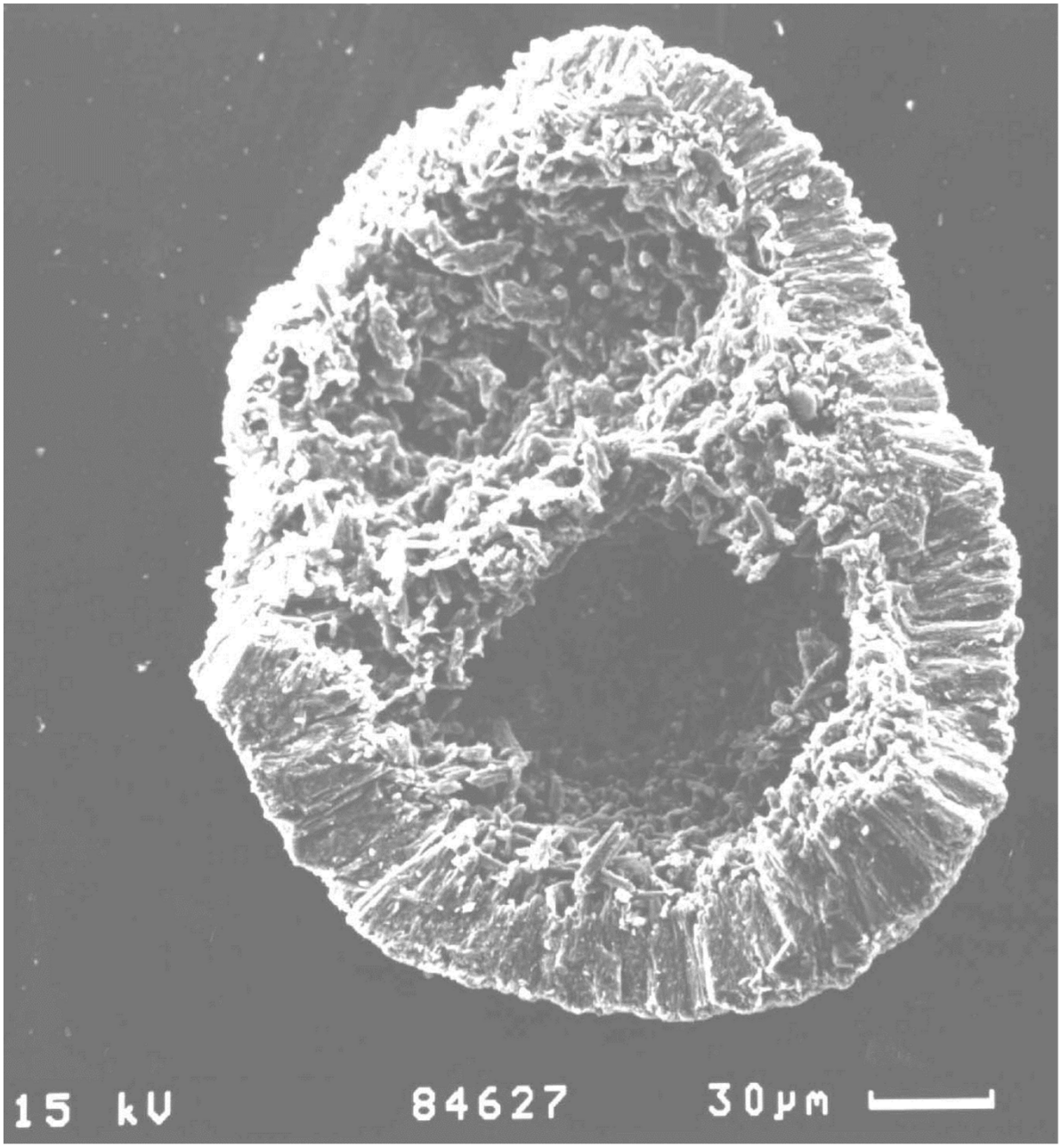

15 kV
84635
10µm

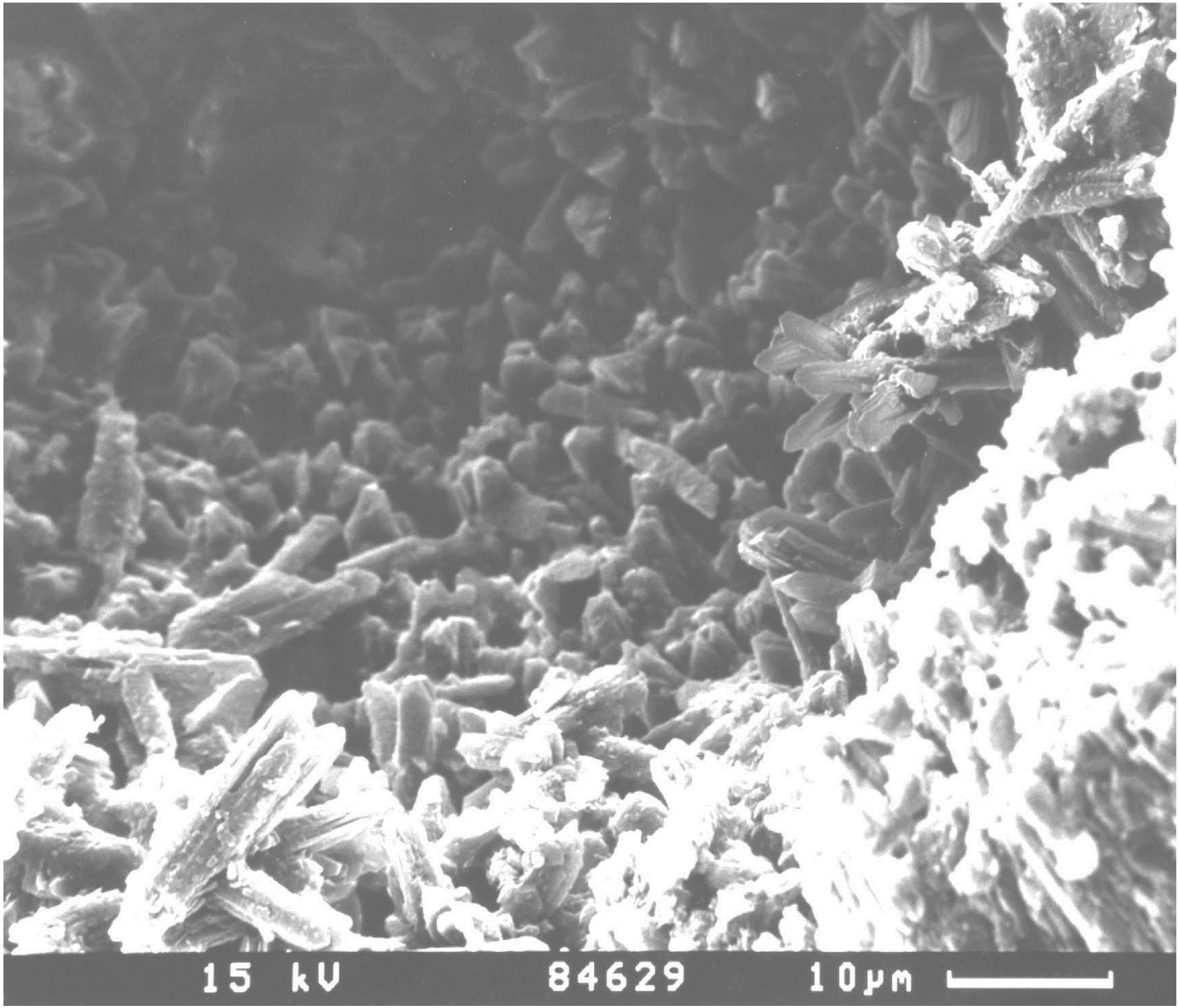

15 kV
84633
30µm

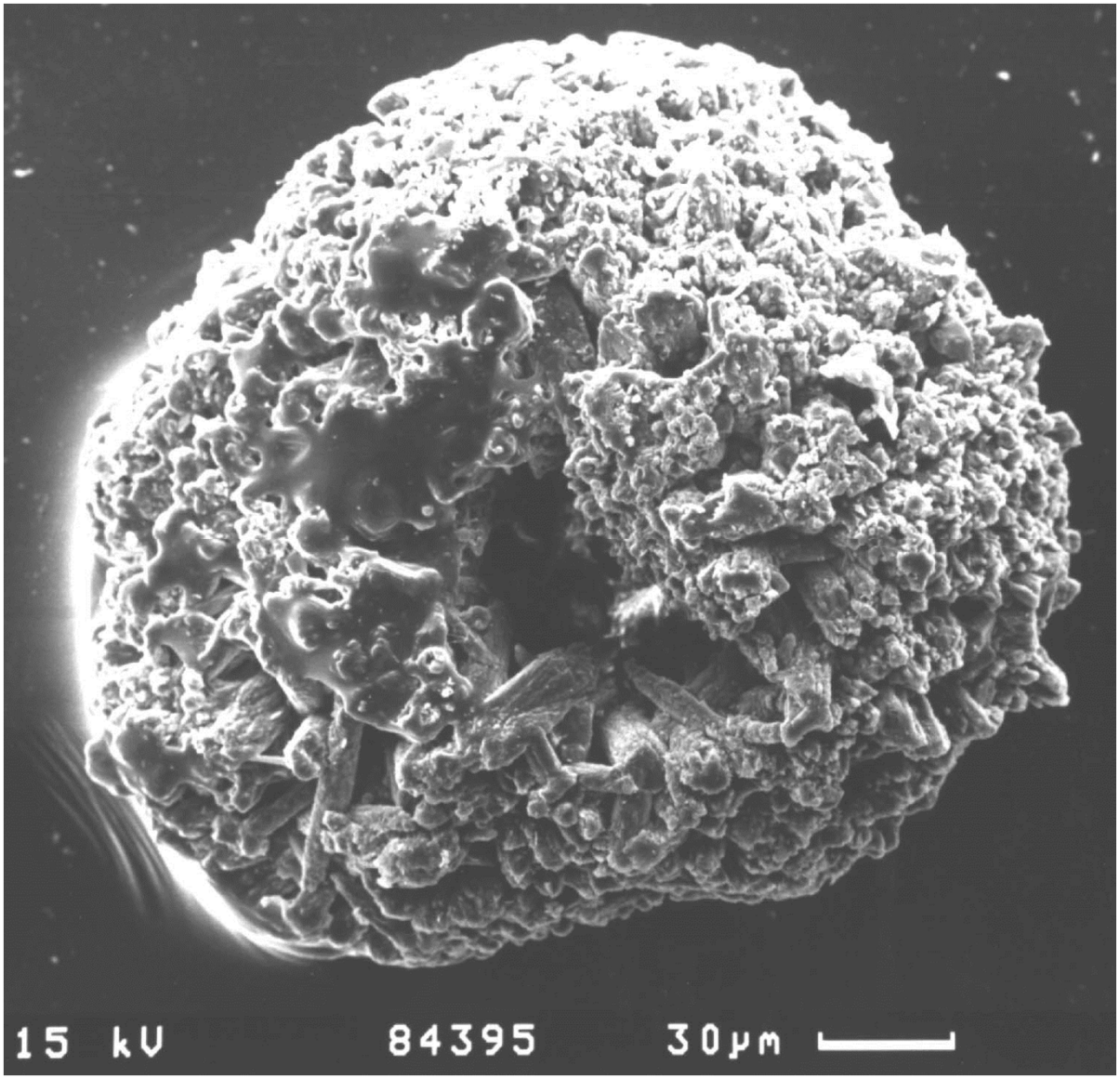
N. pachyderma sinistral
$\delta^{13}C$ = - 3.9 per mil



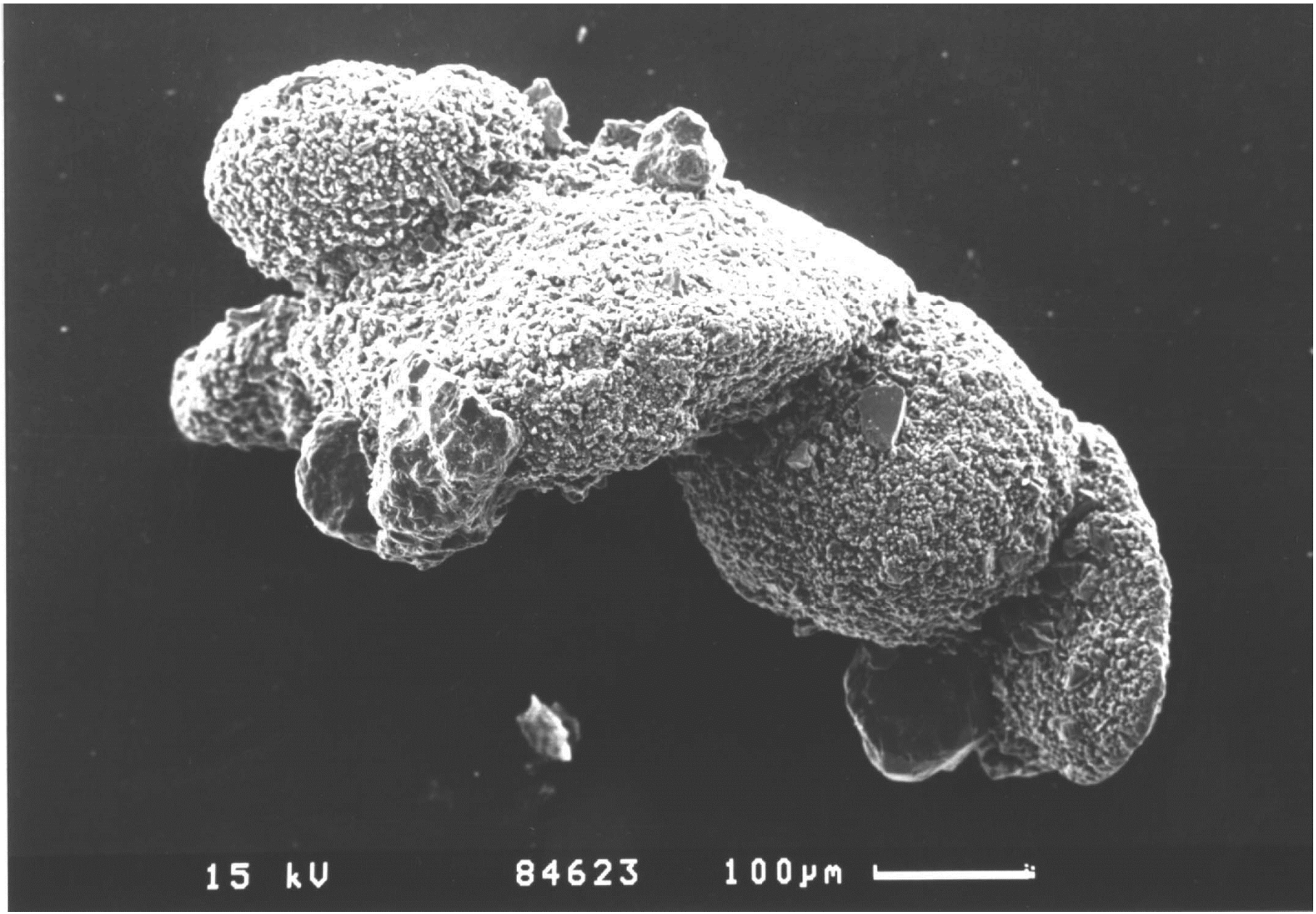

15 kV
84120
10µm


15 kV
84627
30µm

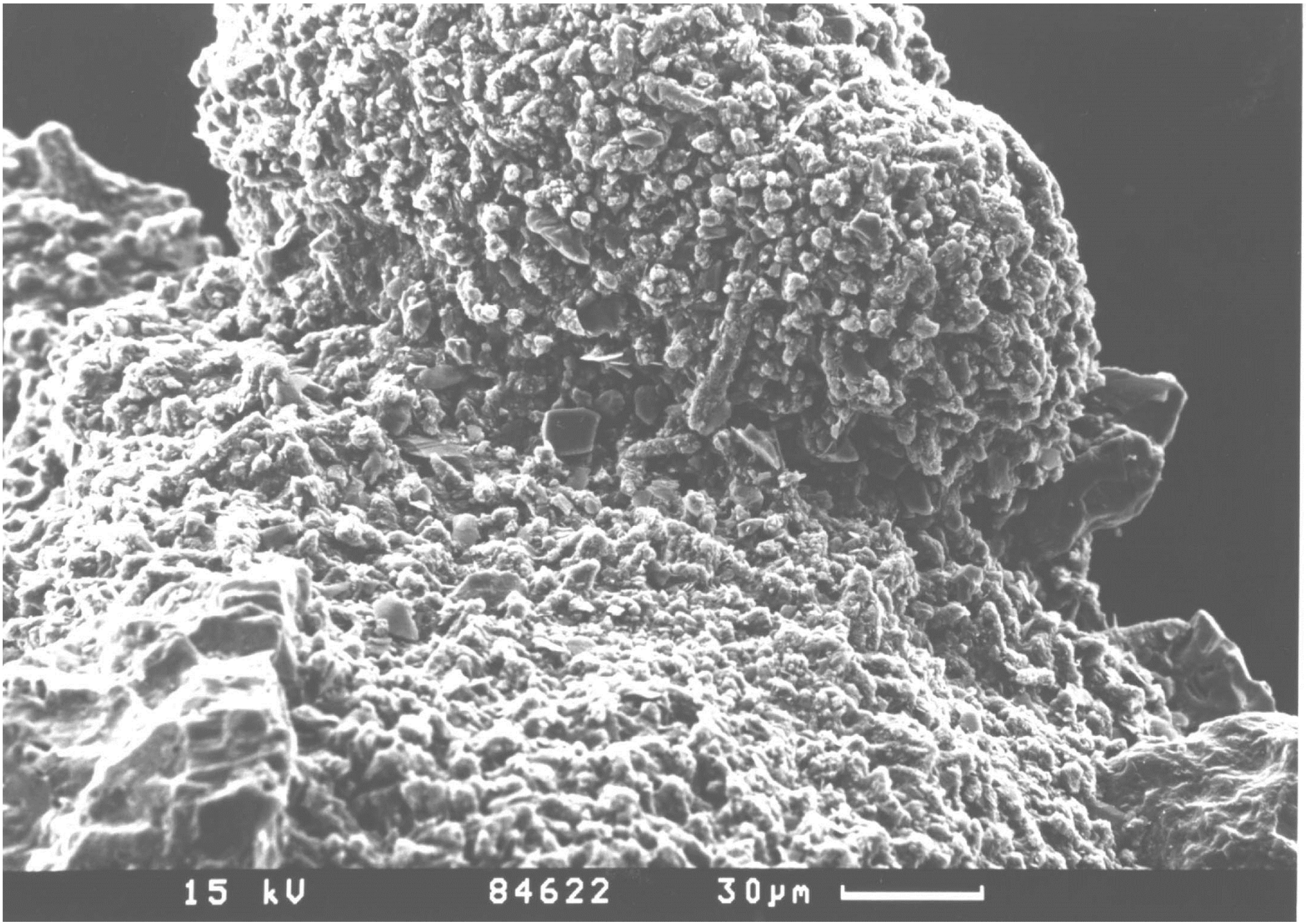

15 kV
84629
10µm


15 kV
84395
30µm

Precipitation of secondary carbonate



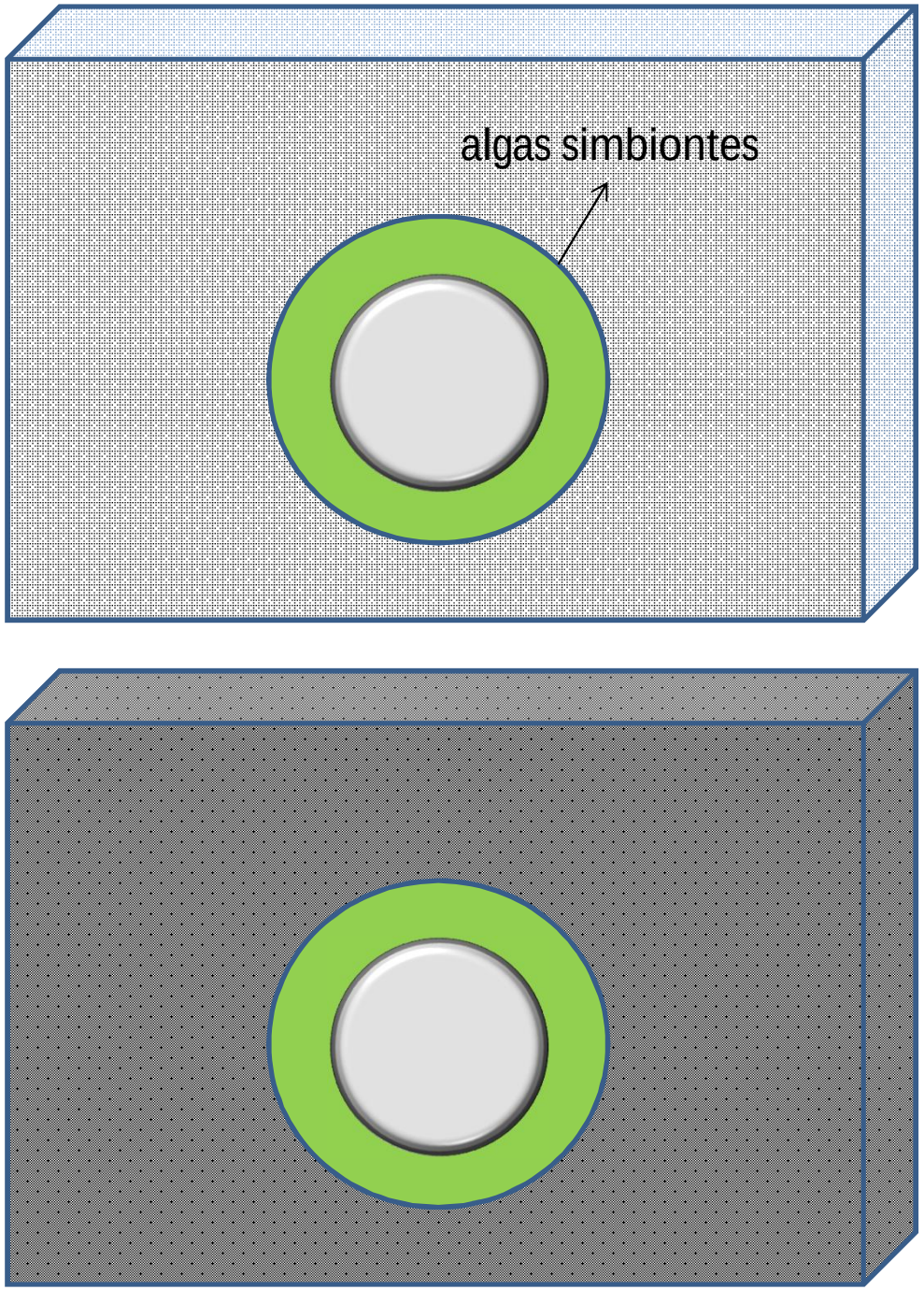
Precipitation of secondary carbonate



Efeitos vitais

Orbulina universa

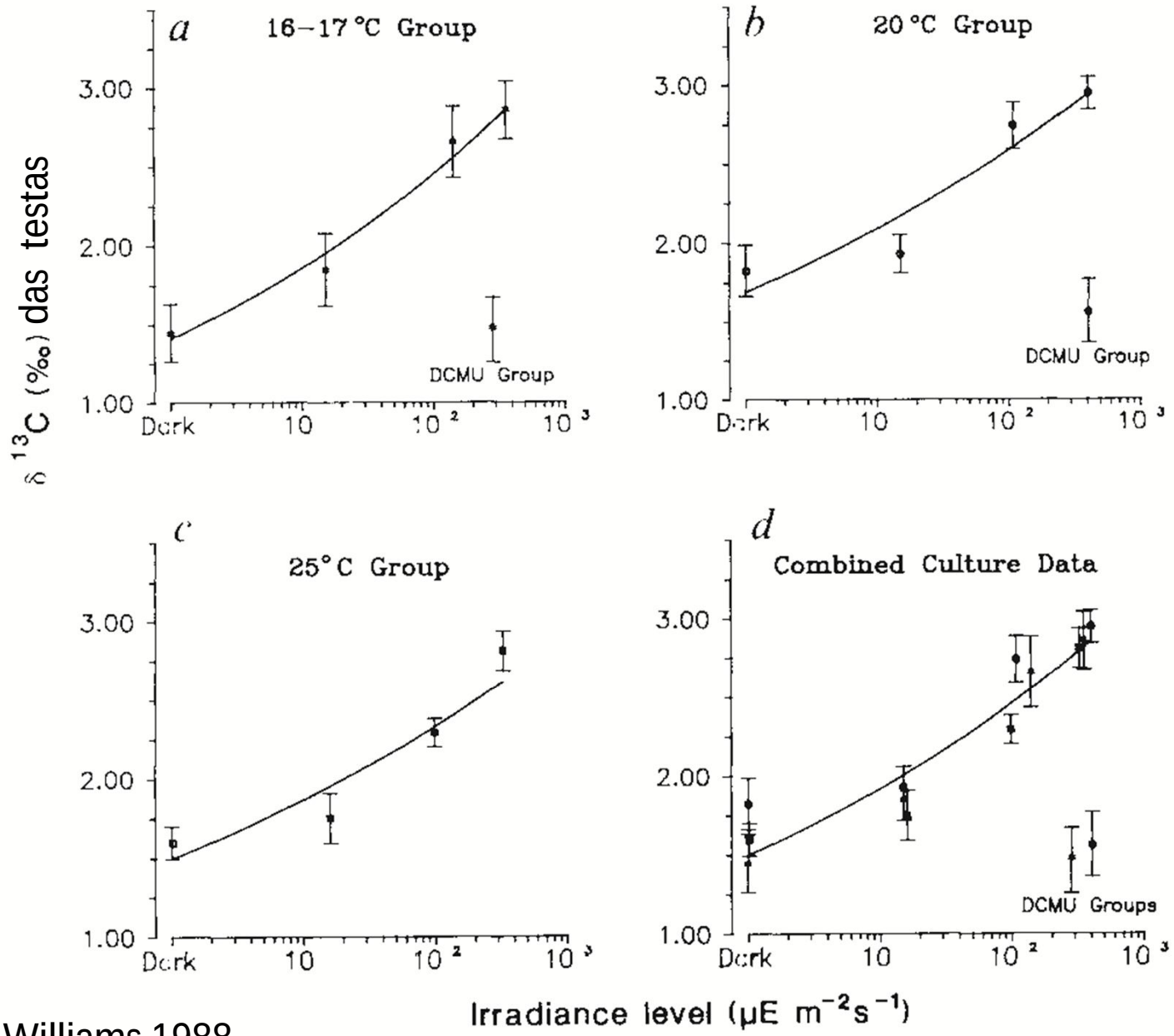
vive em simbiose
com algas



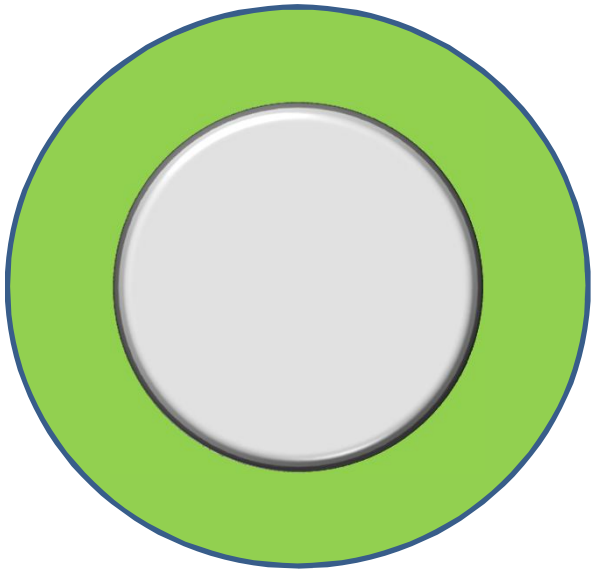
Experimento em cultura

Spero & Williams (1988)

Experimento em cultura

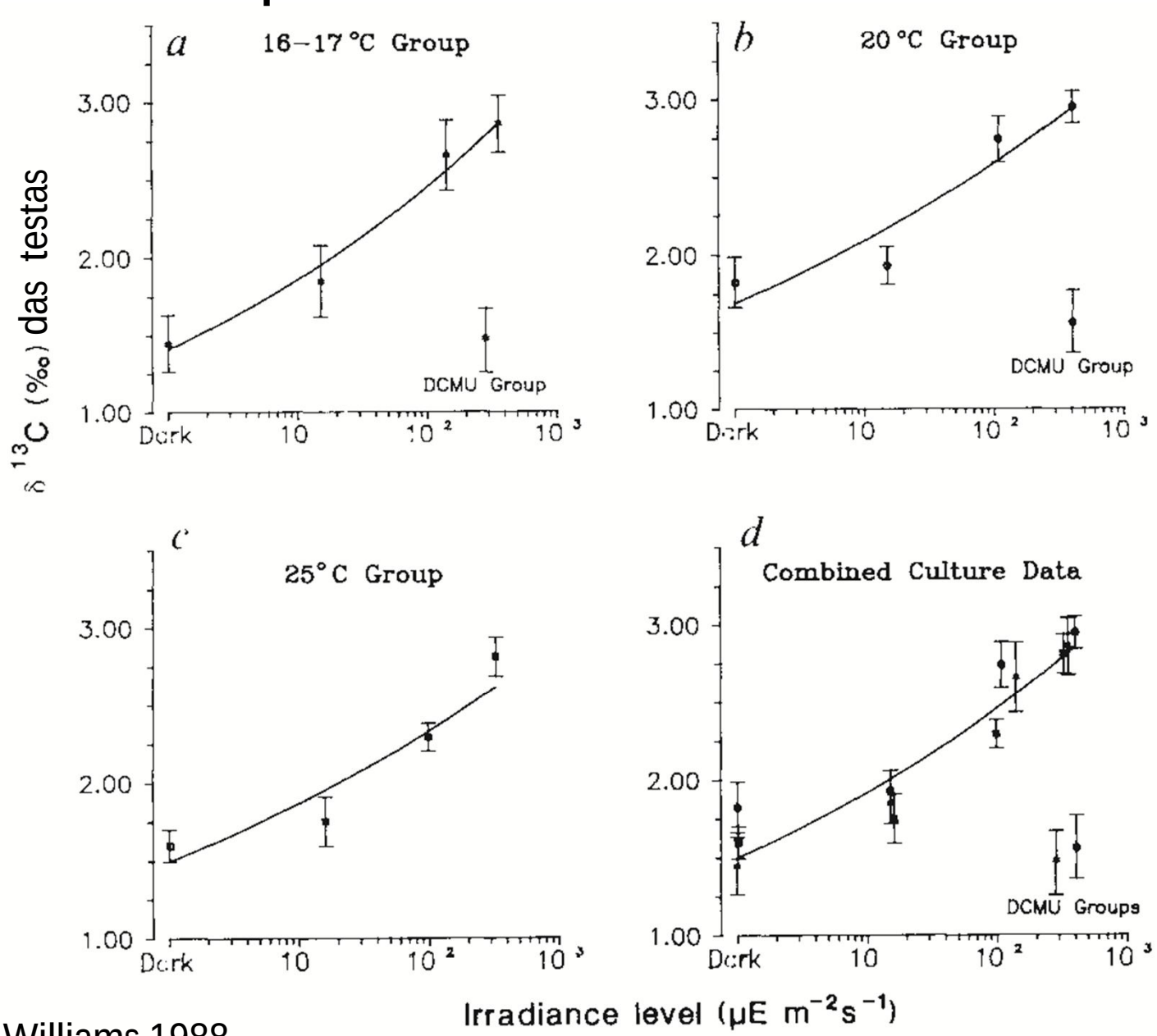


Spero & Williams 1988



“LIGHT”:
as algas simbiontes usam **mais ^{12}C** durante a fotossíntese.
O microambiente ao redor do foraminífero fica **enriquecido em ^{13}C**

Experimento em cultura

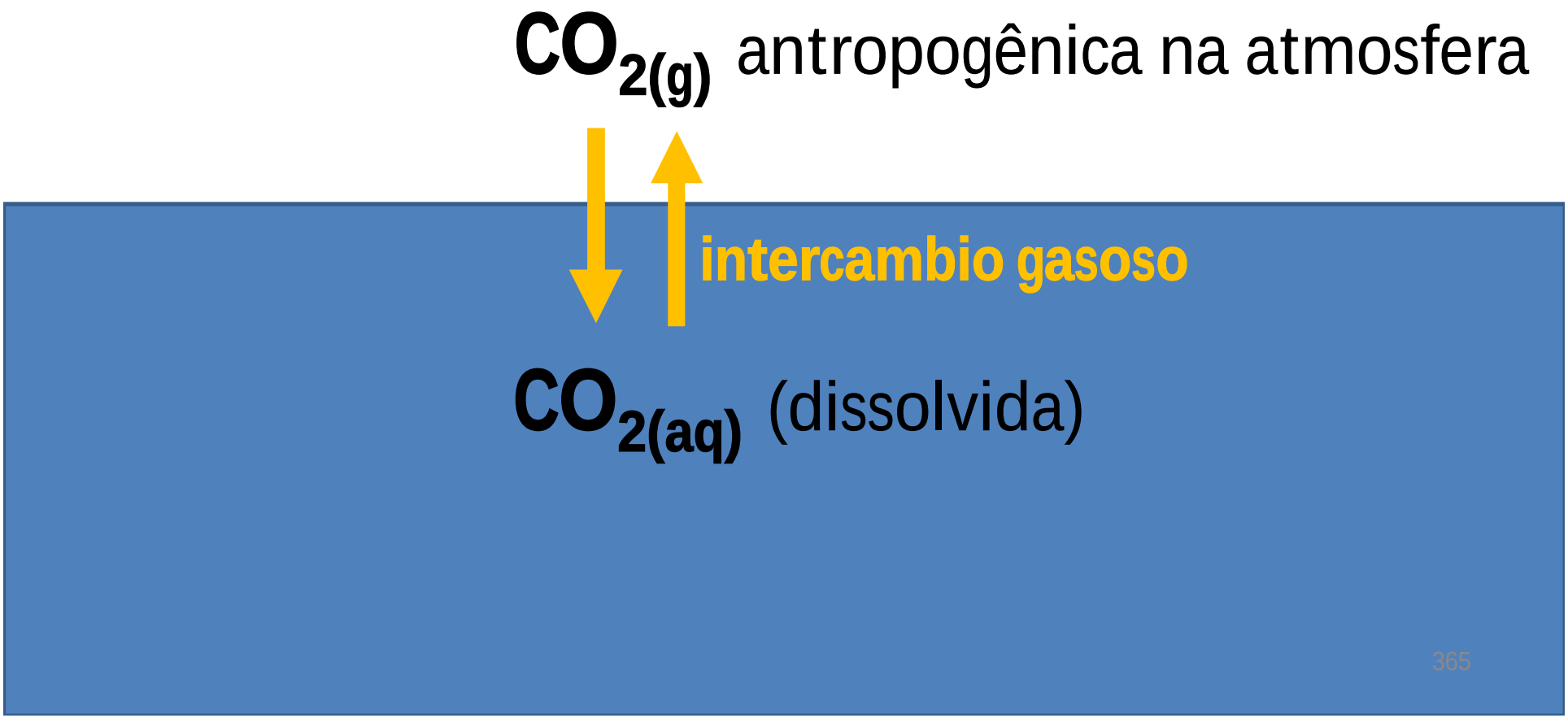


Spero & Williams 1988

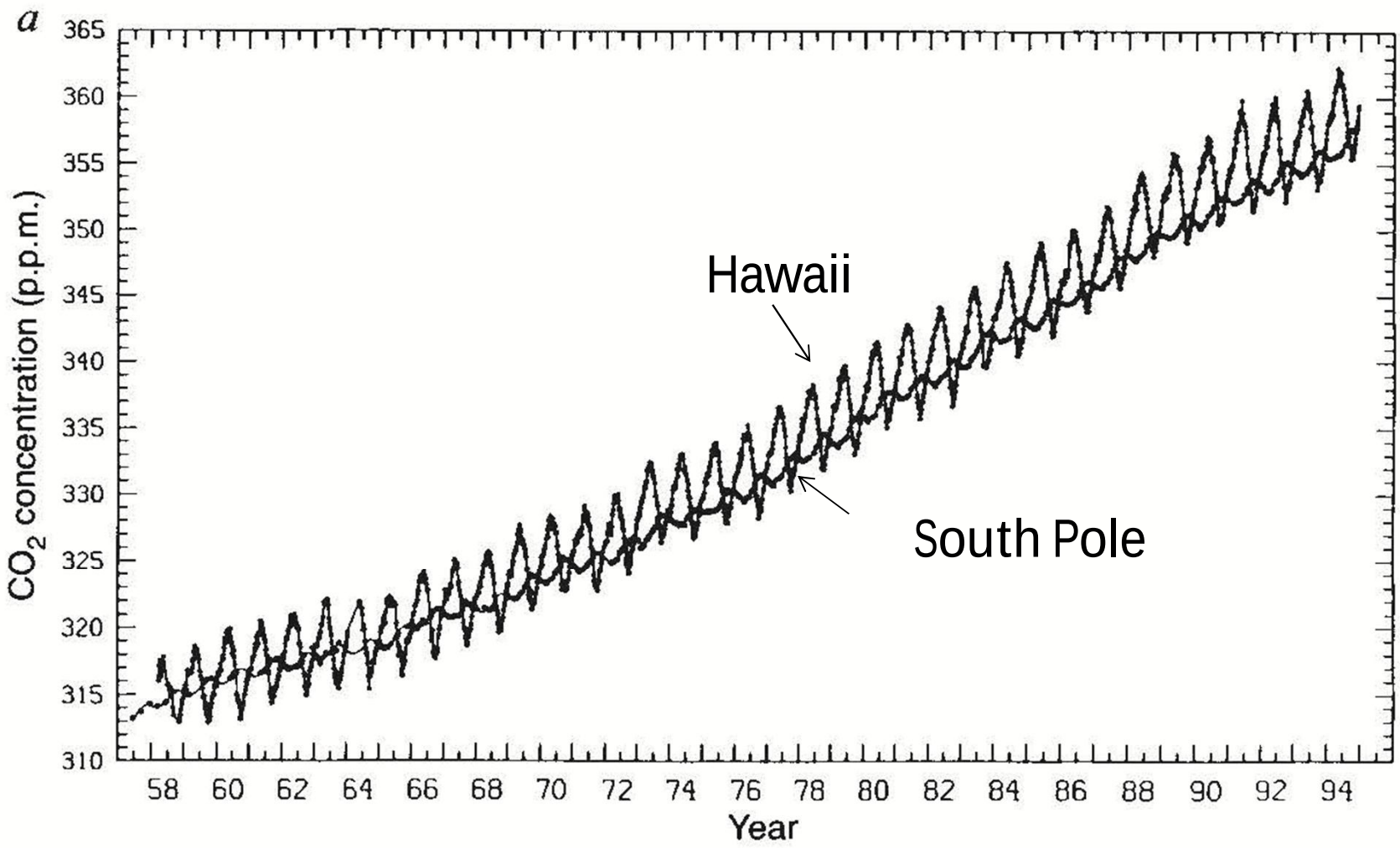
Perspectivas:

Como a CO_2 antropogênica poderia mudar o $\delta^{13}C$ do DIC oceânico ?

Como a CO_2 antropogênica poderia mudar o $\delta^{13}C$ do DIC oceânico ?

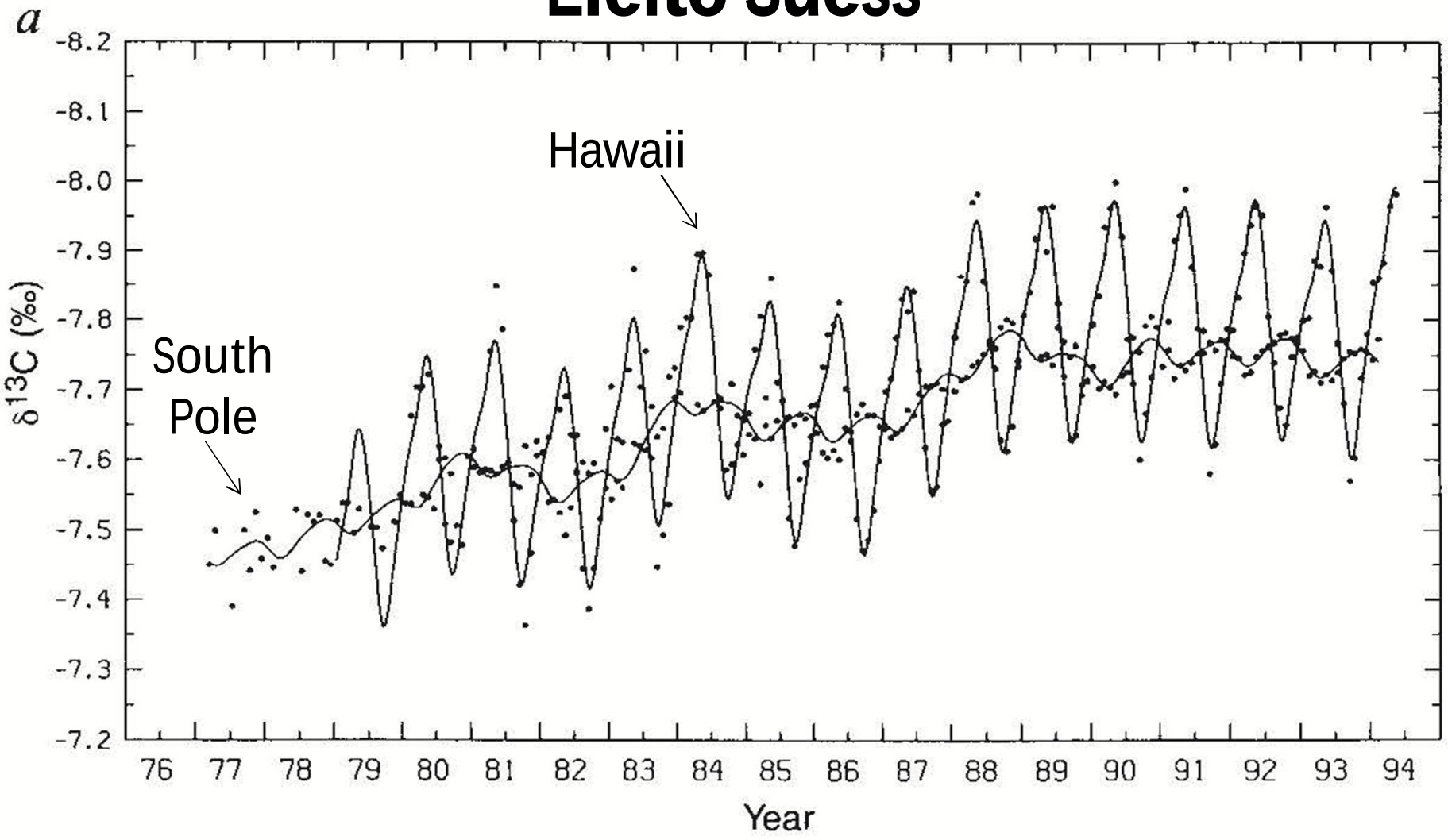


Concentração de CO_2 na atmosfera



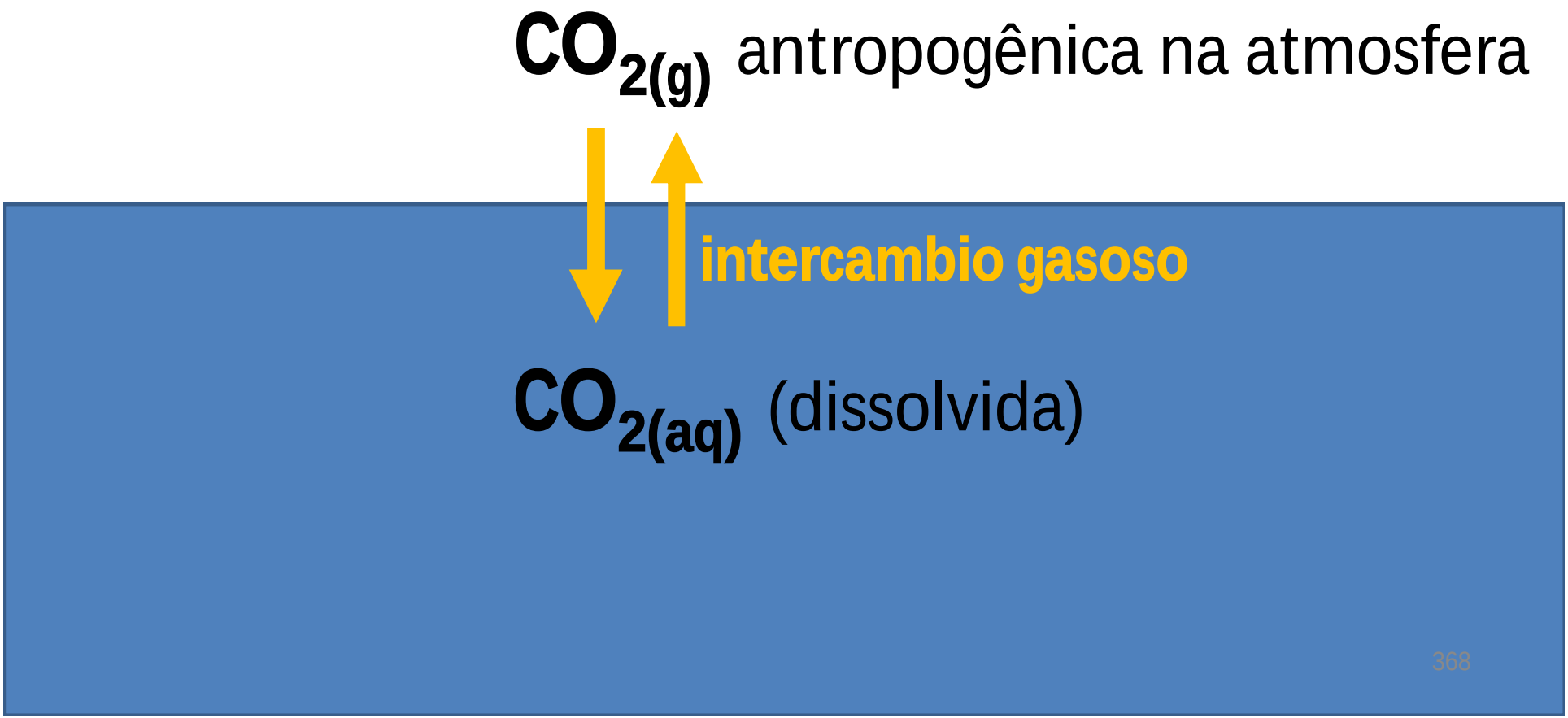
Keeling et al. (1995)

Valor de $\delta^{13}C$ do CO_2 atmosférico "Efeito Suess"

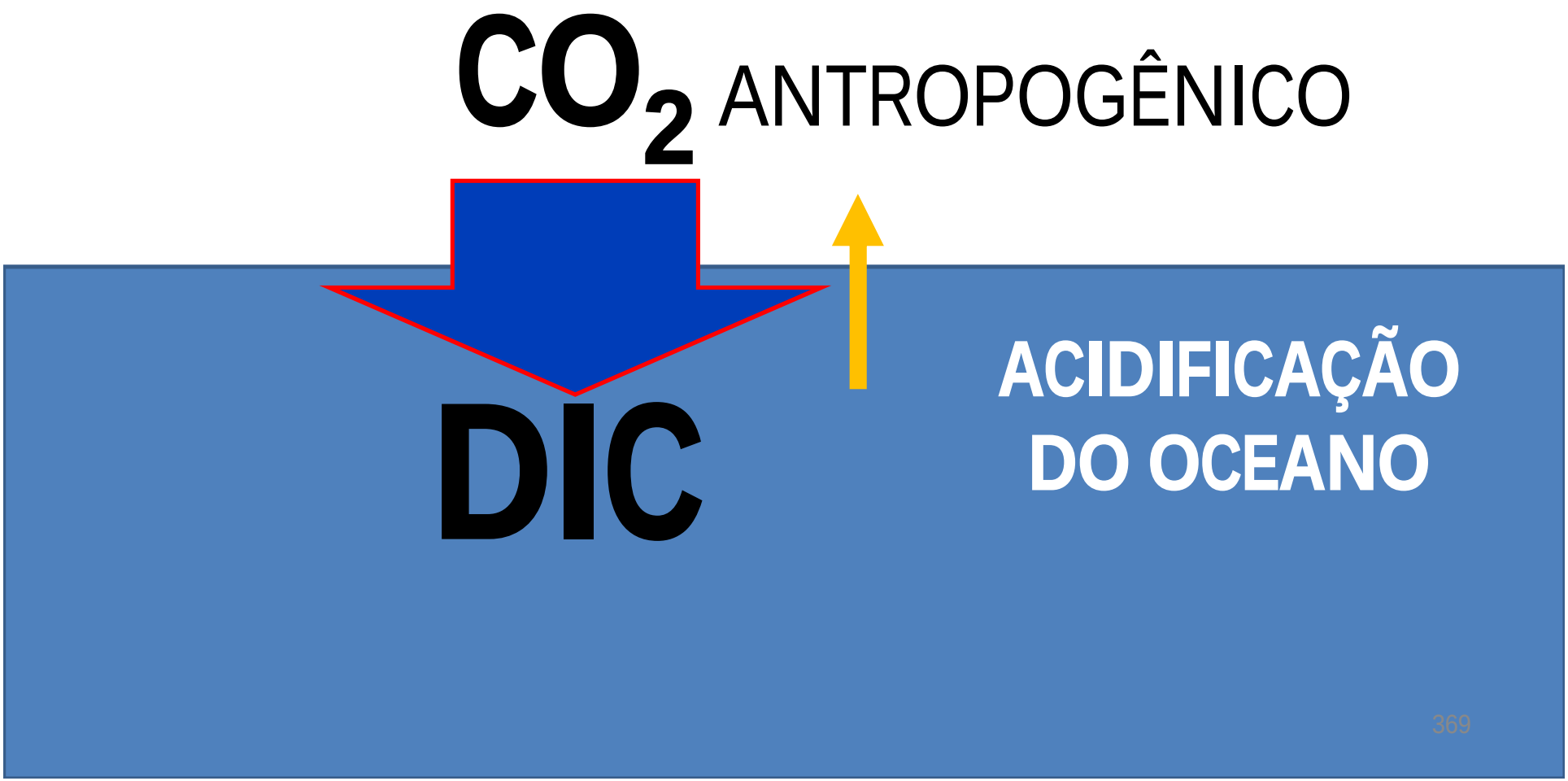


Keeling et al. (1995)

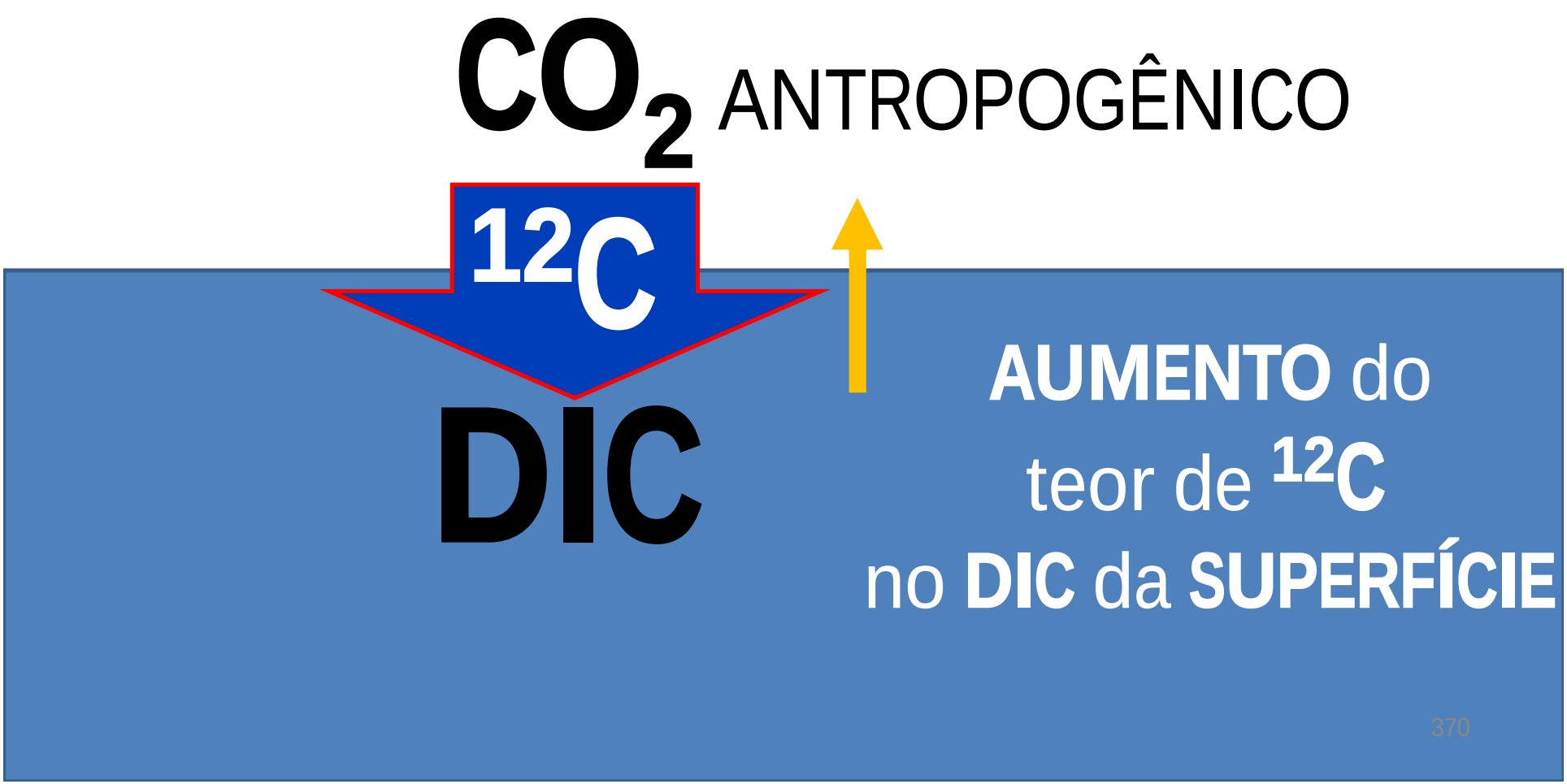
Como a CO_2 antropogênica poderia mudar o $\delta^{13}C$ do DIC oceânico ?



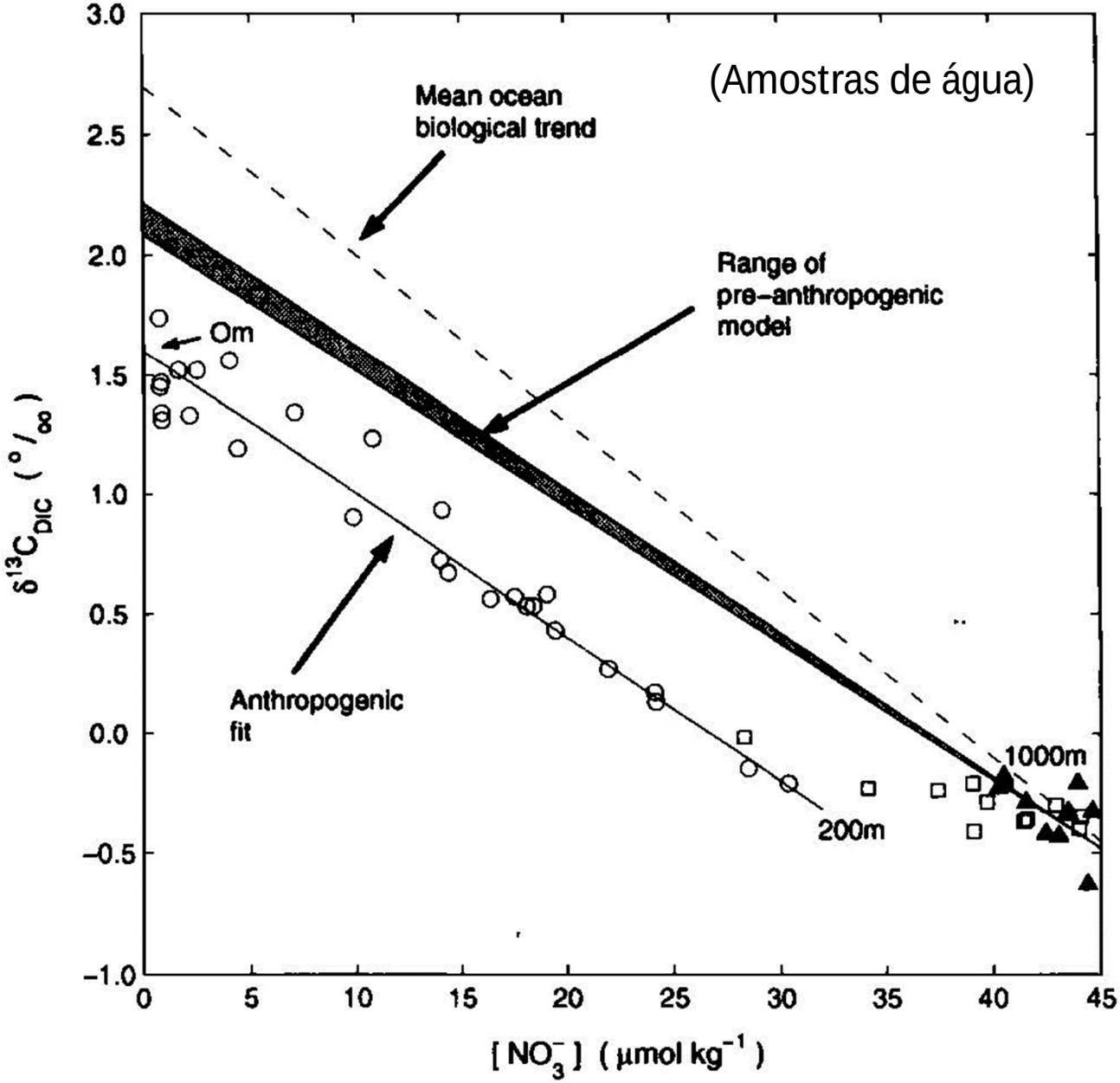
Efeito da absorção de CO_2 antropogênico no oceano



Efeito da absorção de CO2 antropogênico no oceano



Exemplo da literatura: Pacifico Norte



Ortiz et al. (2000)

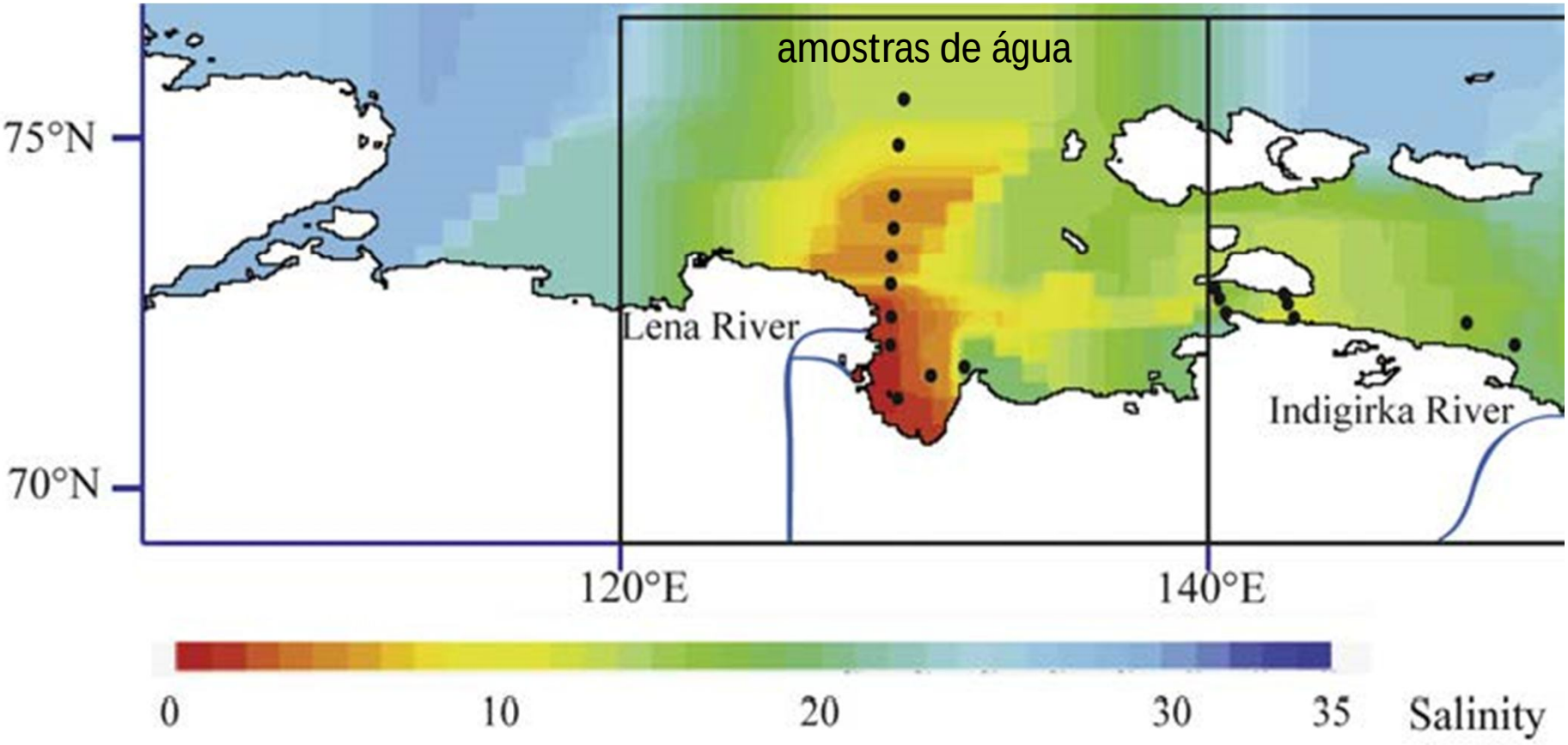
Razão $^{13}C/^{12}C$ no DIC

como marcador de **produção primaria**

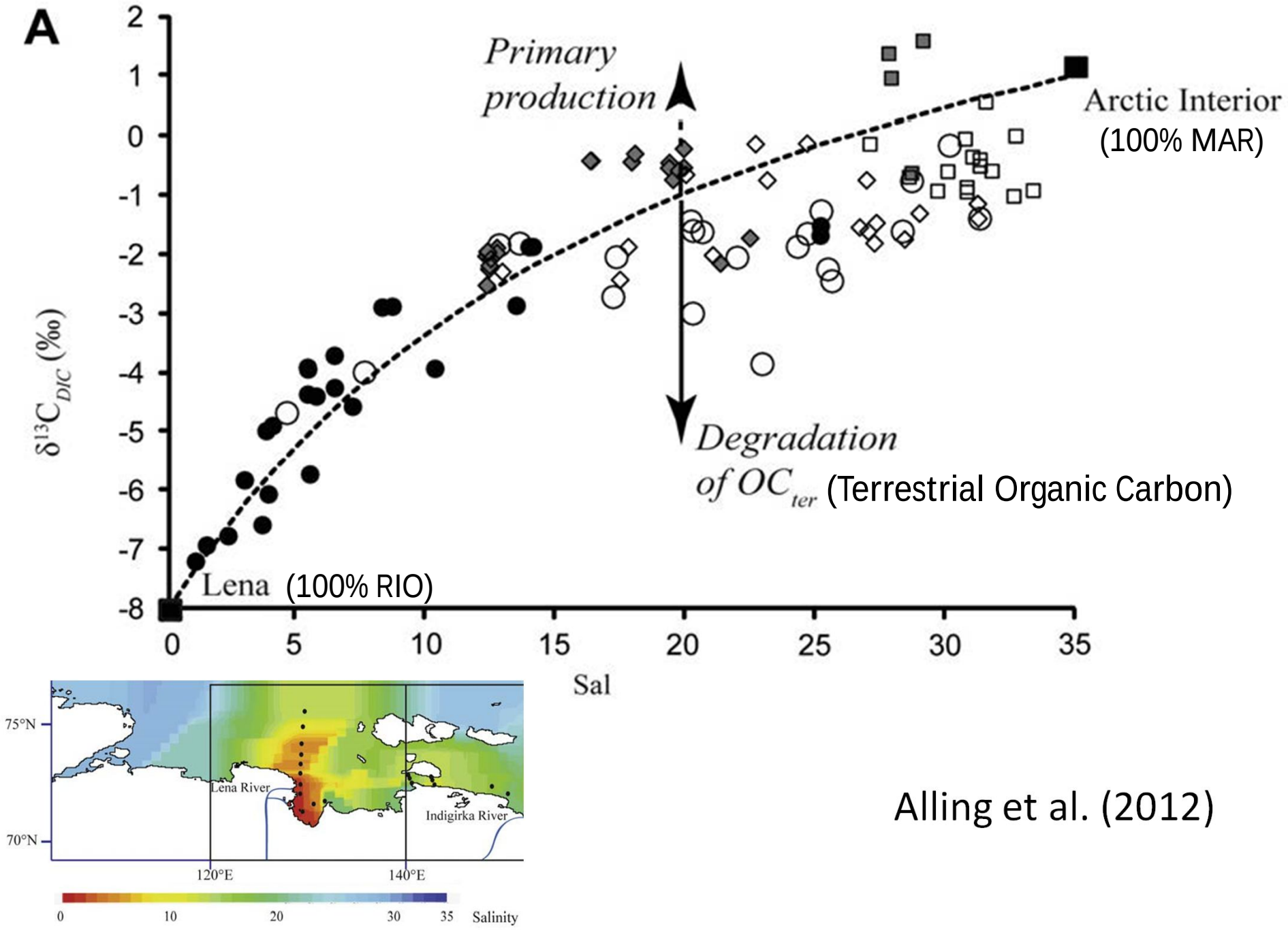
e de **degradação da materia orgânica**

em **ambiente estuarino**

Exemplo da literatura: Mar da Sibéria

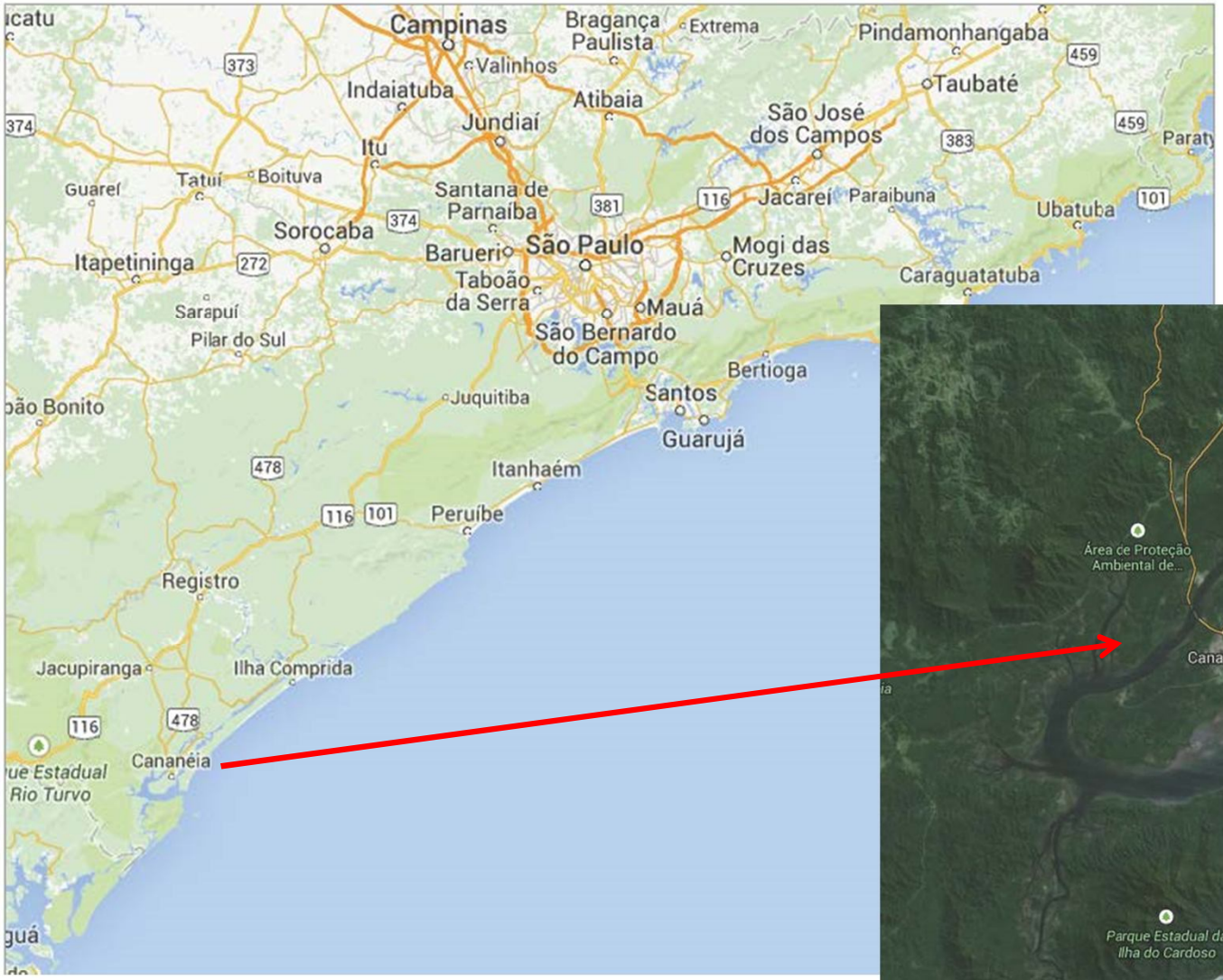


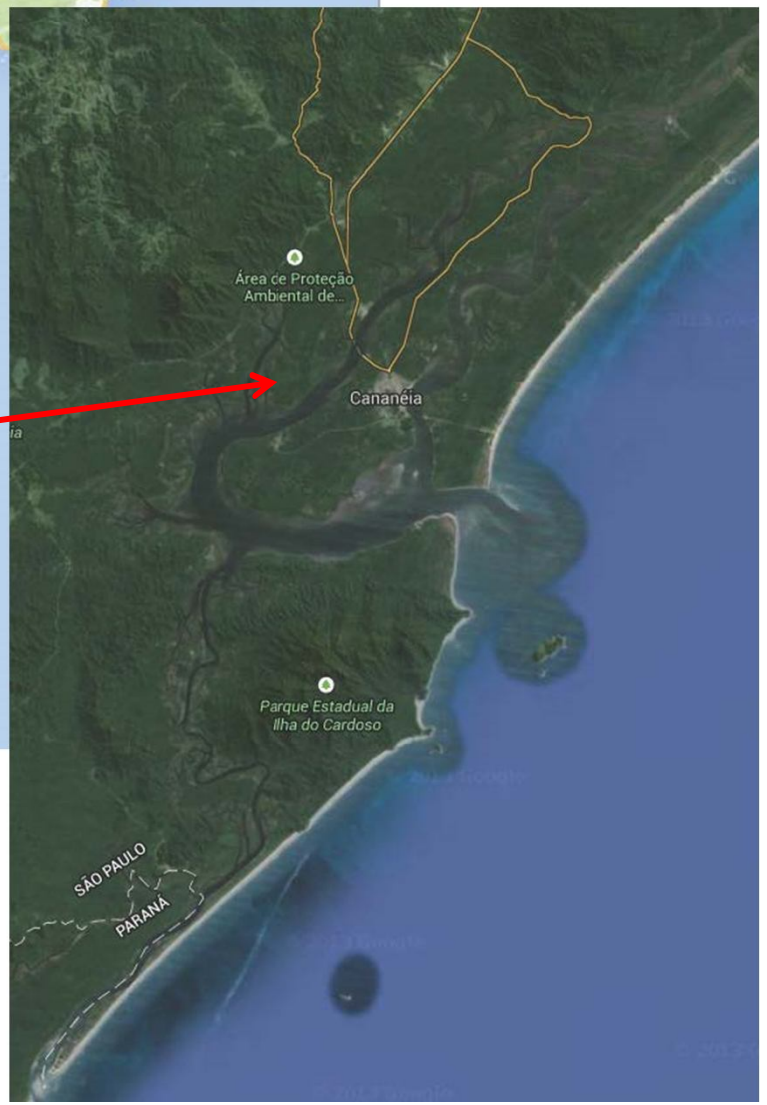
Alling et al. (2012)

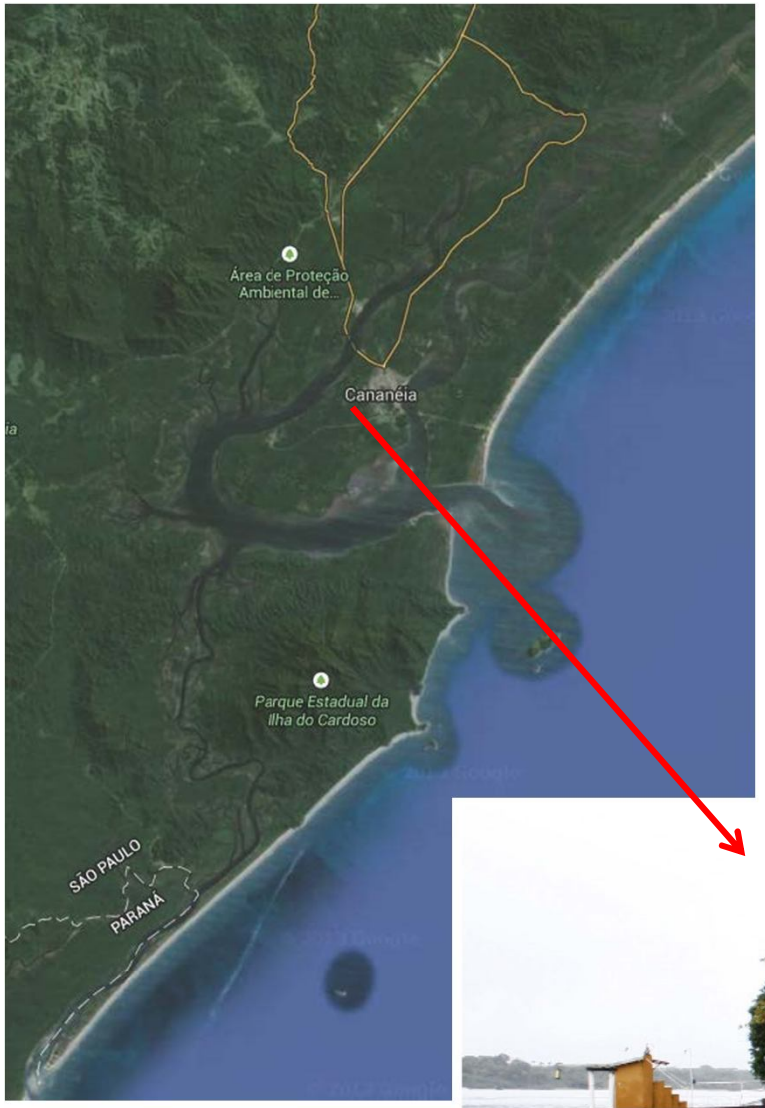


Alling et al. (2012)

Cananeia

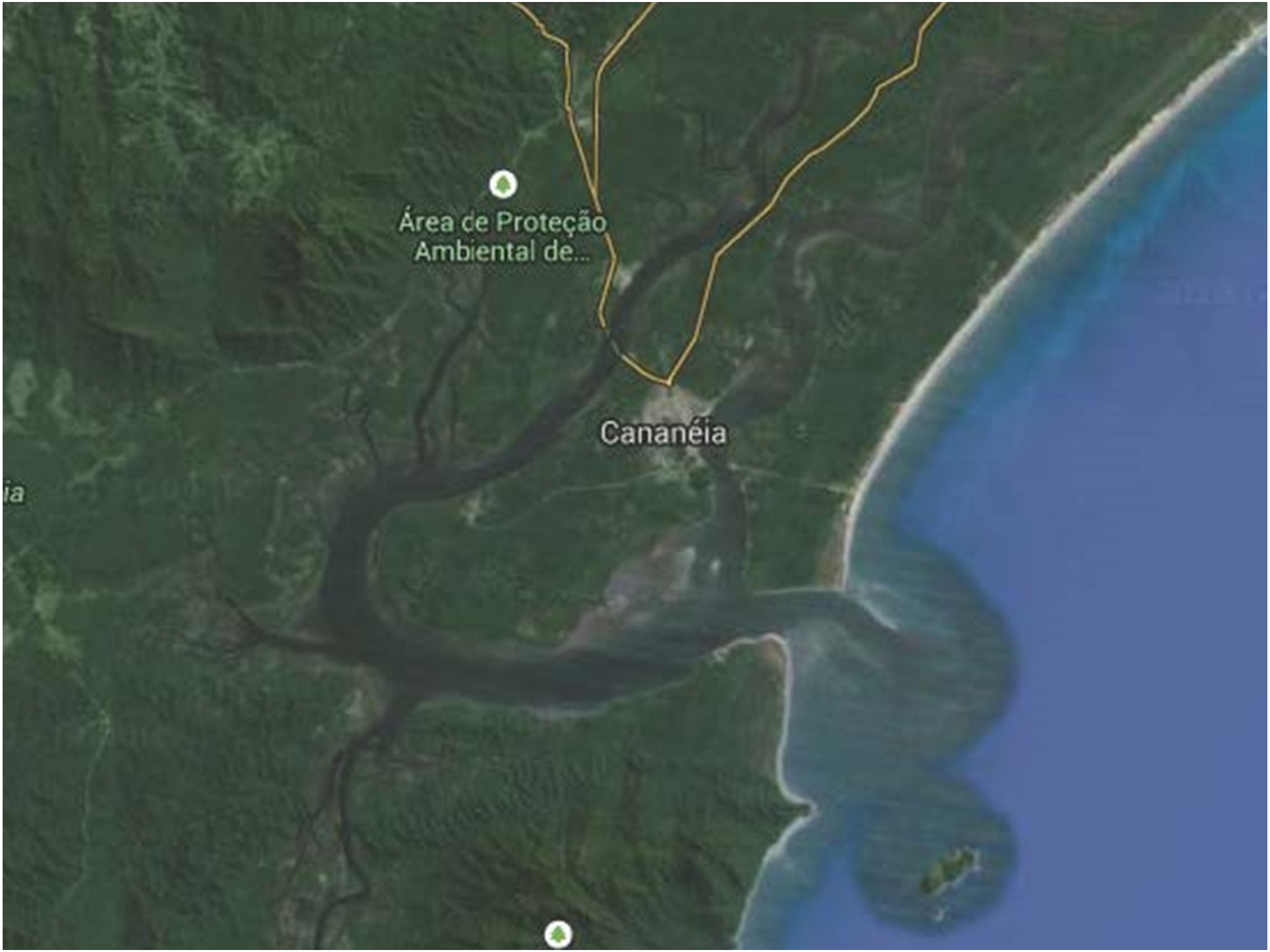

Campinas
Bragança Paulista
Extrema
Pindamonhangaba
Valinhos
Taubaté
Indaiatuba
Atibaia
Jundiaí
São José dos Campos
Itu
Tatuí
Boituva
Guareí
Santana de Parnaíba
Jacareí
Paraibuna
Ubatuba
Sorocaba
Barueri
São Paulo
Mogi das Cruzes
Itapetininga
Taboão da Serra
Caraguatatuba
Sarapuí
Mauá
Pilar do Sul
São Bernardo do Campo
Bertioga
Juquitiba
Santos
Guarujá
Itanhaém
Peruíbe
Registro
Jacupiranga
Ilha Comprida
Cananéia
Parque Estadual Rio Turvo

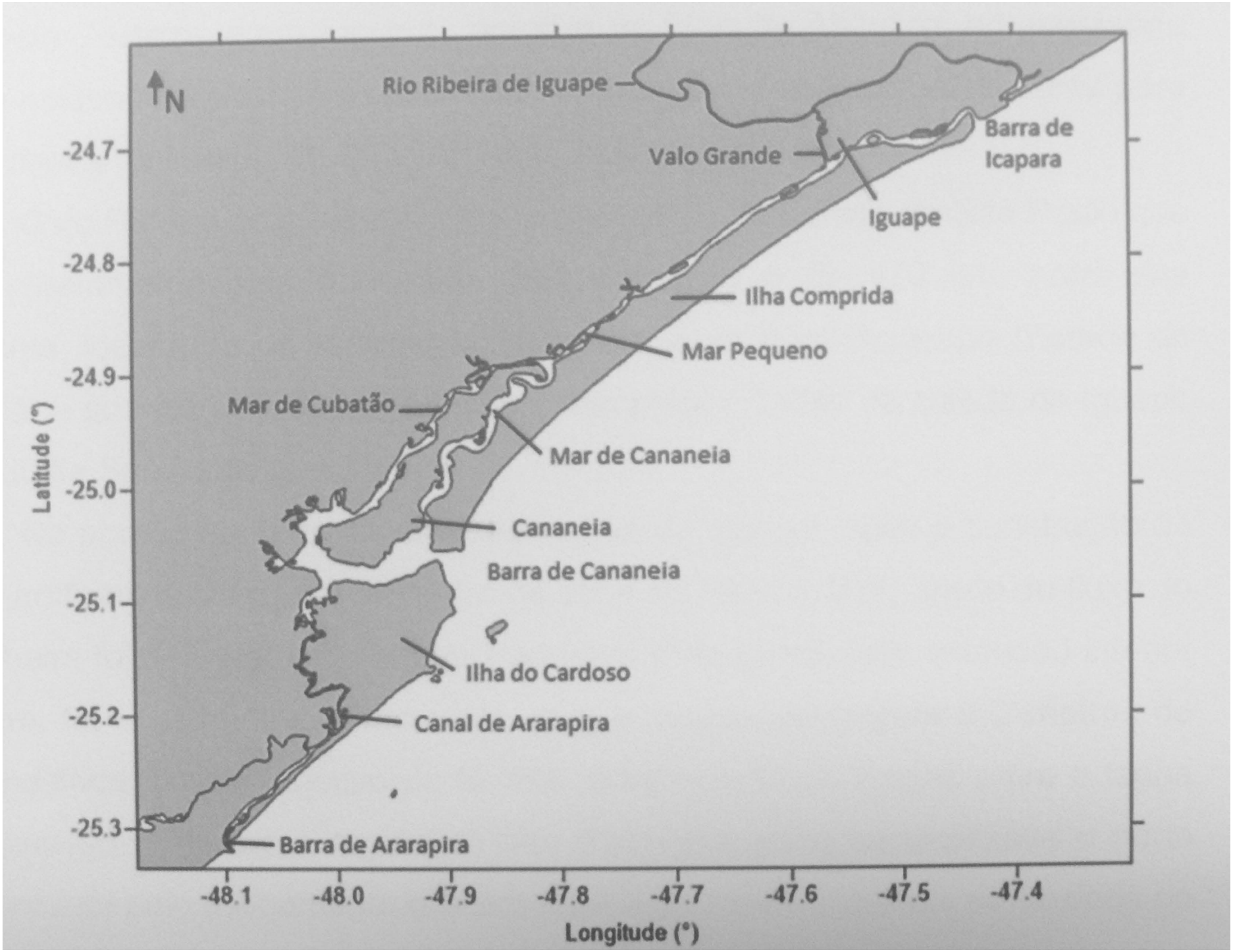

Área de Proteção Ambiental de...
Cananéia
Parque Estadual da Ilha do Cardoso
SÃO PAULO
PARANÁ

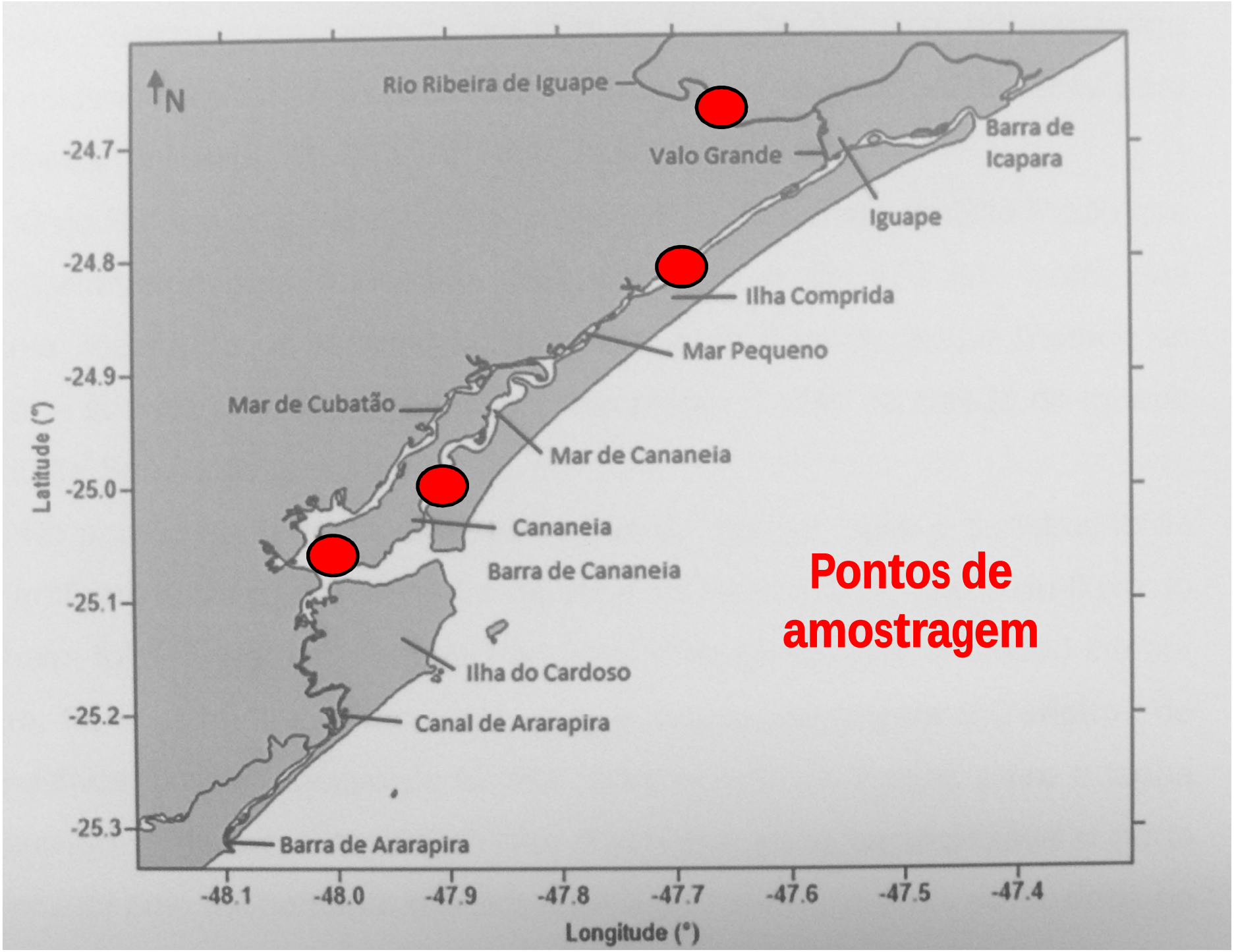


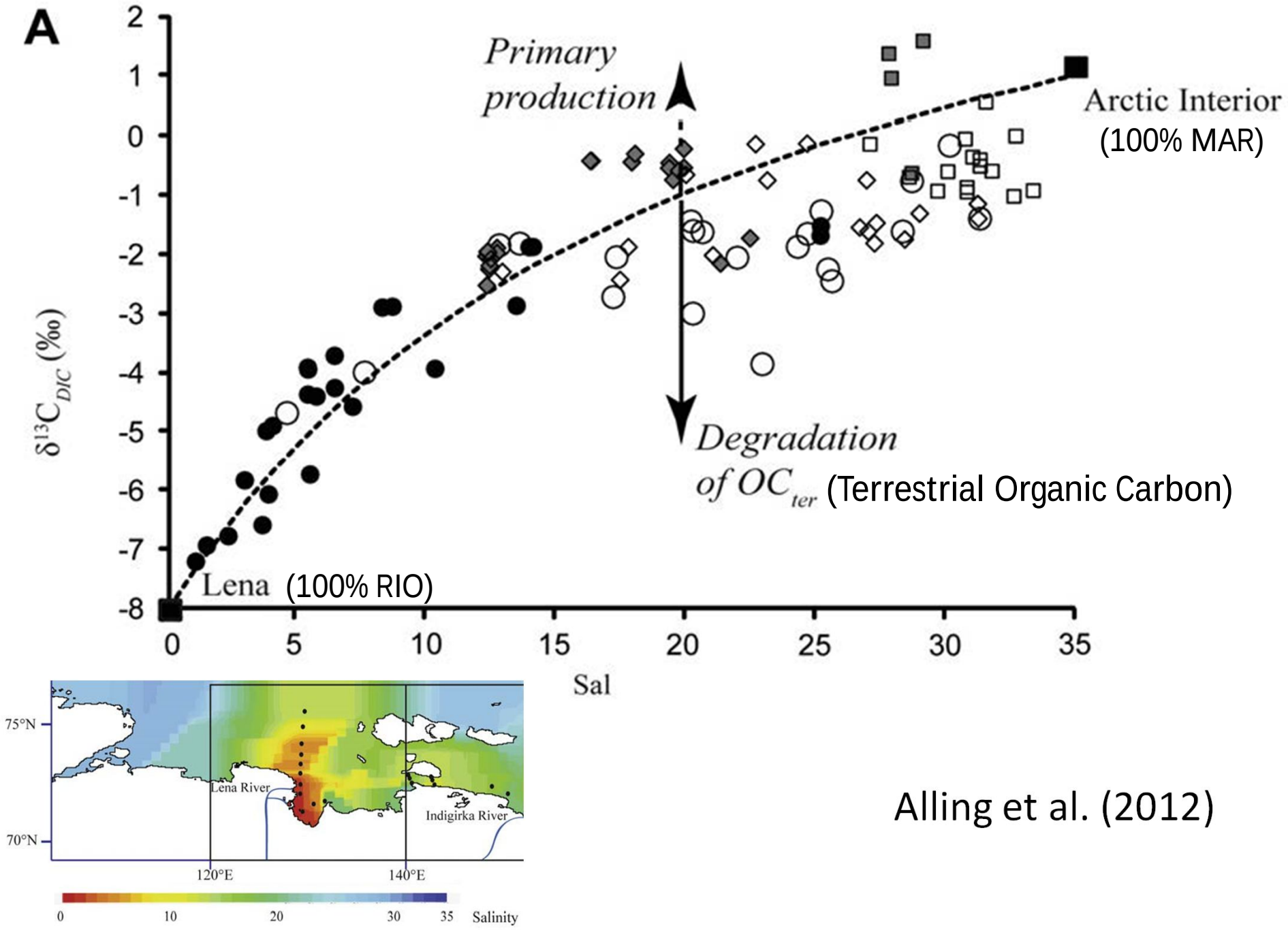
Instituto Oceanográfico

Base de Cananeia

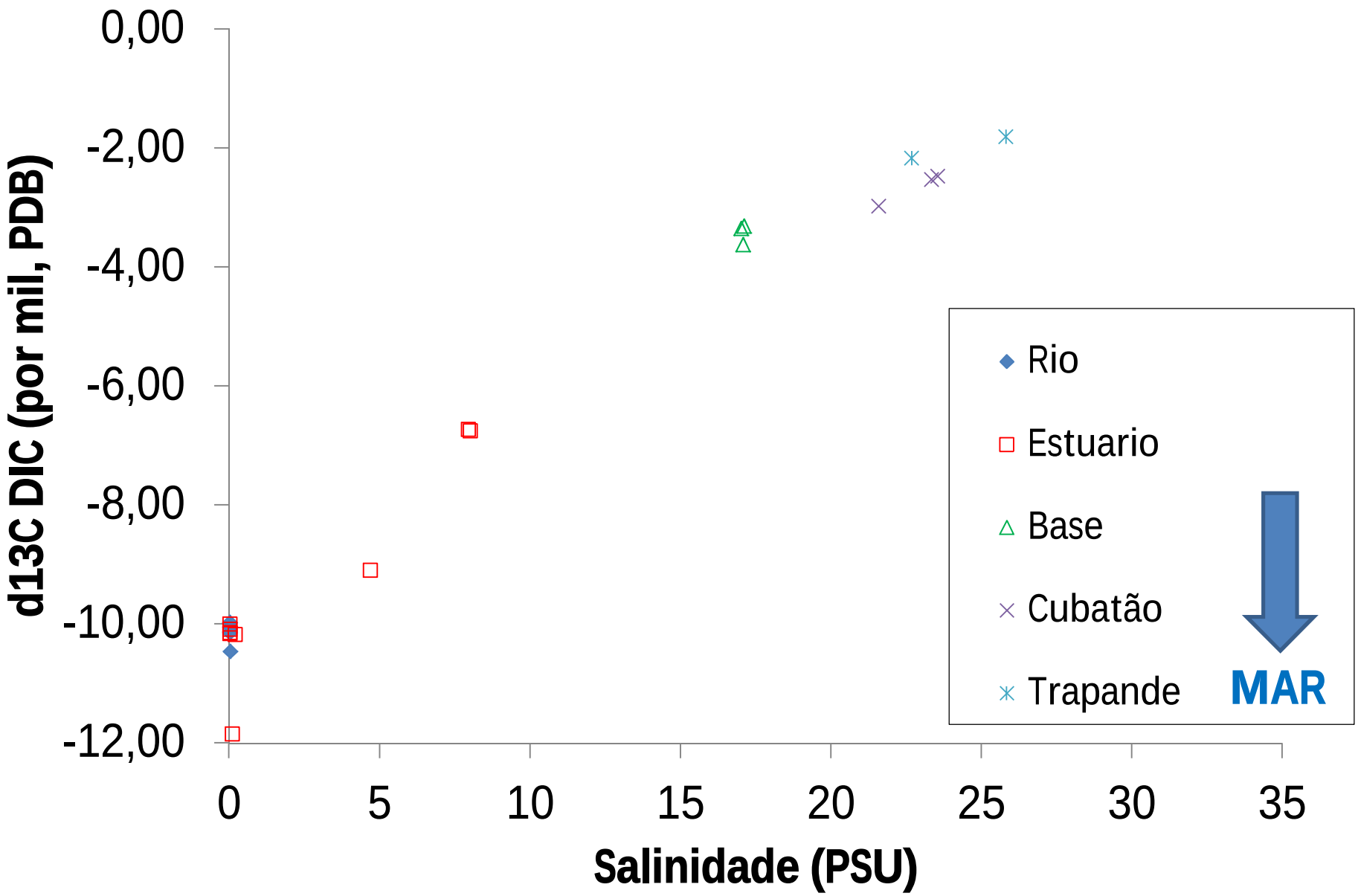

Área de Proteção
Ambiental de...
Cananéia


N
Rio Ribeira de Iguape
Barra de Icapara
Valo Grande
Iguape
Ilha Comprida
Mar Pequeno
Mar de Cubatão
Mar de Cananeia
Cananeia
Barra de Cananeia
Ilha do Cardoso
Canal de Ararapira
Barra de Ararapira
-24.7
-24.8
-24.9
-25.0
-25.1
-25.2
-25.3
Latitude (°)
-48.1
-48.0
-47.9
-47.8
-47.7
-47.6
-47.5
-47.4
Longitude (°)


N
Rio Ribeira de Iguape
Valo Grande
Barra de Icapara
Iguape
Ilha Comprida
Mar Pequeno
Mar de Cubatão
Mar de Cananeia
Cananeia
Barra de Cananeia
Ilha do Cardoso
Canal de Ararapira
Barra de Ararapira
Pontos de amostragem
-24.7
-24.8
-24.9
-25.0
-25.1
-25.2
-25.3
Latitude (°)
-48.1
-48.0
-47.9
-47.8
-47.7
-47.6
-47.5
-47.4
Longitude (°)



Alling et al. (2012)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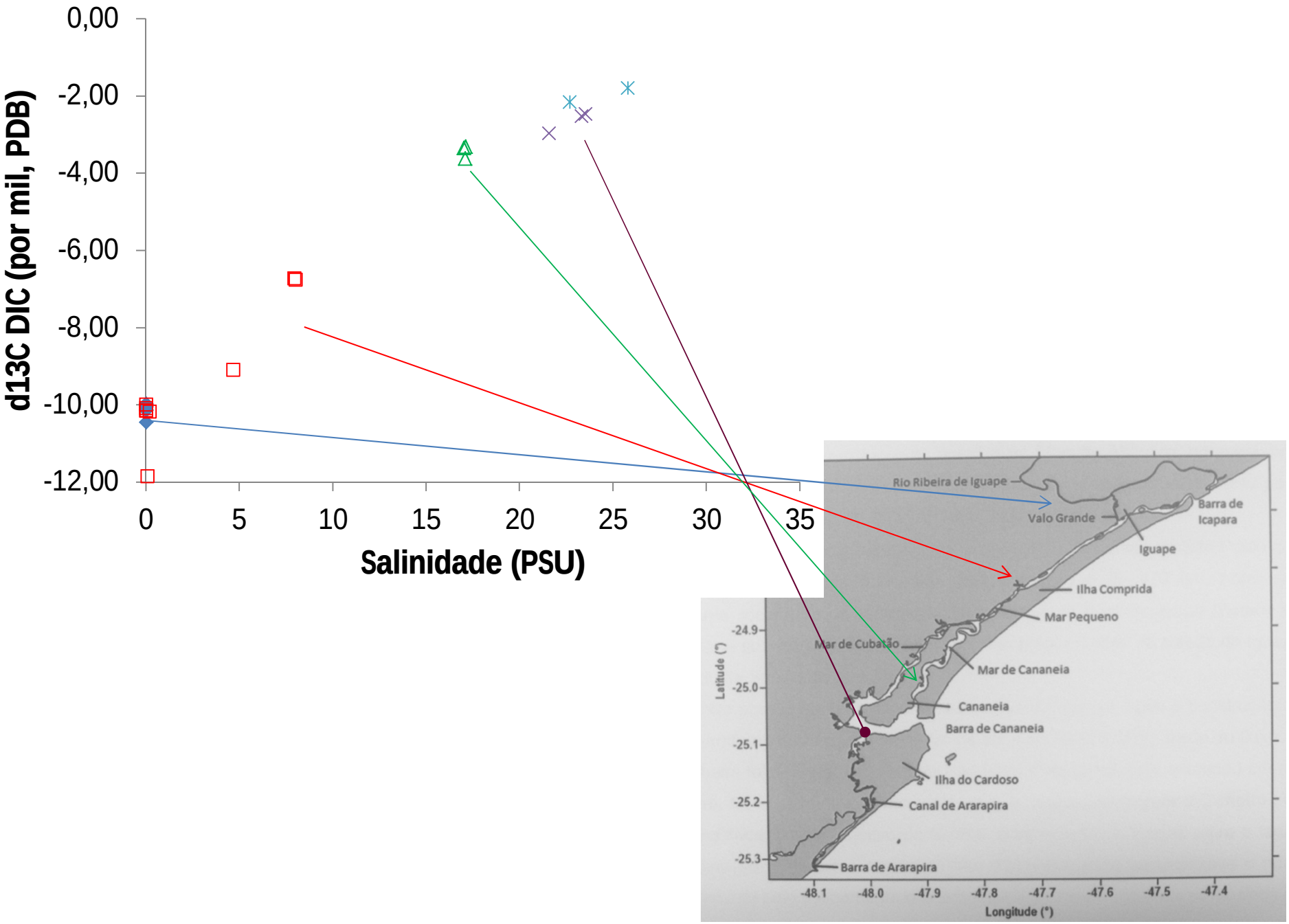
-2,00
-4,00
-6,00
-8,00
-10,00
-12,00
d13C DIC (por mil, PDB)
0
5
10
15
20
25
30
35
Salinidade (PSU)
Rio
Estuario
Base
Cubatão
Trapande
MAR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-2,00
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-6,00
-8,00
-10,00
-12,00
d13C DIC (por mil, PDB)
0
5
10
15
20
25
30
35
Salinidade (PSU)
Rio Ribeira de Iguape
Valo Grande
Barra de Icapara
Iguape
Ilha Comprida
Mar Pequeno
Mar de Cubatão
Mar de Cananeia
Cananeia
Barra de Cananeia
Ilha do Cardoso
Canal de Ararapira
Barra de Ararapira
Latitude (°)
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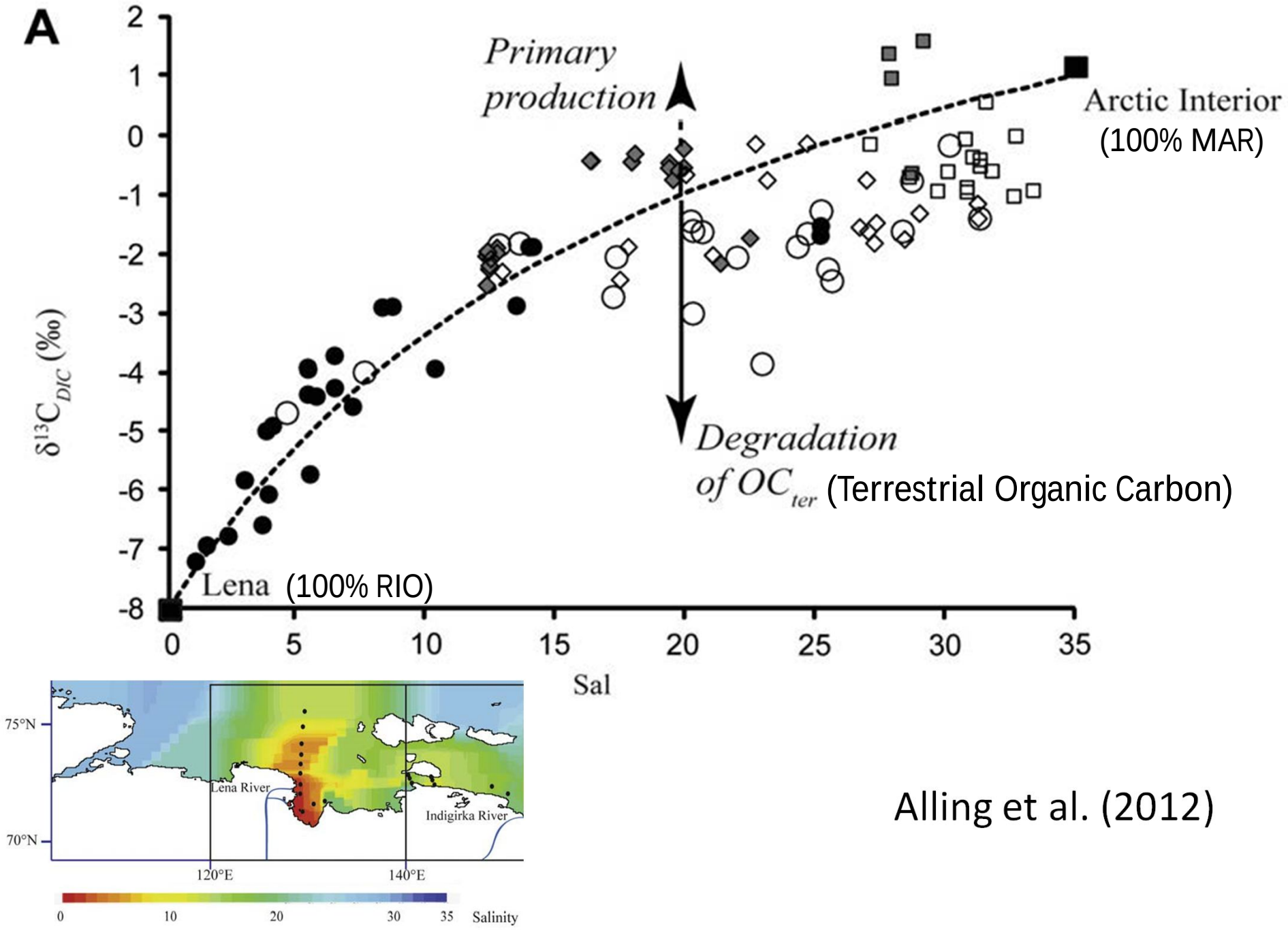
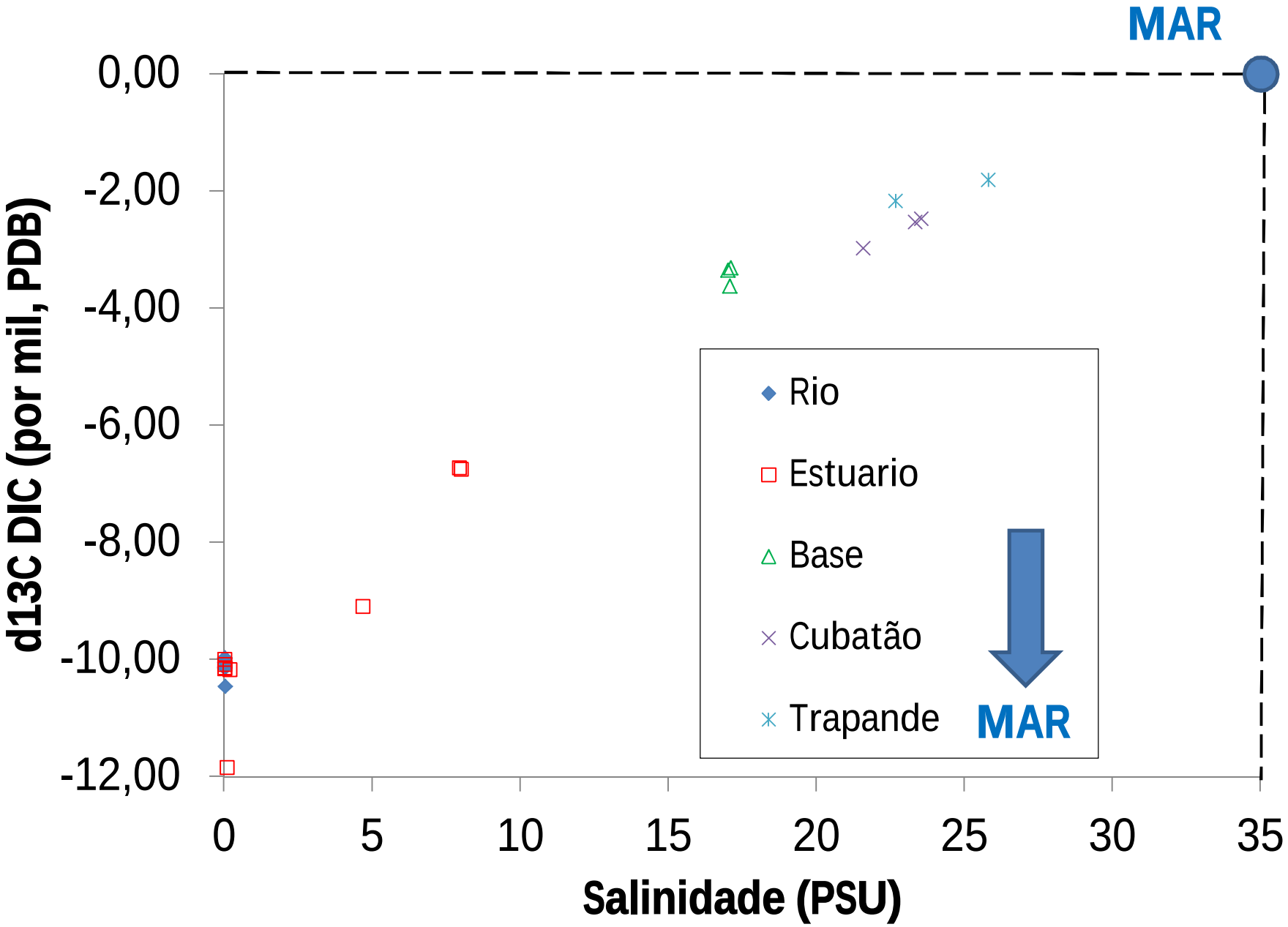
-25.0
-25.1
-25.2
-25.3
-48.1
-48.0
-47.9
-47.8
-47.7
-47.6
-47.5
-47.4
Longitude (°)



Alling et al. (2012)


MAR
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-2,00
-4,00
-6,00
-8,00
-10,00
-12,00
d13C DIC (por mil, PDB)
0
5
10
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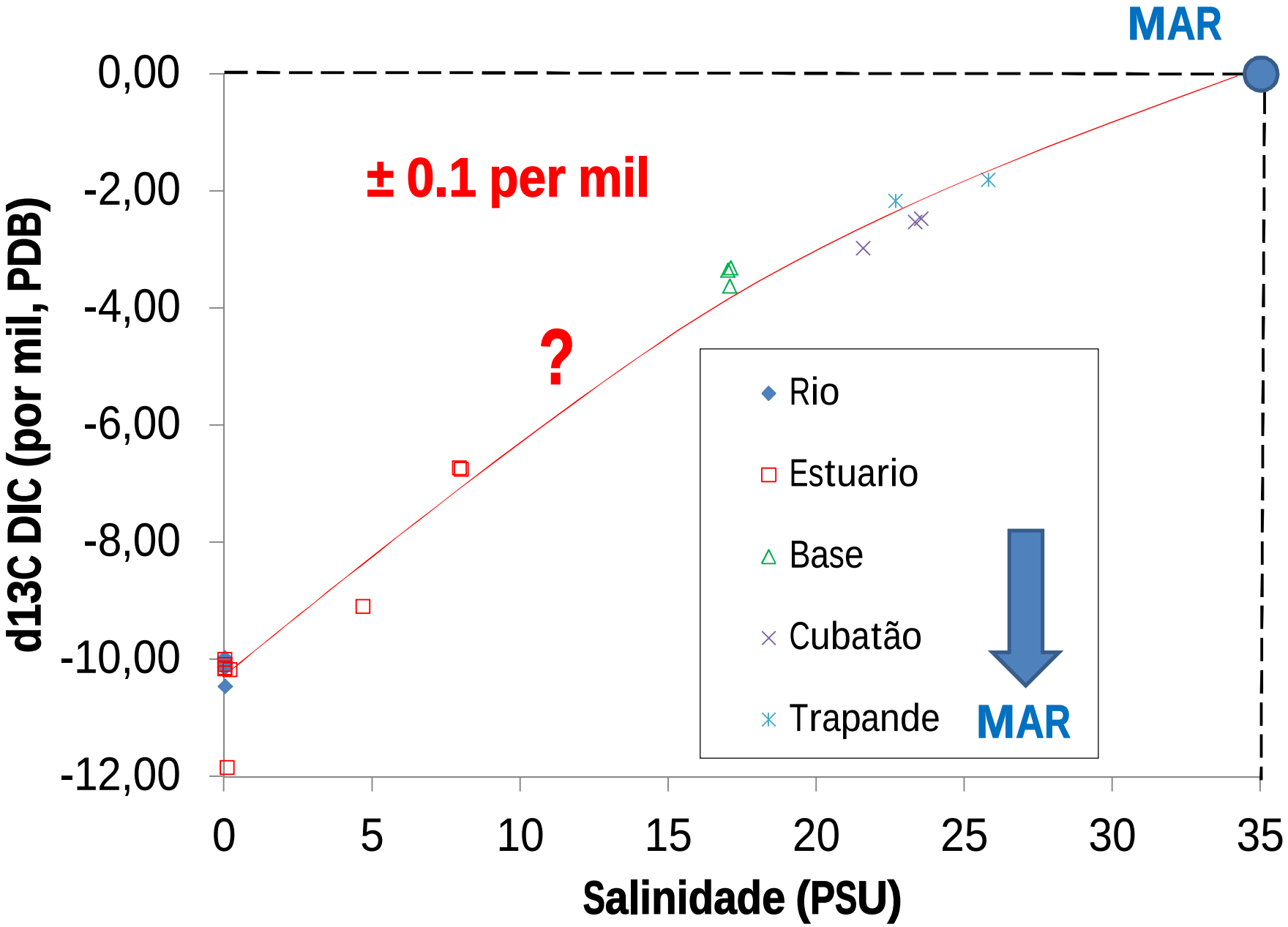
35
Salinidade (PSU)
Rio
Estuario
Base
Cubatão
Trapande
MAR


MAR
0,00
-2,00
-4,00
-6,00
-8,00
-10,00
-12,00
± 0.1 per mil
?
Rio
Estuario
Base
Cubatão
Trapande
MAR
0
5
10
15
20
25
30
35
d13C DIC (por mil, PDB)
Salinidade (PSU)

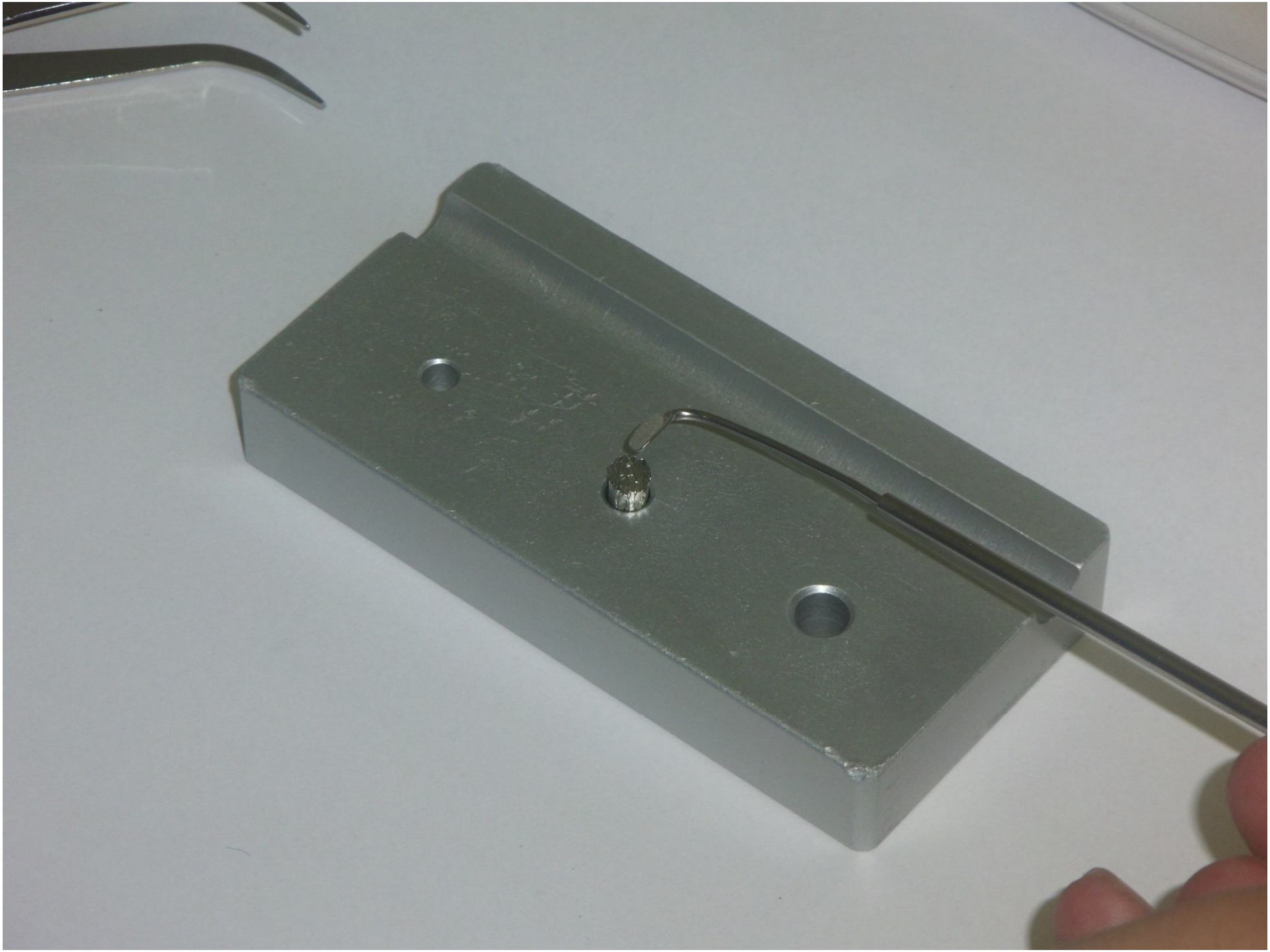
Sumario do curso

1. Analise isotópica de oxigênio em água do mar
2. Analise isotópica de hidrogênio em água do mar
3. Analise isotópica de carbonato de cálcio
4. Analise isotópica de carbono inorgânico dissolvido (DIC)
5. **Analise isotópica de material orgânico**

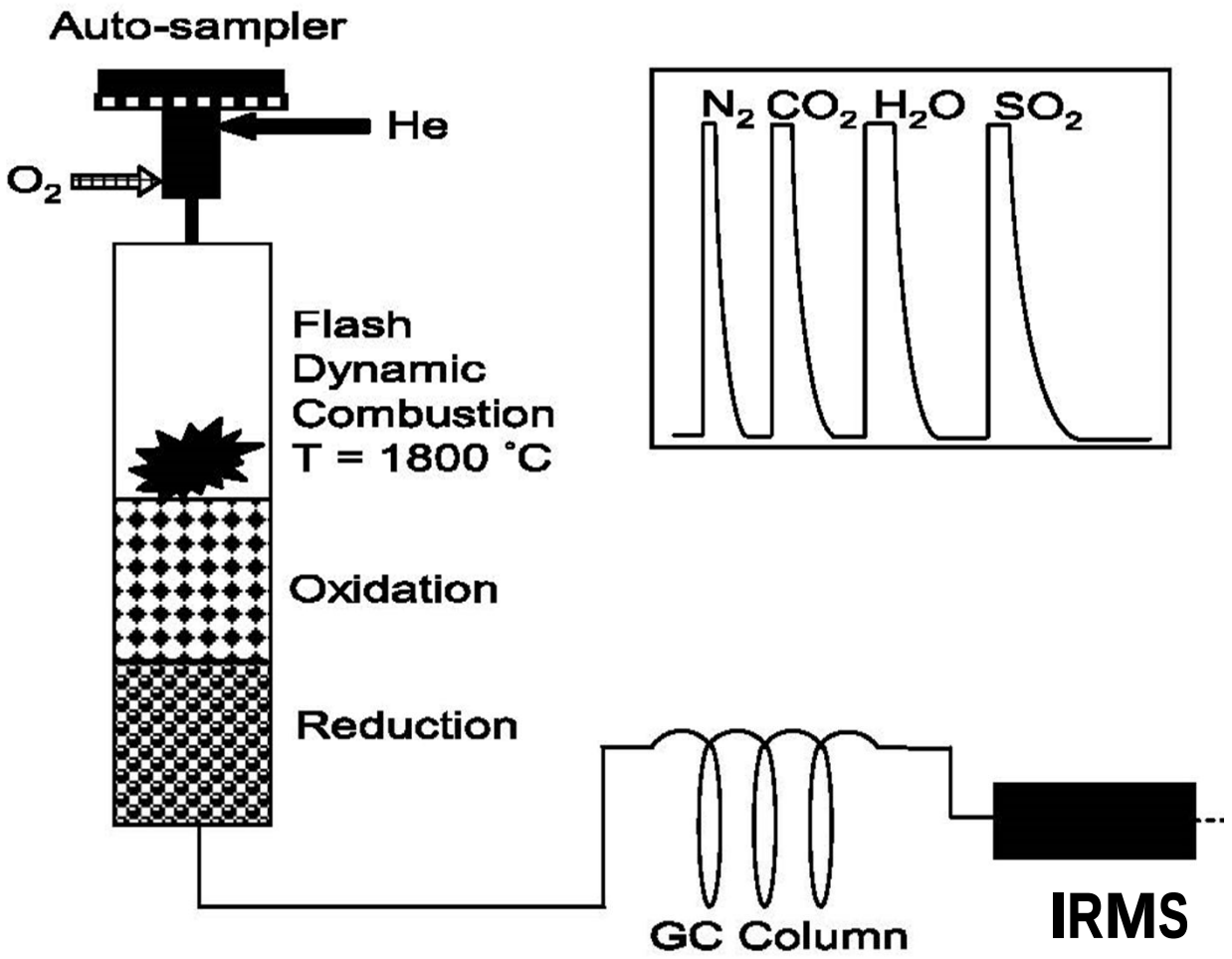
Elemental Combustion System
CHNS-O

Fluxo contínuo de He

Autosampler



CHNS Elemental Analysers



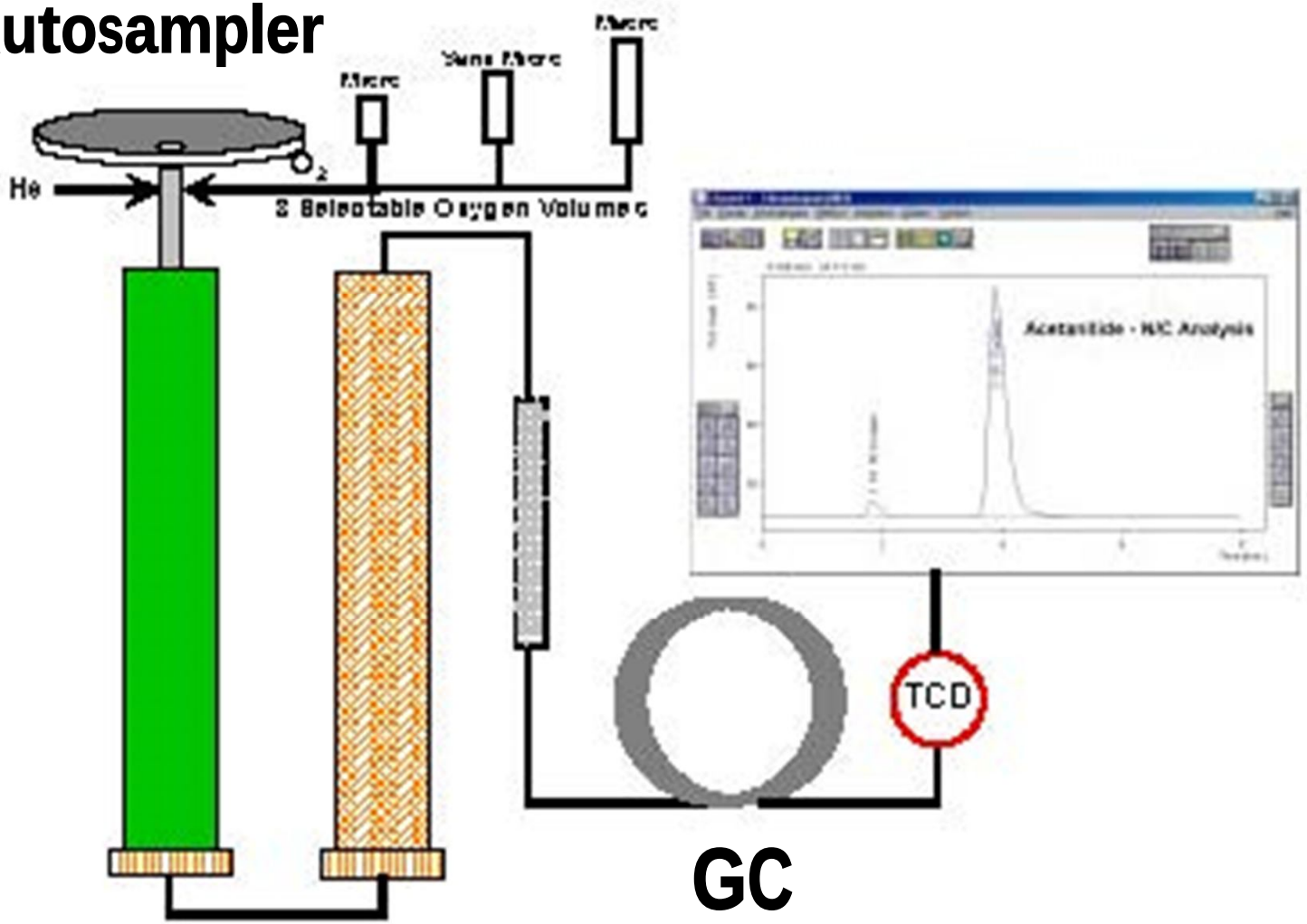
Oxidação (O_2 1800 °C)

$$M.O. \rightarrow NOx + CO_2 + H_2O + SO_2$$

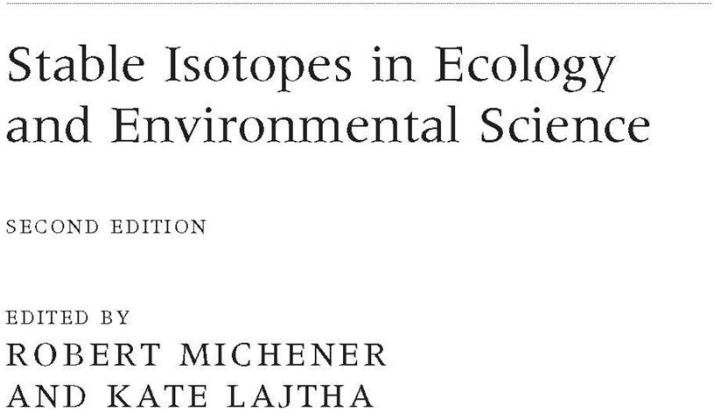
Redução (Cr 680 °C)

$$NOx \rightarrow N_2$$

Autosampler



Aplicação de isótopos estáveis para estudos de teia trófica



- **Gut content is not always a good reflection of what the animal might have been eating on a regular basis.**

- **Potential overestimation of prey with hard body parts, due to differential digestion of different types of prey.**

- **Gut content analysis requires the researcher to have a good taxonomic knowledge of nearly all the macroorganisms present. In aquatic ecosystems, predators can be remarkably undiscriminating in what they eat.**

ISÓTOPOS ESTÁVEIS DO NITROGÊNIO

^{14}N (99.6%)

^{15}N (0.4%)

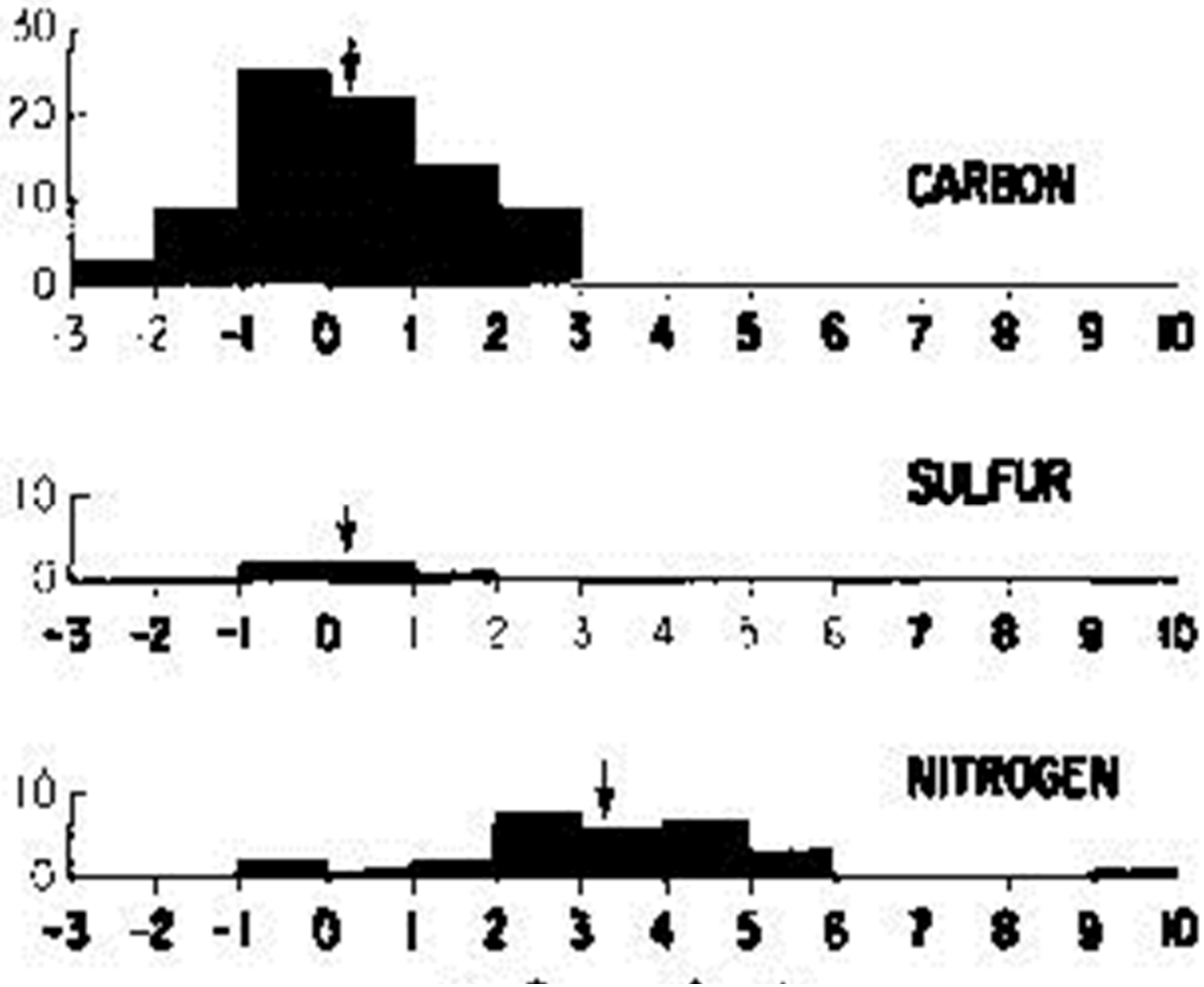
$\delta^{15}N$

ISÓTOPOS ESTÁVEIS DO NITROGÊNIO

$\delta^{15}N$

Padrão internacional = ar atmosférico
(0 por mil, por definição)

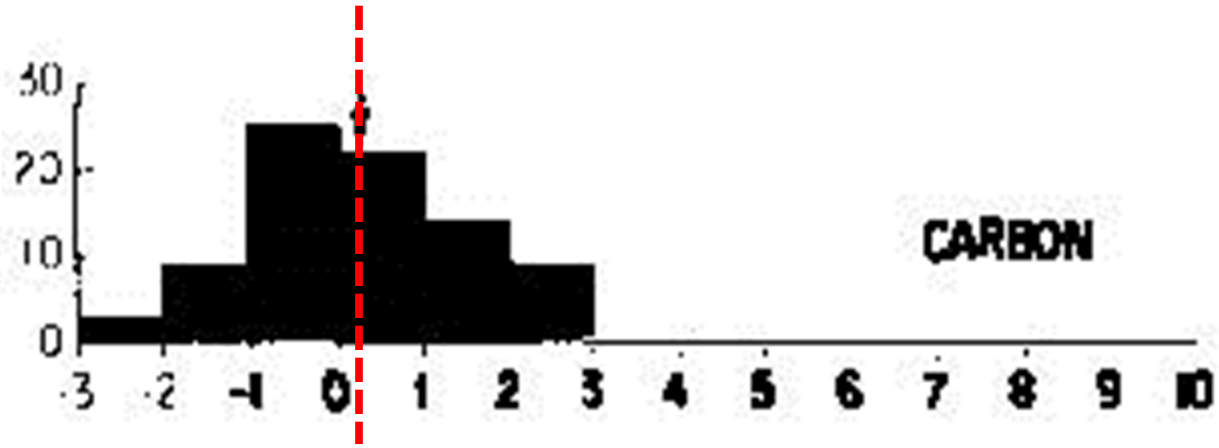
N amostral



$$\Delta (\delta_{animal} - \delta_{diet})$$

Peterson and Fry, 1985

N amostral



$$\delta^{13}C_{animal} - \delta^{13}C_{diet}$$

"Somos o que comemos"

Peterson and Fry, 1985

Oecologia (1991) 86:598–600

Oecologia
© Springer-Verlag 1991

Polar bears make little use of terrestrial food webs: evidence from stable-carbon isotope analysis

M.A. Ramsay[1,] * **and K.A. Hobson**[2]

[1] Freshwater Institute, Department of Fisheries and Oceans, 501 University Crescent, Winnipeg, Manitoba R3T 2N6, Canada
[2] Department of Zoology, University of Manitoba, Winnipeg, Manitoba R3T 2N2, Canada

Received April 9, 1990 / Accepted in revised form January 28, 1991

Summary. The mean stable-carbon isotope ratios ($\delta^{13}C$) for polar bear *(Ursus maritimus)* tissues (bone collagen −15.7‰, muscle −17.7‰, fat −24.7‰) were close to those of the same tissues from ringed seals *(Phoca hispida)* (−16.2‰, −18.1‰, and −26.1‰, respectively), which feed exclusively from the marine food chain. The $\delta^{13}C$ values for 4 species of fruits to which polar bears have access when on land in summer ranged from −27.8 to −26.2‰, typical of terrestrial plants in the Arctic. An animal's $\delta^{13}C$ signature reflects closely the $\delta^{13}C$ signature of it's food. Accordingly, the amount of food that polar bears consume from terrestrial food webs appears negligible, even though some bears spend 1/3 or more of each year on land during the seasons of greatest primary productivity.

Key words: Polar bears – $\delta^{13}C$ – Fasting – Arctic food web – *Ursus maritimus*

cides with the annual season of greatest primary productivity. Although anecdotal accounts show that some polar bears eat limited amounts of sedges (*Carex* spp.) and berries while on land (Russell 1975; Knudsen 1978; Lunn and Stirling 1985), the importance of terrestrial foods in their annual diet is not known.

A test to determine whether an animal has been feeding on a terrestrial or marine food web is to measure the stable-carbon isotopic composition of its tissues. The basis for this method is that the isotopic signature of carbon incorporated by photosynthetic pathways differs characteristically between terrestrial (C–3) and marine ecosystems and this difference is maintained through successive trophic levels in both food webs (Chisholm et al. 1982). Animal tissues such as bone collagen, muscle, and fat turn over carbon at different rates because of differences in their metabolic activity. Isotopic signatures in these tissues may thus vary in such a way as to allow long-term or recent-past dietary information (Tieszen et al. 1983). Stable-carbon isotope signatures are expressed as parts per mil (‰) relative to the Pee Dee Belemnite

Polar bears make little use of terrestrial food webs: evidence from stable-carbon isotope analysis

M.A. Ramsay[1, *] **and K.A. Hobson**[2]

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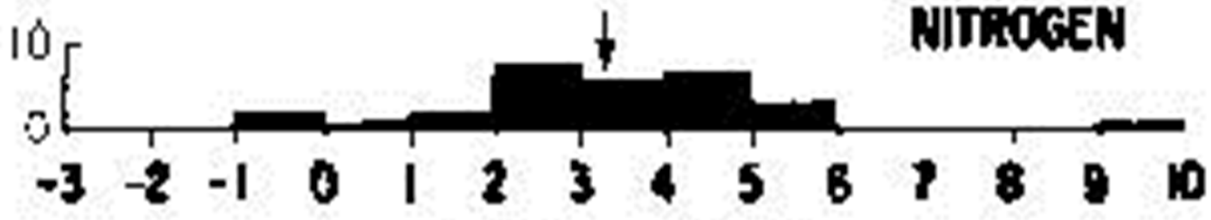
N amostral

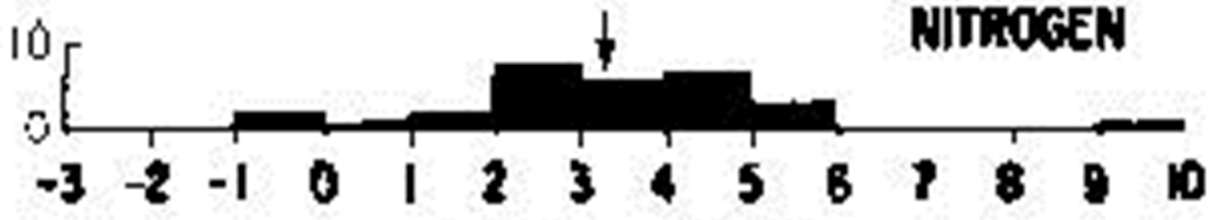


$$\delta^{15}N_{animal} - \delta^{15}N_{diet}$$

A pele e a gordura do animal tendem ser **enriquecidos em ^{15}N**, pois o ^{14}N é eliminado junto á ureia

Peterson and Fry, 1985

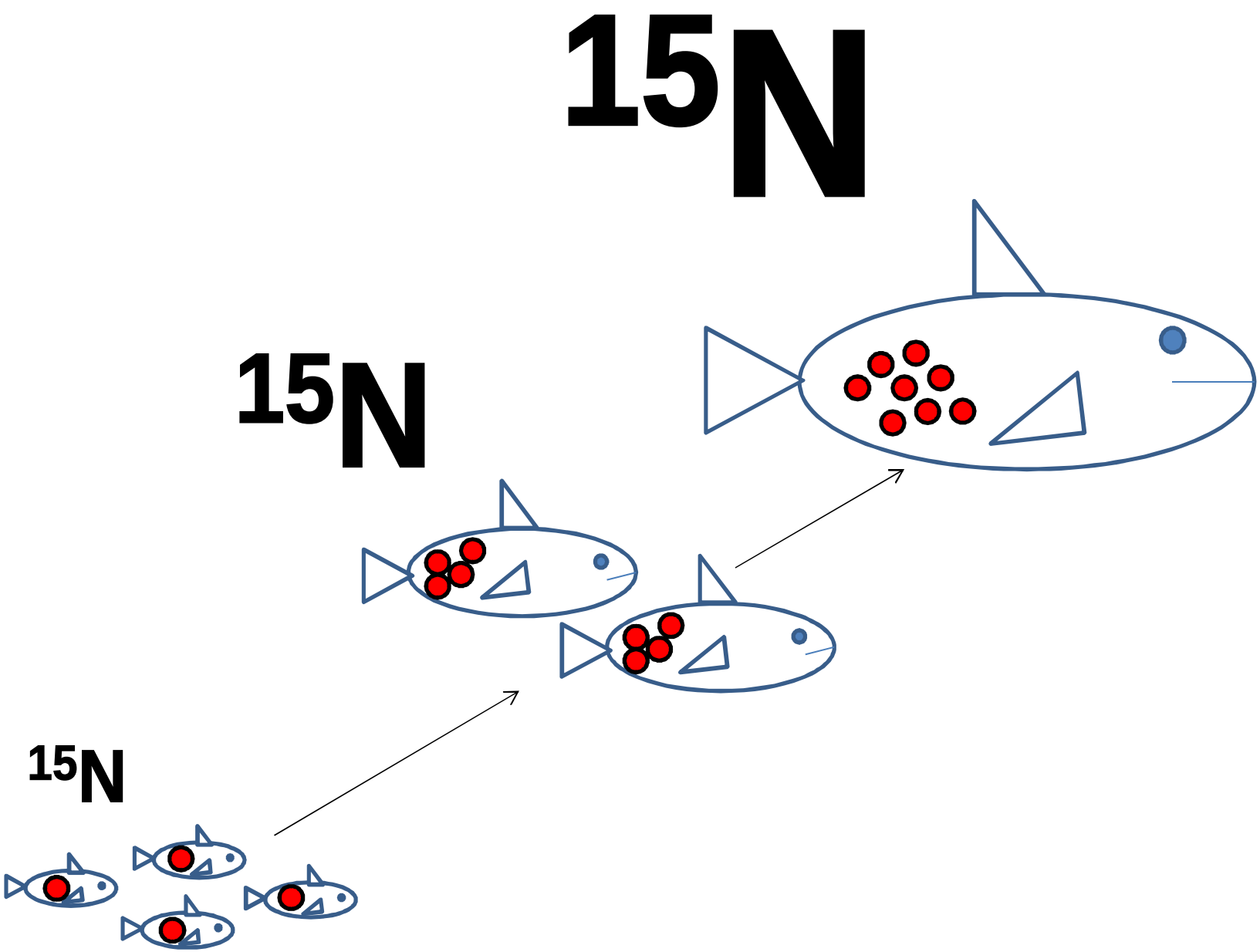
N amostral



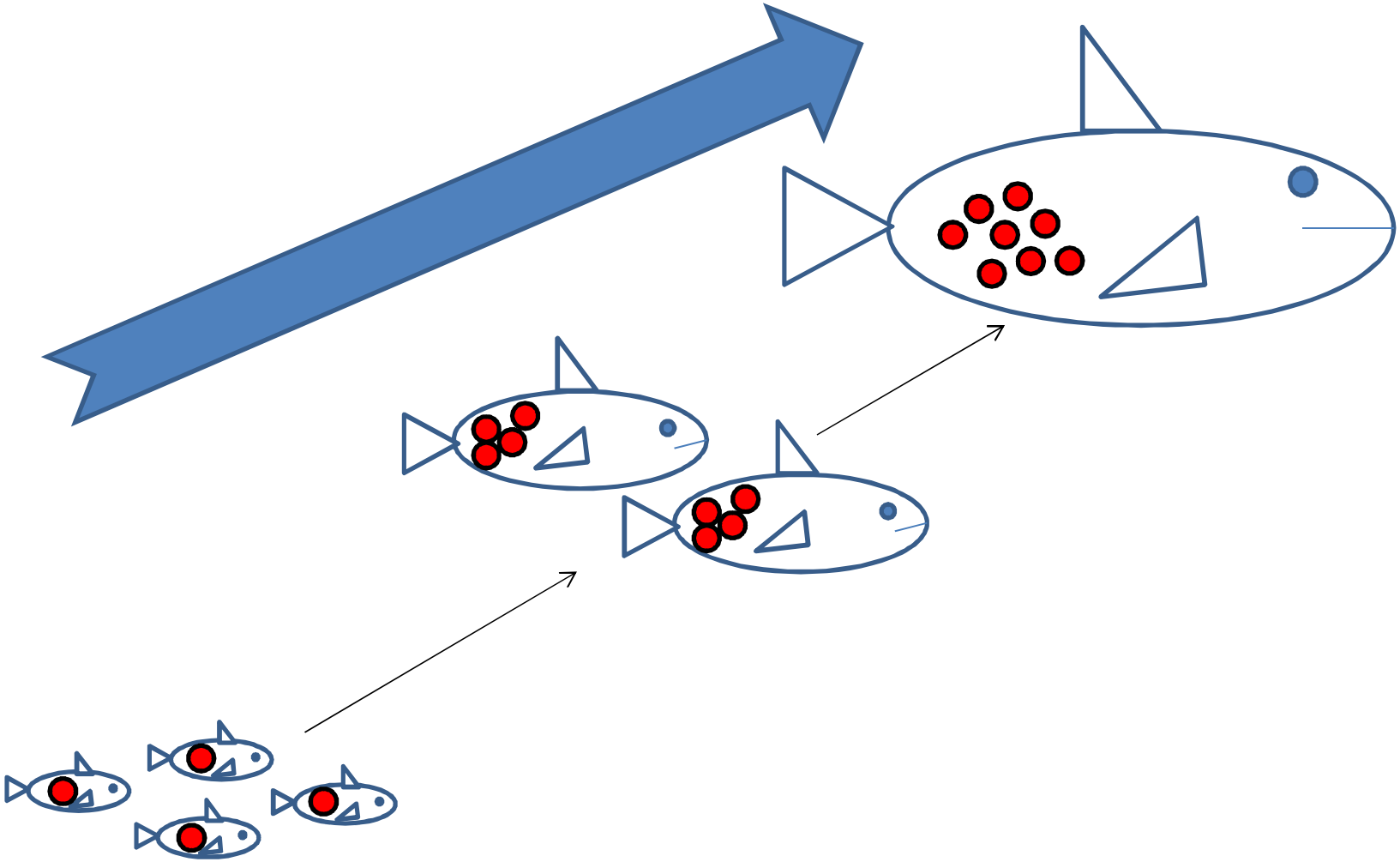
$$\delta^{15}N_{animal} - \delta^{15}N_{diet}$$

"Somos o que comemos, menos o que excretamos"

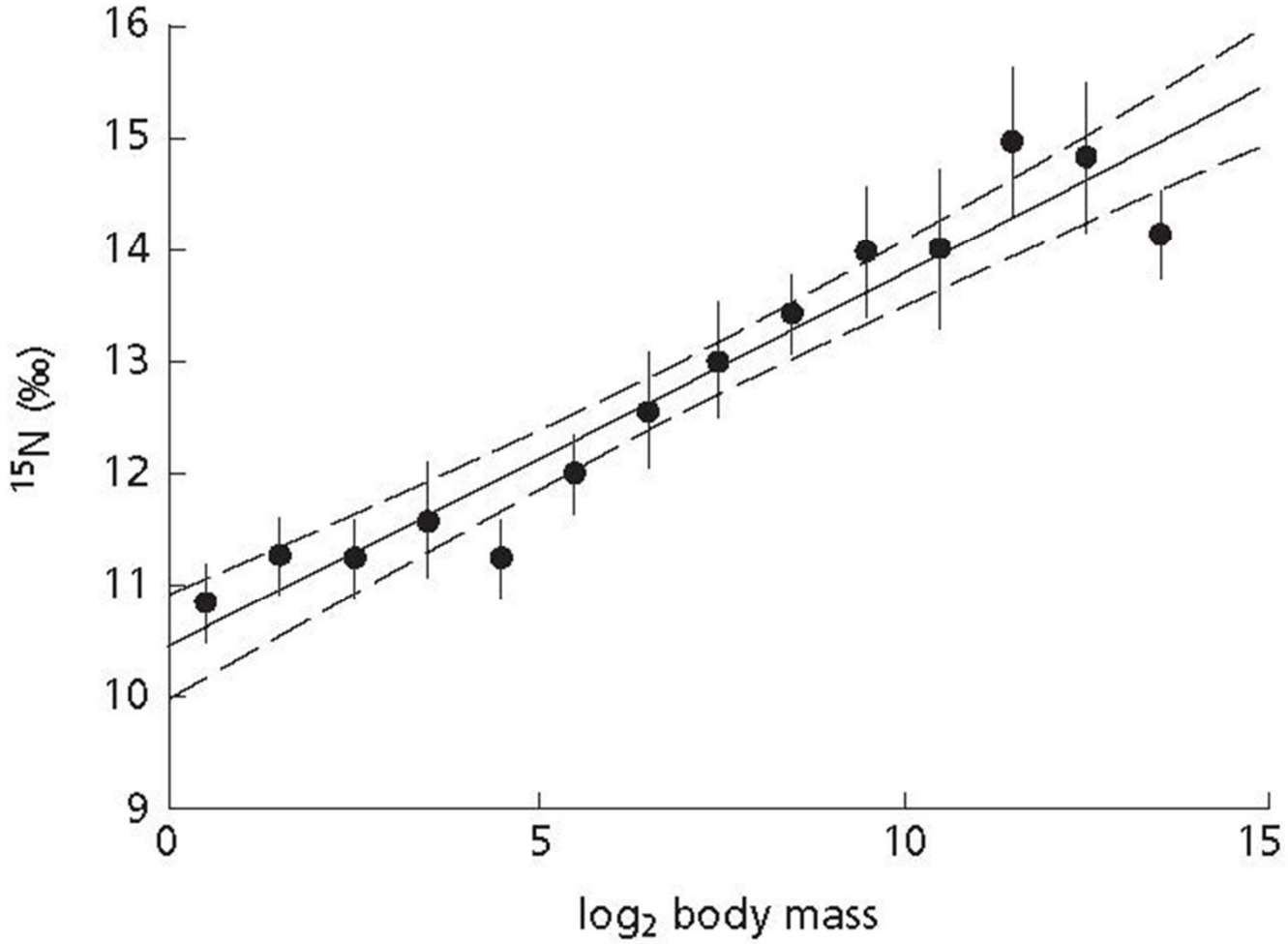
Peterson and Fry, 1985


15N
15N
15N

$\delta^{15}N$ aumenta



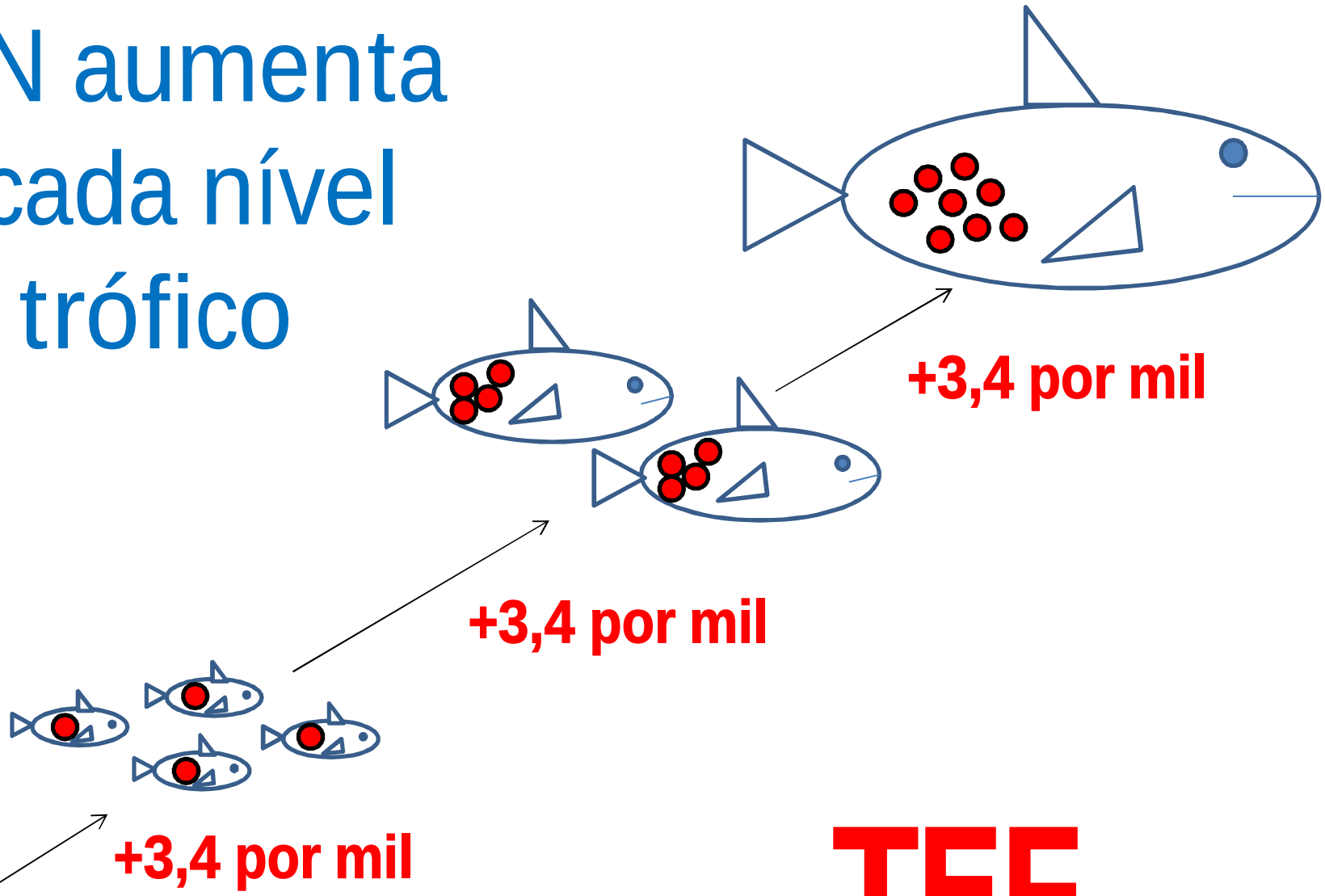
Relação entre $\delta^{15}N$ e tamanho em uma população de peixes



Michener & Lajtha Eds. Stable Isotopes in Ecology and Environmental Science

Trophic Enrichment Factor
TEF

$\delta^{15}N$ aumenta a cada nível trófico



TEF

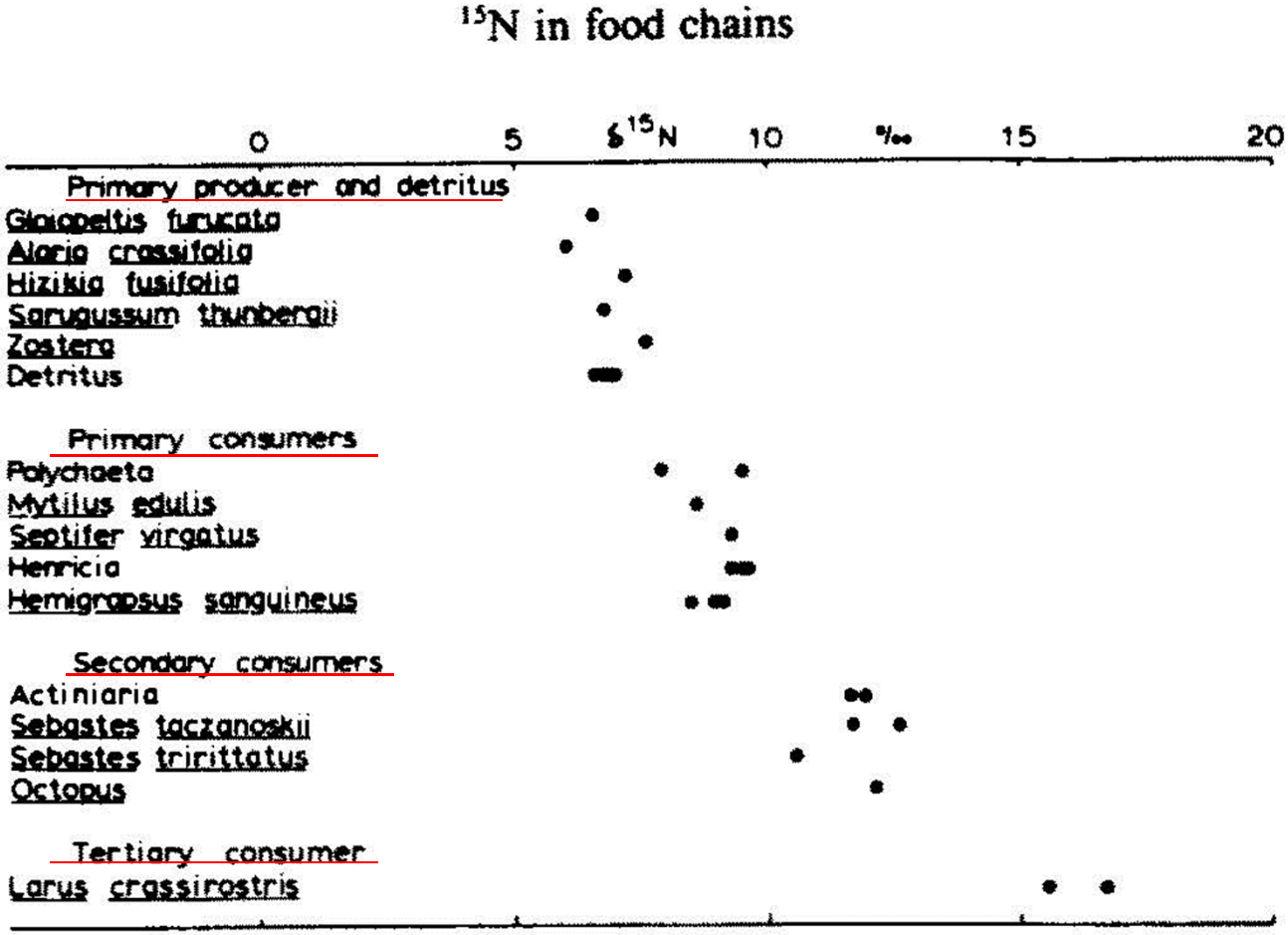
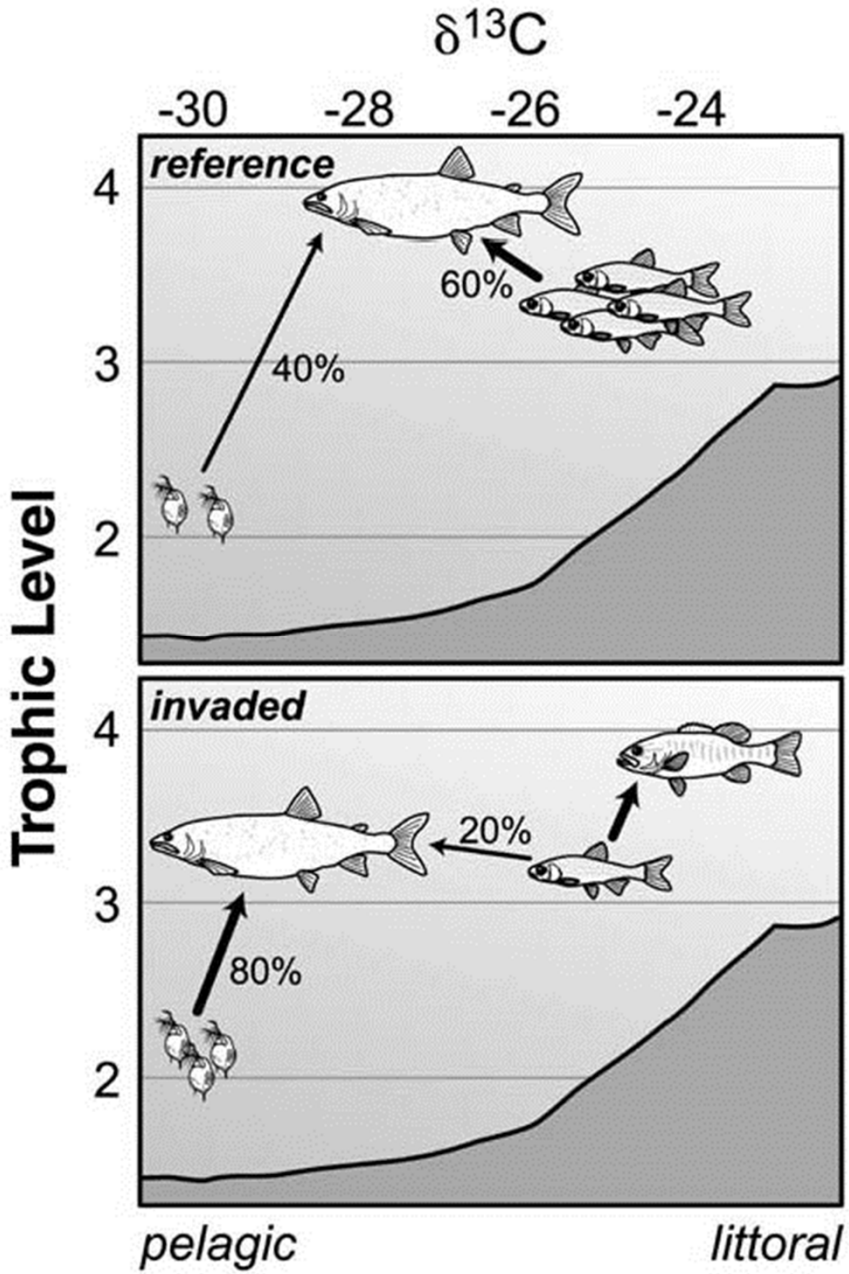


FIG. 2. The δ^{15}N of biota from Usujiri intertidal zone. The average for adults is shown for *Mytilus edulis* and *Septifer virgatus*.

Minagawa & Wada 1984



B. Fry - Stable Isotope Ecology (Springer)

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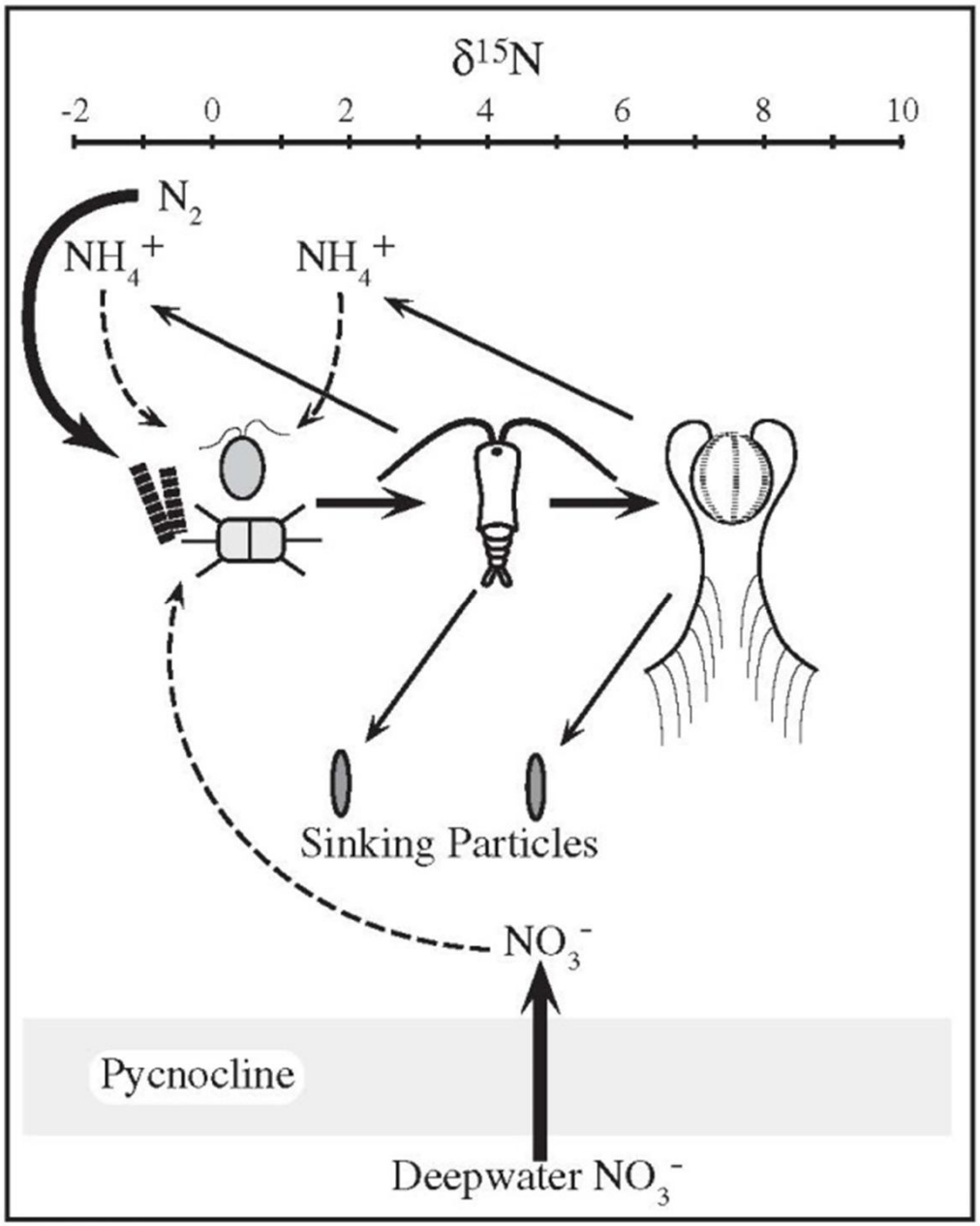


DENITRIFICATION

Efeito isotópico de denitrificação do nitrato sobre o valor de $\delta^{15}N$ do próprio nitrato

Denitrificação do nitrato

$$NO_3^- \longrightarrow N_2$$

$$2\,NO_3^- + 10\,e^- + 12\,H^+ \rightarrow N_2 + 6\,H_2O$$

Denitrificação do nitrato

$$NO_3^- \longrightarrow N_2$$

$$2\ {}^{14}NO_3^- + 10\ e^- + 12\ H^+ \rightarrow N_2 + 6\ H_2O$$

$$2\ {}^{15}NO_3^- + 10\ e^- + 12\ H^+ \rightarrow N_2 + 6\ H_2O$$

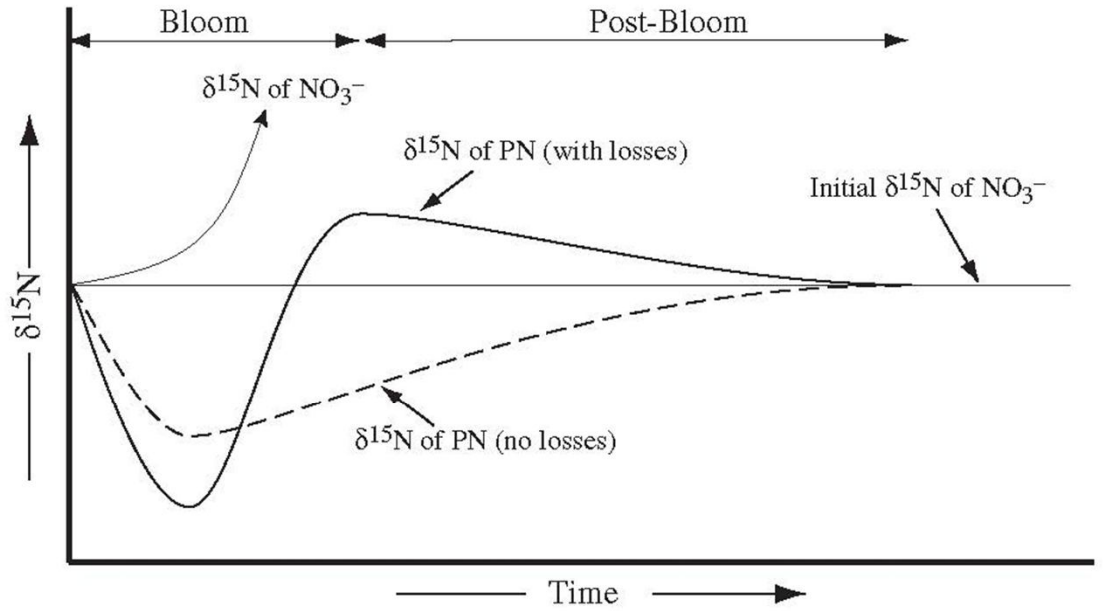
Denitrificação do nitrato

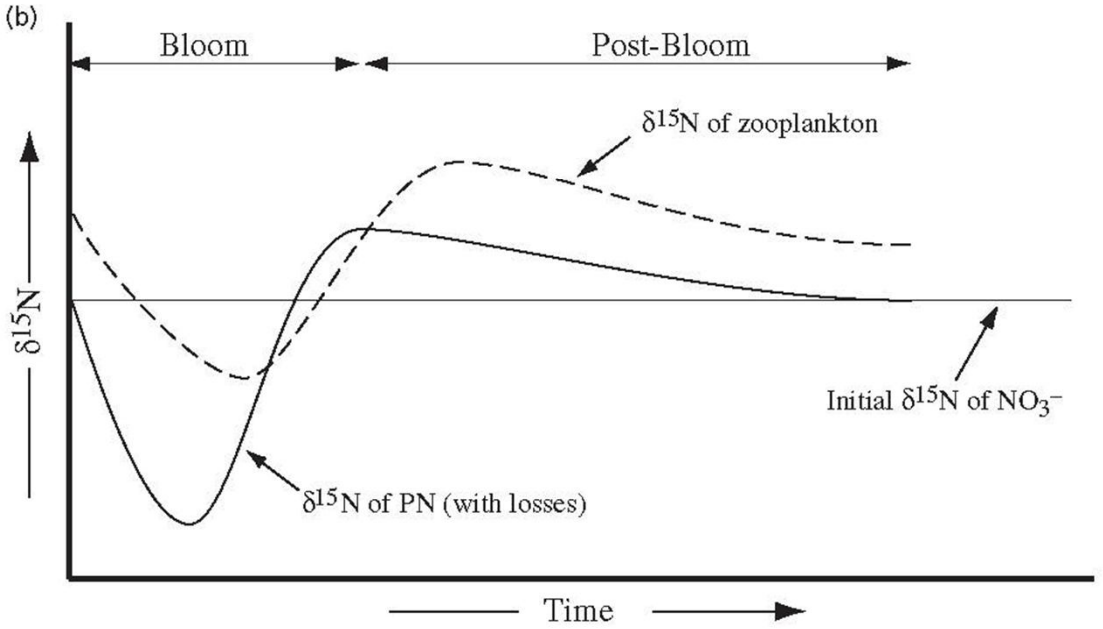
SUBSTRATO **PRODUTO**

NO_3^- ⟶ N_2

Enriquecido em ^{15}N **Enriquecido em ^{14}N**

Efeitos ligados a proliferação de produtores primários (*blooms*)





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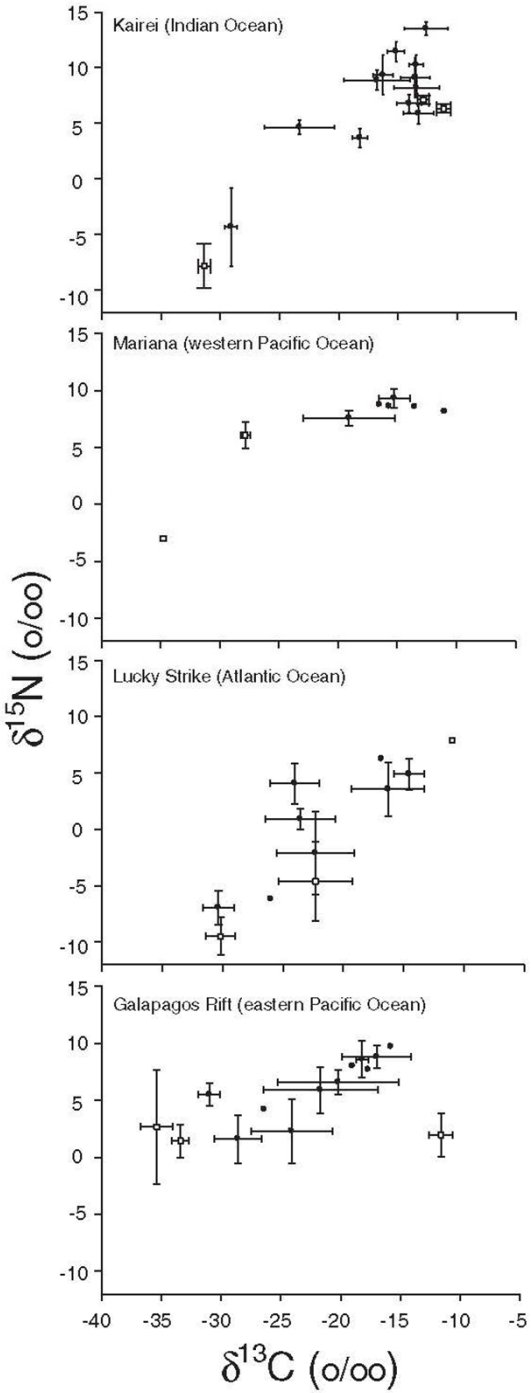
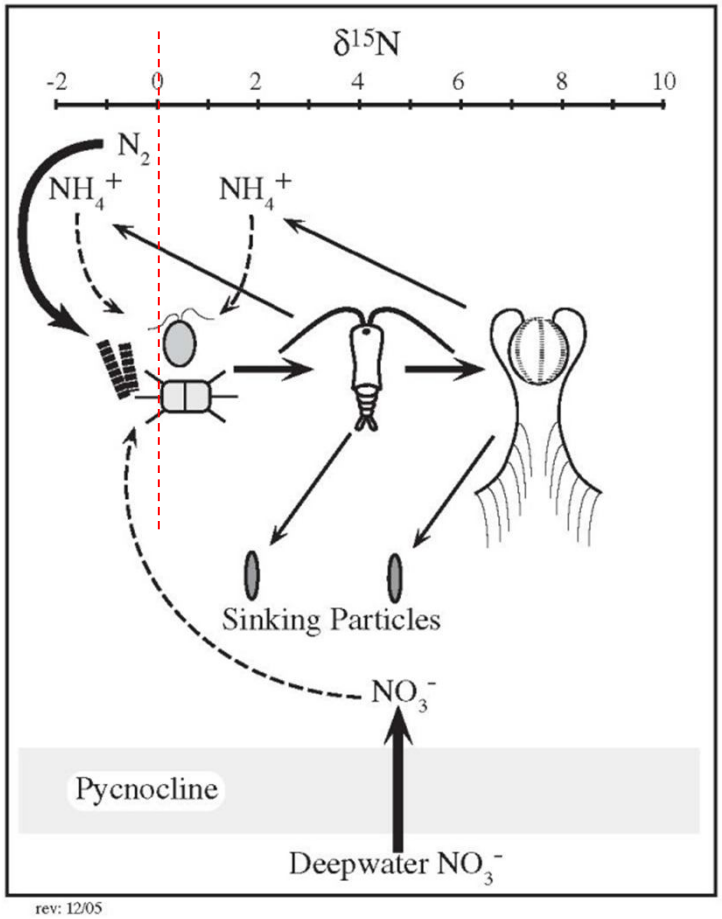
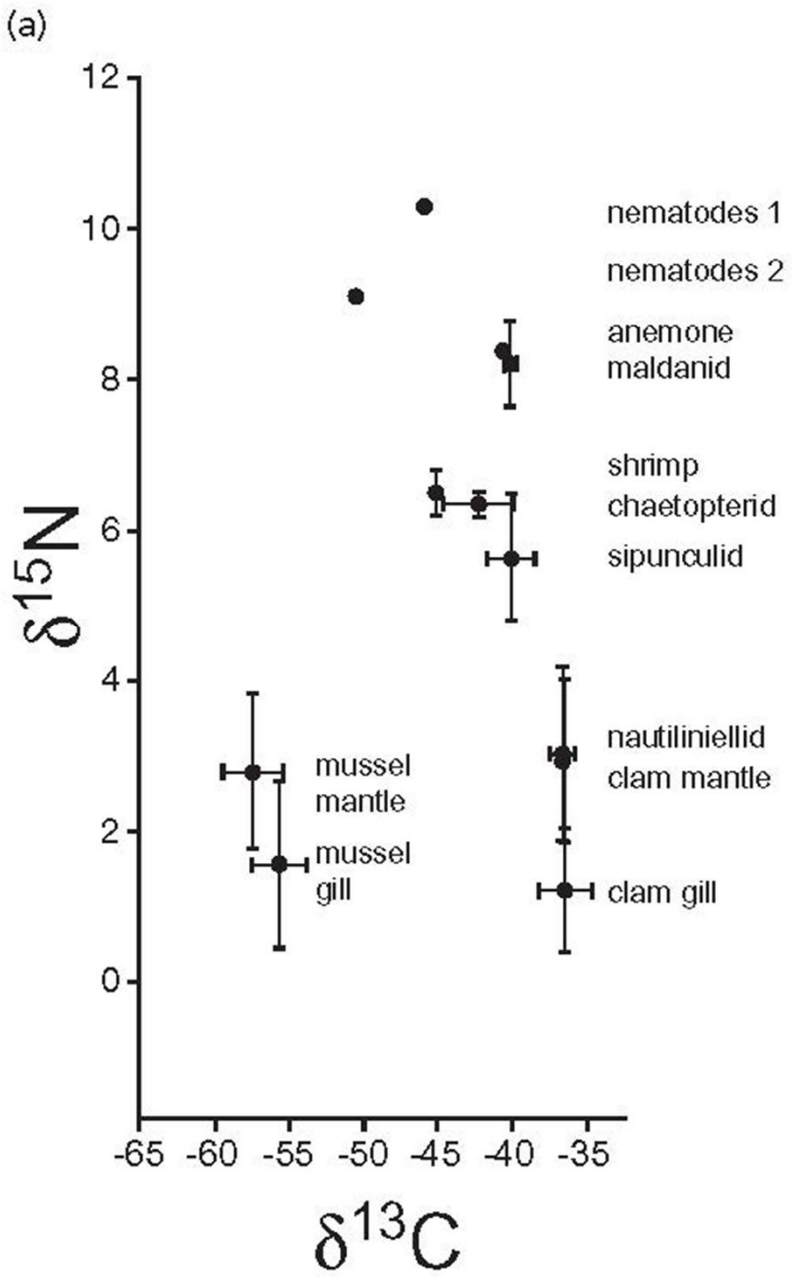




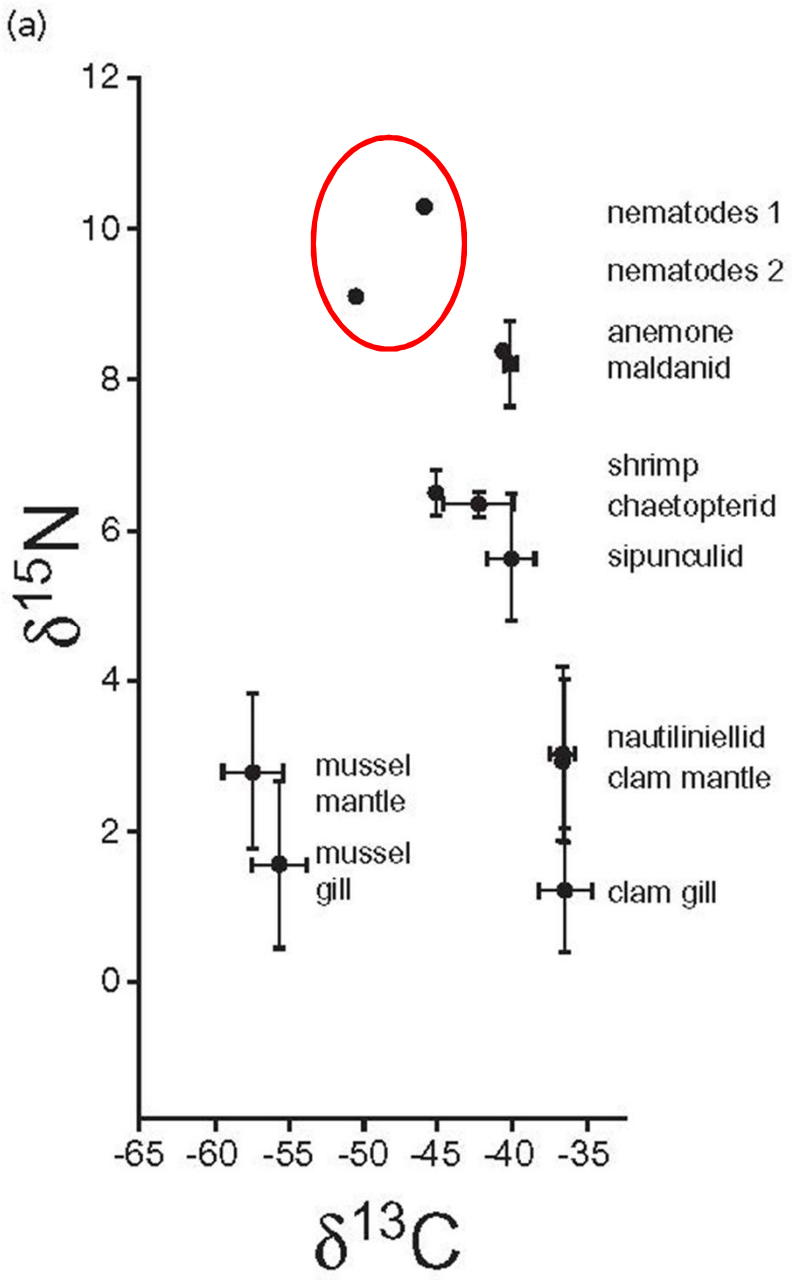
Sistemas hidrotermais: comportamentos peculiares

Michener & Lajtha Eds. Stable Isotopes in Ecology and Environmental Science



Sistemas hidrotermais: comportamentos peculiares

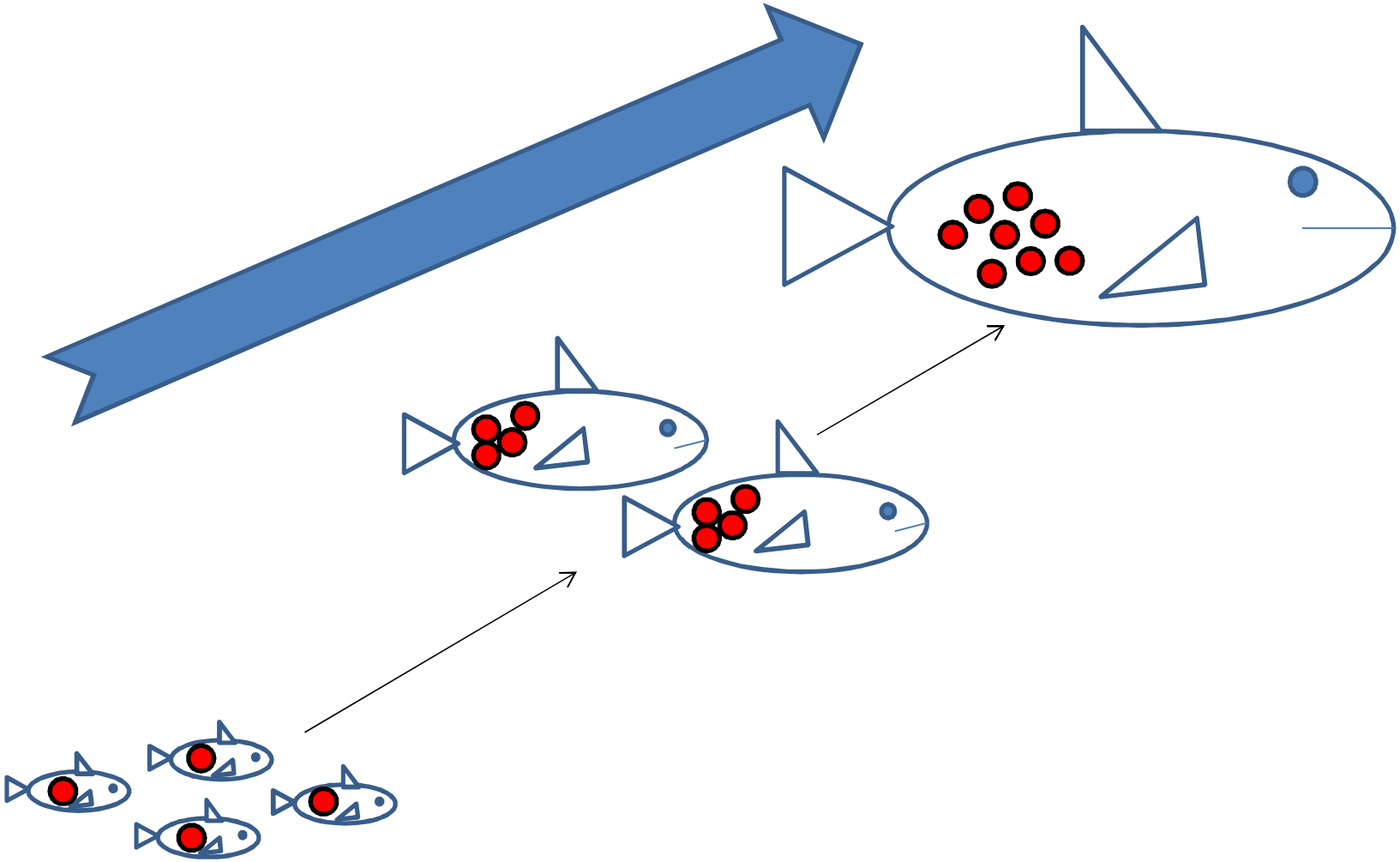
Michener & Lajtha Eds. Stable Isotopes in Ecology and Environmental Science



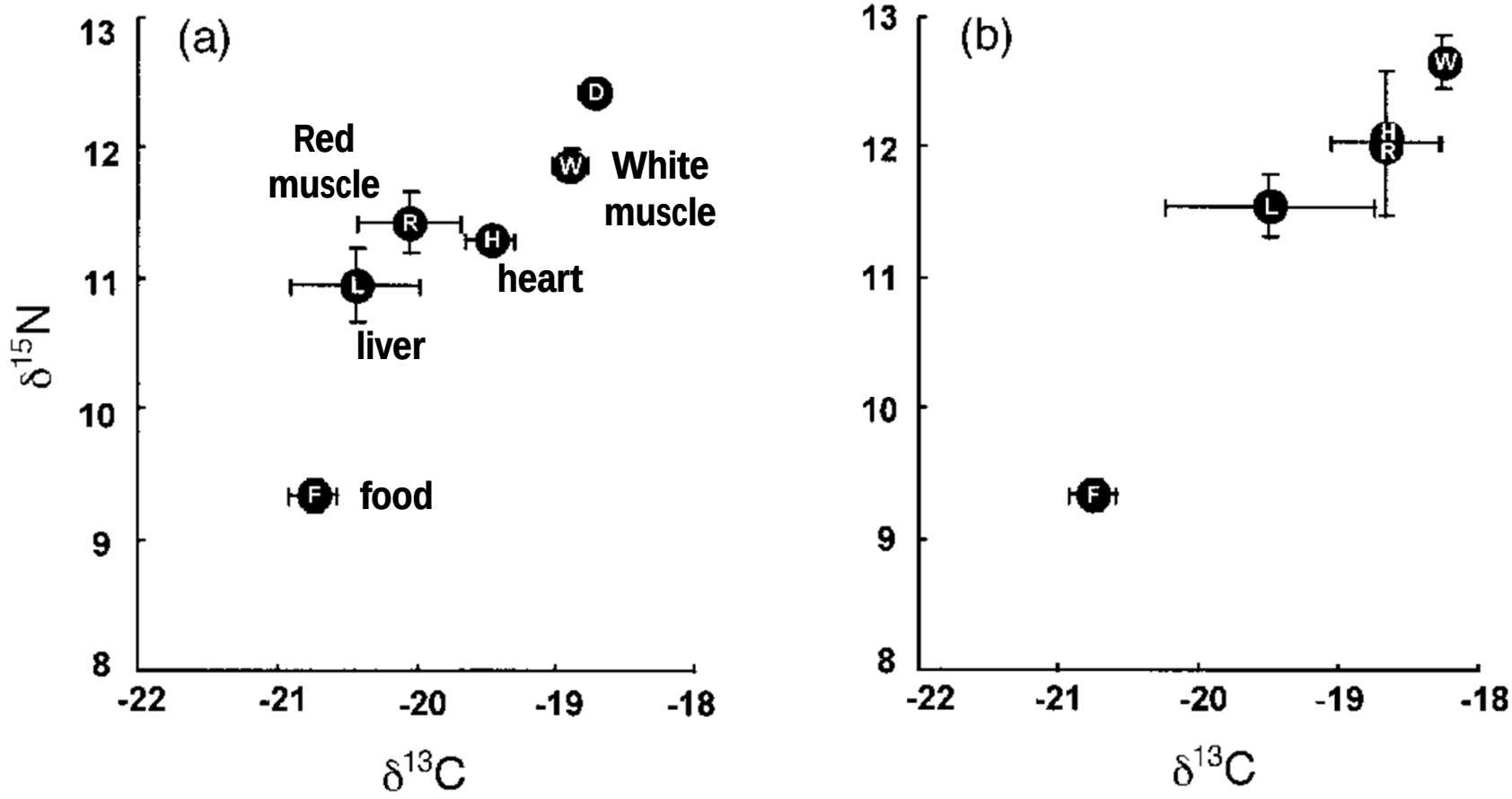
Sistemas hidrotermais: comportamentos peculiares

Conceito de *turn-over time* em tecidos de organismos

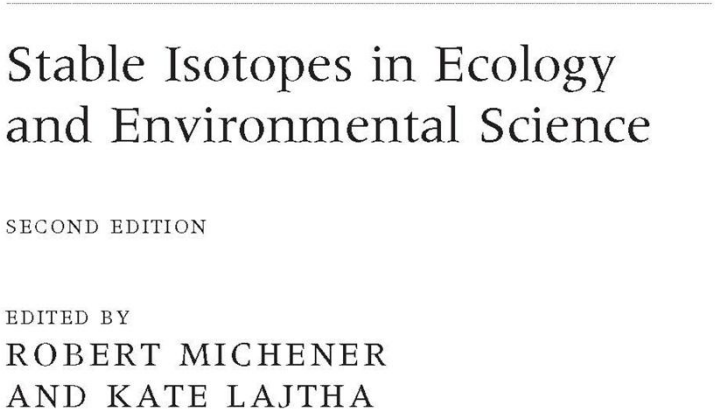
$\delta^{15}N$ aumenta



Isotopic variations among different tissues and variation in turnover time among different tissues



Pinnegar & Polunin (1999)



Given these problems, it would seem prudent to

- **either sample the entire organism to get an integrated isotopic value,**
- **to sample several tissue types covering a range of tissue turnover times,**
- **or consistently sample a particular tissue, as this is likely to show the least bias due to heterogeneity (e.g., lipid composition).**

Estudos de caso

Mammalian Biology 81 (2016) 571–578

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Mammalian Biology

journal homepage: www.elsevier.com/locate/mambio

Original investigation

What are you eating? A stable isotope insight into the trophic ecology of short-beaked common dolphins in the Southwestern Atlantic Ocean

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Stable isotopes
Trophic ecology
Argentina

ABSTRACT

As the feeding habits of marine mammals are particularly difficult to observe, stable isotope analysis has become an essential tool to investigate several aspects of dolphin's trophic ecology. Short-beaked common dolphins (*Delphinus delphis*) occur off Argentina coastline from 36°S to 42°S, but nothing is known about latitudinal changes in diet. A stable isotope analysis has been performed on three short-beaked common dolphins groups: Buenos Aires group (n = 20), Golfo San Matías inshore group (n = 24), and Golfo San Matías offshore group (n = 16). Highly significant differences in average $\delta^{15}N_{skin}$ values were found among the three groups of dolphins considered. Differences were also statistically significant for the average $\delta^{13}C_{skin}$ and Suess corrected $\delta^{13}C_{skin}$ values. Furthermore, no significant differences were found in the $\delta^{13}C_{skin}$ values of the dolphins from the inshore group sampled during summer and winter seasons at Golfo San Matías, although the $\delta^{15}N_{skin}$ values changes seasonally, suggesting that probably dolphins forage at a higher trophic level in summer. The SIAR model indicated that juvenile Argentine hake (*Merluccius hubbsi*) had the highest mean feasible contribution to the diet of short-beaked common dolphins inhabiting the continental shelf off Buenos Aires and northern Patagonia, whereas the Argentine anchovy (*Engraulis anchoita*) did it for short-beaked common dolphins inhabiting the Golfo San Matías The results obtained in this study increase the knowledge of trophic ecology for high trophic level predators such as short-beaked common dolphins along the Argentina coast in support of effective conservation and management programs.

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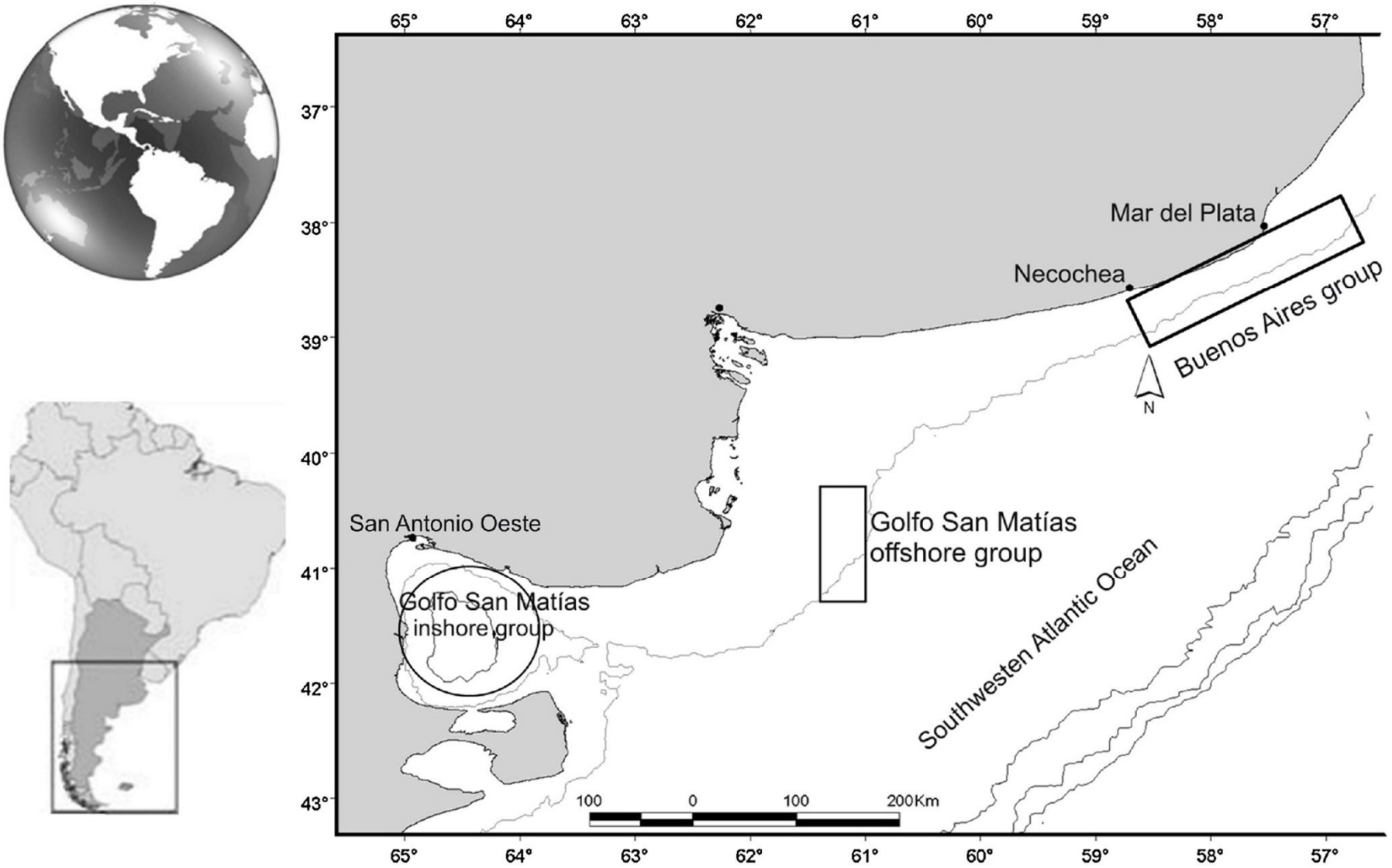
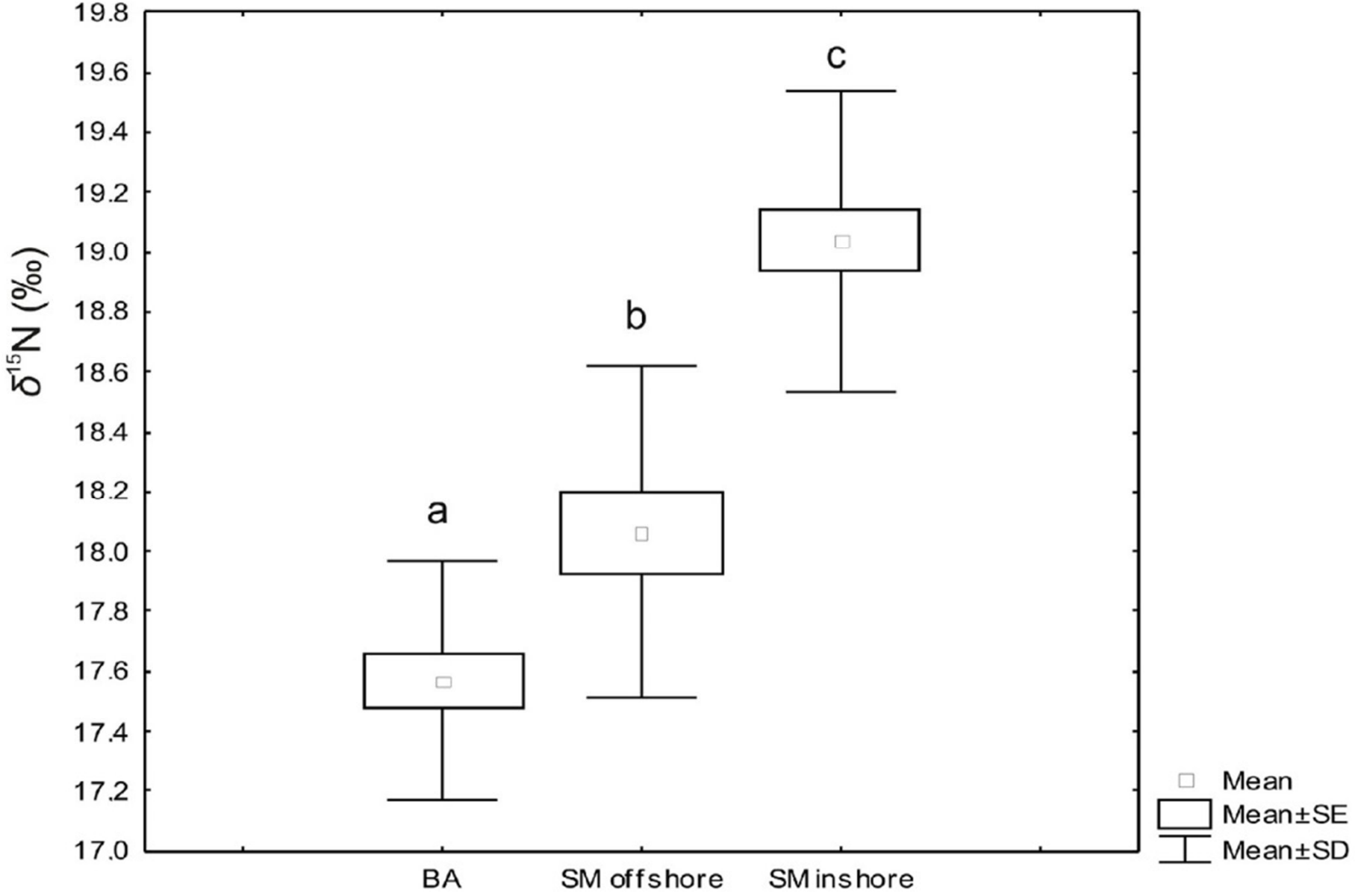
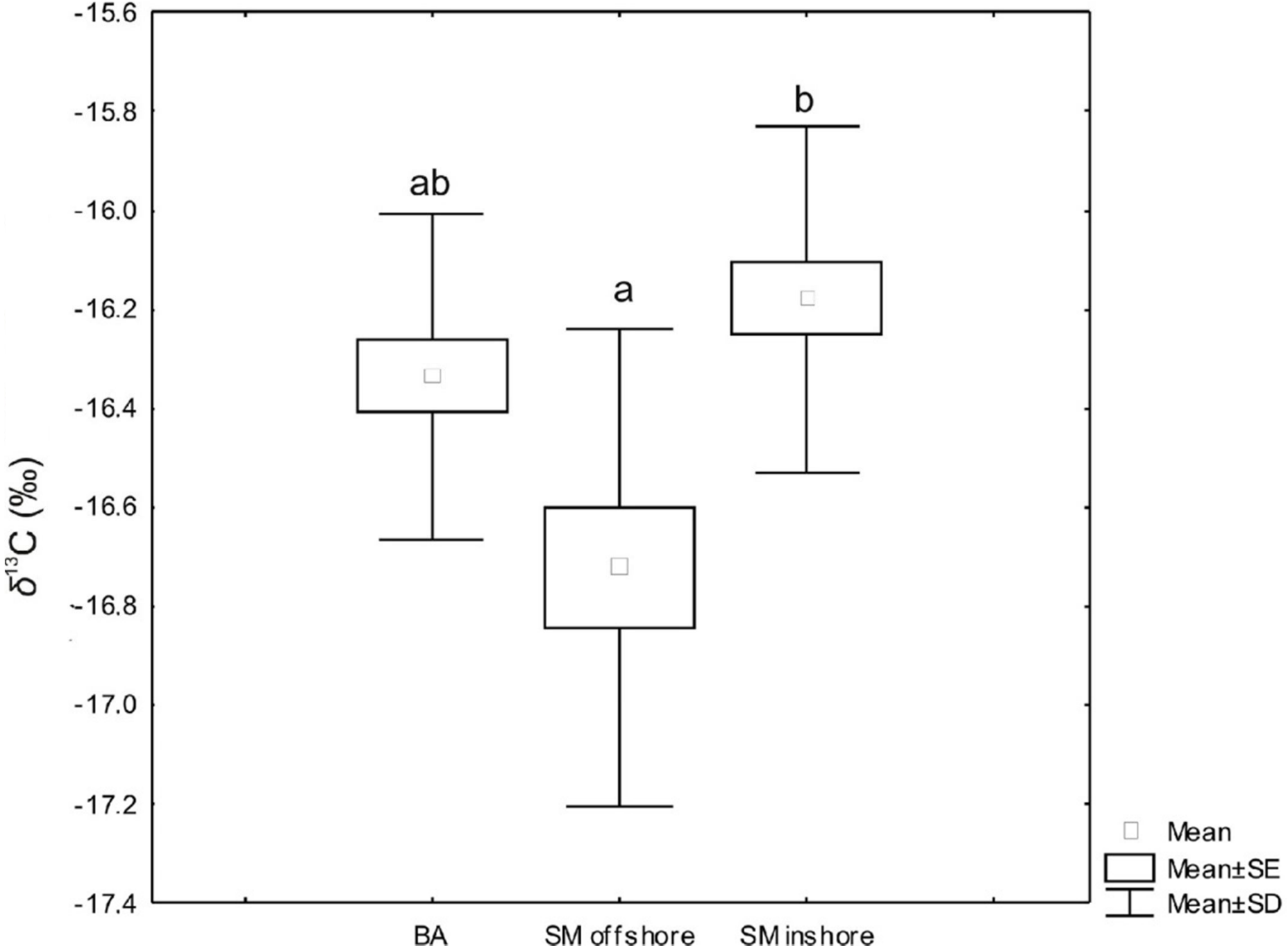
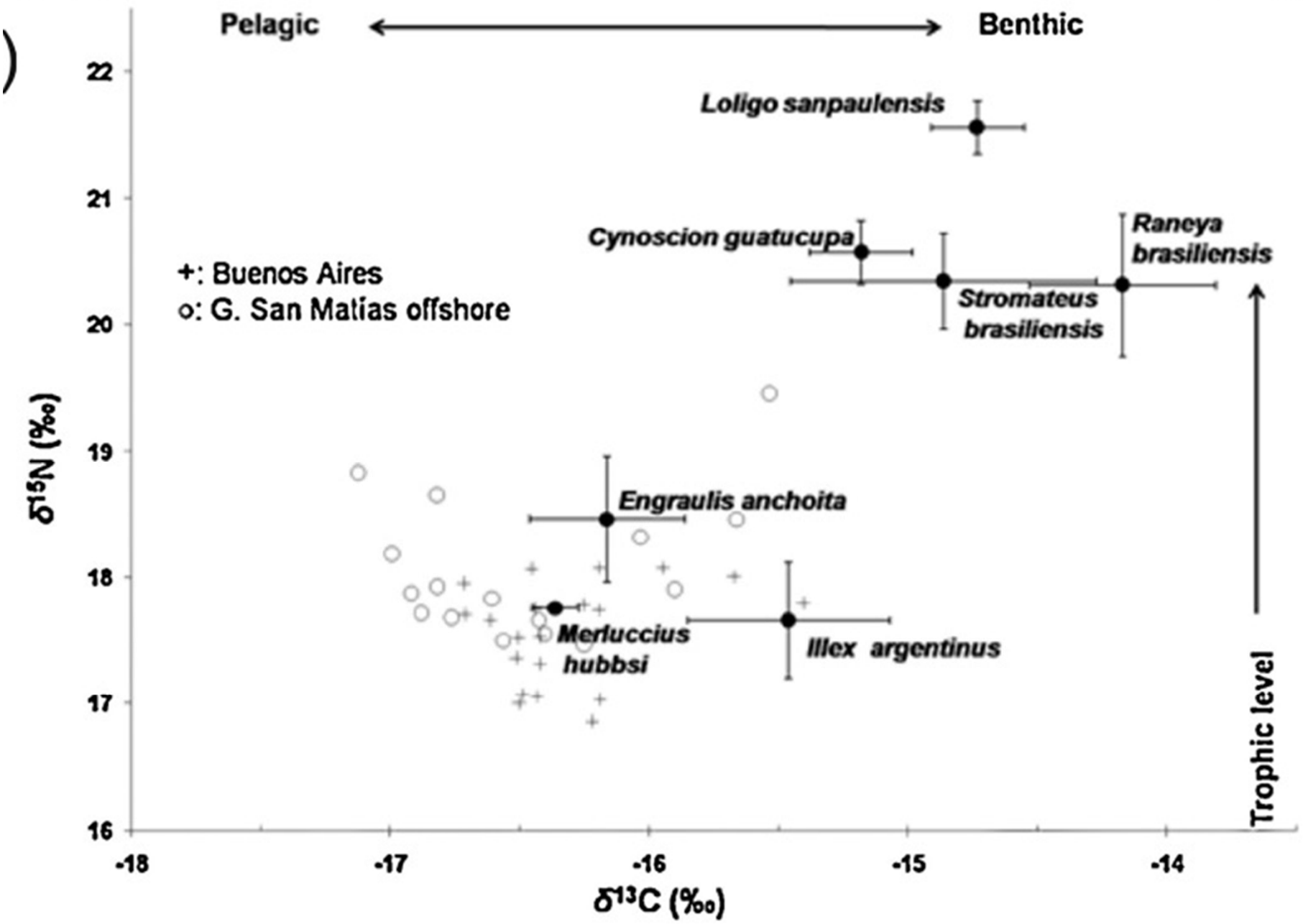
Questão cientifica:

o que comem os golfinhos ?


65°
64°
63°
62°
61°
60°
59°
58°
57°
37°
38°
39°
40°
41°
42°
43°
Mar del Plata
Necochea
Buenos Aires group
N
San Antonio Oeste
Golfo San Matías
inshore group
Golfo San Matías
offshore group
Southwesten Atlantic Ocean
100
0
100
200Km


19.8
19.6
19.4
19.2
19.0
18.8
18.6
18.4
18.2
18.0
17.8
17.6
17.4
17.2
17.0
δ15N (‰)
a
b
c
BA
SM offshore
SM inshore
Mean
Mean±SE
Mean±SD


-15.6
-15.8
-16.0
-16.2
-16.4
-16.6
-16.8
-17.0
-17.2
-17.4
δ13C (‰)
ab
a
b
BA
SM offshore
SM inshore
Mean
Mean±SE
Mean±SD


Pelagic
Benthic
Loligo sanpaulensis
Cynoscion guatucupa
Ranoya brasiliensis
Stromateus brasiliensis
+: Buenos Aires
o: G. San Matías offshore
Engraulis anchoita
Merluccius hubbsi
Illex argentinus
Trophic level
δ15N (‰)
δ13C (‰)
22
21
20
19
18
17
16
-18
-17
-16
-15
-14

Research Article

Rapid Communications in Mass Spectrometry

Received: 24 September 2014 Revised: 12 February 2015 Accepted: 15 February 2015 Published online in Wiley Online Library

Rapid Commun. Mass Spectrom. **2015**, *29*, 855–863
(wileyonlinelibrary.com) DOI: 10.1002/rcm.7173

Variation in $\delta^{15}N$ and $\delta^{13}C$ stable isotope values in common dolphins (*Delphinus* spp.) worldwide, with particular emphasis on the eastern North Atlantic populations

A. M. Pinela*, A. Borrell and A. Aguilar

Department of Animal Biology and Biodiversity Research Institute (IrBio), University of Barcelona, Barcelona, Spain

RATIONALE: Distinguishing population units of small cetaceans continuously distributed in a widespread area is challenging but critical for their conservation and management. The use of chemical markers allows the investigation of foraging ecology and inter-specific variability, in order to detect population structure and niche segregation in the common dolphin (*Delphinus* spp.).

METHODS: The stable isotope ratios of carbon ($\delta^{13}C$ values) and nitrogen ($\delta^{15}N$ values) were measured in the bone tissue of common dolphins accidentally by-caught or stranded along the north-eastern and eastern Subtropical Atlantic Ocean, by means of continuous flow elemental analyser/isotope ratio mass spectrometry. Trophic positions were determined and compared, taking into account the local ecosystem trophic baseline for each study area. Data obtained for the study areas were qualitatively compared with those for common dolphin species/populations distributed worldwide.

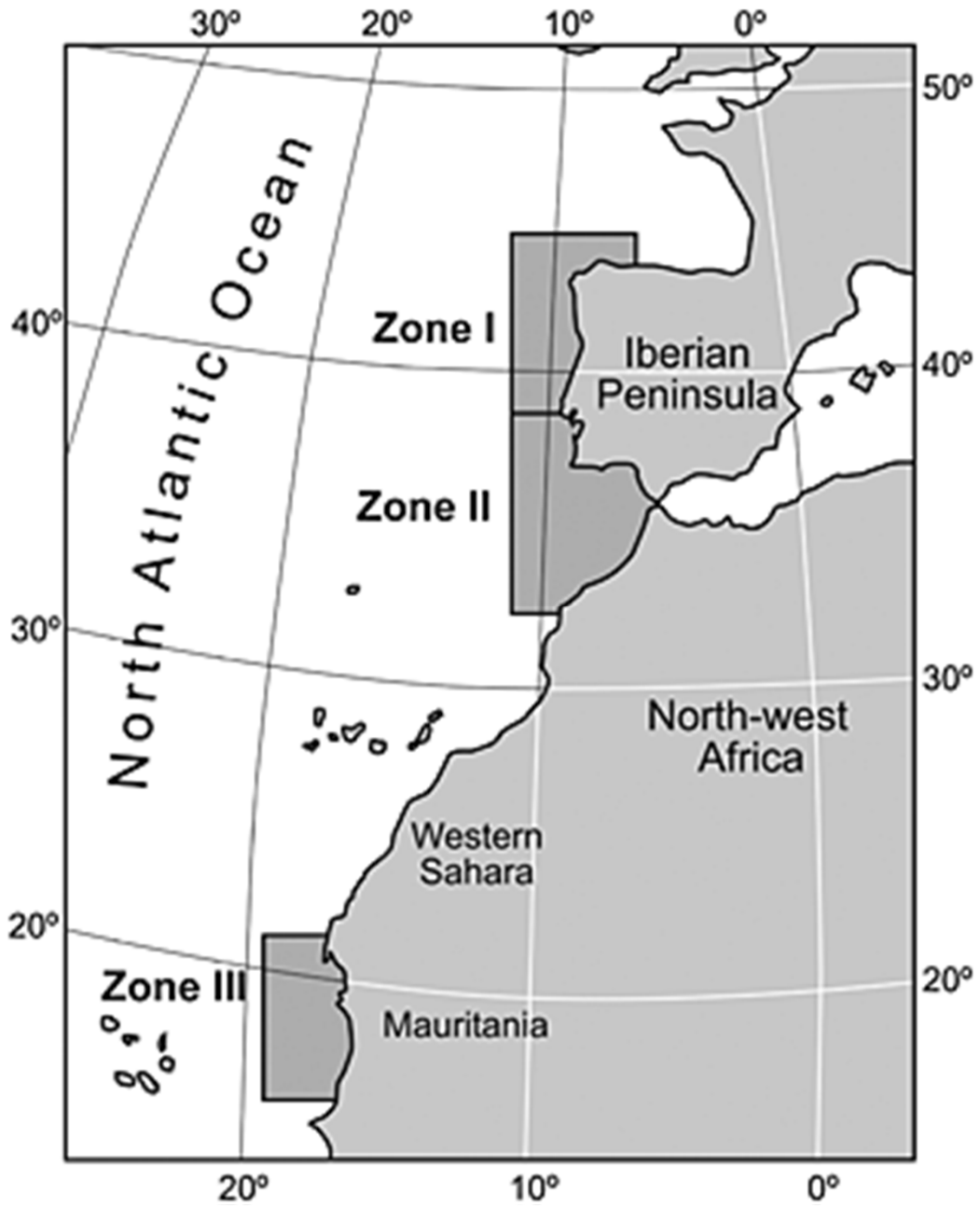
RESULTS: The $\delta^{13}C$ and $\delta^{15}N$ values were higher in the eastern Subtropical Atlantic as a consequence of the coexistence in the area of the common dolphin short- and long-beaked morphotypes. Individuals from the north-eastern Atlantic displayed lower $\delta^{15}N$ values, reflecting dissimilarities in diet and variation in local isotopic baselines. Comparisons with other areas around the world suggest that the species is extremely adaptive and feeds at different trophic levels to adapt to local variations.

CONCLUSIONS: Stable isotopes are a useful tool to investigate population structure and trophic niche segregation. The trophic behaviour of worldwide populations of common dolphins was fruitfully analysed and revealed substantial differences, probably reflective of both adaptive strategies of the genus and dissimilarities in the structure of the ecosystems. Copyright © 2015 John Wiley & Sons, Ltd.

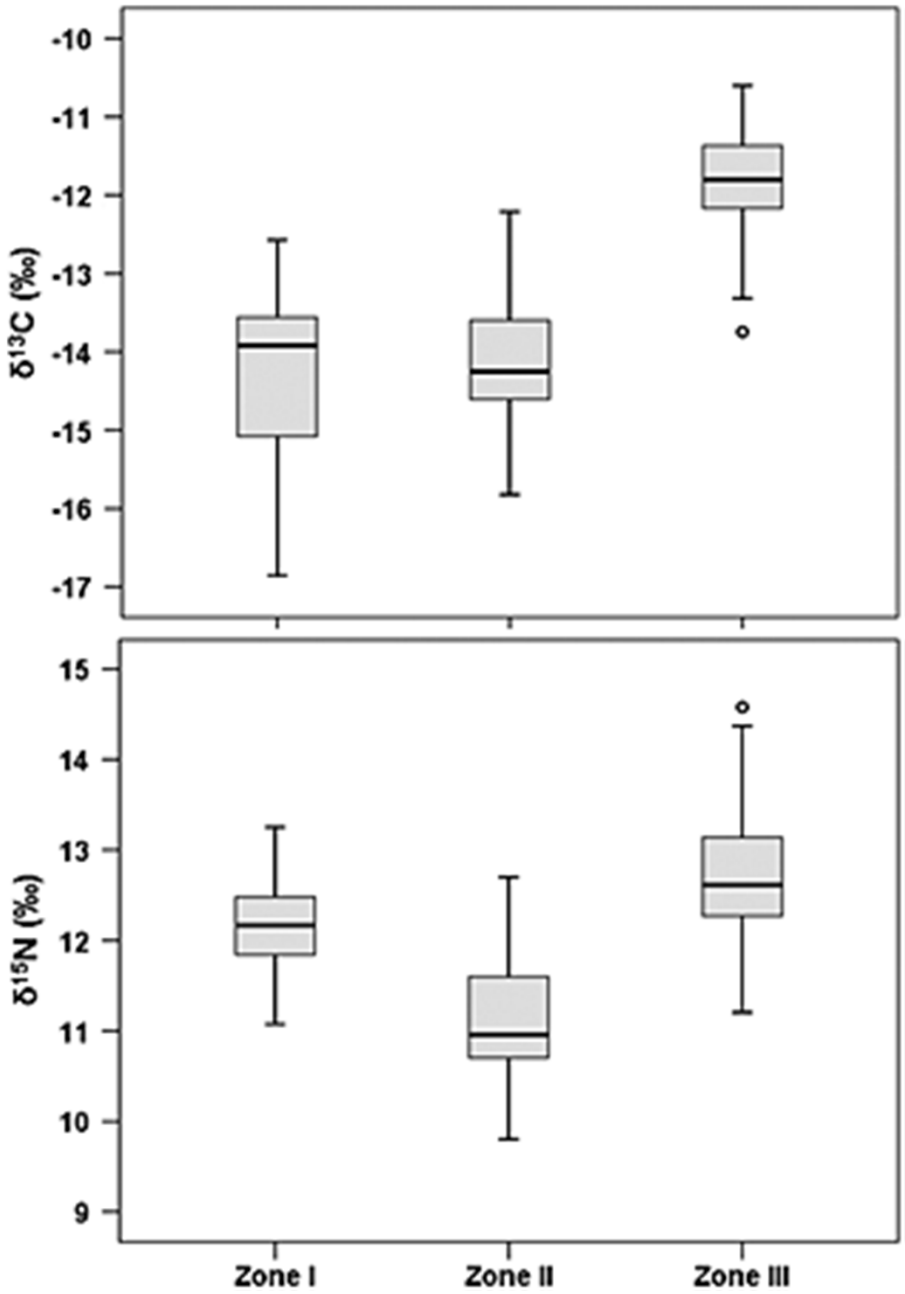
Questão cientifica:

Golfinhos tem distribuição espacial vasta e contínua.

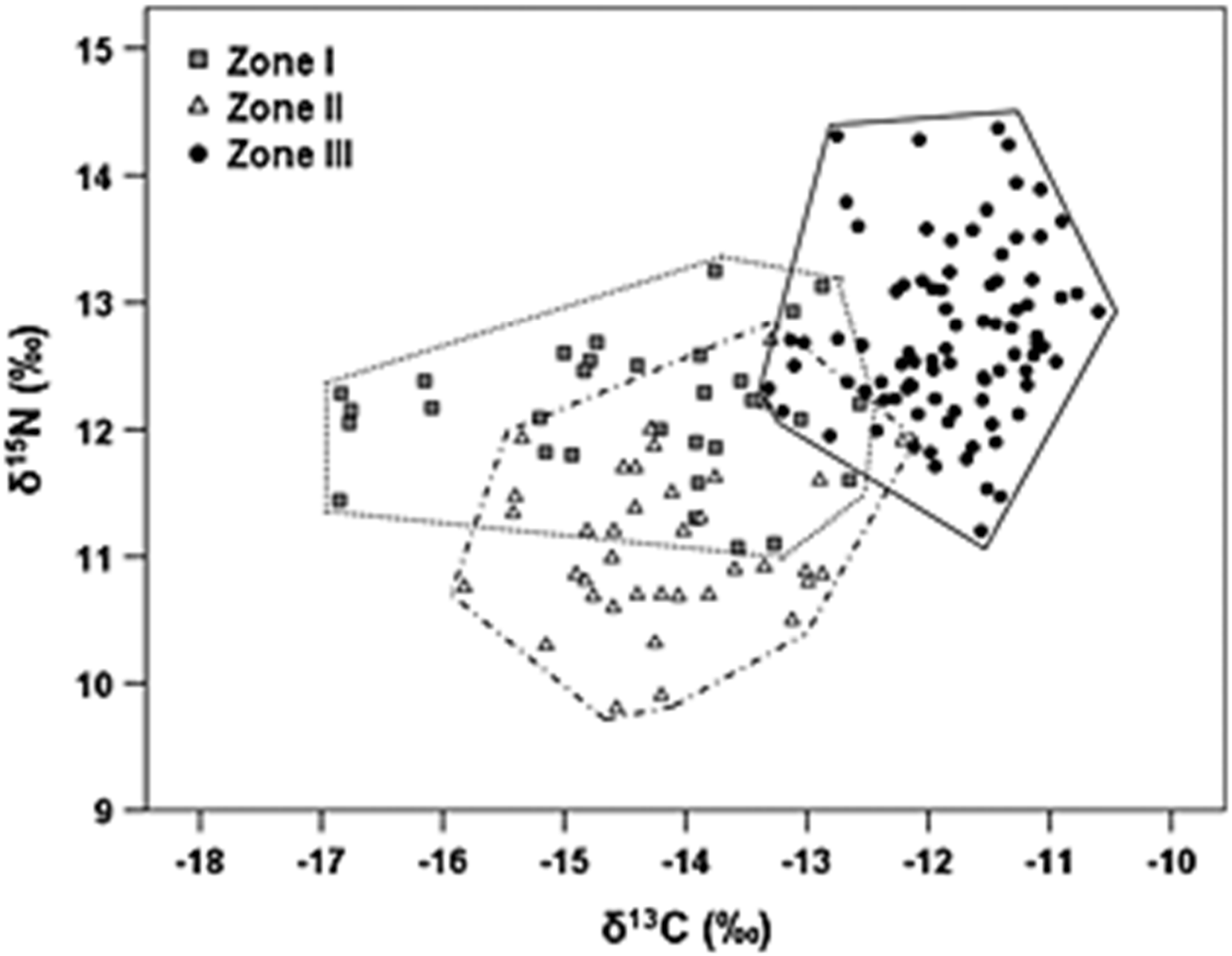
Isto complica o estudo das populações



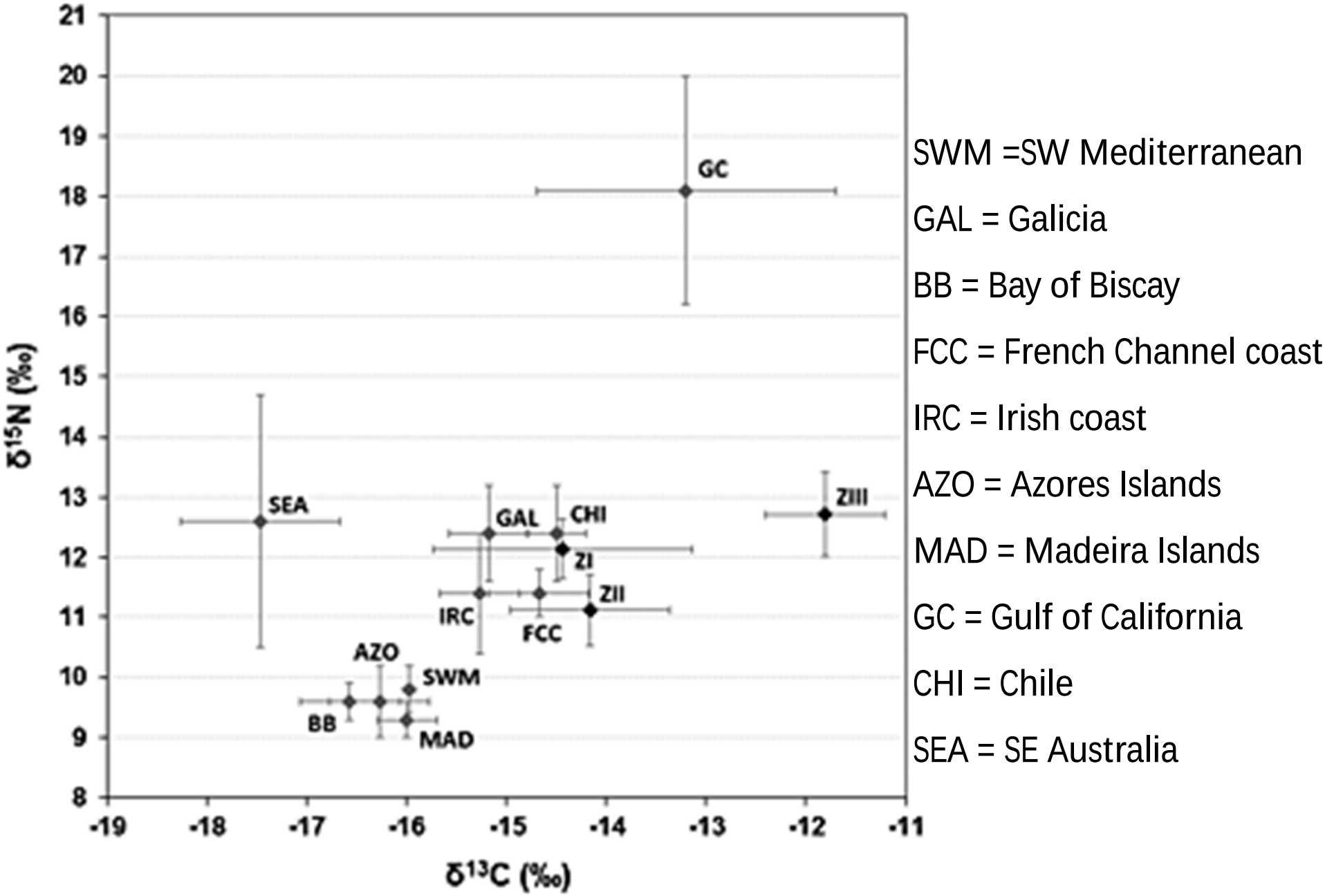
Rapid Communications in Mass Spectrometry
Volume 29, Issue 9, pages 855-863, 25 MAR 2015 DOI: 10.1002/rcm.7173
http://onlinelibrary.wiley.com/doi/10.1002/rcm.7173/full#rcm7173-fig-0001



Rapid Communications in Mass Spectrometry
Volume 29, Issue 9, pages 855-863, 25 MAR 2015 DOI: 10.1002/rcm.7173
http://onlinelibrary.wiley.com/doi/10.1002/rcm.7173/full#rcm7173-fig-0002



Rapid Communications in Mass Spectrometry
Volume 29, Issue 9, pages 855-863, 25 MAR 2015 DOI: 10.1002/rcm.7173
http://onlinelibrary.wiley.com/doi/10.1002/rcm.7173/full#rcm7173-fig-0003



SWM =SW Mediterranean

GAL = Galicia

BB = Bay of Biscay

FCC = French Channel coast

IRC = Irish coast

AZO = Azores Islands

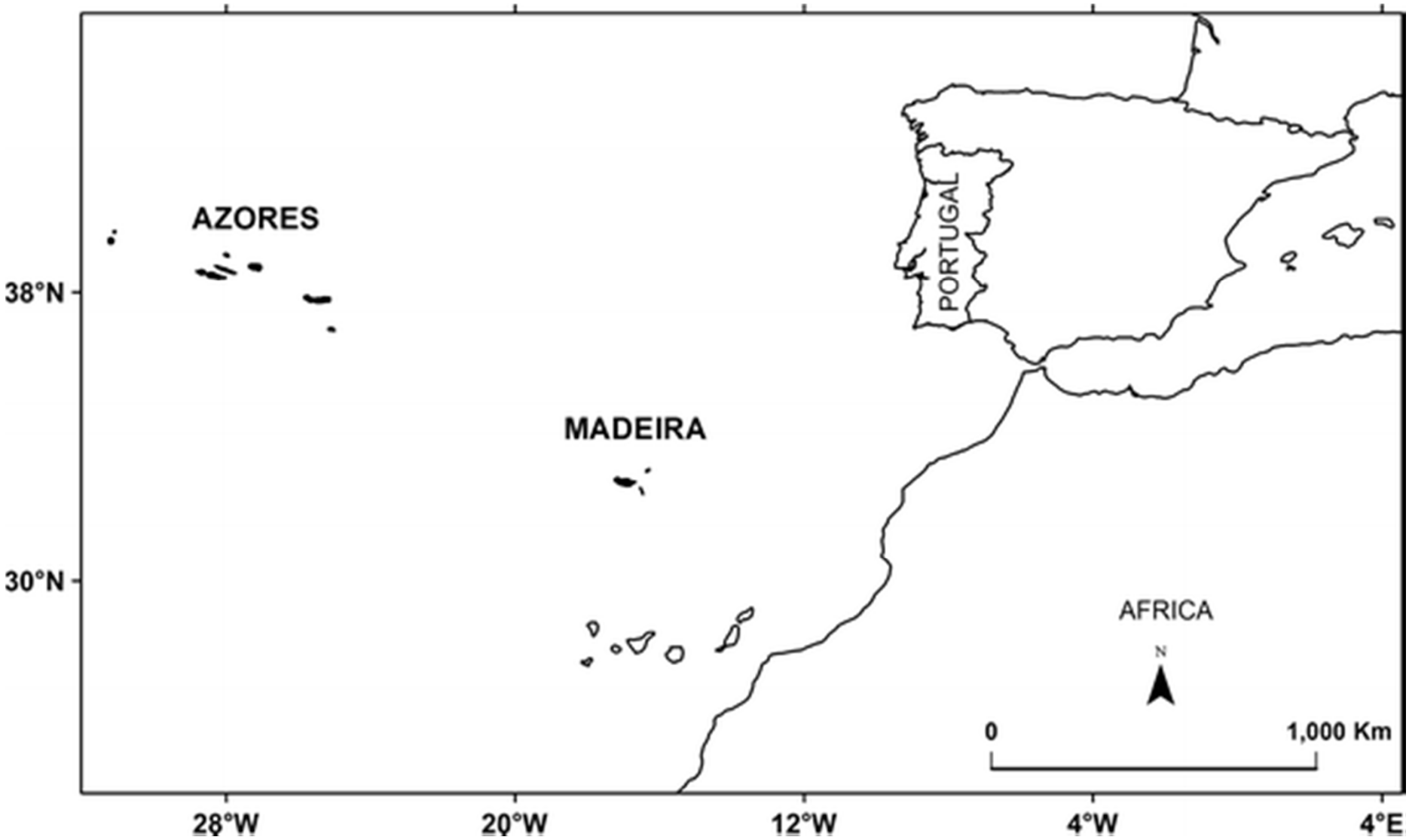
MAD = Madeira Islands

GC = Gulf of California

CHI = Chile

SEA = SE Australia

Rapid Communications in Mass Spectrometry
Volume 29, Issue 9, pages 855-863, 25 MAR 2015 DOI: 10.1002/rcm.7173
http://onlinelibrary.wiley.com/doi/10.1002/rcm.7173/full#rcm7173-fig-0004

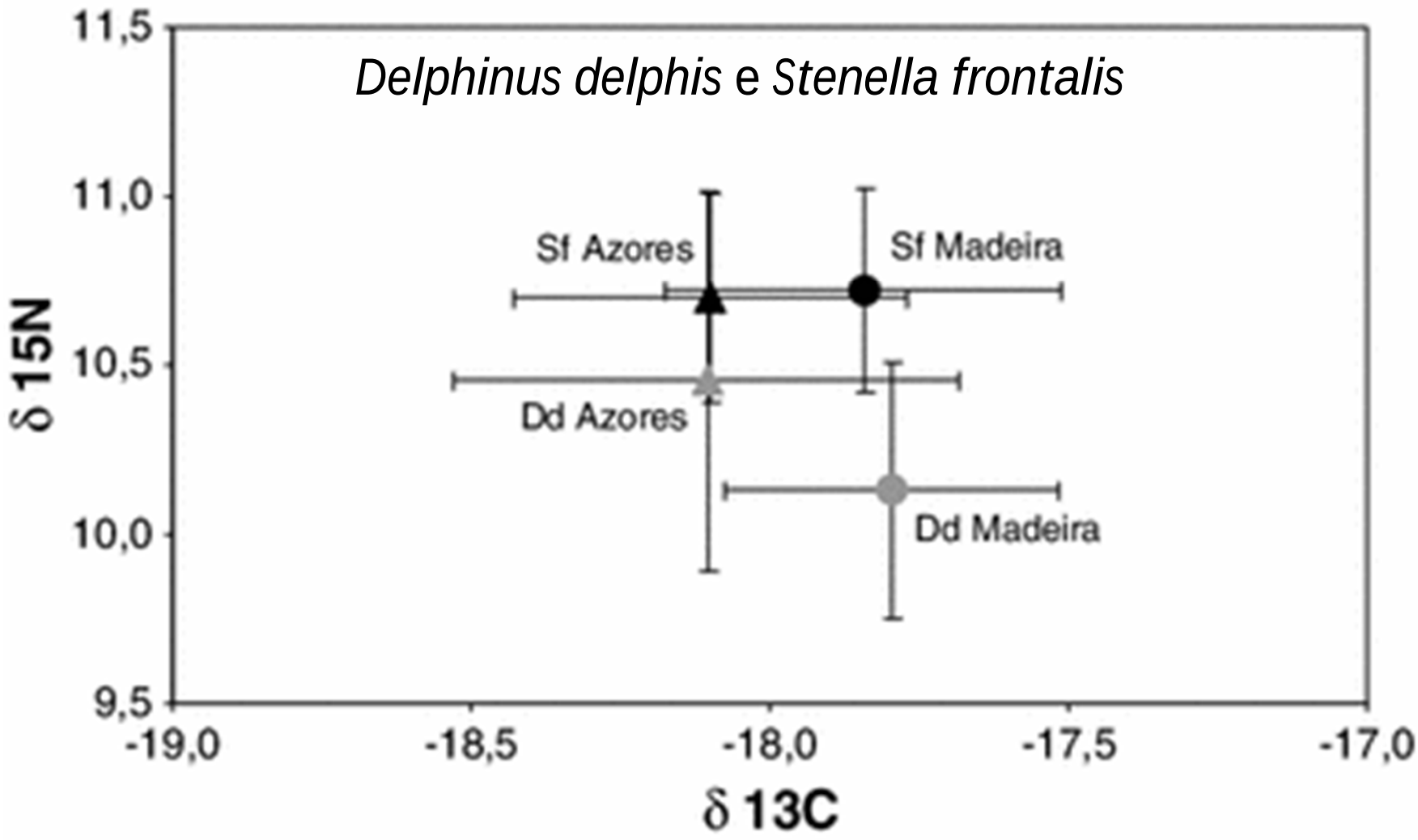

AZORES
MADEIRA
PORTUGAL
AFRICA
N
0
1,000 Km
38°N
30°N
28°W
20°W
12°W
4°W
4°E

Delphinus delphis e *Stenella frontalis* sou as espécies mais abundantes entre Madeira e Açores. As duas espécies tem habitat parecido e são observadas caçar junto.

Questões cientificas:

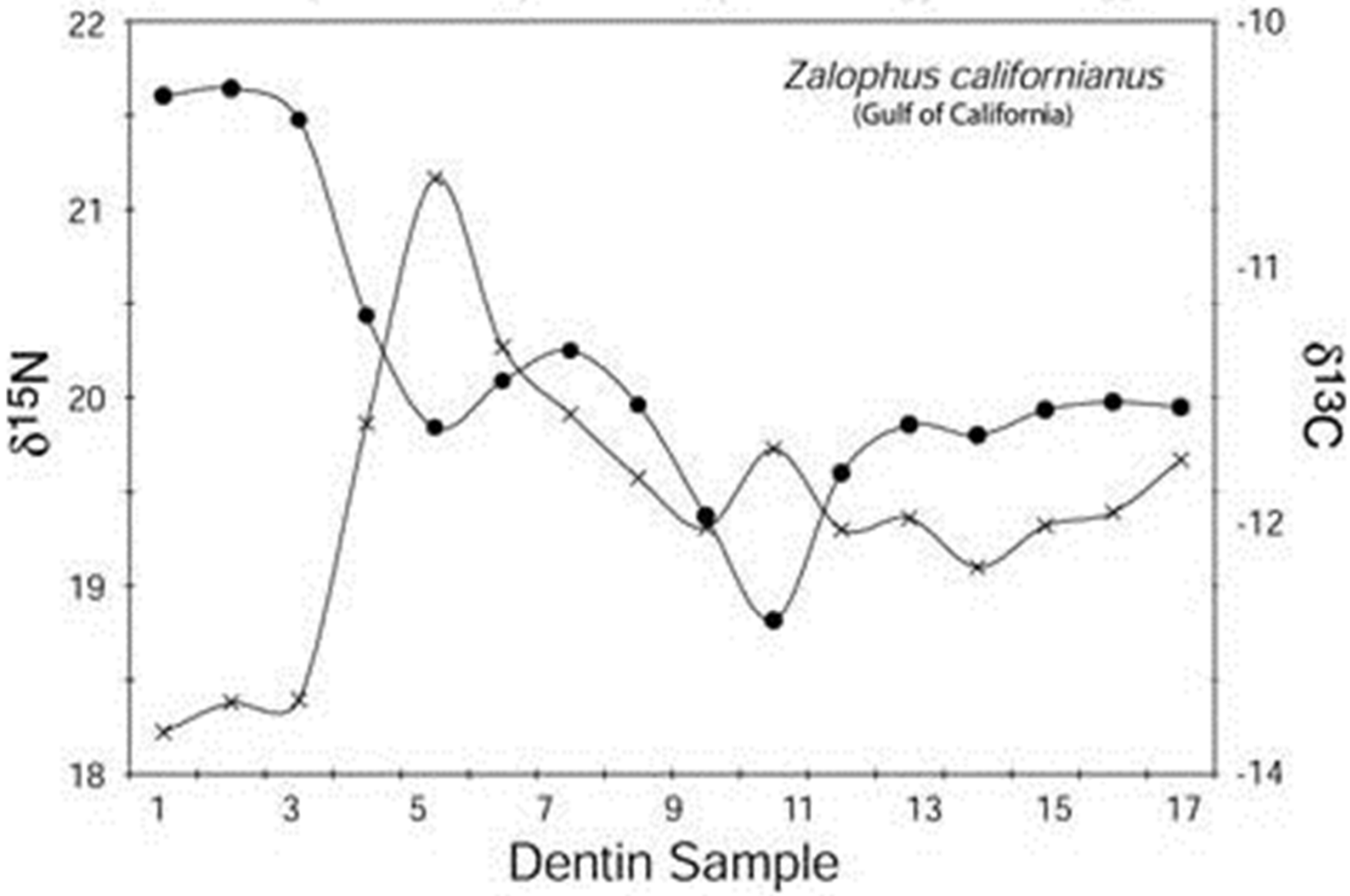
1) Indivíduos nos dois sítios pertencem ao mesmo grupo ecológico?

2) Eles têm o mesmo habito alimentar?


Delphinus delphis e *Stenella frontalis*
Sf Azores
Sf Madeira
Dd Azores
Dd Madeira
11,5
11,0
10,5
10,0
9,5
δ 15N
-19,0
-18,5
-18,0
-17,5
-17,0
δ 13C

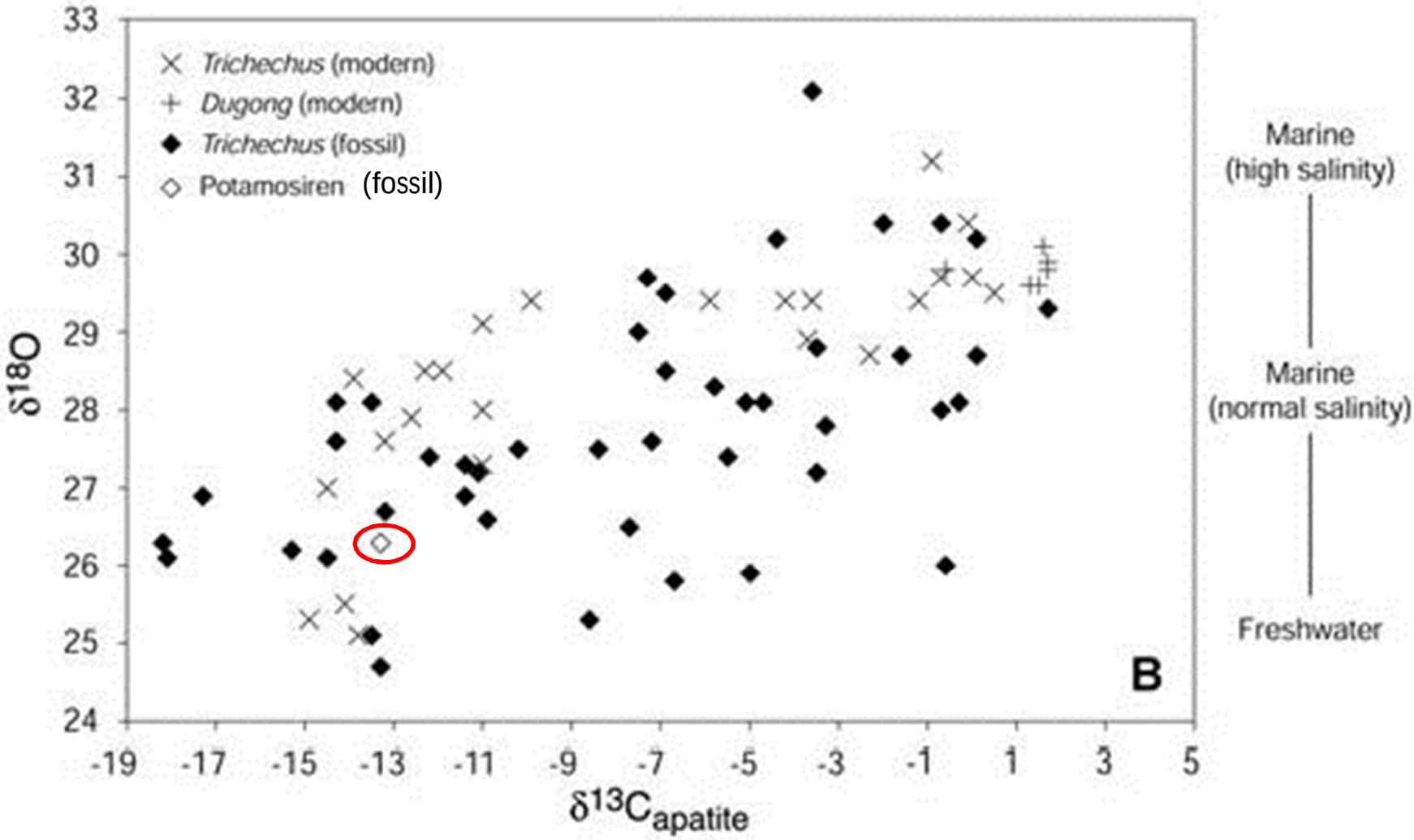
Mamamento x desmamamento

Leão marinho da Califórnia



Newsome et al. 2010

Evolução de mamíferos marinhos: Transição de habitat da água doce para água marinha



Newsome et al. 2010

Isótopos do nitrogênio em estudos ambientáis

Efeito isotópico de denitrificação do nitrato

Denitrificação do nitrato

$$NO_3^- \longrightarrow N_2$$

$$2\,NO_3^- + 10\,e^- + 12\,H^+ \rightarrow N_2 + 6\,H_2O$$

Denitrificação do nitrato

$$NO_3^- \longrightarrow N_2$$

$$2\ {}^{14}NO_3^- + 10\ e^- + 12\ H^+ \rightarrow N_2 + 6\ H_2O$$

$$2\ {}^{15}NO_3^- + 10\ e^- + 12\ H^+ \rightarrow N_2 + 6\ H_2O$$

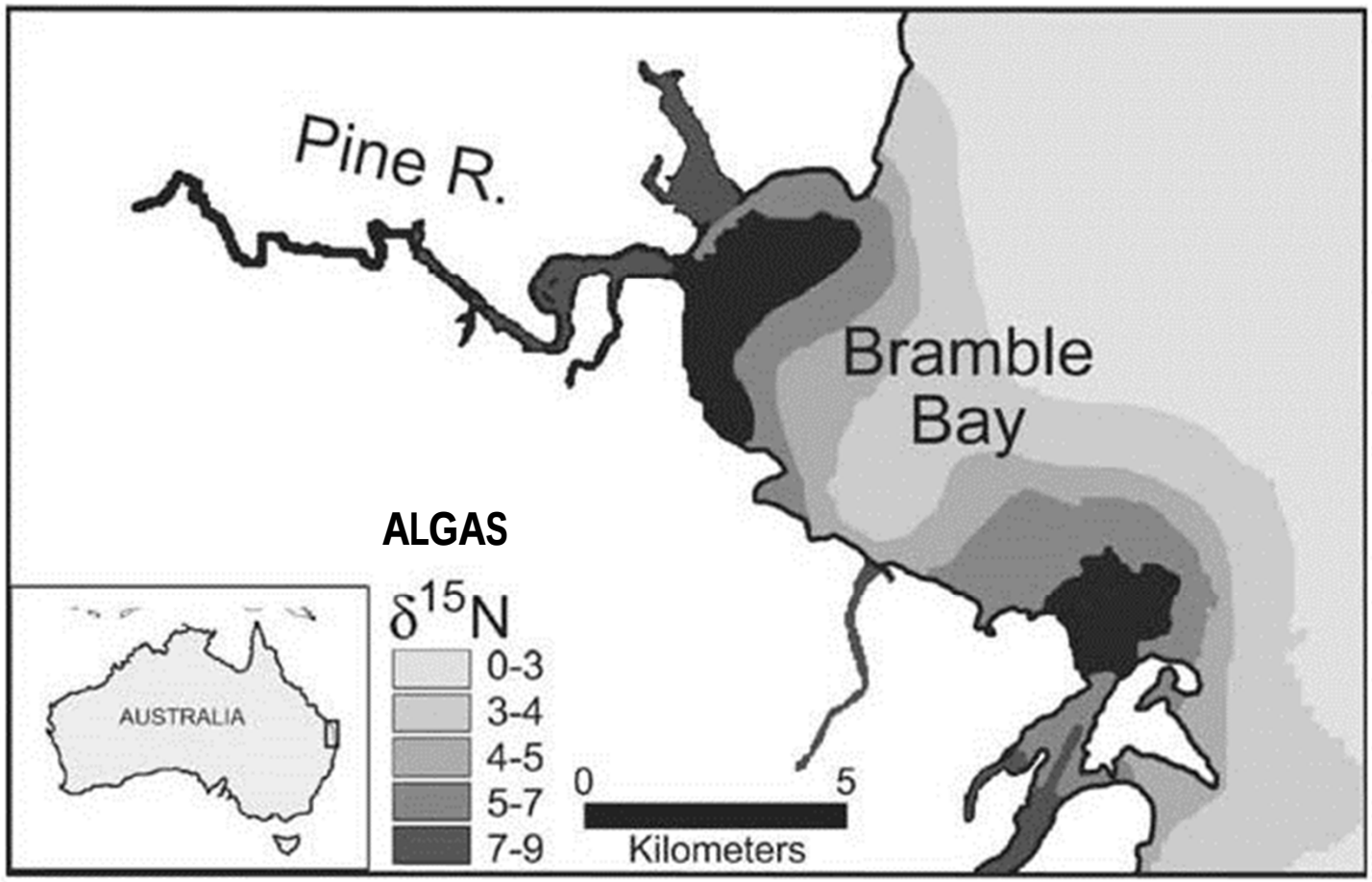
Denitrificação do nitrato

SUBSTRATO **PRODUTO**

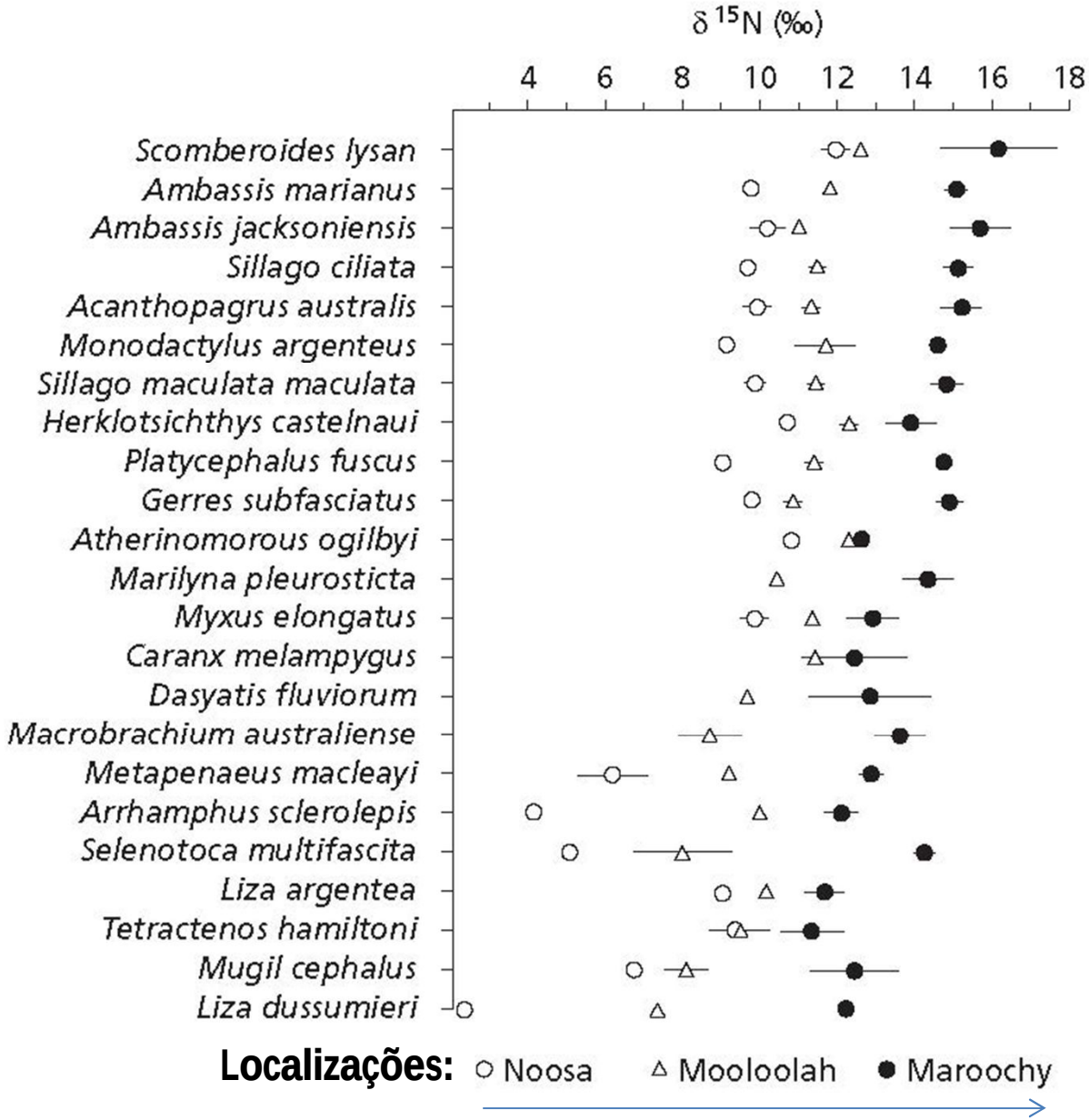
NO_3^- ⟶ N_2

Enriquecido em ^{15}N **Enriquecido em ^{14}N**

Poluição de estuários



B. Fry - Stable Isotope Ecology (Springer)



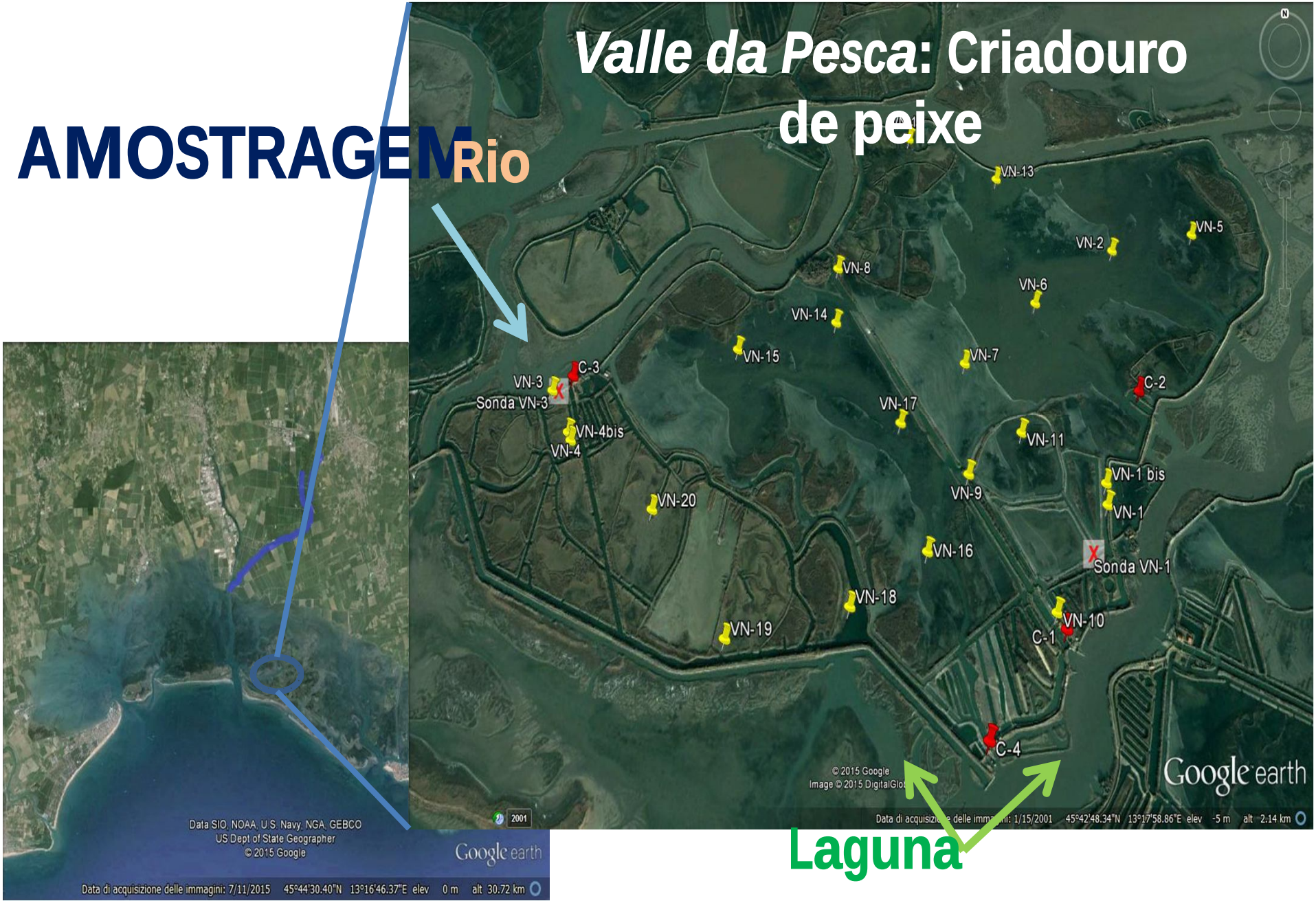
Michener & Lajtha Eds. Stable Isotopes in Ecology and Environmental Science

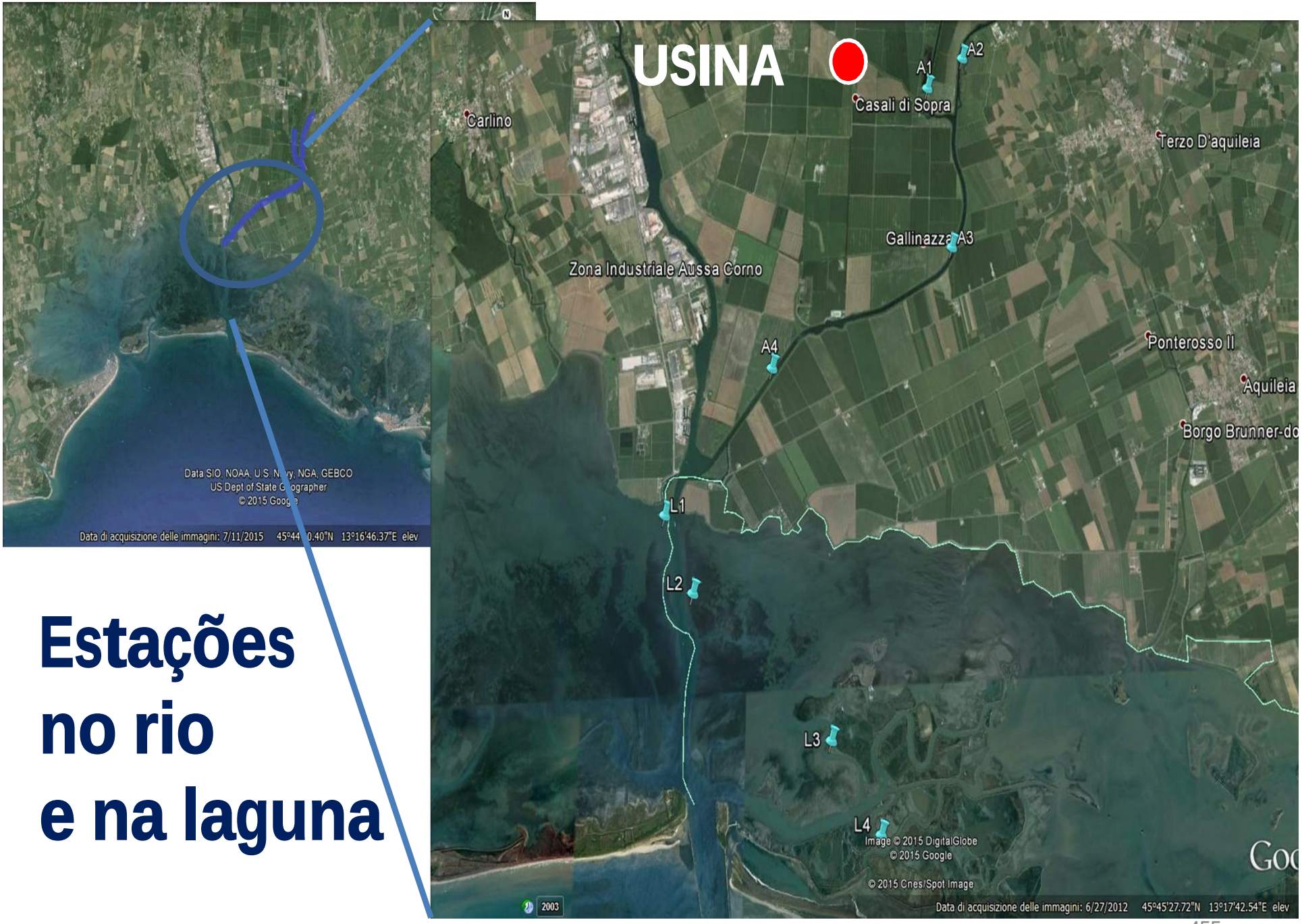
United Kingdom
Isle of Man
Dublin
Ireland
Éire
Manchester
London
Danmark
Denmark
Hamburg
Berlin
Nederland
Netherlands
Deutschland
Germany
België
Belgique
Belgium
Frankfurt am Main
Praha
Česká republika
Czech Republic
Paris
München
Wien
Österreich
Austria
Schweiz
Suisse
Svizzera
Switzerland
France
Slovenija
Slovenia
Venezia
Hrvatska
Croatia
Bay of Biscay
Monaco
Firenze
Italia
Italy
Andorra
Barcelona
Porto
Madrid
España
Spain
Valencia
Portugal
Lisboa
Tyrrhenian Sea
Sevilla
Granada
Malaga
Gibraltar
Algiers
Tunis
Malta

FRIULI-VENEZIA
GIULIA

ÁREA DE ESTUDO

AMOSTRAGEM

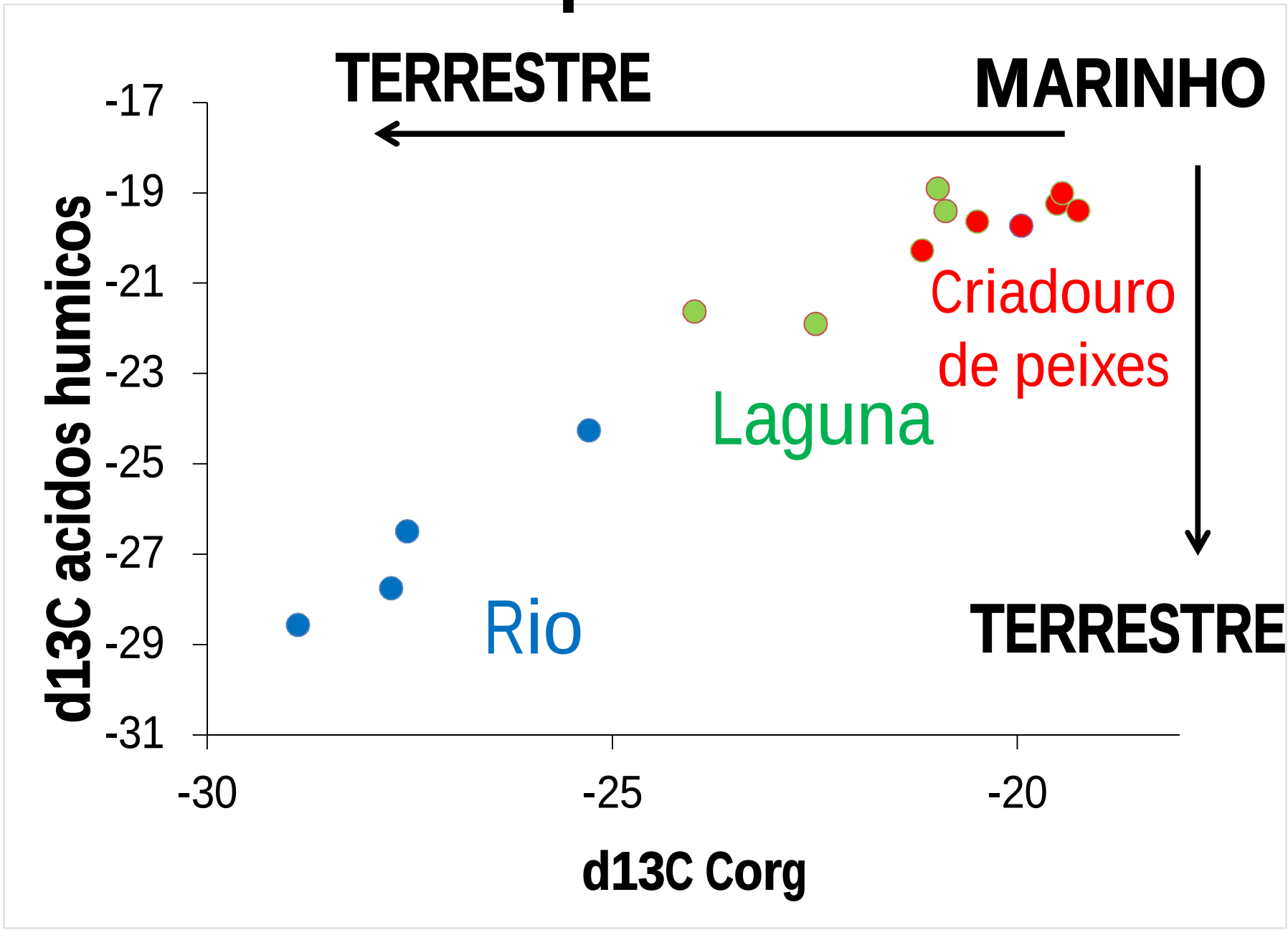

Rio
Valle da Pesca: Criadouro de peixe
VN-13
VN-5
VN-2
VN-8
VN-6
VN-14
VN-15
VN-7
C-3
VN-3
Sonda VN-3
C-2
VN-17
VN-4bis
VN-4
VN-11
VN-1 bis
VN-9
VN-20
VN-1
VN-16
Sonda VN-1
VN-18
VN-10
VN-19
C-1
C-4
Laguna
Google earth
Data SIO, NOAA, U.S. Navy, NGA, GEBCO
US Dept of State Geographer
© 2015 Google
Data di acquisizione delle immagini: 7/11/2015 45°44'30.40"N 13°16'46.37"E elev 0 m alt 30.72 km



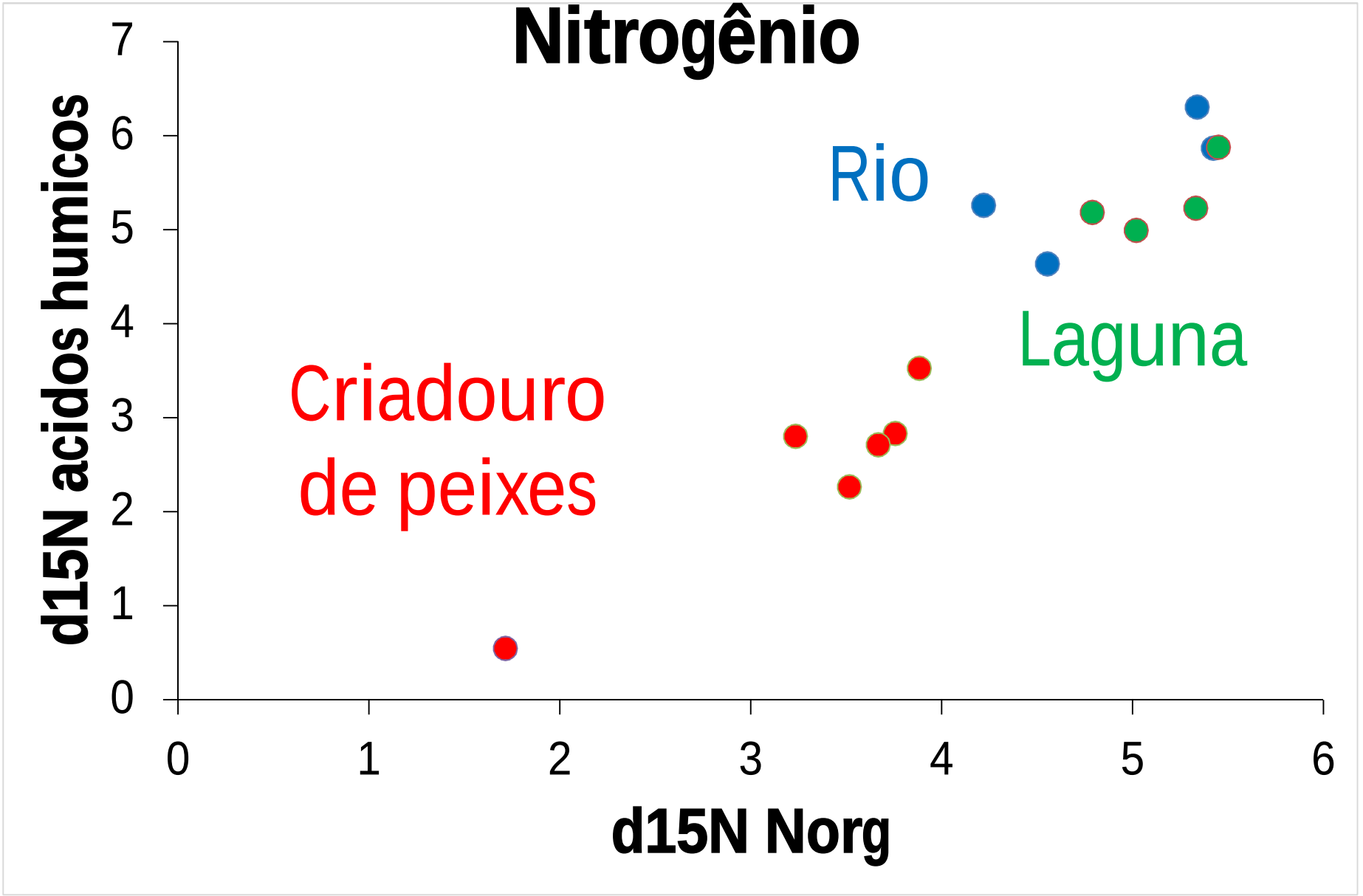
Estações no rio e na laguna

Resultados

Isótopos de Carbono



Isótopos de Nitrogênio



Fracionamento isotópico

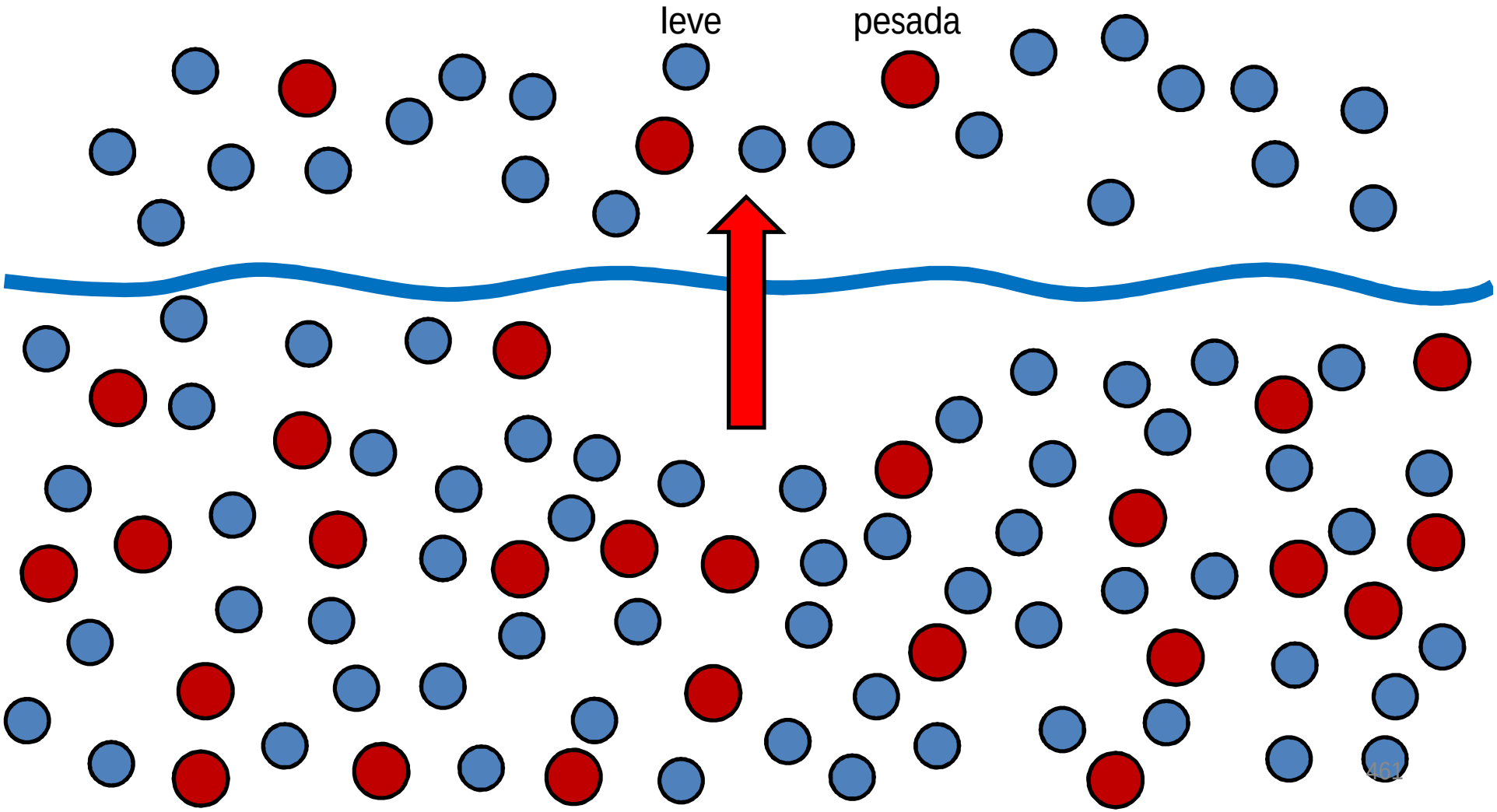
Tipos de fracionamento isotópicos:

Fracionamento cinético:
deslocamento unidirecional de isótopos

Fracionamento ao equilíbrio:
intercambio isotópico (bidirecional)

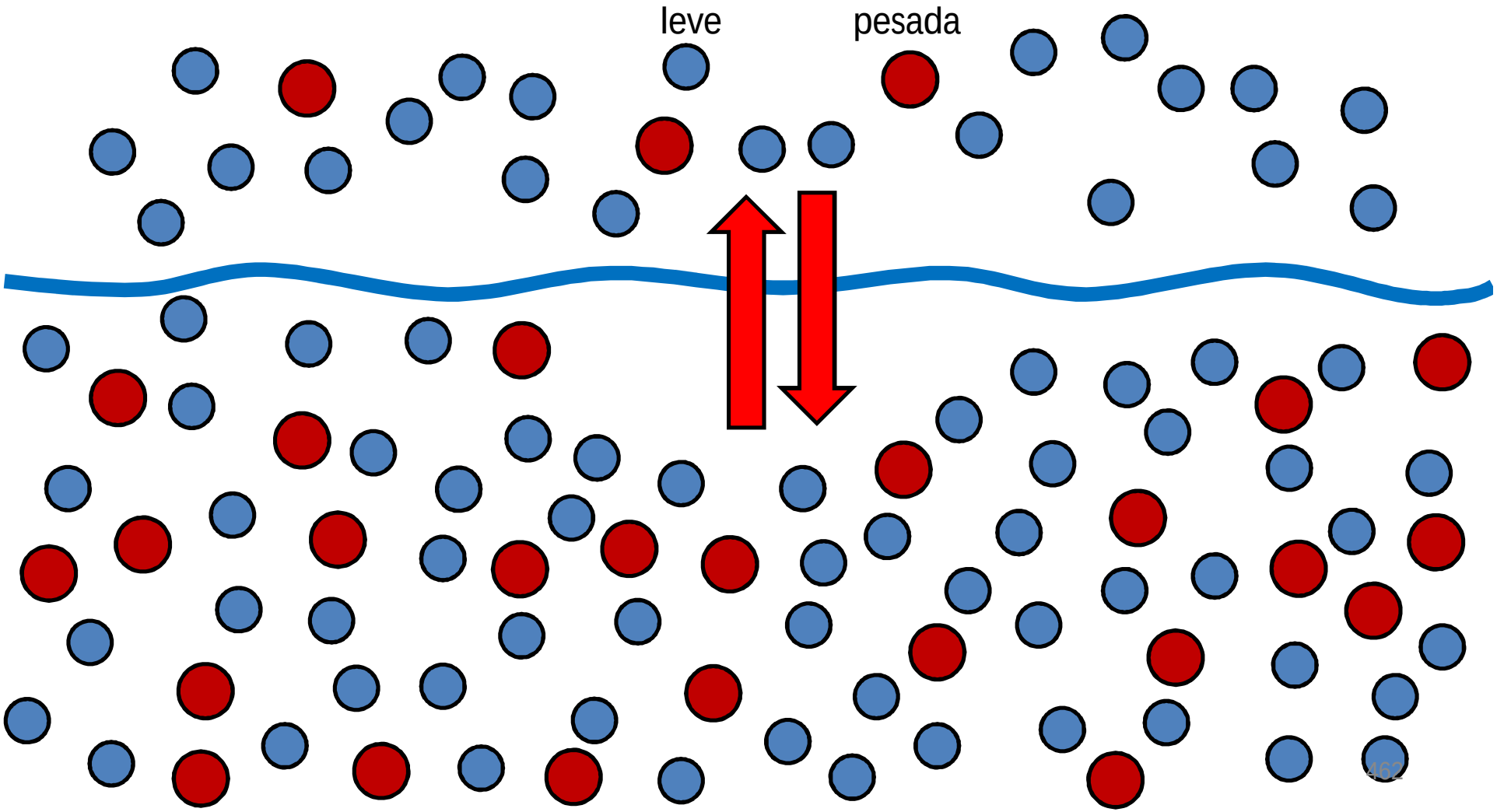
Fracionamento cinético:

por ex.: processo de evaporação



Fracionamento ao equilíbrio:

por ex.: evaporação e re-condencação

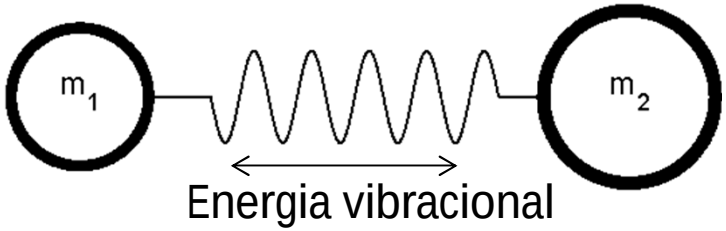


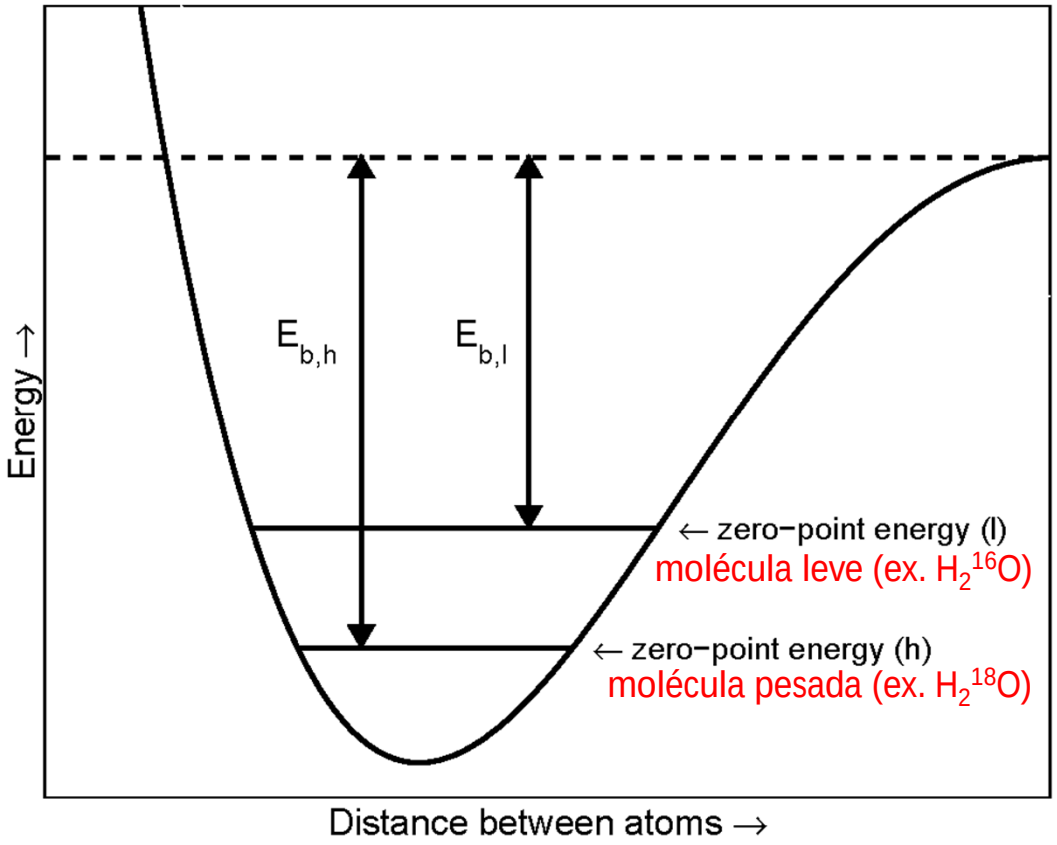
Fracionamento isotópico ao equilíbrio

Massas diferentes = comportamento físico diferente

Porque ?

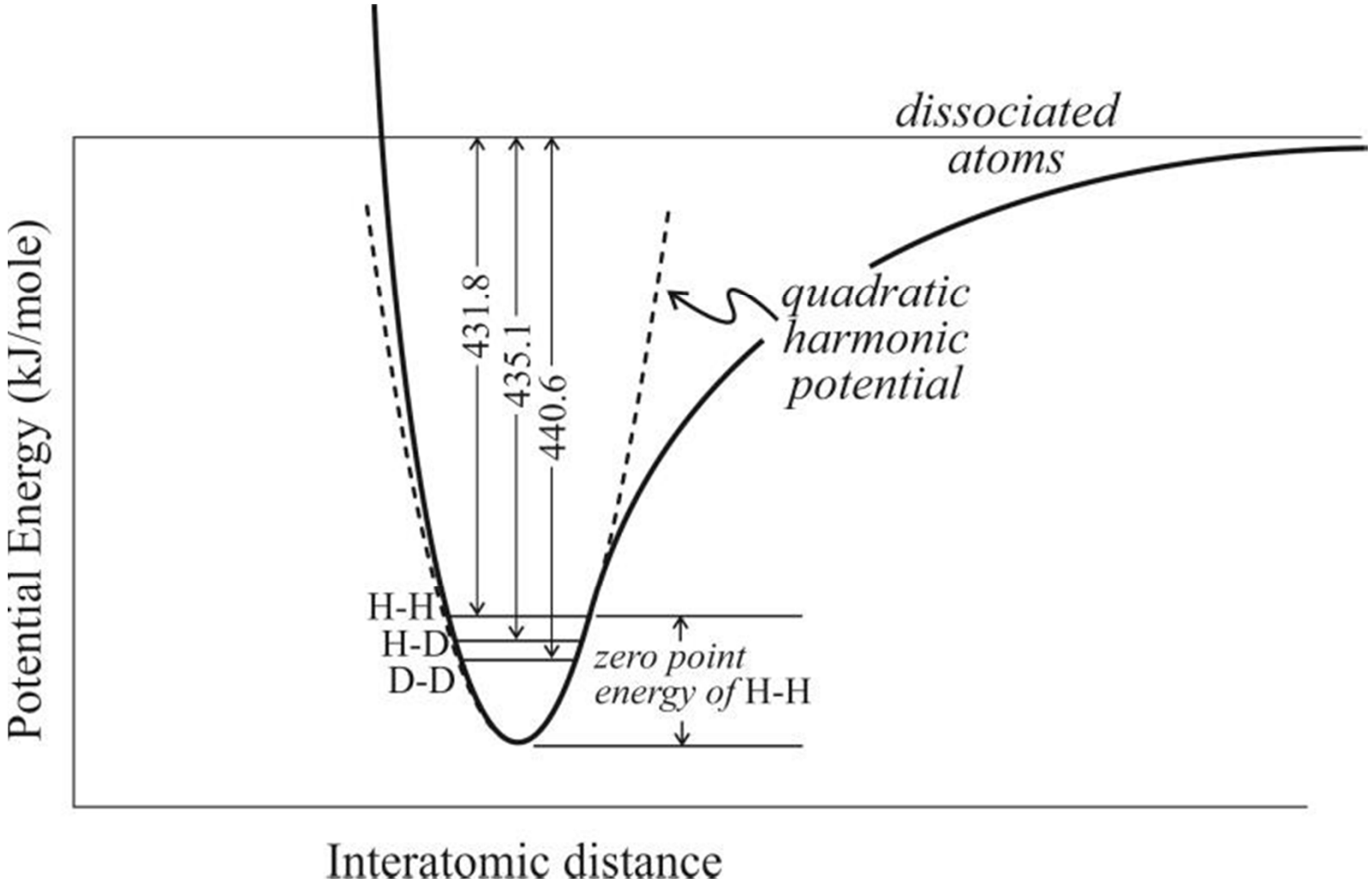
Modelo de molécula





IMPORTANTE: Moléculas isotopicamente leves têm energia maior

Zeebe & Wolf-Gladrow (2001) *CO_2 in Seawater: Equilibrium, Kinetics, Isotopes*.



Sharp (2007) *Principles of Stable Isotope Geochemistry*

Como se exprime o fracionamento isotópico

- **Fator de fracionamento:**

$$\alpha_{(agua\text{-}vapor)} \equiv \frac{\left(\frac{^{18}O}{^{16}O}\right)_{agua}}{\left(\frac{^{18}O}{^{16}O}\right)_{vapor}} \cong 1.009\ ‰\ (25\ °C)$$

$$\alpha_{(agua\text{-}vapor)} = \frac{\delta^{18}O_{agua} + 10^3}{\delta^{18}O_{vapor} + 10^3}$$

- **Fator de enriquecimento:**

$$\varepsilon_{(agua\text{-}vapor)} \equiv (\alpha_{(agua\text{-}vapor)} - 1) \times 10^3 \cong 9\ ‰$$

- **delta maiúsculo:**

$$\Delta^{18}O_{(agua\text{-}vapor)} = \delta^{18}O_{agua} - \delta^{18}O_{vapor}$$

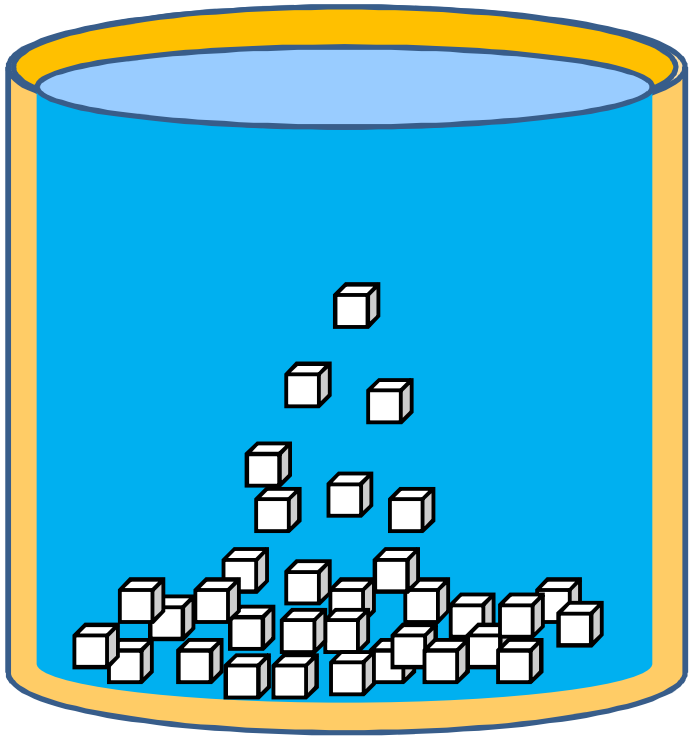
Matematicamente se demostra que:

$$\Delta^{18}O_{(agua\text{-}vapor)} \cong \varepsilon_{(agua\text{-}vapor)} \cong 10^3 \ln\alpha_{(agua\text{-}vapor)}$$

$$\cong 9\ ‰\ (25\ °C)$$

Fator de fracionamento de isótopos do oxigênio entre H_2O e $CO_{2(g)}$

T constante (25 °C)



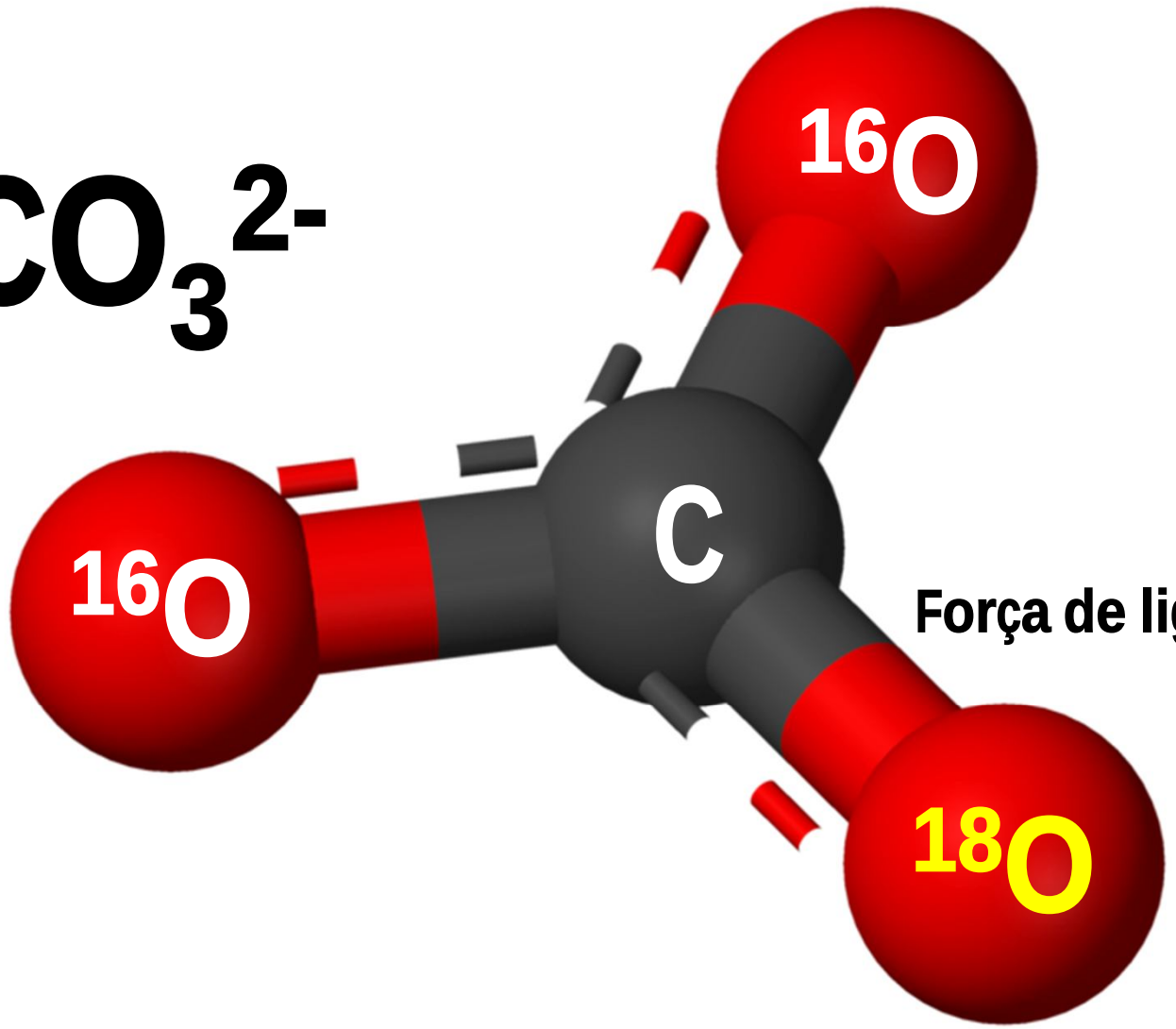
Equilíbrio isotópico

H_2O $CaCO_3$

$H_2{}^{16}O$ $CaC^{16}O^{16}O^{18}O$

íon carbonato

CO_3^{2-}



Força de ligação maior

Fracionamento isotópico ao equilíbrio entre $CaCO_3$ e H_2O

$$CO_2 + H_2O \rightleftharpoons H^+ + H\overset{18}{C}O_3^-$$

$$CO_2 + OH^- \rightleftharpoons H\overset{18}{C}O_3^-$$

$$CO_3^{2-} + H^+ \rightleftharpoons HCO_3^-$$

$$\downarrow$$

$$CO_3^{2-} + Ca^{2+} \rightarrow Ca\overset{18}{C}O_3(s)$$

Fracionamento de ^{18}O entre água e carbonato de cálcio não é o mesmo para calcita e aragonita

Equation of Kim & O'Neil (1997)
(carbonato inorgânico precipitado em laboratório)

$$10^3 \ln \alpha_{calcite\text{-}water} = 18.03\,(10^3\,T^{-1}) - 32.42$$

$$(\sim \delta^{18}O_{cal} - \delta^{18}O_{water})$$

Essa relação vale **só** se a precipitação acontece

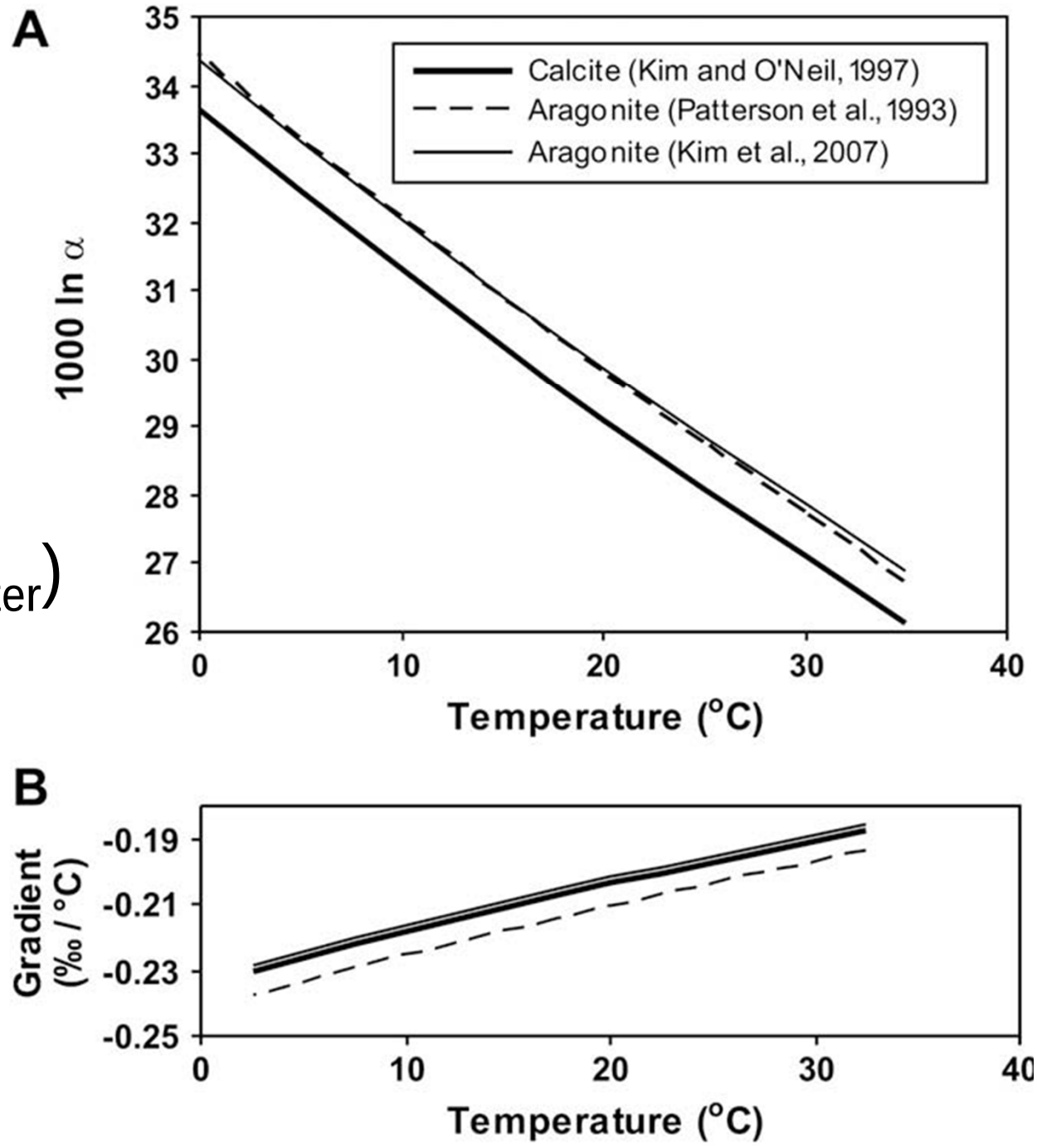
ao **equilíbrio isotópico** entre $CaCO_3$ e agua

Kim & O'Neil , 1997
Geochim. Cosmochim. Acta (61), 3461-3475.

EFEITO DA TEMPERATURA

$10^3 \ln \alpha_{CaCO3\text{-}water}$

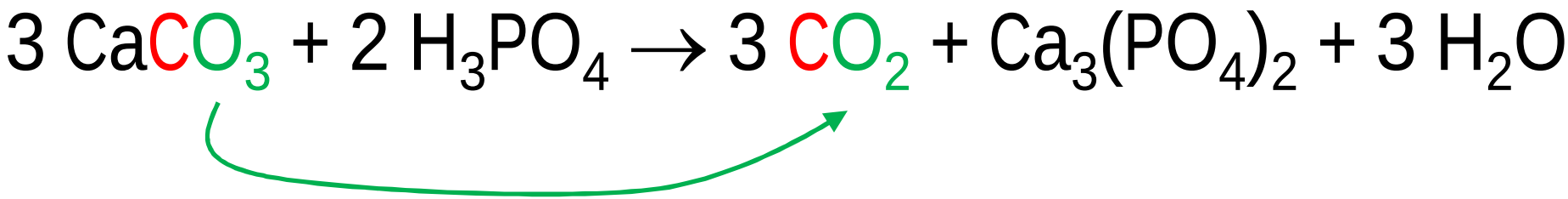
$(\sim \delta^{18}O_{carb} - \delta^{18}O_{water})$



Lachniet et al. 2009

Fator de fracionamento devido ao ácido (*acid fractionation factor*)

Método de McCrea (1950)



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Chemical Geology 246 (2007) 135–146
CHEMICAL GEOLOGY
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www.elsevier.com/locate/chemgeo

Phosphoric acid fractionation factors for calcite and aragonite between 25 and 75 °C: Revisited

Sang-Tae Kim [a,*], Alfonso Mucci [a], Bruce E. Taylor [b]

[a] *Earth & Planetary Sciences and GEOTOP-UQAM-McGill, McGill University, Montreal, QC, Canada H3A 2A7*
[b] *Geological Survery of Canada, Ottawa, Ontario, Canada K1A OE8*

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Editor: D. Rickard

Abstract

This paper reports the results of an investigation to re-determine the phosphoric acid fractionation factor for calcite and aragonite in order to improve the accuracy and limit the uncertainty of this very important quantity. The $^{18}O/^{16}O$ ratio of 100% of the oxygen in calcite and aragonite was determined directly by combining data from a two-step procedure involving thermal decomposition of the carbonate, followed by fluorination of the residual CaO using BrF_5.

The oxygen isotope composition of the CO_2 released upon the thermal decarbonation of calcite and aragonite is enriched in ^{18}O relative to that of the remaining CaO. Reproducibility of the oxygen isotope composition of the CO_2 derived from each of the decarbonation and fluorination steps was 0.55‰ (1σ). Nevertheless, statistically reliable acid fractionation factors for calcite ($s_e=0.07$) and aragonite ($s_e=0.10$) were determined on the basis of a large set of combined data from conventional phosphoric acid reactions at three different temperatures, plus the decarbonation/fluorination procedure. The acid fractionation factors ($\alpha_{CO_2(ACID)\text{-carbonate}}$) determined in this study for calcite ($\alpha_{CO_2(ACID)\ Calcite}=1.01030$) and aragonite ($\alpha_{CO_2(ACID)\ Aragonite}=1.01063$) at 25 °C, differ slightly from previously published values.

On the basis of our new determinations at 25, 50, and 75 °C, revised expressions for the temperature dependence of the acid

$$1000\ln\alpha_{CO_2(ACID)-Calcite} = 3.59\left(10^3/T\right) - 1.79$$

$$1000\ln\alpha_{CO_2(ACID)-Aragonite} = 3.39\left(10^3/T\right) - 0.83$$

$$10^3 \ln\alpha_{(CO2\text{-}CaCO3)} \cong \delta^{18}O_{CO2} - \delta^{18}O_{CaCO3}$$

Chemical Geology 277 (2010) 160–166

Contents lists available at ScienceDirect

Chemical Geology

journal homepage: www.elsevier.com/locate/chemgeo

Oxygen isotope fluctuations in a modern North Sea oyster (*Crassostrea gigas*) compared with annual variations in seawater temperature: Implications for palaeoclimate studies

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North Sea
Palaeoclimate

ABSTRACT

A total of 181 oxygen isotope values from sequential samples of the left shell of a modern Pacific Oyster (*Crassostrea gigas*) that lived on a sub-tidal oyster bank in the List Basin (North Sea, Germany) shows periodically varying values between +1.3‰ and −2.5‰. In order to test whether these $\delta^{18}O$ fluctuations reflect seawater temperature changes, the isotope values of the shell were compared to actual seawater temperature variations from the region. *C. gigas* serves as an excellent proxy for temperature of palaeoseawater and the results show that the examined oyster precipitated its shell in $\delta^{18}O$ equilibrium with the ambient seawater. A cessation of the oyster shell calcification starts at water temperatures below 6 °C, at lower temperatures than previously thought for *Crassostrea*. For palaeoclimate investigations the termination of shell production is important because the lowest temperatures might not be represented in the shell. Higher water temperatures are therefore potentially over-represented in time spans when the shell grows faster and mean oxygen isotope values may potentially bias the mean seawater temperatures of annual variations towards higher temperatures.

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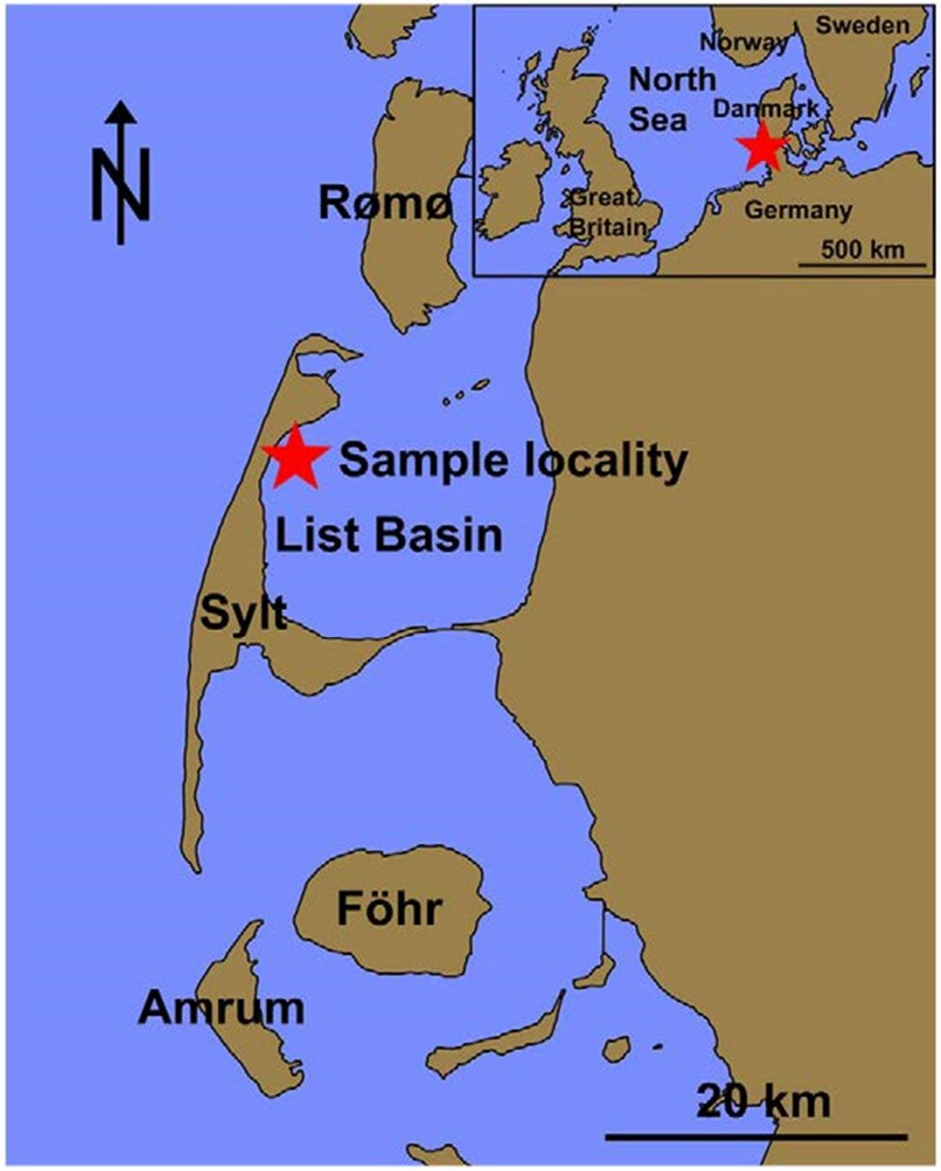
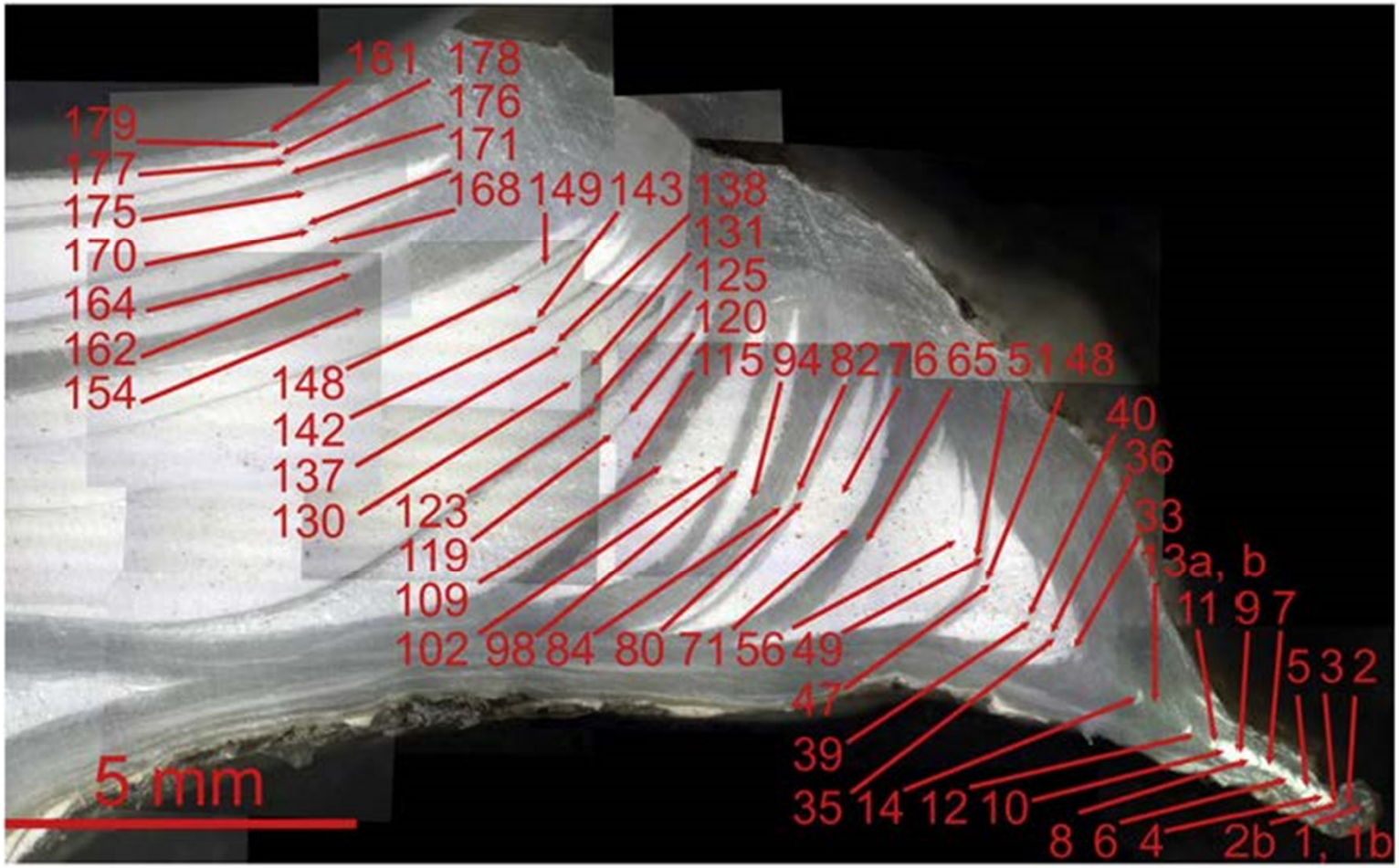
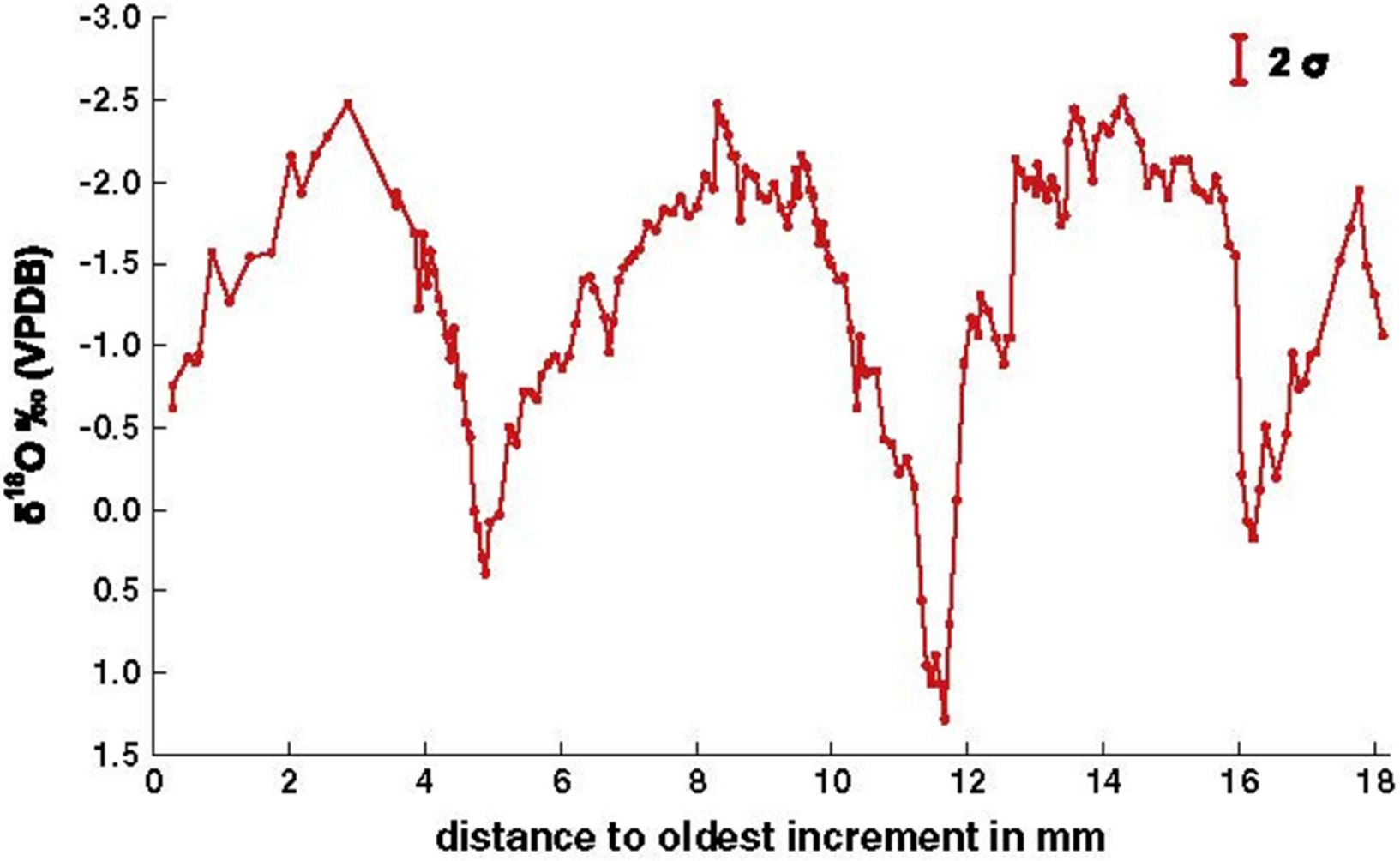
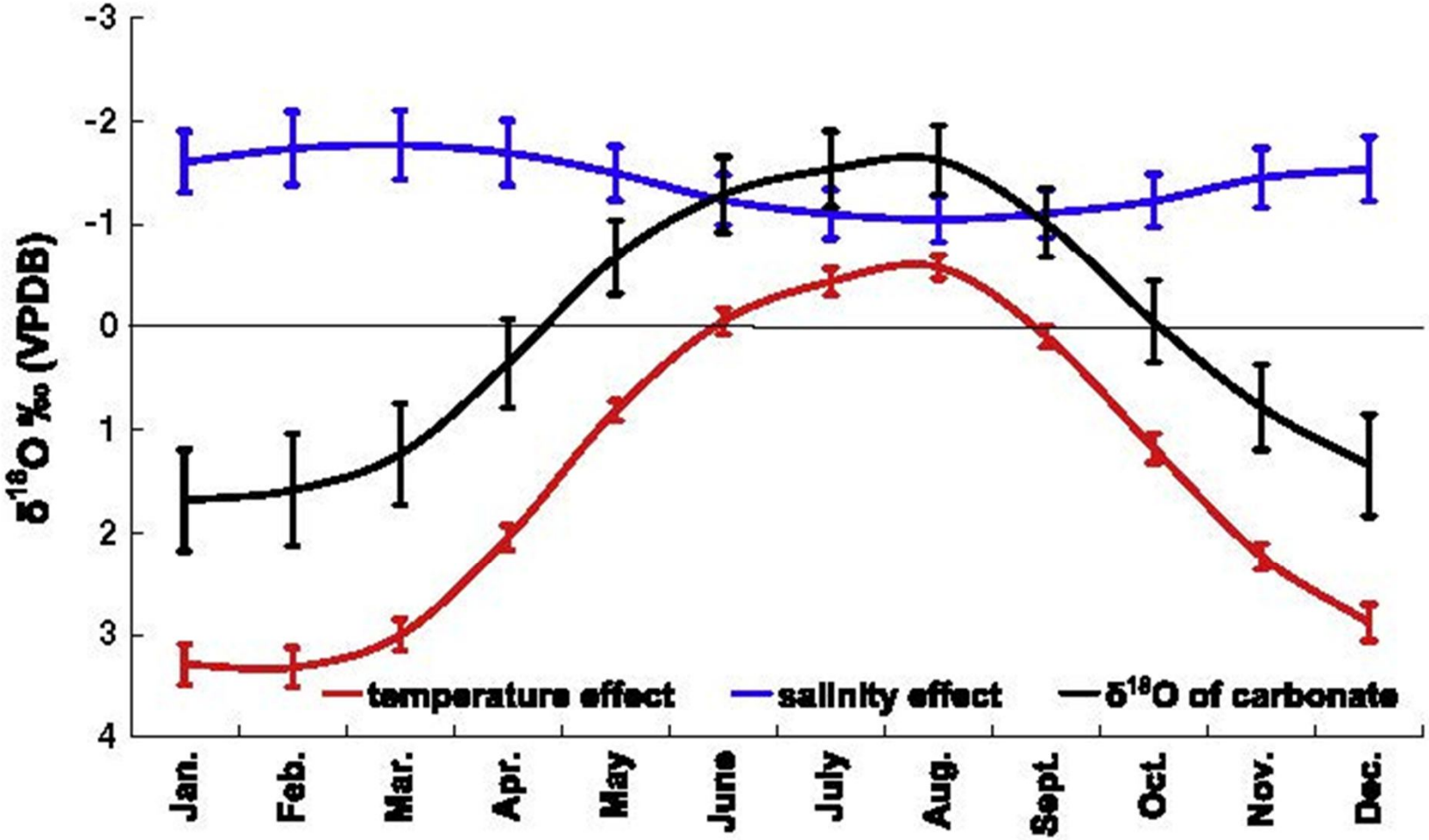
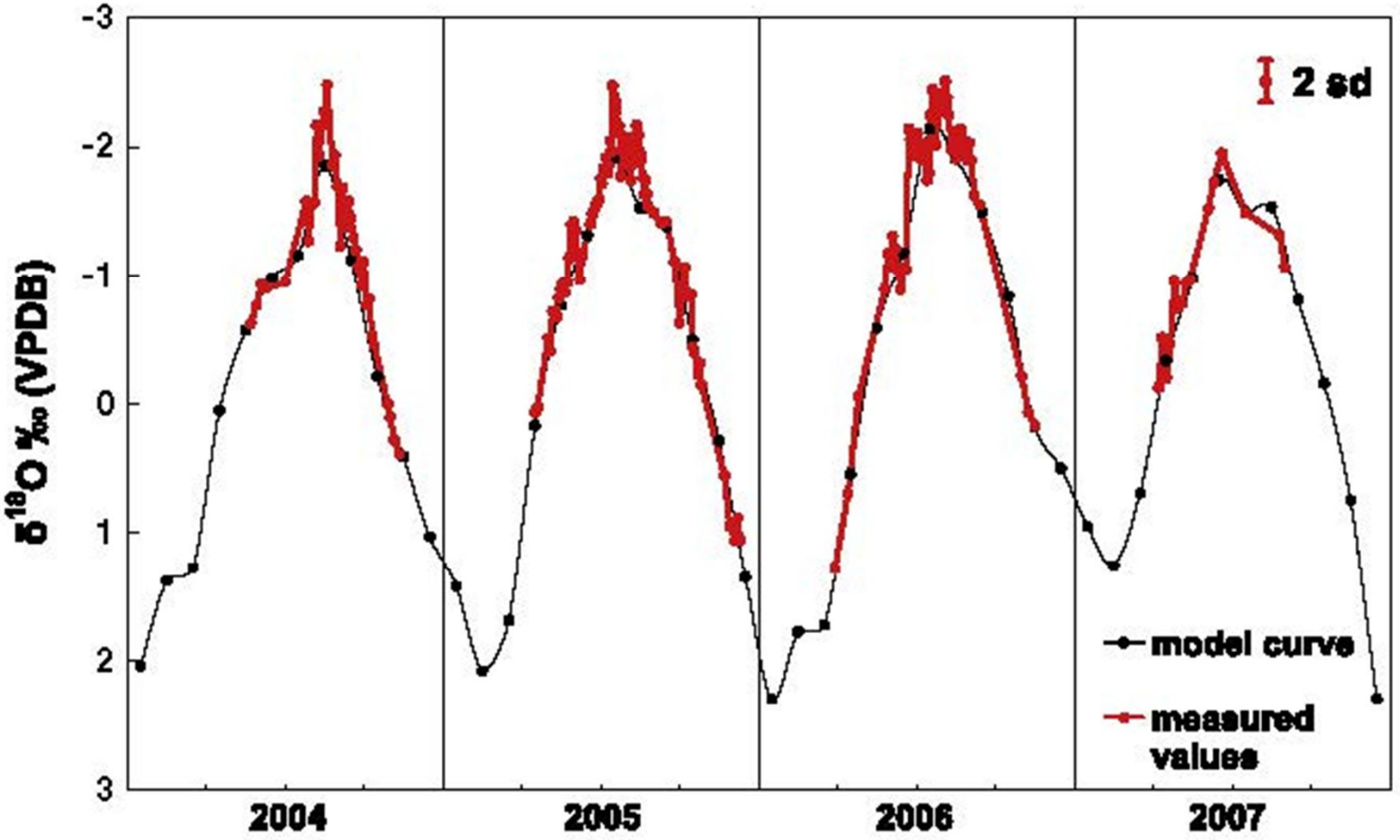


Fig. 1. Geographic map showing sample locality in the List Basin.


181
178
179
176
177
171
175
168
149
143
138
170
131
164
125
162
120
154
115
94
82
76
65
51
48
148
40
142
36
137
33
130
123
13a, b
119
11
9
7
109
5
3
2
102
98
84
80
71
56
49
47
39
35
14
12
10
8
6
4
2b
1, 1b
5 mm


-3.0
-2.5
-2.0
-1.5
-1.0
-0.5
0.0
0.5
1.0
1.5
δ18O ‰ (VPDB)
2σ
0
2
4
6
8
10
12
14
16
18
distance to oldest increment in mm

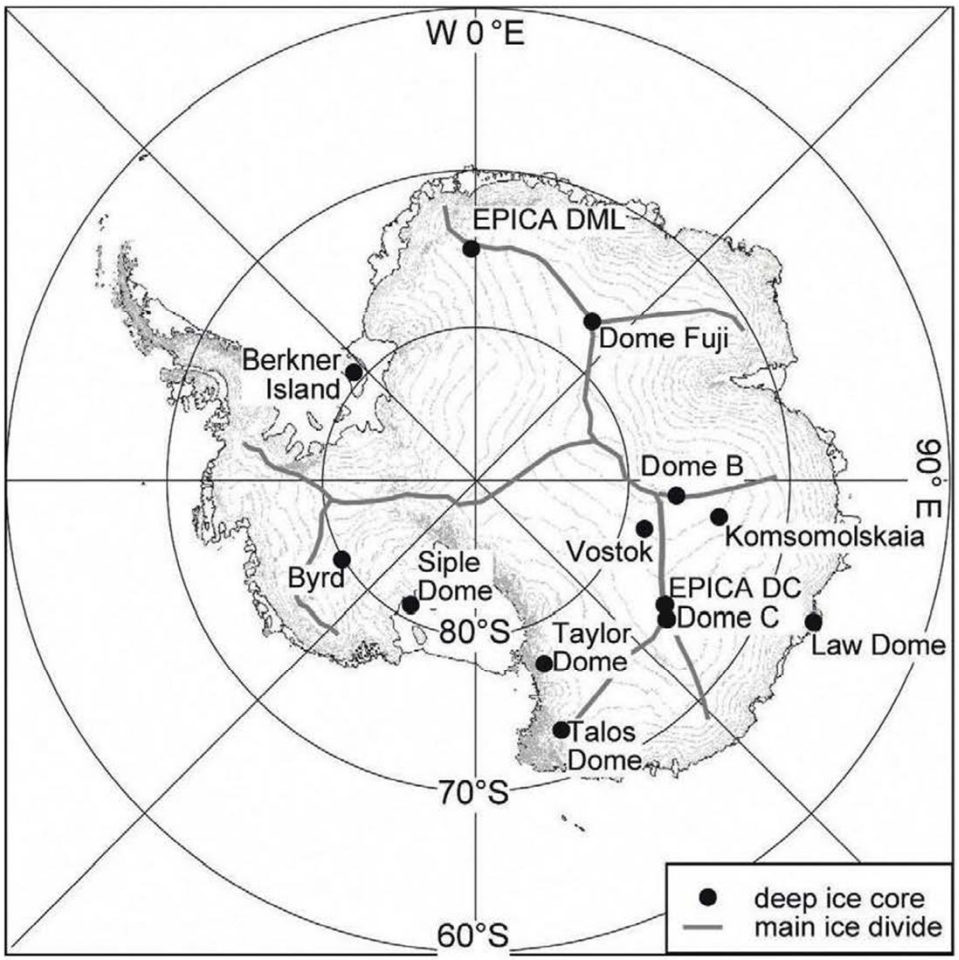

-3
-2
-1
0
1
2
3
4
δ18O ‰ (VPDB)
temperature effect
salinity effect
δ18O of carbonate
Jan.
Feb.
Mar.
Apr.
May
June
July
Aug.
Sept.
Oct.
Nov.
Dec.

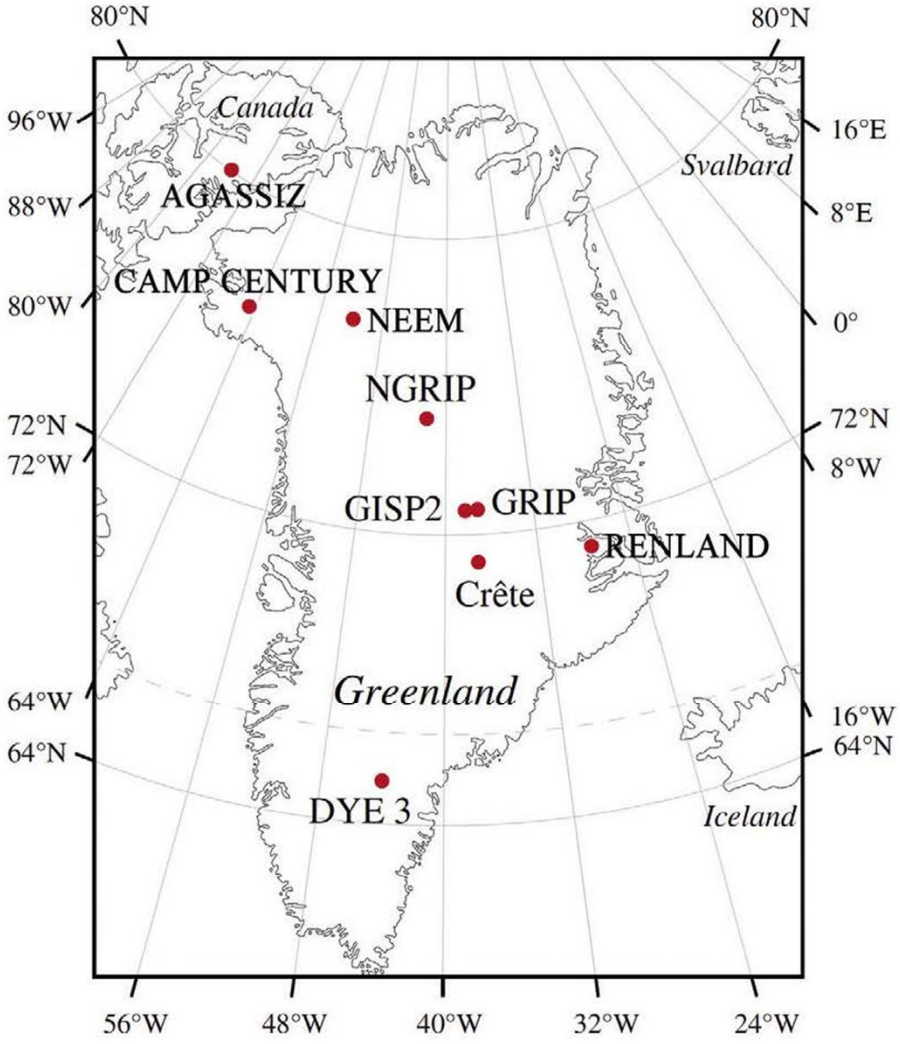

-3
-2
-1
0
1
2
3
δ18O ‰ (VPDB)
2004
2005
2006
2007
2 sd
model curve
measured values

TESTEMUNHOS DE GELO

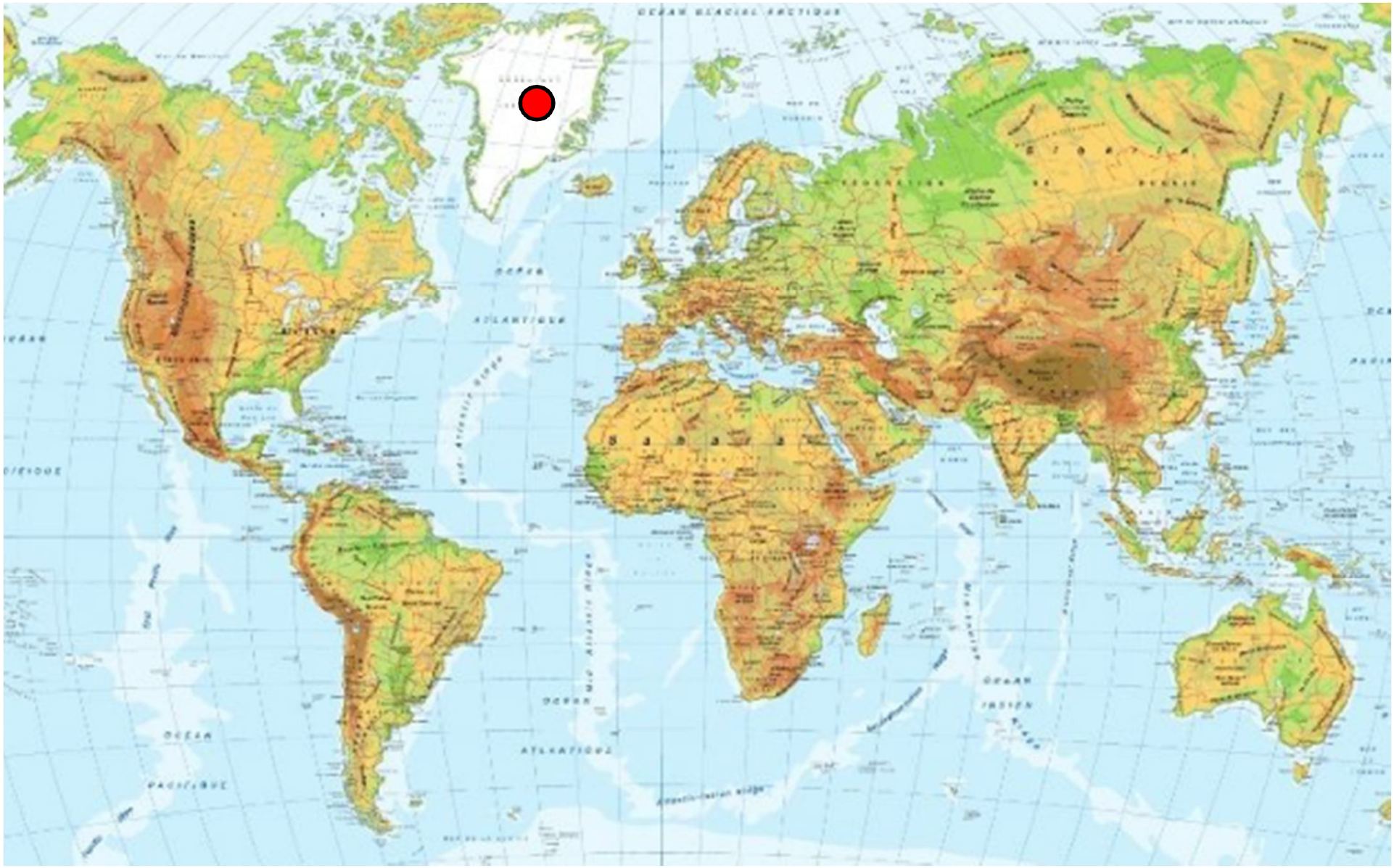
Earthobservatory.nasa.gov

ICE CORES


W 0 °E
90°E
EPICA DML
Dome Fuji
Berkner Island
Dome B
Komsomolskaia
Vostok
Byrd
Siple Dome
EPICA DC
Dome C
80°S
Taylor Dome
Law Dome
Talos Dome
70°S
60°S
deep ice core
main ice divide

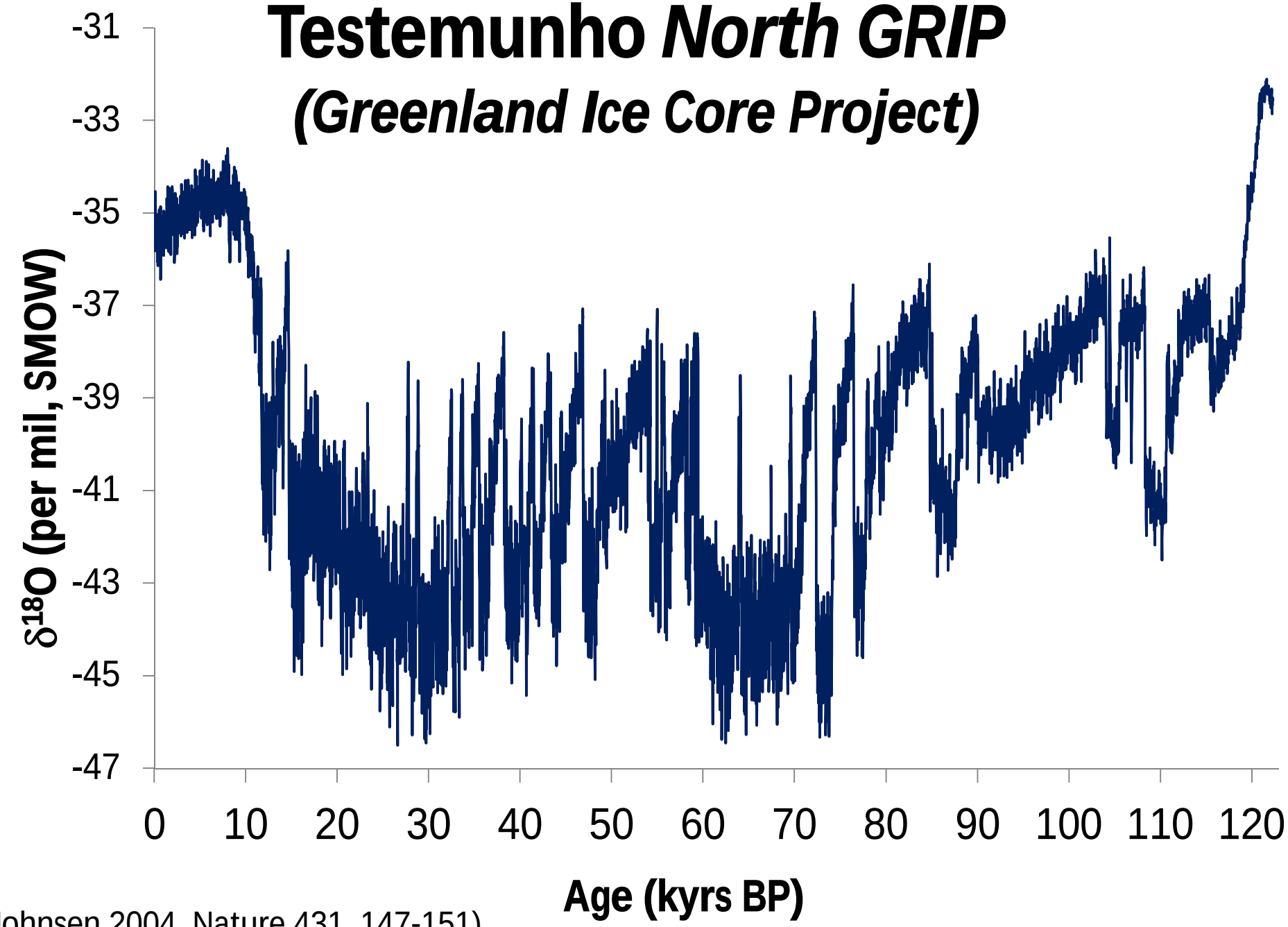

80°N
80°N
96°W
16°E
Canada
Svalbard
AGASSIZ
88°W
8°E
CAMP CENTURY
80°W
NEEM
0°
NGRIP
72°N
72°N
72°W
8°W
GISP2
GRIP
RENLAND
Crête
64°W
Greenland
16°W
64°N
64°N
DYE 3
Iceland
56°W
48°W
40°W
32°W
24°W

Testemunhos da Groenlândia



Testemunho *North GRIP*

(Greenland Ice Core Project)

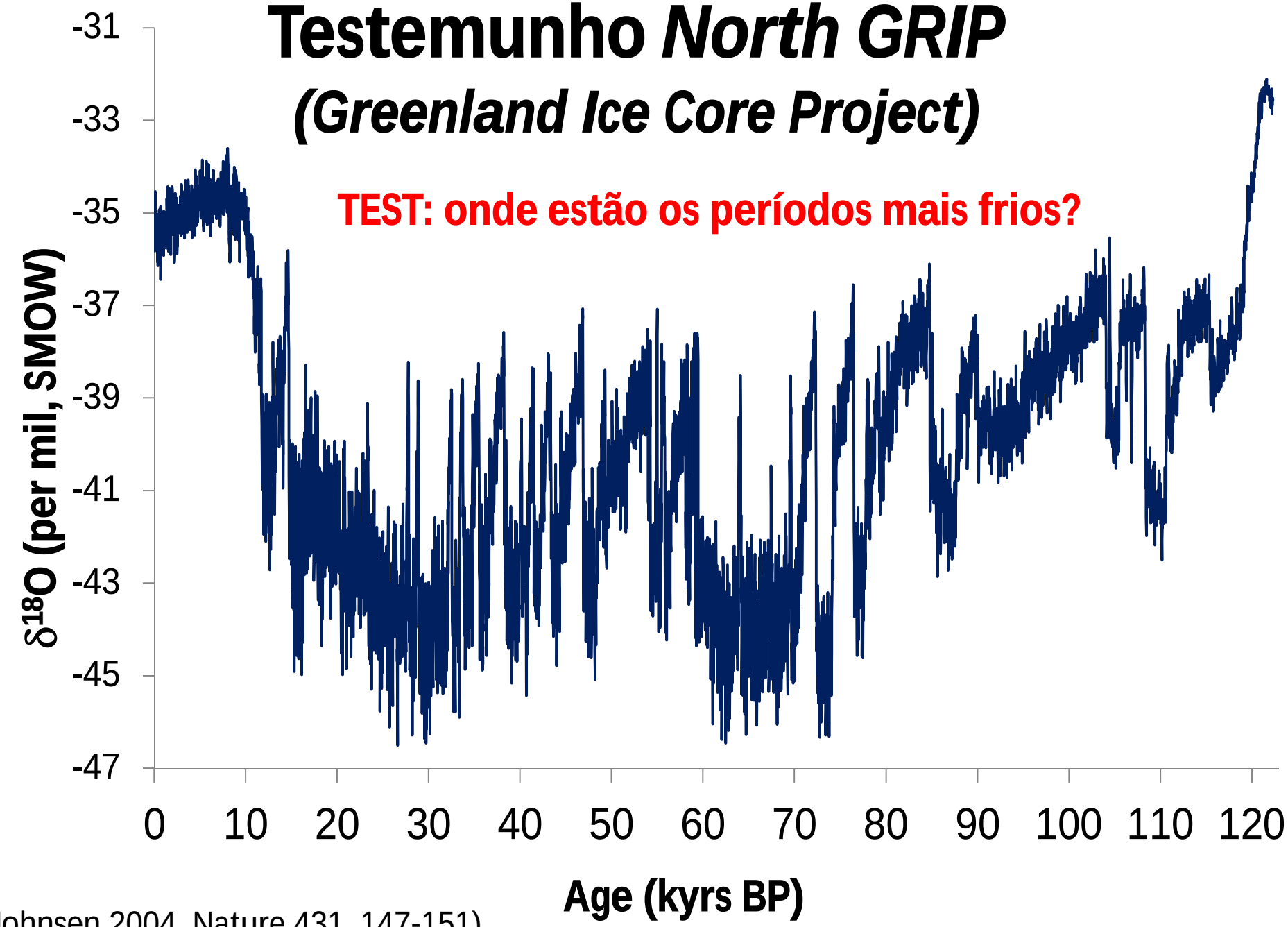


(Johnsen 2004, Nature 431, 147-151)

Testemunho *North GRIP*

(Greenland Ice Core Project)

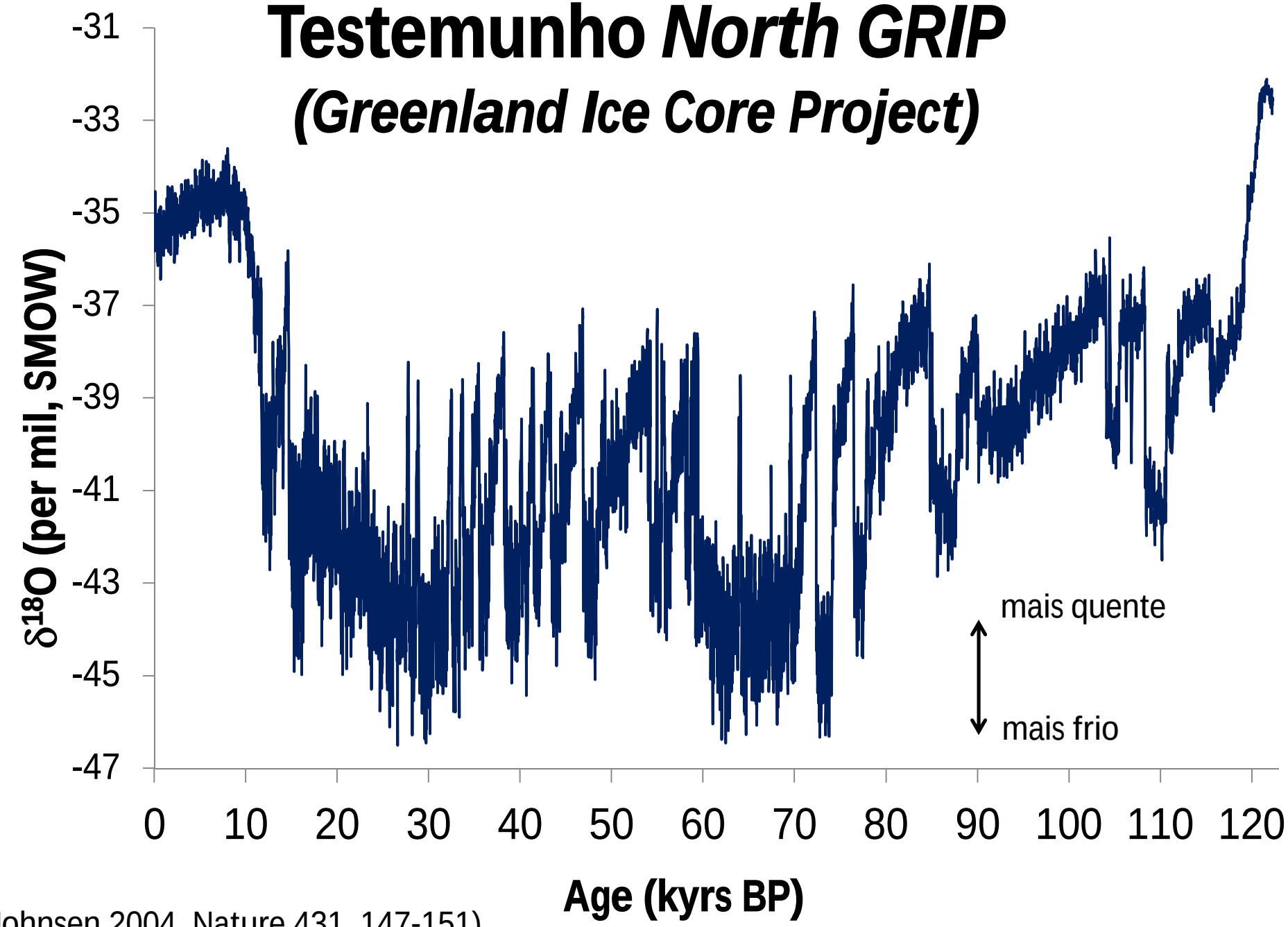
TEST: onde estão os períodos mais frios?



(Johnsen 2004, Nature 431, 147-151)

Testemunho *North GRIP*

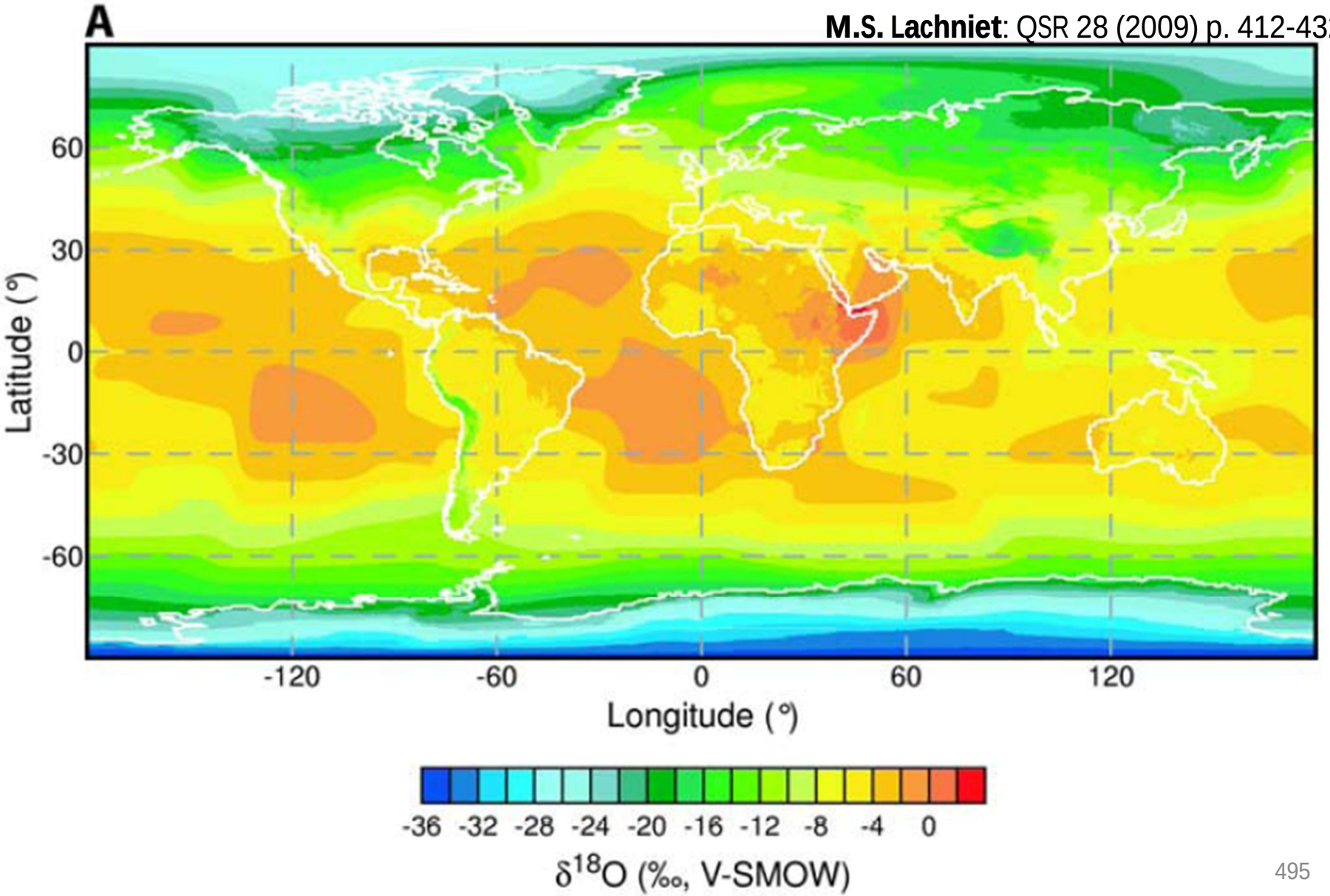
(Greenland Ice Core Project)

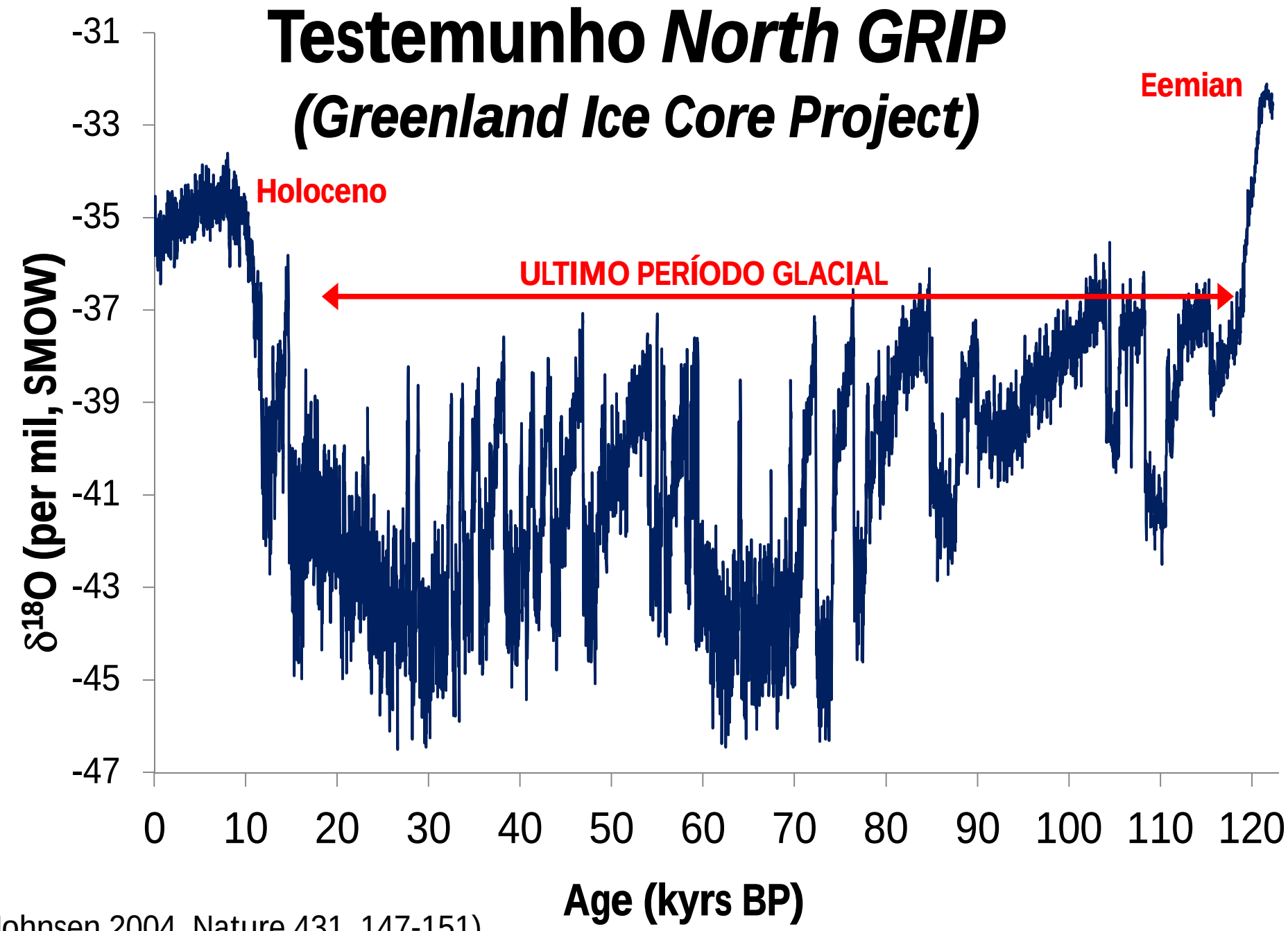


(Johnsen 2004, Nature 431, 147-151)

Mapa global da composição isotópica de chuva e neve

M.S. Lachniet: QSR 28 (2009) p. 412-432.



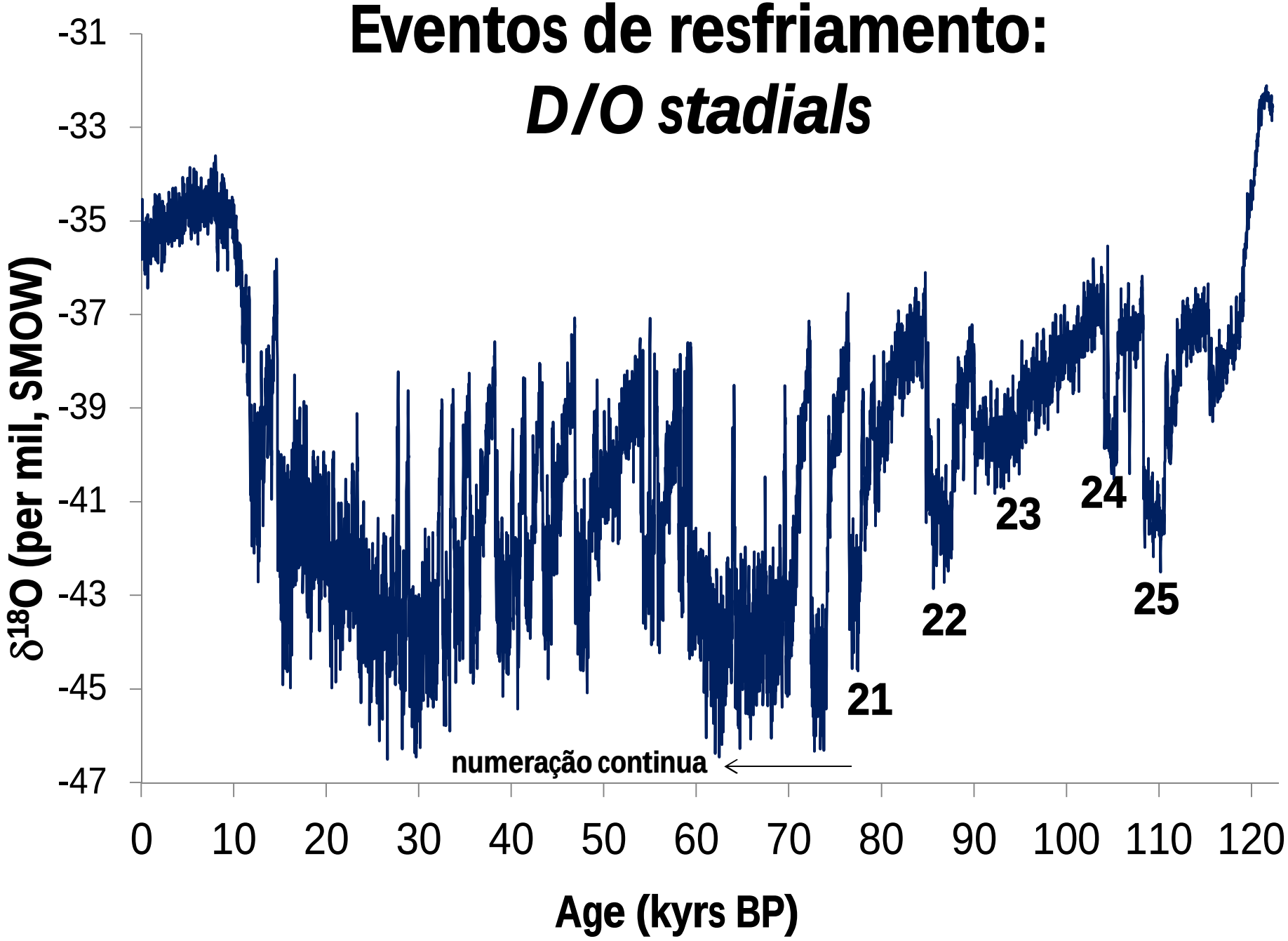


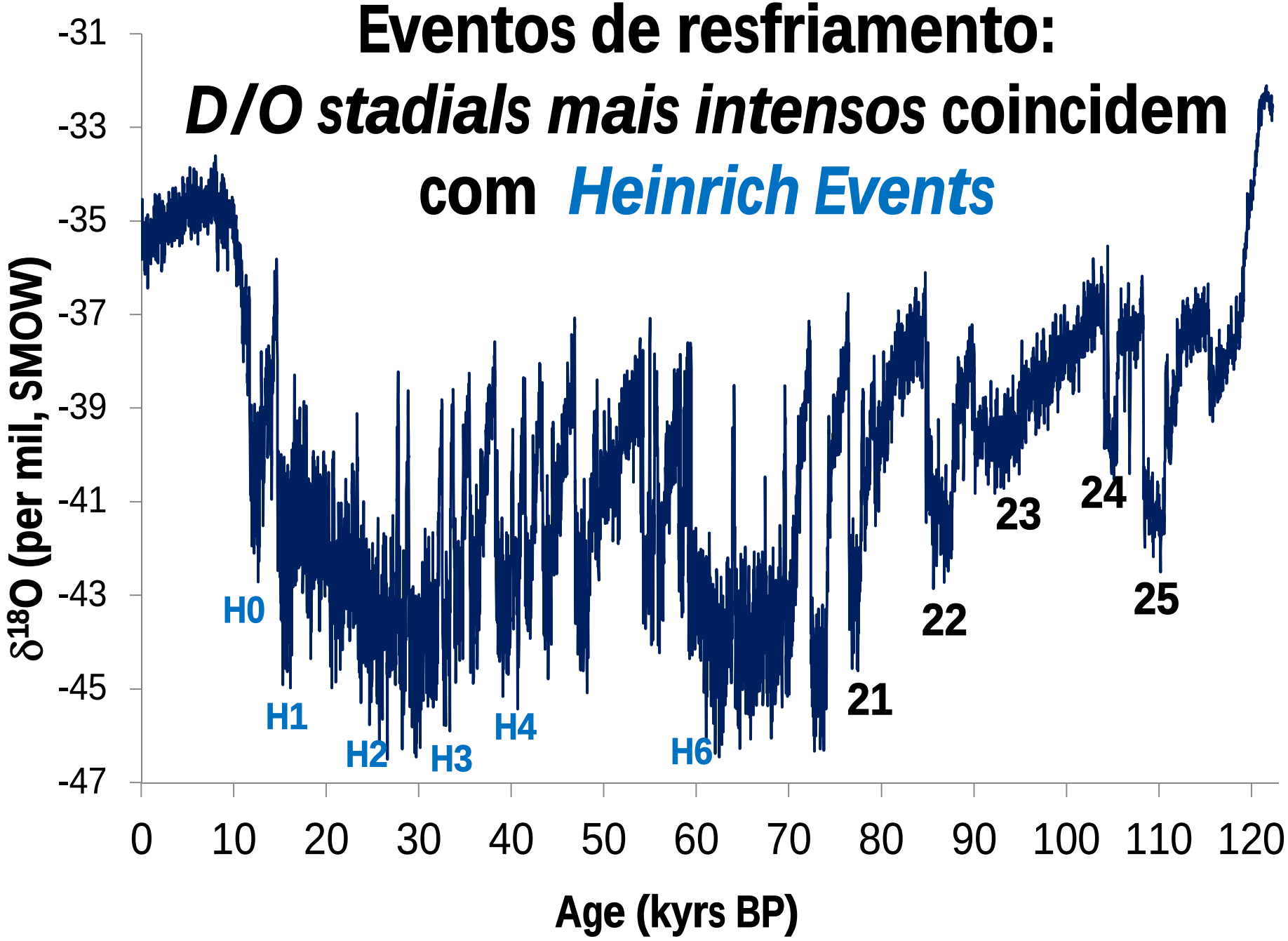
(Johnsen 2004, Nature 431, 147-151)

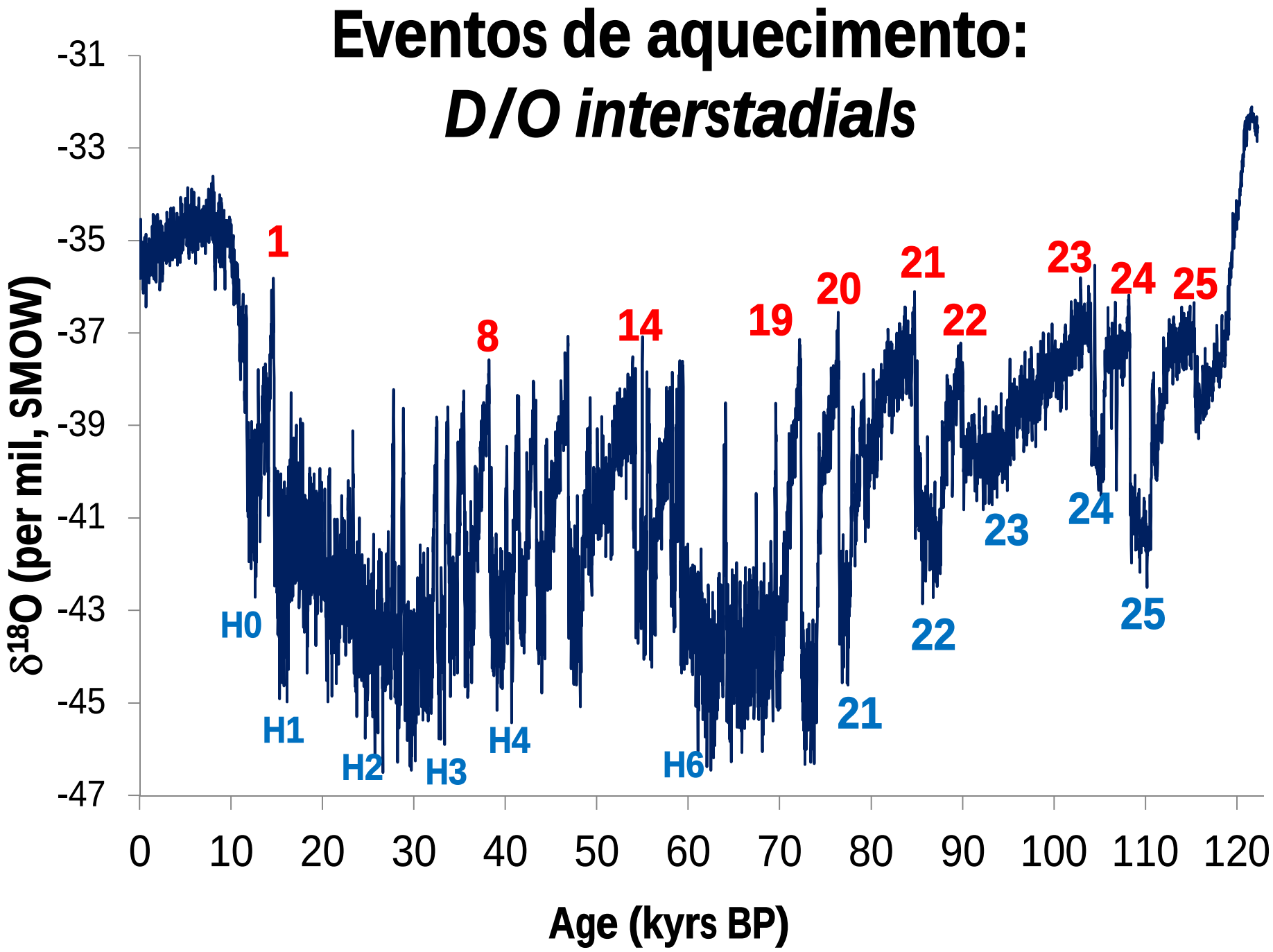
EVENTOS CLIMÁTICOS DURANTE O ULTIMO CICLO GLACIAL

Eventos Dansgaard – Oeschger

(Eventos D/O)


Eventos de resfriamento:
D/O stadials
-31
-33
-35
-37
-39
-41
-43
-45
-47
δ18O (per mil, SMOW)
0
10
20
30
40
50
60
70
80
90
100
110
120
Age (kyrs BP)
21
22
23
24
25
numeração continua


Eventos de resfriamento:
D/O stadials mais intensos coincidem
com Heinrich Events
-31
-33
-35
-37
-39
-41
-43
-45
-47
δ18O (per mil, SMOW)
H0
H1
H2
H3
H4
H6
21
22
23
24
25
0
10
20
30
40
50
60
70
80
90
100
110
120
Age (kyrs BP)


Eventos de aquecimento:
D/O interstadials
-31
-33
-35
-37
-39
-41
-43
-45
-47
δ18O (per mil, SMOW)
0
10
20
30
40
50
60
70
80
90
100
110
120
Age (kyrs BP)
1
8
14
19
20
21
22
23
24
25
H0
H1
H2
H3
H4
H6
21
22
23
24
25

Eventos Dansgaard – Oeschger:

- Oscilações climáticas na escala milenar (período de 1470 a.) durante o ultimo ciclo glacial (os últimos 120 mil anos)
- Identificados nos testemunhos de gelo da Groenlândia
- Eventos de resfriamento: D/O stadials
- Eventos de aquecimento: D/O interstadials
- Alguns D/O stadials coincidem com os Heinrich Events do Atlântico Norte

Heinrich Events

Sedimento lítico: *Ice Rafted Debris* (IRD)

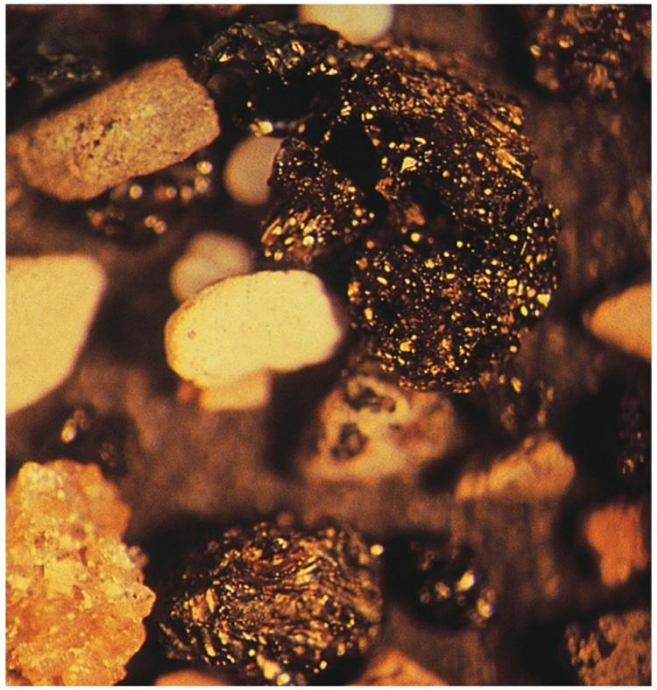


Figure 14-4a
Earth's Climate: Past and Future, Second Edition
© 2008 W. H. Freeman and Company

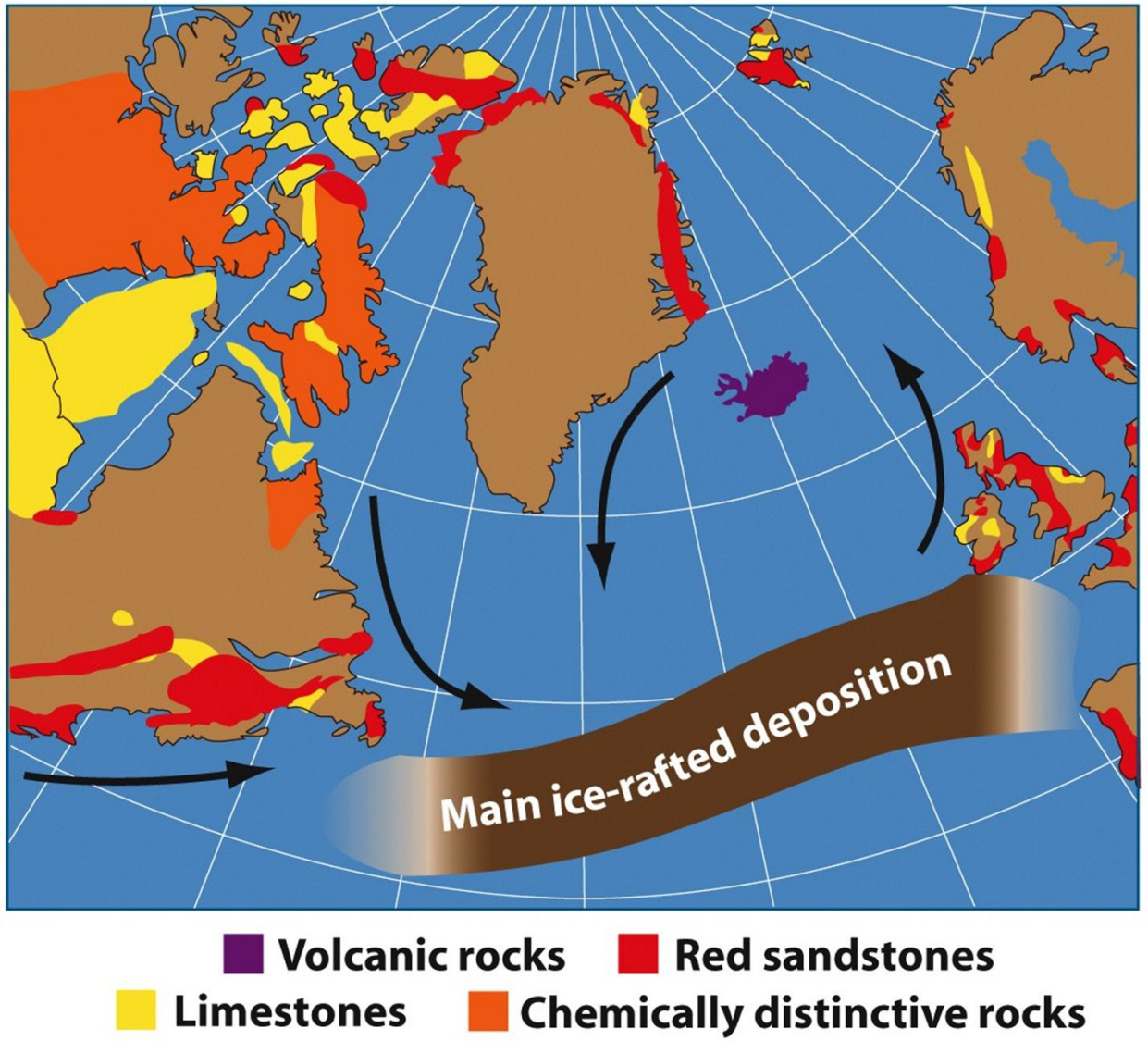


Figure 14-3
Earth's Climate: Past and Future, Second Edition
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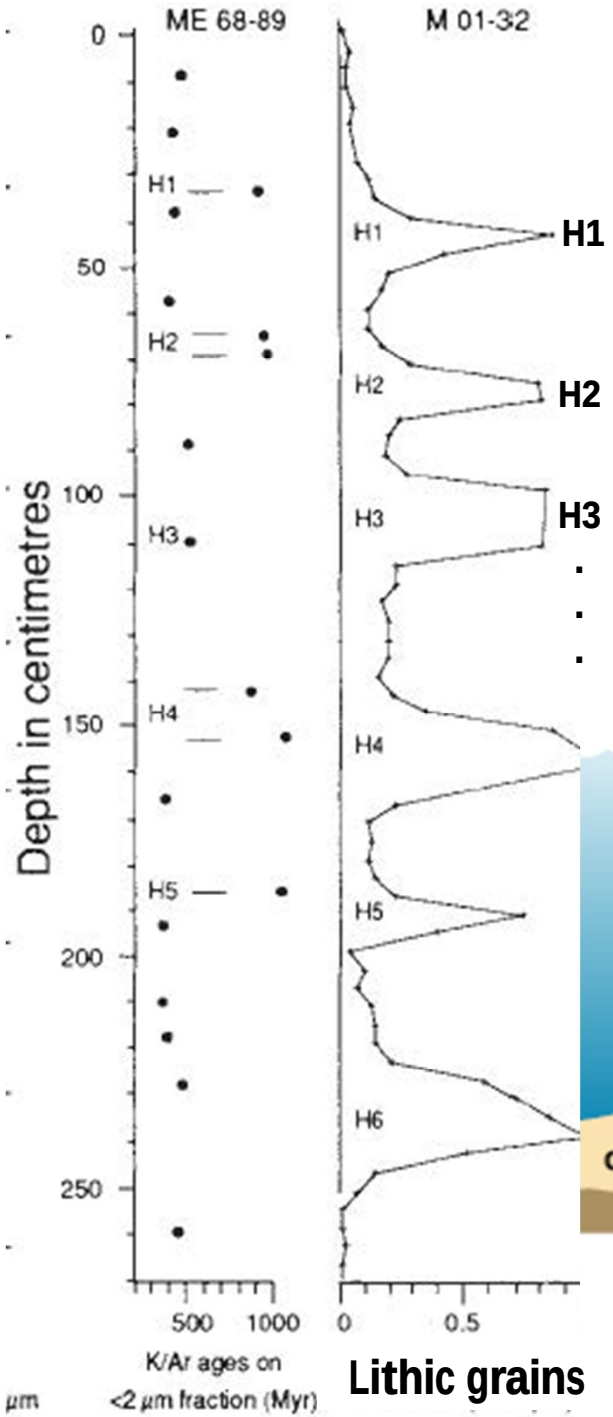
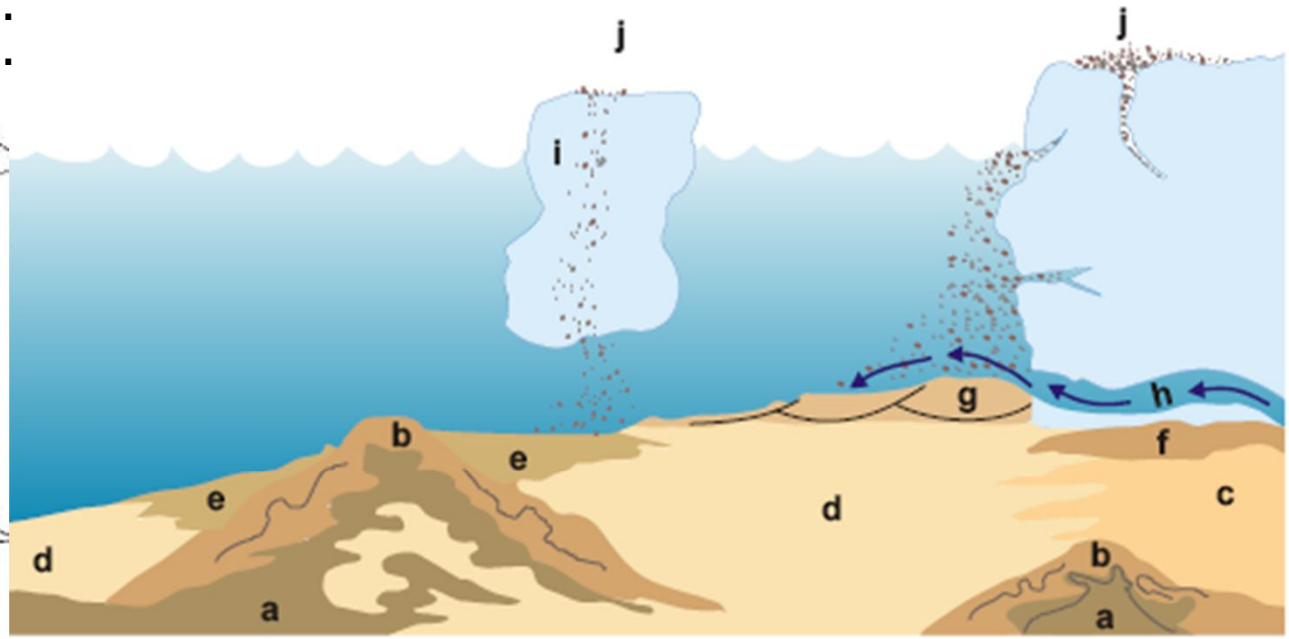
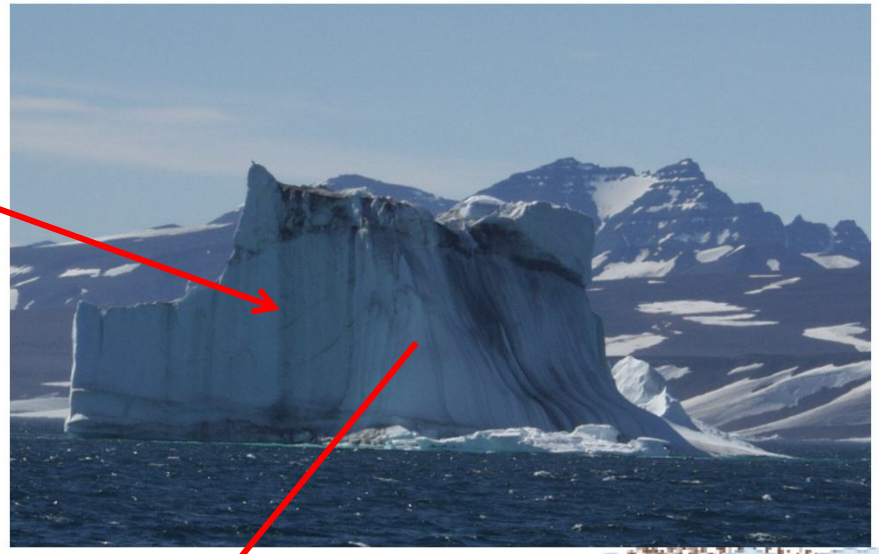
Deposition of ***Ice Rafted Debris*** **(IRD)** during **Heinrich Events** in the North Atlantic

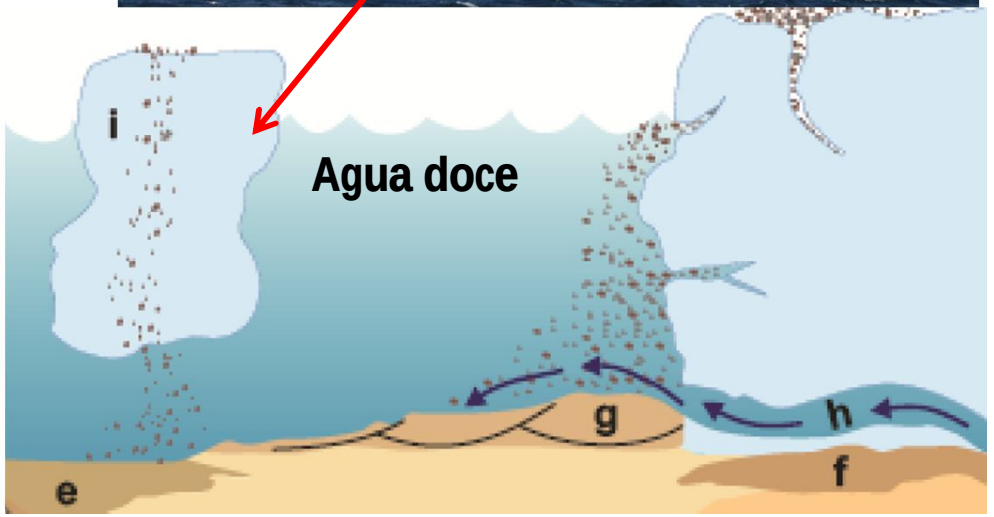




O que acontece durante os Eventos Heinrich:

www.eos.ubc.ca



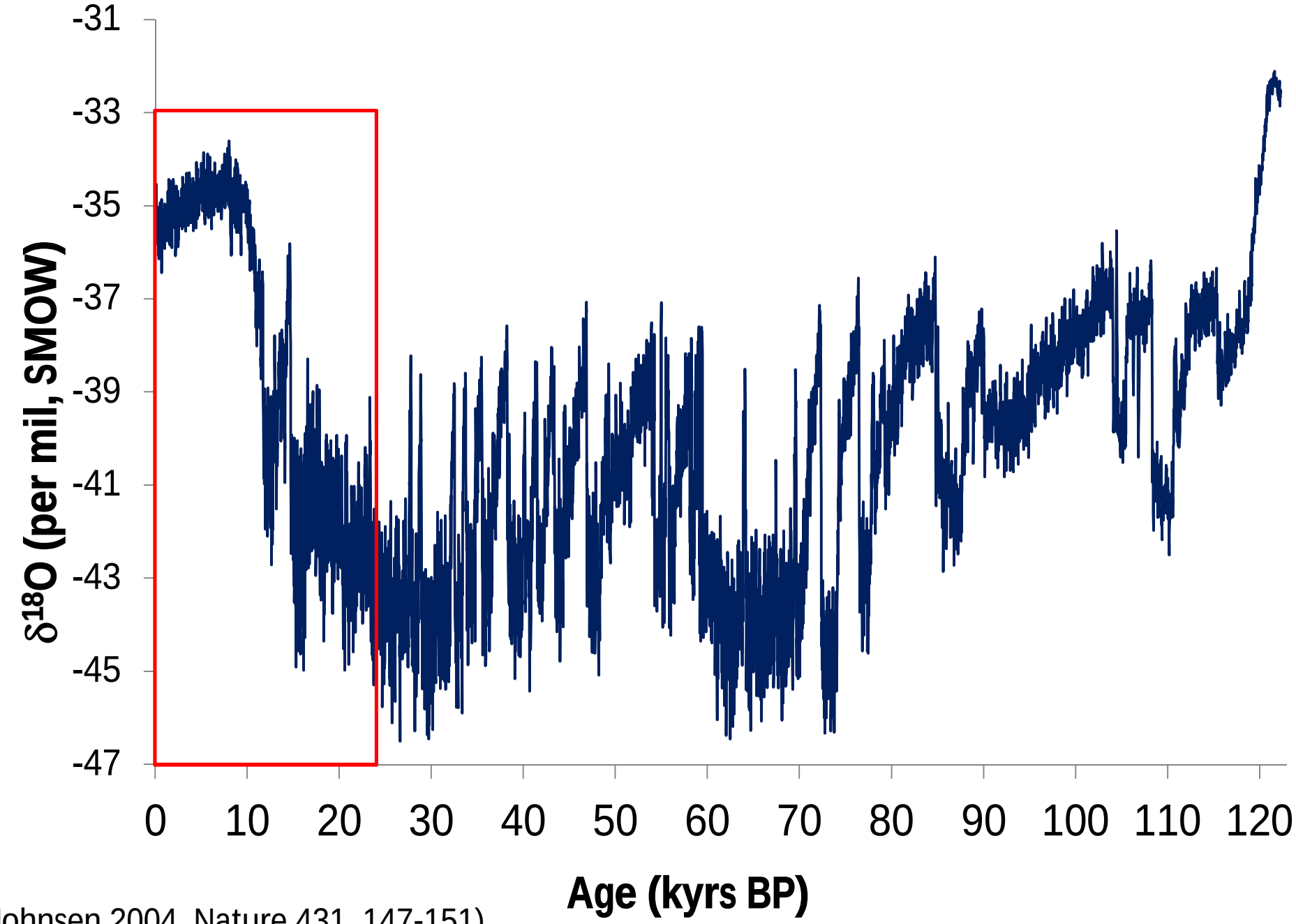


Eventos Heinrich:

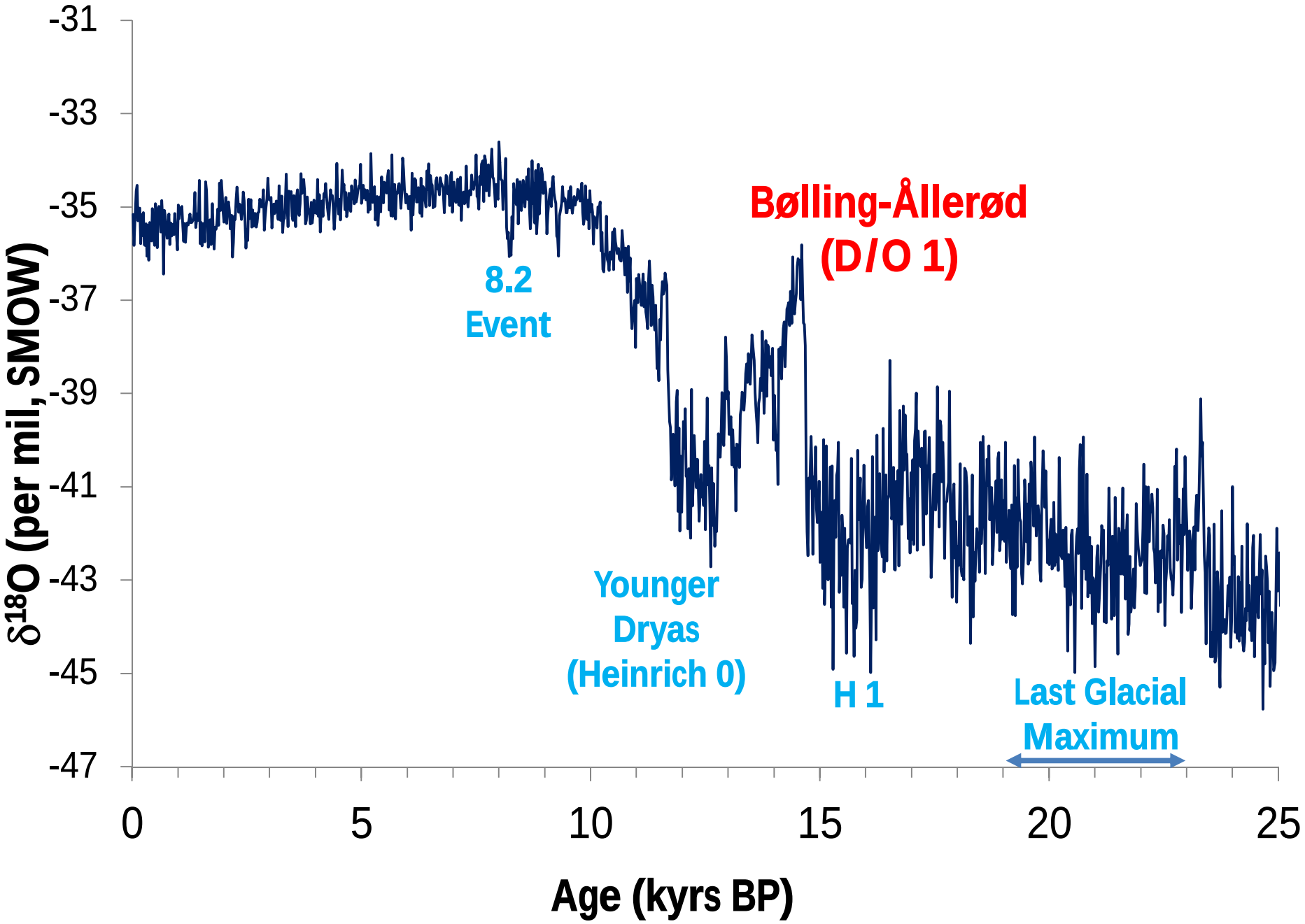
- Mudanças climáticas na escala milenar (período ~10 mil a.) durante o ultimo ciclo glacial (os últimos 120 mil anos)
- Identificados nos testemunhos de sedim. do Atlântico Norte
- O termo denota só eventos de resfriamento
- Alguns D/O stadials coincidem com alguns Heinrich Events

Fim do ultimo período glacial

Termination I



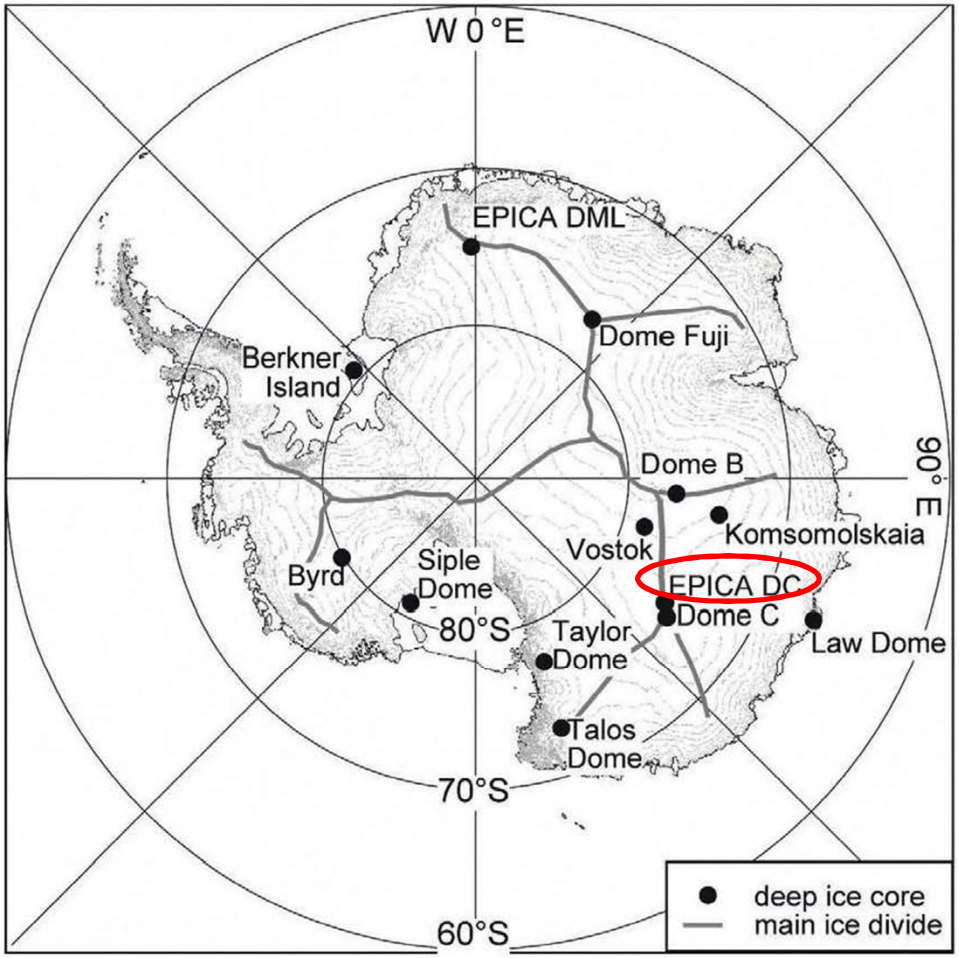
(Johnsen 2004, Nature 431, 147-151)

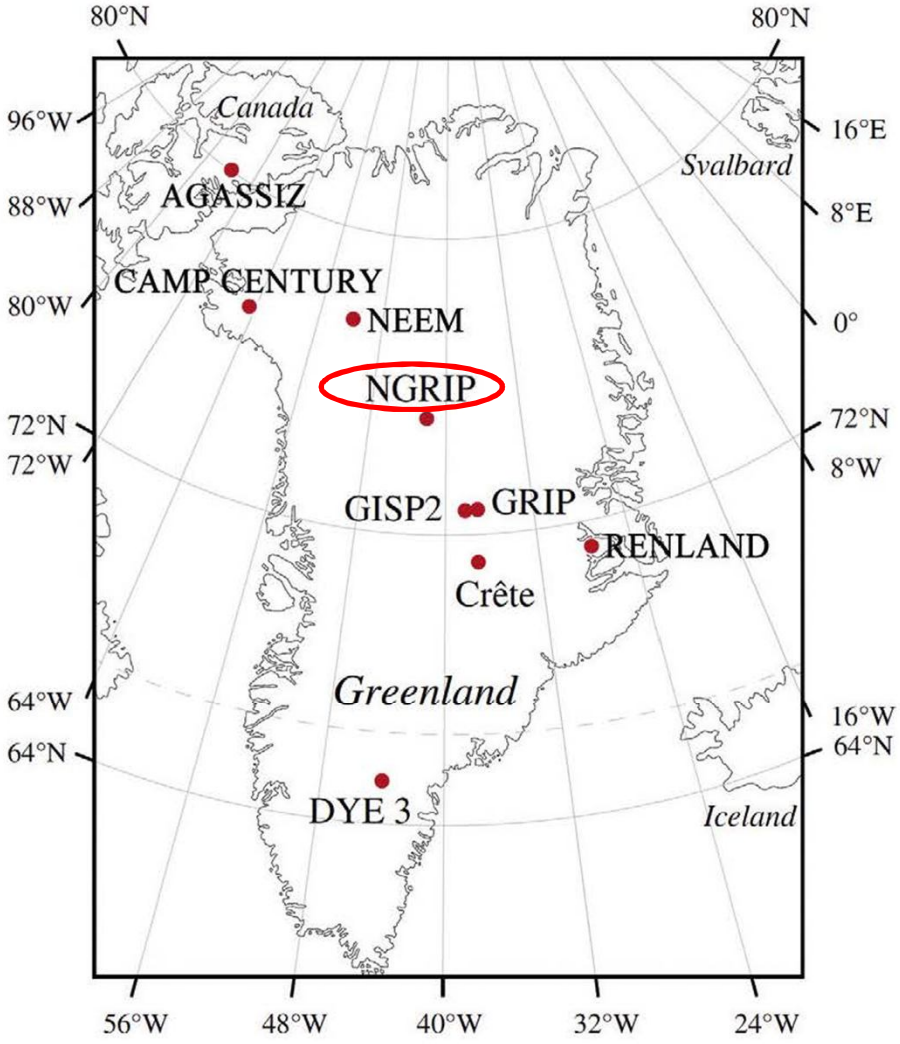

-31
-33
-35
-37
-39
-41
-43
-45
-47
δ18O (per mil, SMOW)
0
5
10
15
20
25
Age (kyrs BP)
8.2
Event
Younger
Dryas
(Heinrich 0)
Bølling-Ållerød
(D/O 1)
H 1
Last Glacial
Maximum

Comparação entre testemunhos de gelo da Groenlândia e da Antártica

BIPOLAR SEESAW

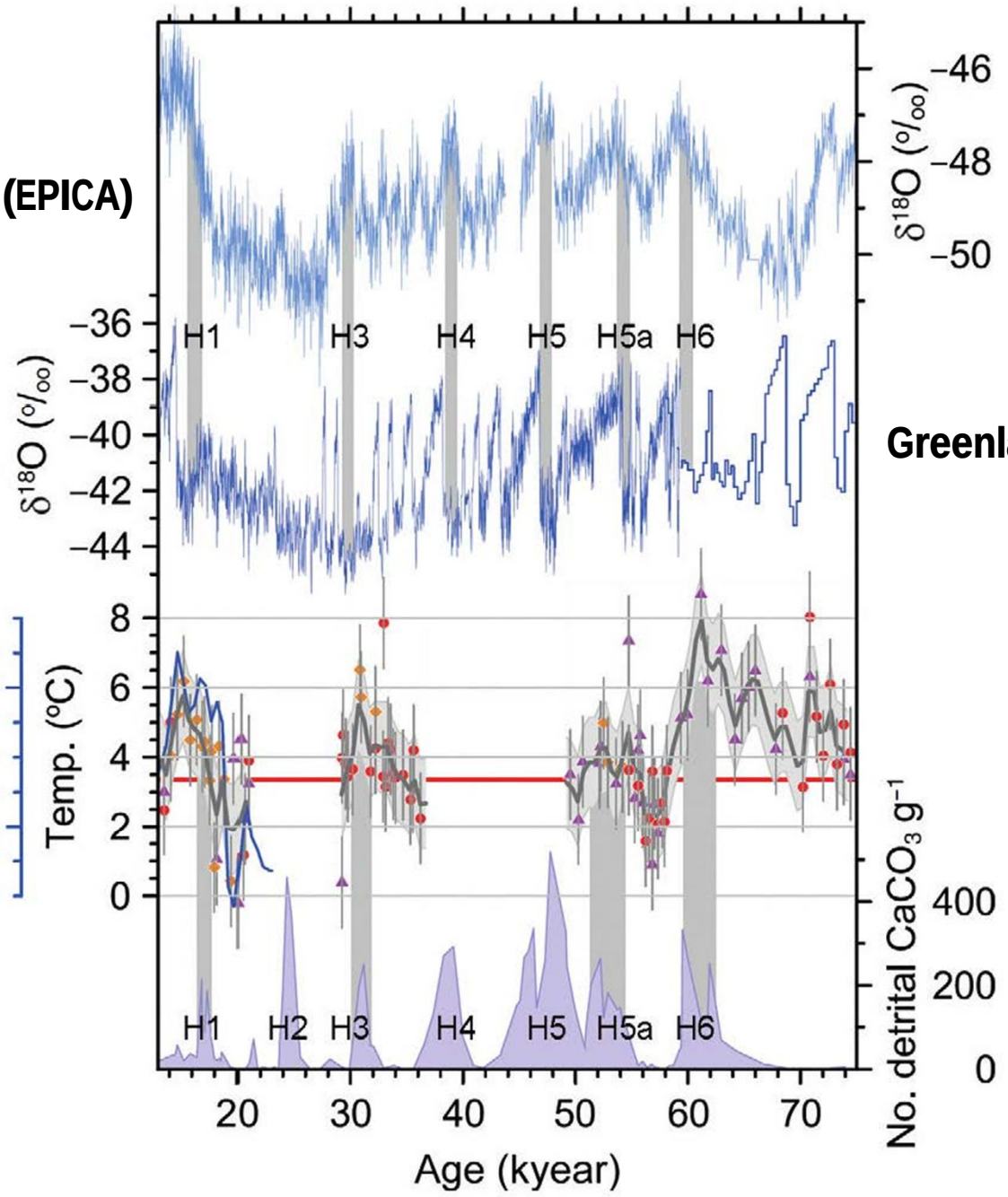
ICE CORES


W 0 °E
90°E
EPICA DML
Dome Fuji
Berkner Island
Dome B
Komsomolskaia
Vostok
Byrd
Siple Dome
EPICA DC
Dome C
80°S
Taylor Dome
Law Dome
Talos Dome
70°S
60°S
deep ice core
main ice divide


80°N
80°N
96°W
16°E
Canada
Svalbard
AGASSIZ
88°W
8°E
CAMP CENTURY
80°W
NEEM
0°
NGRIP
72°N
72°N
72°W
8°W
GISP2
GRIP
RENLAND
Crête
64°W
Greenland
16°W
64°N
64°N
DYE 3
Iceland
56°W
48°W
40°W
32°W
24°W

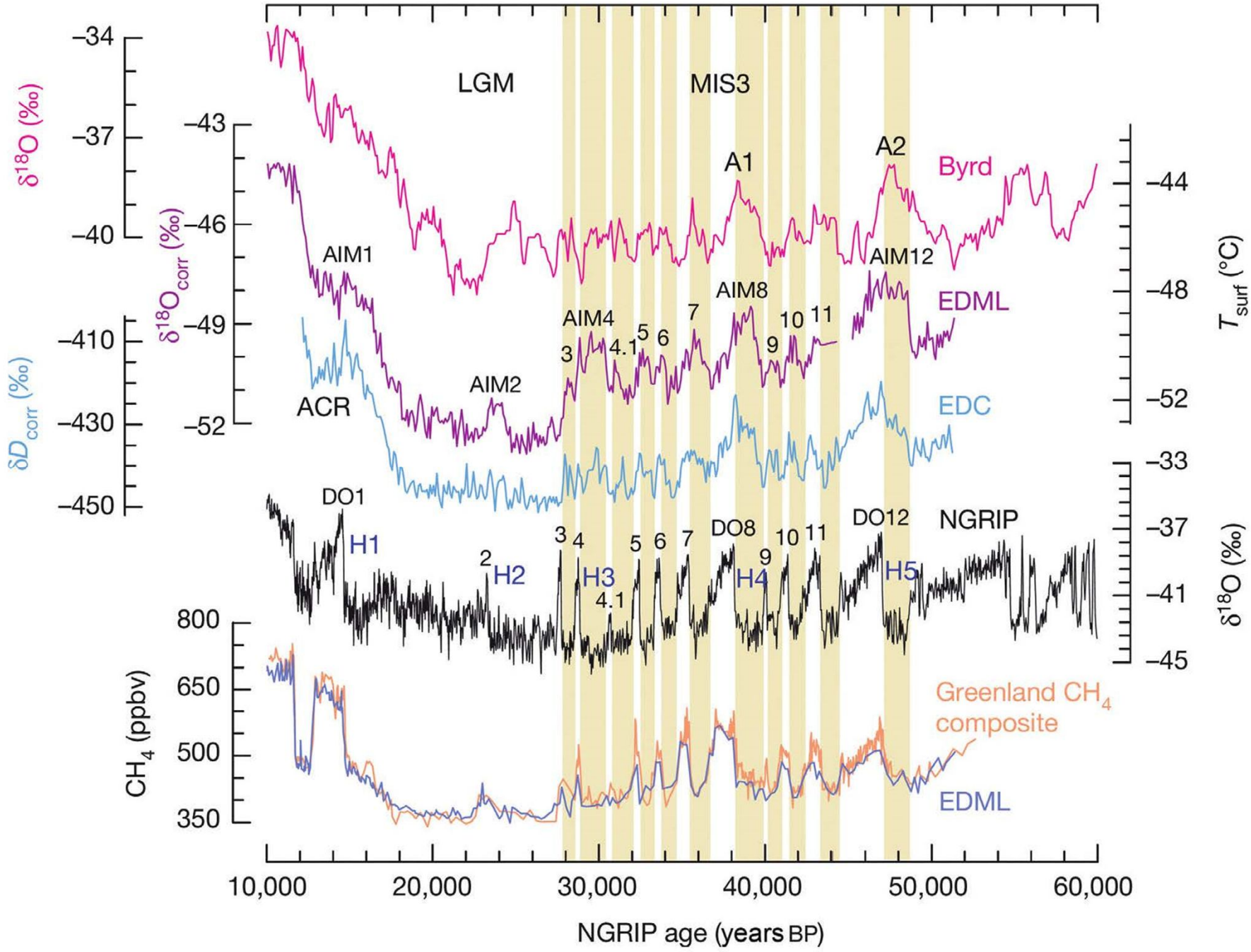
BIPOLAR SEESAW

Antarctica (EPICA)



Greenland (NGRIP)

BIPOLAR SEESAW



Como se explica o *Bipolar Seesaw*

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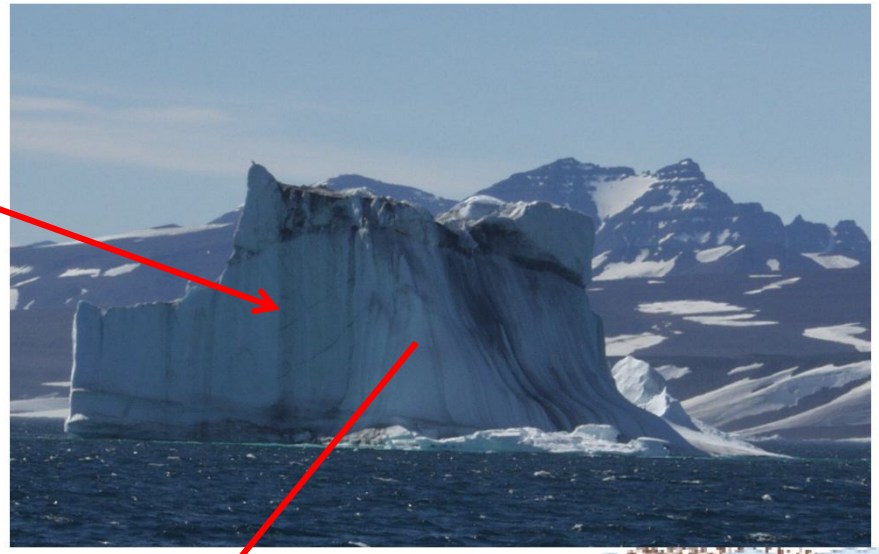
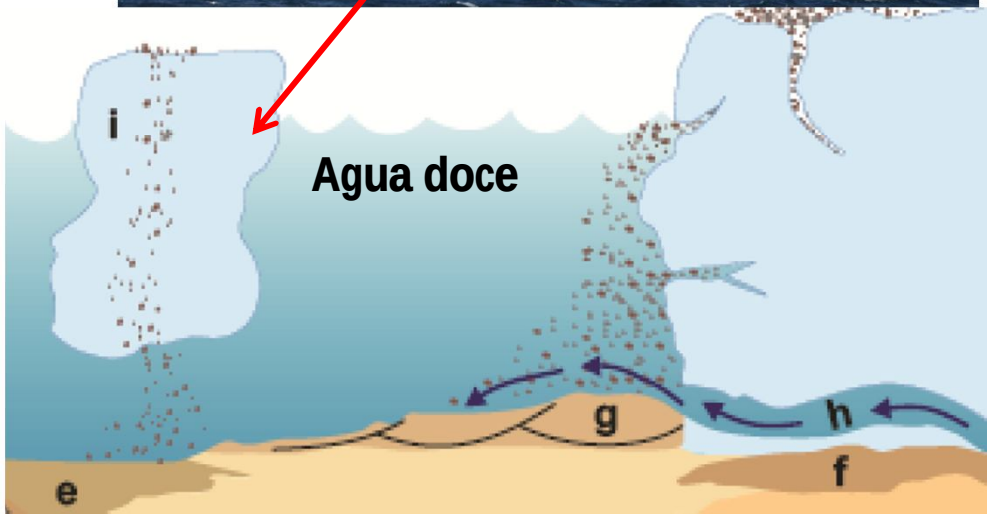
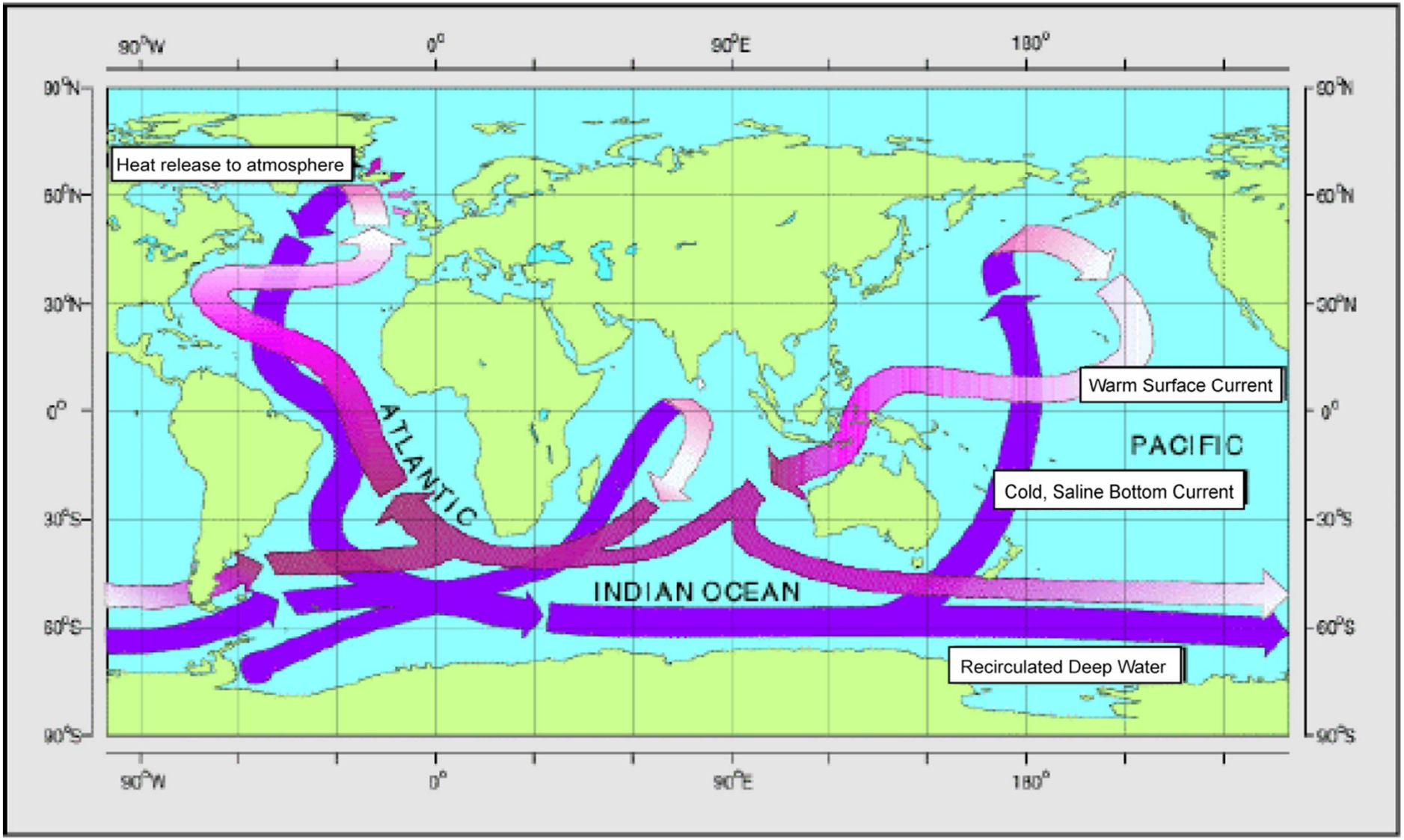


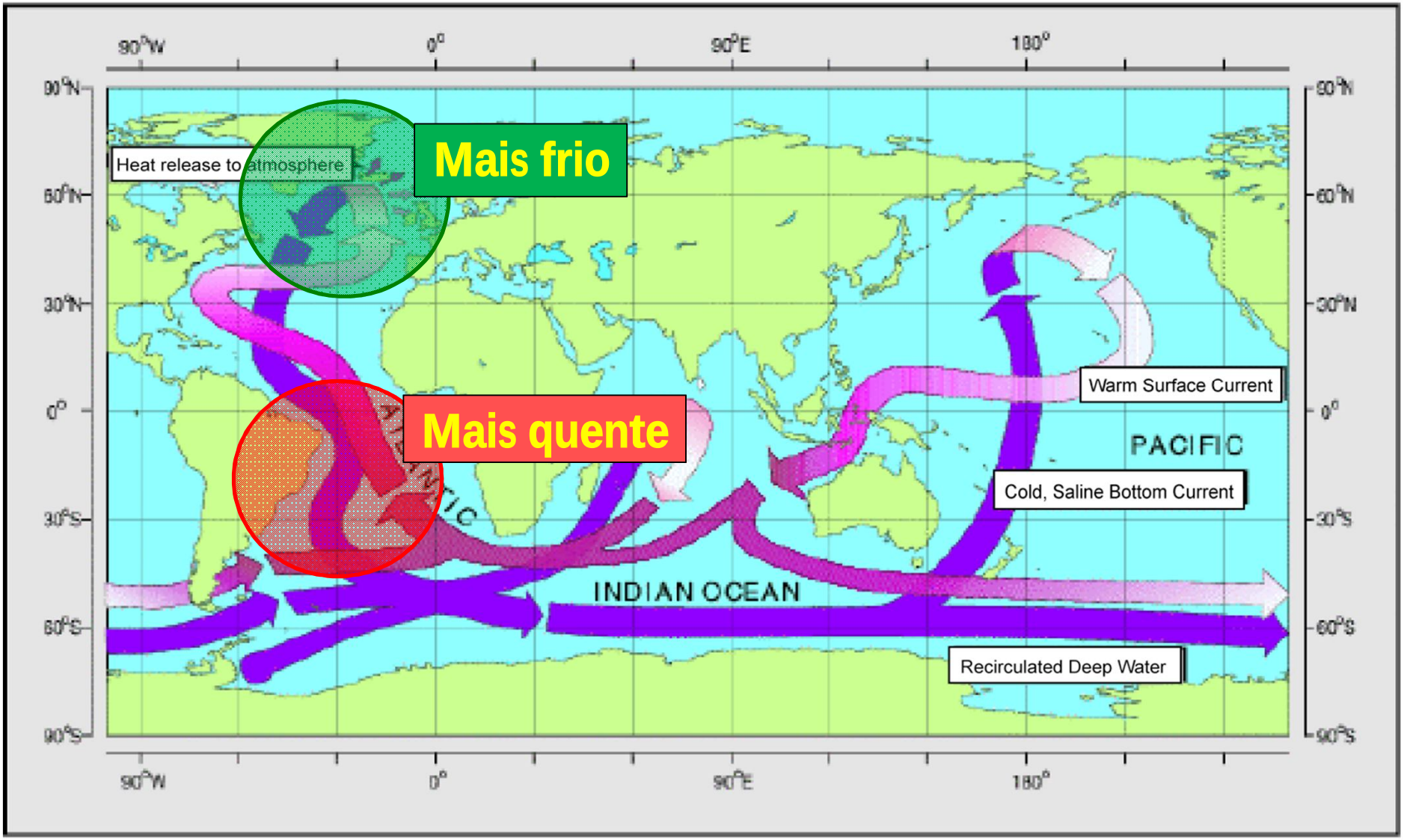


A Circulação Oceânica Hoje



after W. Broecker, modified by E. Maier-Reimer

Durante os eventos Heinrich...



after W. Broecker, modified by E. Maier-Reimer

CONCLUSÃO GERAL

Ao resfriamento nas altas latitudes

do **Hemisfério Norte**

durante os *Heinrich Events* e *D/O stadials*

corresponde

aquecimento nas altas latitudes

do **Hemisfério Sul (e vice-versa)**

graça ao mecanismo do ***Bipolar Seesaw***